

DULAC INDIVIDUAL FINITENESS BY A COMPLETE ÉCALLE ROUTE

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ABSTRACT. We prove that every real planar polynomial vector field has only finitely many isolated periodic orbits. The proof follows Écalle’s formalization and actualization architecture by reconstructing complete physical return maps, including divergent normalizers and compatible nonreal transition data, and controlling the entire remainder after all cancellations. The main result of this paper was obtained using generative AI, particularly gpt-6-astra and the Danus system.

CONTENTS

1. Introduction	1
2. Notation and physical return maps	4
3. Global reduction and resolved limits	4
4. Analytic families and original source norms	17
5. Prepared ordinary saddle passages	33
6. Canonical equations and physical matching	47
7. Coverage of actual elementary itineraries	64
8. Root sources and ordered comparisons	69
9. Initial source spaces and the first comparison level	92
10. Retained source forests	125
11. Mixed sources and extensions from an arbitrary current	159
12. Literal finite words and complete actualization	200
13. Closed formal comparison of the physical return	232
14. Finiteness of isolated periodic orbits	240
Appendix A. Details of the use of generative AI	241
References	245

1. INTRODUCTION

Dulac finiteness. The finiteness problem for a fixed polynomial vector field asks whether its isolated periodic orbits can form an infinite collection. This is the individual finiteness theorem associated with Dulac’s problem. Écalle’s monograph develops a constructive approach through formalization and analytic reconstruction of local return maps [Eca92, Avant-propos, pp. 3–4]. Ilyashenko’s account distinguishes individual finiteness from a bound depending only on the polynomial degree and describes a proof of the former [Ily93, Abstract and Chapter 1, Section 1.1, pp. 279–280]. We give a proof of individual finiteness following the formalization and actualization architecture.

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A periodic orbit means the image of a nonconstant periodic integral curve. It is isolated if some open neighborhood contains no other periodic orbit. The theorem proved in Section 14 is the following.

Theorem 1.1. *Every real planar polynomial vector field has only finitely many isolated periodic orbits.*

The field is fixed throughout the argument. The conclusion does not give a uniform bound in its degree. The local construction treats the complete physical return, including repeated visits to a singularity, divergent normalizing series, and compatible nonreal sectorial transition data. The precise itinerary data and the meaning of a complete leading term are given in Definitions 2.2 and 2.3.

Complete comparison and actualization. The analytic part of the proof compares the entire displacement after composition. It must distinguish an identity return from a nonzero germ even when earlier terms cancel. For each fixed physical itinerary, Theorem 13.1 constructs its formal displacement and identifies its actual value. A zero formal displacement gives the identity. A nonzero displacement has a nonzero real coefficient times a positive comparison scale, with a relative remainder tending to zero for the complete germ. Thus the return has one strict displacement sign on a sufficiently small positive interval unless it is the identity.

The formal objects are constructed from normed spaces of original analytic data. We call these spaces source presentations; their elements retain the coefficients, integration data, and inverse families used to define the actual functions. Independently defined bounded comparison maps on these spaces identify all finite terms of a given action before any test of the full analytic germ. The equality between their common kernel and the analytic kernel is proved in Theorems 8.23 and 9.17 and Lemma 11.23. Only afterwards do we pass to the closed comparison quotient, with its original infimum norm. The formal zero condition therefore has an analytic reconstruction theorem behind it.

Two estimates carry this reconstruction. For finite collections of residual exponential phases, normalization and differentiation reduce the number of terms; positive primitive or tail integrals then control the full cancellation. The original absolutely summed coefficient norm bounds the entire later-action tail. For the sectorial integral factors, the proof retains the full angular integrals and their complete errors. The finite-sector estimate of Lemma 8.20 and the reconstruction theorem work on their actual complex domains. The strictly positive action reserves and the lower-stage domain hypotheses are part of each application.

From original equations to finite physical words. The source construction must remain available after each physical passage. A comparison result for one fixed expression alone does not provide the next composition. The prepared equation and its correlated inverse supply specific distributions with finite action records in Proposition 5.2. The sectorial normalizers and their genuine inverses are constructed in Proposition 6.7 on the stated original domains. These are the local analytic inputs to the physical factorization of Theorem 7.5.

A particular complete model supplies the initial source presentation before the physical word is executed. Its construction in Propositions 9.27, 10.31, and 12.15 retains the original input, unspecialized coefficient data, and entire inverse families. When a coordinate is left and later revisited, its old source data remain distinct from newly created data even if the current scalar values agree. This retained information is needed for later returns and is not erased by the model's scalar identities.

Section 11 constructs the subsequent mixed operations. Its comparison estimates keep whole comparable-parent kernels until the total action is cut, including every derivative of the cutoff. The exact horizontal law and its transport to a faster coordinate are proved before the next exponential factors are adjoined. The weighted difference maps of Lemma 11.24 retain the entire first removed factor together with every later factor. Their original coefficient and inclusion costs

are paid before a fresh receiving radius is chosen. This gives the source-norm estimates needed for subsequent operations, as well as the complete scalar remainders.

Proposition 12.16 then receives the original equations with arbitrary positive real current exponents. Its weak coordinate construction retains the basal logarithmic term and uses genuine inverse charts. Matching of selected sectorial normalizers is proved on full fibers in Lemma 12.11 before evaluating endpoint graphs or inverse formulas. Consequently, reality is a property of the matched physical return; the individual selected normalizers need not be real. Theorem 12.18 constructs every finite physical word and retains both its original input and its final current. The quantitative joint comparison in Section 13 applies to their full difference.

The geometric reduction. The global argument uses compactification, elementary resolution, and the limit-graphic theorem on the sphere. The external geometric inputs are stated with their scopes in Section 3; in particular, Roussarie’s result is used for a constant analytic family on the sphere with isolated approximating cycles [Rou93, Proposition 1.1, p. 350]. Proposition 3.17 selects an actual first return from the approximating periodic leaves, retaining their physical sides and all necessary visits.

Two local points in this reduction require care. A regular periodic leaf on a nonorientable resolved surface may have a Möbius neighborhood. Lemma 3.6 composes the reversing one-circuit holonomy on explicit domains and uses its increasing two-circuit return. For singular limits, the periodic lifts avoid the entire exceptional divisor. Lemma 3.7 first excludes regular points transverse to any divisor component. The completeness and singular endpoints of lifted edges are established only after this exclusion in Proposition 3.16, with every original singular fiber covered by the divisor. The identity or strict-sign alternative then excludes an accumulating sequence of isolated periodic orbits.

Structure of the paper. Section 2 fixes the physical return data and notation. Section 3 gives the geometric reduction. Section 4 proves the whole-family analytic estimates. Section 5 constructs prepared passages and their correlated inverses. Section 6 constructs the canonical sectorial maps. Section 7 identifies the local physical factors. Section 8 develops original root sources, finite-sector flatness, and phase comparison. Section 9 proves initial reconstruction and constructs the scalar model inlet. Section 10 constructs the full retained model. Section 11 establishes the mixed receiving operations. Section 12 constructs arbitrary finite physical words. Section 13 proves formalization and the complete displacement alternative. Section 14 completes the global finiteness proof. Appendix A records the use of generative AI.

Remark 1.2. The main result of this paper was obtained using generative AI, particularly gpt-6-astra and the Danus system, a specialized agent built on the Rethlas system and substantially more capable of conducting fundamental mathematical research. Automated manuscript reviews and polishing were done afterwards. Complete independent human verification of the proof is not recorded. See [Liu⁺26] and [Ju⁺26] for detailed introductions to the Danus system and the Rethlas system, respectively. Due to the limitation of generative AI, it is possible that we have missed some related references in the literature, and we welcome any comments from experts. The details of the use of generative AI are documented in Appendix A for transparency. The author is fully responsible for the correctness of the paper.

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2. NOTATION AND PHYSICAL RETURN MAPS

Set-up 2.1. A real planar polynomial vector field is $V = P(x, y)\partial_x + Q(x, y)\partial_y$, with $P, Q \in \mathbb{R}[x, y]$. A periodic orbit is the image of a nonconstant periodic integral curve. It is isolated if it has an open neighborhood containing no other periodic orbit. A germ at 0^+ is an equivalence class under restriction to a smaller interval $(0, \epsilon)$. Equality of germs always means equality on such an interval, not equality of an asymptotic expansion.

Definition 2.2. A *finite one-sided elementary itinerary* Γ consists of a finite cyclic list of occurrences of isolated elementary singularities of a real analytic foliation on a surface, chosen incoming and outgoing analytic half-transversals A_i, B_i , a positive coordinate on each half-transversal, an actual intervening singular-sector passage $S_i : A_i \rightarrow B_i$, and an actual regular holonomy $R_i : B_i \rightarrow A_{i+1}$. Elementary means that the linear part of a local reduced generator has at least one nonzero eigenvalue. At each singular occurrence q_i , an actual singular-sector passage includes an incoming regular orbit branch γ_i^- tending to q_i in positive time, an outgoing regular orbit branch γ_i^+ tending to q_i in negative time, and a germ of a local sector between them on the selected side. The zero endpoints of A_i and B_i are regular points of γ_i^- and γ_i^+ , respectively. Every sufficiently near interior leaf meeting A_i meets B_i exactly once before leaving the selected sector; its prescribed segment is the passage represented by S_i . These segments fill the sector germ between the cuts, and their endpoint map tends to the outgoing boundary point as the incoming point tends to its boundary point. The actual passage segments are part of the data. Restricting a representative of the sector germ retains the prescribed initial and terminal regular pieces outside that representative.

Each S_i is positive and injective, analytic for positive input, and extends continuously to zero with value zero. Each R_i is an analytic diffeomorphism at zero, fixes zero and has positive derivative there. Every prefix is defined on a common positive source interval. The final half-transversal and coordinate are the original A_1 and its coordinate. The word follows the prescribed leaf direction and has no intermediate intersection with that physical half-transversal. Its *physical return* is

$$(2.1) \quad \Delta_\Gamma = R_m \circ S_m \circ \cdots \circ R_1 \circ S_1.$$

Repeated singularities, edges and equation germs are permitted. Occurrence indices do not make an intermediate visit to A_1 disappear. Repeated equation data retain their common parameters; sides, analytic branches and full laps are part of the data.

Definition 2.3. A real germ f has a *positive leading term with complete remainder* if there exist a constant $a \in \mathbb{R} \setminus \{0\}$, a positive germ Θ , and a real germ ϵ tending to zero at 0^+ such that $f = a\Theta(1 + \epsilon)$ as an equality of actual germs. The word complete means that the equality includes the entire germ after all cancellations; it does not describe a fixed truncation. The analytic sections construct Θ from their comparison scales.

3. GLOBAL REDUCTION AND RESOLVED LIMITS

The reduction has two distinct compactness arguments. We first place the polynomial foliation on the sphere, where a limit of isolated periodic orbits has a finite description. After resolving its singularities, we recover a finite description upstairs without applying a theorem about the sphere to the resolved surface. The argument upstairs uses avoidance of the exceptional divisor and the actual passages of the approximating periodic leaves.

We use ordinary elementary resolution in the following form [IY08, Theorem 8.14, p. 120]. The divisor condition needed here will be obtained by a further explicit modification.

Theorem 3.1 (Elementary resolution). *Let $\mathcal{F}$ be a germ of a holomorphic foliation at an isolated singularity of $\mathbb{C}^2$, represented by a holomorphic one-form whose coefficients have no common factor. A finite composition π of point blow-ups resolves this germ to a holomorphic foliation*

with only elementary singularities on the exceptional divisor. The map is an isomorphism away from the original singular point, and its exceptional divisor is a finite union of smooth projective lines with simple normal crossings.

For a local real analytic generator $a\partial_x + b\partial_y$, finite colength of its complexified coefficient ideal means $\dim_{\mathbb{C}} \mathbb{C}\{x, y\}/(a, b) < \infty$, where the coefficients on the right are their holomorphic extensions. This condition concerns the complex zero set, not merely the real zero set.

Lemma 3.2. *Let $\mathcal{F}$ be a saturated real analytic foliation on a compact real analytic surface M , with finite singular set S . Suppose the complexified coefficient ideal of every local reduced generator at each point of S has finite colength. There exists a finite composition $\pi : \widehat{M} \rightarrow M$ of real projective point blow-ups, every center lying over S , which is an analytic isomorphism over $M \setminus S$, such that $D = \pi^{-1}(S)$ is a finite compact simple-normal-crossing divisor and the saturated transformed foliation has only isolated elementary singularities. At every regular foliation point of D , some incident branch is transverse to the foliation or a smooth incident branch is locally invariant. No invariance of every divisor branch through a singularity is asserted.*

Proof. We first obtain a real resolution over every point of S , then show that only finitely many regular divisor points need correction, and finally compute one blow-up at each of those points in both charts.

Step 1. We construct a finite real resolution supported over S . If S is empty, use the identity map. Otherwise complexify the reduced generator in a real coordinate neighborhood of each point of S . Finite colength makes its complex zero isolated and excludes a common holomorphic coefficient factor. Theorem 3.1 therefore applies to the complexified germ.

First blow up each original point once, even if it is already elementary. This puts every point of S below an exceptional curve. In each complex chart, pull back a local defining one-form and divide its coefficients by their maximal common exceptional factors. Off the exceptional set the map is an isomorphism and the original foliation is regular, so a common factor can have its zero curve only in the exceptional set. After this division no common curve of zeros remains. The transformed singularities are isolated and hence finite on the compact exceptional curves. Apply ordinary resolution to the remaining nonelementary germs; no claim that this first blow-up preserves elementarity is needed.

The construction of the resolution simultaneously blows up the entire finite set of current nonelementary singularities and terminates after finitely many stages [IY08, pp. 137–138]. Starting with a real germ, that set is preserved by complex conjugation at every stage. Blowing up the whole set lifts conjugation to the new surface; saturation also commutes with it. A real center restricts to a real projective point blow-up. A conjugate pair of nonreal centers contributes no real exceptional point: the image of a conjugation-fixed point must itself be fixed. Its blow-up thus restricts to an isomorphism on the real locus. Omitting these isomorphisms from the finite list yields a finite sequence of real point blow-ups with elementary singularities at every real point of the final exceptional set. The real charts are $(u, v) \mapsto (u, uv)$ and $(s, t) \mapsto (st, t)$, as in [IY08, Definition 8.4 and Remark 8.5, p. 115].

Perform this construction in charts over the finitely many original points. Point blow-up agrees with the original surface away from its center, so the local modifications glue; a global complexification of M is unnecessary. Denote the resulting real map by $\pi_0 : M_1 \rightarrow M$ and its total divisor by $E_0 = \pi_0^{-1}(S)$. All centers map to S . Each real blow-up is proper and inserts a compact curve isomorphic to $\mathbb{RP}^1$, so M_1 is compact.

The divisor has simple normal crossings by induction. Blowing up a smooth point of a divisor branch makes its strict transform meet the new component transversely. At a crossing the two old strict transforms separate and meet the new component transversely at two different points. Consequently the components remain smooth and no triple intersection is created. Their number

is finite. The saturated foliation is regular off E_0 and has isolated elementary singularities on E_0 , hence finitely many singularities in total.

Step 2. We show that the regular tangencies are finite in number. Let C be a connected component of one of the smooth divisor curves. Choose a local reduced vector field X and a local defining equation $h = 0$ for C , with $dh \neq 0$ on C . The normal component of the foliation along C is represented by $(Xh)|_C$. If X is replaced by uX and h by vh , for analytic units u, v , this representative is multiplied by $uv|_C$. Thus these representatives define an analytic section of a real line bundle on the whole compact curve C .

If this section vanishes identically, analytic division by the smooth equation h gives $Xh = hk$ locally. Uniqueness of the local differential equation makes C invariant at every regular foliation point. If the section is not identically zero, its zeros are isolated by analytic continuation in one real variable and are finite by compactness. This argument takes place on the entire curve, including its points where the foliation is singular and its crossings with other divisor components. It excludes accumulation of regular tangencies at such points.

At a regular corner, the nonzero foliation direction cannot be tangent to both of the two distinct branch tangent lines. At least one branch is transverse, so the desired condition already holds there. The only points to modify form the finite set B of smooth divisor points where the foliation is regular and tangent to a noninvariant branch. Choose disjoint neighborhoods of these points containing no old foliation singularity and no old divisor corner.

Step 3. We analyze the chart containing the tangent direction. Fix $q \in B$. An analytic flow box gives coordinates with foliation $dy = 0$ and the unique divisor branch

$$(3.1) \quad y = f(x) = x^m a(x), \quad m \geq 2, \quad a(0) \neq 0.$$

The finite order follows because an identically zero f would make the branch invariant. Shorten the neighborhood so that both $a(x)$ and $ma(x) + xa'(x)$ are nonzero.

In the blow-up chart $x = u, y = uv$, the pulled-back form and a reduced generator are

$$(3.2) \quad \omega_A = v du + u dv, \quad X_A = u\partial_u - v\partial_v.$$

The coefficients u, v have no common factor, so this form is already saturated. In particular, no factor u can be divided out. Its only singularity is $(0, 0)$, with eigenvalues $1, -1$. Write H for the new exceptional curve and C_1 for the strict old branch. Their equations in this chart are

$$(3.3) \quad H = \{u = 0\}, \quad C_1 = \{v = u^{m-1}a(u)\}.$$

The tangent to C_1 has nonzero u component, so C_1 is smooth and transverse to H , including when $m = 2$. The new divisor has a normal crossing at the new saddle. Its regular points on H lie on an invariant branch, since $X_A u = u$. For $g = v - u^{m-1}a(u)$,

$$(3.4) \quad (X_A g)|_{C_1} = -u^{m-1}(ma(u) + ua'(u)).$$

It is nonzero for small $u \neq 0$. Every punctured point of the strict old branch is therefore transverse to the foliation. At the saddle itself C_1 need not be invariant; the assertion of the lemma imposes no regular-point condition there.

Step 4. We analyze the complementary chart. In the other chart, $x = st, y = t$, the pulled-back form is dt . It is saturated and nonsingular, with generator ∂_s . The exceptional branch $H = \{t = 0\}$ is invariant throughout this chart, including the point representing the vertical direction. The old branch equation pulls back to

$$(3.5) \quad t(1 - s^m t^{m-1} a(st)) = 0.$$

Its strict transform therefore has equation $1 - s^m t^{m-1} a(st) = 0$ wherever it is present. It misses every finite point of H in this chart, since this expression equals 1 on $t = 0$. Off H the blow-up is an isomorphism, so the transversality already established for the punctured old branch is preserved. On the overlap $u = st, v = 1/s$, the form $v du + u dv$ becomes exactly dt . The two

charts therefore account for the entire new exceptional curve and give only the one elementary saddle in the tangent direction.

Step 5. We assemble the final modification. Blow up each point of the finite set B once. The chosen neighborhoods are disjoint; outside them the foliation and divisor are unchanged. Each modification adds an invariant exceptional curve and one elementary saddle, and its old branch is transverse at every new regular point. It introduces no further bad regular tangency. Existing elementary singularities and old corners are untouched. Thus the composite $\pi: \widehat{M} \rightarrow M$ has all the asserted properties. Every new center lay on E_0 , so all centers still map to S , and its reduced total divisor is exactly $\pi^{-1}(S)$. $\square$

The following elementary portraits give the local sector information [DLA06, Theorem 2.15(i)–(iii), pp. 71–73], [DLA06, Theorem 2.19 and Remark 2.20, pp. 74–75].

Theorem 3.3 (Elementary local portraits). *An isolated real analytic planar singularity with nonzero real eigenvalues of opposite signs has analytic stable and unstable separatrices and the oriented saddle portrait. If both eigenvalues have nonzero real parts of matching signs, it is a local source or sink. If its eigenvalues are 0 and $\lambda \neq 0$, then, after a possible reversal of time, its oriented orbits are locally topologically equivalent to one of*

$$(3.6) \quad (\dot{u}, \dot{v}) = (-u, v), \quad (\dot{u}, \dot{v}) = (u, v), \quad (\dot{u}, \dot{v}) = (u^2, v).$$

The last model has two hyperbolic sectors and one parabolic sector. These are topological orbit equivalences; no analytic weak separatrix or analytic center manifold is asserted.

For the semihyperbolic assertion, the isolated-singularity hypothesis supplies the nonflatness required in the cited formulation. In coordinates with linear part $\text{diag}(0, \lambda)$, write the equations as $\dot{x} = A(x, y)$, $\dot{y} = \lambda y + B(x, y)$, where A, B have vanishing first jets. The implicit function theorem gives an analytic solution $y = f(x)$ of $\lambda y + B(x, y) = 0$. If $A(x, f(x))$ vanished identically, this curve would consist of singularities. Hence $A(x, f(x))$ has a finite order at least two and a nonzero leading coefficient, as required. Reversing time treats $\lambda < 0$.

Definition 3.4. A finite orbit graph of an oriented foliation is a compact set consisting of finitely many singular points and finitely many complete regular orbits. Each regular orbit has a single singular point as its limit in each time direction. These points are the vertices and these orbits are the directed edges. A cyclic graphic is such a graph whose edges can all be visited by one finite directed closed walk. A walk may repeat a vertex or an edge.

We use the following sphere theorem with a constant analytic family [Rou93, Proposition 1.1, p. 350]. The approximating orbits in this application are isolated; the terminology of limit periodic sets and graphics is that of [Rou93, Definitions 1.1, 1.3–1.4, p. 349].

Theorem 3.5 (Roussarie). *Let Y_λ be a real analytic family of vector fields on an open subset of the two-sphere. Suppose isolated periodic orbits of Y_{λ_n} converge in Hausdorff distance to a nonempty compact set K , where $\lambda_n \rightarrow \lambda_0$, and the singular points of Y_{λ_0} in K are isolated. Then K is one singular point, one periodic orbit, or a finite graphic consisting of cyclically arranged singular-point occurrences and regular connecting orbits with their indicated singular endpoints.*

We first treat regular compact leaves, including those with a Möbius neighborhood. Orienting the leaves does not orient their transverse line.

Lemma 3.6. *Let C be a compact regular leaf of a real analytic foliation with oriented leaf direction on a possibly nonorientable surface. Let P be its analytic one-circuit holonomy germ on a signed transversal at $0 \in C$. If distinct compact periodic leaves converge to C in Hausdorff distance, then $P = \text{id}$ when $P'(0) > 0$, and $P \circ P = \text{id}$ when $P'(0) < 0$. In the second case the composition is formed on an interval J containing zero with $P(J)$ in the domain of P ; it is the first return to a chosen punctured half-transversal. A neighborhood of C is filled by periodic*

leaves, closing after one circuit in the first case and, away from the core, two circuits in the second case. All these leaves are nonisolated.

Proof. We construct the relevant return on a signed interval, use the nearby periodic leaves to identify its germ, and then choose an invariant interval on which its suspension describes a neighborhood of C .

Step 1. We construct the signed return. Choose a tubular neighborhood of C with its projection to C . The positive tangent direction to the leaves has positive projection after the neighborhood is shortened. Cutting along one fiber gives a strip of flow boxes. Following these boxes once around C defines an analytic diffeomorphism $P: I \rightarrow P(I)$ between open intervals containing zero, with $P(0) = 0$. This construction does not require $P(I) \subset I$. If $P'(0) < 0$, choose an open interval J containing zero such that $J \subset I$ and $P(J) \subset I$. On J the composition $Q = P \circ P$ is defined, analytic, and increasing. It extends analytically across zero. A positive point first returns to the positive half of the fiber after two circuits, provided the point is sufficiently close to zero.

Step 2. We obtain fixed points from compact leaves. Every sufficiently close compact leaf lies inside the tubular neighborhood. Its projection to C is a local diffeomorphism from a circle, hence a finite covering. Its intersections with the cut fiber consequently form a finite cycle of P . After a further restriction of the neighborhood, all these intersections lie in I , and in J when P is decreasing. If P is increasing, any such cycle is a fixed point: $P(x) > x$ forces every subsequent iterate to increase, and $P(x) < x$ forces every subsequent iterate to decrease. If P is decreasing, apply this argument to the finite cycle of the increasing map Q on either half of the fiber. Distinct leaves have disjoint intersections with the fiber. They therefore supply distinct zeros tending to zero of $P(x) - x$ or of $Q(x) - x$, respectively. These functions are analytic on an interval containing zero, so the identity theorem gives the asserted identity.

Step 3. We construct an invariant suspension. If $P = \text{id}$, shorten its representative until it is the identity throughout the interval; its suspension is an annulus of periodic leaves. Suppose $P'(0) < 0$ and $P^2 = \text{id}$. Choose $b > 0$ so small that both applications of P are defined near the closed interval $[P(b), b]$, with $P^2(b) = b$. The interval $I_* = (P(b), b)$ is invariant under P . On this interval $h(x) = (x - P(x))/2$ has positive derivative and satisfies $h(P(x)) = -h(x)$. Thus the suspension is a Möbius neighborhood, conjugate in its transverse coordinate to the suspension of reflection. The core closes after one circuit, and every other leaf closes after two circuits. In both cases the transverse intervals give neighboring periodic leaves arbitrarily close to each leaf. Hence none is isolated. $\square$

The next observation uses the entire exceptional divisor. An individual noninvariant component need not itself be preserved by the flow.

Lemma 3.7. *Let Z be a C^1 vector field on a surface and let compact invariant sets K_n converge in Hausdorff distance to K . Let D be a closed union of finitely many smooth curve components and suppose $K_n \cap D = \emptyset$ for every n . Then K contains no regular point of Z at which Z is transverse to any component of D through that point. This conclusion also holds at a crossing of components whenever one component is transverse.*

Proof. Suppose q is such a point. In a flow box centered at q , write $Z = \partial_t$ and represent the transverse component by $t = a(s)$, where a is C^1 and $a(0) = 0$. Choose a smaller open rectangle U about q such that the vertical segment from any $(s, t) \in U$ to $(s, a(s))$ remains in the original flow box. This segment is a piece of one orbit, in forward or backward time. If K_n met U , its invariance would imply that it met D . Thus $K_n \cap U = \emptyset$ for every n . Hausdorff convergence now gives $q \notin K$. At a crossing we apply the argument to the one transverse component; the other components impose no additional requirement. $\square$

Definition 3.8. Fix finitely many small vertex neighborhoods and regular edge tubes about a finite orbit graph. A channel state consists of an incoming analytic crosscut at a regular point of an edge, one punctured half of that crosscut, and the oriented direction of travel. It is a passing state if every point of a sufficiently short half-crosscut follows one specified sector to one specified outgoing half-crosscut attached to an edge tube of the graph. A terminal state has an orbit end at the vertex. An escaping state leaves the union of vertex neighborhoods and edge tubes through a boundary crosscut disjoint from the graph. Neither alternative can be used by a periodic orbit contained in this union. The successor of a passing state is the incoming half-crosscut reached through its outgoing edge tube. States refer to physical half-crosscuts, including their sides, rather than to positions in an arbitrarily chosen graph walk.

Lemma 3.9. *An isolated analytic planar singularity with eigenvalues $\pm i\beta$, where $\beta \neq 0$, has a punctured neighborhood filled by periodic orbits, or a neighborhood in which every nonconstant orbit converges to the singularity in one fixed time direction.*

Proof. After a real linear change and reversal of time if necessary, write $x = r \cos \theta$, $y = r \sin \theta$. The equations give $\dot{\theta} = \beta + O(r) > 0$ and $dr/d\theta = rH(r, \theta)$ with H analytic for signed r near zero and periodic in θ . The return over an angular increment 2π is therefore an analytic increasing map F on an interval containing zero, with $F(0) = 0$. If $F = \text{id}$, all sufficiently small positive inputs close and fill a punctured neighborhood. Otherwise the first nonzero Taylor coefficient of $F(r) - r$ gives it one strict sign on a sufficiently small positive interval. For the negative sign, choose $b > 0$ with $F(b) < b$. Then $F((0, b)) \subset (0, b)$ and every positive iterate tends to zero: any positive limiting value would be a fixed point. Bounds on the flow for $0 \leq \theta \leq 2\pi$ show that the intervening arcs also tend to the singularity. The saturation of a suitably shortened radial segment provides a forward trapping neighborhood. For the positive sign apply this reasoning to F^{-1} and reversed time. A full turn takes time bounded below because $\dot{\theta}$ is bounded above in this neighborhood, so these convergences occur in infinite time. $\square$

Lemma 3.10. *For a finite orbit graph of a real analytic foliation with elementary vertices, one can choose finitely many channel states as in Definition 3.8. Every sufficiently small passing half-crosscut has one successor on its whole germ. Its singular passage S is positive, injective, analytic at every positive input, and satisfies $S(s) \rightarrow 0$ as $s \rightarrow 0^+$. Its incoming and outgoing boundary branches tend to the vertex in positive and negative time, respectively. The crosscut endpoints are regular points of these branches, and the prescribed passage segments fill the sector germ between the cuts. Every sufficiently near interior leaf entering through the incoming crosscut reaches the outgoing crosscut exactly once before leaving this sector. The intervening regular passage R is an analytic diffeomorphism at zero with $R(0) = 0$ and $R'(0) > 0$, in coordinates positive on the selected sides. Every sufficiently close periodic leaf that stays in the union of the vertex neighborhoods and edge tubes uses passing states only. A nonperiodic orbit approaching the graph either converges to a vertex or eventually follows these states.*

Proof. We first examine passage across a sector, and then join the sectors by edge tubes. Crosscuts are taken at regular points, even when a bounding separatrix is not analytic at its singular endpoint.

Step 1. We analyze the ordinary saddle sectors. Straighten the two analytic separatrices supplied by Theorem 3.3. On a sufficiently small quadrant the equations have the form

$$(3.7) \quad \dot{x} = xA(x, y), \quad \dot{y} = -yB(x, y), \quad 0 < a_0 \leq A \leq a_1, \quad 0 < b_0 \leq B.$$

Take the incoming crosscut (s, r) and the outgoing crosscut (r, y) , with $0 < s < r$. The orbit stays in the rectangle until it crosses $x = r$, and does so exactly once. Its crossing time $T(s)$ satisfies

$$a_1^{-1} \log(r/s) \leq T(s) \leq a_0^{-1} \log(r/s).$$

Consequently its outgoing coordinate satisfies

$$(3.8) \quad 0 < S(s) \leq r(s/r)^{b_0/a_1} \longrightarrow 0.$$

The positive y -axis is the incoming boundary branch and the positive x -axis is the outgoing boundary branch. Their orbit ends at the origin occur in positive and negative time, respectively. The union of these segments, together with the two boundary pieces and the vertex, represents the sector germ between the cuts: every interior orbit sufficiently near the vertex extends backward to $y = r$ and forward to $x = r$ by the signs in (3.7). Shortening the incoming half-crosscut just selects a smaller such germ. Transversality and analytic dependence of the flow give analyticity at every $s > 0$. Uniqueness of an orbit crossing the two crosscuts gives injectivity. Reflections of coordinates and reversal of time give the other sectors with their prescribed directions.

Step 2. We analyze the semihyperbolic sectors. Use the oriented sector portrait of Theorem 3.3. In the saddle model, a sector is foliated by arcs joining two adjacent crosscuts and tending, as the input tends to its endpoint, to the two bounding half-orbits and the vertex. In the saddle-node model the two hyperbolic sectors have this description; in the parabolic sector an orbit has the vertex as one of its ends. These assertions can also be read directly from $\dot{u} = u^2$, $\dot{v} = v$: on the saddle side $u < 0$, u increases to zero while v moves away from zero. A node has only terminal behavior in one time direction.

Pull back closed, shortened sector boxes by the orbital equivalence. Their incoming and outgoing edges are topological crosscuts. At each regular endpoint replace a short piece by an analytic transverse segment inside a flow box. Sliding along the flow identifies the old and new crosscuts homeomorphically, including the endpoint. Shorten the incoming segment until all its points meet the chosen outgoing segment. This produces a passage on an entire half-interval, and the model passage and the two endpoint-continuous identifications prove $S(s) \rightarrow 0$. The two boundary half-orbits pull back to an incoming regular branch and an outgoing regular branch of the original foliation. Their oriented ends approach the vertex in opposite time directions. For a local smooth generator vanishing there, uniqueness prevents either branch from reaching the vertex in finite time. Their selected cut points are regular. The pulled-back model segments fill the sector between the cuts and cross each cut once; sliding the cuts in regular flow boxes preserves these properties. Each positive input has a compact regular orbit segment up to its transverse exit. Analytic dependence and the implicit function theorem therefore give an analytic exit map near that input. Two inputs cannot have one exit, since the corresponding orbit crosses the selected entrance only once. This proves the stated properties without an analytic change of coordinates at the singularity.

In both constructions retain the actual orbit segment for every input. When a smaller neighborhood of the vertex represents the sector germ, restrict its middle portion and retain the initial and terminal regular pieces joining it to the original cuts. This restriction leaves the specified passage unchanged.

Step 3. We obtain finitely many states with uniform successors. Choose disjoint small vertex neighborhoods. Cut each of the finitely many edges just outside these neighborhoods and cover the remaining compact regular arc by a thin flow tube. Its holonomy is an analytic local diffeomorphism fixing the edge coordinate zero. On the chosen incoming and outgoing sides it is positive, hence its derivative at zero is positive. Each crosscut has two sides, and each elementary portrait has finitely many passing, terminal, or escaping possibilities. If a sector's outgoing branch does not belong to the graph, a short crosscut on that branch forms an exit disjoint from the graph; shortening the entrance makes this an escaping state on the entire half-germ. The sector construction gives an open half-interval with one exit, not merely an exit chosen for each orbit separately. Its outgoing side specifies the next tube and therefore one successor state on the whole germ.

An elementary vertex having both an incoming and an outgoing edge must be of saddle or saddle-node type. Sources and sinks have only one direction of singular ends. For a singularity with eigenvalues $\pm i\beta$, Lemma 3.9 gives either a center, which has no incident edge, or a trapping neighborhood in one time direction. Thus it too has only terminal alternatives. A periodic orbit contained in the chosen graph neighborhood cannot use a terminal or escaping alternative. An orbit approaching the graph cannot repeatedly use an exit separated from that graph by a positive distance. Unless its tail remains in one vertex neighborhood and converges to that vertex, it crosses edge crosscuts indefinitely and uses the passing alternatives just constructed. $\square$

Lemma 3.11. *Let $F: (0, a) \rightarrow (0, \infty)$ be continuous and injective, with $F(s) \rightarrow 0$ as $s \rightarrow 0^+$. Suppose an orbit of F is defined for all nonnegative iterates and has distinct iterates $s_j \rightarrow 0$. Then there exists $b > 0$ such that $0 < F(s) < s$ for $0 < s \leq b$ and $F((0, b)) \subset (0, b)$. If F is a finite composition of the physical passages of a directed channel cycle, no periodic leaf meets the corresponding short base half-crosscut.*

Proof. A continuous injective map on an interval is strictly monotone. Positivity and its zero endpoint limit exclude a decreasing map. Thus F is increasing. The iterates in the hypothesis must decrease strictly; an increasing pair would force all later iterates to increase. Put $b = s_j$ for an iterate lying in the domain. If $F(c) = c$ for some $0 < c < b$, monotonicity would give $s_{j+k} > c$ for every k , a contradiction. Hence $F(s) - s$ has no zero in $(0, b]$ and is negative there. This also proves invariance of $(0, b)$.

For a physical channel cycle, all sufficiently short prefixes are defined by their zero endpoint limits. Every orbit through the resulting base half-crosscut follows the cycle repeatedly and has strictly decreasing return coordinates. These returns cannot be intersections of a periodic leaf, which meets a fixed compact transverse segment in finitely many points. The repeated routes are defined for unbounded time: a fixed compact regular subarc of an edge takes a uniformly positive time to cross in a smaller tube for any smooth generator. Thus no finite-time accumulation of circuits interrupts this conclusion. $\square$

Lemma 3.12. *Let L be a Hausdorff limit of compact periodic orbits of a smooth complete generator of an analytic foliation with finitely many elementary singularities. An orbit contained in L cannot have as a proper forward or backward limit a regular periodic leaf or a finite orbit graph containing an edge.*

Proof. Suppose a nonperiodic orbit $\gamma \subset L$ has such a forward limit. For a regular periodic limit, use a signed one-circuit holonomy, or its square on a suitable interval in the transverse-orientation-reversing case, constructed as in Lemma 3.6. The approaching tail of γ gives distinct iterates tending to zero on a fixed half-crosscut. Lemma 3.11 applies.

For a graph limit containing an edge, use Lemma 3.10. The tail cannot stay in a single vertex neighborhood, since its limit contains a regular edge point. It therefore has infinitely many channel visits, whose coordinates tend to zero. The finitely many states have deterministic successors. After a finite initial string their sequence is a directed cycle. Composition around this state cycle is continuous, injective, positive, and has zero endpoint limit. The return coordinates of γ are distinct and tend to zero. Lemma 3.11 again gives a short base half-crosscut through which no periodic leaf passes.

In either case choose a crossing of γ in the interior of that half-crosscut. A smaller flow box around the crossing has the property that every orbit meeting it crosses the forbidden interval. Thus this flow box misses every approximating periodic orbit. It cannot contain a point of their Hausdorff limit L , contrary to $\gamma \subset L$. Replacing the generator by its negative proves the backward assertion. $\square$

Lemma 3.13. *Let ϕ_t be a continuous flow on a compact metric space. A Hausdorff limit L of periodic orbits is compact, connected, invariant, and internally chain transitive. Here internal*

chain transitivity means that for any $x, y \in L$ and $\epsilon, T > 0$ there exist points $x = x_0, x_1, \dots, x_k = y$ in L and times $t_i \geq T$ such that $\text{dist}(\phi_{t_i}(x_i), x_{i+1}) < \epsilon$.

Proof. Compactness and connectedness pass to a Hausdorff limit of nonempty compact connected sets. For fixed t , approximate $x \in L$ by points on the periodic orbits and use continuity of ϕ_t to obtain $\phi_t(x) \in L$; using $-t$ proves invariance.

Fix $\epsilon, T > 0$. Uniform continuity of $(t, z) \mapsto \phi_t(z)$ for $T \leq t \leq 2T$ gives $\eta > 0$ such that points within η have images within $\epsilon/2$ for all these times. Take $\eta < \epsilon/2$ and a periodic orbit within Hausdorff distance η of L . Choose on it points x', y' within η of x, y . Following the periodic orbit from x' to y' , add complete laps until the travel time A is at least $2T$. Put $k = \lfloor A/T \rfloor$ and divide this route into k intervals of length $A/k \in [T, 2T]$. Replace every division point by a point of L within η , using x, y for the two ends. The flow error and the final replacement error sum to less than ϵ at every link. This is the required chain. $\square$

Lemma 3.14. *An internally chain transitive finite orbit graph containing an edge is a cyclic graphic.*

Proof. Consider the strongly connected components of its finite directed graph. If the graph has more than one component, choose a terminal component A . Enlarge A within the orbit graph by short terminal portions of all edges entering it, to obtain a compact neighborhood U of A relative to the graph. Choose shorter terminal portions to obtain U_0 with $U_0 \subset \text{int}_G U$. Both sets are forward invariant. Travel from the finitely many outer cut points to the inner cut points takes bounded time, so there exists $T > 0$ with $\phi_t(U) \subset U_0$ for all $t \geq T$. Compactness gives $\text{dist}(U_0, G \setminus \text{int}_G U) > 0$. An internal chain with smaller errors, starting at a vertex of A , can never leave U . It therefore cannot reach a vertex of another component, contradicting internal chain transitivity. Isolated vertices are covered by taking a relative neighborhood with no incident edges.

The graph is thus strongly connected. List its finitely many edges. Starting at the initial vertex of the first edge, traverse that edge, use a directed path to the initial vertex of the next edge, and continue through the list. Finally return to the starting vertex by a directed path. This is a finite directed closed walk visiting all edges, as required by Definition 3.4. $\square$

Lemma 3.15. *Let a saturated real analytic foliation on a compact surface be obtained by finitely many point blow-ups of an analytic vector field. The line bundle of its local reduced generators has an unbranched orientation cover extending across the singular points. On this cover the foliation has a smooth complete generator with exactly its prescribed singularities. A regular compact leaf has two degree-one lifts to this cover. Blow-up away from its center and passage to this cover preserve isolation of regular compact leaves in both directions.*

Proof. Local reduced generators Z_i differ on overlaps by nowhere zero analytic functions: $Z_i = u_{ij}Z_j$. Removing their common factors is essential here. The functions u_{ij} define a line bundle over the whole surface, including the isolated zeros of the generators. Its orientation cover is defined by the signs of these units and is therefore unbranched over the singularities. On the cover choose compatible positive local generators and a smooth partition of unity. Their weighted sum is a smooth generator; away from a singularity it is a positive multiple of any local generator and cannot vanish. The surface is compact, so its flow is complete.

On a regular compact leaf the generator line maps isomorphically to its tangent line, which is orientable. A tubular neighborhood retracts to that leaf, including when the neighborhood is a Möbius band. The line bundle's orientation cover is consequently trivial on this neighborhood, with two sheets mapping diffeomorphically onto it. This proves the assertion about lifts and isolation under the cover. A point blow-up is an analytic diffeomorphism off the exceptional divisor. A compact leaf disjoint from the center has a neighborhood there on which the blow-up is a diffeomorphism, so it too preserves isolation. Iterate over the finitely many blow-ups. The

smooth global generator is used only for compactness and time arguments; every local holonomy is computed with the analytic local generators. $\square$

Proposition 3.16. *Let Y be analytic on a compact surface with finite singular set S , and let distinct regular periodic orbits converge to one point of S or to a finite graphic with complete connecting orbits and singular endpoints. Let $\Pi : M' \rightarrow M$ be a finite composition of point blow-ups over S , with saturated transform having finitely many isolated elementary singularities. Require every point of S to have been blown up, so that the total exceptional divisor is exactly $D = \Pi^{-1}(S)$. Assume D has normal crossings and, at each point of this divisor where the transformed foliation is regular, either an incident component is transverse to the foliation or a smooth incident branch is locally invariant. Take the unbranched cover orienting the generator line. The strict periodic lifts avoid the entire lifted divisor. Every Hausdorff limit of a subsequence of these lifts is one elementary singular point, one regular periodic leaf, or a finite elementary cyclic graphic. Each strict lift of a base connecting orbit contained in the limit is a complete orbit with a single elementary singular point at each end.*

Proof. Write $p: \widehat{M} \rightarrow M'$ for the orientation cover and put $f = \Pi \circ p$. The entire lifted exceptional divisor is $\widehat{D} = p^{-1}(D) = f^{-1}(S)$, using the hypothesis $D = \Pi^{-1}(S)$. By Lemma 3.15 use a smooth complete generator Z on $\widehat{M}$. Choose one periodic lift C_n of each base orbit, and let a subsequence converge to L . We first exclude transverse divisor points, then establish completeness and endpoints of the lifted edges, and finally identify the finite graph.

Step 1. We establish divisor avoidance before addressing completeness. Every base periodic orbit misses S . Hence $C_n \cap \widehat{D} = \emptyset$, including all components introduced at later blow-ups. Its image under f is the original periodic orbit. Continuity on compact spaces gives $f(L) = K$, the base limit. Lemma 3.13 makes L connected and invariant. Lemma 3.7 excludes from L every regular point of $\widehat{D}$ with a transverse incident component. A regular point at a normal-crossing corner has a nonzero tangent vector, which cannot be tangent to both distinct component directions. Thus all regular corners are excluded too. The remaining regular divisor points in L lie on a smooth locally invariant branch by the adaptedness hypothesis.

Step 2. We prove that the strict edges are complete. Off $f^{-1}(S)$, the map f is a finite local diffeomorphism and takes the transformed foliation to the original one. Each base edge has at most two strict lifts. If a point of such a lift lies in L , invariance and continuation along finite regular subarcs show that the whole lifted edge lies in L .

If the maximal time interval for a trajectory to remain in this lifted edge had a finite endpoint, completeness of the ambient flow would give a limiting point $q \in L$ at that time. It must lie over a base vertex: away from S the local covering continues the lift across every regular interior point of the base edge. Thus $q \in f^{-1}(S) = \widehat{D}$. If q is singular, uniqueness for the smooth field prevents a nonconstant orbit from reaching it in finite time. If q is regular, the prior transverse-point exclusion and adaptedness put it on a smooth locally invariant branch of $\widehat{D}$. Uniqueness then prevents an off-divisor orbit from reaching that branch at a finite time. Regular corners have already been excluded. These cases exhaust the fiber because $\widehat{D} = f^{-1}(S)$; no point over a base singularity lies outside the divisor. The lifted edge is therefore a complete orbit of Z .

Step 3. We identify a single singular endpoint at each end. The forward and backward limit sets of this complete orbit are nonempty, compact, connected, and invariant. On a connected strict edge $\tilde{e}$, the local orbital diffeomorphism satisfies $df(Z) = hY \circ f$ for a continuous nowhere zero function h . Its sign is constant on $\tilde{e}$ but need not be positive. The restriction of f identifies $\tilde{e}$ with the base edge and therefore identifies their two ends, possibly interchanging the forward and backward ends. A tail of the complete lifted orbit leaves every compact subarc and projects to one of the prescribed singular ends of the base edge. Each of its limit sets is consequently contained in the fiber over that endpoint. This uses no equality of time parametrizations or positivity of h .

Every fiber over a point of S is a finite union of compact exceptional curves, since $D = \Pi^{-1}(S)$. Its inverse image under p is again a finite union of compact smooth curves in $\widehat{D}$. Remove their finitely many singular points and corners. If a remaining component meets one of these limit sets, invariance includes its regular orbit. A locally invariant branch is invariant along its analytic component: the analytic determinant of its tangent and the local generator vanishes on an open arc and hence along its continuation. A transverse component cannot contribute a regular point. It follows that a limit set in the fiber is a single singularity, a regular periodic component, or a finite orbit graph of divisor arcs with singular endpoints. Only finitely many such arcs occur because the components and their singularities are finite. Disjoint regular periodic components cannot be joined to other pieces without meeting a singular point, so connectedness separates the periodic alternative.

The lifted base edge is disjoint from its endpoint fiber. Its limit in that fiber is thus a proper limit. Lemma 3.12 excludes both a regular periodic component and any orbit graph containing an edge. A finite connected set of singularities is one point. This proves the required single singular endpoint at each end.

Step 4. We assemble the limit graph. Outside the fibers, L consists of some of the finitely many strict base edges. Inside the fibers it consists of some of the finitely many invariant divisor arcs, singular points, and regular periodic components just described. Every divisor arc has singular endpoints. The single-endpoint argument establishes this also for every strict base edge. If L contains a regular periodic component, the finite decomposition and connectedness force L to equal that component. If it contains no regular orbit, it is one singular point. Otherwise it is a finite orbit graph. Its singularities are elementary by hypothesis. Internal chain transitivity follows from Lemma 3.13, and Lemma 3.14 makes it a cyclic graphic. $\square$

Proposition 3.17. *For a real analytic foliation with isolated elementary singularities and oriented leaf direction on a compact surface, suppose distinct regular periodic leaves converge to a finite elementary graphic. Then a subsequence supplies one itinerary Γ of Definition 2.2 and distinct positive points $x_n \rightarrow 0$ on its base half-transversal with $\Delta_\Gamma(x_n) = x_n$. Its closed routes are the corresponding entire periodic leaves. If $\Delta_\Gamma = \text{id}$, every sufficiently late such leaf is nonisolated.*

Proof. We select a physical first return from the periodic leaves and then check the domains of its singular and regular factors.

Step 1. We select the return from actual periodic leaves. Choose the finite vertex neighborhoods and tubes of Lemma 3.10, and shorten the crosscuts so that their closures meet the limiting graph only at their origins. All sufficiently late periodic leaves lie in their union. A crossing coordinate on any one of these crosscuts tends uniformly to zero along the sequence: otherwise a subsequence of crossing points in the compact crosscut, separated from zero, would converge to a point of the graph outside its origin.

Choose a regular graph point on an incoming crosscut. Hausdorff convergence and a flow box imply that each sufficiently late leaf crosses that crosscut. Pass to a subsequence with one fixed side, denoted A_1 . No crossing can equal its endpoint, since that endpoint lies on an edge with a singular end. All the crossings use passing states; a terminal state cannot belong to an entire periodic leaf in the graph neighborhood. The deterministic successor map on the finite set of physical states therefore describes their routes. Since a periodic leaf returns to its initial physical state, that state lies on a cycle of the successor map. Follow it until its first recurrence. The word obtained is independent of the chosen leaf after this restriction. It contains no earlier intersection with A_1 : such an intersection is a recurrence of that physical state, regardless of how its occurrence might be named.

Step 2. We specify its domains and fixed points. For every visit to a vertex q_i , retain the incoming boundary branch γ_i^- and outgoing boundary branch γ_i^+ of its selected sector.

Lemma 3.10 gives their limits q_i in positive and negative time, respectively. Choose A_i and B_i at their regular cut points. The sector construction supplies the actual segment from every sufficiently small positive input on A_i to its unique exit on B_i . These segments fill the sector germ between the cuts. Keep the segments, including the initial and terminal regular pieces retained when its representative is shrunk, as part of the data. Their exit points tend to the zero endpoint of B_i as their entrance points tend to the zero endpoint of A_i .

Let $S_i: A_i \rightarrow B_i$ be this singular passage and let $R_i: B_i \rightarrow A_{i+1}$ be the actual edge holonomy. Lemma 3.10 gives positivity, injectivity, analyticity for positive inputs, and the zero endpoint limit of each S_i . Each R_i is analytic at zero with positive derivative. Start at the last domain and shorten the preceding source interval until its image lies there. Repeating this finitely many times gives a common positive interval on which every prefix is defined. The last half-crosscut is physically A_1 , with its original coordinate. Thus these data satisfy Definition 2.2 and their composition is its stated physical first return.

Every positive continuous injective factor with zero endpoint limit is increasing, so the return Δ_Γ is increasing. Its intersections with any selected periodic leaf form a finite cycle, entirely in its domain for late leaves. A strictly increasing interval map has no finite cycle of length greater than one. Hence each such intersection is a fixed point x_n of Δ_Γ . Distinct leaves give distinct points, and $x_n \rightarrow 0$. Starting from x_n , the route now closes at the first physical return. It is therefore the entire periodic leaf. Every necessary repeated vertex, edge, sheet, and lap was retained in the physical state cycle; a graph walk alone would not provide this conclusion.

Step 3. We treat identity returns. If $\Delta_\Gamma = \text{id}$, all sufficiently small positive inputs close by this finite route. Near one fixed positive input the route is a compact regular orbit segment, and its constituent flow maps depend continuously on that input. The resulting compact periodic leaves converge to the leaf at that input as the input varies. Each has a regular annular neighborhood obtained by suspending this positive-input identity return. Every sufficiently late selected leaf is consequently nonisolated. This argument does not require an annular neighborhood of the singular graphic itself. $\square$

Lemma 3.18. *Distinct isolated periodic leaves of an analytic foliation cannot converge in Hausdorff distance to one isolated elementary singularity.*

Proof. The source, sink, saddle, and saddle-node portraits of Theorem 3.3 have neighborhoods containing no complete regular periodic orbit. For eigenvalues $\pm i\beta$, Lemma 3.9 gives either a neighborhood with no regular periodic orbit or a punctured neighborhood filled by periodic orbits. In the center case each positive radial input gives a closed orbit depending continuously on the input, so no sufficiently small periodic orbit is isolated. Both alternatives exclude the sequence in the statement. $\square$

Lemma 3.19. *For a nonzero real planar polynomial vector field V , there exists an analytic vector field on the two-sphere with finitely many singular points such that every regular periodic orbit of V is represented by a regular periodic orbit in an open hemisphere. Distinct orbits remain distinct, and isolation is preserved in a neighborhood of each such orbit. The foliation of the sphere field is saturated, and at every singular point its complexified coefficient ideal has finite colength.*

Proof. We remove the common polynomial factor, prove finite complex colength for a saturated homogeneous one-form, and then obtain a global generator on the sphere.

Step 1. We remove the common factor. Write $V = gV_0$, where $V_0 = P_0\partial_x + Q_0\partial_y$ and P_0, Q_0 have no common nonconstant real polynomial factor. A nonconstant periodic orbit of V cannot meet $\{g = 0\}$, since every point of $\{g = 0\}$ is an equilibrium. On a neighborhood of such a compact orbit g is nowhere zero. The foliations of V and V_0 agree there and the time change is bounded away from zero. Thus periodicity and isolation agree locally. This reasoning also

covers a zero component of V ; after removal of the common factor the other component is a nonzero constant.

Step 2. We saturate the field on the projective plane. Put $d = \max(\deg P_0, \deg Q_0)$ and homogenize both polynomials to degree d , obtaining $P_h(x, y, z), Q_h(x, y, z)$. Set

$$(3.9) \quad A = zQ_h, \quad B = -zP_h, \quad C = yP_h - xQ_h, \quad \Omega = A dx + B dy + C dz.$$

These homogeneous coefficients satisfy $xA + yB + zC = 0$. Divide them by their greatest common homogeneous factor G and denote the resulting coefficients by A_s, B_s, C_s . In the chart $z = 1$, the polynomial $G(x, y, 1)$ divides P_0 and Q_0 and is therefore a nonzero constant. The saturated foliation in that chart is consequently the foliation of V_0 .

Step 3. We establish finite complex colength. The greatest common divisor of A_s, B_s, C_s over $\mathbb{C}$ is constant. For if a complex irreducible factor divided all three real polynomials, its conjugate would too; their product, or a real scalar normalization when they are associates, would give a nonconstant real common factor. Choose a nonzero coefficient U . If it is constant, the simultaneous zero set is empty. Otherwise it has finitely many irreducible homogeneous factors. For each factor, one of the other coefficients is not divisible by it. A constant linear combination W of the other coefficients can therefore be chosen outside the finitely many proper linear conditions for divisibility by these factors. Then U, W are coprime over $\mathbb{C}$.

Fix an affine projective chart. The dehomogenizations $\tilde{U}, \tilde{W}$ remain coprime: a nonconstant common factor would homogenize to a common factor of U, W . If either dehomogenization is a nonzero constant, the simultaneous coefficient zero set in this chart is empty. Otherwise choose affine linear coordinates s, t for which both polynomials have positive degree in either variable. Their resultants eliminating t and s are nonzero one-variable polynomials $r(s)$ and $q(t)$. The determinant construction of a resultant, by the adjugate identity for its coefficient matrix, expresses each as a polynomial linear combination of $\tilde{U}, \tilde{W}$. Thus

$$(3.10) \quad r(s), q(t) \in (\tilde{U}, \tilde{W}) \subset (A_s, B_s, C_s)$$

in the coordinate ring of this chart. Coprimality ensures that these resultants are not identically zero, since a zero resultant would give a common positive-degree factor over the corresponding rational function field and hence a common polynomial factor.

The simultaneous coefficient zeros lie in the finite set $\{r(s) = q(t) = 0\}$. At one such point (s_0, t_0) , write the local factorizations $r(s) = (s - s_0)^k r_0(s)$ and $q(t) = (t - t_0)^\ell q_0(t)$, where r_0, q_0 are units. In the local ring of convergent complex power series, the coefficient ideal therefore contains $(s - s_0)^k$ and $(t - t_0)^\ell$. Its quotient is spanned by the $k\ell$ monomials $(s - s_0)^i (t - t_0)^j$ with $i < k$ and $j < \ell$. It has finite colength. Three charts cover the complex projective plane, so the simultaneous projective zero set is finite.

In a chart $z = 1$, the restricted one-form has coefficients A_s, B_s , and $C_s = -xA_s - yB_s$. Its coefficient ideal is thus precisely the three-coefficient ideal used in (3.10). The other two charts have the corresponding identity. The local forms differ by analytic units under projective changes of chart, by their homogeneity and the radial identity. Their finite complex colength excludes any local common analytic coefficient factor. They define a saturated analytic foliation, with an isolated complexified singularity at each real singular point.

Step 4. We construct a global field on the sphere. Restrict $A_s dx + B_s dy + C_s dz$ to the unit sphere to obtain the analytic one-form α . The coefficient vector is orthogonal to the radius by the homogeneous identity. Hence α vanishes on the tangent plane exactly when all three coefficients vanish. Its zero set is finite. For the analytic area form ω of the sphere, define Y by $\iota_Y \omega = \alpha$. This is a global analytic vector field with precisely those zeros. The projection of the upper hemisphere to the chart $z = 1$ is an analytic diffeomorphism, under which the kernel foliation of α agrees with that of V_0 . On a compact regular leaf the two nonzero generators

differ by a nowhere zero factor. Such a leaf is a periodic orbit for either generator, and the local correspondence preserves isolation and distinctness.

Finally, the projection from the real sphere to the real projective plane is a local analytic diffeomorphism. Its holomorphic extension near a real point is still a local biholomorphism: on the complex quadric $x^2 + y^2 + z^2 = 1$, the radial line is transverse to the tangent plane. Homogeneity and the radial identity identify the pulled-back projective one-form with α up to a nonvanishing analytic factor. Contraction with the local nonvanishing area form changes the coefficient vector by an invertible analytic matrix. Finite complex colength and saturation therefore pass to the local coefficients of Y . $\square$

Proposition 3.20. *Suppose every itinerary in Definition 2.2 has a physical return which is either the identity or has displacement of one strict nonzero sign on a sufficiently small positive interval. Then every real planar polynomial vector field has finitely many isolated periodic orbits.*

Proof. We compactify an assumed infinite sequence, resolve its singular limit, and apply the stated return-map dichotomy only after selecting a physical first return.

Step 1. We obtain a limit on the sphere. The zero vector field has no nonconstant periodic orbit. For a nonzero field with infinitely many isolated periodic orbits, choose a sequence of distinct such orbits and apply Lemma 3.19. This gives distinct isolated periodic orbits of a fixed analytic vector field Y on the sphere, whose singular set S is finite. Compactness of the space of nonempty compact subsets of the sphere gives a Hausdorff convergent subsequence with limit K . The constant family $Y_\lambda = Y$ satisfies the hypotheses of Theorem 3.5. Thus K is a singular point, a periodic orbit, or a finite cyclic graphic. The periodic-orbit alternative is excluded by Lemma 3.6, since the approximating isolated orbits eventually lie in the resulting periodic neighborhood.

Step 2. We pass to a resolved limit. Lemma 3.19 supplies saturation and finite complex colength at each point of S . Lemma 3.2 therefore supplies a finite resolution Π over S in which every point of S is blown up and the entire exceptional divisor is $\Pi^{-1}(S)$. The transformed foliation has only isolated elementary singularities and satisfies the regular-divisor condition required by Proposition 3.16. Thus both the coverage of every singular fiber and the regular-point adaptedness hold. No invariance of every divisor branch at a singular point is required. Pass to the unbranched orientation cover and choose one lift of each periodic orbit. These lifts are distinct and isolated by Lemma 3.15. They avoid the entire lifted divisor, which is the full inverse image of S under the resolution composed with the cover. Compactness gives a Hausdorff limit L . Lemma 3.7 excludes its transverse regular divisor points. Proposition 3.16, with both hypotheses on the divisor verified, now shows that the strict edges are complete and that L is one elementary singularity, one regular periodic leaf, or a finite elementary cyclic graphic. Lemma 3.18 excludes the first alternative, and Lemma 3.6 excludes the second, using the two-circuit holonomy if the periodic core is one-sided.

Step 3. We apply the physical return alternative. For the remaining graphic alternative, Proposition 3.17 supplies one physical itinerary Γ and distinct positive fixed points $x_n \rightarrow 0$ of its actual return. A displacement of one strict nonzero sign cannot have these zeros. The assumed dichotomy forces $\Delta_\Gamma = \text{id}$. The last assertion of Proposition 3.17 then makes every sufficiently late selected leaf nonisolated. This contradicts isolation of the lifts. $\square$

4. ANALYTIC FAMILIES AND ORIGINAL SOURCE NORMS

The estimates in this section keep the norm of the input family fixed. Restriction, differentiation and substitution have separate receiving domains. We first establish the coefficient calculus, then treat Laplace transforms and local analytic differential equations.

For $h \geq 1$ and $\nu = (\nu_1, \dots, \nu_h) \in \mathbb{N}^h$, write $|\nu| = \sum_i \nu_i$, $\nu! = \prod_i \nu_i!$ and $\rho^\nu = \prod_i \rho_i^{\nu_i}$. Inequalities between radius vectors are componentwise. Write $D_\rho = \{z \in \mathbb{C}^h : |z_i| < \rho_i \text{ for all } i\}$.

For every Banach space E and closed linear subspace K , equip E/K with $\|x + K\|_{E/K} = \inf_{k \in K} \|x + k\|_E$. For a unital commutative complex Banach algebra A , equip $\mathcal{W}_\rho(A)^d$ with the product norm $\|u\| = \max_j \|u_j\|_{\mathcal{W}_\rho(A)}$; use this convention also after adjoining a time variable.

Definition 4.1. For a complex Banach space E and positive radii $\rho = (\rho_1, \dots, \rho_h)$, $\mathcal{W}_\rho(E)$ is the space of series $F(z) = \sum_{\nu \in \mathbb{N}^h} F_\nu z^\nu$ with norm $\sum_\nu \|F_\nu\|_E \rho^\nu < \infty$. The radii in this norm are input data. A receiving restriction may decrease the range of evaluation; it does not redefine the input norm.

4.1. Coefficient spaces and substitution. Absolute summability supplies both completeness and the permission to regroup the series used below. We include these facts with their norm bounds.

Lemma 4.2. *Let E and H be complex Banach spaces and let ρ be positive. Then $\mathcal{W}_\rho(E)$ is a Banach space, its polynomials are dense, and deleting any collection of coefficients defines a projection of norm at most one. A bounded linear map $L : E \rightarrow H$ acts coefficientwise with operator norm $\|L\|$.*

If A is a complex Banach algebra with submultiplicative norm, the Cauchy product makes $\mathcal{W}_\rho(A)$ a Banach algebra and

$$(4.1) \quad \|FG\|_{\mathcal{W}_\rho(A)} \leq \|F\|_{\mathcal{W}_\rho(A)} \|G\|_{\mathcal{W}_\rho(A)}.$$

The space

$$(4.2) \quad \mathcal{W}_\rho^1(E) = \left\{ F : \sum_\nu (1 + |\nu|) \|F_\nu\|_E \rho^\nu < \infty \right\}$$

is Banach for the displayed norm. For algebra-valued series this norm is submultiplicative. Moreover,

$$(4.3) \quad \|z_i \partial_i F\|_{\mathcal{W}_\rho(E)} \leq \|F\|_{\mathcal{W}_\rho^1(E)}, \quad \|\partial_i F\|_{\mathcal{W}_\rho(E)} \leq \rho_i^{-1} \|F\|_{\mathcal{W}_\rho^1(E)}.$$

Proof. Multiplication of the ν th coefficient by ρ^ν identifies the first space isometrically with $\ell^1(\mathbb{N}^h; E)$. To prove completeness, let $(a^{(n)})$ be Cauchy in this sequence space. Each coordinate converges in E , say to a_ν . For every finite set $J \subset \mathbb{N}^h$,

$$\sum_{\nu \in J} \|a_\nu - a_\nu^{(n)}\|_E = \lim_{m \rightarrow \infty} \sum_{\nu \in J} \|a_\nu^{(m)} - a_\nu^{(n)}\|_E.$$

The right side is uniformly small for large n , independently of J , because the sequence is Cauchy. Taking the supremum over finite J proves $a \in \ell^1$ and convergence in norm. Truncating an absolutely summable sequence proves density; deleting coordinates decreases its norm. This proof applies to any positive coefficient weights, including those in (4.2).

An absolutely summable family in a Banach space has a sum independent of its enumeration: outside a suitable finite set, the sum of the norms is arbitrarily small. Applying this observation to the family $F_\mu G_\nu$, with weight $\rho^{\mu+\nu}$, gives

$$\sum_{\mu, \nu} \|F_\mu G_\nu\|_A \rho^{\mu+\nu} \leq \left(\sum_\mu \|F_\mu\|_A \rho^\mu \right) \left(\sum_\nu \|G_\nu\|_A \rho^\nu \right).$$

Regrouping by $\mu + \nu$ proves (4.1). For the first weight, use $1 + |\mu + \nu| \leq (1 + |\mu|)(1 + |\nu|)$. The coefficientwise action of L has norm at most $\|L\|$; constant series give the reverse inequality. Finally, the coefficients of $z_i \partial_i F$ are $\nu_i F_\nu$. The inequality $\nu_i \leq 1 + |\nu|$ proves the first bound in (4.3); shifting the i th exponent proves the second. $\square$

Lemma 4.3. *Let E be a complex Banach space, $F \in \mathcal{W}_\rho(E)$, and $0 < r_i < \rho_i$. The series for F converges uniformly on $\overline{D}_\rho$, defines a holomorphic function on D_ρ , and determines its coefficients uniquely. For $\kappa \in \mathbb{N}^h$,*

$$(4.4) \quad \|F_\kappa\|_E \leq \rho^{-\kappa} \|F\|_{\mathcal{W}_\rho(E)}, \quad \partial^\kappa F(0) = \kappa! F_\kappa,$$

$$(4.5) \quad \|\partial^\kappa F\|_{\mathcal{W}_r(E)} \leq \kappa! \prod_{i=1}^h (\rho_i - r_i)^{-\kappa_i} \|F\|_{\mathcal{W}_\rho(E)}.$$

If $F_0 = 0$ and $0 < \theta < 1$, then

$$(4.6) \quad \|F\|_{\mathcal{W}_{\theta\rho}(E)} \leq \theta \|F\|_{\mathcal{W}_\rho(E)}.$$

For $s > 0$, a series $G(\eta) = \sum_\nu a_\nu (s\eta)^\nu$ in $\mathcal{W}_{(1,\dots,1)}(E)$ satisfies, for each fixed κ ,

$$(4.7) \quad \|a_\kappa\|_E \leq s^{-|\kappa|} \|G\|_{\mathcal{W}_{(1,\dots,1)}(E)}, \quad \|\kappa! a_\kappa\|_E \leq \kappa! s^{-|\kappa|} \|G\|_{\mathcal{W}_{(1,\dots,1)}(E)}.$$

Proof. Every evaluation on $\overline{D}_\rho$ is dominated by $\sum_\nu \|F_\nu\|_E \rho^\nu$. This proves uniform convergence and the coefficient bound. To exhibit a local power series at any $z^0 \in D_\rho$, choose positive δ_i with $|z_i^0| + \delta_i < \rho_i$. Expand each monomial $\prod_i (z_i^0 + w_i)^{\nu_i}$ by the binomial formula. The sum of the norms of all resulting terms, weighted at radius δ in w , is at most

$$\sum_\nu \|F_\nu\|_E \prod_i (|z_i^0| + \delta_i)^{\nu_i} \leq \|F\|_{\mathcal{W}_\rho(E)}.$$

Absolute regrouping therefore gives a norm-convergent power series for $F(z^0 + w)$ on a neighborhood of $w = 0$.

Put $t_i = r_i/\rho_i$. For $\nu_i \geq \kappa_i$, the binomial expansion of $(t_i + (1 - t_i))^{\nu_i}$ gives

$$\binom{\nu_i}{\kappa_i} t_i^{\nu_i - \kappa_i} (1 - t_i)^{\kappa_i} \leq 1.$$

Multiplying these inequalities yields

$$\frac{\nu!}{(\nu - \kappa)!} r^{\nu - \kappa} \rho^{-\nu} \leq \kappa! \prod_i (\rho_i - r_i)^{-\kappa_i}.$$

Summation gives (4.5) for the formally differentiated series. These series are the derivatives of F . To check this directly, place a compact polydisc inside a larger polydisc strictly contained in D_ρ . The estimate just proved bounds the second derivative series on the larger polydisc. The monomial identity

$$(w + u)^n - w^n - nw^{n-1}u = u^2 \sum_{j=2}^n \binom{n}{j} w^{n-j} u^{j-2}$$

then bounds the remainder after a coordinate increment, with the segment of that increment in the larger polydisc of radius R , by

$$\frac{|u|^2}{2} \sum_{\nu_i \geq 2} \nu_i(\nu_i - 1) \|F_\nu\|_E R^{\nu - 2e_i},$$

where e_i is the i th coordinate vector. This sum is finite by the derivative estimate already proved. Dividing by u proves coordinate differentiability in the Banach norm. Repetition proves the derivative formula for every κ , and evaluation at zero proves uniqueness. Finite coordinate increments also give the joint differential by telescoping.

If $F_0 = 0$, every remaining term has $|\nu| \geq 1$, so $(\theta\rho)^\nu \leq \theta\rho^\nu$. This proves (4.6). The κ th coefficient of G is $s^{|\kappa|} a_\kappa$; applying (4.4) proves (4.7). $\square$

The factors in (4.7) belong to the specified coefficient or derivative. The estimate does not assert a sum over all recovered derivatives with these factors removed.

Lemma 4.4. *Let E be a complex Banach space and suppose a formal series $F(z) = \sum_{\nu \in \mathbb{N}^h} F_\nu z^\nu$ satisfies $\|F_\nu\|_E \leq BR^{-\nu}$ for positive radii R and $B \geq 0$. For $0 < \rho_i < R_i$ it belongs to $\mathcal{W}_\rho(E)$, with*

$$(4.8) \quad \|F\|_{\mathcal{W}_\rho(E)} \leq B \prod_{i=1}^h (1 - \rho_i/R_i)^{-1}.$$

In two variables with $R_1 = R_2 = R$, $\rho_1 = \rho_2 = r$ and $t = r/R$,

$$(4.9) \quad \|F\|_{\mathcal{W}_{(r,r)}^1(E)} \leq B \frac{1+t}{(1-t)^3}.$$

Proof. The sum of $\prod_i (\rho_i/R_i)^{\nu_i}$ is the product of the h geometric series, which proves (4.8). In two variables the number of monomials of total degree n is $n+1$. The first weighted sum is therefore bounded by $B \sum_{n \geq 0} (n+1)^2 t^n = B(1+t)/(1-t)^3$. The final identity follows by differentiating the scalar geometric series twice on $|t| < 1$, as justified by Lemma 4.3. $\square$

Lemma 4.5. *Let E be a complex Banach space and A a unital complex Banach algebra with submultiplicative norm and $\|1_A\|_A = 1$. Let $L : E \rightarrow A$ be bounded linear, $F \in \mathcal{W}_\rho(E)$, and let $b_1, \dots, b_h \in A$ commute pairwise. If $\|b_i\|_A \leq \theta \rho_i$ for $0 < \theta < 1$, then, with products written in increasing order of indices,*

$$(4.10) \quad F_L(b) = \sum_{\nu} L(F_{\nu}) b_1^{\nu_1} \cdots b_h^{\nu_h}, \quad \|F_L(b)\|_A \leq \|L\| \|F\|_{\mathcal{W}_\rho(E)},$$

$$(4.11) \quad \left\| F_L(b) - \sum_{|\nu| \leq M} L(F_{\nu}) b^{\nu} \right\|_A \leq \|L\| \theta^{M+1} \|F\|_{\mathcal{W}_\rho(E)} \quad (M \in \mathbb{N}).$$

If $c_i \in A$ also commute pairwise and $\|c_i\|_A \leq \theta \rho_i$, then

$$(4.12) \quad \|F_L(b) - F_L(c)\|_A \leq \frac{\|L\|}{1-\theta} \|F\|_{\mathcal{W}_\rho(E)} \max_i \frac{\|b_i - c_i\|_A}{\rho_i}.$$

When $E = A$, $L = \text{id}$, and the b_i commute with all coefficients, substitution also respects Cauchy products.

Proof. The norm of the ν th term is at most $\|L\| \|F_{\nu}\|_E \rho^{\nu} \theta^{|\nu|}$. Summing over all ν , or only over $|\nu| > M$, proves (4.10) and (4.11). This also proves absolute convergence before evaluation in any additional parameters of A .

For the difference estimate, replace one factor at a time in each ordered monomial. The identity $x^n - y^n = \sum_{j=0}^{n-1} x^j (x-y) y^{n-1-j}$ holds without a commutativity assumption on x, y . Consequently, if $|\nu| \geq 1$,

$$\|b^{\nu} - c^{\nu}\|_A \leq \rho^{\nu} \theta^{|\nu|-1} \sum_i \nu_i \frac{\|b_i - c_i\|_A}{\rho_i}.$$

For every integer $n \geq 1$, $n\theta^{n-1} \leq \sum_{j=0}^{n-1} \theta^j \leq (1-\theta)^{-1}$. Summing the preceding bound against $\|L\| \|F_{\nu}\|_E$ proves (4.12). If the substituted elements commute with the coefficients, products can be reordered. Absolute summability, proved above, then identifies the substituted Cauchy product with the product of the substituted series. $\square$

One may take $A = \mathcal{W}_\sigma(B)$ for a unital commutative Banach algebra B . The conclusion is then an estimate for the complete σ -parameter family. In particular, a bounded coefficient map contributes its norm once in (4.11).

Lemma 4.6. *Let A be a unital complex Banach algebra with submultiplicative norm and $\|1_A\|_A = 1$. If $c \in \mathbb{C} \setminus \{0\}$ and $v \in A$ satisfy $\|v\|_A < |c|$, then*

$$(4.13) \quad (c1_A + v)^{-1} = c^{-1} \sum_{n \geq 0} (-v/c)^n, \quad \|(c1_A + v)^{-1} - c^{-1}1_A\|_A \leq \frac{\|v\|_A}{|c|(|c| - \|v\|_A)}.$$

Proof. The sum of the norms is bounded by the convergent scalar geometric series. Multiplication of its partial sums by $c1_A + v$ leaves a remainder whose norm tends to zero. Summing the terms of positive degree gives the bound in (4.13). $\square$

4.2. Closed quotients. For a Banach space E and a closed linear subspace K , the quotient norm is $\|x + K\|_{E/K} = \inf_{k \in K} \|x + k\|_E$. The maps used on quotients below are bounded linear maps, or bounded multilinear maps with the kernel condition in the next lemma.

Lemma 4.7. *Let $E_1, \dots, E_m, H$ be complex Banach spaces, let $K_i \subset E_i$ and $N \subset H$ be closed linear subspaces, and let $B : E_1 \times \dots \times E_m \rightarrow H$ be multilinear with*

$$(4.14) \quad \|B(x_1, \dots, x_m)\|_H \leq C \prod_{i=1}^m \|x_i\|_{E_i}.$$

Suppose $B(x_1, \dots, x_m) \in N$ whenever at least one x_i lies in K_i . Then B induces a multilinear map on the quotients with

$$(4.15) \quad \|\bar{B}(\bar{x}_1, \dots, \bar{x}_m)\|_{H/N} \leq C \prod_i \|\bar{x}_i\|_{E_i/K_i}.$$

These quotients are Banach spaces. In particular, the quotient of a Banach algebra by a closed two-sided ideal has submultiplicative quotient norm. For a complex Banach space E and a closed subspace K , the coefficientwise quotient map induces an isometric isomorphism

$$(4.16) \quad \mathcal{W}_\rho(E)/\mathcal{W}_\rho(K) \rightarrow \mathcal{W}_\rho(E/K).$$

Proof. We prove completeness, the bound for descended maps, and the assertion for families.

Step 1. We prove completeness of the quotient. Let $(\bar{x}_n)$ be Cauchy in E/K . Choose a subsequence whose successive differences have quotient norm less than 2^{-j-1} . Choose representatives u_j of these differences with $\|u_j\|_E < 2^{-j}$. The partial sums of the u_j , added to a representative of the first subsequence term, converge in E . Their quotient classes are precisely the chosen subsequence. Continuity of the quotient map gives a limit for that subsequence, and the Cauchy condition gives a limit for the original sequence. Closedness of K ensures that the quotient seminorm vanishes only on the zero class.

Step 2. We prove independence of representatives. Changing one representative changes B by an element of N . Changing all representatives successively therefore leaves its class unchanged. For each i choose a representative with norm at most $\|\bar{x}_i\|_{E_i/K_i} + \epsilon$. Apply the bound for B and let ϵ decrease to zero. This proves (4.15). Multiplication and a closed two-sided ideal satisfy exactly this kernel condition.

Step 3. We lift a whole family. The coefficientwise map $\mathcal{W}_\rho(E) \rightarrow \mathcal{W}_\rho(E/K)$, before passing to the quotient in (4.16), is contractive and has kernel $\mathcal{W}_\rho(K)$. Given $U = \sum_\nu \bar{u}_\nu z^\nu \in \mathcal{W}_\rho(E/K)$, choose positive numbers ϵ_ν with $\sum_\nu \epsilon_\nu < \epsilon$, and representatives u_ν such that

$$\|u_\nu\|_E \rho^\nu \leq \|\bar{u}_\nu\|_{E/K} \rho^\nu + \epsilon_\nu.$$

The resulting lift lies in $\mathcal{W}_\rho(E)$ and has norm at most $\|U\|_{\mathcal{W}_\rho(E/K)} + \epsilon$. Taking the infimum over lifts and then letting ϵ decrease to zero proves both surjectivity and isometry. $\square$

For a nonlinear map, descent requires equality of output classes on each entire input coset. Merely mapping K into N does not impose that condition. The linear and multilinear hypotheses in Lemma 4.7 make the condition explicit.

4.3. Weighted measures and Laplace transforms. The weight of a measure is part of its source norm. The distance of a Laplace argument from the boundary of convergence provides the derivative and substitution margins.

Definition 4.8. For $a \in \mathbb{R}$, let $\mathcal{M}_a$ be the space of complex Radon measures on $[0, \infty)$ that are finite on compact sets and satisfy

$$(4.17) \quad \|\mu\|_a = \int_{[0, \infty)} e^{-a\beta} d|\mu|(\beta) < \infty.$$

Here $|\mu|$ denotes total variation. For $\operatorname{Re} T \geq a$, define

$$(4.18) \quad \mathcal{L}_T \mu = \int_{[0, \infty)} e^{-T\beta} d\mu(\beta).$$

Convolution is the image under $(\beta, \gamma) \mapsto \beta + \gamma$ of the product measure. The unit point mass at zero is denoted by δ_0 .

Lemma 4.9. *For every $a \in \mathbb{R}$, the space $\mathcal{M}_a$ is a commutative unital Banach algebra under convolution, with*

$$(4.19) \quad \|\mu * \lambda\|_a \leq \|\mu\|_a \|\lambda\|_a, \quad \|\delta_0\|_a = 1.$$

*If $\operatorname{supp} \mu \subset [g, \infty)$ and $\operatorname{supp} \lambda \subset [j, \infty)$ for $g, j \geq 0$, then $\operatorname{supp}(\mu * \lambda) \subset [g + j, \infty)$. The map $\mathcal{L}_T : \mathcal{M}_a \rightarrow \mathbb{C}$ has norm at most one for $\operatorname{Re} T \geq a$ and respects convolution products. For $n \in \mathbb{N}$ and $\delta > 0$, set*

$$(4.20) \quad c_0(\delta) = 1, \quad c_n(\delta) = \left(\frac{n}{e\delta}\right)^n \quad (n \geq 1).$$

Then

$$(4.21) \quad \|\beta^n \mu\|_{a+\delta} \leq c_n(\delta) \|\mu\|_a.$$

Proof. Multiplication by $e^{-a\beta}$ identifies $\mathcal{M}_a$ isometrically with the finite complex Radon measures in total variation norm. This latter space is complete. To see the relevant convergence explicitly, a series of measures $\sum_j \lambda_j$ with $\sum_j |\lambda_j|([0, \infty)) < \infty$ defines a measure by summing on Borel sets. Countable additivity follows by interchanging absolutely convergent sums over disjoint sets. Its variation is bounded by $\sum_j |\lambda_j|$, and the tails converge in variation norm. Regularity follows by approximating a finite initial sum by compact and open sets and bounding the remaining variation uniformly. A Cauchy sequence has a subsequence with summable successive differences, so it converges by this construction. Undoing the weight gives completeness of $\mathcal{M}_a$.

The absolute integral controlling convolution is

$$(4.22) \quad \iint e^{-a(\beta+\gamma)} d|\mu|(\beta) d|\lambda|(\gamma) = \|\mu\|_a \|\lambda\|_a.$$

The integrand in (4.22) is a product of a function of β and a function of γ . The equality holds first for nonnegative simple factors, by the measure of a product set, and then for the displayed factors by increasing approximation. Absolute integrability permits the corresponding complex integrals. The variation of a pushforward is bounded by the pushforward of the variation. This proves (4.19). Regrouping the absolutely integrable triple product proves associativity; interchanging the factors proves commutativity. The support assertion follows because $\beta + \gamma \geq g + j$ on the product of the two supports. The point mass δ_0 is a unit of norm one.

For $\operatorname{Re} T \geq a$, the absolute value of the Laplace kernel is at most $e^{-a\beta}$, proving boundedness. The identity $e^{-T(\beta+\gamma)} = e^{-T\beta} e^{-T\gamma}$, together with (4.22), proves multiplicativity. Finally, the maximum of $\beta^n e^{-\delta\beta}$ on $[0, \infty)$ equals $c_n(\delta)$: for $n > 0$ its derivative changes sign at $\beta = n/\delta$, and for $n = 0$ the maximum is attained at zero. Multiplying this bound by the integrand in (4.17) proves (4.21). $\square$

Lemma 4.10. *Let ρ be positive, $a \in \mathbb{R}$, and $U(z) = \sum_\nu \mu_\nu z^\nu \in \mathcal{W}_\rho(\mathcal{M}_a)$. For $\operatorname{Re} T > a$, the series*

$$(4.23) \quad \mathcal{L}_T U(z) = \sum_\nu (\mathcal{L}_T \mu_\nu) z^\nu$$

is holomorphic in T with values in $\mathcal{W}_\rho(\mathbb{C})$. For $\delta > 0$, $\operatorname{Re} T \geq a + \delta$, and $n \in \mathbb{N}$,

$$(4.24) \quad \|\partial_T^n \mathcal{L}_T U\|_{\mathcal{W}_\rho(\mathbb{C})} \leq c_n(\delta) \|U\|_{\mathcal{W}_\rho(\mathcal{M}_a)}, \quad \partial_T^n \mathcal{L}_T \mu_\nu = \int (-\beta)^n e^{-T\beta} d\mu_\nu(\beta).$$

For $0 < r < \rho$ and $\kappa \in \mathbb{N}^h$, the joint estimate is

$$(4.25) \quad \|\partial_z^\kappa \partial_T^n \mathcal{L}_T U\|_{\mathcal{W}_r(\mathbb{C})} \leq \kappa! \prod_i (\rho_i - r_i)^{-\kappa_i} c_n(\delta) \|U\|_{\mathcal{W}_\rho(\mathcal{M}_a)}.$$

If every μ_ν is supported in $[g, \infty)$ for $g \geq 0$, then

$$(4.26) \quad \|\mathcal{L}_T U\|_{\mathcal{W}_\rho(\mathbb{C})} \leq e^{-g(\operatorname{Re} T - a)} \|U\|_{\mathcal{W}_\rho(\mathcal{M}_a)} \quad (\operatorname{Re} T \geq a).$$

Proof. For a single measure, all displayed derivatives are absolutely integrable by (4.21). The exponential power series gives

$$|e^{-u\beta} - 1 + u\beta| \leq \frac{1}{2}|u|^2 \beta^2 e^{|u|\beta}.$$

If $\operatorname{Re} T \geq a + \delta$ and $|u| \leq \delta/2$, the norm of the difference between the first difference quotient and the proposed first derivative is at most

$$\frac{1}{2}|u|c_2(\delta/2)\|\mu\|_a.$$

Thus the derivative exists as a bounded functional on $\mathcal{M}_a$. Inserting β^n in this calculation replaces c_2 by c_{n+2} and proves all higher derivative formulas. Multiplying each estimate by ρ^ν and summing proves differentiation in the whole $\mathcal{W}_\rho$ norm and (4.24). Lemma 4.3 then proves (4.25). On the asserted supports, $e^{-(\operatorname{Re} T - a)\beta} \leq e^{-g(\operatorname{Re} T - a)}$; integrating coefficientwise proves (4.26). $\square$

Lemma 4.11. *Let $U \in \mathcal{W}_\rho(\mathcal{M}_a)$, $\delta > 0$, and $q \in \mathcal{W}_\rho(\mathbb{C})$ satisfy $\|q\|_{\mathcal{W}_\rho(\mathbb{C})} \leq \theta\delta$ for $0 < \theta < 1$. For $\operatorname{Re} T \geq a + \delta$, set*

$$(4.27) \quad H(T, z) = \sum_{n \geq 0} \frac{(-q(z))^n}{n!} \sum_\nu z^\nu \int \beta^n e^{-T\beta} d\mu_\nu(\beta), \quad U(z) = \sum_\nu \mu_\nu z^\nu.$$

The series converges absolutely in $\mathcal{W}_\rho(\mathbb{C})$ and satisfies

$$(4.28) \quad \|H(T, \cdot)\|_{\mathcal{W}_\rho(\mathbb{C})} \leq \|U\|_{\mathcal{W}_\rho(\mathcal{M}_a)},$$

$$(4.29) \quad \left\| \sum_{n > M} \frac{(-q)^n}{n!} \sum_\nu z^\nu \int \beta^n e^{-T\beta} d\mu_\nu \right\|_{\mathcal{W}_\rho(\mathbb{C})} \leq \theta^{M+1} \|U\|_{\mathcal{W}_\rho(\mathcal{M}_a)} \quad (M \in \mathbb{N}).$$

For $z \in \overline{D}_\rho$ its evaluation equals $\mathcal{L}_{T+q(z)}(U(z))$. It is holomorphic in T on $\operatorname{Re} T > a + \|q\|_{\mathcal{W}_\rho(\mathbb{C})}$, and on the stated half-plane

$$(4.30) \quad \|\partial_T^j H(T, \cdot)\|_{\mathcal{W}_\rho(\mathbb{C})} \leq c_j((1 - \theta)\delta) \|U\|_{\mathcal{W}_\rho(\mathcal{M}_a)} \quad (j \in \mathbb{N}).$$

Proof. Write $M_n(T, z)$ for the inner series in (4.27). Nonnegative summation under the variation integrals gives

$$\begin{aligned} \sum_{n \geq 0} \frac{\delta^n}{n!} \|M_n(T, \cdot)\|_{\mathcal{W}_\rho(\mathbb{C})} &\leq \sum_\nu \rho^\nu \int e^{-\operatorname{Re} T\beta} \sum_{n \geq 0} \frac{(\delta\beta)^n}{n!} d|\mu_\nu|(\beta) \\ &= \sum_\nu \rho^\nu \int e^{-(\operatorname{Re} T - \delta)\beta} d|\mu_\nu|(\beta) \leq \|U\|_{\mathcal{W}_\rho(\mathcal{M}_a)}. \end{aligned}$$

Multiplication by q^n contributes at most $(\theta\delta)^n$. Keeping all terms proves (4.28); keeping $n > M$ and using $\theta^n \leq \theta^{M+1}$ proves (4.29).

Evaluation has norm at most one. The preceding absolute majorant therefore permits evaluation, summation and integration in either order. The sum of the exponential series under the integral is $e^{-(T+q(z))\beta}$, proving the claimed equality.

For $\operatorname{Re} T > a + \|q\|_{\mathcal{W}_\rho(\mathbb{C})}$, put $d = \operatorname{Re} T - a - \|q\|_{\mathcal{W}_\rho(\mathbb{C})} > 0$ and define, for $j \in \mathbb{N}$,

$$(4.31) \quad J_j(T, z) = \sum_{n \geq 0} \frac{q(z)^n}{n!} \sum_\nu z^\nu \int (-\beta)^{n+j} e^{-T\beta} d\mu_\nu(\beta).$$

The sum of the norms of its terms is bounded by

$$\sum_{\nu} \rho^{\nu} \int \beta^j e^{-(\operatorname{Re} T - \|q\|_{\mathcal{W}_\rho(\mathbb{C})})\beta} d|\mu_{\nu}|(\beta) \leq c_j(d) \|U\|_{\mathcal{W}_\rho(\mathcal{M}_a)}.$$

Thus every J_j converges in the whole family norm and $J_0 = H$. For $0 < |u| \leq d/2$, the exponential series gives $|e^{-u\beta} - 1 + u\beta| \leq |u|^2 \beta^2 e^{|u|\beta}/2$. Insert this bound in the difference quotient of (4.31) and sum the absolute majorants. The result is

$$\begin{aligned} \left\| \frac{J_j(T + u, \cdot) - J_j(T, \cdot)}{u} - J_{j+1}(T, \cdot) \right\|_{\mathcal{W}_\rho(\mathbb{C})} &\leq \frac{|u|}{2} \sum_{\nu} \rho^{\nu} \int \beta^{j+2} e^{-(a+d-|u|)\beta} d|\mu_{\nu}|(\beta) \\ (4.32) \qquad \qquad \qquad &\leq \frac{|u|}{2} c_{j+2}(d/2) \|U\|_{\mathcal{W}_\rho(\mathcal{M}_a)}. \end{aligned}$$

Consequently $\partial_T J_j = J_{j+1}$ in $\mathcal{W}_\rho(\mathbb{C})$. Expanding $e^{-u\beta}$ under the absolute majorant also yields $H(T + u, \cdot) = \sum_{j \geq 0} J_j(T, \cdot) u^j / j!$ in that norm for $|u| < d$: the sum of absolute majorants is bounded by $\sum_{\nu} \rho^{\nu} \int e^{-(a+d-|u|)\beta} d|\mu_{\nu}|$. This proves holomorphy on the asserted open half-plane. On $\operatorname{Re} T \geq a + \delta$, we have $d \geq (1 - \theta)\delta$, so the bound for J_j proves (4.30). $\square$

The estimates in this subsection concern absolute convergence and operator bounds. A conclusion about leading terms of a signed measure's Laplace transform would require additional hypotheses on that measure.

4.4. A local moment convolution. The next construction is a local coefficient space for a fixed positive integer k . Its integral product has an explicit normalization.

Definition 4.12. For real $x, y > 0$, write

$$(4.33) \qquad \Gamma(x) = \int_0^\infty t^{x-1} e^{-t} dt, \qquad B(x, y) = \int_0^1 t^{x-1} (1-t)^{y-1} dt,$$

where the integrals are improper integrals of nonnegative functions. For an integer $k \geq 1$, set $C_k = B(1/k, 1/k)$. For $r > 0$, let $\mathcal{W}_r^0(\mathbb{C})$ be the closed subspace of $\mathcal{W}_r(\mathbb{C})$ consisting of series with zero constant coefficient.

Lemma 4.13. *The integrals in (4.33) are finite and positive. For $x, y > 0$,*

$$(4.34) \qquad \Gamma(1) = 1, \qquad \Gamma(x+1) = x\Gamma(x), \qquad B(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}.$$

Proof. Near zero the integrand defining $\Gamma(x)$ is bounded by t^{x-1} . For $t \geq 1$, choose an integer $m > x$. The positive exponential series gives $e^{t/2} \geq (t/2)^m / m!$, and consequently

$$t^{x-1} e^{-t} \leq 2^m m! e^{-t/2}.$$

These bounds prove convergence at both endpoints. On $(0, 1/2)$ the second factor in the integrand defining $B(x, y)$ is bounded; on $(1/2, 1)$ its first factor is bounded. Comparison with the powers t^{x-1} and $(1-t)^{y-1}$ proves convergence. Positivity follows by integration over any compact interval in the interior. Direct integration gives $\Gamma(1) = 1$. Integration by parts on a compact interval, followed by the endpoint limits $t^x e^{-t} \rightarrow 0$, gives the recurrence for Γ .

For the product identity we use positive double integrals only. Interchanging the order for a continuous nonnegative integrand on $(0, \infty)^2$ follows by integration on compact rectangles and taking the supremum over the rectangles. On a compact rectangle the equality of the two iterated integrals follows from their common rectangular Riemann sums. The two increasing suprema commute, proving the assertion also for the improper integrals below.

First substitute $v = ut$ in the inner integral defining the product $\Gamma(x)\Gamma(y)$ and then interchange the positive integrations. The one-variable changes of variables can first be made on compact intervals and then passed to increasing endpoint limits. They give

$$\begin{aligned}\Gamma(x)\Gamma(y) &= \int_0^\infty \int_0^\infty u^{x-1} v^{y-1} e^{-u-v} dv du \\ &= \int_0^\infty \int_0^\infty u^{x+y-1} t^{y-1} e^{-u(1+t)} du dt \\ &= \Gamma(x+y) \int_0^\infty \frac{t^{y-1}}{(1+t)^{x+y}} dt.\end{aligned}$$

The last equality uses $w = u(1+t)$ in the inner integral. Finally $s = t/(1+t)$ transforms the remaining integral into $\int_0^1 s^{y-1} (1-s)^{x-1} ds = B(x, y)$, after replacing s by $1-s$. Division by the positive number $\Gamma(x+y)$ proves (4.34). $\square$

Lemma 4.14. *Let $k \geq 1$ be an integer and $r > 0$. For $\varphi, \psi \in \mathcal{W}_r^0(\mathbb{C})$ define, for $|\xi| \leq r$,*

$$(4.35) \quad (\varphi *_k \psi)(\xi) = \int_0^1 \varphi(\xi(1-t)^{1/k}) \psi(\xi t^{1/k}) \frac{dt}{t(1-t)},$$

where the roots of t and $1-t$ are the positive real roots. This integral defines a commutative associative product on $\mathcal{W}_r^0(\mathbb{C})$ and satisfies

$$(4.36) \quad \|\varphi *_k \psi\|_{\mathcal{W}_r(\mathbb{C})} \leq C_k \|\varphi\|_{\mathcal{W}_r(\mathbb{C})} \|\psi\|_{\mathcal{W}_r(\mathbb{C})}.$$

The unitization $\mathbb{C} \oplus \mathcal{W}_r^0(\mathbb{C})$, with product

$$(4.37) \quad (a, \varphi)(b, \psi) = (ab, a\psi + b\varphi + \varphi *_k \psi)$$

and norm $|a| + C_k \|\varphi\|_{\mathcal{W}_r(\mathbb{C})}$, is a unital commutative Banach algebra.

Proof. For positive integers p, q , integration of two monomials gives

$$(4.38) \quad \xi^p *_k \xi^q = B(p/k, q/k) \xi^{p+q} = \frac{\Gamma(p/k)\Gamma(q/k)}{\Gamma((p+q)/k)} \xi^{p+q}.$$

For $0 < t < 1$, $p, q \geq 1$ imply

$$t^{q/k-1} (1-t)^{p/k-1} \leq t^{1/k-1} (1-t)^{1/k-1}.$$

Consequently $B(p/k, q/k) \leq C_k$. If $\varphi = \sum_{p \geq 1} c_p \xi^p$ and $\psi = \sum_{q \geq 1} d_q \xi^q$, the sum of the absolute integral majorants, weighted at radius r , is at most

$$\sum_{p, q \geq 1} |c_p| |d_q| r^{p+q} B(p/k, q/k) \leq C_k \|\varphi\|_{\mathcal{W}_r(\mathbb{C})} \|\psi\|_{\mathcal{W}_r(\mathbb{C})}.$$

Thus the endpoint integral exists, its monomial expansion may be integrated term by term, and the resulting series has the asserted norm. This proves (4.36). The coefficient multiplying $\xi^{p+q+\ell}$ in either bracketing of three monomials is

$$\frac{\Gamma(p/k)\Gamma(q/k)\Gamma(\ell/k)}{\Gamma((p+q+\ell)/k)}.$$

Hence the product is associative on polynomials. Its bound and density of the zero-constant polynomials extend associativity to the complete space. Symmetry of (4.38) proves commutativity. Finally,

$$|ab| + C_k \|a\psi + b\varphi + \varphi *_k \psi\|_{\mathcal{W}_r(\mathbb{C})} \leq (|a| + C_k \|\varphi\|_{\mathcal{W}_r(\mathbb{C})}) (|b| + C_k \|\psi\|_{\mathcal{W}_r(\mathbb{C})}).$$

Completeness follows from Lemma 4.2, and $(1, 0)$ is a unit of norm one. $\square$

Definition 4.15. For an integer $k \geq 1$ and $r > 0$, let $\mathcal{G}_{k,r}$ consist of the formal series $f(u) = \sum_{n \geq 0} a_n u^n$ with

$$(4.39) \quad \|f\|_{k,r} = |a_0| + C_k \sum_{n \geq 1} \frac{|a_n| r^n}{\Gamma(n/k)} < \infty.$$

Define the normalized coefficient transform by

$$(4.40) \quad \mathcal{B}_k f = \left(a_0, \sum_{n \geq 1} \frac{a_n}{\Gamma(n/k)} \xi^n \right).$$

Lemma 4.16. *With ordinary Cauchy multiplication of formal series, $\mathcal{G}_{k,r}$ is a unital commutative Banach algebra, and $\mathcal{B}_k$ is an isometric algebra isomorphism onto the unitization in Lemma 4.14. If $\mathcal{B}_k f = (a_0, \varphi)$, then*

$$(4.41) \quad \mathcal{B}_k(u^{k+1} f'(u)) = (0, k \xi^k \varphi), \quad \|u^{k+1} f'(u)\|_{k,r} \leq k r^k \|f\|_{k,r}.$$

If $|a_n| \leq C A^n \Gamma(1 + n/k)$ for $n \geq 1$, with $C \geq 0$, $A > 0$, and $r < A^{-1}$, then

$$(4.42) \quad \|f\|_{k,r} \leq |a_0| + \frac{C_k C}{k} \frac{A r}{(1 - A r)^2}.$$

Proof. The transform is an isometric linear bijection by its coefficient formula and the positivity of $\Gamma(n/k)$. Dividing (4.38) by $\Gamma(p/k)\Gamma(q/k)$ proves that it respects products on monomials. The bound in Lemma 4.14 and coefficientwise finite sums for each formal product extend this identity to every pair of series. Completeness and submultiplicativity follow through this isometry.

The coefficient of u^{n+k} in $u^{k+1} f'(u)$ is $n a_n$ for $n \geq 1$. By (4.34),

$$\frac{n a_n}{\Gamma((n+k)/k)} = \frac{k a_n}{\Gamma(n/k)}.$$

This proves (4.41), including its norm estimate. Under the last hypothesis,

$$\frac{|a_n| r^n}{\Gamma(n/k)} \leq \frac{C}{k} n (A r)^n.$$

Summing the differentiated geometric series proves (4.42). $\square$

This construction provides a convergent transform on $|\xi| < r$. It imposes no continuation or growth condition outside that disc.

4.5. Analytic differential equations with parameters. We construct solutions in a space of convergent series. This gives a bound on the complete parameter family as part of the existence argument. For a unital commutative complex Banach algebra A , we equip $\mathcal{W}_\rho(A)^d$ with the product norm $\|u\| = \max_{1 \leq j \leq d} \|u_j\|_{\mathcal{W}_\rho(A)}$. This convention also applies when a time variable is adjoined.

Lemma 4.17. *Let A be a unital commutative complex Banach algebra with submultiplicative norm and $\|1_A\|_A = 1$. Let $d \geq 1$, $R > 0$, and ρ be positive. Suppose*

$$(4.43) \quad f(y, z) = \sum_{\alpha \in \mathbb{N}^d} f_\alpha(z) y^\alpha, \quad f_\alpha \in \mathcal{W}_\rho(A)^d, \quad B = \sum_{\alpha} \|f_\alpha\| R^{|\alpha|} < \infty.$$

Let $0 \leq r < s < R$, let $u_0 \in \mathcal{W}_\rho(A)^d$ satisfy $\|u_0\| \leq r$, and choose $\tau > 0$ such that

$$(4.44) \quad r + \tau B \leq s, \quad q = \frac{\tau B}{R - s} < 1.$$

Then the equation

$$(4.45) \quad \partial_t u = f(u, z), \quad u(0, z) = u_0(z)$$

has a unique solution in the closed ball of radius s in $\mathcal{W}_{(\tau,\rho)}(A)^d$. It satisfies

$$(4.46) \quad \|u - u_0\| \leq \tau B.$$

If $u^{(0)} = u_0$ and $u^{(n+1)} = u_0 + \int_0^t f(u^{(n)}(v, z), z) dv$, where integration is coefficientwise, then

$$(4.47) \quad \|u - u^{(n)}\| \leq \frac{\tau B q^n}{1 - q} \quad (n \in \mathbb{N}).$$

All these norms use the parameter radii ρ in (4.43). Parameter and time derivatives on strictly smaller polydiscs satisfy (4.5) with input radius (τ, ρ) .

Proof. Let $\mathcal{B} = \mathcal{W}_{(\tau,\rho)}(A)$ and regard time-independent series as elements of $\mathcal{B}$. Coefficientwise integration

$$I \left(\sum_{n,\nu} a_{n,\nu} t^n z^\nu \right) = \sum_{n,\nu} \frac{a_{n,\nu}}{n+1} t^{n+1} z^\nu$$

has norm at most τ . For $u \in \mathcal{B}^d$ with $\|u\| \leq s$, absolute multiplication gives $\|f(u, z)\| \leq B$. Telescoping each state monomial, as in Lemma 4.5, gives, for u, v in this ball,

$$\begin{aligned} \|f(u, z) - f(v, z)\| &\leq \sum_{|\alpha| \geq 1} \|f_\alpha\| |\alpha| s^{|\alpha|-1} \|u - v\| \\ &\leq \frac{B}{R - s} \|u - v\|. \end{aligned}$$

The last bound follows from $n(s/R)^{n-1} \leq (1 - s/R)^{-1}$. Thus $u \mapsto u_0 + If(u, z)$ preserves the ball by (4.44) and decreases distances by a factor at most q .

The first successive difference of the iterates has norm at most τB ; the n th successive difference has norm at most $\tau B q^{n-1}$. Completeness and summation of the geometric series give a limit u and (4.47). Continuity of the iteration map gives its fixed-point equation, which also yields (4.46). If v is another fixed point in the ball, then $\|u - v\| \leq q\|u - v\|$ and hence $u = v$. Differentiation on the open time disc, justified by Lemma 4.3, identifies the fixed-point equation with (4.45). Conversely, comparison of time coefficients in that differential equation gives the fixed-point equation for any solution in the asserted ball. Finally, (4.5) applies to each component with the already established (τ, ρ) norm. $\square$

Lemma 4.18. *Let G be a holomorphic map from a neighborhood of $p \in \mathbb{C}^d$ to $\mathbb{C}^d$, represented there by convergent power series, and suppose $DG(p)$ is invertible. Given neighborhoods U_* of p and V_* of $G(p)$, there exist open neighborhoods $U \subset U_*$ and $V \subset V_*$ such that $G : U \rightarrow V$ has a holomorphic inverse. The inverse can be obtained as a convergent series in a coefficient space. Its derivative is $(DG)^{-1}$ at corresponding points. The assertion holds for real analytic maps with real analytic inverses.*

If $H(x, y)$ is analytic, $H(p, q) = 0$, and $D_y H(p, q)$ is invertible, then, on sufficiently small neighborhoods, its zero set is exactly a graph $y = \varphi(x)$ with φ analytic.

Proof. We first construct the inverse, then choose the source and target neighborhoods so that they correspond exactly.

Step 1. We construct the solution in a coefficient space. Translations and multiplication of the target by $DG(p)^{-1}$ reduce the problem to $G(x) = x + N(x)$, where $N(0) = 0$ and $DN(0) = 0$. Write $N_j(x) = \sum_{|\alpha| \geq 2} n_{j,\alpha} x^\alpha$. Choose $R > 0$ such that all these series converge absolutely at radius R . For sufficiently small $r \in (0, R)$,

$$(4.48) \quad L(r) = \max_j \sum_{|\alpha| \geq 2} |n_{j,\alpha}| |\alpha| r^{|\alpha|-1} < 1.$$

To justify this choice from the fixed R -norm, take $r \leq R/2$. The summands are bounded by a fixed constant times $R^{-1}|n_{j,\alpha}|R^{|\alpha|}$ and tend to zero for each α as r tends to zero. A finite initial sum and its uniformly small summable tail therefore show $L(r) \rightarrow 0$.

Choose $\epsilon > 0$ with $\epsilon + rL(r) < r$. On the ball of radius r in $\mathcal{W}_{(\epsilon, \dots, \epsilon)}(\mathbb{C})^d$, the map $v \mapsto y - N(v)$ preserves the ball and decreases distances by a factor at most $L(r)$. These statements follow by telescoping monomials; the bound for $N(v)$ is $rL(r)$, and each coordinate function y_j has coefficient norm ϵ . Set $v^{(0)} = 0$ and $v^{(n+1)} = y - N(v^{(n)})$. Every iterate lies in the ball, and

$$\|v^{(n+1)} - v^{(n)}\| \leq \epsilon L(r)^n, \quad \sum_{j \geq n} \|v^{(j+1)} - v^{(j)}\| \leq \frac{\epsilon L(r)^n}{1 - L(r)}.$$

Completeness from Lemma 4.2 therefore gives a limit v in the ball. The contraction estimate implies $N(v^{(n)}) \rightarrow N(v)$, so passing to the limit in the iteration gives $v = y - N(v)$. It is holomorphic by Lemma 4.3 and satisfies $G(v(y)) = y$.

Step 2. We specify the exact neighborhoods. For x, x' in the radius r polydisc, telescoping gives

$$\|N(x) - N(x')\|_\infty \leq L(r)\|x - x'\|_\infty.$$

Thus G is injective there. Make r small enough that this polydisc lies in the prescribed source neighborhood. Choose a target polydisc V of radius less than ϵ contained in the prescribed target neighborhood, and put $U = D_{(r, \dots, r)} \cap G^{-1}(V)$ in the normalized coordinates. The constructed inverse takes V into $D_{(r, \dots, r)}$ because its norm is at most $\epsilon + rL(r) < r$. Consequently $G : U \rightarrow V$ is bijective with inverse v . Differentiating both inverse identities by the power-series chain rule gives the derivative formula. The chain rule here follows by substituting the linear increments and the quadratic remainders of the convergent series; the latter are bounded on a smaller polydisc by Lemma 4.3.

Real coefficients are preserved by every iterate, giving the real analytic assertion. Finally apply the inverse construction to the map $(x, y) \mapsto (x, H(x, y))$, whose derivative is invertible at (p, q) . The inverse preserves the first coordinate. On a sufficiently small product neighborhood of $(p, 0)$ in its target, write it as $(x, w) \mapsto (x, \psi(x, w))$. Then $\varphi(x) = \psi(x, 0)$ has the claimed property, including uniqueness of every nearby zero. $\square$

Lemma 4.19. *Let Y be a real analytic vector field near $p \in \mathbb{R}^d$. For sufficiently small real t and x near p , there exists a jointly real analytic solution $\Phi(t, x)$ of*

$$(4.49) \quad \partial_t \Phi(t, x) = Y(\Phi(t, x)), \quad \Phi(0, x) = x.$$

Whenever the indicated local trajectories are defined in a sufficiently small common neighborhood,

$$(4.50) \quad \Phi(t, \Phi(s, x)) = \Phi(t + s, x), \quad \Phi(-t, \Phi(t, x)) = x.$$

The matrix $A(t, x) = D_x \Phi(t, x)$ satisfies

$$(4.51) \quad \partial_t A(t, x) = DY(\Phi(t, x))A(t, x), \quad A(0, x) = \text{id}.$$

In dimension two, suppose $Y(p) \neq 0$ and an analytic arc σ satisfies $\sigma(0) = p$ with $\sigma'(0)$ and $Y(p)$ linearly independent. There exists an analytic coordinate map

$$(4.52) \quad \Psi(s, t) = \Phi(t, \sigma(s))$$

from a small rectangle onto its image. If its inverse coordinates are (S, Θ) , then

$$(4.53) \quad DS \cdot Y = 0, \quad D\Theta \cdot Y = 1.$$

Within this coordinate neighborhood, an orbit segment that stays in the neighborhood meets each slice $\Theta = \tau$ at most once.

Proof. Translate p to zero and complexify the convergent power series of Y . Choose a state radius R at which its coefficient norm B is finite. Choose a parameter radius $\rho_1 = \dots = \rho_d = r > 0$ and a number s with $r < s < R$. In Lemma 4.17, take $A = \mathbb{C}$, $u_0(z) = z$, and the coefficients of $f = Y$ independent of z . A sufficiently small $\tau > 0$ satisfies (4.44). The solution furnished by that lemma has real coefficients and proves joint real analyticity.

For completeness, this analytic solution is unique among local differentiable real solutions with the prescribed initial point. Write $Y(y) = \sum_{\alpha} Y_{\alpha} y^{\alpha}$, so that $B = \sum_{\alpha} \|Y_{\alpha}\|_{\infty} R^{|\alpha|}$. For $\|v\|_{\infty}, \|w\|_{\infty} \leq s$, telescoping each monomial and using $n(s/R)^{n-1} \leq (1 - s/R)^{-1}$ for $n \geq 1$ give

$$\|Y(v) - Y(w)\|_{\infty} \leq \left(\sum_{|\alpha| \geq 1} \|Y_{\alpha}\|_{\infty} |\alpha| s^{|\alpha|-1} \right) \|v - w\|_{\infty} \leq \frac{B}{R - s} \|v - w\|_{\infty}.$$

Put $K = B/(R - s)$. If two solutions agree at a time t_0 , their difference on an interval of length ℓ with $\ell K < 1$ has supremum norm at most ℓK times that supremum norm, by integrating their differential equations. It therefore vanishes. Subdivision proves uniqueness on every common sufficiently small time interval.

The two sides of the first identity in (4.50), as functions of t , solve the equation with initial point $\Phi(s, x)$; uniqueness proves equality. Putting $t = -s$ gives the inverse identity. Joint analyticity permits differentiating (4.49) with respect to x and interchanging the derivatives, which proves (4.51).

The derivative of Ψ at $(0, 0)$ has columns $\sigma'(0)$ and $Y(p)$, so Lemma 4.18 applies. Restrict its source to a small rectangle on which it is a coordinate map. Differentiating Ψ in t gives $Y \circ \Psi$. Differentiating the identities $S \circ \Psi = s$ and $\Theta \circ \Psi = t$ now gives (4.53). Along an orbit segment inside this neighborhood, S is constant and Θ increases at unit rate in the flow parameter. Thus a fixed slice can be met at most once by that segment. $\square$

The local flow construction also shows that the slices $\Theta = \tau$ are transverse to Y . The last assertion of Lemma 4.19 is confined to a single coordinate neighborhood.

Proposition 4.20. *Let ρ be a positive radius vector, let E, H be complex Banach spaces, and let A be a unital complex Banach algebra with submultiplicative norm and $\|1_A\|_A = 1$. Every bounded linear map $L : E \rightarrow H$ acts coefficientwise from $\mathcal{W}_{\rho}(E)$ to $\mathcal{W}_{\rho}(H)$ with operator norm $\|L\|$. For $F, G \in \mathcal{W}_{\rho}(A)$, $\|FG\|_{\mathcal{W}_{\rho}(A)} \leq \|F\|_{\mathcal{W}_{\rho}(A)} \|G\|_{\mathcal{W}_{\rho}(A)}$. If $b_1, \dots, b_h \in A$ commute pairwise and $\|b_i\|_A \leq \theta \rho_i$ for $0 < \theta < 1$, then $\sum_{\nu} F_{\nu} b^{\nu}$ converges absolutely and, for $M \in \mathbb{N}$,*

$$(4.54) \quad \left\| \sum_{|\nu| > M} F_{\nu} b^{\nu} \right\|_A \leq \theta^{M+1} \|F\|_{\mathcal{W}_{\rho}(A)}.$$

Let $m \geq 1$, let E_i, H_0 be complex Banach spaces, and let $K_i \subset E_i$, $N \subset H_0$ be closed linear subspaces. A bounded multilinear map $B : E_1 \times \dots \times E_m \rightarrow H_0$ with $\|B(x_1, \dots, x_m)\|_{H_0} \leq C \prod_i \|x_i\|_{E_i}$ descends to the quotients whenever its value lies in N if any input lies in its corresponding K_i ; the descended map satisfies

$$(4.55) \quad \|\overline{B}(\overline{x}_1, \dots, \overline{x}_m)\|_{H_0/N} \leq C \prod_i \|\overline{x}_i\|_{E_i/K_i}.$$

The coefficient and derivative estimates of Lemma 4.3, the Laplace estimates of Lemmas 4.10 and 4.11, and the ODE estimates of Lemma 4.17 hold in the complete parameter spaces and the original input norms specified there.

Proof. Lemma 4.2 proves the operator and product assertions. The substitution assertion is Lemma 4.5 with $E = A$ and $L = \text{id}$. The multilinear map B satisfies the kernel condition of Lemma 4.7, so (4.15) proves descent with the asserted bound, and (4.16) identifies the family quotient without changing the input radii.

Coefficient and parameter derivative bounds are (4.4)–(4.7). The Laplace estimates are (4.24)–(4.26), and strict substitution of a whole perturbation in its argument is (4.28)–(4.30). Lemma 4.17 constructs the local ODE solution in a complete parameter space and gives its norm and derivative bounds. Each estimate is established before evaluation of its parameters. $\square$

4.6. Supremum bounds and coefficient norms. A bound on a holomorphic map supplies a coefficient norm on every strictly smaller polydisc. We prove this assertion directly for Banach values. A contour integral with values in a Banach space means the norm Riemann integral of the parametrized integrand, summed over the continuously differentiable pieces of the contour.

Lemma 4.21. *Let E be a complex Banach space and $I = \prod_{j=1}^m [a_j, b_j]$, where $m \geq 1$ and $a_j < b_j$. Every continuous map $v : I \rightarrow E$ has a norm Riemann integral, and all orders of iterated integration exist and agree with it. Moreover,*

$$(4.56) \quad \left\| \int_I v(t) dt \right\|_E \leq \prod_{j=1}^m (b_j - a_j) \sup_{t \in I} \|v(t)\|_E.$$

If $\gamma : [0, 1] \rightarrow \mathbb{C}$ is piecewise continuously differentiable and $u : \gamma([0, 1]) \rightarrow E$ is continuous, then the norm contour integral exists and

$$(4.57) \quad \left\| \int_{\gamma} u(\zeta) d\zeta \right\|_E \leq \text{length}(\gamma) \sup_{\zeta \in \gamma} \|u(\zeta)\|_E.$$

Reversing a path negates its integral, and concatenating finitely many paths adds their integrals. Uniform limits of continuous integrands commute with each of these rectangle and contour integrals.

Proof. Uniform continuity makes the Riemann sums for v Cauchy. Comparing two tagged rectangular partitions with their common refinement bounds the difference of their sums by the volume of I times the sum of the moduli of continuity at the two mesh sizes. Completeness of E supplies the limit, and the triangle inequality for the sums proves (4.56).

In one variable the triangle inequality also gives $\|\int_a^b v(t) dt\|_E \leq \int_a^b \|v(t)\|_E dt$. An integral in one coordinate depends continuously on the remaining coordinates: its change is bounded by the interval length times the uniform change of its integrand. Iterating this observation proves existence of every iterated integral. To compare them, approximate v uniformly by functions constant on the cells of a rectangular grid. For each such function every order gives its finite sum of cell values times cell volumes. The error in the rectangle integral and in every iterated integral is at most the volume of I times the uniform approximation error. Passing to the limit identifies all these integrals. The displayed bound also proves their compatibility with uniform limits.

On each continuously differentiable piece of γ , integrate $u(\gamma(t))\gamma'(t)$ in the Banach norm. The one-variable inequality bounds the sum over the pieces by $\sup_{\gamma} \|u\|_E \int_0^1 |\gamma'(t)| dt$, which proves (4.57). Reversal and finite concatenation follow from the defining sums. The contour bound applied to a difference proves compatibility with uniform limits. $\square$

Lemma 4.22. *Let E be a complex Banach space, $\Omega \subset \mathbb{C}$ open, and $g : \Omega \rightarrow E$ holomorphic in the sense of local norm-convergent power series. If $r > 0$ and $\{\zeta : |\zeta| \leq r\} \subset \Omega$, then, for $|w| < r$,*

$$(4.58) \quad g(w) = \frac{1}{2\pi i} \int_{|\zeta|=r} \frac{g(\zeta)}{\zeta - w} d\zeta.$$

The circle is oriented counterclockwise and the integral exists in the norm of E .

Proof. Lemma 4.21 supplies the norm integrals and their bounds. We construct local primitives from power series and compare the outer circle with a small circle about w by a finite subdivision.

Step 1. We construct local primitives. Suppose $v(\zeta) = \sum_{n \geq 0} a_n(\zeta - c)^n$ is norm convergent on a disc about c . Choose $b > 0$ such that the series converges at $\zeta = c + b$. Its terms are bounded, so $\|a_n\|_E b^n \leq C$ for some C . For each $0 < d < b$,

$$\sum_{n \geq 0} \|a_n\|_E d^n \leq C \sum_{n \geq 0} (d/b)^n < \infty.$$

Hence on the smaller disc the series and the integrated series

$$P(\zeta) = \sum_{n \geq 0} \frac{a_n}{n+1} (\zeta - c)^{n+1}$$

converge absolutely in norm, and $P' = v$ by Lemma 4.3. Along a path with compact image in this disc, integration term by term and the scalar polynomial chain rule give

$$\int_{\gamma} v(\zeta) d\zeta = P(\gamma(1)) - P(\gamma(0)).$$

In particular, every closed path in such a disc has zero integral.

Step 2. We perform a finite contour deformation. Fix w as in the statement and put $v_w(\zeta) = g(\zeta)/(\zeta - w)$ on $\Omega \setminus \{w\}$. Near any point different from w , the reciprocal factor has a convergent geometric series. Multiplying it by a local series for g shows that v_w has local norm-convergent series, and hence the local primitives just constructed.

Choose $0 < \epsilon < r - |w|$ so small that the series for g about w converges absolutely on $|\zeta - w| \leq \epsilon$. Consider the smooth map

$$(4.59) \quad H(s, t) = sw + ((1-s)r + s\epsilon)e^{it}, \quad (s, t) \in [0, 1] \times [0, 2\pi].$$

Its image stays in $\Omega \setminus \{w\}$. More precisely,

$$\begin{aligned} |H(s, t)| &\leq (1-s)r + s(|w| + \epsilon) \leq r, \\ |H(s, t) - w| &\geq (1-s)(r - |w|) + s\epsilon > 0. \end{aligned}$$

Cover this compact image by discs on which v_w has a primitive. There exists a finite rectangular subdivision of the parameter rectangle whose every cell maps into one such disc. To verify the needed fineness, suppose that there existed cells of arbitrarily small diameter not contained in any inverse image of a primitive disc. Choose a point in each of these cells and pass to a convergent subsequence. An open inverse image containing the limiting point contains each sufficiently late cell, which is a contradiction.

The integral of v_w along the image of each cell boundary is zero by its local primitive. Sum these finitely many equalities. The two orientations of every interior edge cancel. The edges at $t = 0$ and $t = 2\pi$ also cancel, since their image paths agree. The remaining edges give

$$\int_{|\zeta|=r} v_w(\zeta) d\zeta = \int_{|\zeta-w|=\epsilon} v_w(\zeta) d\zeta,$$

with both circles counterclockwise.

Step 3. We evaluate the integral on the small circle. Write $g(w + \eta) = \sum_{n \geq 0} b_n \eta^n$ there. Parametrizing $\eta = \epsilon e^{it}$ and integrating the absolutely convergent series gives

$$\int_{|\zeta-w|=\epsilon} \frac{g(\zeta)}{\zeta - w} d\zeta = i \sum_{n \geq 0} b_n \epsilon^n \int_0^{2\pi} e^{int} dt = 2\pi i b_0.$$

Since $b_0 = g(w)$, the contour comparison proves (4.58). $\square$

Lemma 4.23. *Let E be a complex Banach space, R a positive radius vector, and $f : D_R \rightarrow E$ holomorphic, with $\sup_{D_R} \|f\|_E \leq M$. Holomorphy means local norm-convergent power series. Its Taylor coefficients at zero satisfy, for $0 < r < R$,*

$$(4.60) \quad f_\nu = (2\pi i)^{-h} \int_{|\zeta_1|=r_1} \cdots \int_{|\zeta_h|=r_h} \frac{f(\zeta)}{\prod_j \zeta_j^{\nu_j+1}} d\zeta_h \cdots d\zeta_1, \quad \|f_\nu\|_E \leq MR^{-\nu}.$$

The integrals are norm integrals on the counterclockwise circles. For every $0 < \rho < R$,

$$(4.61) \quad \|f\|_{\mathcal{W}_\rho(E)} \leq M \prod_j (1 - \rho_j/R_j)^{-1}.$$

Proof. We apply the norm contour formula in each coordinate, expand its kernels on a smaller polydisc, and then let the integration radii increase to the given radii R .

Step 1. We derive the iterated formula. Fix $0 < r < R$ and $z \in D_r$. Holding the other variables fixed, each coordinate slice of f has local norm-convergent power series on its disc of radius R_j . Lemma 4.22 applies on the circle of radius r_j . Apply its formula successively to the h variables. All the resulting integrands are continuous on the product of these compact circles. Parametrizing each circle, Lemma 4.21 gives the iterated norm integrals and permits their interchange. Thus

$$(4.62) \quad f(z) = (2\pi i)^{-h} \int_{|\zeta_1|=r_1} \cdots \int_{|\zeta_h|=r_h} \frac{f(\zeta)}{\prod_j (\zeta_j - z_j)} d\zeta_h \cdots d\zeta_1.$$

Step 2. We expand the formula and recover its coefficients. For $0 < t < r$ and $z \in \overline{D}_t$, the kernel in (4.62) has the expansion

$$(4.63) \quad \prod_{j=1}^h (\zeta_j - z_j)^{-1} = \sum_{\nu \in \mathbb{N}^h} z^\nu \prod_j \zeta_j^{-\nu_j-1}.$$

On the circles its sum of absolute values is bounded uniformly by $\prod_j (r_j - t_j)^{-1}$. After multiplication by $f(\zeta)$, the norm majorant is at most $M \prod_j (r_j - t_j)^{-1}$. The geometric tails are uniform on the whole product of circles and $\overline{D}_t$, so Lemma 4.21 permits integration term by term in norm. It follows that f is represented on D_r by the power series with coefficients

$$a_\nu(r) = (2\pi i)^{-h} \int_{|\zeta_1|=r_1} \cdots \int_{|\zeta_h|=r_h} \frac{f(\zeta)}{\prod_j \zeta_j^{\nu_j+1}} d\zeta_h \cdots d\zeta_1.$$

Using the lengths $2\pi r_j$ of the circles in (4.57) gives

$$(4.64) \quad \|a_\nu(r)\|_E \leq Mr^{-\nu}.$$

This series belongs to $\mathcal{W}_t(E)$ for every $0 < t < r$ by Lemma 4.4. Its derivatives at zero therefore recover its coefficients by Lemma 4.3. Since the represented function is f , these coefficients are its Taylor coefficients f_ν ; in particular they are independent of r .

Step 3. We return to the original outer radii. For each fixed ν , let every r_j increase to R_j in (4.64). The coefficient on the left is fixed, and the right side tends to $MR^{-\nu}$. This proves both assertions in (4.60). For any $0 < \rho < R$, applying Lemma 4.4 with this coefficient bound gives (4.61). The power series equals f throughout D_R , because every point lies in some D_r with $r < R$ and (4.62) was proved on each such polydisc. Thus the estimate is a norm bound for the given function on its receiving coefficient space, with M and R unchanged. $\square$

5. PREPARED ORDINARY SADDLE PASSAGES

We construct a hyperbolic saddle passage directly from its differential equation and construct its inverse from the resulting measures. The estimates will control the full expansions on fixed analytic input norms. A point jet is a finite linear combination of derivatives of a Dirac measure. A point-jet source is a locally finite sum of point jets.

Theorem 5.1. *An actual hyperbolic saddle passage with positive eigenvalue ratio λ admits analytic incoming and outgoing coordinate changes after which*

$$(5.1) \quad D(w) = cw^\lambda \exp U(-\log w), \quad c > 0.$$

The whole function U is obtained from the differential equation as

$$(5.2) \quad U(T) = \sum_{K \geq 1} T^K \int e^{-\beta T} d\nu_K(\beta),$$

with dominating positive measures μ_K , constants $\delta > 0, a > 0, |\nu_K| \leq \mu_K, \text{supp } \mu_K \subset [\delta K, \infty)$, and $\sum_K \int (1 + \beta)^2 e^{-a\beta} d\mu_K < \infty$. The generated point-jet source is $T_U = \sum_{K \geq 1} \partial_\beta^K \nu_K$. The inverse time equation $V = T - U(T)/\lambda$ has a unique small solution $T = V + J(V)$ whose entire source is

$$(5.3) \quad T_J = \sum_{n \geq 1} \frac{(-\beta)^{n-1}}{n! \lambda^n} T_U^{*n}.$$

It gives the actual inverse $D^{-1}(z) = (z/c)^{1/\lambda} \exp[-J((- \log z + \log c)/\lambda)]$. These constructions retain the original analytic input norms, the complete convolution tails and all derivatives of cutoffs in total action.

Proposition 5.2. *For the equation-generated forward bank in Theorem 5.1, the original source can be chosen with $\delta = \min\{1, \lambda\}$ and retains its complete absolutely summed labelled tree presentation. Put*

$$(5.4) \quad \Gamma_\lambda = \{M + \lambda N : M, N \in \mathbb{N}, M + N > 0\}.$$

This positive additive set is locally finite. On each component of $\mathbb{R} \setminus \Gamma_\lambda$, the original ν_K has a polynomial density of degree at most $K - 1$; all its atoms belong to Γ_λ . Its full local distribution is

$$(5.5) \quad T_U = \sum_{\omega \in \Gamma_\lambda} \sum_{j=0}^{\lfloor \omega/\delta \rfloor} c_{\omega j} \partial_\beta^j \delta_\omega.$$

The correlated original inverse has the full ordered source of Lemma 5.9 and is a point-jet distribution on Γ_λ with this degree bound. These assertions concern the generated original distributions and retain their original whole norms. They do not assert injectivity of arbitrary measure presentations.

Definition 5.3. Let E be a complex Banach space and let $a, r, d > 0$. A source in $\mathcal{M}_{a,r,d}(E)$ is a sequence $m = (m_K)_{K \geq 0}$ of locally finite E -valued measures on $[0, \infty)$ such that

$$(5.6) \quad \text{supp } m_K \subset [dK, \infty), \quad \|m\|_{a,r,d} := \sum_{K \geq 0} r^K \int_0^\infty e^{-a\beta} d|m_K|(\beta) < \infty.$$

For a general E , we require the measures to admit Bochner integrable densities with respect to positive measures on compact intervals. This condition holds for every measure constructed below. The scalar case $E = \mathbb{C}$ includes all complex measures of locally finite variation. The variable β is called the action and K the logarithmic degree. A source has gap $g > 0$ if every

component is supported in $[g, \infty)$. For integers $0 \leq j \leq K$, write $(K)_j = K!/(K-j)!$. Its realization and its associated distribution are, respectively,

$$(5.7) \quad \mathcal{S}m(T) = \sum_{K \geq 0} T^K \int e^{-\beta T} dm_K(\beta), \quad \mathcal{T}m = \sum_{K \geq 0} \partial_\beta^K m_K.$$

We extend the measures by zero to the real line when taking distributional derivatives. Thus $\langle \partial_\beta^K m_K, \varphi \rangle = (-1)^K \int \varphi^{(K)} dm_K$ for a smooth compactly supported test function φ . The second sum in (5.7) is locally finite in K .

A point jet at ω is a finite linear combination of $\partial_\beta^j \delta_\omega$, where δ_ω denotes unit mass at ω . A point-jet source is a distribution which, on every compact interval, is a finite sum of point jets. For sources with coefficients in a commutative Banach algebra, convolution adds both action and logarithmic degree:

$$(5.8) \quad (m * m')_K = \sum_{i+j=K} m_i * m'_j.$$

The convolution of distributions supported in $[0, \infty)$ is defined by addition of their variables, locally on each compact interval.

5.1. Weighted sources and analytic preparation.

Lemma 5.4. *The space $\mathcal{M}_{a,r,d}(E)$ is complete. If E is a unital commutative Banach algebra with submultiplicative norm and unit of norm one, (5.8) makes this space a unital Banach algebra. The unit has component δ_0 in degree zero and zero in positive degrees. For a source with gap $g > 0$, put*

$$(5.9) \quad v_{a,r,d}(T) = \Re T - a - d^{-1} \log_+(|T|/r), \quad \log_+ x = \max\{0, \log x\} \quad (x > 0).$$

Whenever $v_{a,r,d}(T) \geq 0$, the realization converges absolutely and

$$(5.10) \quad \|\mathcal{S}m(T)\|_E \leq \|m\|_{a,r,d} e^{-g v_{a,r,d}(T)}.$$

The realization is holomorphic where the margin is positive and is multiplicative. For each integer $q \geq 1$ and $\eta > 0$, differentiating the realization q times has a source $m^{[q]}$ satisfying

$$(5.11) \quad \|m^{[q]}\|_{a+\eta,r,d} \leq \left(1 + \frac{1}{dr}\right)^q \left(\frac{q}{e\eta}\right)^q \|m\|_{a,r,d}.$$

The gap is preserved. The maps $\mathcal{T}$ commute with convolution.

Proof. Multiplication by $e^{-a\beta}$ identifies each component with a space of finite measures in variation norm. To check completeness with the density convention of Definition 5.3, take an absolutely summable series of such weighted measures. The sum of their variation measures is a finite positive dominating measure. Relative to this measure their densities form an absolutely summable series in $L^1(E)$, whose sum represents the limit. Completeness of the weighted sum over K follows by summing component norms. The support requirements are closed under these limits.

For convolution, the variation is bounded by convolution of the positive variation measures. Using

$$e^{-a(\beta+\beta')} r^{K+K'} = e^{-a\beta} r^K e^{-a\beta'} r^{K'}$$

in the absolute integral gives the product norm bound. Bochner densities can be multiplied on the product measure space and then pushed forward by addition, so the density convention is preserved. Addition preserves $\beta \geq dK$ and adds the gaps of two sources.

Since $K \leq \beta/d$ on the support of m_K ,

$$|T|^K e^{-\beta \Re T} \leq r^K e^{-a\beta} e^{-\beta v_{a,r,d}(T)}.$$

Integration proves (5.10). The estimate is uniform on compact subsets with positive margin. Absolute integration also identifies realization of a convolution with the product of the realizations.

The derivative of a single integrand is given by the finite formula

$$(5.12) \quad \partial_T^q (T^K e^{-\beta T}) = \sum_{j=0}^{\min(K,q)} \binom{q}{j} (K)_j T^{K-j} (-\beta)^{q-j} e^{-\beta T}.$$

At logarithmic radius r , the sum of the absolute factors in this formula is at most

$$r^K (\beta + K/r)^q \leq r^K \beta^q (1 + 1/(dr))^q.$$

The maximum of $\beta^q e^{-\eta\beta}$ for $\beta \geq 0$ is $(q/(e\eta))^q$. This proves (5.11) and justifies differentiation under the full sum. Each new degree is $K - j \leq K$, so its action bound and gap persist. Finally, on a compact interval only finitely many degrees enter $\mathcal{T}m$ and $\mathcal{T}m'$. Pairing with a test function on the compact region $\beta + \beta' \leq B$ shows that distributional differentiation commutes with convolution. This gives $\mathcal{T}(m * m') = (\mathcal{T}m) * (\mathcal{T}m')$. $\square$

Lemma 5.5. *Let an analytic real vector field have a hyperbolic saddle at the origin with eigenvalues $\alpha > 0$ and $-\gamma < 0$. Put $\lambda = \gamma/\alpha$. On a chosen hyperbolic sector, analytic coordinates and a positive analytic change of time put its equation into*

$$(5.13) \quad \dot{X} = X, \quad \dot{Y} = Y(-\lambda + A(X, Y)), \quad A(X, Y) = \sum_{m,n \geq 1} a_{mn} X^m Y^n.$$

Both boundary half-axes and their positive sides are preserved. Passage between sufficiently small prescribed transverse half-sections differs from passage between $Y = v$ and $X = \epsilon$, for sufficiently small $v, \epsilon > 0$, by analytic coordinate diffeomorphisms with positive derivative. The construction uses convergent analytic series on strict receiving radii, with bounds in the original coefficient norms.

Proof. We construct the two invariant axes, normalize the equation on each axis, and identify the coordinates on the physical sections.

Step 1. We construct the invariant axes. After a real linear change, write the equation as $\dot{x} = \alpha x + f(x, y)$, $\dot{y} = -\gamma y + g(x, y)$, where f, g vanish to order two. Fix a positive radius R_0 at which the full absolute coefficient norms of f and g are finite. For series $\phi(x) = \sum_{k \geq 2} \phi_k x^k$, use

$$\|\phi\|_{r,1} = \sum_{k \geq 2} (1+k) |\phi_k| r^k, \quad E_x \phi = x \phi'(x).$$

The diagonal map $\alpha E_x + \gamma$ has inverse bounded from the unweighted coefficient space to this space, with norm at most $C_u = \sup_{k \geq 2} (1+k)/(\alpha k + \gamma) < \infty$. An invariant graph $y = \phi(x)$ is therefore a fixed point of

$$(5.14) \quad \phi = (\alpha E_x + \gamma)^{-1} \left[g(x, \phi) - (E_x \phi) \frac{f(x, \phi)}{x} \right].$$

Every quotient by x in this formula is analytic and has no constant term, because ϕ starts in degree two.

Here are the estimates that give a convergent solution. On the ball $\|\phi\|_{r,1} \leq Hr^2$, choose r so that $Hr \leq 1$ and $r < R_0/2$. Absolute substitution and telescoping of powers give, with C depending only on the original coefficient norms and R_0 ,

$$\begin{aligned} \|g(x, \phi)\|_r &\leq Cr^2, & \|f(x, \phi)/x\|_r &\leq Cr, \\ \|g(x, \phi) - g(x, \tilde{\phi})\|_r &\leq Cr \|\phi - \tilde{\phi}\|_{r,1}, & \|(f(x, \phi) - f(x, \tilde{\phi}))/x\|_r &\leq C \|\phi - \tilde{\phi}\|_{r,1}. \end{aligned}$$

For example, a term of total degree $p + q \geq 2$ is bounded by $|f_{pq}| r^{p+q}$; its difference under substitution of two graphs is bounded by $q |f_{pq}| r^{p+q-1} \|\phi - \tilde{\phi}\|_{r,1}$. The sums of these bounds are

controlled at R_0 , since the factors $(r/R_0)^{p+q-2}$ and $q(r/R_0)^{p+q-2}$ are bounded for $r < R_0/2$. Division by x multiplies the coefficient norm by r^{-1} . Also $\|E_x\phi\|_r \leq \|\phi\|_{r,1}$. Consequently the right side of (5.14) has norm at most $C_u C(r^2 + Hr^3)$ and Lipschitz constant at most $C_u C(r + Hr^2)$. Choose $H > 2C_u C$, then decrease r until the ball is invariant and the Lipschitz constant is less than $1/2$. Successive iterates starting at zero have differences bounded by a geometric series. Their limit in the complete coefficient space solves (5.14) and is the unique solution in the ball.

For the stable graph $x = \psi(y)$, the equation is instead

$$(5.15) \quad \psi = (\gamma E_y + \alpha)^{-1} \left[-f(\psi, y) + (E_y \psi) \frac{g(\psi, y)}{y} \right].$$

Its diagonal inverse has bound $C_s = \sup_{k \geq 2} (1+k)/(\gamma k + \alpha)$. The four estimates just displayed, with $f(\psi, y)$ in place of $g(x, \phi)$ and $g(\psi, y)/y$ in place of $f(x, \phi)/x$, follow by replacing the factor q by p in the coefficient difference. Taking $H > 2 \max(C_u, C_s)C$ and imposing both radius conditions gives convergent iterates for (5.15) as well. The coefficients of both graphs are real, since all iterates are real.

The coordinate map $(x, y) \mapsto (x - \psi(y), y - \phi(x))$ has derivative the identity at the origin. To see that its inverse is analytic, write it as $z \mapsto z + R(z)$ and take a disk on which $\|DR\| < 1/2$. For targets in a smaller disk the iteration $z_{k+1} = w - R(z_k)$ stays in the first disk and converges uniformly. The iterates are holomorphic in w ; their limit is a holomorphic inverse. The coordinate map sends the two invariant graphs to the coordinate axes. The transformed field thus has the form $xa(x, y)\partial_x - yb(x, y)\partial_y$, with $a(0, 0) = \alpha$, $b(0, 0) = \gamma$.

Step 2. We normalize the equation on the axes. Divide time by the positive analytic factor a and put $C(x, y) = b(x, y)/a(x, y) - \lambda$. Define

$$H_0(x) = \int_0^x \frac{C(s, 0)}{s} ds, \quad y_1 = ye^{H_0(x)}.$$

The integrand is analytic. The equation for y_1 is $\dot{y}_1 = -y_1(\lambda + C_1(x, y_1))$, where $C_1(x, y_1) = C(x, y_1 e^{-H_0(x)}) - C(x, 0)$ vanishes when $y_1 = 0$. Put $c_1(y_1) = C_1(0, y_1)$ and define

$$H_1(y_1) = \int_0^{y_1} \frac{\lambda/(\lambda + c_1(s)) - 1}{s} ds, \quad Y = y_1 e^{H_1(y_1)}.$$

This integrand is analytic because $c_1(0) = 0$. We obtain

$$\frac{\dot{Y}}{Y} = -\lambda + \frac{\lambda(c_1(y_1) - C_1(x, y_1))}{\lambda + c_1(y_1)}.$$

The correction vanishes at $x = 0$ and at $Y = 0$. Its convergent Taylor series is therefore divisible by xY , proving (5.13) with $X = x$. Both exponential coordinate factors are positive on real points near the origin and have derivative one on the axes at the origin. They preserve the chosen sector.

Step 3. We choose the physical sections and control their norms. The portions of a boundary orbit between two small transverse sections are regular compact orbit segments. The local integral equation for the flow is a contraction for a sufficiently short time interval, uniformly for initial points in a small complex disk: its Lipschitz constant is the time length times a bound for the derivative of the field. Uniform limits of its holomorphic iterates give an analytic flow. Finitely many such intervals cover each compact segment. An analytic defining function of the target section has nonzero time derivative along the segment. The local inverse construction used for $z + R(z)$, after dividing by this nonzero derivative, solves its hitting equation analytically. Uniqueness of integral curves gives the inverse hitting map. Choosing coordinates positive toward the interior makes its derivative positive. These hitting maps give the asserted incoming and outgoing coordinate changes.

All coefficient sums used in the graph estimates are bounded by the original norms at R_0 . Reciprocal and exponential series are used on strict disks, and division by an axis variable has the explicit cost r^{-1} . Integration divides positive Taylor indices by positive integers. These estimates also hold for coefficients in a commutative Banach algebra. In particular, one may take the coefficient algebra to be an original parameter space $\mathcal{W}_\rho$, keeping its radii fixed, by Proposition 4.20. The required spatial receiving radii can depend on that full norm and on the fixed positive eigenvalues; no norm at a smaller parameter radius replaces it. $\square$

5.2. The full tree expansion.

Lemma 5.6. *Let $\lambda > 0$, set $\delta = \min\{1, \lambda\}$, and let A be as in (5.13). Fix radii $R_X, R_Y > 0$ for which $M_A = \sum_{m,n \geq 1} |a_{mn}| R_X^m R_Y^n < \infty$. Choose $R > 0$, $\tau > \max\{1, 1/(\epsilon\delta)\}$, and positive real section sizes ϵ, v . If $0 < \epsilon < R_X$, $0 < v\epsilon^R < R_Y$, and*

$$(5.16) \quad \tau \sum_{m,n \geq 1} |a_{mn}| \epsilon^m v^n (1+m+\lambda n)^2 e^{nR} \leq R,$$

then the actual passage from $(X, Y) = (\epsilon w, v)$ to $X = \epsilon$ has the form $D(w) = vw^\lambda \exp U(-\log w)$ for $0 < w < 1$. It admits real measures ν_K and positive measures μ_K such that

$$(5.17) \quad U(T) = \sum_{K \geq 1} T^K \int e^{-\beta T} d\nu_K(\beta), \quad |\nu_K| \leq \mu_K, \quad \text{supp } \mu_K \subset [\delta K, \infty),$$

and

$$(5.18) \quad \sum_{K \geq 1} \tau^K \int (1+\beta)^2 d\mu_K(\beta) \leq R.$$

The measures retain an absolutely summable presentation by all labeled trees. Filtering any of these labels or their frequencies by factors of modulus at most one is dominated by this presentation, and increasing exhaustions converge in its norm. Condition (5.16) can always be achieved by decreasing the physical section sizes, with M_A unchanged.

Proof. Put $X = \epsilon x$, $w = e^{-T}$, and $Y(t) = v e^{-\lambda t} \exp u(t; T)$. Since $x(t) = e^{-(T-t)}$, the equation for u , with $u(0; T) = 0$, is

$$(5.19) \quad u(t; T) = \sum_{m,n \geq 1} b_{mn} \int_0^t e^{-m(T-s)-\lambda ns} \exp(nu(s; T)) ds, \quad b_{mn} = a_{mn} \epsilon^m v^n.$$

We prove convergence of its expansion before identifying the expansion with the solution.

Step 1. We construct the tree integrals and their action records. Use finite rooted trees whose children at each vertex are ordered. A vertex p has a label (m_p, n_p) with $m_p, n_p \geq 1$ and k_p children. Assign the tree the scalar

$$c_F = \prod_{p \in F} b_{m_p n_p} \frac{n_p^{k_p}}{k_p!}.$$

Each vertex also has a time s_p , with $0 \leq s_{\text{child}} \leq s_{\text{parent}} \leq t$. Expansion of each exponential in (5.19) gives exactly these coefficients: its k_p factors are ordered before division by $k_p!$. For a tree with K vertices, rescale $s_p = T x_p$. At $t = T$ let P_F be its order polytope, defined by the resulting inequalities in the unit cube. Define

$$(5.20) \quad \beta_F(x) = \sum_{p \in F} (m_p(1-x_p) + \lambda n_p x_p), \quad \rho_F = (\beta_F)_*(\mathbf{1}_{P_F} dx).$$

The tree integral equals $T^K \int e^{-\beta T} d\rho_F(\beta)$. The measure ρ_F is positive, has mass at most one, and satisfies

$$(5.21) \quad \delta K \leq \beta_F(x) \leq \sum_{p \in F} (m_p + \lambda n_p).$$

For $t < T$, the polytope is restricted by $x_p \leq t/T$; the bounds in (5.21) continue to hold.

Step 2. We obtain a convergent majorant. Define $\Phi(z) = \sum_{m,n \geq 1} |b_{mn}|(1+m+\lambda n)^2 e^{nz}$. Starting from $W_0(z) = 0$, iterate $W_{j+1}(z) = z\Phi(W_j(z))$ as series with nonnegative coefficients. At $z = \tau$, condition (5.16) implies $0 \leq W_j(\tau) \leq R$. The coefficients increase, and every finite tree occurs in a finite iterate. Their limits define $W(z) = \sum_{K \geq 1} w_K z^K$, with

$$\sum_{K \geq 1} w_K \tau^K \leq R.$$

The coefficient w_K is the sum of $|c_F|$ times $\prod_{p \in F} (1+m_p+\lambda n_p)^2$ over all trees of size K . By (5.21) and $1 + \sum_p a_p \leq \prod_p (1+a_p)$ for $a_p \geq 0$,

$$\int (1+\beta)^2 d\rho_F \leq \prod_{p \in F} (1+m_p+\lambda n_p)^2.$$

It follows that

$$\nu_K = \sum_{|F|=K} c_F \rho_F, \quad \mu_K = \sum_{|F|=K} |c_F| \rho_F$$

converge in variation with the moment weight and satisfy (5.18). Each summand remains part of this absolutely convergent presentation.

For real $T > 0$, the absolute value of a tree integral is bounded by $T^K e^{-\delta K T} |c_F|$. Since $\sup_{T \geq 0} T e^{-\delta T} = 1/(e\delta) < \tau$, the sum of all tree integrals is bounded by R , uniformly for $0 \leq t \leq T$. Decomposing trees at their root and integrating the absolutely convergent series proves (5.19). The exponential series converges because $|u| \leq R$ and $ve^R < R_Y$. This produces a solution of the original equation which stays in its analytic domain. If two such solutions are compared on a finite time interval, a bound L for the derivative of the integrand in u gives $|u_1(t) - u_2(t)| \leq L \int_0^t |u_1(s) - u_2(s)| ds$. Iteration bounds this difference by $\sup |u_1 - u_2| (Lt)^j / j!$ for every j , so it vanishes. The tree sum is therefore the actual solution, and $U(T) = u(T; T)$ gives the passage in the statement.

Step 3. We bound the original coefficient norms and tails. Put $\theta_X = \epsilon/R_X$ and $\theta_Y = ve^R/R_Y$. The sum in (5.16), without τ , is at most

$$(5.22) \quad M_A \sup_{m,n \geq 1} (1+m+\lambda n)^2 \theta_X^m \theta_Y^n.$$

For $\theta_X, \theta_Y \leq 1/2$, the supremum divided by $\theta_X \theta_Y$ is bounded independently of the section sizes. Hence (5.22) tends to zero with those sizes. This proves the smallness assertion using the original M_A . All filtered sums are dominated term by term by (5.18). The norm of the omitted terms tends to zero in an increasing exhaustion, which proves the asserted full tail and filtering statements. $\square$

Lemma 5.7. *For the measures in Lemma 5.6, set*

$$(5.23) \quad \Gamma_\lambda = \{M + \lambda N : M, N \in \mathbb{N}, M + N > 0\}.$$

This set is locally finite and is closed under addition. On every component of $\mathbb{R} \setminus \Gamma_\lambda$, the measure ν_K has a polynomial density of degree at most $K-1$; all its atoms lie in Γ_λ . Consequently

$$(5.24) \quad T_U = \sum_{K \geq 1} \partial_\beta^K \nu_K = \sum_{\omega \in \Gamma_\lambda} \sum_{j=0}^{\lfloor \omega/\delta \rfloor} c_{\omega j} \partial_\beta^j \delta_\omega$$

is a locally finite point-jet source with real coefficients.

Proof. For $M + \lambda N \leq B$, both nonnegative integers are bounded, so Γ_λ is locally finite. Adding two represented numbers gives another representation of this form.

We first examine one tree measure. Partition the order polytope P_F by all total orders of its K coordinates consistent with the tree order. These pieces are simplices; their intersections

have K -dimensional volume zero and all their vertices have coordinates zero or one. The action at each vertex is therefore a sum of positive integers m_p and numbers λn_p , so it belongs to Γ_λ .

If the affine action is constant on a simplex, its pushforward is a point mass. Otherwise its level sets have volume zero. Between two successive action values at the vertices, the simplex clipped by $\beta_F \leq b$ has a fixed face structure. Its vertices consist of fixed vertices and intersections with edges whose endpoint action values straddle b . An intersection on an edge with action values $a_i < a_j$ has barycentric coefficients $(a_j - b)/(a_j - a_i)$ and $(b - a_i)/(a_j - a_i)$, hence depends affinely on b . Triangulate the clipped polytope by successively coning from the first vertex in a fixed order to triangulations of the opposite faces. The incidence list is fixed throughout this interval. Each simplex volume is the absolute value of a determinant polynomial of degree at most K in b , with constant sign on the interval. Thus the cumulative pushforward mass is polynomial of degree at most K , and its density has degree at most $K - 1$. Summing the finitely many simplices proves this assertion for ρ_F .

Now fix K and a bounded open interval whose closure avoids Γ_λ . The sum of the variations of all tree summands on this interval is finite by (5.18). Their polynomial densities form a Cauchy sequence in L^1 in the finite dimensional space of polynomials of degree at most $K - 1$. The limit is a polynomial of that degree bound. On overlapping intervals the two polynomials agree almost everywhere and hence are identical. Exhausting a component of $\mathbb{R} \setminus \Gamma_\lambda$ gives one polynomial density there. Atom coefficients converge absolutely on each compact interval.

The K th distributional derivative of this density vanishes off Γ_λ ; boundary jumps and atoms give only point jets of order at most K . Moreover, ν_K is supported in $[\delta K, \infty)$, so a compact interval $[0, B]$ involves only $K \leq B/\delta$. At a point ω , only $K \leq \omega/\delta$ can occur. This proves (5.24), including its order bound and local finiteness. $\square$

5.3. The inverse from the forward measures.

Lemma 5.8. *Let $\lambda, H > 0$ and $A \in \mathbb{R}$. Suppose U is holomorphic on $\{T : \Re T > A, |\Im T| < H + 1\}$ and $U(T), U'(T)$ tend to zero uniformly there as $\Re T \rightarrow \infty$. On a sufficiently far strip $\{V : \Re V > A', |\Im V| < H\}$, the equation $V = T - U(T)/\lambda$ has a unique holomorphic solution $T = V + J(V)$ with $J(V) \rightarrow 0$, and*

$$(5.25) \quad J(V) = \sum_{n \geq 1} \frac{1}{n! \lambda^n} \partial_V^{n-1} (U(V)^n).$$

The series converges uniformly on every sufficiently far closed substrip.

Proof. Fix $0 < r_0 < 1$ and put $g = U/\lambda$. On each disk $|z - V| \leq r_0$ sufficiently far to the right, arrange $|g(z)| < r_0/2$ and $|g'(z)| < 1/2$. The iteration $z_{k+1} = V + g(z_k)$, starting at $z_0 = V$, stays in the disk and converges uniformly to its unique fixed point. Uniform convergence on slightly larger parameter neighborhoods proves holomorphy in V . Its displacement $J(V)$ tends to zero. Any other solution whose displacement tends to zero eventually lies in this disk and coincides with it. This specifies the small solution in the statement.

For completeness, its exact expansion can be obtained by integration around $|z - V| = r_0$. Let $F(z) = z - V - g(z)$ and let z_* be its zero. The zero is simple and is the only zero in the disk. Factoring $F(z) = (z - z_*)h(z)$ with h holomorphic and nonvanishing gives

$$J(V) = \frac{1}{2\pi i} \int_{|z-V|=r_0} (z - V) \frac{F'(z)}{F(z)} dz = -\frac{1}{2\pi i} \int_{|z-V|=r_0} \log \left(1 - \frac{g(z)}{z - V} \right) dz.$$

For the first equality the h'/h term integrates to zero and the simple pole contributes $z_* - V$. The second equality is integration by parts along the circle; the logarithm is given there by its uniformly convergent power series. Its term of degree n contributes

$$\frac{1}{n} \frac{1}{2\pi i} \int_{|z-V|=r_0} \frac{g(z)^n}{(z - V)^n} dz = \frac{1}{n!} \partial_V^{n-1} (g(V)^n).$$

This proves (5.25) for the actual inverse. On the contour the logarithm's ratio has modulus less than 1/2, so the n th integral is bounded by $r_0/(n2^n)$, uniformly on the asserted strips. This also proves the stated uniform convergence. $\square$

Lemma 5.9. *Let $\lambda, \delta, S > 0$ and let $(\nu_K)_{K \geq 1}$ be scalar measures dominated by positive measures $(\mu_K)_{K \geq 1}$, with $\text{supp } \mu_K \subset [\delta K, \infty)$ and*

$$(5.26) \quad N_S = \sum_{K \geq 1} \int e^{-S\beta} d\mu_K(\beta) < \infty.$$

Put $U(T) = S\nu(T)$. The equation $V = T - U(T)/\lambda$ has, on every sufficiently far right strip of fixed bounded width, a unique holomorphic solution $T = V + J(V)$ with $J(V) \rightarrow 0$. The inverse correction has the expansion

$$(5.27) \quad J(V) = \sum_{n \geq 1} \frac{1}{n! \lambda^n} \partial_V^{n-1} (U(V)^n) = \sum_{m \geq 1} V^m \int e^{-\beta V} d\nu_m^{\text{inv}}(\beta).$$

More precisely, for every ordered tuple $K_1, \dots, K_n \geq 1$, write $K = K_1 + \dots + K_n$. The component of degree $m = K - j$ retains the measure

$$(5.28) \quad \frac{\binom{n-1}{j} (K)_j}{n! \lambda^n} (-\beta)^{n-1-j} (\nu_{K_1} * \dots * \nu_{K_n})(d\beta), \quad 0 \leq j \leq \min(K, n-1).$$

Here β is the sum of the n original action variables. Let μ_m^{inv} be the sum of the positive measures obtained by taking absolute values in (5.28) and replacing each ν_{K_i} by μ_{K_i} . For $r > 0$, put

$$(5.29) \quad c_r = \delta^{-1} \log_+ r, \quad C_r = 1 + (\delta r)^{-1}, \quad q > \frac{C_r N_S}{\lambda}, \quad S_{\text{inv}} = S + c_r + q, \quad \vartheta = \frac{C_r N_S}{\lambda q} < 1.$$

Then

$$(5.30) \quad \sum_{m \geq 1} r^m \int e^{-S_{\text{inv}} \beta} d\mu_m^{\text{inv}}(\beta) \leq \frac{N_S}{\lambda} \sum_{n \geq 1} \frac{\vartheta^{n-1}}{n}.$$

The part with $n > N$, for any integer $N \geq 0$, has norm at most

$$(5.31) \quad \frac{N_S}{\lambda} \frac{\vartheta^N}{(N+1)(1-\vartheta)}.$$

If $N_S = 0$, every measure and correction is zero and all tail bounds are interpreted as zero. Each retained measure satisfies $\beta \geq \delta K \geq \delta m$ and $\beta \geq \delta n$. Finally, as an identity of distributions,

$$(5.32) \quad \sum_{m \geq 1} \partial_\beta^m \nu_m^{\text{inv}} = \sum_{n \geq 1} \frac{(-\beta)^{n-1}}{n! \lambda^n} \left(\sum_{K \geq 1} \partial_\beta^K \nu_K \right)^{*n}.$$

For the equation-generated measures of Theorem 5.1, this is a point-jet source on Γ_λ , of order at most $\lfloor \omega/\delta \rfloor$ at action ω .

Proof. We first identify the analytic inverse and then prove convergence with all original measures retained. The distribution identity will follow from the finite product rule, after that convergence has been proved.

Step 1. We construct the analytic inverse. Lemma 5.4, applied with $r = 1$, gives $U(T), U'(T) \rightarrow 0$ uniformly on strips of fixed bounded width as $\Re T \rightarrow \infty$. To use its derivative bound, increase the weight by any fixed positive amount. The margin in (5.9) still tends to infinity. Lemma 5.8 now gives the unique small solution and the first equality of (5.27).

Step 2. We retain every convolution leaf and bound its majorant. For fixed n , expand $U(V)^n$ into ordered tuples of its original measures. In a tuple the polynomial factor is V^K and the exponential factor is $e^{-(\beta_1 + \dots + \beta_n)V}$. Formula (5.12) with $q = n - 1$ gives exactly (5.28). Since

$K \geq n$ and $j \leq n-1$, its degree $m = K-j$ is positive. Addition of supports gives $\beta \geq \delta K$ and $\beta \geq \delta n$.

The sum of all absolute polynomial factors of a fixed tuple, evaluated at radius r , is bounded by

$$\begin{aligned} \sum_j \binom{n-1}{j} (K)_j r^{K-j} \beta^{n-1-j} &\leq r^K (\beta + K/r)^{n-1} \\ &\leq e^{cr\beta} (C_r \beta)^{n-1}. \end{aligned}$$

Retaining the $(n-1)$ st term in $e^{q\beta}$ gives $\beta^{n-1} e^{-q\beta} \leq (n-1)! q^{-(n-1)}$. After division by $n! \lambda^n$, integration at weight S_{inv} therefore gives at most

$$\frac{C_r^{n-1}}{n \lambda^n q^{n-1}} \prod_{i=1}^n \int e^{-S\beta_i} d\mu_{K_i}(\beta_i).$$

Summing over every K_i yields the n th term of (5.30). Summing $n > N$ and using $1/n \leq 1/(N+1)$ proves (5.31). These estimates justify all integrations, all sums over tree and coefficient labels, and the second equality of (5.27). Additional fixed powers of β can be included by increasing S_{inv} by a fixed positive amount. Thus the complete inverse also has the moment bounds needed for differentiation and cutoffs.

Step 3. We construct the distribution of the inverse. For a measure η and nonnegative integers K, p , the product rule for distributions gives

$$(5.33) \quad (-\beta)^p \partial_\beta^K \eta = \sum_{j=0}^{\min(K,p)} \binom{p}{j} (K)_j \partial_\beta^{K-j} [(-\beta)^{p-j} \eta].$$

To verify the signs, pair both sides with a test function, differentiate $(-\beta)^p$ inside the finite Leibniz expansion, and use $\partial_\beta^j (-\beta)^p = (-1)^j (p)_j (-\beta)^{p-j}$. With $p = n-1$ and $\eta = \nu_{K_1} * \dots * \nu_{K_n}$, the right side of (5.33) is precisely the contribution of (5.28) after its m th distributional derivative. Derivatives commute with convolution by Lemma 5.4. On $[0, B]$ the bounds $n \leq B/\delta$ and $K \leq B/\delta$ leave only finitely many indices in this distribution calculation. This proves (5.32).

For the prepared tree measures, (5.24) is supported on the locally finite additive set Γ_λ . Convolution adds its point-jet orders and locations, and multiplication by $(-\beta)^{n-1}$ cannot increase those orders. The order of any contributing tuple is at most $K \leq \omega/\delta$ at action ω . This proves the final assertion. The point-jet assertion concerns the sum over inverse degrees in (5.32); the finite product rule preserves the cancellations between those degrees. $\square$

Proof of Proposition 5.2. Take the original bank of Lemma 5.6, with $\delta = \min\{1, \lambda\}$. Its actual passage identity is (5.17), and this is the bank used in Theorem 5.1. The action bound (5.21) holds for every labelled tree. The full presentation and its positive majorant satisfy (5.18); the physical section sizes are chosen using the original coefficient norm in (5.22).

The set in (5.23) is exactly the set in (5.4). Lemma 5.7 proves its local finiteness and additivity, the polynomial density assertion for every original ν_K , and the forward distribution identity (5.24). That identity gives (5.5) with the asserted order bound. This distributional record is formed from the absolutely summed tree bank, whose full presentation and bound (5.18) are retained.

For any $S > 0$, that moment bound supplies the finite N_S required by Lemma 5.9. Apply the lemma to these original measures. Its ordered summands (5.28) retain every original tree label and action variable, have $\beta \geq \delta K \geq \delta m$, and converge with the full norm and tail bounds (5.30)–(5.31). Their full distribution is (5.32). On $[0, B]$, only $n \leq B/\delta$ and total degree $K \leq B/\delta$ contribute to this identity. A convolution of jets at $\omega_1, \dots, \omega_n$ has location $\omega = \sum_i \omega_i \in \Gamma_\lambda$ and order at most $\sum_i \lfloor \omega_i/\delta \rfloor \leq \lfloor \omega/\delta \rfloor$. Multiplication by $(-\beta)^{n-1}$ cannot increase that order. Thus the full inverse distribution has the claimed support and order bound. Its identification

uses the complete ordered inverse and its absolute majorant, including the cancellations between logarithmic degrees in (5.32). $\square$

5.4. Total-action cuts and original families.

Lemma 5.10. *Let $m \in \mathcal{M}_{a,r,d}(E)$ have a positive gap. Choose $0 < B < B_+$ and a smooth real function χ on $\mathbb{R}$, with $0 \leq \chi \leq 1$, equal to one on $(-\infty, B]$ and zero on $[B_+, \infty)$. Put*

$$(5.34) \quad J_\chi = \lfloor B_+/d \rfloor, \quad C_\chi = \max \left\{ 1, \max_{0 \leq j \leq J_\chi} \|\chi^{(j)}\|_\infty \right\}, \quad a_\chi = a + d^{-1} \log(1 + 1/r).$$

The sources

$$(5.35) \quad m_l^\chi = \sum_{K \geq l} (-1)^{K-l} \binom{K}{l} \chi^{(K-l)} m_K, \quad m^{\text{tail}} = m - m^\chi$$

satisfy

$$(5.36) \quad \mathcal{T}m^\chi = \chi \mathcal{T}m, \quad \mathcal{T}m^{\text{tail}} = (1 - \chi) \mathcal{T}m.$$

Their norms and supports obey

$$(5.37) \quad \|m^\chi\|_{a_\chi, r, d}, \|m^{\text{tail}}\|_{a_\chi, r, d} \leq C_\chi \|m\|_{a, r, d}, \quad \text{supp } m_l^\chi \subset [0, B_+], \quad \text{supp } m_l^{\text{tail}} \subset [B, \infty).$$

In particular, with $v = v_{a_\chi, r, d}(T) \geq 0$,

$$(5.38) \quad \|\mathcal{S}m(T) - \mathcal{S}m^\chi(T)\|_E \leq C_\chi \|m\|_{a, r, d} e^{-Bv}.$$

If $\mathcal{T}m = \sum_{\omega, j} c_{\omega j} \partial_\beta^j \delta_\omega$ is a point-jet source on a locally finite set Γ , and $[B, B_+] \cap \Gamma = \emptyset$, then

$$(5.39) \quad \mathcal{S}m^\chi(T) = \sum_{\omega < B} \sum_j c_{\omega j} T^j e^{-\omega T}.$$

The constants a_χ, r, d can be fixed independently of B, B_+ and χ . These assertions apply to the full inverse source of Lemma 5.9 and to every dominated filtering of its original convolution leaves.

Proof. For a measure η and a nonnegative integer K , pairing with test functions gives the finite identity

$$(5.40) \quad \chi \partial_\beta^K \eta = \sum_{j=0}^K (-1)^j \binom{K}{j} \partial_\beta^{K-j} (\chi^{(j)} \eta).$$

This is the distributional product rule in a form which specifies the underlying measures. Its application to m_K , followed by $l = K - j$, gives (5.35) and (5.36). Every component still satisfies the action bound, since $\beta \geq dK \geq dl$.

For $j \geq 1$, the factor $\chi^{(j)}(\beta)$ is supported in $[B, B_+]$, so any nonzero term has $K \leq B_+/d$. The bound C_χ therefore includes every derivative which actually occurs. Summing absolute degree factors gives

$$\sum_{j=0}^K \binom{K}{j} r^{K-j} |\chi^{(j)}(\beta)| \leq C_\chi (r+1)^K \leq C_\chi r^K e^{(a_\chi - a)\beta}.$$

After integration this proves the first bound in (5.37). Replacing χ by $1 - \chi$ gives the second bound with the identical constant: its zeroth derivative is bounded by one, and its positive derivatives are the negatives of those of χ . These terms represent exactly $m - m^\chi$, since all positive cutoff derivatives cancel in the sum of the two sources. The support assertions follow directly from these formulas, and Lemma 5.4 gives (5.38).

Only finitely many degrees enter the compact source m^χ . Its realization is the pairing of $\mathcal{T}m^\chi$ with $e^{-\beta T}$, defined using an additional compact test function equal to one near its support. If the transition interval avoids Γ , all positive derivatives of χ vanish at the point jets, where χ equals either zero or one. Formula (5.36) then proves (5.39).

To specify the cutoff on the inverse, write $\eta = \nu_{K_1} * \cdots * \nu_{K_n}$ and $K = \sum_i K_i$. Combining (5.33) and (5.40) gives

$$(5.41) \quad \chi(-\beta)^{n-1} \partial_\beta^K \eta = \sum_{j=0}^{\min(K, n-1)} \binom{n-1}{j} (K)_j \sum_{k=0}^{K-j} (-1)^k \binom{K-j}{k} \cdot \partial_\beta^{K-j-k} [\chi^{(k)}(\beta) (-\beta)^{n-1-j} \eta].$$

Each tuple carries the additional factor $1/(n!\lambda^n)$. The cutoff argument is $\beta = \beta_1 + \cdots + \beta_n$ throughout (5.41). Thus both the inverse derivative terms and the cutoff derivative terms are retained at the total action of the original leaves. Their absolute sum is controlled by (5.30) followed by (5.37). Any filtering dominated by those positive measures has these bounds. Increasing exhaustions converge in the weighted norm. After any fixed number of derivatives in T , the full tail satisfies (5.11) with gap B and any fixed additional weight $\eta > 0$. $\square$

Lemma 5.11. *Let E be a unital commutative Banach algebra with submultiplicative norm and unit of norm one, and let $S, \delta, g > 0$. Let $\nu = (\nu_K)_{K \geq 0}$ be an E -valued source dominated by positive measures μ_K , with gap $g > 0$, action bound $\beta \geq \delta K$, and $N_S = \sum_K \int e^{-S\beta} d\mu_K < \infty$. Fix $\alpha > 0$, $b \in \mathbb{R}$, $h \in E$, and $r > 0$. Put $c_h = |b| + \|h\|_E$ and choose $a > 0$ such that*

$$(5.42) \quad a\alpha \geq S + c_h + \delta^{-1} \log_+(\alpha r + c_h).$$

The source

$$(5.43) \quad m_l = \sum_{K \geq l} (\beta \mapsto \alpha\beta)_* \left[\binom{K}{l} \alpha^l (b+h)^{K-l} \exp_E(-\beta(b+h)) \nu_K(d\beta) \right]$$

belongs to $\mathcal{M}_{a,r,\alpha\delta}(E)$, has gap αg , and has norm at most N_S . Here $\exp_E$ denotes the convergent exponential series in E , and b means b times the unit. For any contractive character $e : E \rightarrow \mathbb{C}$, its realization satisfies

$$(5.44) \quad e(\mathcal{S}m(T)) = \sum_K (\alpha T + b + e(h))^K \int e^{-\beta(\alpha T + b + e(h))} d(e\nu_K)(\beta).$$

The distribution is

$$(5.45) \quad \mathcal{T}m = (\beta \mapsto \alpha\beta)_* [\exp_E(-\beta(b+h)) \mathcal{T}\nu].$$

All expressions use the entire exponential and the original measures.

Proof. The finite sum over degrees at a fixed (K, β) is bounded by

$$\sum_{l=0}^K r^l \binom{K}{l} \alpha^l \|(b+h)^{K-l} \exp_E(-\beta(b+h))\|_E \leq (\alpha r + c_h)^K e^{\beta c_h}.$$

Since $K \leq \beta/\delta$, condition (5.42) bounds its weighted integral by $\int e^{-S\beta} d\mu_K$. Summing K gives the full source norm bound. The new action $t = \alpha\beta$ satisfies $t \geq \alpha\delta K \geq \alpha\delta l$ and $t \geq \alpha g$.

Expand only the finite polynomial $(\alpha T + b + e(h))^K$ in powers of T . A character commutes with the norm-convergent exponential series, and the norm bound just proved justifies every integral and sum. This proves (5.44) on a sufficiently far right strip. Pairing with test functions gives

$$\exp_E(-\beta(b+h)) \partial_\beta^K \nu_K = \sum_{l=0}^K \binom{K}{l} \partial_\beta^l [(b+h)^{K-l} \exp_E(-\beta(b+h)) \nu_K].$$

Pushforward by $\beta \mapsto \alpha\beta$ multiplies a derivative of order l by α^l . Local finiteness of degrees then proves (5.45). Each measure in (5.43) has a Bochner density with respect to a positive variation measure, as required in Definition 5.3. $\square$

Proposition 5.12. *Suppose the analytic coefficients in the constructions of this section belong to an original parameter algebra $\mathcal{W}_\rho(\mathbb{C})$, with finite full norms on the stated spatial coefficient radii. Let p be the number of parameters. Fix the positive eigenvalues, the action scalings, and the strict inequalities needed for preparation. Then the tree, inverse, affine substitution, and cutoff estimates hold before specialization, with coefficient absolute values replaced by their full $\mathcal{W}_\rho$ norms. Every resulting source family has an absolutely convergent parameter series on the original radii. On $|z_i| \leq \theta \rho_i$, $0 < \theta < 1$, its complete parameter tail of total degree greater than an integer $M \geq 0$ is bounded by θ^{M+1} times its full source-family norm.*

If an input family F is instead supplied as a bounded holomorphic map on the open polydisk $\Omega_\rho = \{z : |z_i| < \rho_i\}$ into its Banach source space, its supremum norm gives uniform source bounds on that full polydisk, provided the required strict inequalities hold uniformly. Conversion to an absolute Taylor norm on radii $\rho'_i < \rho_i$ has the explicit bound

$$(5.46) \quad \sum_{\kappa \in \mathbb{N}^p} (\rho')^\kappa \|F_\kappa\| \leq \sup_{z \in \Omega_\rho} \|F(z)\| \prod_{i=1}^p (1 - \rho'_i/\rho_i)^{-1}.$$

No assertion of absolute Taylor summability at the boundary follows from bounded holomorphy alone.

Proof. For the first assertion use $E = \mathcal{W}_\rho(\mathbb{C})$ as the coefficient algebra throughout. Proposition 4.20 makes multiplication submultiplicative at precisely those radii. The graph iterations in Lemma 5.5 are contractions in that coefficient norm. Their limits are full parameter families. In (5.16) use $\|a_{mn}\|_{\mathcal{W}_\rho}$; the tree majorant then bounds the full coefficient sum before any integration. In (5.28), expand every product as the product in E . The proof of (5.30) uses only submultiplicativity and positive integration, so it bounds these complete E -valued measures. Specialization is a bounded character and identifies their realization with the unique scalar inverse of Lemma 5.9. No coefficient is divided by in forming that inverse.

Lemmas 5.10 and 5.11 already give estimates in E . Absolute integration interchanges the coefficient sum and the source norm sum. Hence all resulting sources lie in the corresponding $\mathcal{W}_\rho$ space of scalar sources. The parameter tail estimate is then Proposition 4.20, applied to that whole family. A varying nonzero leading constant enters the affine formula by its prescribed logarithm in E , with the norm of that logarithm included in c_h .

For a bounded holomorphic family, first carry out each construction as a map between its Banach input and output spaces. Products and strict power series are holomorphic maps there. Graph contractions converge geometrically with a uniform contraction constant. The inverse sum has the uniform geometric bound (5.31), with q chosen from the full input supremum. For the affine formula choose a fixed positive reserve in (5.42). Expansion on a small h -increment disk introduces factors $\beta^j/j!$, whose full sum is bounded by the exponential paid by that reserve. Thus the affine map is holomorphic in h in its output source norm. These arguments give holomorphy and bounds in the full original supremum norm, without interchanging a parameter supremum with an unprovided sum of coefficient majorants.

Integrating F on a parameter torus of radii strictly below ρ gives $\|F_\kappa\| \leq B_F \rho^{-\kappa}$ after increasing the torus radii to ρ , where $B_F = \sup_{\Omega_\rho} \|F\|$. Multiplication by $(\rho')^\kappa$ and summation of the geometric series gives (5.46). If this conversion is used for an input, its displayed factor is included before choosing the tree and inverse weights. If the supplied family is holomorphic on a neighborhood of the closed original polydisk, a fixed larger polydisk is available and the conversion pays for the entire original Taylor radius. This proves the stated distinction between the two input norms. $\square$

Lemma 5.13. *The inverse construction of Lemma 5.9 also holds for a scalar bank $(\nu_K)_{K \geq 0}$ with positive majorants $(\mu_K)_{K \geq 0}$ satisfying*

$$(5.47) \quad \text{supp } \mu_K \subset [g, \infty) \cap [\delta K, \infty), \quad N_S = \sum_{K \geq 0} \int e^{-S\beta} d\mu_K < \infty, \quad g, \delta, S > 0.$$

In the ordered inverse formula allow $K_i \geq 0$ and $m = K - j \geq 0$. Each inverse summand has total action $\beta \geq gn$ and $\beta \geq \delta m$. The whole norm and inverse-degree tail bounds of that lemma hold with these degree ranges and unchanged c_r, C_r, S_{inv} and ϑ . The distributional formula holds with all degree sums starting at zero. If the original distribution is point-jet supported on a locally finite positive additive monoid with finitely many positive generators, so is its inverse. The entire bank $(\nu_K)_{K \geq 0}$ admits the affine transport and total-action cuts of Lemmas 5.11 and 5.10, including their whole coefficient-valued bounds. In particular these conclusions apply to the original degree-zero bank

$$(5.48) \quad U_*(T) = \sum_{m \geq 1} e^{-mT}/m, \quad \nu_0 = \sum_{m \geq 1} m^{-1} \delta_m, \quad \nu_K = 0 \ (K > 0).$$

Proof. The positive action gap controls the number of convolution factors even when a factor has degree zero. We check the inverse estimates with this gap, then identify the inverse of the displayed original bank.

Step 1. We construct the inverse while retaining degree zero. The input belongs to $\mathcal{M}_{S,1,\delta}(\mathbb{C})$ with gap g . Lemma 5.4 gives absolute convergence of U and U' and their decay to zero on sufficiently far strips. Lemma 5.8 therefore gives the unique small analytic inverse and its convergent derivative expansion (5.25).

Expanding its n th term now uses every ordered tuple $K_1, \dots, K_n \geq 0$. Put $K = \sum_i K_i$. Differentiation of $V^K e^{-\beta V}$ still gives (5.28) for $0 \leq j \leq \min(K, n-1)$, with $m = K - j \geq 0$. Each original action is at least g and at least δK_i . Consequently $\beta \geq gn$ and $\beta \geq \delta K \geq \delta m$. In particular, the degree-zero terms retain a positive action gap.

For any logarithmic radius $r > 0$, the finite derivative sum satisfies

$$\sum_j \binom{n-1}{j} (K)_j r^{K-j} \beta^{n-1-j} \leq r^K (\beta + K/r)^{n-1} \leq e^{c_r \beta} (C_r \beta)^{n-1}.$$

This also holds at $K = 0$: only $j = 0$ remains. Multiplying by $e^{-S_{\text{inv}} \beta}$ and using $\beta^{n-1} e^{-q\beta} \leq (n-1)! q^{-(n-1)}$ gives, for the absolute integral of this tuple after division by $n! \lambda^n$,

$$\frac{C_r^{n-1}}{n \lambda^n q^{n-1}} \prod_{i=1}^n \int e^{-S\beta_i} d\mu_{K_i}(\beta_i).$$

The sum over all $K_i \geq 0$ is $(N_S/\lambda) \vartheta^{n-1}/n$. Summing over n gives (5.30) with the source degrees starting at zero, and summing $n > N$ gives (5.31). Hence the full degree-zero terms and every original convolution label converge in $\mathcal{M}_{S_{\text{inv}},r,\delta}(\mathbb{C})$. Realization of that source is the analytic inverse already obtained from the contour formula.

On a compact action interval $[0, B]$, a contributing term must satisfy $n \leq B/g$ and $K \leq B/\delta$. For fixed n, K , only finitely many nonnegative tuples have sum K . Formula (5.33) therefore proves the distribution identity by finite operations on that interval, with all degree sums starting at zero. If the original distribution is supported by point jets on the stated positive additive monoid, convolution adds their locations and orders, and multiplication by $(-\beta)^{n-1}$ preserves point support. The bounds on n and K make the sum locally finite. This proves the inverse point-jet assertion.

The inverse belongs to $\mathcal{M}_{S_{\text{inv}},r,\delta}(\mathbb{C})$ with gap g . Lemmas 5.10 and 5.11 permit degree zero in their hypotheses. Their formulas therefore apply to this entire inverse; in particular (5.41) retains both derivative sums with the present range of K . For coefficient-valued measures, replace scalar

absolute values by the coefficient norm before taking these product integrals. Submultiplicativity gives the displayed majorant, with the full input norm, and Proposition 5.12 controls the original parameter families. Thus the coefficient-valued transport and cutoff bounds retain the complete inverse bank.

Step 2. We retain the original logarithmic bank. For (5.48), take $g = \delta = 1$ and any $S > 0$. It has positive majorant $\mu_0 = \nu_0$ and

$$N_S = \sum_{m \geq 1} \frac{e^{-Sm}}{m} \leq \frac{e^{-S}}{1 - e^{-S}} < \infty.$$

Its distribution is the original atomic measure on the positive integers. The convergent power series for the logarithm gives $U_*(T) = -\log(1 - e^{-T})$ on $\Re T > 0$, with its determination tending to zero as $\Re T \rightarrow \infty$. All the conclusions above therefore apply for each $\lambda > 0$.

When $\lambda = 1$, the time equation can also be solved directly:

$$(5.49) \quad V = T + \log(1 - e^{-T}) = \log(e^T - 1), \quad T = V + \log(1 + e^{-V}), \quad J_*(V) = \log(1 + e^{-V}).$$

These identities first hold on the far positive ray and then on the far strips with the determinations just specified. The positive quantity $e^T - 1$ fixes the determination of $\log(e^T - 1)$ on that ray. Uniqueness of the small inverse identifies this J_* with the inverse constructed from the original bank.

In its ordered source expansion every $K_i = 0$, so $K = j = m = 0$. In particular its full measure is

$$(5.50) \quad \nu_0^{\text{inv}} = \sum_{n \geq 1} \frac{(-\beta)^{n-1}}{n!} \nu_0^{*n}, \quad \nu_m^{\text{inv}} = 0 \quad (m > 0).$$

At action $M \in \mathbb{N} \setminus \{0\}$ the first sum has the finite coefficient

$$\sum_{n=1}^M \frac{(-M)^{n-1}}{n!} \sum_{\substack{m_1 + \dots + m_n = M \\ m_i \geq 1}} \frac{1}{m_1 \cdots m_n}.$$

The whole weighted bound just proved makes these coefficients an absolutely convergent power series in e^{-V} near zero. Its value is $J_*(V)$, so uniqueness of power-series coefficients identifies the coefficient at M with $(-1)^{M+1}/M$. This identifies the collected atoms of the measure defined in (5.50); the ordered presentation and its positive majorant remain available for every total-action cut, affine transport, and complete tail estimate. $\square$

Proof of Theorem 5.1. Lemma 5.5 gives the prepared equation and the analytic coordinates on the prescribed physical half-sections. Choose ϵ, v by Lemma 5.6. In its normalized coordinates $c = v > 0$; a positive constant change of outgoing coordinate changes this constant to any corresponding $c > 0$. Formula (5.17) is the exact passage identity, and (5.18) gives the required second moment bound for any $a > 0$. Lemma 5.7 gives T_U .

Taking logarithms of the actual passage yields $(-\log z + \log c)/\lambda = T - U(T)/\lambda$. Lemma 5.9 supplies its unique small solution and the distribution in (5.32). Exponentiating $-T = -V - J(V)$ gives the displayed formula for D^{-1} . Reality follows either from the real measures or from the real fixed-point iterates. Since $U'(T) \rightarrow 0$, the derivative of $\log D(w)$ with respect to $\log w$ is $\lambda - U'(-\log w) > 0$ for sufficiently small positive w , so this inverse is the positive physical branch.

The full convolution tails are bounded by (5.30) and (5.31). Lemma 5.10 retains every derivative in (5.41) and bounds the entire omitted part by (5.38). The inverse logarithmic change of variable is included by Lemma 5.11, with $\alpha = 1/\lambda$, $b = \log c/\lambda$, and $h = 0$. Proposition 5.12 gives these bounds in the original analytic input norms. Every argument uses only $\lambda > 0$, with $\delta = \min\{1, \lambda\}$. $\square$

6. CANONICAL EQUATIONS AND PHYSICAL MATCHING

We construct the normalizing maps from the equation and then identify their values on the prescribed transverse sections. The estimates apply to the whole dependence on the transverse coordinate. This is needed when a transverse section is a graph rather than a coordinate line.

The equations are fixed. Uniform family estimates below concern finitely many such equations and analytic section graphs or recipes on their specified input disks. The multiplicity, nonzero eigenvalue and leading drift of each equation remain fixed in these estimates.

Theorem 6.1. *Every actual hyperbolic-sector passage of an isolated semihyperbolic real analytic singularity admits finite analytic preparation and sectorial normalization derived from its own differential equation. In a suitable positive coordinate s its direct passage has the form*

$$(6.1) \quad D(s) = k(ce^{-Z}Z^aG(s)), \quad Z = \beta s^{-p}, \quad p \in \mathbb{N} \setminus \{0\}, \quad \beta, c > 0,$$

where k is an analytic germ with $k(0) = 0$, $k'(0) = 1$, and G is the whole sectorial normalizing factor, with physical limit one. The construction includes the true inverse on the specified outgoing section, and the full matching between sectorial labels. It does not require a convergent formal normalizer, an analytic weak separatrix, or real Martinet–Ramis transition data. The independently variable weak coordinate, every native normalizing source and all lower-height terms are retained before their physical evaluation.

Definition 6.2. A planar singularity is *semihyperbolic* if its linear part has one zero eigenvalue and one nonzero real eigenvalue. Its strong branch is an invariant analytic curve tangent to the nonzero eigenspace. An *actual hyperbolic-sector passage* here has a strong boundary and an off-axis boundary with opposite approach and departure directions, chosen sides, analytic transverse half-sections, and the actual leaf segments between them. The passage is positive, injective and endpoint-continuous as in Definition 2.2. We call the passage from the strong section to the off-axis section the *direct passage*; the other direction uses its actual inverse. This convention is independent of the sign of a chosen generator of the foliation.

A *compatible analytic sectorial label* specifies an actual holomorphic normalizer Φ_σ of the fixed prepared equation, satisfying $X\Phi_\sigma = (1 + \lambda x^p)\Phi_\sigma$, with its formal normalization, root and logarithm determinations and selected leaf paths. Its sector contains the physical tail, and its maps tend to the identity uniformly on each prescribed smaller fixed fiber disk. Retain its genuine fiber inverse Ψ_σ and strict composition and inverse-image domains. Labels are compared on a connected matched physical component with compatible continued lifts. A finite list may include narrow sectors; no native root source is inferred from formal asymptotics. For one label the uniform identity limit and Cauchy on a larger fiber disk give $\partial_y(\Phi_\sigma - y) \rightarrow 0$, so the inverse is also obtained by a uniform contraction on smaller fixed disks. For whole parameter families, the uniform original norms, nonvanishing leading data and inverse-image margins are part of the supplied family data, not consequences of pointwise limits.

A *source-equipped reference label* additionally retains the whole forward and inverse root sources and their actual synthesis identities in Definition 6.5. For each fixed equation and physical core choose a common nonreal summation direction before constructing such reference sources by Proposition 6.7. A proved restriction or continuation of those actual source identities is also source-equipped. An arbitrary compatible analytic label is instead represented by the full weak replay of Proposition 6.15.

We first record the analytic operations used in the preparation. In particular, no separatrix or normalization theorem is being imported.

Lemma 6.3. *A holomorphic map with invertible derivative has a holomorphic local inverse. An analytic vector field has a unique local analytic flow, jointly analytic in time and initial point. Holonomy along a fixed compact regular leaf segment between transverse analytic arcs is an analytic local diffeomorphism; its inverse is holonomy along the reversed segment. For*

finitely many such maps and segments, all compositions and inverses can be restricted to common neighborhoods with strictly contained images.

Proof. We construct the inverse and flow by uniform iteration and then solve the transverse hitting equation.

Step 1. We construct analytic inverses. After linear changes, the inverse problem is $z = w - R(z)$, where $R(0) = 0$ and $DR(0) = 0$. On a sufficiently small closed ball, $\sup \|DR\| < 1/2$. For smaller w , iteration starting at zero stays in that ball, and consecutive iterates have geometrically decreasing differences. They converge uniformly on smaller complex balls to a holomorphic solution. The derivative bound proves uniqueness and both inverse identities. Applying this argument to $(x, y) \mapsto (x, F(x, y))$ also proves the analytic implicit-function assertion when F_y is invertible.

Step 2. We construct the local flow. On a closed complex neighborhood of an initial point let M and L bound the field and its derivative. Choose a time disk of radius T with $TL < 1/2$ and TM smaller than the distance to the boundary. The operator $z(t) \mapsto z_0 + \int_0^t X(z(\tau)) d\tau$ on holomorphic paths has successive differences bounded by a geometric series. Its limit is the unique solution, holomorphic also in z_0 by uniform convergence of the iterations. Uniqueness gives the flow law and the inverse flow. A finite chain of such neighborhoods covers a compact regular segment.

Step 3. We construct transverse holonomy. Write the flow as $\Phi(t, z)$ and a transverse source arc as $\alpha(u)$. If the target is $f = 0$, transversality gives $df(X) \neq 0$ at the endpoint. The implicit-function assertion supplies the hitting time $t = \tau(u)$. The map $\Phi(\tau(u), \alpha(u))$ has nonzero transverse derivative: with $A = D_z\Phi$, the identities $AX = X \circ \Phi$ and $\det A \neq 0$ give

$$(6.2) \quad v'(u) \det(\beta'(v(u)), X(\beta(v(u)))) = \det A \det(\alpha'(u), X(\alpha(u))) \neq 0,$$

where β parametrizes the target. Reversing the selected segment therefore gives the true inverse. Each compact image has a positive distance from the complement of a slightly larger neighborhood; taking finitely many smaller neighborhoods gives the asserted strict margins. The construction concerns the selected segment and makes no assertion about other returns to the target. $\square$

Proposition 6.4. *Let Y be a real analytic planar vector field with an isolated semihyperbolic zero. There exist a positive integer p , finite analytic coordinate changes with analytic inverses, and a nonvanishing time factor which transform Y into*

$$(6.3) \quad X = x^{p+1}\partial_x + B(x, y)\partial_y, \quad B(0, y) = y.$$

Its transverse polynomial can be reduced to $1 + \lambda x^p$, with $\lambda \in \mathbb{R}$. More precisely, for any fixed integer $h > p$ one can achieve

$$(6.4) \quad B(x, y) = (1 + \lambda x^p)y + R(x, y), \quad R(x, y) = O(x^h),$$

on a product of disks. All these changes are finite analytic changes; they do not assert an analytic invariant weak graph.

On a hyperbolic side there exists a positive coordinate s and a fixed $\zeta \in \mathbb{C} \setminus \{0\}$ with $x = \zeta s$ and

$$(6.5) \quad Z = -\frac{1}{px^p} = \beta s^{-p}, \quad \beta > 0.$$

The base change to this coordinate retains the full lower singular powers in the primitive of the transverse linear equation.

Proof. We straighten the strong branch, remove finitely many null-direction powers, and then normalize the transverse polynomial and its primitive.

Step 1. We construct the strong branch. Real linear coordinates put the field in the form $\dot{u} = P(u, v)$, $\dot{v} = \nu v + S(u, v)$, where P, S vanish to order two and $\nu \neq 0$. For an invariant graph $u = F(v) = v^2 G(v)$ the equation $F'(v)(\nu v + S(F(v), v)) = P(F(v), v)$ becomes $vG' = -2G + \tilde{P}(v, G)/(\nu + v\tilde{S}(v, G))$. At $v = 0$ the last quotient is constant in G . Choose its constant solution G_0 and write $H = G - G_0$. The resulting equation is

$$(6.6) \quad vH' + 2H = vL(v, H), \quad (n+2)[v^n]H = [v^{n-1}]L(v, H).$$

Here L is analytic near $(0, 0)$. Replace its Taylor coefficients by their absolute values to obtain a convergent majorant L_+ . The analytic implicit solution $K = vL_+(v, K)$ exists by Lemma 6.3; its coefficients are nonnegative by recursion. (6.6) shows inductively that they dominate the absolute coefficients of H . Thus F converges and gives the strong branch. The change $u \mapsto u - F(v)$ makes the first component divisible by u .

Step 2. We identify the finite null-direction order. The transverse component has nonzero v derivative and hence an analytic nullcline. The first component restricted to it is not identically zero, since the zero of Y is isolated. Let its exact order be $m \geq 2$. After straightening the strong branch write $\dot{u} = \sum_{j \geq 1} a_j(v)w^j$ and $\dot{v} = C(u, v)$, with $C(0, v) = c_{\text{ax}}(v) = v\ell(v)$ and $\ell(0) \neq 0$. The change $u = s_1(v)w$, where $s_1(v) = \exp \int_0^v a_1(t)/c_{\text{ax}}(t) dt$, removes the coefficient of w . The integrand is analytic because $a_1(0) = 0$.

Suppose the first remaining power is $a_j(v)w^j$, with $2 \leq j < m$. On the current transverse nullcline its coefficient of w^j is $a_j(0)$, so $a_j(0) = 0$. Under a triangular change $u = \Theta(w, v)$ the restriction to that nullcline satisfies $\dot{u} = \Theta_w \dot{w}$; the reduced order m is preserved by its analytic coordinate and unit. The change $w = z + d_j(v)z^j$, with $d_j(v) = \int_0^v a_j(t)/c_{\text{ax}}(t) dt$, removes this coefficient. Only the finitely many indices $j < m$ occur. We obtain $\dot{u} = u^m A(u, v)$, with $A(0, 0) \neq 0$.

Translate the current transverse nullcline and divide by the positive unit $\text{sign}(A(0, 0))A$. The field becomes

$$(6.7) \quad \sigma u^{p+1} \partial_u + (f(u) + b(u)v + v^2 Q(u, v)) \partial_v, \quad p = m - 1, \quad \sigma \in \{-1, 1\}, \quad b(0) \neq 0.$$

Initially $f = O(u^{p+1})$. A translation $v = w + \alpha u^{p+1}$, where $\alpha = -[u^{p+1}]f/b(0)$, makes $f = O(u^{p+2})$: the differentiated translation has order $2p + 1 \geq p + 2$. All inverses here follow from Lemma 6.3. If χ is the composite coordinate map up to this stage and Y_r is the displayed real prepared field, then $\chi_* Y = a_{\text{time}} Y_r$, where a_{time} is the positive analytic unit on the image of χ obtained from $\text{sign}(A(0, 0))A$ by transporting it through the nullcline and final translations. We keep this analytic function, not just its sign.

Step 3. We construct the transverse polynomial. Put $b_* = b(0)$ and let $C_*(v)$ be the transverse component at $u = 0$. The analytic coordinate

$$(6.8) \quad f_*(v) = v \exp \int_0^v \left(\frac{b_*}{C_*(t)} - \frac{1}{t} \right) dt$$

satisfies $C_* f'_* = b_* f_*$. With a chosen root $\rho^p = \sigma/b_*$, set $x = \rho u$ and $y = f_*(v)$, and divide the field by b_* . This gives (6.3), with $B(x, 0) = O(x^{p+2})$. For $j = 1, \dots, p$, the nonlinear part of the coefficient of x^j is removed by $y = z + x^j v_j(z)$. The required homological operator is $Lv = zv' - v$; on $\sum_{n \geq 2} c_n z^n$ its inverse divides c_n by $n - 1$. This inverse preserves convergence and produces an actual analytic change. We obtain

$$(6.9) \quad x^{p+1} \partial_x + (P(x)y + x^{p+1} R_0(x, y)) \partial_y, \quad P(x) = 1 + \sum_{j=1}^p P_j x^j, \quad R_0(x, 0) = O(x).$$

If $r_1(x) = \partial_y R_0(x, 0)$, the change $y = \exp(\int_0^x r_1(t) dt)w$ removes this remaining linear term. It leaves the constant forcing term in place.

Step 4. We make the exact base change. Set $\lambda = P_p$. The implicit equation

$$(6.10) \quad v(x)^{-p} - 1 - p \sum_{j=1}^{p-1} \frac{P_j x^j}{p-j} - p \lambda x^p \log v(x) = 0, \quad v(0) = 1$$

has an analytic solution: its derivative with respect to v at $(0, 1)$ is $-p$. Put $g(x) = xv(x)$ and $q_*(x) = v(x)^{p+1}/(v(x) + xv'(x))$. Direct differentiation gives

$$(6.11) \quad -\frac{1}{pg(x)^p} + \lambda \log g(x) = -\frac{1}{px^p} - \sum_{j=1}^{p-1} \frac{P_j x^{j-p}}{p-j} + \lambda \log x,$$

$$(6.12) \quad q_*(x)P(x) = 1 + \lambda g(x)^p.$$

Changing the base to $g(x)$ and multiplying the field by q_* gives the desired transverse polynomial. In particular the new height is

$$(6.13) \quad -\frac{1}{pg(x)^p} = -\frac{1}{px^p} - \sum_{j=1}^{p-1} \frac{P_j x^{j-p}}{p-j} - \lambda \log v(x).$$

The displayed lower singular powers need not be bounded. None has been discarded or replaced by a small reciprocal-height correction.

For completeness we can raise the remainder to any fixed order $h > p$. The formal invariant graph is determined recursively by $x^{p+1}H' = B(x, H)$, since the coefficient of its newest unknown is 1. Use only its polynomial jet through degree $h-1$ to translate the fiber. The new constant term is $O(x^h)$. If its linear coefficient is $b_h(x)$, scale the fiber by $\exp \int_0^x (b_h(t) - P(t))/t^{p+1} dt$; the integrand is analytic. Successively remove the nonlinear fiber terms at orders $1, \dots, h-1$ by the operator L above. No new constant or linear term of lower order is introduced. This gives remainder $O(x^h)$ with polynomial P . The base change (6.10) preserves that order and gives (6.4).

Step 5. We fix the real side and time direction. Complex conjugation in the scaled base is $x \mapsto \omega \bar{x}$, where $\omega = \rho/\bar{\rho}$ and $\omega^p = 1$. Each coefficient operation just used commutes with this involution. Thus $P_j \omega^j = \bar{P}_j$, and $\lambda = P_p$ is real. The units v, q_* and the fiber scaling are positive on the real slice after restriction, since their values at zero are one. The total time factor has the sign of b_* . More explicitly, let T be the complete coordinate map to the final base x and fiber y , and let $a_* = g^{-1}$ be the actual base inverse. The relation to the original field, including the subsequent finite fiber changes, is

$$(6.14) \quad T_*Y = U_T X, \quad U_T(x, y) = \frac{b_* a_{\text{time}}(\chi(T^{-1}(x, y)))}{q_*(a_*(x))}.$$

Every factor is analytic and nonzero on the chosen neighborhood. The reverse coordinate construction uses T^{-1} and the reciprocal time factor on its literal domain.

On the chosen side write $u = \epsilon s_0$, $s_0 > 0$, $\epsilon = \pm 1$. Opposite strong and off-axis directions give $\sigma \epsilon^p = -\text{sign}(b_*)$. Consequently $\zeta = \epsilon \rho$ satisfies $\beta = -1/(p\zeta^p) > 0$. After the base change the positive coordinate is $s = s_0 v(\zeta s_0)$, proving (6.5). The direct passage follows decreasing normalized time; whether this is forward or reverse original time is determined by the retained time factor. $\square$

We next construct the normalizer, including its inverse, on an algebra which keeps every transverse coefficient. We use $\Gamma(t) = \int_0^\infty e^{-u} u^{t-1} du$ for $t > 0$.

Definition 6.5. Fix $p \geq 1$, a root $\kappa^p = -1/p$, and the prescribed lifted relation $x = \kappa \exp(-\text{Log } Z/p)$. Let q be a nonreal complex number with $|q| = 1$. Write Log_+ for the

logarithm on the right half-plane with imaginary part in $(-\pi/2, \pi/2)$. On the initial Laplace domain $\operatorname{Re}(qZ) > A > 0$, choose $\operatorname{Log} q$ compatibly with the prescribed lift, so that

$$(6.15) \quad \operatorname{Log}_+(qZ) = \operatorname{Log} q + \operatorname{Log} Z, \quad q^{n/p} = \exp((n/p) \operatorname{Log} q).$$

These logarithms and roots are continued together along the retained paths. Choose $\epsilon > 0$ and $\rho_0 > 0$ such that $\rho_0^p < 1/2$ and qz^p stays angularly separated from both real rays on each sector below. Set

$$(6.16) \quad \mathcal{D} = \{|z| < \rho_0\} \cup \bigcup_{j=0}^{p-1} \{z \neq 0 : |\arg z - 2\pi j/p| < \epsilon/p\}.$$

For $A > 0$, put $\delta = 1/p$ and

$$(6.17) \quad N_A(\phi) = 8pA^{-\delta} \sup_{z \in \mathcal{D} \setminus \{0\}} e^{-A|z|^p} (1 + A|z|^p)^{1+\delta} \frac{|\phi(z)|}{|z|}.$$

The space $\mathcal{B}_A$ consists of pairs (c, ϕ) , where $c \in \mathbb{C}$, ϕ is holomorphic on $\mathcal{D}$, $\phi(0) = 0$, and $N_A(\phi) < \infty$, with norm $|c| + N_A(\phi)$. Constants are represented by $(c, 0)$. Multiplication of minors is

$$(6.18) \quad (\phi \star \psi)(z) = \int_0^1 \phi(z(1-t)^{1/p}) \psi(zt^{1/p}) \frac{dt}{t(1-t)};$$

the product of pairs includes the two constant-minor products.

A *native normalizing source* is the constant and continued root minor of a specified source-equipped equation reference, with this domain, norm and actual synthesis identity. It may also be an independently supplied source with a proved actual identity on its retained determination, as in Definition 8.4. A compatible analytic sectorial label alone does not assert existence of such a source. Its formal and analytic realizations are respectively

$$(6.19) \quad c + \sum_{n \geq 1} f_n x^n \longleftrightarrow \left(c, \sum_{n \geq 1} \frac{f_n \kappa^n q^{n/p}}{\Gamma(n/p)} z^n \right),$$

$$(6.20) \quad \mathcal{S}_q(c, \phi)(Z) = c + \int_0^\infty e^{-qZt} \phi(t^{1/p}) \frac{dt}{t}.$$

For an endpoint radius $r > 0$ the *whole endpoint norm* is $\sum_{j \geq 0} r^j \|(c_j, \phi_j)\|_{\mathcal{B}_A}$. Additional analytic inputs use the original radii of Definition 4.1.

Lemma 6.6. *The space $\mathcal{B}_A$ is a commutative unital Banach algebra. The map (6.20) is an algebra map of norm at most one on $\operatorname{Re}(qZ) \geq A$. An analytic germ $T(x, y) = O(x^d)$ on larger product disks, with $d \geq 1$, has an entire root minor t satisfying, at any fixed smaller endpoint radius r ,*

$$(6.21) \quad \sum_j r^j N_A(t_j) \leq C_d A^{-d/p} \quad (A \geq A_0).$$

If $d > p$, the minor t/z^p has whole norm at most $C_d A^{-(d-p)/p}$. Constants and all radii in these assertions are chosen before coefficient extraction.

Proof. We prove the product estimate, its moment identity, and the estimate for analytic germs.

Step 1. We define the product and prove completeness. For $u > 0$ put $w_A(u) = u^{\delta-1}/(1 + Au)^{1+\delta}$. If $0 < v \leq u/2$, then $w_A(u-v)/w_A(u) \leq 4$. Splitting convolution at $u/2$ and using $\int_0^\infty w_A(v) dv = pA^{-\delta}$ gives

$$(6.22) \quad (w_A * w_A)(u) \leq 8pA^{-\delta} w_A(u).$$

On each root ray, $\phi(z)/z^p$ is bounded by a constant times $e^{A|z|^p} w_A(|z|^p)$. The substitution $v = ut$ converts (6.18) divided by z^p to ordinary convolution. Thus (6.22) gives $N_A(\phi \star \psi) \leq$

$N_A(\phi)N_A(\psi)$. The endpoint singularities are bounded by $t^{\delta-1}$ and $(1-t)^{\delta-1}$, which are integrable. They justify holomorphy and associativity by absolute integration. A norm-Cauchy sequence converges locally uniformly, including after division by z at zero; the weighted bound passes to the limit. This proves completeness.

Step 2. We construct moments and synthesis. Integration by parts gives $\Gamma(a+1) = a\Gamma(a)$. Changing variables $(u, v) = (s(1-t), st)$ in the product of two positive Gamma integrals gives

$$(6.23) \quad \int_0^1 (1-t)^{a-1} t^{b-1} dt = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)} \quad (a, b > 0).$$

Hence the normalized monomials in (6.19) multiply exactly as powers of x . The estimate

$$(6.24) \quad \left| \int_0^\infty e^{-qZt} \phi(t^{1/p}) \frac{dt}{t} \right| \leq \frac{N_A(\phi)A^\delta}{8p} \int_0^\infty w_A(t) dt = \frac{N_A(\phi)}{8}$$

proves bounded synthesis. Absolute double integration proves its product identity. For $n \geq 1$, the Gamma integral on the positive real axis and holomorphic continuation throughout $\text{Re}(qZ) > 0$ give the monomial identity

$$(6.25) \quad \begin{aligned} \mathcal{S}_q \left(0, \frac{\kappa^n q^{n/p} z^n}{\Gamma(n/p)} \right) (Z) &= \kappa^n \exp \left(\frac{n}{p} \text{Log } q - \frac{n}{p} \text{Log}_+(qZ) \right) \\ &= \kappa^n \exp \left(-\frac{n}{p} \text{Log } Z \right) = x^n. \end{aligned}$$

The second equality uses (6.15). Consequently synthesis realizes the prescribed analytic monomials and the original equation on its fixed lift.

Step 3. We admit analytic input germs. For the last assertion, Cauchy's coefficient estimate on the larger product gives $|[x^n y^j]T| \leq MR_x^{-n} R_y^{-j}$. Sum the geometric endpoint series at $r < R_y$ first. Split the remaining sum into the p congruences $n = pl + k$. The Gamma recurrence bounds $1/\Gamma(l + k/p)$ by a constant times a geometric multiple of $1/(l-1)!$ for $l \geq 1$; only finitely many initial terms occur. It follows that

$$(6.26) \quad \sum_j r^j |t_j(z)| \leq C|z|^d e^{b|z|^p}.$$

In (6.17) substitute $u = A|z|^p$ and take $A \geq 2b$. The resulting bounded function of u leaves the factor $A^{-d/p}$. Division by z^p gives the second estimate when $d > p$. All sums used in this proof are absolute in the whole endpoint norm. $\square$

Proposition 6.7. *Let X be the prepared (6.4), with $h > p$, and fix the data of Definition 6.5. On fixed smaller endpoint disks there exist sources for holomorphic maps $\Phi(x, y)$ and $\Psi(x, Y)$ such that*

$$(6.27) \quad X\Phi = (1 + \lambda x^p)\Phi,$$

$$(6.28) \quad x^{p+1}\Psi_x + (1 + \lambda x^p)Y\Psi_Y = B(x, \Psi),$$

$$(6.29) \quad \Phi(x, \Psi(x, Y)) = Y, \quad \Psi(x, \Phi(x, y)) = y.$$

Both maps tend to the identity in the endpoint variable. Their nonconstant source norms tend to zero as $A \rightarrow \infty$, on disks fixed before A is chosen. Their formal expansions are independent of the nonreal summation direction, with remainders bounded in the whole endpoint norm by

$$(6.30) \quad CL^N \Gamma(1 + N/p) |x|^N \quad (N \geq 1)$$

on every strictly smaller closed sector of their domains. These domains have opening greater than π/p . On such a domain the maps are uniquely specified by their formal expansion and their identity limit. In the chosen final base coordinate, the conclusions hold for (6.3) before its further finite fiber changes, whose restrictions to $x = 0$ are the identity, are made.

Proof. We solve the full normalizing equation in an endpoint Wiener space, construct the inverse in that space, and then control the sector and the asymptotic remainder.

Step 1. We solve the whole equation. Write $P(x) = 1 + \lambda x^p$, $\Phi = y + u$, and note that $x^{p+1}\partial_x = \partial_Z$. The exact equation for u is

$$(6.31) \quad \partial_Z u + (y\partial_y - 1)u = -(P - 1)(y\partial_y - 1)u - R(1 + \partial_y u).$$

Let τ be the entire minor of $P - 1$, let $\mathfrak{r}_i$ be that of $[y^i]R$, and let ϕ_j denote the unknown minor of $[y^j]u$. The moment rule in Lemma 6.6 sends ∂_Z to multiplication by $-qz^p$. Thus

$$(6.32) \quad (j - 1 - qz^p)\phi_j = -(j - 1)(\tau \star \phi_j) - \mathfrak{r}_j - \sum_{i+l=j+1, l \geq 1} l(\mathfrak{r}_i \star \phi_l).$$

Fix an endpoint radius r strictly inside the equation's disk and use $\|\phi\|_{A,r} = \sum_j r^j N_A(\phi_j)$. For $j \neq 1$, the disk and angular separation in (6.16) give constants independent of j such that

$$(6.33) \quad \frac{1}{|j - 1 - qz^p|} \leq \frac{C}{j + 1 + |z|^p}, \quad \left| \frac{j - 1}{j - 1 - qz^p} \right| \leq C.$$

In the sum in (6.32), $l \leq j + 1$. The first bound absorbs the endpoint derivative, and $r^j = r^{-1}r^i r^l$ gives the whole bound $Cr^{-1}\|\mathfrak{r}\|_{A,r}\|\phi\|_{A,r}$.

Step 2. We determine the resonant row. When $j = 1$, the polynomial term vanishes, and we divide the equation by $-qz^p$. The forcing is divisible by z^h . In the remaining sum only $l = 1, 2$ can occur, and

$$(6.34) \quad \frac{(\mathfrak{r}_i \star \phi_l)(z)}{z^p} = \int_0^1 (1 - t) \left(\frac{\mathfrak{r}_i}{z^p} \right) (z(1 - t)^{1/p}) \phi_l(z t^{1/p}) \frac{dt}{t(1 - t)}.$$

The extra factor has modulus at most one. The algebra estimate therefore bounds this row by $2r^{-1}\|\mathfrak{r}/z^p\|_{A,r}\|\phi\|_{A,r}$. Solving (6.32) for ϕ gives an affine operator whose linear part has norm at most

$$(6.35) \quad C\|\tau\|_A + Cr^{-1}\|\mathfrak{r}\|_{A,r} + 2r^{-1}\|\mathfrak{r}/z^p\|_{A,r}.$$

By Lemma 6.6, this tends to zero. The forcing tends to zero as well. Choose A so that (6.35) is less than $1/2$. The geometric series of iterates converges in the complete whole endpoint space. It gives all ϕ_j simultaneously and has norm at most twice the forcing norm.

The formal recursion is unique. Outside $j = 1$, the reciprocal in (6.33) preserves root order, while τ raises it. In the resonant row, multiplication by the divided remainder raises root order by at least $h - p > 0$. Consequently each root coefficient is determined by earlier ones. Different sufficiently large weights agree as germs at zero and hence as continued minors on $\mathcal{D}$. The whole norm tends to zero by the forcing estimate. Synthesis, with a strict endpoint margin for differentiation, turns (6.32) into (6.27).

Step 3. We construct the inverse in the source algebra. Choose $0 < r_2 < r_1 < r$ before increasing A . In $\mathcal{W}_{r_2}(\mathcal{B}_A)$ solve

$$(6.36) \quad v(Y) = -u(Y + v(Y)).$$

Take a ball of radius η with $r_2 + \eta < r_1$. The whole norm of u tends to zero, and its derivative on r_1 is bounded by $\|u\|_{A,r}/(r - r_1)$ times a fixed geometric constant. Thus the right side preserves this ball and has Lipschitz constant less than $1/2$ when A is large. Its iterates converge in the whole space. The synthesis $\Psi = Y + v$ satisfies the first inverse identity. Since $\sup |\partial_y u| < 1/2$ on the larger endpoint disk, $y \mapsto y + u(x, y)$ is injective there: integrate its derivative along the segment between two points. This gives the second identity on its literal inverse image. Differentiating these identities proves (6.28). The formal inverse is the coefficientwise inverse of the uniquely determined formal forward map.

Step 4. We control the sector opening and remainder. The Laplace ray can be rotated through $|\arg t| < \epsilon$ in (6.20). On a common strict half-plane the integral on the connecting circle at radius a is $O(a^{1/p})$ as $a \downarrow 0$, and the circle at radius b tends to zero exponentially as $b \rightarrow \infty$. Contour deformation therefore identifies these integrals. Their domains form a far sector in Z of opening greater than π , hence one in x of opening greater than π/p .

The Taylor remainder of the root minor on a fixed smaller root disk has whole norm at most $CL^N|z|^N$. Along each strict ray the exponential bound extends this to $CL^N|t|^{N/p}e^{b|t|}$, after increasing C, L, b independently of N : outside the root disk use the bound on the whole minor and the finite geometric sum of its Taylor polynomial. Integrating the resulting bound against $e^{-qZ}dt/t$ gives $CL^N\Gamma(N/p)|Z|^{-N/p}$ on every strict subsector. Increasing C, L gives (6.30). The inverse has this estimate because its whole minor belongs to this algebra.

For uniqueness, compose two maps with identical formal expansions to obtain a fiber map $T(x, Y)$ commuting with $x^{p+1}\partial_x + P(x)Y\partial_Y$. Its coefficient of Y^j satisfies

$$(6.37) \quad x^{p+1}T'_j + (j-1)P(x)T_j = 0, \quad T_j = c_j E(x)^{1-j}, \quad E(x) = x^\lambda e^{-1/(px^p)}.$$

A sector of opening greater than π/p contains a ray on which E grows exponentially and another on which it decays exponentially. Boundedness of $T - Y$ on a fixed endpoint disk forces $c_j = 0$ for $j \neq 1$, and its identity limit forces $c_1 = 1$. Thus $T = \text{id}$.

Finally, each of the finitely many analytic fiber changes and its inverse has an entire root minor by Lemma 6.6. Choose strict intermediate endpoint disks and compose their whole sources using Proposition 4.20. This undoes the preparation without extracting a transverse coefficient first. For a changed base coordinate one applies the construction to the exact transformed equation; no bounded substitution on an arbitrary completed native source space is asserted. $\square$

Lemma 6.8. *Let E be the fixed Banach space of a whole endpoint or analytic-input family. Suppose that the two lateral realizations of a specified root minor glue to an E -valued holomorphic function F on $|Z| > R_0$, $|\arg Z + d| < \alpha$, where $\alpha > \pi/2$, and that $\|F(Z)\|_E \leq C|Z|^{-\delta}$, $\delta = 1/p$. The constant part is assumed to have been separated. Then the specified root minor continues across this direction with an exponential bound in the original whole norm. On a smaller sector, its divided density $m(t)$ satisfies, for every fixed integer $k \geq 0$,*

$$(6.38) \quad \|t^k m^{(k)}(t)\|_E \leq C_k C |t|^{\delta-1} e^{A_1|t|}.$$

In particular, for $B > A_1$ its radial weighted integral is at most $C_0 C \Gamma(\delta)/(B - A_1)^\delta$. The whole weighted supremum in (6.17), taken on this continued root sector together with the original germ disk, is finite and tends to zero at large weight. The supremum here adds only the specified root lift; it does not require or assert continuation across the corresponding directions on the other lifts.

Proof. We estimate a contour inverse and identify it with the original density by its Laplace transform.

Step 1. We construct the contour inverse. Rotate first to $d = 0$, retaining the Jacobian in the density. Choose $\pi/2 < \theta < \alpha$ and a small $\eta > 0$ with $\theta - \eta > \pi/2$. For $R > \max(R_0, 1)$, let Γ_R run from infinity on the ray $-\theta$ to $Re^{-i\theta}$, then along the circle through the right half-plane to $Re^{i\theta}$, and then to infinity on the ray θ . Define

$$(6.39) \quad m(t) = \frac{1}{2\pi i} \int_{\Gamma_R} e^{tZ} F(Z) dZ, \quad |\arg t| < \eta.$$

On the two rays, $\text{Re}(tZ) \leq -b|t||Z|$ for some $b > 0$. The integral and each of its derivatives therefore converge absolutely on compact subsets of this sector. Analytic contour deformation shows that it is independent of R .

For the estimate choose instead the inner radius $\max(R, 1/|t|)$. On a ray, the bound for the k th derivative, multiplied by $|t|^k$, is $C|t|^k \int_{\max(R, 1/|t|)}^\infty e^{-b|t|r} r^{k-\delta} dr$. Substitution $u = |t|r$ bounds

this by a constant times $C|t|^{\delta-1}$. On the finite circular part, the derivative bound is bounded by $C_k C|t|^{\delta-1} e^{2R|t|}$: for $|t|R \leq 1$ its radius is $1/|t|$, and for $|t|R \geq 1$ the remaining polynomial is absorbed by one further factor $e^{R|t|}$. This proves (6.38) with $A_1 = 2R$ after enlarging constants. Integrating the bound on the positive ray gives the stated Gamma bound.

Step 2. We identify it with the original minor. For real Z sufficiently large, absolute integration gives

$$(6.40) \quad \int_0^\infty e^{-Zt} m(t) dt = \frac{1}{2\pi i} \int_{\Gamma_R} \frac{F(W)}{Z - W} dW = F(Z).$$

For the last equality, close the contour through the right of Z . The outer arc has bound $O(L^{-\delta})$ at radius L . The closed contour is clockwise, and the pole of $F(W)/(Z - W)$ has residue $-F(Z)$, giving the sign in (6.40). The bound (6.38) justifies rotation to neighboring Laplace rays.

Here is the uniqueness argument needed to compare with the old lateral densities. If an exponentially weighted, continuous density has zero Laplace transform, multiply it by a sufficiently strong exponential and push its finite measure forward by $t \mapsto e^{-t}$. Every moment on $[0, 1]$ then vanishes, including the zeroth by a limit. Polynomials approximate continuous functions there: the Bernstein polynomials converge uniformly, since uniform continuity controls the summands with $|j/n - x| < a$, and the remaining binomial mass is at most $1/(4na^2)$. Thus integration against every continuous test function is zero. Continuous compactly supported tests on the original half-line show that the density is zero pointwise. Apply this scalar argument after each continuous linear functional on E ; these separate points. It follows that (6.39) agrees with the given densities on their lateral sectors. Thus $z^p m(z^p)$ agrees with the original root germ near zero and extends that germ across the specified direction.

Step 3. We bound the whole source norm. The local germ bound controls a fixed root disk, and (6.38) controls the exterior of the specified continued root sector. Substitution in the weight of (6.17), using $A > A_1$, proves finiteness on their union. On the disk rescale $u = A|z|^p$; outside a smaller fixed disk exponential slack makes the norm tend to zero. This proves the large-weight claim. All estimates were in E . Cauchy's formula for additional parameters therefore keeps their original outer radii; at inner radii $r_j < \rho_j$ the coefficient sum is bounded by $\prod_j (1 - r_j/\rho_j)^{-1}$ times the original bound.

Finally, undoing the initial rotation changes a density by its Jacobian. In the convention (6.20), the density on the ray qt is characterized by

$$(6.41) \quad q m(qt) = \frac{\phi(t^{1/p})}{t}.$$

This is continuation of the prescribed source, rather than a choice of a different germ specified only by those formal coefficients. $\square$

Definition 6.9. For a zero-constant formal series $f(x) = \sum_{n \geq 1} f_n x^n$, its ordinary congruence decomposition and minors are

$$(6.42) \quad f(x) = \sum_{k=0}^{p-1} x^k \left(f_k + \sum_{l \geq 1} f_{pl+k} \xi^l \right), \quad \xi = x^p, \quad f_0 = 0, \quad H_k(\eta) = \sum_{l \geq 1} \frac{f_{pl+k} \eta^{l-1}}{\Gamma(l)}.$$

The constants f_k with $1 \leq k < p$ are retained separately.

Lemma 6.10. *The ordinary minors in (6.42) are recovered from all p rotated lifts of the whole root minor. If all these lifts continue across a direction with whole exponential bounds, then the ordinary minors continue there with whole exponential bounds. A single removable lift supplies the root reconstruction of Lemma 6.8 without providing the other lifts used in this projection.*

Proof. Use the unscaled root germ $\chi(w) = \sum_{n \geq 1} f_n w^n / \Gamma(n/p)$, obtained from (6.19) by $w = \kappa q^{1/p} z$. For $\omega = e^{2\pi i/p}$, put $\chi_k(w) = p^{-1} \sum_{j=0}^{p-1} \omega^{-jk} \chi(\omega^j w)$. When $\eta = w^p$, $H_0(\eta) = \chi_0(w)/\eta$. For $1 \leq k < p$, write $a = k/p$ and $F_k(\eta) = \chi_k(w)/w^k$. Then

$$(6.43) \quad B_k(\eta) = \frac{1}{\Gamma(1-a)} \int_0^1 (1-t)^{-a} t^{a-1} F_k(\eta t) dt, \quad H_k = B'_k.$$

The beta identity (6.23) gives $B_k = \sum_{l \geq 0} f_{p l + k} \eta^l / \Gamma(l+1)$, and in particular $B_k(0) = f_k$. The integral has integrable endpoints and preserves whole exponential bounds. Differentiation uses a strictly smaller sector and a fixed Cauchy margin. These are operations on the whole endpoint family, so coefficient extraction can follow them.

The constants cannot be suppressed during reconstruction. If $m_k(\eta) = \eta^{a-1} F_k(\eta)$, formula (6.43) is convolution with $\eta^{-a} / \Gamma(1-a)$. Taking Laplace transforms at $1/\xi$ gives

$$(6.44) \quad \mathcal{L}m_k = \xi^a (f_k + \mathcal{L}H_k).$$

The identity follows either by absolute convolution or by integration by parts in $\mathcal{L}B_k = \xi(f_k + \mathcal{L}H_k)$. Together with the $k=0$ term it reconstructs the original root source. The projection defining χ_k uses every rotated lift. Its continuation assertion therefore requires all the specified lifts. $\square$

Definition 6.11. On a simply connected lifted base sector put

$$(6.45) \quad X_0 = x^{p+1} \partial_x + (1 + \lambda x^p) y \partial_y, \quad E(x) = x^\lambda e^{-1/(p x^p)}, \quad H(x, y) = y/E(x).$$

The logarithm in x^λ is the specified one. A fiber map $(x, y) \mapsto (x, g(x, y))$ is an *isotropy* of this model if it is locally invertible and preserves its leaves. A domain $\{x \in U : |H(x, y)| < r_H\}$, with U connected, is called saturated in the H coordinate.

Lemma 6.12. On a saturated domain an isotropy of (6.45) has the form

$$(6.46) \quad g(x, y) = E(x) h(y/E(x))$$

for one holomorphic function h . In particular, transitions between sectorial normalizers have this form wherever the displayed expression is defined. If such a transition is asymptotic to the identity uniformly on fixed smaller y disks, then on an overlap where $E \rightarrow 0$ it is $y + a_0 E(x)$, and on an overlap where $E \rightarrow \infty$ its function h is a germ with $h(0) = 0$, $h'(0) = 1$. The coefficients of h can be complex. Holonomy in the original equation is exactly the normalizer, model holonomy along the retained path, and the true inverse normalizer at the outgoing section, in that order.

Proof. Since the base component is unchanged, preservation of leaves gives $X_0 g = (1 + \lambda x^p) g$. In coordinates (x, H) this says that g/E has zero x derivative. Connectedness gives (6.46). Analyticity of h follows by restricting to any fixed base point. This argument uses only the given saturated domain.

For a transition defined instead on a product with a fixed y disk, expand $g(x, y) = \sum_{j \geq 0} g_j(x) y^j$. The coefficient equation is (6.37), so $g_j(x) = c_j E(x)^{1-j}$. Cauchy estimates on a smaller fixed y disk apply to $g - y$. If $E \rightarrow 0$, they force $c_j = 0$ for $j \geq 2$ and $c_1 = 1$; the constant c_0 remains free. If $E \rightarrow \infty$, they force $c_0 = 0$ and $c_1 = 1$. At one fixed base point the convergent Taylor series defines h near zero, with these coefficients. This proves the assertion without postulating a uniform saturated H disk inside the entire growing- E product. Conversely, a convergent tangent-to-identity h gives $g - y = O(y^2/E)$ on each smaller y disk when $E \rightarrow \infty$.

If Φ_σ, Φ_τ are two normalizers, their transition is $\Phi_\tau \circ \Psi_\sigma$, so the preceding equations apply. Both normalizers and their inverses map actual leaves to actual leaves. The first integral in (6.45) gives model holonomy $y \mapsto E(x_{\text{out}})y/E(x_{\text{in}})$, with its prescribed analytic continuation along the path. Uniqueness of integral curves in Lemma 6.3 identifies the resulting composition

with the selected physical holonomy. Its inverse reverses the path and every arrow. No reality condition on h was used. $\square$

Proposition 6.13. *Fix an actual passage as in Definition 6.2, its original cuts, and finitely many sectorial labels whose domains contain its physical tail. There exist inner transverse sections and analytic regular holonomies to the original sections, with their true inverses, such that the direct inner passage is*

$$(6.47) \quad P_{\text{phys}}(s) = k(v(s)G(s)), \quad v(s) = ce^{-Z}Z^a, \quad Z = \beta s^{-p}, \quad a = \lambda/p, \quad c > 0.$$

For each label the function G is the whole normalizer evaluated on the incoming graph, divided by its nonzero limiting value; it tends to one. The germs h and k on the outgoing section are actual analytic inverses with $h(0) = k(0) = 0$ and $h'(0) = k'(0) = 1$. The inverse of (6.47) is the actual reverse passage on the specified sections.

Proof. We choose the inner cuts on the actual leaves, center the outgoing coordinate at its actual limit, and use the model first integral.

Step 1. We choose the cuts and domains. Make the finite preparation of Proposition 6.4. Its physical ray is $x = \zeta s$, $s > 0$, and $Z = \beta s^{-p} > 0$. Choose the source root and logarithm by

$$(6.48) \quad \kappa = \zeta\beta^{1/p}, \quad \text{Log } Z = \log Z \quad (Z > 0), \quad x = \kappa Z^{-1/p} = \zeta s,$$

where $\beta^{1/p}$ is positive. Then $\kappa^p = -1/p$. For a new core, choose a nonreal q in the first quadrant sufficiently near the positive direction; its Laplace sector contains a physical tail. If $q = e^{i\theta}$ with $0 < \theta < \pi/2$, use $\text{Log } q = i\theta$ and the compatible powers in (6.15). Construct one source-equipped reference by Proposition 6.7 on this q . Every previously prescribed compatible analytic label retains its actual map, true inverse, roots and logarithms; this construction does not reconstruct that arbitrary label as a pure native source. Choose the cuts in their finite common matched actual domains and fix the larger endpoint and inverse-image disks. These choices precede any further weight increase used for graph or weak-coordinate substitution. Such an increase leaves the previously constructed actual maps unchanged. A finite family of prescribed labels imposes only finitely many strict restrictions.

In the real prepared coordinates the incoming cut is a graph $y = b(x)$ with $b(0) = b_0 \neq 0$, and the outgoing cut can be taken at $x = x_w = \zeta s_w$, $s_w > 0$. These are obtained by regular holonomy along the chosen strong and off-axis boundaries. To check the whole inner leaf segment, choose a small real rectangle $0 < s \leq s_w$, $|y| \leq r_*$. Since $B(0, y) = y$, the reversed normalized field points inward at $y = r_*$ and $y = -r_*$ once s_w is small. Its s component is $-\zeta^p s^{p+1} > 0$. An incoming graph strictly inside this rectangle reaches $s = s_w$ in finite time for each positive input and remains inside the larger endpoint disk throughout. No uniform bound on these times is claimed as the input tends to zero.

The finite regular segments joining old and new cuts are covered by Lemma 6.3. They retain the original path and its inverse. If an old core has already been prescribed, use its own strict equation, graph and inverse margins; smaller endpoint disks of a newly constructed core are not assigned to that old core.

Step 2. We identify the actual outgoing center. Let y_* be the limiting endpoint of the actual passage at x_w and write the positive outgoing coordinate as $y = y_* + \kappa_w t$, with real $\kappa_w \neq 0$ chosen for that side. For a selected label Φ , the first integral gives

$$(6.49) \quad \Phi(x_w, y_* + \kappa_w P_{\text{phys}}(s)) = \frac{E(x_w)}{E(\zeta s)} \Phi(\zeta s, b(\zeta s)).$$

The numerator on the right tends to $E(x_w)b_0$, whereas $E(\zeta s) = (\zeta s)^\lambda e^Z$ grows exponentially. Endpoint continuity therefore gives $\Phi(x_w, y_*) = 0$, or $y_* = \Psi(x_w, 0)$. In particular, the outgoing center need not be the coordinate value zero before this translation.

Put $d = \kappa_w \partial_y \Phi(x_w, y_*) \neq 0$ and define

$$(6.50) \quad h(t) = \frac{\Phi(x_w, y_* + \kappa_w t)}{d}, \quad k(w) = \frac{\Psi(x_w, dw) - y_*}{\kappa_w}.$$

The true fiber inverse identities prove $h \circ k = k \circ h = \text{id}$, and both derivatives at zero equal one. Define

$$(6.51) \quad G(s) = \frac{\Phi(\zeta s, b(\zeta s))}{b_0}, \quad \ell(s) = -\log G(s),$$

where the logarithm is the branch tending to zero. The full endpoint normalizer tends to the identity, so $G \rightarrow 1$. Since $a = \lambda/p$, there exists a fixed nonzero constant c such that

$$(6.52) \quad \frac{E(x_w)b_0}{dE(\zeta s)} = ce^{-Z}Z^a.$$

(6.49)–(6.52) give (6.47). Moreover $P_{\text{phys}}(s)/(e^{-Z}Z^a) \rightarrow c$. The quotient is positive real on the physical interval; its nonzero limit implies $c > 0$, even when Φ , h and k are not real.

Step 3. We construct the actual inverse. Regard ℓ as a function of Z and set

$$(6.53) \quad A_*(Z) = Z - a \log Z - \log c + \ell(Z), \quad B_*(t) = -\log h(t).$$

The logarithms are chosen by continuation from the physical identity $B_*(P_{\text{phys}}(s)) = A_*(\beta s^{-p})$. For the reference label the whole asymptotic estimate and a strict Cauchy margin give $\ell'_0(Z) = O(Z^{-1-1/p})$. For any compatible analytic label, Lemma 12.11 (proved below) uses only the normalizing equation, the supplied genuine fiber inverse pairs and uniform fixed-fiber identity limits, the compatible lift, and the common inner cuts and endpoint continuity already available in **Step 1**. It requires neither the passage inverse constructed in this step nor any receiving or weak-replay result. Its common endpoint derivative and positive physical constant are obtained directly from those data. It gives $\ell_\sigma = \ell_0 - \log H_\sigma(V_{\text{phys}}G_0)$, where $H_\sigma = C_\sigma/\text{id}$ and $H_\sigma(0) = 1$. Its difference from ℓ_0 and its derivative are exponentially small on a strict physical subsector by the convergent gauge series and Cauchy on that sector. Thus $\ell'_\sigma(Z) = O(Z^{-1-1/p})$ there as well. On a far convex sector about the positive ray, $|A'_*(Z) - 1| < 1/2$. Integration along a segment in this sector proves that A_* is injective there. On its literal image the inverse is

$$(6.54) \quad P_{\text{phys}}^{-1}(t) = \left(\frac{\beta}{A_*^{-1}(B_*(t))} \right)^{1/p},$$

with the positive physical root. Equivalently it is obtained by reversing the actual leaf segment and the arrows Φ , model holonomy, and Ψ . These descriptions give both inverse identities on their actual domains. They do not replace an inverse by a reciprocal coefficient. $\square$

Definition 6.14. Let $G_{\text{src}} \in \mathcal{B}_A$ be the whole graph source of a source-equipped reference in Definition 6.2. An *independent weak coordinate* is a variable V with original Wiener radius $R_V > 0$. The whole passage source is $k(VG_{\text{src}})$ on a strict disk of k , before its physical evaluation $V = ce^{-Z}Z^a$. Another compatible analytic label is represented by its complete weak replay below, not by an automatically existing native constant term. A separately supplied native presentation retains only its proved actual source identity and its original domain and norm. For a tangent-to-identity germ f , put $D_f(w) = -\log(f(w)/w)$ on the branch zero at zero.

Proposition 6.15. *For a fixed equation and physical core, a source-equipped reference σ has the whole sources of Definition 6.14, with complete endpoint and weak tails before physical evaluation. Let τ be any compatible analytic label of Definition 6.2. Their positive physical leading constants coincide. Put $C = h_\tau \circ k_\sigma$. In the reference source algebra define*

$$(6.55) \quad G_\tau^{[\sigma]} = G_\sigma(C/\text{id})(VG_\sigma),$$

$$(6.56) \quad \ell_\tau^{[\sigma]} = \ell_\sigma + D_C(VG_\sigma).$$

These entire weak families have the actual τ values on the physical diagonal and satisfy the source identity

$$(6.57) \quad k_\tau(VG_\tau^{[\sigma]}) = k_\sigma(VG_\sigma) \quad \text{in } \mathcal{W}_{R_V}(\mathcal{B}_A).$$

The full forward fiber source and its genuine inverse are also received by the formulas and original-norm estimates of Lemma 6.16 below. The transition maps have their true analytic inverses and ordered cocycles. At zero transition degree the reference forward and inverse sources and graph unit are recovered. The original endpoint germs remain in the endpoint identity; this erasure does not identify their gauge with the reference endpoint gauge. No identification or bounded transfer of arbitrary free native banks on different rays is asserted.

Proof. We construct whole graph substitutions in the reference algebra, then use the analytic matching identity before weak evaluation.

Step 1. We perform whole graph and weak substitutions. Let σ be the source-equipped reference and let b_{src} be the source of the entire incoming graph, including b_0 . Choose a fixed endpoint radius $r_e > |b_0|$ inside the normalizer's disk. By Lemma 6.6, increasing A makes $\|b_{\text{src}}\|_{\mathcal{B}_A} < r_e$. Put $\rho = \|b_{\text{src}}\|_{\mathcal{B}_A}/r_e < 1$. Write $\Phi = y + \sum_{j \geq 0} u_j y^j$ and let $\epsilon_A = \sum_j r_e^j \|u_j\|_{\mathcal{B}_A}$. The complete sum of endpoint degrees $j > N$ after graph substitution is bounded by

$$(6.58) \quad \sum_{j > N} \|u_j b_{\text{src}}^j\|_{\mathcal{B}_A} \leq \epsilon_A \frac{\rho^{N+1}}{1 - \rho} \quad (N \geq -1).$$

Also $(b_{\text{src}} - b_0)/b_0$ tends to zero in norm. Thus $G_{\text{src}} - 1$ tends to zero. Its logarithm and inverse are the convergent Banach-algebra series, including all their terms. Choose R_V with $R_V \|G_{\text{src}}\|$ strictly inside the common radii of k_σ, k_τ, C and their needed inverse germs. Proposition 4.20 then gives all weak substitutions, with the full tail bound $\theta^{M+1} \|F\|$ whenever the relevant inner norm is at most $\theta < 1$ times the original outer radius. In particular the logarithmic mismatch is the whole source

$$(6.59) \quad \ell_{\text{src}} + D_k(VG_{\text{src}}).$$

The variable V has not yet been assigned a physical value.

Step 2. We prove the complete matching identity. Lemma 12.11 applies to the reference and the compatible analytic label, using the actual equation, uniform identity limit, genuine inverses and connected matched domains. It gives the full fiber identity with their common endpoint derivative and hence their common $c > 0$. On the physical diagonal,

$$(6.60) \quad vG_\tau = C(vG_\sigma), \quad \ell_\tau = \ell_\sigma + D_C(vG_\sigma).$$

Independently of that evaluation, the convergent source series gives $VG_\tau^{[\sigma]} = C(VG_\sigma)$. The analytic germ identity $k_\tau \circ C = k_\sigma$ proves (6.57) by whole analytic functional calculus, not by a uniqueness theorem for the physical diagonal. For three labels the exact identities are $(h_v \circ k_\tau) \circ (h_\tau \circ k_\sigma) = h_v \circ k_\sigma$ and $(h_\tau \circ k_\sigma)^{-1} = h_\sigma \circ k_\tau$. They hold on the finite common strict disks.

Step 3. We distinguish native inputs from weak replay. Every positive degree of V remains in the source. At $V = 0$ the graph replay equals the reference G_σ , not the arbitrary original G_τ . A separately source-equipped τ retains its own proved native identity, but this calculation constructs no bounded operator on its arbitrary completed native bank. Original nonanalytic windows, complements and norms remain in their original spaces. For a scalar analytic germ $D(w) = \sum_{n \geq 1} d_n w^n$, the special substitution $D(e^{-T})$ has the discrete source $\sum_{n \geq 1} d_n \delta_n$. This does not turn the ordinary root densities of $G_{\text{src}} - 1$ or ℓ_{src} into point masses. $\square$

Lemma 6.16 (Whole label replay and original norms). *Fix a source-equipped reference with whole mutually inverse endpoint sources F_0, J_0 , graph constant $g_0 \neq 0$, and the compatible analytic label σ of Lemma 12.11. Put $C_\sigma = h_\sigma \circ k_0$ and define the removable germs $H_\sigma = C_\sigma / \text{id}$, $H_\sigma^- = C_\sigma^{-1} / \text{id}$. On strict original composition domains the whole source formulas are*

$$(6.61) \quad F_\sigma^{[0]}(z, V) = F_0(z)H_\sigma(VF_0(z)/g_0),$$

$$(6.62) \quad J_\sigma^{[0]}(w, V) = J_0(wH_\sigma^-(Vw/g_0)).$$

They satisfy both source inverse identities, including $V = 0$, and have the original label's actual maps at $V = V_{\text{phys}}$. At $V = 0$ they give F_0, J_0 . For $G_0 = F_0(g)/g_0$ they retain $G_\sigma^{[0]} = G_0H_\sigma(VG_0)$ and $k_\sigma(VG_\sigma^{[0]}) = k_0(VG_0)$ as whole source identities.

If $\|F_0\| \leq M$ in its original endpoint Wiener norm and $\|H_\sigma\|_R = \sum_j |h_j| R^j < \infty$, choose a receiving radius with $r_V M / |g_0| \leq \theta R$, $0 < \theta < 1$. Then

$$(6.63) \quad \|F_\sigma^{[0]}\| \leq M \|H_\sigma\|_R, \quad \|\text{Tail}_{>N,V} F_\sigma^{[0]}\| \leq M \theta^{N+1} \|H_\sigma\|_R.$$

Write $H_\sigma^-(w) = \sum_j h_j^- w^j$ and let r_- be the original buffered radius of J_0 . Choose $r_V^+ > r_V$ and $r_w > 0$ with

$$(6.64) \quad r_w \sum_j |h_j^-| (r_V^+ r_w / |g_0|)^j < r_-.$$

Then the inverse composition has whole norm at most $\|J_0\|_{r_-}$ at r_V^+ , and its weak tail at r_V is at most $(r_V / r_V^+)^{N+1} \|J_0\|_{r_-}$. These assertions hold in an original coefficient Banach algebra with g_0 invertible and $\|g_0^{-1}\|$ retained, replacing $|g_0|^{-1}$ by that norm. Bounded holomorphic parameter data on larger original polydisks give the corresponding smaller-disk Wiener bounds with factor $\prod_i (1 - \rho'_i / \rho_i)^{-1}$, provided the stipulated uniform nonvanishing and inverse-image margins hold there.

Proof. For $\alpha = V/g_0 \neq 0$ the forward map is $\alpha^{-1}C_\sigma(\alpha F_0(z))$, whose inverse is $J_0(\alpha^{-1}C_\sigma^{-1}(\alpha w))$. This proves both identities by convergent analytic substitution in the reference algebra. The displayed formulas have only nonnegative powers of V , so the identities extend to zero, where the reference pair is recovered. The fiber identity (12.41) identifies their physical values. The endpoint identity follows directly from $k_\sigma C_\sigma = k_0$. Rescaling an analytic germ by $C \mapsto \alpha^{-1}C(\alpha \cdot)$ respects composition and inversion, including its removable value at zero. Thus the ordered analytic cocycle also holds for the whole weak recipes.

The j th forward term has norm at most $M|h_j|(r_V M / |g_0|)^j$. Summing gives (6.63); for $j > N$ keep the additional θ^{N+1} . Inequality (6.64) puts the entire inverse argument strictly inside the original J_0 disk at the larger weak radius. Absolute substitution then gives its whole bound there; restriction of the full nonnegative weak degrees gives the stated ratio tail. The inverse uses a compositional inverse, not $1/H_\sigma$. Original graph substitution and its complete endpoint tail are bounded before these operations. Since $H_\sigma - 1$ vanishes at zero, a smaller weak radius also keeps its full logarithm on the original branch after the reference graph logarithm disk is fixed.

For parameters work in the complete original coefficient algebra, including the inverse norm of g_0 . A scalar pointwise inequality $g_0 \neq 0$ is not a bound for that Banach inverse. For bounded holomorphic data on a larger polydisk, uniform nonvanishing bounds $1/g_0$ and the endpoint derivative inverse there; Cauchy gives their Wiener inverse norms on prescribed smaller disks. The full family bound precedes extraction. Cauchy's coefficient estimate summed in all parameters gives the displayed product factor, and the tail in which some degree exceeds N_i has the additional factor $\sum_i (\rho'_i / \rho_i)^{N_i+1}$. Fixed derivatives retain their original factorial and radius losses. No original norm is changed after specialization. These statements concern the fixed matched equation sources and their complete analytic recipe families, not an arbitrary free native bank on another ray. $\square$

Lemma 6.17. *For a finite collection of equation-generated sources as above, the following operations preserve the whole source before coefficient extraction: restriction with fixed strict margins; passage to a common root order; and the analytic graph and parameter substitutions whose norms lie strictly inside the original input radii. Passage to a common root order changes neither the radial density nor any retained radial window. A change of an original section coordinate retains the exact pullback of its native height.*

Proof. We compare the radial densities under root refinement, bound analytic substitutions on their original disks, and retain the exact section pullbacks.

Step 1. We refine the root coordinate. For a common multiple $P = mp$ of the finitely many root orders, define $\tilde{\phi}(z) = \phi(z^m)$, with the inherited lifted sectors. Then for $t > 0$,

$$(6.65) \quad \frac{\tilde{\phi}(t^{1/P})}{t} = \frac{\phi(t^{1/p})}{t}.$$

Thus the actual density, and any function multiplying it on the radial variable, is literally unchanged. The product integral in (6.18) is unchanged by this substitution, as is its moment identity. With unchanged exponential weight the norms in the two orders satisfy

$$(6.66) \quad N_{A,P}(\tilde{\phi}) \leq \frac{P}{p} N_{A,p}(\phi).$$

At $t = |z|^P$, the ratio of the defining pointwise weights is $\frac{P}{p}(At/(1+At))^{1/p-1/P} \leq P/p$. This also proves the bound in the whole endpoint and parameter norms.

Step 2. We perform whole analytic substitutions. For strict analytic substitutions, Proposition 4.20 applies on the original input radii. All endpoint estimates were made before any coefficient or parameter was selected. For a parameter family bounded on $|u_j| \leq \rho_j$, Cauchy's formula gives at $r_j < \rho_j$ the factor $\prod_j (1 - r_j/\rho_j)^{-1}$. A union bound for the tail indices $\nu_j > N_j$ gives the additional factor $\sum_j (r_j/\rho_j)^{N_j+1}$. A specified mixed derivative with orders b_j is bounded by the original bound times $\prod_j b_j! \rho_j / (\rho_j - r_j)^{b_j+1}$. These constants concern the original larger family, not a newly assigned norm on a specialized value.

Step 3. We continue angularly in the intrinsic coordinate. Let J be a compact interval of directions strictly inside one nonreal half-plane. Work first at native order p , with the original κ and a continuous compatible choice of $\text{Log } q$ for $q = e^{i\theta}$, $\theta \in J$. The resolvent bounds (6.33) have a common angular constant. To compare the locally constructed minors, write their direction explicitly as ϕ_q and remove its factor $q^{1/p}$:

$$(6.67) \quad \phi_q^{\text{int}}(w) = \phi_q(q^{-1/p}w) = \sum_{n \geq 1} \frac{f_n \kappa^n}{\Gamma(n/p)} w^n.$$

The coefficients f_n are the formal coefficients of the fixed equation. Their uniqueness in Proposition 6.7 therefore gives equality of the intrinsic germs on a common original disk. The domain for ϕ_q^{int} is $q^{1/p} \mathcal{D}_q$, where $\mathcal{D}_q$ denotes its original domain (6.16). Each radial component of an overlap of two such domains attaches to their common disk. The holomorphic identity theorem identifies the intrinsic continuations on that component. Finitely many angular neighborhoods cover J , so these functions glue to one ϕ^{int} on their union. This argument applies in the whole endpoint norm and separately to the true inverse in its own endpoint variable.

The rotations have modulus one. Thus the defining weighted supremum is unchanged when a native domain is written in the w coordinate. Choose a common weight above the finitely many original weights A_i . For $t = |w|^p$, the ratio of its weight to the weight at A_i is

$$(6.68) \quad \left(\frac{A_i}{A}\right)^{1/p} e^{-(A-A_i)t} \left(\frac{1+At}{1+A_i t}\right)^{1+1/p},$$

which is bounded on $t \geq 0$. The whole norm on the finite union is at most the sum of the piece norms. Each nonconstant piece norm tends to zero at large weight, so their finite sum does also. All endpoint and analytic-input radii stay fixed in this argument.

The corresponding Laplace parametrization is

$$(6.69) \quad \int_0^\infty e^{-qZt} \phi_q(t^{1/p}) \frac{dt}{t} = \int_{q(0,\infty)} e^{-Zu} \phi^{\text{int}}(u^{1/p}) \frac{du}{u}, \quad u^{1/p} = q^{1/p} t^{1/p} \quad (u = qt).$$

For nearby directions, these intrinsic integrals agree on a common far Z sector by contour deformation. The small connecting arc is $O(r^{1/p})$ at radius r ; on the large arc the common strict Laplace margin dominates the whole exponential bound. Both arcs vanish in the whole endpoint norm. A finite chain of overlaps gives angular continuation along J . This identifies the specified equation-generated continuations in the intrinsic coordinate. Each original ϕ_q , its labelled domain, and its native norm are retained. Root refinement is then performed by (6.65) on each of these original presentations.

Step 4. We transport the sources under translations and dilations. At refined order $P = mp$, a height translation $Z \mapsto Z + T$ multiplies the native minor $\tilde{\phi}_q(z)$ by e^{-qTz^P} . For $|T| \leq r_T$, its modulus is at most $e^{r_T|z|^P}$, so a fixed weight increase gives the bound for the whole analytic T family. For the dilation $x \mapsto x(1 + T)$, apply the native-order rule $\phi_q(z) \mapsto \phi_q((1 + T)z)$ before refinement. Equivalently, on the refined presentation it is

$$(6.70) \quad \tilde{\phi}_q(z) \mapsto \tilde{\phi}_q((1 + T)^{1/m}z) = \phi_q((1 + T)z^m), \quad (1 + T)^{1/m} = \exp\left(\frac{1}{m} \log(1 + T)\right),$$

where $\log(1 + T)$ is zero at $T = 0$. A native monomial of degree n has refined degree mn ; the multiplier in (6.70) is exactly $(1 + T)^n$.

Choose $r_T < 1$ so that $|\arg(1 + T)|$ is smaller than a retained native angular margin. At refined order the angular displacement is $|\arg(1 + T)|/m$. Restrict the receiving root disk so that all images stay in the original germ disk. If the original exponential weight is A , choose a receiving weight greater than $A(1 + r_T)^p$:

$$(6.71) \quad |(1 + T)^{1/m}z|^P = |1 + T|^p|z|^P \leq (1 + r_T)^p|z|^P.$$

The remaining polynomial factors in the weighted supremum have a uniform bound using this weight slack. These bounds precede endpoint and T -coefficient extraction and use the original source norms. For Cauchy extraction choose a fixed smaller T disk inside this larger disk of analytic dependence.

Step 5. We return to the original section coordinates. If a positive regular map changes an incoming coordinate by $s_{\text{new}} = I(s_{\text{old}})$, its native height is $\beta_{\text{new}}/I(s_{\text{old}})^p$. It is this exact expression that is used in every evaluation and inverse. Neither (6.66) nor analytic restriction replaces it by $\beta_{\text{old}}/s_{\text{old}}^p$. In particular, the lower terms in (6.13) remain part of the original data. $\square$

The distinction between a single root lift and its ordinary congruence minors occurs already for a polynomial equation. This example also shows why convergence of the formal normalizer cannot be assumed.

Example 6.18. Put $\gamma = \Gamma(3/2)$ and consider the polynomial vector field

$$(6.72) \quad Y = -s^3 \partial_s + (v + (\gamma s^3 - \sqrt{2}s^4)v^2) \partial_v.$$

There exists a divergent formal series U for which the maps $N(s, v) = v/(1 + U(s)v)$ and $M(s, V) = V/(1 - U(s)V)$ are formal inverse normalizers. Their positive root sums are true sectorial inverse maps. The positive root minor is nonsingular on its whole positive ray, whereas an ordinary congruence minor has a pole on the corresponding ordinary ray.

Proof. We compute the root sum, compare its two congruence lifts, and then identify the actual passage.

Step 1. We construct an actual root sum for a divergent normalizer. The normalizing equation is

$$(6.73) \quad s^3 U' - U = -\gamma s^3 + \sqrt{2} s^4, \quad U_n = \Gamma(n/2)(-\sqrt{2})^{n-3} \quad (n \geq 3).$$

Direct coefficient comparison proves this formula; coefficients below three vanish. Since $U_{n+2}/U_n = n$, the series diverges. Its unscaled root minor is $w^3/(1 + \sqrt{2}w)$. For $Z = 1/(2s^2)$ the positive root sum is

$$(6.74) \quad U(s) = \int_0^\infty e^{-Zt} \frac{t^{1/2}}{2\sqrt{2}(1 + \sqrt{t})} dt.$$

The root minor $z^3/(2\sqrt{2}(1 + z))$ is holomorphic around the entire positive ray with polynomial growth. Rotating its Laplace ray through a small angle gives a Z sector of opening greater than π . Since $s^3 \partial_s = -\partial_Z$, applying $s^3 \partial_s - 1$ to this integral replaces its density by $(t-1)t^{1/2}/(2\sqrt{2}(1 + \sqrt{t})) = (t - \sqrt{t})/(2\sqrt{2})$. The Gamma integrals of these two terms are $\sqrt{2}s^4$ and γs^3 , respectively. This proves (6.73) for the actual function. Substituting N into the vector field gives $YN = N$, and the displayed rational formulas give both inverse identities wherever their denominators are nonzero. Fixed smaller fiber disks provide such domains when s is small.

Step 2. We identify the ordinary projection. To compare the ordinary minors, set $x = is$, so that the base component becomes $x^3 \partial_x$, and put $u(x) = U(-ix)$. Its unscaled root minor is $\chi_u(w) = iw^3/(1 - i\sqrt{2}w)$. Its even projection is $-\sqrt{2}w^4/(1 + 2w^2)$. The coefficient of v^2 in $N - v$ is $-u$, so its congruence minor with $k = 0$ is

$$(6.75) \quad H_{0,2}(\eta) = \frac{\sqrt{2}\eta}{1 + 2\eta}.$$

It has a pole at $\eta = -1/2$, on the ray $\eta = -t/2$ corresponding to physical $x = is$. The positive physical root minor in (6.74) has no such pole. Thus removal on that root lift does not justify continuation of the ordinary projection which also uses the other lift.

Step 3. We identify the physical passage. For the field $-Y$, take the incoming cut $v = b > 0$ and the outgoing cut $s = a > 0$, both small. If $Z_a = 1/(2a^2)$, conservation of the model first integral gives

$$(6.76) \quad v_{\text{out}} = \frac{be^{Z_a} e^{-Z}}{1 + bU(s) - be^{Z_a} U(a)e^{-Z}}.$$

The denominator stays positive for a sufficiently short physical tail. This exact passage retains the divergent normalizer through its entire sectorial sum, rather than replacing it by a convergent formal series. $\square$

Proof of Theorem 6.1. We assemble the equation construction on inner cuts and then restore the original sections and their sources.

Step 1. We identify the equation and sectorial maps. Proposition 6.4 supplies the finite analytic preparation, its true inverse, and the signed time factor. It also identifies the positive height and retains its full primitive. Proposition 6.7 constructs the whole normalizer and the true fiber inverse from this prepared equation. Lemma 6.8 covers a removable root direction by continuation of its original whole source. Lemma 6.12 identifies the sectorial transitions and the actual leaf maps without imposing reality on the transition data.

Step 2. We retain the original cuts and inverse. Proposition 6.13 gives the asserted formula on the inner sections, with $a = \lambda/p \in \mathbb{R}$. Let I be the positive analytic holonomy from the original incoming section to the inner one and O that from the inner outgoing section to the

original outgoing section. The original direct passage is $O \circ P_{\text{phys}} \circ I$. Use the positive incoming coordinate given by the inner coordinate composed with I . If $r = O'(0) > 0$, replace c by rc and replace the inner k by $w \mapsto O(k(w/r))$. This new k fixes zero and has derivative one, so it proves the displayed formula on the specified original outgoing section. The reverse passage is exactly $I^{-1} \circ P_{\text{phys}}^{-1} \circ O^{-1}$, with P_{phys}^{-1} given by (6.54) on its literal image.

Step 3. We establish the whole matching. Proposition 6.15 supplies matching with the weak coordinate still independent and proves the entire analytic cocycle, including its inverse. Lemma 6.17 retains the original input norms, common root refinements, and exact heights under the regular section changes. The finite occurrence list uses common strict restrictions while keeping each original label and each repeated equation parameter. The weak-coordinate replay is used only in the source-equipped reference algebra and retains every original compatible analytic label. Every genuinely supplied native source keeps its proved actual identity, full tail and original domain. The existence of a native constant term is not asserted for an arbitrary analytic label; its entire forward and inverse are the weak recipes of Lemma 6.16. These constructions prove all the assertions. $\square$

7. COVERAGE OF ACTUAL ELEMENTARY ITINERARIES

The incoming and outgoing boundary orbits in Definition 2.2 determine which elementary singularities can contribute a passage. We first identify those passages, then retain the analytic transports between their local preparation cuts and the original sections to factor the physical return.

Lemma 7.1. *A real analytic vector field has a jointly real analytic local flow. An analytic map with invertible differential has a local analytic inverse. If an analytic equation $F(z, w) = 0$ has invertible differential in w at a solution, its nearby solution $w = w(z)$ is analytic.*

Proof. We give the local constructions, including their dependence on the initial point. Complexify the convergent real Taylor series of a vector field Y on a smaller polydisk. On a closed polydisk inside its domain, choose bounds M for Y and L for its differential. Restrict the initial point to a smaller polydisk and choose a complex time disk of radius τ with τM less than the distance to the boundary and $\tau L < 1$. The iterations

$$u_0(t, z) = z, \quad u_{n+1}(t, z) = z + \int_0^t Y(u_n(\zeta, z)) d\zeta$$

remain in the larger polydisk. Their successive differences are bounded in supremum norm by a geometric sequence with ratio τL . They converge locally uniformly to a holomorphic function of (t, z) solving the integral equation. The contraction estimate gives uniqueness. Conjugating the iterations shows that the solution is real on real data. Uniqueness also gives the local flow law and makes opposite-time maps mutually inverse.

For the implicit assertion, let (z_0, w_0) be the specified solution and put $A = F_w(z_0, w_0)$. The equation is equivalent to

$$w = w - A^{-1}F(z, w).$$

The differential of the right side in w vanishes at (z_0, w_0) . On small nested polydisks it has norm less than $1/2$, and its value at w_0 stays sufficiently close to w_0 for the map to preserve a closed ball. Its iterations converge uniformly and holomorphically in z . They give the unique solution. Taking $F(z, w) = H(w) - z$ proves the inverse assertion for H ; uniqueness gives both inverse identities. $\square$

Lemma 7.2. *Let γ be a compact regular orbit segment of a real analytic foliation. Fix analytic transverse arcs at its endpoints. Then transport along γ defines an analytic diffeomorphism germ between these arcs, and transport along the reversed segment is its inverse. In coordinates*

vanishing at the endpoints and positive on transported half-sides, the map fixes zero and has positive derivative there.

Proof. Choose local analytic generators along γ . By Lemma 7.1, their local flows are analytic. For an analytic transverse arc parametrized by $\sigma(s)$, the map $(s, t) \mapsto \Phi(t, \sigma(s))$ has independent differential columns $\sigma'(0)$ and $Y(\sigma(0))$. The inverse assertion of that lemma makes it an analytic coordinate chart. The first coordinate is constant on each local leaf. Its restrictions to two transverse arcs therefore give inverse analytic transition maps. Finitely many such charts cover γ ; composing their transitions gives the required transport. Flow uniqueness identifies the reverse composite with transport along the reversed segment. The derivative is nonzero, and its sign is positive because positive inputs have positive outputs. $\square$

Lemma 7.3. *Let Y be a real analytic vector field with an isolated equilibrium at the origin and eigenvalues 0 and $\nu \neq 0$. Then there exist real analytic local coordinates (u, v) , an integer $p \geq 1$, a sign $\sigma \in \{-1, 1\}$, and real analytic functions c, h, b, η such that $c > 0$, $b(0) \neq 0$, and*

$$(7.1) \quad Y = c(u, v) \left(\sigma u^{p+1} \partial_u + (h(u) + b(u)v + v^2 \eta(u, v)) \partial_v \right), \quad h(u) = O(u^{p+2}).$$

The coordinate map has an actual analytic inverse. The axis $u = 0$ is the analytic strong separatrix. The construction requires only finitely many analytic coordinate changes after constructing this strong branch.

Proof. We construct the strong branch, remove finitely many powers of the transverse coordinate from the first component, and retain the forcing in the second component. In real linear coordinates the original equation is

$$\dot{x} = f(x, y), \quad \dot{y} = \nu y + g(x, y), \quad f, g = O((|x| + |y|)^2).$$

Step 1. We construct the strong branch. Its graph $x = F(y)$ must satisfy

$$F'(y)(\nu y + g(F(y), y)) = f(F(y), y).$$

Put $F = y^2 G$. Taylor division gives analytic functions $\tilde{f}, \tilde{g}$ with $f(y^2 G, y) = y^2 \tilde{f}(y, G)$ and $g(y^2 G, y) = y^2 \tilde{g}(y, G)$. Thus

$$yG' = \frac{\tilde{f}(y, G)}{\nu + y\tilde{g}(y, G)} - 2G.$$

The value $\tilde{f}(0, G)$ is independent of G . There exists a unique constant G_0 for which the right side vanishes at $y = 0$, and, after writing $G = G_0 + H$, the equation becomes

$$yH' + 2H = yL(y, H), \quad H(0) = 0,$$

with L analytic. If $H = \sum_{n \geq 1} h_n y^n$, its coefficient recursion is $(n+2)h_n = [y^{n-1}]L(y, H)$. Here $[y^j]$ denotes Taylor coefficient extraction. Let $L^\#(z, w)$ be the convergent series obtained by replacing the coefficients of L by their absolute values. The analytic implicit assertion of Lemma 7.1 gives a convergent solution $K = zL^\#(z, K)$ near zero. Its coefficients are nonnegative and dominate $|h_n|$ inductively, since $(n+2)^{-1} \leq 1$. Consequently H , and hence F , converges. Its graph is invariant and tangent to the eigendirection of ν .

The triangular change $x \mapsto x - F(y)$ has an analytic inverse and straightens this branch. Rename the resulting coordinates (x, y) . The equation now has the form

$$\dot{x} = xA(x, y), \quad \dot{y} = C(x, y), \quad A(0, 0) = 0, \quad C(0, y) = \nu y + O(y^2).$$

There exists an analytic nullcline $y = \psi(x)$ with $C(x, \psi(x)) = 0$. Isolation implies that $xA(x, \psi(x))$ is not identically zero. Let $m \geq 2$ be its exact order. This order is preserved by an analytic triangular base change $x = \Theta(u, y)$ with $\Theta_u(0, 0) \neq 0$: on the transverse nullcline the chain rule reads $\dot{x} = \Theta_u \dot{u}$, and $u \mapsto \Theta(u, \psi(u))$ is an analytic coordinate there. This applies also to the initial straightening, since the original nullcline has order at least two.

Step 2. We perform the finite elimination. Write $\dot{x} = \sum_{j \geq 1} a_j(y)x^j$ and $C_0(y) = C(0, y) = y\ell(y)$, where $\ell(0) = \nu$. Since $a_1(0) = 0$, the quotient a_1/C_0 is analytic. Set

$$a(y) = \exp\left(\int_0^y \frac{a_1(t)}{C_0(t)} dt\right), \quad x = a(y)u.$$

The coefficient of u in its new equation is $a_1 - C_0 a'/a = 0$. Suppose the powers below j , with $2 \leq j < m$, have been removed. The coefficient $a_j(0)$ must vanish because the reduced scalar order remains m . Hence a_j/C_0 is analytic. The change $u = w + d_j(y)w^j$, with $d_j(0) = 0$ and $d'_j = a_j/C_0$, removes the coefficient of w^j . The other terms in its chain rule have order at least $j+1$ in w . All these changes have nonzero Jacobian at zero. After this finite list the first component is $x^m A_*(x, y)$ with $A_*(0, 0) \neq 0$; otherwise the reduced scalar order would exceed m .

Step 3. We identify the forcing and time unit. Translate the current transverse nullcline to $v = 0$. The first component is still $x^m A_1(x, v)$ with $A_1(0, 0) \neq 0$, and the constant term of the second component is $O(x^m)$. Put $\sigma = \text{sign } A_1(0, 0)$ and divide by the positive analytic unit σA_1 . The equation becomes

$$\dot{x} = \sigma x^m, \quad \dot{v} = h_0(x) + b_0(x)v + v^2 \eta_0(x, v), \quad b_0(0) \neq 0.$$

If $h_0(x) = e_0 x^m + O(x^{m+1})$, use $v = z - e_0 x^m / b_0(0)$. Its new constant term has zero coefficient at x^m ; the extra differentiated term has order $2m - 1 \geq m + 1$. Taylor division gives the remaining analytic linear and quadratic terms. Set $p = m - 1$ and transport the positive time unit through this last change. This gives (7.1) and its actual inverse. No step constructs a convergent weak invariant graph. $\square$

Lemma 7.4. *Let q be an isolated elementary singularity of a real analytic foliation. Suppose it admits an actual singular-sector passage with the boundary orbits and sector data of Definition 2.2. Then its eigenvalues are either two real numbers of opposite signs or $0, \nu$ with $\nu \neq 0$. In the second case the passage is a hyperbolic sector with one strong boundary and one off-axis boundary. In the coordinates of Lemma 7.3, its side $u = \epsilon r$, where $r > 0$ and $\epsilon \in \{-1, 1\}$, satisfies*

$$(7.2) \quad \sigma \epsilon^p = -\text{sign } b(0).$$

Nodes, foci, centers and terminal parabolic sectors do not admit such a passage.

Proof. Choose a local analytic generator respecting the prescribed leaf direction. Multiplication by a positive analytic unit preserves that direction and multiplies the linearization at q by a positive scalar. We exclude the spectral cases that cannot have both boundary orbits, then identify the semihyperbolic sector.

Case 1. The eigenvalues have real parts of one strict sign. Suppose first that the real parts of both eigenvalues of the linear part L are negative. Its real Jordan form gives exponential decay of $\exp(tL)$, up to a polynomial factor. Thus

$$Q = \int_0^\infty \exp(tL^\top) \exp(tL) dt$$

is a positive definite matrix and $L^\top Q + QL = -I$. Along the nonlinear field, for z near zero,

$$\frac{d}{dt}(z^\top Q z) = -|z|^2 + O(|z|^3) < 0 \quad \text{if } z \neq 0.$$

A nonconstant local orbit tending to zero as $t \rightarrow -\infty$ would therefore have its positive value of $z^\top Q z$ bounded above by zero. This is impossible. Replacing the generator by its negative excludes an incoming branch when both real parts are positive. These arguments include real nodes and foci with nonzero real part. If the real parts have opposite signs, a real two-dimensional matrix has two real eigenvalues of opposite signs, giving the saddle case.

Case 2. The eigenvalues form a nonzero purely imaginary pair. After a real linear change and a choice of angular direction, the polar equation is

$$\dot{r} = r^2 a_1(\theta, r), \quad \dot{\theta} = \omega + r a_2(\theta, r), \quad \omega > 0,$$

where a_1, a_2 are analytic in (θ, r) and periodic in θ . For small $|r|$, the angular velocity is at least $\omega/2$. The equation with independent variable θ is analytic across $r = 0$ and has right side of order r^2 . By Lemma 7.1, continued over the compact angular interval $[0, 2\pi]$, it has an analytic one-turn map H with $H(0) = 0$ and $H'(0) = 1$. Its reverse is an analytic inverse.

If H is not the identity germ, its first nonzero Taylor coefficient in $H(r) - r$ gives one strict sign for $r > 0$ sufficiently small. When $H(r) > r$, successive forward intersections cannot tend to zero, so no incoming branch exists. When $H(r) < r$, successive backward intersections cannot tend to zero, so no outgoing branch exists. An orbit tending to the singularity would have infinitely many such intersections because the angular velocity stays bounded away from zero. If H is the identity, each sufficiently small initial radius gives a periodic orbit. Finite-time continuous dependence over one turn shows that these orbits fill a punctured neighborhood of zero. None tends to zero in either time direction. This exclusion uses only an ordinary analytic map on the polar cylinder.

Case 3. One eigenvalue is zero. The remaining possibility is $0, \nu$ with real $\nu \neq 0$. Use Lemma 7.3. On $u = 0$ both strong half-branches depart when $b(0) > 0$ and approach when $b(0) < 0$. On a fixed side $u = \epsilon r$, the exact first equation gives

$$\dot{r} = c(\epsilon r, v) \sigma \epsilon^p r^{p+1}.$$

Every off-axis branch tending to the singularity on this side therefore departs if $\sigma \epsilon^p > 0$ and approaches if $\sigma \epsilon^p < 0$. A strictly increasing positive r cannot tend to zero in positive time; reversing time gives the other case.

Two strong boundary branches have equal orientations and cannot be the required pair. Two off-axis branches on one side also have equal orientations. A passage joining opposite sides would cross the invariant strong axis, contrary to uniqueness. The strong axis cannot be an interior passage leaf, since its orbits terminate at the singularity in one time direction. Consequently precisely one boundary is strong, the other is off-axis, and their opposite orientations give (7.2).

This is the hyperbolic sector of the semihyperbolic portrait. One can also check directly that its off-axis boundary is unique on that side. Express two candidate boundary orbits as graphs $v_1(r), v_2(r)$. After cancelling the positive time unit, their difference w satisfies

$$w'(r) = \frac{B(r)}{\sigma \epsilon^p r^{p+1}} w(r), \quad B(r) \rightarrow b(0).$$

Here $B(r)$ is the divided difference of $h(\epsilon r) + b(\epsilon r)v + v^2 \eta(\epsilon r, v)$ at the two graphs. By (7.2), the coefficient is at most $-c_1 r^{-p-1}$ for some $c_1 > 0$. A nonzero w would grow without bound as $r \rightarrow 0$, contradicting $v_1(r), v_2(r) \rightarrow 0$. The supplied interior passage leaves lie between this boundary and the strong branch. Where the radial and strong orientations agree, no opposite boundary pair exists; these are the terminal sides. Both parities of p and both signs of the nonzero eigenvalue remain allowed. The argument imposes no analyticity at zero on the off-axis boundary. $\square$

Theorem 7.5. *Every itinerary Γ in Definition 2.2 admits an exact finite literal factorization of its return into regular positive analytic diffeomorphisms, prepared saddle passages and their actual inverses, and canonical semihyperbolic passages and their actual inverses as in Theorems 5.1 and 6.1. All factors are constructed from the original equation, paths, sections and branches. Repeated occurrences retain correlated equation data and their own paths. The factorization applies on orientation covers and retains every lap.*

Proof. Each singular occurrence has the boundary orbits and sector germ of Definition 2.2. Thus Lemma 7.4 identifies each singular passage. We place it between inner sections where the

appropriate local construction applies and retain the regular transports to the original sections. More precisely, we construct positive analytic diffeomorphism germs I_i, O_i , a direct local passage C_i of one of the two stated types, and a sign $\varepsilon_i \in \{-1, 1\}$ such that

$$(7.3) \quad S_i = O_i \circ C_i^{\varepsilon_i} \circ I_i.$$

Here C_i^{-1} denotes the actual inverse on the specified section and branch. We then choose a common source interval and compose these exact equalities in the original order.

Step 1. We choose inner sections and actual section maps. At an occurrence choose inner transverse cuts inside its preparation neighborhood, on its original boundary branches. Between an outer cut and an inner cut the relevant boundary segment is compact and regular. Lemma 7.2 gives its analytic transport and inverse. The coordinate restrictions of the analytic preparation also have analytic inverses. Compose the incoming restrictions and transports to obtain I_i , and compose the outgoing restrictions and transports to obtain O_i . These maps use the given positive sides, so they have positive derivatives at zero. They retain the outer section coordinates.

For a saddle with opposite real eigenvalues, Theorem 5.1 constructs the entire passage

$$C_i(w) = c_i w^{\lambda_i} \exp U_i(-\log w), \quad c_i, \lambda_i > 0.$$

The function U_i and all of its source measures are those constructed from this equation. A reversed occurrence uses the actual inverse provided by that theorem, including its full convolution source from U_i . Choosing the inverse requires no preceding forward occurrence in the itinerary.

For a semihyperbolic occurrence, the boundary classification supplies the hyperbolic-sector hypothesis of Theorem 6.1. Its direct contracting correspondence is the entire map

$$C_i(s) = k_i(c_i e^{-Z_i} Z_i^{a_i} G_i(s)), \quad Z_i = \beta_i s^{-p_i}, \quad c_i, \beta_i > 0.$$

We use its actual endpoint and inverse on the specified weak section. In particular, centering that section uses the physical limiting point, which can have nonzero transverse coordinate in the prepared chart. The incoming graph meets the strong branch at a nonzero regular point. All analytic transports relating these cuts to the original ones remain in I_i, O_i .

The orientation in this formula is determined by the equation. In (7.1) the strong branch attracts if $b(0) < 0$ and repels if $b(0) > 0$. Together with (7.2), this makes the strong-to-off-axis contracting correspondence follow the original direction in the first case and its reverse in the second. The sign ε_i records that choice. The chosen generator and its positive time unit therefore determine the recorded direction. The root and logarithm branches are transported from the original side. The construction uses the full G_i and k_i , including the matching between the prescribed sectorial labels; it does not replace C_i by its leading expression.

Each inner leaf segment and its two outer regular segments are parts of the original prescribed leaf. Uniqueness at every regular intersection therefore proves (7.3) as an equality of actual maps. In particular its reverse is exactly $I_i^{-1} \circ C_i^{-\varepsilon_i} \circ O_i^{-1}$ on the matched inverse domains.

Step 2. We retain common domains and correlated data. Choose the preparation once at each physical equation germ and use its restrictions at every repeated occurrence. Distinct paths and section endpoints retain their own labels. A repeated original parameter enters through its original value at every use. For a prepared inverse the forward measures remain those of that equation; for a canonical inverse and changes of sectorial labels the full normalizers and endpoint inverse germs remain those constructed by Theorem 6.1. Thus repeated occurrences do not create independently variable copies of their equation data.

All analytic inverse arrows are restricted to their actual image neighborhoods. Each singular passage and its inverse sends positive arguments tending to zero to positive arguments tending to zero; this follows for the inverse from continuity and injectivity of the original passage. The set of arrows is finite. Continuity at zero therefore permits one source interval on which every successive image lies in the domain of the next arrow. These restrictions preserve the whole local

constructions of the two passage theorems, with their original input norms, all inverse terms and all retained source tails. They introduce no family of independently varying vector fields.

Step 3. We assemble the physical word and its covers. Substitute (7.3) at each original occurrence in Definition 2.2. This gives

$$(7.4) \quad \Delta_\Gamma = R_m \circ O_m \circ C_m^{\varepsilon_m} \circ I_m \circ \cdots \circ R_1 \circ O_1 \circ C_1^{\varepsilon_1} \circ I_1.$$

Every R_i remains between its original neighbors. The first input and the final output retain the original coordinate on A_1 . The concatenated path is precisely the prescribed return path, so its absence of an intermediate intersection with that physical half-transversal is preserved. Repetitions and all required circuits remain in their original order.

On an unbranched analytic cover, an evenly covered coordinate disk projects by an analytic diffeomorphism. A lifted generator is its coordinate pullback, possibly multiplied by a nonvanishing analytic unit. At a singularity its derivative is conjugate to a nonzero scalar multiple of the original derivative, so it is elementary. Perform the preceding constructions with the actual lifted generator, sections and chosen direction, retaining any time reversal. No projection identifies the initial half-transversal with a different lift or shortens a circuit. Orienting the leaf direction need not orient the surface; this proof uses local surface charts only. The argument establishes local coverage and the equality of the finite physical word without any finiteness theorem for periodic orbits. $\square$

8. ROOT SOURCES AND ORDERED COMPARISONS

A root-Laplace presentation records an analytic germ at the origin of the source variable, its continuation along a specified ray, and the entire integral on that ray. The origin germ supplies comparison coefficients. The continued source supplies the actual function. We first relate these two operations for one root order and one selected ray. We then treat finite action fibers and total-action cuts. We also prove the sectorial flatness and finite-phase comparison results on their specified domains. The estimates are made on complete source families before taking their coefficients.

Definition 8.1. Fix an integer $P \geq 1$, put $\delta = 1/P$, and fix $q \in \mathbb{C}$ with $|q| = 1$ and $\Re q > 0$. Use $\text{Log } q = i \arg q$ with $-\pi/2 < \arg q < \pi/2$. Let $D \subset \mathbb{C}$ be a connected open set containing a disk about zero and the positive ray, such that $u\zeta \in D$ for $\zeta \in D$ and $0 \leq u \leq 1$. For $A > 0$ set

$$(8.1) \quad M_A(\phi) = \sup_{\zeta \in D} e^{-A|\zeta|^P} (1 + A|\zeta|^P)^{1+\delta} \frac{|\phi(\zeta)|}{|\zeta|}, \quad C_A = 8PA^{-\delta}.$$

The quotient at zero is interpreted by continuity. The root source space $\mathcal{B}_A(D)$ consists of pairs $f = (c, \phi)$, where $c \in \mathbb{C}$, ϕ is holomorphic on D , $\phi(0) = 0$, and $\|f\|_A = |c| + C_A M_A(\phi) < \infty$. For $f = (c, \phi)$ and $g = (d, \psi)$ define

$$(8.2) \quad fg = (cd, c\psi + d\phi + \phi * \psi), \quad (\phi * \psi)(\zeta) = \int_0^1 \frac{\phi(\zeta u^\delta) \psi(\zeta(1-u)^\delta)}{u(1-u)} du.$$

The actual and formal maps are

$$(8.3) \quad \mathcal{S}_q(c, \phi)(Z) = c + \int_0^\infty e^{-qZt} \phi(t^\delta) \frac{dt}{t}, \quad \Re(qZ) > A,$$

$$(8.4) \quad \mathcal{T}_q(c, \phi)(x) = c + \sum_{j \geq 1} \Gamma(j\delta) q^{-j\delta} \phi_j x^j, \quad \phi(\zeta) = \sum_{j \geq 1} \phi_j \zeta^j.$$

Here $\Gamma(v) = \int_0^\infty e^{-t} t^{v-1} dt$ for $v > 0$, and $x = Z^{-\delta}$ on the branch positive for $Z > 0$. The scalar component represents the atom at action zero.

Lemma 8.2. *The space $\mathcal{B}_A(D)$ is a unital commutative Banach algebra with unit $(1, 0)$ and submultiplicative norm. The map $\mathcal{S}_q$ is a contractive algebra homomorphism for the supremum norm on $\{\Re(qZ) > A\}$. Every coefficient of $\mathcal{T}_q$ is continuous, and $\mathcal{T}_q$ is an algebra homomorphism into $\mathbb{C}[[x]]$. Both maps are injective. For each integer $N \geq 0$ and each fixed $f = (c, \phi)$, as $Z \rightarrow +\infty$,*

$$(8.5) \quad \mathcal{S}_q f(Z) = c + \sum_{j=1}^N \Gamma(j\delta) q^{-j\delta} \phi_j Z^{-j\delta} + O_f(Z^{-(N+1)\delta}).$$

All bounded linear maps in this assertion act on $\mathcal{W}_\rho(\mathcal{B}_A(D))$ with the original input radii ρ .

Proof. We prove the norm estimate, identify the two products, and then compare the actual integral with its origin coefficients.

Step 1. We define the source norm. Put

$$(8.6) \quad w_A(t) = \frac{t^{\delta-1}}{(1+At)^{1+\delta}}, \quad \int_0^\infty w_A(t) dt = PA^{-\delta}.$$

For $0 < u < t/2$, $w_A(t-u) \leq 4w_A(t)$: the two ratios are bounded by $2^{1-\delta}$ and $2^{1+\delta}$. Splitting the convolution integral at $t/2$ gives

$$(8.7) \quad (w_A * w_A)(t) \leq 8PA^{-\delta} w_A(t) = C_A w_A(t).$$

In (8.2), use $t = |\zeta|^P$ in this estimate. The exponential factors multiply to e^{At} , so $M_A(\phi * \psi) \leq C_A M_A(\phi) M_A(\psi)$. The integrand is locally uniformly integrable at both endpoints, because each minor vanishes at zero. This also proves holomorphy of the product. The weighted supremum space of minors is complete by locally uniform convergence; adding the scalar component preserves completeness.

Step 2. We establish actual and formal multiplication. The density $\phi(t^\delta)/t$ is bounded in absolute value by $M_A(\phi)e^{At}w_A(t)$. Thus

$$(8.8) \quad |\mathcal{S}_q(c, \phi)(Z)| \leq |c| + PA^{-\delta} M_A(\phi) \leq \|(c, \phi)\|_A.$$

Absolute integration on the positive quadrant identifies the product of two such integrals with convolution of their densities. Substituting $t_1 = tu$ and $t_2 = t(1-u)$ gives (8.2). This proves multiplicativity of $\mathcal{S}_q$. Associativity follows either from the absolutely convergent integral over three positive variables or from the formal calculation below and injectivity of $\mathcal{T}_q$.

On any fixed circle contained in D , the coefficient integral bounds $|\phi_j|$ by a constant times $M_A(\phi)$. For monomials of positive degrees j, k , the coefficient of their product in (8.2) is multiplied by

$$(8.9) \quad \int_0^1 u^{j\delta-1} (1-u)^{k\delta-1} du = \frac{\Gamma(j\delta)\Gamma(k\delta)}{\Gamma((j+k)\delta)}.$$

The identity follows by the change of variables $t = t_1 + t_2$ in the product of the two defining gamma integrals. It proves multiplicativity of $\mathcal{T}_q$. If $\mathcal{T}_q f = 0$, every Taylor coefficient of ϕ and the scalar c vanish. Connectedness of D gives $f = 0$.

Step 3. We prove actual asymptotics and injectivity. For (8.5), choose $t_0 > 0$ so that $|\zeta| \leq t_0^\delta$ lies in the origin disk. There the Taylor remainder after degree N has absolute value at most $C_{f,N} t^{(N+1)\delta}$. Its integral over $(0, t_0)$ is $O_f(Z^{-(N+1)\delta})$. On $[t_0, \infty)$, the whole minor is bounded by (8.6), and its integral is exponentially small in Z . The tails of the finitely many subtracted monomials are exponentially small as well. Finally,

$$(8.10) \quad \int_0^\infty e^{-qZt} t^{j\delta-1} dt = \Gamma(j\delta) q^{-j\delta} Z^{-j\delta}.$$

This identity holds first for positive qZ by substitution, and then throughout its right half-plane by holomorphy. It proves the expansion. If $\mathcal{S}_q f$ vanishes, the expansion successively makes every

coefficient of $\mathcal{T}_q f$ zero. The injectivity already proved for $\mathcal{T}_q$ gives $f = 0$. Proposition 4.20 now applies to the bounded maps and products, without changing their input coefficient norms. $\square$

Lemma 8.3. *Let finitely many source spaces have root orders p_i , a fixed common q , and domains D_i as in Definition 8.1. Choose a common multiple P of the p_i and a connected radial domain D containing an origin disk and the positive ray such that $\zeta^{P/p_i} \in D_i$ for every i . At a common weight A the maps*

$$(8.11) \quad (c, \phi) \longmapsto (c, \phi(\zeta^{P/p_i}))$$

are bounded algebra homomorphisms of norm at most P/p_i . They preserve the complete actual integrals and their selected value of q . They do not change the determination of an actual sectorial map.

Proof. Write $\delta_i = 1/p_i$. The ratio of the new minor weight to the old one, after substitution, is bounded by

$$(8.12) \quad A^{\delta-\delta_i} \left(\frac{A|\zeta|^P}{1+A|\zeta|^P} \right)^{\delta_i-\delta} \leq A^{\delta-\delta_i}.$$

Multiplying by the constants $8PA^{-\delta}$ and $8p_iA^{-\delta_i}$ gives the asserted norm bound. At carrier t , the new minor has value $\phi(t^{1/p_i})$, so the entire density and integral are unchanged. The convolution formula has this identical substitution in both factors. No contour or ray is moved. $\square$

Definition 8.4. A selected normalizing input on a strict sector consists of actual holomorphic maps $\Phi(Z, w)$ and $\Psi(Z, w)$, their specified domains and branches, and source series $F \in \mathcal{W}_{r+}(\mathcal{B}_A(D))$ and $G \in \mathcal{W}_{r-}(\mathcal{B}_A(D))$ satisfying the actual identities

$$(8.13) \quad \Phi(Z, w) = \mathcal{S}_q(F(w))(Z), \quad \Psi(Z, w) = \mathcal{S}_q(G(w))(Z).$$

If Ψ is to be used as a fiber inverse, its specified domains must also give $\Psi(Z, \Phi(Z, w)) = w$ and $\Phi(Z, \Psi(Z, w)) = w$ wherever these compositions are used. The data include the original coefficient norms of F, G , a common strict receiving sector, and the physical matching which identifies the selected maps in (8.13). A common-ray input is a finite list of these data with one q ; each identity concerns its selected actual determination.

Lemma 8.5. *Let $F \in \mathcal{W}_r(\mathcal{B}_A(D))$, let $b \in \mathcal{B}_A(D)$, and suppose $\|b\|_A \leq \theta r$ with $0 < \theta < 1$. Then the whole source $F(b)$ converges in $\mathcal{B}_A(D)$, and*

$$(8.14) \quad \left\| F(b) - \sum_{j=0}^N F_j b^j \right\|_A \leq \theta^{N+1} \|F\|_{\mathcal{W}_r(\mathcal{B}_A(D))}.$$

Its actual value at Z is the full series $\sum_j \mathcal{S}_q F_j(Z) (\mathcal{S}_q b(Z))^j$. For a selected input in Definition 8.4, this gives the selected graph values of both maps on their respective strict disks. If $c \neq 0$ and $\|v\|_A < |c|$, the reciprocal of $c+v$ exists in the source algebra and has norm at most $(|c| - \|v\|_A)^{-1}$.

If a complete source family $H(\omega)$ is holomorphic and bounded by M on the parameter polydisk $|\omega_i| < \rho_i$, then, for $0 < \rho'_i < \rho_i$,

$$(8.15) \quad \sum_{\nu \in \mathbb{N}^h} \|H_\nu\|_A (\rho')^\nu \leq M \prod_{i=1}^h (1 - \rho'_i/\rho_i)^{-1}.$$

Each fixed parameter derivative is bounded on a further strict polydisk. If an original Wiener norm is supplied instead, bounded source maps preserve that full norm at its original radii.

Proof. Apply Proposition 4.20 with the submultiplicative norm of Lemma 8.2. Continuity and multiplicativity of $\mathcal{S}_q$ identify the sum with the actual substitution. The identities in (8.13) then identify the selected graph values. The convergent geometric series $c^{-1} \sum_{j \geq 0} (-v/c)^j$ proves the

reciprocal assertion. This operation is scalar inversion in a Banach algebra; a fiber inverse uses the separately supplied map Ψ .

Integrating the complete family on a parameter torus of radii below ρ and increasing those radii to ρ gives $\|H_\nu\|_A \leq M\rho^{-\nu}$. Summing the resulting geometric series proves (8.15). For a derivative of multi-order α , the coefficient sum on a smaller polydisk acquires the finite factor $\sup_\nu (1 + |\nu|)^{|\alpha|} \vartheta^{|\nu|}$ for some $0 < \vartheta < 1$, together with fixed inverse radii. Thus the derivative is bounded after a fixed radius loss. The bounded family is constructed before either coefficient integration or differentiation. Bounded holomorphy alone does not give an absolute coefficient sum on the boundary ρ . $\square$

Lemma 8.6. *Let $u \in \mathcal{W}_r(\mathcal{B}_A(D))$ have full norm at most ϵ . Choose $0 < r_2 < r_1 < r$ and $\eta > 0$ with $r_2 + \eta < r_1$, and suppose*

$$(8.16) \quad \epsilon < \eta, \quad \frac{\epsilon}{r(1 - r_1/r)^2} < \frac{1}{2}.$$

The source map $F(w) = w + u(w)$ has a source inverse $G(Y) = Y + v(Y)$ with $v \in \mathcal{W}_{r_2}(\mathcal{B}_A(D))$ and $\|v\|_{\mathcal{W}_{r_2}} \leq \epsilon$. Their actual values are mutual fiber inverses on the indicated strict source and image disks. The construction uses the whole original norm of u and fixed margins before any coefficient extraction.

Proof. Work in the complete algebra $\mathcal{W}_{r_2}(\mathcal{B}_A(D))$, where the element Y has norm r_2 . For $\|v\| \leq \eta$, the norm of $Y + v$ is less than r_1 . Proposition 4.20 gives $\|u(Y + v)\| \leq \epsilon < \eta$. For two such elements V, W , use

$$(8.17) \quad V^j - W^j = \sum_{k=0}^{j-1} V^k (V - W) W^{j-1-k}$$

to obtain

$$(8.18) \quad \|u(V) - u(W)\| \leq \frac{\epsilon}{r} \sum_{j \geq 1} j(r_1/r)^{j-1} \|V - W\| < \frac{1}{2} \|V - W\|.$$

Iteration of $v \mapsto -u(Y + v)$ from zero stays in the closed η -ball, and its successive differences decrease geometrically. Completeness gives its unique fixed point, whose equation also gives $\|v\| \leq \epsilon$. This equation is exactly $F(G(Y)) = Y$ in the source algebra.

After synthesis, the derivative of the correction u on $|w| < r_1$ has modulus less than $1/2$, by the bound in (8.18). Integration along a straight segment in that disk proves injectivity of the actual F . The image of the actual G on $|Y| < r_2$ lies in $|w| < r_1$. Thus $G(F(w)) = w$ whenever $|w| < r_1$ and $|F(w)| < r_2$. If a complete parameter Wiener algebra is included in the coefficient algebra from the outset, these norm estimates apply to that full algebra before parameters are specialized. $\square$

Lemma 8.7. *For $r > 0$ and $0 < \vartheta < \pi$, put*

$$(8.19) \quad D(r, \vartheta) = \{|\zeta| < r\} \cup \{\zeta \neq 0 : |\arg \zeta| < \vartheta/P\}.$$

Choose $0 < \rho_0 < \rho_1 < 1$, $r' > 0$, and $\vartheta' > 0$ such that

$$(8.20) \quad \begin{aligned} r' &< r(1 - \rho_1)^\delta, \\ \vartheta' + \sup_{|\eta| \leq \rho_1} |\arg(1 + \eta)| &< \vartheta, \\ |\arg q| + \sup_{|\eta| \leq \rho_1} |\arg(1 + \eta)| &< \pi/2. \end{aligned}$$

Fix $B > A/(1 - \rho_1)$. The maps

$$(8.21) \quad \mathcal{D}_\eta(c, \phi) = (c, \phi((1 + \eta)^{-\delta} \zeta)) = \sum_{n \geq 0} \eta^n \mathcal{D}_n(c, \phi)$$

are holomorphic bounded operators from $\mathcal{B}_A(D(r, \vartheta))$ to $\mathcal{B}_B(D(r', \vartheta'))$ for $|\eta| < \rho_1$. With $C = B(1 + \rho_1)/A$ they satisfy $\|\mathcal{D}_n\| \leq C\rho_0^{-n}$ and, on the common actual domain,

$$(8.22) \quad \mathcal{S}_q(\mathcal{D}_\eta f)(Z) = \mathcal{S}_q f(Z(1 + \eta)).$$

All target constants and domains are fixed before the coefficient index n or any parameter coefficient is selected.

Proof. The two domain inequalities in (8.20) place every transformed argument in the original domain. Write $T = |1 + \eta|$ and $t = |\zeta|^P$. In the minor norm the ratio to the original bound is at most

$$(8.23) \quad T^{-\delta} e^{-(B-A/T)t} \left(\frac{1+Bt}{1+(A/T)t} \right)^{1+\delta} \leq T^{-\delta} (BT/A)^{1+\delta}.$$

After the norm constants $8PB^{-\delta}$ and $8PA^{-\delta}$ are included, the operator bound is $BT/A \leq C$. Integrating the pointwise holomorphic function on $|\eta| = \rho_0$ gives coefficient operators with norm at most $C\rho_0^{-n}$. On each smaller parameter disk the integral formula also bounds the remaining geometric series in operator norm. Thus the operator series is holomorphic and has the asserted coefficients.

For real η with $|\eta| < \rho_1$, the change of variables $t = (1 + \eta)u$ in the entire Laplace integral proves (8.22). For all sufficiently large positive Z , its two sides are holomorphic throughout $|\eta| < \rho_1$, by the last inequality in (8.20). The identity therefore holds for complex η there. Holomorphic continuation in Z gives it on the common connected receiving domain. In particular, no agreement of formal series is used to choose an actual branch. $\square$

Definition 8.8. Fix $h \geq 1$, $c_i > 0$, $a_i \in \mathbb{R}$, and original radii $R_i > 0$. For $F = \sum_{\nu \in \mathbb{N}^h} f_\nu z^\nu$ in $\mathcal{W}_R(\mathcal{B}_A(D))$, define its actual weak-coordinate value by

$$(8.24) \quad \mathcal{E}(F)(Z) = \sum_{\nu \in \mathbb{N}^h} c^\nu e^{-|\nu|Z} Z^{a \cdot \nu} \mathcal{S}_q f_\nu(Z), \quad Z > 0 \text{ sufficiently large.}$$

For each integer $n \geq 0$, partition $\{\nu : |\nu| = n\}$ into classes E by $\nu \sim \nu'$ if $a \cdot \nu - a \cdot \nu' \in \delta\mathbb{Z}$. Put $d_E = \max_{\nu \in E} a \cdot \nu$ and $j_\nu = (d_E - a \cdot \nu)/\delta \in \mathbb{N}$. Define

$$(8.25) \quad u_0 = (1, 0), \quad u_j = \left(0, \frac{q^{j\delta} \zeta^j}{\Gamma(j\delta)} \right) \quad (j \geq 1), \quad g_{n,E} = \sum_{\nu \in E} c^\nu u_{j_\nu} f_\nu.$$

The collected row at $(n, d_E - k\delta)$ is the coefficient of x^k in $\mathcal{T}_q g_{n,E}$. Rows from distinct classes at degree n have disjoint exponent sets.

Lemma 8.9. The rows in Definition 8.8 are continuous linear maps on the original space $\mathcal{W}_R(\mathcal{B}_A(D))$. Their joint kernel equals the kernel of $\mathcal{E}$ as a map to actual germs. If a row is nonzero, there exist $n_* \in \mathbb{N}$, $d_* \in \mathbb{R}$, and $b_* \in \mathbb{C} \setminus \{0\}$ such that

$$(8.26) \quad \mathcal{E}(F)(Z) = b_* e^{-n_* Z} Z^{d_*} (1 + o(1)), \quad Z \rightarrow +\infty.$$

The remainder includes every degree of F . If $\mathcal{E}(F)$ is real on the physical positive tail, then b_* is real, and the scale $e^{-n_* Z} Z^{d_*}$ is positive there.

Proof. We collect the finite fibers before identifying a first nonzero term, and bound every higher degree using the original coefficient norm.

Step 1. We identify the finite fibers. Every fiber $|\nu| = n$ is finite. Each row is a finite sum of continuous coefficient maps applied to bounded multiplication by the fixed elements u_{j_ν} . Moreover, $\mathcal{S}_q u_j(Z) = Z^{-j\delta}$, by (8.10). Hence the complete degree n contribution to (8.24) equals

$$(8.27) \quad e^{-nZ} \sum_E Z^{d_E} \mathcal{S}_q g_{n,E}(Z).$$

If all its rows vanish, injectivity of $\mathcal{T}_q$ makes each $g_{n,E}$ zero. If every degree has zero rows, absolute summation of (8.27) gives $\mathcal{E}(F) = 0$.

Step 2. We identify the first nonzero degree. Suppose instead that n_* is the first degree with a nonzero row. For each nonzero $g_{n_*,E}$ let k_E be its first nonzero formal degree. Lemma 8.2 gives its actual leading coefficient and a complete relative error tending to zero. The exponents $d_E - k_E\delta$ are distinct for distinct classes E . The largest one, denoted by d_* , therefore has a nonzero coefficient b_* in the finite sum (8.27).

Step 3. We control the complete remainder. Set

$$(8.28) \quad \epsilon(Z) = \max_{1 \leq i \leq h} \frac{c_i}{R_i} e^{-Z} Z^{a_i}.$$

For all large positive Z , $\epsilon(Z) < 1$ and $\Re(qZ) > A$. Contractivity of $\mathcal{S}_q$ gives

$$(8.29) \quad \left| \sum_{|\nu| > n_*} c^\nu e^{-|\nu|Z} Z^{a \cdot \nu} \mathcal{S}_q f_\nu(Z) \right| \leq \epsilon(Z)^{n_*+1} \|F\|_{\mathcal{W}_R(\mathcal{B}_A(D))}.$$

After division by $e^{-n_*Z} Z^{d_*}$ this tends to zero, since an exponential factor e^{-Z} remains and every other factor has at most fixed power growth. This proves (8.26) and the converse kernel inclusion. For a real actual germ, the quotient by the positive scale is real and tends to b_* ; hence $b_* \in \mathbb{R} \setminus \{0\}$. $\square$

Definition 8.10. Let $\mathcal{C}$ be a unital commutative complex Banach algebra with unit norm one. Suppose $\varepsilon_L : \mathcal{C} \rightarrow \mathbb{C}$ are contractive characters, holomorphic in L on a specified common domain. Fix $g, \delta_0, S > 0$ and complex measures ν_K , $K \geq 0$, with positive majorants μ_K satisfying

$$(8.30) \quad |\nu_K| \leq \mu_K, \quad \text{supp } \mu_K \subset [\max(g, \delta_0 K), \infty), \quad N_S = \sum_{K \geq 0} \int e^{-S\beta} d\mu_K(\beta) < \infty.$$

Write $U(X) = \sum_K X^K \int e^{-\beta X} d\nu_K(\beta)$ wherever this full sum converges. For $b_0 \in \mathbb{R}$, $\rho, d > 0$, let $\mathcal{M}_{b_0, \rho, d}(\mathcal{C})$ consist of sequences of $\mathcal{C}$ -valued measures m_l such that

$$(8.31) \quad \text{supp } m_l \subset [dl, \infty), \quad \|m\|_{\mathcal{M}} = \sum_{l \geq 0} \rho^l \int e^{-b_0 t} d|m_l|(t) < \infty.$$

Multiplication uses convolution in t , addition of l , and multiplication in $\mathcal{C}$. Its synthesis and the remaining exponential weight are

$$(8.32) \quad \mathcal{V}(m)(L) = \sum_{l \geq 0} L^l \int e^{-tL} \varepsilon_L(dm_l(t)), \quad v(L) = \Re L - b_0 - d^{-1} \log_+(|L|/\rho),$$

where $\log_+ x = \max(0, \log x)$.

Lemma 8.11. *The space in (8.31) is a unital Banach algebra. Its synthesis is a contractive homomorphism where $v(L) \geq 0$. If all its measures are supported in $[B, \infty)$, then*

$$(8.33) \quad |\mathcal{V}(m)(L)| \leq \|m\|_{\mathcal{M}} e^{-Bv(L)}.$$

Fix $\alpha > 0$, $b \in \mathbb{R}$, and $h \in \mathcal{C}$, and put $c_h = |b| + \|h\|$ and $d = \alpha\delta_0$. Choose b_0 so that

$$(8.34) \quad b_0\alpha \geq S + c_h + \delta_0^{-1} \log_+(\alpha\rho + c_h + \alpha).$$

For $0 \leq l \leq K$ define

$$(8.35) \quad m_{K,l} = (\beta \mapsto \alpha\beta)_* \left[\binom{K}{l} \alpha^l (b+h)^{K-l} \exp_{\mathcal{C}}[-\beta(b+h)] d\nu_K(\beta) \right], \quad m_l = \sum_{K \geq l} m_{K,l}.$$

Then $\|m\|_{\mathcal{M}} \leq N_S$, every carrier is at least αg , and

$$(8.36) \quad \mathcal{V}(m)(L) = U(\alpha L + b + \varepsilon_L h)$$

on the common receiving domain. The bound retains the complete original measures in (8.30).

Proof. The variation norm of a convolution is bounded by the product of the variation norms. Its exponential weight factors, and its supports obey $t+t' \geq d(l+l')$. These facts give completeness, bounded multiplication, and the unit at $(t, l) = (0, 0)$. For $t \geq dl$,

$$(8.37) \quad |L|^l e^{-t\Re L} \leq \rho^l e^{-b_0 t} e^{-tv(L)}.$$

Integration proves contractivity and (8.33); absolute convolution proves multiplicativity.

The norm of the complete Banach exponential in (8.35) is at most $e^{\beta c_h}$. Summing its finite polynomial factors gives

$$(8.38) \quad \sum_{l=0}^K \rho^l \binom{K}{l} \alpha^l c_h^{K-l} e^{\beta c_h} = e^{\beta c_h} (\alpha \rho + c_h)^K.$$

Use $K \leq \beta/\delta_0$ and (8.34) before summing K or integrating β . This bounds the full source norm by N_S . Evaluation of the complete exponential and the finite binomial formula then give (8.36). The majorant just proved justifies all interchanges. It also proves the assertions uniformly for any restriction dominated by the original μ_K . $\square$

Lemma 8.12. *Under the hypotheses of Lemma 8.11, fix $0 < \kappa < \Lambda$ and a real smooth function χ with $0 \leq \chi \leq 1$, $\chi = 1$ on $(-\infty, \kappa]$, and $\chi = 0$ on $[\Lambda, \infty)$. Put*

$$(8.39) \quad J = \left\lfloor \frac{\Lambda}{\alpha \delta_0} \right\rfloor, \quad H_\chi = \max \left(1, \max_{0 \leq j \leq J} \|\chi^{(j)}\|_\infty \right).$$

The cut source m_χ has, at tag l , the sum over $K \geq 0$, $0 \leq j \leq K$, and $0 \leq l \leq K-j$ of

$$(8.40) \quad (\beta \mapsto \alpha\beta)_* \left[(-1)^j \binom{K}{j} \binom{K-j}{l} \alpha^{j+l} \chi^{(j)}(\alpha\beta) (b+h)^{K-j-l} \exp_C[-\beta(b+h)] d\nu_K(\beta) \right].$$

At the fixed source stage chosen in (8.34),

$$(8.41) \quad \|m_\chi\|_{\mathcal{M}} + \|m - m_\chi\|_{\mathcal{M}} \leq 2H_\chi N_S, \quad \text{supp } m_\chi \subset [0, \Lambda], \quad \text{supp}(m - m_\chi) \subset [\kappa, \infty).$$

Consequently the entire complementary function satisfies

$$(8.42) \quad |U(\alpha L + b + \varepsilon_L h) - \mathcal{V}(m_\chi)(L)| \leq 2H_\chi N_S e^{-\kappa v(L)} \quad (v(L) \geq 0).$$

Proof. Let D_β denote distributional differentiation. Pairing with a compactly supported smooth test function and expanding the finite product derivatives proves

$$(8.43) \quad \chi(\alpha\beta) D_\beta^K \nu_K = \sum_{j=0}^K (-1)^j \binom{K}{j} D_\beta^{K-j} [\alpha^j \chi^{(j)}(\alpha\beta) \nu_K].$$

This formula includes atoms. Taking its compact Laplace transform gives a factor X^{K-j} in the j th term. Substitute $X = \alpha L + b + \varepsilon_L h$ and expand that finite polynomial. The result is (8.40).

For $j \geq 1$ a nonzero derivative term has $\kappa \leq \alpha\beta \leq \Lambda$. Its original power satisfies $K \leq \beta/\delta_0 \leq \Lambda/(\alpha\delta_0)$, so only derivatives through J occur. The $j = 0$ term is bounded by one. At each original (K, β) , summing first l and then j bounds the norm by

$$(8.44) \quad H_\chi e^{\beta c_h} \sum_{j=0}^K \binom{K}{j} \alpha^j (\alpha\rho + c_h)^{K-j} = H_\chi e^{\beta c_h} (\alpha\rho + c_h + \alpha)^K.$$

The fixed weight (8.34) therefore gives $\|m_\chi\|_{\mathcal{M}} \leq H_\chi N_S$. Replacing χ by $1 - \chi$ gives the other bound. Linearity of (8.43) identifies that source with $m - m_\chi$, including the opposite signs of all positive derivatives. Its support is at least κ , and the compact source is supported below Λ . Apply (8.33) to obtain (8.42). $\square$

Lemma 8.13. *In Definition 8.10, the distribution $T_U = \sum_K D_\beta^K \nu_K$ is well defined locally. Suppose, in addition, that a specified set $\mathcal{G} \subset (0, \infty)$, locally finite in $[0, \infty)$, and coefficients $c_{\omega,r}$ give*

$$(8.45) \quad T_U = \sum_{\omega \in \mathcal{G}} \sum_{0 \leq r \leq \lfloor \omega/\delta_0 \rfloor} c_{\omega,r} D_\beta^r \delta_\omega.$$

Here δ_ω is the unit point mass. If $[\kappa, \Lambda] \cap \alpha\mathcal{G} = \emptyset$, the cut of Lemma 8.12 has actual value

$$(8.46) \quad \mathcal{V}(m_\chi)(L) = \sum_{\alpha\omega < \kappa} e^{-\alpha\omega L} \sum_{l \geq 0} L^l \varepsilon_L(A_{\omega,l}), \quad A_{\omega,l} = \alpha^l \exp_{\mathcal{C}}[-\omega(b+h)] \sum_{r \geq l} c_{\omega,r} \binom{r}{l} (b+h)^{r-l}.$$

Both sums are finite. The coefficients $A_{\omega,l}$ are constructed in the complete algebra $\mathcal{C}$, before actual evaluation. If $\mathcal{J}(m) = \sum_l D_t^l m_l$, then

$$(8.47) \quad \mathcal{J}(m) = (\beta \mapsto \alpha\beta)_* [\exp_{\mathcal{C}}(-\beta(b+h)) T_U], \quad \mathcal{J}(m_\chi) = \chi \mathcal{J}(m).$$

These are identities of local distributions. They assert neither injectivity of the measure presentation nor equality of its kernel with any further formal comparison kernel.

Proof. On a bounded carrier interval, the inequality $K \leq \beta/\delta_0$ bounds the number of differentiations and powers that occur. It makes T_U a distribution of finite order on that interval. The inequality $l \leq t/d$ gives the corresponding statement for $\mathcal{J}(m)$. For $a = b+h$, the distributional product formula is

$$(8.48) \quad \exp_{\mathcal{C}}(-\beta a) D_\beta^K \nu_K = \sum_{l=0}^K \binom{K}{l} D_\beta^l [a^{K-l} \exp_{\mathcal{C}}(-\beta a) \nu_K].$$

One verifies it by pairing with a test function and expanding its K th derivative. Under $t = \alpha\beta$, a derivative D_β^l acquires the factor $\alpha^l D_t^l$. Thus (8.35) gives the first identity in (8.47); (8.43) gives the second.

The cutoff is constant near every point in (8.45). It keeps the jets below κ/α and deletes the others. Its Laplace transform is therefore the finite sum of $e^{-\omega X} \sum_r c_{\omega,r} X^r$ with $X = \alpha L + b + \varepsilon_L h$. The finite binomial expansion of X^r gives exactly (8.46). The complete exponential remains inside $\mathcal{C}$ throughout. $\square$

Lemma 8.14. *Let the measures in (8.30) be real and fix $\lambda > 0$. On a sufficiently far positive tail the map $T \mapsto T - U(T)/\lambda$ has an actual inverse $T = V + J(V)$ with $J(V) \rightarrow 0$. It has the complete expansion*

$$(8.49) \quad J(V) = \sum_{n \geq 1} \frac{1}{n! \lambda^n} \partial_V^{n-1} (U(V)^n).$$

For ordered $K_1, \dots, K_n \geq 0$, write $K = \sum_i K_i$ and $\nu = \nu_{K_1} * \dots * \nu_{K_n}$. Its contribution at power $m = K - j$, $0 \leq j \leq \min(K, n-1)$, is

$$(8.50) \quad \frac{\binom{n-1}{j} (K)_j}{n! \lambda^n} (-\beta)^{n-1-j} d\nu(\beta),$$

where $(K)_j = K(K-1) \dots (K-j+1)$ and $(K)_0 = 1$. With $C = 1 + \delta_0^{-1}$, choose $Q > C N_S / \lambda$ and put $S' = S + Q$. The positive absolute majorants of these contributions satisfy

$$(8.51) \quad \sum_{m \geq 0} \int e^{-S' \beta} d\mu_m^{\text{inv}}(\beta) \leq \sum_{n \geq 1} \frac{N_S^n}{n \lambda^n} \left(\frac{C}{Q} \right)^{n-1} < \infty.$$

They have $\beta \geq g$ and $\beta \geq \delta_0 m$. The independent local distribution is

$$(8.52) \quad T_J = \sum_{n \geq 1} \frac{(-\beta)^{n-1}}{n! \lambda^n} T_U^{*n}.$$

The cuts in Lemma 8.12 apply to this whole bank with S' in place of S ; their cutoff acts on the total convolution carrier β .

Proof. We first identify the actual inverse, bound its complete ordered source, and then identify the independent local distribution.

Step 1. We construct the actual inverse. On each fixed horizontal strip with $\Re V$ large, the bounds $K \leq \beta/\delta_0$ and $\beta \geq g$ show that U and each fixed derivative tend uniformly to zero. A fixed derivative contributes only finitely many factors bounded by fixed powers of β , whereas

$$(8.53) \quad \Re V - S - \delta_0^{-1} \log \max(1, |V|) \rightarrow +\infty.$$

Any fixed power of β is absorbed by a fixed positive weight slack. A disk of fixed small radius about V consequently has $|U/\lambda|$ less than half its radius and $|U'/\lambda| < 1/2$. The map $T \mapsto V + U(T)/\lambda$ is a contraction there and gives the unique small solution.

On the boundary circle, the series for $-\text{Log}(1 - U(z)/(\lambda(z - V)))$ converges uniformly. Integration by parts in the residue formula for the unique zero of $z - V - U(z)/\lambda$ expresses its displacement as the integral of this logarithm divided by $2\pi i$. Its n th term is $\partial_V^{n-1}(U(V)^n)/(n!\lambda^n)$ by the coefficient integral formula. This proves (8.49) for the actual inverse.

Step 2. We prove the complete source bound. Expanding the n original powers with their ordered source labels, and differentiating $V^K e^{-\beta V}$ exactly $n - 1$ times, gives (8.50). At such a tuple, $\beta \geq ng$ and $K \leq \beta/\delta_0$. The sum of its absolute polynomial derivative factors is bounded by

$$(8.54) \quad \sum_j \binom{n-1}{j} (K)_j \beta^{n-1-j} \leq (\beta + K)^{n-1} \leq (C\beta)^{n-1}.$$

The inequality $\beta^{n-1} e^{-Q\beta} \leq (n-1)! Q^{-(n-1)}$ then leaves the n th term of (8.51) after all the original powers and carriers are summed. Its geometric ratio is less than one. This proves absolute convergence in one fixed source norm, not just convergence of the scalar inverse series. The support inequalities follow from $m \leq K$ and $\beta \geq ng$.

Step 3. We identify the local distribution. For a measure ν , pairing with a compactly supported smooth test function and expanding the product derivative gives

$$(8.55) \quad (-\beta)^r D_\beta^K \nu = \sum_{j=0}^{\min(K,r)} \binom{r}{j} (K)_j D_\beta^{K-j} [(-\beta)^{r-j} \nu].$$

On each bounded carrier interval, $n \leq \beta/g$ and $K \leq \beta/\delta_0$ leave only finitely many orders. Differentiation commutes there with positive-carrier convolution, as follows by integration against a test function on its compact frequency simplex. Apply (8.55) with $r = n - 1$ to obtain (8.52). The bank just bounded meets Definition 8.10, so the full cutoff formula applies to it without truncating the inverse input. $\square$

Lemma 8.15. Suppose two banks satisfy (8.30), with positive gaps g_1, g_2 , linkages δ_1, δ_2 , and finite norms $N_S^{(1)}, N_S^{(2)}$ at a common weight S . Their product has the complete ordered source

$$(8.56) \quad \nu_K = \sum_{K_1+K_2=K} \nu_{K_1}^{(1)} * \nu_{K_2}^{(2)}.$$

It satisfies (8.30) with $g = g_1 + g_2$, $\delta_0 = \min(\delta_1, \delta_2)$, and norm at most $N_S^{(1)} N_S^{(2)}$. Its local distribution is $T_{U_1 U_2} = T_{U_1} * T_{U_2}$. A total-action cut of this product is given by (8.40), with $\beta = \beta_1 + \beta_2$ in the original convolution.

Proof. The inequalities $\beta_1 + \beta_2 \geq g_1 + g_2$ and $\beta_1 + \beta_2 \geq \min(\delta_1, \delta_2)(K_1 + K_2)$ give the support claims. The weight factors as $e^{-S(\beta_1 + \beta_2)} = e^{-S\beta_1} e^{-S\beta_2}$, so absolute convolution and summation give the norm bound. On a compact carrier interval only finitely many powers occur. Pairing

with a test function there gives $D^{K_1}\nu_{K_1}^{(1)} * D^{K_2}\nu_{K_2}^{(2)} = D^{K_1+K_2}(\nu_{K_1}^{(1)} * \nu_{K_2}^{(2)})$. This proves the distribution identity. Applying (8.43) to that fully convolved measure proves the cut assertion. In particular its cutoff is $\chi(\alpha(\beta_1 + \beta_2))$, including the derivatives in (8.40). $\square$

Lemma 8.16. *Let $A_1, \dots, A_m > 0$, $m \geq 1$, and collect numerical coincidences in the additive monoid*

$$(8.57) \quad \mathcal{G}_0 = \left\{ \sum_{i=1}^m n_i A_i : n_i \in \mathbb{N} \right\}.$$

Fix $d > 0$. There exist constants $C_0, C_1 > 0$, depending only on d and the generators, such that for every $R \geq 1$ there exists a smooth cutoff χ_R , equal to one up to κ_R and zero from Λ_R onward, with

$$(8.58) \quad R \leq \kappa_R < \Lambda_R \leq 2R, \quad [\kappa_R, \Lambda_R] \cap \mathcal{G}_0 = \emptyset.$$

For $J_R = \lfloor \Lambda_R/d \rfloor$ it satisfies

$$(8.59) \quad 2 + 2^{J_R} \max_{1 \leq j \leq J_R} \|\chi_R^{(j)}\|_\infty \leq C_0 \exp(C_1 R \log(eR)),$$

where the maximum is zero if $J_R = 0$. For a source satisfying Lemma 8.12 with $d = \alpha\delta_0$ and $\alpha\mathcal{G} \subset \mathcal{G}_0$, these cutoffs therefore give

$$(8.60) \quad |U(\alpha L + b + \varepsilon_L h) - \mathcal{V}(m_{\chi_R})(L)| \leq 2C_0 N_S \exp(C_1 R \log(eR) - Rv(L)) \quad (v(L) \geq 0).$$

The estimate is on the fixed receiving domain and in the original norm N_S ; its constants do not depend on R .

Proof. We construct a smooth profile with controlled derivatives and place it inside a sufficiently long gap.

Step 1. We choose a fixed smooth profile. Define $\psi(t) = e^{-1/t}$ for $t > 0$ and $\psi(t) = 0$ for $t \leq 0$. On $|z - t| \leq t/2$, $\Re(1/z) \geq 2/(9t)$. The coefficient integral for its j th derivative gives

$$(8.61) \quad |\psi^{(j)}(t)| \leq j!(2/t)^j e^{-2/(9t)} \leq 9^j (j!)^2.$$

The bound also proves smooth extension at zero. Set $\varphi(t) = \psi(t)\psi(1-t)$ and $I = \int_0^1 \varphi(t) dt$. On $[1/3, 2/3]$, $\varphi \geq e^{-6}$, so $I \geq e^{-6}/3$. Leibniz' formula gives $\|\varphi^{(j)}\|_\infty \leq 18^j (j!)^2$ for $j \geq 1$, and $\|\varphi\|_\infty \leq 1$. Thus $\chi_*(t) = I^{-1} \int_t^\infty \varphi(u) du$ has

$$(8.62) \quad \|\chi_*^{(j)}\|_\infty \leq 3e^6 18^j (j!)^2 \quad (j \geq 1).$$

Step 2. We control the gap and its derivative costs. Put $C_G = \prod_i (1 + 2/A_i)$. The number of numerical monoid points in $[0, 2R]$ is at most $C_G R^m$, since each n_i is bounded by $2R/A_i$. Subdividing $[R, 2R]$ by its monoid points leaves an interval of length

$$(8.63) \quad \ell \geq (C_G + 3)^{-1} R^{1-m}$$

with no monoid point in its interior. Choose the middle third as $[\kappa_R, \Lambda_R]$ and rescale χ_* to this interval. Using (8.62) with (8.63) gives

$$(8.64) \quad \|\chi_R^{(j)}\|_\infty \leq 3e^6 [54(C_G + 3)R^{m-1}]^j (j!)^2.$$

Let $D_0 = \max(1, 2/d)$ and $B_0 = 54(C_G + 3)D_0^2$. If $J_R = 0$, the left side of (8.59) is 2. For $J_R \geq 1$, use $J_R \leq D_0 R$ and $J_R! \leq J_R^{J_R}$ to bound that left side by

$$(8.65) \quad (2 + 3e^6) \exp(D_0 [\log(2B_0) + m + 1] R \log(eR)).$$

This specifies admissible C_0, C_1 . Finally $H_{\chi_R} \leq C_0 \exp(C_1 R \log(eR))$ and $\kappa_R \geq R$. Substitution in (8.42) proves (8.60). Taking $R = 2^n$ gives a cofinal sequence of cuts. $\square$

Remark 8.17. The compact records of Lemma 8.13 need not be a zero test when a new source is attached to an old coefficient algebra. For example, let $b > 0$ and take the convergent Wiener algebra in z at radius r with $br < 1$. On a sufficiently far horizontal strip let $\varepsilon_L(z) = Le^{-L}$; these characters are contractive. The complete old coefficient $f(z) = -\text{Log}(1 - bz)$ and the new power-measure bank

$$(8.66) \quad U(L) = \sum_{n \geq 1} \frac{b^n}{n} L^n e^{-nL} = -\text{Log}(1 - bLe^{-L})$$

have exactly equal actual values. The bank has $\nu_n = (b^n/n)\delta_n$, so it satisfies (8.30) for $g = \delta_0 = 1$ and all sufficiently large S . In the new record of $U - f$, the carrier-zero coefficient is $-f \neq 0$, and the positive carriers have nonzero jets. Its actual value is nevertheless identically zero. The comparison in Lemma 8.9 avoids this problem by collecting the exact fibers before testing them. The distributional identity (8.47) alone does not perform that collection in a general old coefficient algebra.

Corollary 8.18. *For a finite list of actual semihyperbolic passage occurrences, choose one nonreal q with $|q| = 1$ and $\Re q > 0$ before constructing their reference normalizers. The construction of Proposition 6.7, on each specified physical root lift, supplies selected input data of Definition 8.4 for the normalizers and their true fiber inverses. Their endpoint and parameter buffers precede the source weights. A common multiple of the root orders and common strict angular and receiving restrictions give a finite common-ray list, with bounded maps from each original source norm.*

The original physical passage is recovered with its actual section adapters as in Proposition 6.13. If other labels were previously selected, their passage formulas are replayed by the whole weak-coordinate identity of Proposition 6.15, on its exact physical diagonal and strict disks. A previously supplied source-equipped label keeps its own native bank and proved synthesis identity in parallel. A general compatible analytic label is retained by its full forward, inverse and graph weak recipes of Lemma 6.16. Independently free banks on another ray are not inputs to the common- q comparison. No bounded rotation or identification of their completions is asserted.

Proof. We construct the common-ray reference list first and then recover the original labels by their exact passage identities. Choose, for example, $q = \exp(i\pi/4)$. This choice precedes every normalizer. For occurrence i , Proposition 6.4 supplies its integer p_i , its final analytic base coordinate, and its positive physical height $Z_i = \beta_i s_i^{-p_i}$. Fix the lift $x_i = \kappa_i Z_i^{-1/p_i}$ induced by that coordinate, with $\kappa_i^{p_i} = -1/p_i$, and the logarithm real on $Z_i > 0$. These choices fix the factors $\kappa_i^n q^{n/p_i}$ in (6.19).

Step 1. We retain the complete selected tuples. Use Definition 6.5 with this q and each fixed lift. Its root domain contains an origin disk and the positive ray and is invariant under radial contraction. Its norm and product are those of Definition 8.1, with $P = p_i$. Proposition 6.7 gives the whole forward and inverse endpoint sources and their actual Laplace identities. In (8.13) interpret $\Phi(Z_i, w)$ as $\Phi(x_i(Z_i), w)$ and interpret Ψ through that identical physical lift. The actual identities (6.29) supply both inverse compositions. Thus these are selected inputs, including their actual determinations.

Choose the strict endpoint, inverse image, and parameter buffers before the weight. The nonconstant whole source norms tend to zero with the weight by Proposition 6.7. They can therefore meet (8.16) and the strict graph bounds after the buffers have been fixed. The complete graph and parameter substitutions are the operations in Lemma 6.17. A supplied parameter Wiener norm is retained throughout; if only bounded holomorphy is supplied, the explicit strict-radius factor is (8.15), applied to the whole family.

Let A_i be the resulting source weights and take $A \geq \max_i A_i$. At an unchanged root order and domain, the weight comparison is

$$(8.67) \quad \|(c, \phi)\|_A \leq \frac{A}{A_i} \|(c, \phi)\|_{A_i}.$$

To prove it, put $t = |\zeta|^{p_i}$. The quotient of the two minor weights, including their norm constants, is bounded by $(A_i/A)^{1/p_i} (A/A_i)^{1+1/p_i} = A/A_i$; the exponential factor is at most one. The scalar component satisfies this bound too. Let P be a common multiple of the p_i . A sufficiently small origin disk together with a sufficiently narrow sector about the positive root ray is mapped into every original domain by $\zeta \mapsto \zeta^{P/p_i}$. Finiteness of the list permits these restrictions to be chosen together. Lemma 8.3 gives the norm factor P/p_i after (8.67), and preserves each whole radial density. Intersecting the finitely many strict Laplace sectors leaves a receiving sector about each physical positive height; the physical tails can be shortened together. None of these operations rotates a minor or changes a lift.

Step 2. We recover the original physical passages. Proposition 6.13 recovers each passage with its actual incoming and outgoing section adapters and their true inverses. Suppose σ is the new reference label and τ was previously selected. Proposition 6.15 constructs the full replay $G_\tau^{[\sigma]}$ on strict independent weak-coordinate disks and gives

$$(8.68) \quad k_\tau(VG_\tau^{[\sigma]}) = k_\sigma(VG_\sigma).$$

Only after this source identity is formed do we set $V = ce^{-Z}Z^a$. On that exact physical diagonal the left side is the old passage formula, including its entire weak tail. Finite minima of the strict disks allow all the occurrences to be replayed. At $V = 0$ this replay gives the reference σ source. It does not recover a pure native source for an arbitrary τ . If τ has a separately proved source-equipped presentation, that original presentation and norm remain available in parallel; they are not jointly compared on q by this identity. The full forward and inverse recipes and their original-radius bounds are those of Lemma 6.16. Identity (8.68) preserves actual passage formulas without identifying independent completed native spaces. $\square$

Proposition 8.19. *Let a finite family of selected normalizing inputs have the exact source and physical matching data of Definition 8.4. Their root algebras, bounded root refinements, strict graph substitutions, fiber inverses and relative clocks have the whole original-norm bounds of Lemmas 8.2–8.7, with the respective hypotheses and domains stated there. The explicit weak-coordinate presentation of Definition 8.8 has the independently collected exact kernel and complete head of Lemma 8.9.*

For each action receiver supply the full data of Definition 8.10 and, when a compact point-jet record is invoked, the additional independent identity (8.45). Then whole transport, ordered time inversion, total-action product-rule cuts, and their complete complements obey Lemmas 8.11–8.15. For a finitely generated positive action monoid the cofinal estimates are those of Lemma 8.16, on its specified fixed receiving domain. The local distribution record alone is not an exact comparison kernel for an enlarged coefficient algebra. Distinct native clocks and larger phase fields require the separately specified comparison constructions and their source-domain hypotheses.

Proof. Lemmas 8.2, 8.3, 8.5, 8.6, and 8.7 give the root operations with their full input norms. The independent weak-coordinate rows and their exact kernel are those of Lemma 8.9. For each supplied action receiver, Lemma 8.11 constructs the complete source, Lemma 8.12 bounds its compact part and its entire complement, and Lemma 8.13 identifies the compact record under the stipulated point-jet identity. Lemmas 8.14 and 8.15 retain the original ordered inverse and product sources. Finally, Lemma 8.16 supplies the cofinal derivative costs on that receiver's fixed domain. The limitation on enlarged comparison kernels follows from the explicit equality and nonzero uncollected record in Remark 8.17. $\square$

Lemma 8.20. *Let E be a complex Banach space, $0 < a < b$, and $r > 0$. For a bounded open angular interval J on the logarithmic cover put $S(J, r) = \{u : 0 < |u| < r, \arg u \in J\}$. Let $J_i = (\alpha_i, \beta_i)$, $1 \leq i \leq N$, be an ordered finite chain with strictly increasing endpoints, $\alpha_{i+1} < \beta_i$, and $|\bigcup_i J_i| > \pi/a$. Suppose holomorphic maps $f_i : S(J_i, r) \rightarrow E$ satisfy*

$$(8.69) \quad \|f_i(u)\|_E \leq M e^{-c_a |u|^{-a}}, \quad \|f_i(u) - f_{i+1}(u)\|_E \leq M e^{-c_b |u|^{-b}}$$

on their sectors and consecutive overlaps, with $M \geq 0$ and $c_a, c_b > 0$. For any fixed closed angular buffers $K_i \subseteq J_i$ there exist $r' > 0$ and $C, c' > 0$ such that

$$(8.70) \quad \|f_i(u)\|_E \leq C M e^{-c' |u|^{-b}}, \quad 0 < |u| < r', \quad \arg u \in K_i.$$

The constants depend on the sector geometry, buffers and displayed exponents and decay constants, independently of M , the Banach space and its coefficient indices. A single whole holomorphic function with the first bound on a sector of aperture greater than π/a is zero. The assertions apply to whole analytic parameter families with uniform bounds on their original larger polydisks.

Proof. We transform a finite sector chain into Laplace integrals on an exterior sector, use the complete jumps to extend those transforms to entire functions, and invert them on a contour where the entire growth yields the stronger decay. All integrations below are in the original norm of E . On compact paths they are limits of Riemann sums; the infinite tails are absolutely convergent. Their norm is bounded by the integral of the norm. Integration of norm-convergent local power series supplies local primitives. Subdividing a compact contour deformation into such disks proves the contour identities in E . The coefficient formula on a circle follows by integrating the local series. Absolute double integration is justified by compact double sums and the norm bounds on their discarded tails. A nonzero leading vector in a local Taylor series makes its zeros isolated. A zero Taylor series propagates through overlapping disks, giving the identity theorem on a connected domain. None of these operations uses coordinates in E .

Step 1. We specify the finite sector geometry. Fix an argument in one prescribed K_i . Choose a subinterval of the union of the J_i , containing that argument with a margin, whose length is strictly between π/a and $2\pi/a$. Refine the sector chain on this interval, retaining the desired representative near that argument and switching labels inside the original consecutive overlaps. Repeating a label gives zero jump; any additional overlap comparison is a sum of finitely many original consecutive jumps. Its bound is therefore at most $N M \exp(-c_b |u|^{-b})$. The resulting finite factor is absorbed in the constants below.

On this logarithmic sheet set $y = \omega u^{-a}$, where $|\omega| = 1$ centers its angular image so that it contains both $-\pi/2$ and $\pi/2$ with positive margins. The image interval has length between π and 2π , so y is an injective planar sector coordinate. Reorder the chain after this change. With $\ell = b/a > 1$, its representatives g_i satisfy, for $|y|$ sufficiently large,

$$(8.71) \quad \|g_i(y)\| \leq C M e^{-c_a |y|}, \quad \|g_i(y) - g_{i+1}(y)\| \leq C M e^{-c_b |y|^\ell}.$$

Refine into angular intervals of width less than $\pi/2$ with buffered overlaps. In each such cone choose a fixed far point d_i on an interior ray. For every direction θ in the smaller closed interval being used, the translated ray stays in its original exterior sector and satisfies

$$(8.72) \quad |d_i + t e^{i\theta}| \geq c_i (|d_i| + t), \quad t \geq 0, \quad c_i > 0.$$

For this inequality, take the projection on the cone's bisector; both summands have a positive projection bounded below by a fixed multiple of their modulus. Choose $|d_i|$ large enough that the projection also keeps every ray beyond the forbidden origin disk. All these choices depend only on the angular and radial buffers.

Step 2. We construct entire extensions of the transforms. For an allowed direction θ define

$$(8.73) \quad G_i(z) = \int_{d_i + [0, \infty)e^{i\theta}} g_i(y) e^{-zy} dy.$$

The bounds (8.71) and (8.72) give absolute convergence on a fixed disk about $z = 0$ and on an open neighborhood of $\Re(ze^{i\theta}) \geq 0$. They also give

$$(8.74) \quad \|G_i(z)\| \leq CM \exp(C|z|) \quad (\Re(ze^{i\theta}) \geq 0).$$

On compact subsets, each differentiated integrand is dominated by a polynomial times a remaining exponential decay. The integral is therefore holomorphic. For two allowed source directions, deform the truncated translated wedge first with z in the fixed small disk. The outer arc tends to zero because the source decay dominates $\exp(|z||y|)$. The two integrals agree near zero, and then agree on their connected overlaps. Their domains are unions of half-planes containing zero and a disk about zero, hence are star-shaped. This defines a single G_i on that union.

For consecutive representatives, choose a far point d_* and a ray in their buffered overlap. Moving the two starting points to d_* adds finite path integrals. Thus, near zero,

$$(8.75) \quad G_i - G_{i+1} = E_i,$$

where E_i is the sum of two finite path integrals and the transform of $g_i - g_{i+1}$ on the common translated ray. The finite paths contribute at most $CM \exp(C|z|)$. The jump transform is entire: putting $\rho = \ell/(\ell - 1)$, its integrand is bounded using

$$(8.76) \quad |z|t - \frac{c}{2}t^\ell \leq C|z|^\rho \quad (t \geq 0).$$

For $z \neq 0$, divide by $|z|^\rho$ and substitute $t = |z|^{1/(\ell-1)}v$; the remaining function $v - (c/2)v^\ell$ has a finite maximum. For $z = 0$ the inequality holds directly. The other half of the negative exponential gives an integrable majorant. Consequently

$$(8.77) \quad \|E_i(z)\| \leq CM \exp(C|z|^\rho), \quad z \in \mathbb{C}.$$

The finite path contributions, including the changes of starting point, are retained in this estimate.

An interval containing the direction $-\pi/2$ supplies a transform on a neighborhood of the closed upper half-plane. An interval containing $\pi/2$ supplies a transform on a neighborhood of the closed lower half-plane. The finite sums of the entire E_i make these transforms agree near zero. The identity theorem makes them agree on their full intersections, which are star-shaped and contain a disk about zero. The two half-planes cover $\mathbb{C}$. This gives an entire extension of one transform, and (8.75) extends every other transform, with the uniform global bound

$$(8.78) \quad \|G_i(z)\| \leq CM \exp(C|z|^\rho), \quad z \in \mathbb{C}.$$

If all the representatives are restrictions of one holomorphic function, their common-ray differences vanish. In that case the E_i consist only of finite path integrals, and the stronger bound $CM \exp(C|z|)$ holds throughout $\mathbb{C}$.

Step 3. We prove exact inversion and faster decay. Fix a target direction γ inside one of the narrowed source intervals. Choose $0 < \epsilon < \pi/8$ such that $[\gamma - 2\epsilon, \gamma + 2\epsilon]$ stays in that interval. For y far along a small neighborhood of that direction, set $\zeta = y - d_i$ and ensure $|\arg \zeta - \gamma| < \epsilon/4$. Define

$$(8.79) \quad \begin{aligned} H_i(z) &= e^{d_i z} G_i(z), & \theta_- &= \gamma - \epsilon, & \theta_+ &= \gamma + \epsilon, \\ \phi_+ &= -\gamma + \pi/2 + \epsilon/2, & \phi_- &= -\gamma - \pi/2 - \epsilon/2. \end{aligned}$$

Pair source direction θ_- with transform direction ϕ_+ , and source direction θ_+ with transform direction ϕ_- . For either pair, $v = d_i + te^{i\theta}$ and $z = se^{i\phi}$ satisfy

$$(8.80) \quad \Re(z(v - y)) \geq c_\epsilon s(t + |\zeta|), \quad s, t \geq 0, \quad c_\epsilon > 0.$$

The source and transform angles add to $\pi/2 - \epsilon/2$ or $-\pi/2 + \epsilon/2$, while $\Re(z\zeta) \leq -s|\zeta|\sin(\epsilon/4)$. These two cosine inequalities prove the displayed bound.

Let Γ_0 run inward on the ray ϕ_- to zero and outward on the ray ϕ_+ . The double integral obtained by inserting (8.73) is absolutely convergent: after integration in s , its norm is bounded by a constant times

$$(8.81) \quad \int_0^\infty \frac{\|g_i(d_i + te^{i\theta})\|}{t + |\zeta|} dt < \infty.$$

Integrating the exponential first in the transform variable gives

$$(8.82) \quad \frac{1}{2\pi i} \int_{\Gamma_0} H_i(z) e^{z\zeta} dz = \frac{1}{2\pi i} \left(\int_{d_i + [0, \infty)e^{i\theta_-}} \frac{g_i(v)}{v - y} dv - \int_{d_i + [0, \infty)e^{i\theta_+}} \frac{g_i(v)}{v - y} dv \right) = g_i(y).$$

For the last equality, close the source contour by a far counterclockwise arc from the lower to the upper ray. The resulting contour surrounds y positively, and its far arc tends to zero by (8.71). Excise a small circle about y and integrate the local power series there. Its constant term gives $g_i(y)$ and all higher terms integrate to zero. This also fixes the sign in (8.82).

Since H_i is entire, replace the origin join of Γ_0 by the circular arc of radius R passing through direction $\pi - \gamma$, with the clockwise angular choice. This arc lies to the left relative to ζ and satisfies

$$(8.83) \quad \Re(z\zeta) \leq -cR|\zeta|.$$

On the two remaining infinite rays, the original paired Laplace representations still hold. The factor $e^{d_i z}$ removes their starting-point exponential. Direct integration of the source decay and the factor $\exp(-c|z|t)$ gives

$$(8.84) \quad \|H_i(z)\| \leq \frac{CM}{1 + |z|}.$$

The two tails beyond R are therefore bounded by $CM \exp(-cR|\zeta|)$ for sufficiently large $|\zeta|$. The complete circular arc has norm at most

$$(8.85) \quad CM R \exp(CR^\rho - cR|\zeta|),$$

using (8.78) also for H_i . Choose $R = (c|\zeta|/(2C))^{1/(\rho-1)}$ and shorten the radial tail so that $R \geq 1$. The exponent in (8.85) is then at most $-(c/2)R|\zeta|$. Since $\rho/(\rho-1) = \ell$, the arc and both tails are bounded by $CM \exp(-c'|\zeta|^\ell)$ after the factor R is absorbed by a smaller $c' > 0$. For large y , $|y - d_i|$ and $|y|$ are comparable. Undoing $y = \omega u^{-a}$ gives the required b -order decay near the chosen target direction. A finite cover of all the compact intervals K_i fixes common constants and a common radius.

For a single function, use the global exponential-type bound in place of (8.78). For fixed $|\zeta|$ sufficiently large, the arc bound becomes $CM R \exp((C - c|\zeta|)R)$ and tends to zero as $R \rightarrow \infty$. Both ray tails tend to zero. The exact inversion formula gives $g_i(y) = 0$ there. Vanishing extends to the original connected sector by the identity theorem, also on its original logarithmic sheet.

Step 4. We retain whole parameter families. Every path, angular margin, starting point and constant was chosen from the original geometry and decay constants. All estimates were linear in M and used the norm of the entire E -valued function. They consequently hold uniformly on any original parameter polydisk on which the hypotheses hold uniformly. The absolutely

convergent contour integrals preserve joint holomorphy. For fixed parameter circles $|z_j| = \rho_j < R_j$, coefficient integration gives

$$(8.86) \quad \|[z']f_i(u, z)\|_E \leq CM\rho^{-\nu}e^{-c'|u|^{-b}}.$$

At fixed $s_j < \rho_j$, the full coefficient sum has the additional factor $\prod_j(1 - s_j/\rho_j)^{-1}$. No contour or decay constant depends on ν . On a further angular buffer a fixed disk $|v - u| \leq \eta|u|$ remains in the region of the improved bound. The circle formula gives, simultaneously for all integers $d \geq 0$,

$$(8.87) \quad \|\partial_u^d f_i(u)\|_E \leq C_* M d! (\eta|u|)^{-d} \exp(-c_*|u|^{-b}), \quad c_* = c'(1 + \eta)^{-b} > 0.$$

The parameter and derivative factors multiply when both sets of fixed circles are used. These estimates retain any complete source or marked-variable Banach norm already present in E . $\square$

Set-up 8.21. Let K be a differential field of complex holomorphic germs at a positive real tail in T , containing $\mathbb{C}$, with derivation $D = d/dT$. Suppose it is the fraction field of a directed union of unital commutative Banach algebras E_s , with bounded injective coherent inclusions and submultiplicative norms. Each stage has a faithful unital realization on a connected domain Ω_s containing a real tail, bounded on each compact subset in its original norm and satisfying $|a(T)| \leq C_s \|a\|_s$ on that entire tail. For every s , supply a later stage u , bounded linear maps $B_j : E_s \rightarrow E_u$, and fixed $c_j \in K$ such that

$$(8.88) \quad Da = \sum_{j=1}^{m_s} c_j B_j(a) \quad (a \in E_s).$$

All choices and bounds precede a .

Fix $H \in K$ with $h = \Re H \rightarrow +\infty$ and $\Im H \rightarrow 0$. For every nonzero $a \in K$, supply a complete head $a = d_a \theta_a (1 + \epsilon_a)$, with $d_a \neq 0$, $\theta_a > 0$, $\epsilon_a \rightarrow 0$ and $|\log \theta_a| = o(h)$. Fix $q \geq 1$, positive real numbers τ_i , and $P_i \in K$ with $P_i = o(h)$ and $\Im P_i \rightarrow 0$. Put

$$(8.89) \quad w_i = \exp(-\tau_i H + P_i), \quad \Lambda = \{\tau \cdot n : n \in \mathbb{N}^q\}.$$

For each stage s and positive radius vector r , the raw input is $\mathcal{W}_r(E_s)$ with its full original coefficient norm, and its actualization is the entire series $\sum_n a_n(T)w(T)^n$ on the connected strict evaluation domain containing a far real tail. These are supplied field and source-domain hypotheses, not consequences of a scalar expression for a current.

For arbitrary input data, the hypotheses of Set-up 8.21 remain assumptions. For the sources constructed here, the model folds and Proposition 10.31 supply the base data; the growth-ordered reconstruction and transport in Sections 10 and 11, in particular Lemma 11.23 and Theorem 11.2, propagate them. The additional horizontal domain and ratio hypotheses are treated separately in Theorem 8.25 and Lemma 11.21; they are not included in the phase assumptions alone.

Lemma 8.22 (Derivative limits from complete heads). *Let $\mathcal{K}$ be a differential field of holomorphic germs along $u_x = x + i\eta(x)$, where η is real and continuously differentiable and $\eta(x), \eta'(x) \rightarrow 0$. Assume that $\mathcal{K}$ contains the constants and the coordinate u and is closed under d/du . Assume that every nonzero $g \in \mathcal{K}$ has, along this curve, a complete head $g(u_x) = c_g \theta_g(x)(1 + o(1))$, with $c_g \neq 0$ and θ_g positive and continuous. No limit or monotonicity of θ_g is assumed. Then the following statements hold.*

- (1) *Every $g \in \mathcal{K}$ has either a finite complex limit or a directional escape to infinity. More precisely, if $g'_u(u_x) = c\theta(x)(1 + o(1)) \neq 0$ and $I(x) = \int_{x_0}^x \theta(t) dt$ diverges, then $g(u_x) = cI(x)(1 + o(1))$. If that integral converges, $g(u_x)$ has a finite complex limit. If $g'_u = 0$, then g is constant.*
- (2) *If $f \in \mathcal{K}$ satisfies $f(u_x) = O(u_x)$, then $f(u_x)/u_x$ has a finite limit L and $f'_u(u_x) \rightarrow L$. Thus $f = o(u)$ implies $f'_u \rightarrow 0$, and every bounded f satisfies $f'_u \rightarrow 0$. In particular, $\phi/u \rightarrow \gamma$ implies $\phi'_u \rightarrow \gamma$.*

Proof. Along the curve the exact derivative is

$$\frac{d}{dx}g(u_x) = g'_u(u_x)(1 + i\eta'(x)).$$

The last factor is $1 + o(1)$. If $g'_u \neq 0$, its complete head therefore gives $d[g(u_x)]/dx = c\theta(x)(1 + e(x))$, where $e(x) \rightarrow 0$. If $I(x)$ diverges, integrate this identity. For any $\epsilon > 0$, choose a fixed far point beyond which $|e| \leq \epsilon$. The integral of the earlier error divided by $I(x)$ tends to zero, and the remaining error divided by $I(x)$ has modulus at most ϵ . The initial value divided by $I(x)$ also tends to zero. Hence $g(u_x) = cI(x)(1 + o(1))$. If I converges, the derivative along the curve is absolutely integrable on a far tail, giving a finite limit. A zero derivative gives a constant germ. This proves (1).

For (2), the element f/u belongs to $\mathcal{K}$ and is bounded. By (1) it has a finite limit L . Apply (1) also to the field element f'_u , using its derivative f''_u . If f'_u escaped to infinity, its argument would tend to a fixed direction. After a fixed rotation, the real part of $f'_u(u_x)(1 + i\eta'(x))$ would be at least $|f'_u(u_x)|/2$ on a far tail. Its integral divided by x would tend to infinity, since the average of a positive function tending to infinity tends to infinity. This contradicts $f(u_x) = O(u_x)$ and $u_x/x \rightarrow 1$. Thus $f'_u(u_x)$ has a finite limit, say M ; this includes the case $f''_u = 0$. Integrating $d[f(u_x)]/dx = f'_u(u_x)(1 + i\eta'(x))$ gives $f(u_x)/x \rightarrow M$. Since $u_x/x \rightarrow 1$, we obtain $M = L$. The stated specializations follow.

If a supplied head initially uses a positive scale not declared continuous, on a far zero-free tail replace that scale by $|g(u_x)|/|c_g|$. This positive continuous scale is equivalent to the original scale and preserves the complete head. $\square$

Theorem 8.23. *Under Setup 8.21, define*

$$(8.90) \quad L = \{j \in \mathbb{Z}^q : Db/b = j \cdot DP \text{ for some } b \in K \setminus \{0\}\}, \quad \beta_j = \exp(j \cdot P(T_b)) b/b(T_b),$$

where T_b is any sufficiently far regular point of the witness. Then L is a subgroup, β_j is independent of these choices, and $\beta_{j+k} = \beta_j\beta_k$. The kernel of finite Laurent evaluation over K , with X_i evaluated as $\exp P_i$, is exactly $(X^j - \beta_j : j \in L)$. Every nonzero finite Laurent evaluation has a complete positive head with logarithmic scale $o(h)$.

For $t \in \Lambda$, partition the finite fiber $\tau \cdot n = t$ by $n \sim n'$ if $n - n' \in L$. Choose one n_C in each class and one nonzero common denominator $d_t \in E_{u_t}$ in a later stage clearing all its finitely many β_{n-n_C} . Define the bounded rows before evaluating the variable input by

$$(8.91) \quad R_t\left(\sum_n a_n z^n\right) = \left(d_t \sum_{n \in C} a_n \beta_{n-n_C}\right)_C.$$

Their common kernel equals the entire actual kernel. It is a closed ideal; its quotient in the original infimum norm is a faithful Banach algebra and integral domain. Original coefficient inclusions, later stage inclusions and strict input-radius restrictions induce bounded injective homomorphisms.

For every $0 < \eta < 1$, on a tail fixed before $F = \sum a_n z^n$ and $b \geq 0$,

$$(8.92) \quad \left| \sum_{\tau \cdot n \geq b} a_n(T) w(T)^n \right| \leq C_s \|F\|_{\mathcal{W}_r(E_s)} \exp(-\eta b h(T)).$$

If t_* is the first nonzero row, its complete finite coefficient gives $d_* \neq 0$ and $\Theta_* > 0$ with $|\log \Theta_*| = o(h)$, and the entire value is $d_* \exp(-t_* h) \Theta_*(1 + \epsilon_F)$, where $\epsilon_F \rightarrow 0$. For a real value, d_* is real. No unproved raw tail projection on a quotient is asserted.

The fraction field of the directed union of these quotients is differential, has constant field $\mathbb{C}$, and its nonzero elements have complete ratio heads with logarithmic magnitude $O(h)$. For every fixed raw stage and stricter radius there exist a later stage, one fixed nonzero denominator d , and a bounded linear numerator map G such that $D \text{ Act } F = \text{Act } G(F)/d$. Each fixed finite derivative order has such a representation. All bounded row, whole-tail and numerator maps preserve

absolutely summed original parameter families with their stated operator costs. Iteration at H_{next} additionally requires the entire stage, differentiation and source-domain hypotheses anew, $h = o(\Re H_{\text{next}})$ and the stated new phase conditions.

Proof. We first construct the finite phase relations and the scales that survive finite cancellation. We then define the bounded rows, prove the whole tail and the exact infinite kernel, and finally construct differentiation on complete source stages. All leading-term choices in the argument are for a fixed germ. The bounded row and derivative operators are chosen before the variable source or its parameter coefficients.

Step 1. We choose the real integration coordinate. Every nonzero member of K is eventually nonzero, by its supplied complete head. Its head scale can be chosen continuous: if $a = d_a \theta_a (1 + \epsilon_a)$, replace θ_a on a sufficiently far tail by $\Re(a/d_a) > 0$. The new scale equals $\theta_a(1 + o(1))$, retains its logarithmic bound, and leaves a complete relative error tending to zero. This observation only regularizes the supplied coefficient scales; the finite-sum scales below are obtained by products and positive integrals.

The integral identity for a continuously differentiable complex function follows by telescoping its increments over a partition: uniform differentiability on a compact interval makes the difference between that sum and the Riemann sum of its derivative tend to zero. Thus a zero derivative gives a constant germ, and the constant field of K is $\mathbb{C}$. In particular $DH \neq 0$. Apply the supplied head to this entire derivative, writing $DH = dv(T)(1 + e(T))$, where v is continuous and positive, $d \neq 0$, and $e \rightarrow 0$. The integral $I(T) = \int_{T_0}^T v(t) dt$ must diverge. Otherwise DH would be absolutely integrable on a far tail and H would have a finite limit. Splitting the integrated error at a fixed far point shows that $\int_{T_0}^T v(t)e(t) dt = o(I(T))$. Hence $H(T) = dI(T)(1 + o(1))$. Since $\Im H \rightarrow 0$ and $\Re H \rightarrow +\infty$, the constant d is positive real. It follows that

$$(8.93) \quad h'(T) > 0, \quad \frac{DH(T)}{h'(T)} \rightarrow 1, \quad \nabla = (DH)^{-1}D : K \rightarrow K.$$

The increasing function $h = \Re H$ is now a real coordinate on a far interval. For its inverse $T = T(h)$, the exact chain rule is

$$(8.94) \quad \frac{d}{dh}a(T(h)) = \chi(h)\nabla a(T(h)), \quad \chi(h) = \frac{DH(T(h))}{h'(T(h))} \rightarrow 1.$$

We use ∇ for field operations and h for real integration. Neither h nor χ is asserted to belong to K .

If a continuous positive function $v(h)$ satisfies $|\log v(h)| = o(h)$, then for every $\varepsilon > 0$,

$$(8.95) \quad e^{-\varepsilon h} \leq v(h) \leq e^{\varepsilon h} \quad \text{on a sufficiently far interval.}$$

Its divergent primitive $I(h) = \int_{h_0}^h v(u) du$ has $|\log I| = o(h)$: it is eventually bounded below by a positive constant and above by a fixed constant plus $he^{\varepsilon h}$. If v is integrable at infinity, its positive tail integral $J(h) = \int_h^\infty v(u) du$ also has $|\log J| = o(h)$. It is bounded above, whereas

$$(8.96) \quad J(h) \geq \int_h^{h+1} e^{-\varepsilon u} du \geq e^{-\varepsilon(h+1)}.$$

These comparisons follow directly from the positive Riemann sums. The intervals of length one in this argument are in the coordinate h ; no growth hypothesis on h as a function of T is used.

Step 2. We identify the independent finite relation ideal. If b is a witness for $j \in L$, differentiation of $\exp(j \cdot P)/b$ gives zero. Evaluation at a regular tail point therefore gives

$$(8.97) \quad \beta_j = \exp(j \cdot P) \quad \text{as an equality of actual germs.}$$

This proves independence of the witness and basepoint. Products and inverses of witnesses show that L is a subgroup, and the actual identity gives $\beta_{j+k} = \beta_j \beta_k$. Every proposed generator $X^j - \beta_j$ consequently has zero value.

For a finite Laurent polynomial, partition its exponent set into cosets of L . In each occupied coset choose j_C , and replace $a_j X^j$ by $a_j \beta_{j-j_C} X^{j_C}$. The difference is a Laurent monomial times a generator of the proposed ideal. To finish, suppose there exists a zero evaluation with nonzero coefficients and distinct cosets, and choose one with the least number m of terms. Then $m \geq 2$. Divide by one complete term $a_* \exp(j_* \cdot P)$ and apply ∇ . Its normalized constant term disappears, leaving a relation with at most $m - 1$ distinct-coset terms. Minimality makes each resulting coefficient zero. For a remaining normalized coefficient $b = a_j/a_*$, this says

$$(8.98) \quad \frac{\nabla b}{b} = -(j - j_*) \cdot \nabla P.$$

Multiplying by DH shows that $1/b$ witnesses $j - j_* \in L$, contrary to the distinct-coset choice. The reduced terms are therefore linearly independent over K , proving the asserted Laurent kernel.

Step 3. We obtain complete finite heads after cancellation. We prove the finite Laurent head statement by induction on the number m of nonzero displayed terms. A single term $a \exp(j \cdot P)$ has coefficient d_a , positive scale $\theta_a \exp(j \cdot \Re P)$, and complete error $(1 + \epsilon_a) \exp(ij \cdot \Im P) - 1 \rightarrow 0$. Its logarithmic scale is $o(h)$.

For a nonzero sum L_0 with $m \geq 2$ terms, divide by one complete nonzero term and write $S = L_0/(a_* \exp(j_* \cdot P))$. Then

$$(8.99) \quad \nabla S = \sum_{j \neq j_*} [\nabla(a_j/a_*) + (a_j/a_*)(j - j_*) \cdot \nabla P] \exp((j - j_*) \cdot P).$$

All coefficients are in K , and this sum has at most $m - 1$ terms. Its zero condition is the finite relation ideal just proved. If it is zero, S is a nonzero constant and the one-term case finishes the induction. Otherwise the induction hypothesis and (8.94) give

$$(8.100) \quad \frac{dS}{dh} = dv(h)(1 + e(h)), \quad d \neq 0, \quad v > 0, \quad |\log v| = o(h), \quad e \rightarrow 0.$$

Here the complete error has been multiplied by χ and replaced by $\chi(1 + e_0) - 1$. Thus the imaginary correction in the height has not been discarded.

If $I(h) = \int_{h_0}^h v$ diverges, exact integration gives

$$(8.101) \quad \left| \frac{S(h)}{dI(h)} - 1 \right| \leq \frac{|S(h_0)/d| + \int_{h_0}^h v(u)|e(u)| du}{I(h)} \rightarrow 0.$$

For the limit, first bound $|e|$ by ε past a fixed point. The earlier integral divided by I tends to zero, and the remaining ratio is at most ε . If v is integrable, S has a finite complex limit S_∞ . A nonzero limit gives scale one. A zero limit gives the exact tail expression

$$(8.102) \quad S(h) = -dJ(h)(1 + e_{\text{tail}}(h)), \quad |e_{\text{tail}}(h)| \leq \sup_{u \geq h} |e(u)| \rightarrow 0.$$

The scales I, J , or one have logarithm $o(h)$ by (8.95) and (8.96). Multiplying by the entire term $a_* \exp(j_* \cdot P)$ completes the induction. The products of errors are retained as $(1 + e_1)(1 + e_2) - 1$. This constructs the scale from coefficient scales, positive phase factors and positive integrals, without defining it as $|L_0|$.

Step 4. We bound the rows before the infinite zero test. For any finite B , the inequality $\tau \cdot n \leq B$ bounds each n_i by B/τ_i . Thus Λ is locally finite, each fiber is finite, every nonempty set of actions has a least element, and every action has a next larger action. For fixed s, r, t , the coefficient extraction $F \mapsto a_n$ has norm at most r^{-n} . The finitely many relevant β_j are fractions of elements in the directed coefficient union. Choose a common later stage containing

their numerators and denominators and the image of E_s . The product of the denominators is a nonzero d_t , because each stage embeds in K . This gives the independently chosen rows (8.91).

For example, with the sum norm on their finite target product, an operator bound is

$$(8.103) \quad \|R_t(F)\| \leq \|\iota_{s,u_t}\| \sum_C \sum_{n \in C} r^{-n} \|d_t \beta_{n-n_C}\|_{u_t} \|F\|_{\mathcal{W}_r(E_s)},$$

where ι_{s,u_t} is the specified coefficient inclusion. The maximum product norm is bounded by this sum norm as well. Every quantity in the displayed operator constant precedes F . The finite relation ideal shows that $R_t(F) = 0$ if and only if the complete finite evaluation

$$(8.104) \quad L_t(T) = \sum_{\tau \cdot n = t} a_n(T) \exp(n \cdot P(T))$$

vanishes as a germ. Different finite choices of representatives or clearing denominators have precisely this identical zero condition. The intersection of the row kernels is therefore a closed linear subspace, defined before evaluating the infinite series.

Step 5. We control the entire tail and identify the exact kernel. For fixed $0 < \eta < 1$, the assumptions $P_i = o(h)$ and $\tau_i > 0$ give one tail, before F or b is chosen, on which

$$(8.105) \quad \frac{|w_i(T)|}{r_i} \leq \exp(-\eta \tau_i h(T)) \quad (1 \leq i \leq q).$$

On the supplied coefficient-stage tail,

$$(8.106) \quad |a_n(T) w(T)^n| \leq C_s \|a_n\|_s r^n \exp(-\eta(\tau \cdot n) h(T)).$$

Summing the original absolute norm proves (8.92) for every $b \geq 0$. On each compact subset of the common strict evaluation domain, the coefficient-stage evaluation bound gives locally uniform convergence of the series. Integrating on a small circle within that domain permits termwise coefficient recovery and proves holomorphy of its limit. Absolute products show that actualization is a unital algebra homomorphism.

Suppose all rows vanish as germs. Each finite L_t is holomorphic on the one connected evaluation domain fixed by the original stage and phases. Its vanishing on a tail makes its Taylor series zero near a point of that tail, and overlapping Taylor disks propagate zero throughout that domain. Thus every L_t vanishes on this common domain, even though the tails first used for its denominator witnesses may depend on t . Absolute regrouping by finite fibers now makes the complete series zero.

If a row is nonzero, let t_* be its first action. The corresponding finite evaluation is nonzero and has the complete head $L_{t_*} = d_* \Theta_*(1 + \epsilon_*)$ proved above. Choose b strictly between t_* and the next action, and choose η with $t_*/b < \eta < 1$. All later terms have action at least b , and all earlier fibers vanish identically. Retaining the exact factor $\exp(-t_* H)$ gives

$$(8.107) \quad \begin{aligned} \text{Act } F &= d_* \exp(-t_* h) \Theta_*(1 + \epsilon_F), \\ |\epsilon_F| &\leq |\epsilon_*| + t_* |\Im H| + \frac{C_s \|F\|_{\mathcal{W}_r(E_s)}}{|d_*| \Theta_*} \exp(-(\eta b - t_*) h) \longrightarrow 0. \end{aligned}$$

For the finite contribution, use $|\exp(-it_* \Im H)(1 + \epsilon_*) - 1| \leq |\epsilon_*| + t_* |\Im H|$. The last term in (8.107) tends to zero because $\log \Theta_* = o(h)$ and $\eta b > t_*$. This proves the other kernel inclusion, the complete head, and eventual nonvanishing. If the actual value is real, division by the positive scale $\exp(-t_* h) \Theta_*$ and passage to the limit give $d_* \in \mathbb{R} \setminus \{0\}$.

The independently constructed closed kernel is now the kernel of a multiplicative actualization, and hence an ideal. Lemmas 4.2 and 4.7 give the Banach algebra quotient in its original infimum norm. Every nonzero quotient element is eventually nonzero by (8.107), so the quotient is an integral domain. A constant coefficient evaluates to its original faithful germ. Coefficient inclusions and strict radius restrictions are bounded raw homomorphisms and preserve actual germs; the exact kernel equality makes their quotient maps injective. Taking the infimum over

raw representatives proves their norm bounds. This argument does not descend an arbitrary raw action deletion, which need not preserve the kernel.

Step 6. We justify whole-stage differentiation. Fix a raw stage and $0 < r'_i < r_i$. Set $\lambda_i = r'_i/r_i$. The Euler coefficient map $\mathcal{E}_i F = \sum_n n_i a_n z^n$ satisfies

$$(8.108) \quad \|\mathcal{E}_i F\|_{\mathcal{W}_{r'}(E_s)} \leq M_i \|F\|_{\mathcal{W}_r(E_s)}, \quad M_i = \sup_{k \geq 0} k \lambda_i^k \leq \frac{\lambda_i}{(1 - \lambda_i)^2} < \infty.$$

The other radius ratios are at most one. The geometric-series identity gives the displayed upper bound. Each coefficientwise B_j in (8.88) has its original operator norm on the complete Wiener space, by Proposition 4.20. Differentiating finite polynomial partial sums yields the finite truncations of

$$(8.109) \quad D \text{Act } F = \sum_j c_j \sum_n B_j(a_n) w^n + \sum_{i=1}^q (-\tau_i DH + DP_i) \text{Act}(\mathcal{E}_i F).$$

On a common sufficiently far tail, and on each compact interval there, the original series and this differentiated series converge uniformly by the whole-stage bounds and (8.108). Apply the integral identity to the finite partial sums and pass to their uniformly convergent limits. The integral error is at most the interval length times the uniform error. This proves (8.109) for the entire input.

The multipliers c_j and $-\tau_i DH + DP_i$ form a fixed finite list in K . In a common later coefficient stage, choose a product d of their nonzero denominators, so that their products with d belong to that stage. Include all coefficient source and target stages by their prescribed bounded maps. Multiplying (8.109) by d then gives the actual value of a bounded linear map G on the complete raw input, with

$$(8.110) \quad D \text{Act } F = \text{Act } G(F)/d.$$

The bound for G is the finite sum of the multiplication bounds of dc_j and $d(-\tau_i DH + DP_i)$, the stage inclusion bounds, the norms of the B_j , and the constants M_i . All are fixed before F . No bounded inverse for d in a Banach algebra is used.

If a raw representative has zero actual value, its derivative also vanishes, and the exact kernel statement already proved makes the numerator $G(F)$ zero in the later quotient. Thus this differentiation respects representatives. It puts the derivative of every quotient element in the fraction field of their directed union. The quotient rule proves closure of that field under D ; the zero-derivative integral identity gives constant field $\mathbb{C}$. For each further prescribed derivative, choose another strict radius between the current radius and the final target, apply (8.109), and clear the finitely many new fixed multipliers from the product rule on the previous denominator. Induction gives a bounded linear numerator for every fixed finite derivative order. A product denominator accommodates a fixed finite list of orders.

Every nonzero fraction is a ratio of two nonzero quotient elements in a common later stage. Applying (8.107) to its numerator and denominator gives the exact relative error $(1 + \epsilon_{\text{num}})/(1 + \epsilon_{\text{den}}) - 1 \rightarrow 0$ and a positive ratio scale with logarithm

$$(8.111) \quad (t_{\text{den}} - t_{\text{num}})h + o(h).$$

In particular its logarithmic magnitude is $O(h)$, and the original coefficient field embeds through its numerator and denominator representations.

Step 7. We retain original parameter families through iteration. For any fixed bounded row or numerator map T and an original absolutely summed family, the whole bound is

$$(8.112) \quad \sum_{\alpha} \rho^{\alpha} \|T(F_{\alpha})\| \leq \|T\| \sum_{\alpha} \rho^{\alpha} \|F_{\alpha}\|.$$

The inequality holds for every omitted set of parameter indices. Summing the positive majorant (8.106) over the original parameter indices proves the whole-tail bound for the complete family

as well. The first nonzero row is selected for each fixed parameter value. If $h = o(\Re H_{\text{next}})$, the ratio scales in (8.111) become subexponential at the next height, and the differential field condition has already been proved. The next stage bounds, source domains, differentiation maps and phase conditions remain the additional hypotheses stated in the theorem. $\square$

Definition 8.24. A slowly widening horn is a domain

$$(8.113) \quad H_w = \{x + iy : x > x_0, |y| < w(x)\},$$

where $x_0 > 1$ and w is positive, continuous, nondecreasing and unbounded, with $w(x) \leq \sqrt{x}$. A smaller such width and a farther real threshold are allowed restrictions. Each fixed horizontal strip about the positive tail is eventually contained in H_w .

Theorem 8.25. *In Setup 8.21, take $T = x$ and $H(x) = x$. Suppose every finite list of elements of K and one fixed stage has holomorphic representatives on a slowly widening horn about the positive axis, with all its fixed denominators nonzero there and the whole stage bounded in its original norm. Suppose also that, for every fixed nonzero $a \in K$ and $A > 0$,*

$$(8.114) \quad a(x + iy)/a(x) \rightarrow 1 \quad \text{uniformly for } |y| \leq A.$$

Then every nonzero member of $K(\exp P_1, \dots, \exp P_q)$ has this horizontal ratio property, has either a finite complex limit or magnitude tending to infinity, and satisfies $f'/f \rightarrow 0$. Every nonzero collected finite row in Theorem 8.23 has its complete head on fixed horizontal strips after retaining its exact factor $\exp(-tx)$.

For each entire original stage $\mathcal{W}_r(E_s)$, one slowly widening horn can be fixed before the input and its degrees, on which actualization is bounded in the original norm. For each $0 < \eta < 1$ on a farther fixed tail, its complete action tail obeys

$$(8.115) \quad \left| \sum_{\tau \cdot n \geq B} a_n(z) \exp(-\tau \cdot n z + n \cdot P(z)) \right| \leq C \|F\|_{\mathcal{W}_r(E_s)} \exp(-\eta B \Re z), \quad B \geq 0.$$

If all rows through B vanish, the bound in (8.115) holds for the entire value and descends by original quotient norm infima. These conclusions supply neither the source-domain input (8.114) nor the positive numerical reserve of a subsequently adjoined native root.

Proof. We prove the real limit alternative before using horizontal continuation, then handle finite cancellation by differentiation. The horn for the complete Wiener stage is chosen only from its supplied domain and the fixed finite phase list.

Step 1. We establish limits in the finite exponential field. Write $\mathcal{K} = K(\exp P_1, \dots, \exp P_q)$. Every member of $\mathcal{K}$ is a ratio of finite Laurent evaluations. The finite-head part of Theorem 8.23 therefore gives every nonzero member a complete head $d\theta(1 + e)$ with $|\log \theta| = o(x)$. The field is differential, since $D \exp P_i = (DP_i) \exp P_i$. Thus every nonzero derivative in $\mathcal{K}$ also has a complete head with asymptotically fixed complex argument.

Since $H = x$ belongs to K , the field $\mathcal{K}$ contains the coordinate. Lemma 8.22(1), with $\eta = 0$, gives the finite-limit or directional-escape alternative for every member of $\mathcal{K}$, including logarithmic derivatives.

For $f \neq 0$ choose a continuous logarithm on the real tail from its complete head:

$$(8.116) \quad \log f = \text{Log } d + \log \theta + \text{Log}_0(1 + e) = o(x).$$

Its imaginary part remains bounded. The element $f'/f \in \mathcal{K}$ has the limit alternative just proved. A nonzero finite limit c would give $\log f = cx + o(x)$ after integration. If its magnitude tended to infinity, its complete head would give $f'/f = cv(1 + o(1))$ with $v \rightarrow \infty$, and hence $\log f = c \int v(1 + o(1))$ with $|\log f|/x \rightarrow \infty$. Both contradict (8.116). Consequently $f'/f \rightarrow 0$.

Since $P_i = o(x)$, Lemma 8.22(2) gives $P'_i \rightarrow 0$. For a nonzero P'_i , the original horizontal hypothesis (8.114) makes this convergence uniform on every fixed strip; a zero derivative is an exact zero germ. Vertical integration now yields

$$(8.117) \quad \sup_{|y| \leq A} |P_i(x + iy) - P_i(x)| \rightarrow 0.$$

Together with the original coefficient ratio property, this proves the horizontal ratio assertion for every one-term expression $a \exp(j \cdot P)$.

Step 2. We control finite cancellation on a strip. Induct on the number of nonzero terms in a finite Laurent evaluation L . Divide by one complete term and set $S = L/(a_* \exp(j_* \cdot P))$. Its exact derivative is

$$(8.118) \quad S' = \sum_{j \neq j_*} [(a_j/a_*)' + (a_j/a_*)(j - j_*) \cdot P'] \exp((j - j_*) \cdot P).$$

This expression has one fewer possible nonzero term. If $S' = 0$, then S is a nonzero constant and the one-term result applies. Otherwise the induction hypothesis gives $S'(x + iy)/S'(x) \rightarrow 1$ uniformly for $|y| \leq A$. Every fixed coefficient denominator in (8.118) is nonzero on a far strip by the supplied source-domain hypothesis. The exact vertical integral consequently gives

$$(8.119) \quad \sup_{|y| \leq A} \left| \frac{S(x + iy)}{S(x)} - 1 \right| \leq A \left| \frac{S'(x)}{S(x)} \right| \sup_{|u| \leq A} \left| \frac{S'(x + iu)}{S'(x)} \right| \rightarrow 0.$$

The first factor tends to zero by the already proved logarithmic derivative result for $S \in \mathcal{K}$. Multiplication by the entire chosen term proves the ratio property for L , including all cancellations. A quotient of two such nonzero sums has the ratio property too, since the denominator is uniformly nonzero on each far fixed strip. Any fixed finite list of these functions therefore has a common zero-free widening horn after diagonalizing the strip thresholds; this choice concerns that finite list and does not assert a uniform lower bound for variable coefficients.

Step 3. We choose one horn for the whole original stage. Fix E_s, r and the finite list of phases. Start with the original horn supplied for this stage and the phases and their fixed denominators. Its complete stage evaluation has a bound $|a(z)| \leq C\|a\|_s$ independent of a . The bound (8.117) and $P_i(x) = o(x)$ give $P_i(z)/z \rightarrow 0$ uniformly on every fixed strip. For each integer $N \geq 2$, choose strictly increasing thresholds X_N so that, for $x \geq X_N$, the original horn contains $|y| \leq 2N$, $X_N \geq N^2$, and

$$(8.120) \quad \max_i \sup_{|y| \leq 2N} \frac{|P_i(x + iy)|}{|x + iy|} \leq \frac{1}{N}.$$

Take a continuous nondecreasing width with $w(X_N) = N - 1$, linear between these thresholds. On $[X_N, X_{N+1}]$ its value is at most N , so its horn lies in the original horn, satisfies $w(x) \leq \sqrt{x}$, and has $P_i(z) = o(|z|)$ uniformly at infinity. Also $|z|/\Re z \rightarrow 1$ there. This single domain is fixed before the input F and all its degrees. Restrict its initial threshold once so that the original radius inequalities hold.

For any fixed $0 < \eta < 1$, first reserve an exponential margin strictly between η and one. On a farther part of this horn, the uniform phase bound and then the fixed constants $\log r_i$ give

$$(8.121) \quad \frac{|\exp(-\tau_i z + P_i(z))|}{r_i} \leq \exp(-\eta \tau_i \Re z) \quad (1 \leq i \leq q).$$

Multiplying the inequalities and summing the original absolute coefficient norm proves (8.115), simultaneously for every input F and every $B \geq 0$. The case $B = 0$ gives bounded actualization on the horn. The series is locally uniformly convergent there, and is holomorphic by coefficient integration on small circles. No individual row norms are summed in this estimate.

If all rows through B vanish, their finite evaluations are zero as germs by Theorem 8.23. They are holomorphic on the one connected original horn, so they are identically zero on it. Removing

these finitely many action fibers leaves the whole actual value, with all remaining actions strictly greater than B . The tail estimate therefore bounds the entire value. Every representative of its quotient class has the identical row-zero property and actual value. Taking the infimum over those original raw norms proves the asserted quotient bound; no arbitrary raw tail operator is descended.

Step 4. We obtain the complete first head on fixed strips. For a first nonzero action t_* , its finite evaluation has both the complete real-tail head and the horizontal ratio property:

$$(8.122) \quad L_{t_*}(x + iy) = d_* \Theta_*(x)(1 + o(1)) \quad \text{uniformly for } |y| \leq A.$$

Choose b between t_* and the next action and choose $\eta b > t_*$. The remaining entire tail, divided by the modulus of $\exp(-t_* z)L_{t_*}(x)$, is bounded by

$$(8.123) \quad \frac{C \|F\|_{\mathcal{W}_r(E_s)}}{|L_{t_*}(x)|} \exp(-(\eta b - t_*)x) \longrightarrow 0.$$

The convergence is uniform on each fixed strip, since $\log |L_{t_*}(x)| = o(x)$. The factor $\exp(-t_* z)$ has been kept exact, including its imaginary phase. The whole-family versions follow by summing these bounded source and tail maps in the original absolute parameter norms. Every domain used here was obtained by restriction of the supplied holomorphic horn; no continuation or positive root reserve was deduced from a real head alone. $\square$

9. INITIAL SOURCE SPACES AND THE FIRST COMPARISON LEVEL

We retain complete source coefficients before constructing their comparison rows. The initial physical words use two specified alphabets. Native tensor reconstruction is a separate assertion over a supplied lower presentation; its proof uses the quantitative cofinal bound on the original horn. All completions retain their original input norms.

9.1. Original initial alphabets.

Lemma 9.1. *Let μ be a complex measure on $[0, \infty)$ supported in $[g_0, \infty)$ for some $g_0 > 0$, and suppose $M = \int e^{-A\xi} d|\mu|(\xi) < \infty$. Set $f(Y) = \int e^{-Y\xi} d\mu(\xi)$ on $\Re Y > A$. Fix $R > 0$ and $h > RM$. In the measure norm at weight $A + h$, the complete series*

$$(9.1) \quad \mu_\tau^- = - \sum_{n \geq 1} \frac{\tau^n}{n!} \xi^{n-1} \mu^{*n} \quad (|\tau| \leq R)$$

converges, with bounds

$$(9.2) \quad \|\mu_\tau^-\|_{A+h} \leq -h \log(1 - RM/h), \quad \left\| \sum_{n > N} -\frac{\tau^n}{n!} \xi^{n-1} \mu^{*n} \right\|_{A+h} \leq \frac{h(RM/h)^{N+1}}{(N+1)(1 - RM/h)}.$$

On a sufficiently far right half-plane, its Laplace transform is $(\text{id} + \tau f)^{-1}(Y) - Y$, with the inverse branch tending to Y . The frequency power in (9.1) acts on the total frequency of the ordered convolution.

Proof. For $t \geq 0$, the exponential series gives $t^{n-1}e^{-ht} \leq (n-1)!h^{-(n-1)}$. The convolution norm at weight A is at most M^n . Thus the norm of the n th term in (9.1) is at most $R^n M^n / (nh^{n-1})$. The logarithmic series and its geometric tail give (9.2) before any Laplace integral is taken.

The positive support gap makes f and f' uniformly small on fixed disks centered sufficiently far to the right. Therefore $z = Y - \tau f(z)$ is a contraction on such a disk, uniformly for τ in a slightly larger disk than $|\tau| \leq R$, and defines the required analytic inverse. For completeness, its coefficient of τ^n can be read from

$$\frac{1}{2\pi i} \int \frac{z(1 + \tau f'(z))}{z + \tau f(z) - Y} dz,$$

where the circle encloses the unique small solution. Expand the denominator in its uniformly convergent geometric series. Integrating the derivative of $zf(z)^n/(z-Y)^n$ around the circle reduces the coefficient to

$$\frac{(-1)^n}{n} \frac{1}{2\pi i} \int \frac{f(z)^n}{(z-Y)^n} dz = \frac{(-1)^n}{n!} \partial_Y^{n-1}(f(Y)^n).$$

The full convolution represents f^n . Its $(n-1)$ st derivative multiplies that convolution by $(-\xi)^{n-1}$, giving exactly (9.1). Absolute convergence identifies the whole series with the inverse, including the tail. $\square$

Definition 9.2. Fix finitely many original canonical endpoint passages, represented by common- q source-equipped references and the full weak replays of any other compatible analytic labels. Retain their complete normalizer, graph, weak-gauge and true-inverse recipes, all original input norms and strict source, image and inverse domains. Arbitrary free banks on other rays may be retained as parallel original implementations, but are not included in the tensor to which common- q comparison applies. In their native coordinates write

$$(9.3) \quad P_i((\beta_i/Y)^{1/p_i}) = k_i(e^{-B_i(Y)}), \quad B_i(Y) = Y - a_i \log Y - \log c_i - \log K_i(Y),$$

where p_i is a positive integer, $\beta_i, c_i > 0$, $a_i \in \mathbb{R}$, K_i is the whole original coefficient unit tending to one, and $k_i(0) = 0$, $k'_i(0) = 1$. Retain the selected physical paths, lateral determinations, root and logarithmic lifts, actual matching, and any same-preparation recutting witnesses.

A finite weak core is a literal finite composable chain of original prepared saddle passages, their correlated true inverses and positive analytic weak adapters, with every original source and section datum. It contains no additional saddle-node and no previously completed child source.

The first initial alphabet consists of same-order blocks $P_b^{-1}JP_a$ with $p_a = p_b$, their true inverses, positive analytic strong adapters, and the following original prepared events:

$$(9.4) \quad \begin{aligned} E_i(s) &= F_i^{-1}(F_i(s) + \text{Lap } \mu_i(F_i(s))), \\ F_i(s) &= b_i s^{-r_i} + \sum_{v,h} c_{i,v,h} s^{-v} L^h + \lambda_i L + g_i(V_i(s)), \quad L = -\log s. \end{aligned}$$

Here $b_i, r_i > 0$, the growing coefficients and λ_i are real, the sum is finite, and its indices satisfy either $0 < v < r_i$, $h \in \mathbb{N}$, or $v = 0$, $h \geq 2$. The function g_i is the whole supplied convergent germ in a finite tuple of small positive-power/logarithmic variables, closed downward in logarithmic degree. Write $\text{Lap } \mu(T) = \int_0^\infty e^{-T\xi} d\mu(\xi)$. Each μ_i is the complete normalized density constructed from an original finite forward prepared core with its supplied terminal analytic adapter, or from an original convergent weak-unit recipe in $z^\gamma(-\log z)^h$, $\gamma > 0$, after the specified slowdown $T \mapsto T - \kappa \log T$, $\kappa > \max(h/\gamma)$. Retain its original density construction, positive birth gap, powers and cutoffs. An arbitrary measure having only a positive support gap is not an additional input. A negative occurrence retains the full ordered inverse source of Lemma 9.1. The weak map J in a paired block is either an original positive analytic weak biholomorphism or a finite weak core.

The second initial alphabet consists of the original typed blocks

$$(9.5) \quad S_b P_b^{-1} J P_a S_a^{-1},$$

where the S are original analytic section maps, J is a finite weak core, and p_a, p_b may differ. Its strong connectors and their supplied actual inverse branches have the form

$$(9.6) \quad R(s) = cs^\alpha U(s^{\gamma_1}, \dots, s^{\gamma_l}), \quad c, \alpha, \gamma_j > 0, \quad U(0) = 1,$$

with U the whole original convergent real unit and $l \geq 0$.

An initial word is a fixed finite literally composable signed word in one chosen alphabet, with all forward physical witnesses retained at negative occurrences. The empty word is allowed. Each family retains its original strict parameter polydisks and full input norm. Structural exponents and numerical action labels are fixed before source-window estimates; numerical action collection

is made at the fixed physical specialization. Any varying leading coefficient is retained in its whole clock.

Theorem 9.3. *Fix one original initial presentation and one of the two alphabets in Definition 9.2. Starting with a finite initial prefix, prescribe a finite literally composable sequence of positive analytic or supplied convergent real-power strong adapters, original prepared strong passages and their correlated true inverses, and paired blocks $P_b^{-1}JP_a$ with J a finite weak core. Retain the original sections, branches, matching witnesses, complete source norms and strict inverse domains.*

The generated unphased Wiener presentations and finite exterior modules have bounded whole analytic realizations and independently constructed complete comparison rows. At each positive order the rows collect every finite total-action fiber with its complete lower-order coefficient. Their common kernel equals the analytic kernel. For each algebra presentation this is a closed ideal and its quotient is faithful in the infimum of the original input norm. Every nonzero real realized element has a positive scale and complete remainder in the sense of Definition 2.3.

The construction includes the original ordered inverse sources, all product derivatives of total-action cuts, and the whole cutoff-independent source remainder estimates.

9.2. Complete root coefficients and weak variables.

Set-up 9.4. Fix an integer $P \geq 1$, numbers $g, r > 0$, a unit complex number q with $\Re q > 0$, and a determination of $\log q$ with imaginary part in $(-\pi/2, \pi/2)$. Fix $0 < \vartheta < \pi/2 - |\arg q|$ and put

$$(9.7) \quad D = \{|\zeta| < r\} \cup \{\zeta \neq 0 : |\arg \zeta| < \vartheta/P\}.$$

The root algebra B consists of pairs $v = (v_0, \phi)$, where $\phi \in \text{Hol}(D)$, $\phi(0) = 0$, and

$$(9.8) \quad \|v\|_B = |v_0| + 8Pg^{-1/P} \sup_{\zeta \in D \setminus \{0\}} e^{-g|\zeta|^P} (1 + g|\zeta|^P)^{1+1/P} \frac{|\phi(\zeta)|}{|\zeta|} < \infty.$$

The product of (v_0, ϕ) and (w_0, ψ) is $(v_0w_0, v_0\psi + w_0\phi + \phi * \psi)$, where

$$(9.9) \quad (\phi * \psi)(\zeta) = \int_0^1 \phi(\zeta t^{1/P}) \psi(\zeta(1-t)^{1/P}) \frac{dt}{t(1-t)}.$$

For $\Re(qZ) > g$, define

$$(9.10) \quad S(v)(Z) = v_0 + \int_0^\infty e^{-qZ\xi} \phi(\xi^{1/P}) \frac{d\xi}{\xi}.$$

Write $\Gamma_E(u) = \int_0^\infty e^{-t} t^{u-1} dt$ for $u > 0$. The independent coefficient maps are

$$(9.11) \quad b_0(v) = v_0, \quad b_j(v) = \Gamma_E(j/P) q^{-j/P} [\zeta^j] \phi \quad (j \geq 1).$$

All root and logarithmic determinations are fixed in these formulas. When the bank represents a selected canonical normalizer, its identity with that normalizer is the physical matching supplied by Theorem 6.1.

The presentation in Setup 9.4 is the complete root algebra of Proposition 8.19. In particular its product is bounded in the displayed norm. Its source density is $\phi(\xi^{1/P})/\xi$; the scalar v_0 is separate. The following estimates specify the part of its coefficient comparison that we use.

Lemma 9.5. *Each b_j is bounded on B , and $\bigcap_{j \geq 0} \ker b_j = \{0\}$. For every fixed $n \geq 0$ and every fixed closed angular subsector of $\Re(qZ) > 0$, the complete remainder satisfies*

$$(9.12) \quad \left| S(v)(Z) - \sum_{j=0}^n b_j(v) Z^{-j/P} \right| \leq C_n \|v\|_B |Z|^{-(n+1)/P} \quad (|Z| \geq Z_n).$$

Here the angular subsector has $\Re(qZ) \geq c|Z|$ for some $c > 0$, and C_n, Z_n are independent of v .

Proof. Choose $0 < r_0 < \rho < r$ and put $\xi_0 = r_0^P$. On the circle $|\zeta| = \rho$, the weight in (9.8) has a positive minimum. Cauchy's coefficient formula therefore gives $|\zeta^j \phi| \leq C \rho^{-j} \|v\|_B$. This proves boundedness. If all $b_j(v)$ vanish, $v_0 = 0$ and the Taylor series of ϕ at zero vanishes. Analytic continuation on connected D gives $\phi = 0$.

The complete Taylor remainder on $[0, \xi_0]$ obeys

$$\left| \phi(\xi^{1/P}) - \sum_{j=1}^n [\zeta^j] \phi \xi^{j/P} \right| \leq C_n \|v\|_B \xi^{(n+1)/P}.$$

Its integral against $e^{-qZ\xi} d\xi/\xi$ is bounded by $C_n \|v\|_B \Gamma_E((n+1)/P) (c|Z|)^{-(n+1)/P}$. The defining norm also gives

$$\int_0^\infty e^{-g\xi} |\phi(\xi^{1/P})| \frac{d\xi}{\xi} \leq \frac{1}{8} \|v\|_B,$$

because the integral of $\xi^{1/P-1} (1+g\xi)^{-1-1/P}$ is $Pg^{-1/P}$. Thus the entire original minor above ξ_0 contributes at most $\|v\|_B e^{-(c|Z|-g)\xi_0}/8$. Each of the finitely many subtracted monomials has an exponentially small tail above ξ_0 . Finally,

$$\int_0^\infty e^{-qZ\xi} \xi^{j/P-1} d\xi = \Gamma_E(j/P) (qZ)^{-j/P}.$$

For positive qZ this follows by a change of variable, and then throughout the right half-plane by holomorphy. Combining the small-carrier error, the full minor tail, and the polynomial tails proves (9.12). $\square$

Lemma 9.6. *Use the bank of Setup 9.4. Fix $0 < r_0 < \rho/2 < r/2$. For a real angle α define the initial-disk representative*

$$(9.13) \quad I_\alpha(v; \kappa) = v_0 + P \int_0^{r_0 e^{i\alpha/P}} e^{-q e^\kappa \zeta^P} \phi(\zeta) \frac{d\zeta}{\zeta}.$$

Restrict $\Im \kappa$ to a fixed buffered interval on which $|\arg q + \Im \kappa + \alpha| < \pi/2$. If $|\alpha| < \vartheta$, the complete ray integral along $\arg \xi = \alpha$ is also available. Both representatives are bounded by $C \|v\|_B$ on a far tail. The complete ray representatives continue the original realization to every strict subinterval of $|\arg q + \Im \kappa| < \pi/2 + \vartheta$.

The coefficients of every representative are $b_j(v) e^{-j\kappa/P}$, with the original logarithmic lift. There exist constants $C, D, c, \epsilon_0 > 0$, fixed before n and v , such that

$$(9.14) \quad \left| I_\alpha(v; \kappa) - \sum_{j=0}^n b_j(v) e^{-j\kappa/P} \right| \leq C \|v\|_B \left[D^{n+1} \Gamma_E((n+1)/P) e^{-(n+1)\Re \kappa/P} + e^{-c e^{\Re \kappa}} \right]$$

whenever $0 \leq n \leq \epsilon_0 e^{\Re \kappa}$. Estimate (9.14) also holds for a complete ray representative on its buffered domain. Two initial-disk representatives whose joining angles stay in one strict damping interval differ by at most $C \|v\|_B e^{-c e^{\Re \kappa}}$.

In particular, if n lies between fixed positive multiples of N and $e^{\Re \kappa} \geq K_0 N$ for a sufficiently large fixed K_0 , the whole error is at most $C \|v\|_B e^{-cN}$.

Proof. The coefficient integral $\phi_j = (2\pi i)^{-1} \int_{|\zeta|=\rho} \phi(\zeta) \zeta^{-j-1} d\zeta$ and (9.8) give

$$(9.15) \quad |b_j(v)| \leq C \rho^{-j} \Gamma_E(j/P) \|v\|_B \leq C D^j j^{j/P} \|v\|_B, \quad j \geq 1.$$

For the second estimate, the maximum of $t^u e^{-t/2}$ gives $\Gamma_E(1+u) \leq 2(2u/e)^u$ for $u \geq 1$. The finitely many smaller arguments are absorbed in C .

On each admitted ray the damping satisfies $\Re(q e^\kappa e^{i\alpha}) \geq c_0 e^{\Re \kappa}$ with fixed $c_0 > 0$. The original minor envelope is integrable at zero and is dominated at infinity by $\exp[-(c_0 e^{\Re \kappa} - g)|\xi|]$. Deform one complete ray into another through an admitted angular interval. The small joining arc tends to zero as its radius to the power $1/P$; the large arc tends to zero by this whole

envelope. This proves continuation of the full original integral. For two initial-disk integrals the joining arc remains inside the original root disk and at radius r_0 ; its full integrand gives the asserted exponential difference.

For a monomial ζ^j , extension of its ray integral to infinity gives $\Gamma_E(j/P)q^{-j/P}e^{-j\kappa/P}$. The factor $e^{i\alpha j/P}$ cancels the angle in $\log q + \kappa + i\alpha$. Thus all coefficients retain their original logarithmic determinations.

Put $\xi_0 = r_0^P$. The complete Cauchy remainder below ξ_0 has absolute value at most $C\|v\|_B \rho^{-n-1}|\xi|^{(n+1)/P}$. Its integral gives the gamma term in (9.14). The original minor above ξ_0 has the full exponential bound just established. The tails of every subtracted monomial must also be kept. The function $t^{-1} \log_+(t^{1/P}/\rho)$ is bounded for $t \geq \xi_0$. Consequently, if $n \leq \epsilon_0 e^{\Re \kappa}$ and ϵ_0 is fixed sufficiently small, one part of the damping absorbs $\max\{1, (t^{1/P}/\rho)^n\}$. Another part absorbs the number $n+1$ of terms. The remaining integral is bounded by $C\|v\|_B e^{-ce^{\Re \kappa}}$, uniformly in n . This also pays the polynomial tails for the initial-disk representative, which has no original high-carrier integral. The gamma estimate in (9.15), with K_0 large enough for the fixed multiples of N , proves the last assertion. $\square$

Definition 9.7. Fix $m \geq 1$, $c_i, R_i > 0$ and $a_i \in \mathbb{R}$ for $1 \leq i \leq m$. The weak presentation is the Banach algebra $\mathcal{P} = \mathcal{W}_R(B)$ of Definition 4.1, with independent variables $V_1, \dots, V_m$ and realization

$$(9.16) \quad F = \sum_{\nu \in \mathbb{N}^m} v_\nu V^\nu, \quad \text{Act } F(Z) = \sum_{\nu \in \mathbb{N}^m} S(v_\nu)(Z) c^\nu e^{-|\nu|Z} Z^{a \cdot \nu}.$$

The realization domain is the component containing a far positive interval of $\{\Re(qZ) > g, |c_i e^{-Z} Z^{a_i}| < R_i \text{ for all } i\}$, with the logarithm continued from the positive axis.

For each $n \in \mathbb{N}$, partition the finite set $\{\nu : |\nu| = n\}$ by the equivalence relation $\nu \sim \nu'$ if $a \cdot \nu - a \cdot \nu' \in P^{-1}\mathbb{Z}$. For a class E put $d_E = \max_{\nu \in E} a \cdot \nu$ and $l_\nu = P(d_E - a \cdot \nu) \in \mathbb{N}$. Define

$$(9.17) \quad u_0 = (1, 0), \quad u_l = \left(0, \frac{q^{l/P} \zeta^l}{\Gamma_E(l/P)}\right) \quad (l \geq 1), \quad g_{n,E}(F) = \sum_{\nu \in E} c^\nu u_{l_\nu} v_\nu.$$

The comparison map $\mathcal{R}$ consists of all scalar rows $b_j(g_{n,E}(F))$, indexed by (n, E, j) .

Proposition 9.8. For the presentation in Definition 9.7, the rows are bounded and are defined before realization. Their common kernel is $\ker \text{Act}$, a closed ideal of $\mathcal{P}$. The quotient $\mathcal{A} = \mathcal{P}/\ker \mathcal{R}$ is a faithful Banach algebra in the infimum of its original Wiener norm.

If $\mathcal{R}F \neq 0$, let n_* be the least degree with a nonzero collected coefficient. For each nonzero $g_{n_*,E}(F)$ let j_E be its first nonzero root degree. The number $d_* = \max_E(d_E - j_E/P)$ is attained at one class. With its coefficient $a_* = b_{j_E}(g_{n_*,E}(F)) \neq 0$, the entire value has

$$(9.18) \quad \text{Act } F(Z) = a_* e^{-n_* Z} Z^{d_*} (1 + o(1)) \quad (Z \rightarrow +\infty).$$

The omitted weak degrees satisfy

$$(9.19) \quad |\text{Tail}_{>n_*} \text{Act } F(Z)| \leq \|F\|_{\mathcal{P}} \epsilon(Z)^{n_*+1}, \quad \epsilon(Z) = \max_i \frac{c_i}{R_i} e^{-Z} Z^{\max_i a_i} < 1.$$

If $\text{Act } F$ is real on the positive axis, then a_* is real.

Proof. The root norm gives $|S(v)(Z)| \leq \|v\|_B$ on the realization domain. Proposition 4.20 consequently gives a bounded holomorphic realization of the whole series in (9.16). Each u_l belongs to B and has realization $Z^{-l/P}$, by the gamma integral. At fixed n the exact degree contribution is therefore

$$(9.20) \quad e^{-nZ} \sum_E Z^{d_E} S(g_{n,E}(F))(Z).$$

This is a finite sum of complete root sources. Coefficient extraction $F \mapsto v_\nu$ has norm at most $R^{-\nu}$, so each row in the definition of $\mathcal{R}$ is bounded.

If all rows vanish, Lemma 9.5 makes every $g_{n,E}(F)$ zero. Hence every whole degree in (9.20) vanishes, and absolute convergence gives $\text{Act } F = 0$.

Suppose some row is nonzero. At the first surviving weak degree, Lemma 9.5 gives the complete first term of every nonzero $g_{n*,E}(F)$. Distinct classes cannot have an equal exponent $d_E - j/P$: such an equality would put their representatives in one equivalence class. Thus d_* is attained once. The finitely many other classes and the root remainder in the selected class are smaller by a positive power of Z . For $Z \geq 1$, each $|c_i e^{-Z} Z^{a_i}|/R_i$ is at most $\epsilon(Z)$. Summing the entire original Wiener norm over $|\nu| > n_*$ proves (9.19). Its ratio to $e^{-n_* Z} Z^{d_*}$ tends to zero, since it is bounded by a fixed power of Z times e^{-Z} . This proves (9.18), and excludes an identically zero realization when a row survives.

The row kernel is closed by boundedness. The root product and the absolute Cauchy product make Act an algebra homomorphism; its kernel, now identified independently, is an ideal. The quotient is complete and faithful, with bounds obtained by taking infima over original representatives. Finally the scale in (9.18) is positive. Division by that scale shows that a_* is real for a real realization. $\square$

9.3. Translated inverse equations and total-action cuts.

Lemma 9.9. *Let k be analytic near zero, $k(0) = 0$, $k'(0) = 1$, and let $E(v) = -\log(k(v)/v)$ with its logarithm tending to zero. There exist $T_v, T_z, M_h > 0$ and a bounded holomorphic function $h(v, z)$ on $|v| < T_v$, $|z| < T_z$ such that*

$$(9.21) \quad h + E(v e^{-h}) - E(v) = z, \quad h(v, 0) = 0, \quad |h(v, z)| \leq M_h.$$

For $0 < r_v < T_v$ and $0 < r_z < T_z$, its entire Wiener norm obeys

$$(9.22) \quad \|h\|_{r_v, r_z} \leq M_h (1 - r_v/T_v)^{-1} \frac{r_z/T_z}{1 - r_z/T_z}.$$

If B_1 is an actual injective holomorphic height on a specified domain, and $B_1(Y) + h(v, z)$ lies in its inverse domain, define $I(Y; v, z) = B_1^{-1}(B_1(Y) + h(v, z)) - Y$. At $v = e^{-B_1(Y)}$ this is the exact increment solving

$$(9.23) \quad B_1(Y + I) + E(e^{-B_1(Y+I)}) - B_1(Y) - E(e^{-B_1(Y)}) = z.$$

Proof. Fix a small closed disk $|h| \leq H$. Choose T_v so small that $|v e^{-h} E'(v e^{-h})| \leq 1/4$ on this disk and the chosen v disk. The map $h \mapsto z - E(v e^{-h}) + E(v)$ has Lipschitz constant at most $1/4$. For sufficiently small T_z it maps $|h| \leq H$ into itself. Its uniformly convergent Picard sequence gives a bounded holomorphic solution. Uniqueness at $z = 0$ gives $h(v, 0) = 0$. Cauchy's coefficient bound on circles approaching the defining radii, followed by summation over every v degree and positive z degree, proves (9.22). Finally $B_1(Y + I) = B_1(Y) + h$ and $e^{-B_1(Y+I)} = v e^{-h}$. Substitution in (9.21) proves (9.23) on the stipulated inverse domain. $\square$

The scalar estimate in Lemma 9.9 permits substitution in a Banach algebra only after the complete source norms of v and z lie inside r_v and r_z . A source for the height inverse B_1^{-1} is additional input. Its scalar inverse identity alone does not provide a bound for that source.

Lemma 9.10. *Let $a = (a_1, \dots, a_d)$ be real carrier coordinates, $I \in \mathbb{N}^d$, $\lambda_j > 0$, and $\Omega(a) = \sum_j \lambda_j a_j$. For a distribution μ , a smooth function χ , and $\zeta \in \mathbb{R}$, multiplication of $\partial_a^I \mu$ by the total-action cutoff is*

$$(9.24) \quad \chi(\zeta + \Omega) \partial_a^I \mu = \sum_{J \leq I} (-1)^{|J|} \binom{I}{J} \partial_a^{I-J} (\lambda^J \chi^{(|J|)}(\zeta + \Omega) \mu).$$

Formula (9.24) holds before carrier addition. Suppose $\delta_* > 0$, $\sum_j I_j \leq \Omega/\delta_*$ on $\text{supp } \mu$, and χ is constant outside (k, k^+) . Every nonzero term on the right with $J \neq 0$ then has $|J| \leq \lfloor (k^+ + |\zeta|)/\delta_* \rfloor$.

Proof. Pair both sides with a compactly supported smooth test function. The Leibniz formula for $\partial_a^I(\chi(\zeta + \Omega)\varphi)$ gives (9.24), including its sign. A derivative of positive order is supported where $\Omega \leq k^+ + |\zeta|$. The derivative linkage then gives the asserted bound. The identity applies to an entire source integral whenever its complete lowering majorant is integrable; dominated convergence removes finite source truncations. $\square$

For example, write a prepared bank as $\sum_{K \geq 0} T^K \int e^{-\beta T} d\nu_K(\beta)$, with positive majorants μ_K , $\beta \geq \delta K$, $\beta \geq g_0 > 0$, and $N_S = \sum_K \int e^{-S\beta} d\mu_K < \infty$. At $T = s^{-r}U + \Delta$, let U, Δ have complete norms L, D_* in a supplied Banach source algebra. In the raw presentation that assigns the exterior tag $-rj$ the weight W^{rj} , $W \geq 1$, the polynomial expansion has majorant $(W^r L + D_*)^K$. The source exponential for $e^{-\beta \Delta}$ costs $e^{\beta D_*}$. Including every primary lowering with fixed absolute slope at most Λ replaces this polynomial majorant by $(W^r L + D_* + \Lambda)^K$. Consequently

$$(9.25) \quad \sum_K \int e^{-(A-D_*)\beta} (W^r L + D_* + \Lambda)^K d\mu_K \leq N_S \quad \text{if} \quad A \geq S + D_* + \delta^{-1} \log_+(W^r L + D_* + \Lambda).$$

Here $\log_+ t = \max(0, \log t)$ for $t > 0$, with $\log_+ 0 = 0$. The binomial identity proves this estimate before the K sum. An inverse retains its whole ordered convolution and its total frequency moment as in Lemma 9.1. Thus a cut of an inverse is applied to those complete tensors before their moments; it is not the inverse of the forward compact.

9.4. A criterion for complete lower coefficients.

Set-up 9.11. Let E be a complex Banach space and let $\mathcal{L} : E \rightarrow \text{Hol}(\Omega)$ be linear, where $\Omega = \{Z : |Z| > Z_0, |\arg Z| < \omega\}$ and $0 < \omega < \pi/2$. Fix a finitely generated positive action monoid $\Gamma = \{\sum_{j=1}^d n_j \alpha_j : n_j \in \mathbb{N}\}$, with $d \geq 1$ and $\alpha_j > 0$, and finitely many real shifts ζ_i . Set $\mathcal{B} = \bigcup_i (\zeta_i + \Gamma)$, collecting numerical coincidences.

For $\beta \in \mathcal{B}$, supply a Banach space E_β , a bounded linear row $C_\beta : E \rightarrow E_\beta$, and an injective linear realization A_β of E_β as holomorphic germs at the positive infinite tail. These rows are specified independently of $\mathcal{L}$. Every nonzero $u \in E_\beta$ has an entire head

$$(9.26) \quad A_\beta u(x) = c_u \theta_u(x)(1 + o(1)), \quad c_u \neq 0, \quad \theta_u(x) > 0, \quad |\log \theta_u(x)| = o(x).$$

For every smooth cutoff χ equal to one through k and zero from $k^+ > k$, with its transition in a gap of $\mathcal{B}$, supply a holomorphic compact $\mathcal{L}_\chi F$ on Ω such that, on a sufficiently far positive tail,

$$(9.27) \quad \mathcal{L}_\chi F(x) = \sum_{\substack{\beta \in \mathcal{B} \\ \beta < k}} e^{-\beta x} A_\beta C_\beta F(x).$$

Assume that for fixed $a_0, C > 0$, $C_0 \geq 0$ and $\delta_* > 0$, independently of F and the cutoff,

$$(9.28) \quad |(\mathcal{L} - \mathcal{L}_\chi)F(Z)| \leq C H_\chi \|F\|_{Ee^{(-a_0 k + C_0)\Re Z}}, \quad H_\chi = 2 + 2^{J_\chi} \max_{1 \leq j \leq J_\chi} \|\chi^{(j)}\|_\infty,$$

where $J_\chi = \lfloor (k^+ + \max_i |\zeta_i|)/\delta_* \rfloor$ and an empty maximum is zero. The estimate concerns the whole remainder in the original E norm.

Lemma 9.12. *For the shifted action set in Setup 9.11, there exist constants $T_0, C_1, C_2 > 0$ such that, for every real $T \geq T_0$, a cutoff with transition in a gap can be chosen with*

$$(9.29) \quad T \leq k_T < k_T^+ \leq 2T, \quad H_{\chi_T} \leq C_1 e^{C_2 T \log(eT)}.$$

Proof. The set $\mathcal{B} \cap [T, 2T]$ has at most CT^d numerical elements. Each representation in its i th translate has $n_j \leq (2T + |\zeta_i|)/\alpha_j$, and the number of shifts is finite. Partitioning $[T, 2T]$ by these points gives an empty interval of length at least cT^{1-d} .

We record the derivative cost of a cutoff on that interval. Put $\psi(t) = e^{-1/t}$ for $t > 0$ and zero otherwise. Cauchy's estimate on $|z - t| \leq t/2$ gives $|\psi^{(j)}(t)| \leq j!(2/t)^j e^{-2/(9t)}$. Maximizing

in t gives $\|\psi^{(j)}\|_\infty \leq 9^j(j!)^2$. The pointwise estimate also proves smooth extension at zero. Normalize the integral of $\psi(t)\psi(1-t)$ to obtain a decreasing smooth cutoff χ_* , equal to one on $(-\infty, 0]$ and zero on $[1, \infty)$. For $j \geq 1$, differentiating the product under the integral gives

$$\|\chi_*^{(j)}\|_\infty \leq C9^{j-1} \sum_{\ell=0}^{j-1} \binom{j-1}{\ell} (\ell!)^2 ((j-1-\ell)!)^2 \leq C18^j(j!)^2.$$

Place a rescaled copy of χ_* on the middle third of the empty interval. Its derivatives obey $\|\chi_T^{(j)}\|_\infty \leq C(C'T^{d-1})^j(j!)^2$, whereas $J_{\chi_T} \leq C''T$. Taking logarithms and using $j! \leq j^j$ proves (9.29). $\square$

Proposition 9.13. *Under Setup 9.11,*

$$(9.30) \quad \ker \mathcal{L} = \bigcap_{\beta \in \mathcal{B}} \ker C_\beta.$$

If a row survives, let β_ be the least surviving action and use $c_*\theta_*$ for the head of $A_{\beta_*}C_{\beta_*}F$. Then the entire value satisfies*

$$(9.31) \quad \mathcal{L}F(x) = c_*e^{-\beta_*x}\theta_*(x)(1+o(1)).$$

For a real value the coefficient c_ is real. If E is a Banach algebra and $\mathcal{L}$ a unital homomorphism, the common kernel is a closed ideal and the quotient is faithful in its original infimum norm.*

Proof. We first treat the branch in which every row vanishes, and then bound the complete remainder at the first surviving row.

Step 1. We prove vanishing of the entire value. Every compact in (9.27) vanishes on a positive tail. Its holomorphic continuation to connected Ω therefore vanishes. By Lemma 9.12, for every large T ,

$$|\mathcal{L}F(Z)| \leq CC_1\|F\|_E \exp(C_2T \log(eT) - a_0T\Re Z + C_0\Re Z).$$

Set $x = \Re Z$ and $T = \exp(a_0x/(2C_2))$. For large x , the exponent is at most $-c \exp(bx)$ for fixed $b, c > 0$. These constants and the sector precede F .

We justify uniqueness on this actual domain. The sector Ω contains $\{\Re Z > X, |\Im Z| < 3\pi/(4b)\}$ for sufficiently large X . The function $h(w) = \mathcal{L}F(b^{-1} \log w)$ is holomorphic for far w with $|\arg w| < 3\pi/4$, and $|h(w)| \leq C_F e^{-c|w|}$. Suppose $h(R+1) \neq 0$ for a sufficiently large real R . On the unit disk put $H(z) = h(R + (1+z)/(1-z))$. Factoring its finitely many zeros inside a circle $|z| = \rho < 1$ and taking the mean of the logarithm of the nonvanishing factor gives

$$\log |H(0)| \leq \frac{1}{2\pi} \int_{-\pi}^{\pi} \log |H(\rho e^{it})| dt.$$

Choose radii with no zero on their circles. On $2(1-\rho) \leq t \leq 1/4$, the modulus of $(1+\rho e^{it})/(1-\rho e^{it})$ is at least $2/(3t)$. Hence the mean of $|R + (1+\rho e^{it})/(1-\rho e^{it})|$ diverges as $\rho \uparrow 1$. The exponential bound makes the displayed logarithmic mean tend to $-\infty$, a contradiction. Thus $h(R+1) = 0$ on a real tail, and analytic continuation gives $\mathcal{L}F = 0$ on Ω .

Step 2. We identify the first surviving action. The set $\mathcal{B}$ is locally finite and bounded below, so β_* exists. Choose one cutoff with $k > \beta_*$ and $a_0k - C_0 > \beta_* + 1$. Divide (9.28) on the positive axis by $e^{-\beta_*x}\theta_*(x)$. Its logarithm is bounded above by $-x + o(x) + O(1)$, and its ratio tends to zero. The compact has only finitely many later actions. For such an action β , the quotient of its scale by the selected scale has logarithm $-(\beta - \beta_*)x + o(x)$, which tends to $-\infty$. Earlier rows vanish as whole germs. The selected row has its own complete remainder by (9.26). This proves (9.31) for the full value and proves the reverse inclusion in (9.30).

The quotient assertion follows from boundedness of the rows and the homomorphism property after the kernels have been identified. Reality follows by division by the positive selected scale and taking its nonzero limit. $\square$

9.5. Native tensors over a supplied lower presentation.

Set-up 9.14. Let $x_0 > 1$ and let $w : (x_0, \infty) \rightarrow (0, \infty)$ be continuous, nondecreasing and unbounded, with $w(x) \leq \sqrt{x}$. Set $H_w = \{x + iy : x > x_0, |y| < w(x)\}$ and $\gamma(x) = x + i\eta(x)$, where η is continuous and tends to zero. Supply a unital commutative Banach algebra E and an injective unital homomorphism $A : E \rightarrow \text{Hol}(H_w)$ satisfying $\sup_{H_w} |Af| \leq L\|f\|_E$.

On a monoid Γ generated by finitely many positive real numbers, supply Banach spaces E_λ and bounded independent linear maps $C_\lambda : E \rightarrow E_\lambda$ with zero common kernel. Every nonzero $f \in E$ has a least nonzero row λ_f whose complete lower data specify

$$(9.32) \quad Af(\gamma(x)) = c_f e^{-\lambda_f x} \theta_f(x) (1 + o(1)), \quad c_f \neq 0, \quad \theta_f(x) > 0, \quad |\log \theta_f(x)| = o(x).$$

Assume a whole cofinal estimate, with fixed positive constants,

$$(9.33) \quad [C_\lambda f = 0 \text{ for } \lambda \leq T] \implies |Af(z)| \leq C\|f\|_E \exp(C_1 T \log(eT) - c_1 T \Re z) \quad (T \geq T_0, z \in H_w).$$

Fix an integer $d \geq 1$. For $1 \leq e \leq d$, fix integers $P_e \geq 1$, numbers $k_e > 0$, and whole holomorphic clocks K_e with

$$(9.34) \quad \sup_{\substack{e, z \in H_w \\ \Re z \geq t}} |K'_e(z) - k_e| \longrightarrow 0, \quad \Im K_e(\gamma(x)) \longrightarrow 0.$$

Require elements $t_e \in E$, with $At_e = e^{-K_e/P_e}$, and require each $a_e = k_e/P_e$ to be among the chosen generators of Γ . Require the row reserve

$$(9.35) \quad C_\lambda \left(f \prod_e t_e^{j_e} \right) = 0 \quad \text{if} \quad \lambda < \sum_e a_e j_e, \quad f \in E, j \in \mathbb{N}^d.$$

Supply root banks B_e as in Setup 9.4, with their own $P_e, g_e, r_e, \vartheta_e$ and one compatible prescribed q . All these data precede the tensor input.

Construction 9.15. Under Setup 9.14, form the complete projective tensor product $\mathcal{P} = E \hat{\otimes}_\pi B_1 \hat{\otimes}_\pi \cdots \hat{\otimes}_\pi B_d$. Its norm is the completion of the infimum of $\sum_i \|f_i\|_E \prod_e \|v_{i,e}\|_{B_e}$ over finite tensor representations. The whole actualization and the finite root rectangles on simple tensors are

$$(9.36) \quad \text{Act}(f \otimes v_1 \otimes \cdots \otimes v_d)(z) = Af(z) \prod_e S_e(v_e)(e^{K_e(z)}),$$

$$(9.37) \quad Q_T(f \otimes v_1 \otimes \cdots \otimes v_d) = \sum_{0 \leq j_e \leq \lfloor T/a_e \rfloor} \left(\prod_e b_{e,j_e}(v_e) \right) f \prod_e t_e^{j_e}.$$

Define $R_\lambda = C_\lambda Q_T$ for $T \geq \lambda$.

Proposition 9.16. Under Setup 9.14, the maps in Construction 9.15 extend boundedly to $\mathcal{P}$. The actualization is defined on a fixed sufficiently far strip about the real axis, and R_λ is independent of the chosen $T \geq \lambda$. If some $R_\lambda F$ is nonzero, let λ_* be the least such index. For every fixed $T \geq \lambda_*$, the first lower row of $Q_T F$ supplies $c_* \neq 0$ and $\theta_*(x) > 0$ with

$$(9.38) \quad \text{Act } F(\gamma(x)) = c_* e^{-\lambda_* x} \theta_*(x) (1 + o(1)), \quad |\log \theta_*(x)| = o(x).$$

In particular $\ker \text{Act} \subseteq \bigcap_\lambda \ker R_\lambda$. The remainder in (9.38) includes the entire projective tensor and every root tail.

Proof. Integrating (9.34) on the real axis gives $K_e(x) = k_e x + o(x)$. On a vertical segment the derivative differs uniformly from k_e by $o(1)$; since $w(x) \leq \sqrt{x}$, it follows that $K_e(z) = k_e z + o(\Re z)$ uniformly on the horn. The segment from x to $\gamma(x)$ has length tending to zero, so $\Im K_e(x) \rightarrow 0$. On every fixed strip, $\Im K_e(x + iy) = k_e y + o(1)$. Choose a positive strip width so small that all $qe^{K_e(z)}$ lie uniformly inside the right half-plane for sufficiently large $\Re z$. The full Laplace

bounds then make (9.36) bounded by $L\|f\|_E \prod_e \|v_e\|_{B_e}$. The projective norm extends this map and preserves local holomorphy.

Every rectangle in (9.37) is a finite sum of bounded tensor maps. If $T' > T \geq \lambda$, every added multiindex has some $j_e > \lfloor T/a_e \rfloor$, hence $\sum_e a_e j_e > T$. The reserve (9.35) kills its C_λ row. This proves stabilization before any zero test.

For fixed $T \geq \lambda_*$ set $b_T = \min_e a_e(\lfloor T/a_e \rfloor + 1) > T$. Apply Lemma 9.5 to each factor on the physical curve, using its entire logarithmic lift. The identity $\prod_e A_e - \prod_e B_e = \sum_e (A_e - B_e) \prod_{i < e} A_i \prod_{i > e} B_i$ keeps the other root factors whole. Their full and fixed polynomial values are bounded on the far tail. Multiplication by Af and the projective norm therefore give

$$(9.39) \quad |\text{Act } F(\gamma(x)) - A(Q_T F)(\gamma(x))| \leq C_T \|F\|_\pi \exp(-b_T x + o(x)).$$

All constants and the $o(x)$ bound come from the fixed clocks, independently of the tensor coefficients. Stabilization says that the first lower row of $Q_T F$ is λ_* . Apply (9.32) to this actual element of E . Since $b_T > \lambda_*$ and $|\log \theta_*| = o(x)$, the ratio of (9.39) to its selected scale tends to zero. This proves the head and the kernel inclusion. $\square$

Theorem 9.17. *Under Setup 9.14, use the projective presentation and rows of Construction 9.15. The whole native actualization extends, with its prescribed continued determinations, to a fixed far strip of total width greater than $\pi/\max_e k_e$, and is bounded in the original projective norm. Its kernel is*

$$(9.40) \quad \ker \text{Act} = \bigcap_{\lambda \in \Gamma} \ker R_\lambda.$$

This is a closed ideal, and the quotient is a faithful Banach algebra in the original infimum norm. If the first nonzero row has index λ_ , the complete lower row of any fixed $Q_T F$, $T \geq \lambda_*$, gives*

$$(9.41) \quad \text{Act } F(\gamma(x)) = c_* e^{-\lambda_* x} \theta_*(x) (1 + o(1)), \quad c_* \neq 0, \quad \theta_*(x) > 0, \quad |\log \theta_*(x)| = o(x).$$

The error includes the whole original projective input and every native, lower, inverse and parameter tail. The coefficient c_ is real whenever the whole realized value is real.*

Proof. Proposition 9.16 proves stabilization and the complete head whenever a row survives. We prove that zero rows force the whole native value to vanish. The proof first bounds growing root rectangles, then obtains whole exponential flatness, and finally restores the full integrals through a finite list of angular covers.

Step 1. We sample the whole clocks. Put

$$\epsilon(x) = \sup_{\substack{e, z \in H_w \\ \Re z \geq x}} |K'_e(z) - k_e|, \quad H_e(x) = \Re K_e(x), \quad H_{\min}(x) = \min_e H_e(x), \quad k_{\min} = \min_e k_e.$$

On a far tail ϵ is finite, nonincreasing and tends to zero. The clock calculation in Proposition 9.16 gives $H_e(x) = k_e x + o(x)$ and $H'_e(x) = k_e + o(1)$. Only the labels with minimal speed can attain $H_{\min}$ for large x . For each $\delta > 0$, their increment bounds give

$$(k_{\min} - \delta)(y - x) \leq H_{\min}(y) - H_{\min}(x) \leq (k_{\min} + \delta)(y - x), \quad y \geq x$$

on a far tail. For the lower bound use a minimizing label at y ; for the upper bound use one at x . Thus $H_{\min}$ is strictly increasing, continuous and asymptotic to $k_{\min}x$, even if the minimizing label changes.

Choose a fixed large C_* , to be specified by the coefficient bound below. For every sufficiently large integer N , define x_N by $H_{\min}(x_N) = \log N + C_*$. Define

$$(9.42) \quad V_N = \min\{w(x_N)/4, x_N^{1/4}, \epsilon(x_N)^{-1/2}\}, \quad L_N = C_L \log N.$$

The last entry is infinite when $\epsilon(x_N) = 0$. Then $V_N \rightarrow \infty$ and $x_N \sim (\log N)/k_{\min}$. The closed rectangle $z = x_N + t + iy$, $0 \leq t \leq L_N$, $|y| \leq V_N$ lies strictly in H_w for large N . On its vertical

segments,

$$|\Re K_e(x + iy) - H_e(x)| \leq |y|\epsilon(x_N) \leq \sqrt{\epsilon(x_N)} \leq 1.$$

The term $ik_e y$ has zero real part. Since every H_e is increasing on the far tail, the whole rectangle satisfies

$$(9.43) \quad \Re K_e(z) \geq H_e(x_N) - 1 \geq \log N + C_* - 1.$$

Step 2. We prove two bounds for the complete root rectangle. The coefficient estimate (9.15) and the fixed norms of t_e imply

$$(9.44) \quad \|Q_N\| \leq C \exp(C_Q N \log(eN)).$$

Each root degree is at most a fixed multiple of N , and the number of multiindices is polynomial in N ; both costs are included in this bound. Direct evaluation is much smaller. For $1 \leq j \leq N/a_e$, combining (9.15) with (9.43) gives

$$|b_{e,j}(v_e)(At_e(z))^j| \leq C \|v_e\|_{B_e} \left[D_e a_e^{-1/P_e} e^{(1-C_*)/P_e} \right]^j.$$

Choose C_* so that each bracket is at most $1/2$. Sum all the displayed degrees, keep the scalar coefficient separate, multiply the finite list of root factors, and apply $\sup |Af| \leq L\|f\|_E$. The projective norm gives

$$(9.45) \quad |A(Q_N F)(z)| \leq M_0 \|F\|_\pi$$

on the whole rectangle, with M_0 independent of N and F .

Suppose now that all $R_\lambda F$ vanish. Stabilization gives $C_\lambda Q_N F = 0$ for every $\lambda \leq N$. Apply (9.33) to $Q_N F$, using (9.44) only on the right edge $t = L_N$. A sufficiently large fixed C_L gives

$$|A(Q_N F)(x_N + L_N + iy)| \leq M_0 \|F\|_\pi e^{-2N \log N}, \quad |y| \leq V_N.$$

All choices precede the tensor input.

Step 3. We propagate the estimates through the widening rectangle. Set $a_N = \pi/(2V_N)$ and

$$h_N(t, y) = \frac{\sinh(a_N t) \cos(a_N y)}{\sinh(a_N L_N)}.$$

This harmonic function is zero on the left, top and bottom edges and lies between zero and one on the right edge. Apply the maximum modulus principle to

$$A(Q_N F)(x_N + u) \exp \left[N \log N \frac{\sinh(a_N u)}{\sinh(a_N L_N)} \right].$$

The complete direct bound (9.45) controls three edges; the cofinal bound controls the fourth. Hence

$$(9.46) \quad |A(Q_N F)(x_N + t + iy)| \leq M_0 \|F\|_\pi \exp[-N \log N h_N(t, y)].$$

For a fixed compact set $K \subset \{\Re u > 0\}$, let $t_* = \min_K \Re u > 0$. Since $V_N \rightarrow \infty$, $\cos(a_N \Im u) \geq 1/2$ on K for large N . The inequalities $\sinh(a_N t) \geq a_N t_*$ and $\sinh(a_N L_N) \leq e^{a_N L_N}/2$ yield

$$(9.47) \quad |A(Q_N F)(x_N + u)| \leq C_K \|F\|_\pi \exp \left[-c_K \frac{N \log N}{V_N} \exp \left(-C' \frac{\log N}{V_N} \right) \right], \quad u \in K.$$

The logarithm of the positive decay exponent equals $\log N - o(\log N)$: here $V_N \rightarrow \infty$ and $V_N \leq x_N^{1/4} = O((\log N)^{1/4})$. For every fixed $0 < \alpha < 1$, the right side is therefore at most $C_K \|F\|_\pi e^{-N^\alpha}$ on a sufficiently far tail. No lower rate of widening was used.

Step 4. We prove whole flatness on each fixed angular representative. Fix a bounded interval of imaginary parts and choose $N(x) = \lfloor \delta_0 e^{H_{\min}(x)} \rfloor$, with $\delta_0 > 0$ fixed and sufficiently small.

The whole clock bound on this strip gives $|e^{K_e(x+iy)}| \geq K_0 N(x)$ for the constant required by Lemma 9.6. Also choose $-\log \delta_0 - C_* > 0$. The increment estimates for $H_{\min}$ show

$$x - x_{N(x)} \rightarrow (-\log \delta_0 - C_*)/k_{\min} > 0.$$

Consequently $x + iy - x_{N(x)}$ stays in one compact subset of $\{\Re u > 0\}$.

On a buffered subinterval choose for each root factor either an allowed complete integral or an initial-disk representative with strict damping. Apply the variable-degree error estimate with $n_e = \lfloor N(x)/a_e \rfloor$. The product identity used in Proposition 9.16 keeps all other factors whole. Their uniform norm bounds, the bound for Af , and the projective sum give an entire error at most $C\|F\|_{\pi} e^{-cN(x)}$ between this representative and $A(Q_{N(x)}F)$. Since $\log N(x) = k_{\min}x + o(x)$, (9.47) now yields, for every $0 < \kappa < k_{\min}$,

$$(9.48) \quad |F_{\text{loc}}(x + iy)| \leq C\|F\|_{\pi} \exp[-ce^{\kappa x}].$$

The estimate includes the full high-carrier minor and the subtracted-polynomial tails from Lemma 9.6.

Step 5. We restore the full angular representatives. List the distinct speeds as $k_1 < \dots < k_m$. Choose $0 < \vartheta_* < \min_e \vartheta_e$ also satisfying $\vartheta_* < \pi/2 - |\arg q|$, and set

$$J_j = \left(\frac{-\arg q - \pi/2 - \vartheta_*}{k_j}, \frac{-\arg q + \pi/2 + \vartheta_*}{k_j} \right).$$

These intervals contain zero, are strictly nested, and have length greater than π/k_j . Choose buffered nested intervals $I_j \Subset J_j$ and increasing numbers $0 < \kappa_j < k_j$ close enough to k_j that $|I_j| > \pi/\kappa_j$. Reserve further fixed buffers for the finite applications of Lemma 8.20.

At level j , use the whole root integral for every speed $k_e \leq k_j$. It is available throughout I_j : the number $\arg q + k_e y$ is a convex combination of $\arg q$ and $\arg q + k_j y$, and both lie within the full damping range. The error $\Im K_e(x + iy) - k_e y = o(1)$ preserves the fixed margin on a far tail. For faster speeds use initial-disk integrals, choosing their angles on a finite ordered cover of I_j . A common finite refinement for all factors keeps consecutive overlaps and all damping margins.

Each representative is the whole projective sum of these products with its unchanged lower factor. At level one it has (9.48) with $\kappa = \kappa_1$. On an overlap only faster initial-disk factors change. Their complete joining arcs, followed by the finite product identity, give

$$|\text{consecutive representatives' difference}| \leq C\|F\|_{\pi} \exp[-ce^{\kappa_{j+1}x}].$$

On the next strip I_{j+1} , replacing the next tied-speed initial-disk factors by their full original integrals has this bound as well. The replacement includes their joining arcs and their complete original high-carrier tails.

Use $u = e^{-z}$, with its logarithmic lift, so that $|u| = e^{-x}$ and $\arg u = -y$. The buffered covers are finite sector chains. Lemma 8.20 upgrades the level- j bound from order κ_j to order κ_{j+1} . The whole next-level differences transfer that bound to level $j+1$. After finitely many upgrades, every root factor is a complete original integral. They define one holomorphic function on the final strip, whose width exceeds π/κ_m . The single-function assertion of that lemma makes it zero. If $m = 1$, apply that assertion directly to (9.48). This proves the zero-row implication.

The final full integrals exist and are bounded for arbitrary F , independently of the vanishing argument. Their strip has width greater than $\pi/\max_e k_e$; agreement on overlaps follows from the original root-contour deformation. They therefore give the claimed continuation and preserve the prescribed determinations.

Step 6. We construct the quotient and identify the complete physical head. The converse kernel inclusion and the entire first term are Proposition 9.16. The full actualization is a bounded unital homomorphism, since each complete root realization and A are such maps. Its kernel is closed, so the identified row quotient is a Banach algebra in the infimum of the original projective norm. Faithfulness follows from the kernel equality. For a real whole value, division by its

positive selected scale and passage to the nonzero limit proves that c_* is real. Every estimate preceded the projective sum. Proposition 4.20 therefore passes them through original absolutely summable parameter families; fixed smaller disks retain the full geometric tails and the coefficient and derivative losses of the original input radii. $\square$

Remark 9.18. The linearity hypothesis in Setup 9.14 is essential. For example, take $E = \mathcal{W}_1(\mathbb{C})$ in a variable t , $Af(z) = f(e^{-z})$, $H_w = \{x + iy : x > 1, |y| < \sqrt{x}\}$, and $\Gamma = \mathbb{N}$. For $f = \sum_{n \geq 0} c_n t^n$ define bounded maps into $\mathbb{C}^2$ by

$$(9.49) \quad C_n f = \left(c_n, \frac{c_n c_{n+1}}{1 + |c_{n+1}|} \right).$$

They have zero common kernel and $\|C_n f\|_1 \leq 2\|f\|$. Their first nonzero row gives the whole head $c_n e^{-nx}$, and vanishing through T gives $|Af(z)| \leq \|f\| e^{-T\Re z}$. Set $P = k = 1$, $K(z) = z$, $t_1 = t$, and $\gamma(x) = x$. The row reserve holds because multiplication by t^j kills every coefficient of index less than j . These data satisfy every other requirement of Setup 9.14. For the root bank with $q = 1$ and $v = (1, \zeta)$, however, $Q_0(1 \otimes v) = 1$ and $Q_1(1 \otimes v) = 1 + t$. Their C_0 values are $(1, 0)$ and $(1, 1/2)$. Thus R_0 in Construction 9.15 is not independent of its rectangle threshold without the linearity hypothesis.

9.6. Ordinary coefficients with complete prepared inputs. The comparison of a physical word needs a coefficient class which allows a prepared passage at a logarithmic argument. Its asymptotic series need not satisfy a common Gevrey estimate. We shall approximate the finitely many canonical inputs while keeping those prepared passages whole.

Definition 9.19. Fix a finite list of selected root inputs from Definition 8.4, with their prescribed determinations and original parameter norms. A finite canonical recipe in t is a finite expression formed from their full values $S_i(v_i)(\beta_i t^{-r_i})$, $\beta_i, r_i > 0$, finitely many positive powers of t , constants, and the independent variable $\ell = t^\varepsilon \log t$. Here $\varepsilon > 0$ is chosen below all the finitely many effective orders before the ℓ disk is fixed. The allowed operations are full analytic operations on strict disks, Euler and parameter derivatives, pullback by a positive unit map $t \mapsto tv(t, \ell)$, its actual local inverse, and a fixed normalized remainder $t^{-J}(f - \sum_{\lambda < J} t^\lambda f_\lambda(\ell))$, $J > 0$. The unit pullback lifts the independent variable by

$$(9.50) \quad (t, \ell) \longmapsto (tv, v^\varepsilon(\ell + t^\varepsilon \log v)).$$

All independent parameters and scalar functions of ℓ are retained on their strict disks.

For a finite recipe, its independent record is obtained from the root Taylor coefficients, the monomial exponents and the Taylor series in the independent variables, by these algebraic operations. The records of inverse units are the formal solutions with their specified nonzero limiting derivative.

Lemma 9.20. *Each recipe in Definition 9.19 has a complete source at finitely restricted root domains and increased weights. Its actualization, each fixed record coefficient, and each fixed normalized remainder are bounded in the original input norms. Its record has locally finite support in a monoid generated by finitely many positive real numbers.*

On strict sectors where a choice of full or initial-disk root integrals is defined, the whole recipe has a Gevrey remainder of order the smallest effective root order occurring in it. Changing one choice of initial-disk ray changes the whole recipe by an exponentially small quantity of that root order. The constants are uniform on the fixed original parameter polydisks.

Proof. We construct division by a fixed monomial, then the unit operations, and finally the estimates for growing truncations.

Step 1. We divide by a fixed root power. Let $v = (c, \phi)$ be a root source of order P , and let $b_j(v)$ be its coefficients in (9.11). For an integer $m \geq 1$ put $\alpha = m/P$, choose an integer $k > \alpha$,

and set

$$(9.51) \quad \begin{aligned} \chi_m(\zeta) &= \zeta^{-m} \left(\phi(\zeta) - \sum_{j=1}^m \phi_j \zeta^j \right), \\ I_m(\zeta) &= \frac{1}{\Gamma_E(k-\alpha)} \int_0^1 u^{\alpha-1} (1-u)^{k-\alpha-1} \chi_m(\zeta u^{1/P}) du, \\ \psi_m &= q^{-\alpha} D_0 (D_0 + 1) \cdots (D_0 + k - 1) I_m, \quad D_0 = P^{-1} \zeta \partial_\zeta. \end{aligned}$$

Subtraction makes χ_m holomorphic and zero at the origin. On one smaller disk and sector it has a weighted bound by $C_m \|v\|$. The integral is bounded there because $|\chi_m(\zeta u^{1/P})|$ has the additional factor $u^{1/P}$ at zero. Each of the finitely many Euler derivatives has a bounded source norm on a further strict domain at a larger weight: a Cauchy circle of radius proportional to $|\zeta|$ stays in the original sector, and its exponential growth is absorbed by that weight. These restrictions depend on m , but not on a parameter coefficient of v .

The beta integral gives $(\psi_m)_n = q^{-\alpha} \Gamma_E((n+m)/P) \phi_{n+m} / \Gamma_E(n/P)$ for $n \geq 1$. Thus $N_m v = (b_m(v), \psi_m)$ has actual value

$$(9.52) \quad S(N_m v)(Y) = Y^{m/P} \left(S(v)(Y) - \sum_{j < m} b_j(v) Y^{-j/P} \right).$$

For completeness, this is an identity of sources: multiply $N_m v$ by the root source of $Y^{-m/P}$. Its Taylor record is the record of $v - \sum_{j < m} b_j(v) Y^{-j/P}$, and equality of the holomorphic minors follows coefficient by coefficient on their origin disk and then by continuation. The injective root algebra of Lemma 8.2 gives the actual identity. No divergent physical series is summed in this argument.

To divide at the argument $Y = \beta t^{-r}$, choose $m = \lceil JP/r \rceil$ in (9.52). The remaining factor is the nonnegative power $\beta^{-m/P} t^{rm/P-J}$. For a Wiener series in finitely many positive monomials, partition its tail according to the first multiindex crossing the threshold J . The minimal crossing multiindices are finite: their weights are less than J plus the largest generator. Dividing each such tail introduces only finitely many new positive excesses. For products use the finite identity for a product minus the product of its truncations. This proves bounded normalized division and local finiteness of the enlarged positive monoid.

Step 2. We perform whole unit operations. Start from the projective tensor product of the original root banks and the Wiener algebras of the independent monomials, ℓ , and parameters. If the difference of two presentations has zero actual value and zero record, identify that difference; the closed intersection of these two kernels is an ideal. This auxiliary quotient assumes no equality of the kernels. It permits the monomial identities used in (9.52). The scalar part at $t = 0$ is kept as a whole function of ℓ and the parameters.

The complement of this scalar part can be made small in norm. For a root minor, increase its weight, using its zero at the origin and its fixed old exponential bound. For a positive monomial, decrease its radius. For a finite product use $\prod f_i - \prod c_i = \sum_i (f_i - c_i) \prod_{j < i} f_j \prod_{j > i} c_j$. These operations leave the original parameter radii fixed.

For an independent unit w near a fixed positive constant, the pullback of the i th root factor is the dilation $\phi_i(w^{r_i/P_i} \zeta)$. Lemma 8.7 supplies one strict target domain and an operator family $\sum_{n \geq 0} (w - w_0)^n D_n$, with $\|D_n\| \leq CR^{-n}$. Substitution of a complete source h with $\|h\| < R$ has norm at most $C/(1 - \|h\|/R)$ times the original input norm. The monomials and (9.50) are full analytic families on this fixed unit disk.

For the inverse of $t \mapsto tv(t, \ell)$, write the inverse as $t \mapsto tw(t, \ell)$ and solve $wv(tw, w^\varepsilon(\ell + t^\varepsilon \log w)) = 1$. At $t = 0$ retain the full equation $wv_0(w^\varepsilon \ell) = 1$. Its selected scalar solution and nonzero derivative exist on a smaller ℓ disk. Preconditioning by that derivative makes the derivative of the scalar correction zero at the solution. Choose one fixed unknown disk on which

its derivative is at most $1/4$. The positive source part and its derivative become smaller than $1/8$ at one final source stage. Picard iteration is then a contraction on that stage, with ratio at most $1/2$. The bounded actual homomorphism identifies its limit with the selected actual inverse. Applying a fixed coefficient functional to the iteration identifies its record with the unique formal inverse. Euler and parameter derivatives follow from Cauchy estimates and, for an inverse, differentiation of this full equation.

Step 3. We control physical logarithmic remainders. A fixed remainder after setting $\ell = t^\varepsilon \log t$ also has a complete source. First split the independent record at the finitely many powers $\lambda \leq J$, using the bounded root and monomial division already constructed. At such a power its coefficient is a whole analytic function $f_\lambda(\ell)$. Subtract its Taylor polynomial through $\lfloor (J-\lambda)/\varepsilon \rfloor$ and write the remaining function as $\ell^{n_\lambda} g_\lambda(\ell)$, where $n_\lambda = \lfloor (J-\lambda)/\varepsilon \rfloor + 1$. Cauchy's estimate on its original larger ℓ disk bounds this entire g_λ . Choose a new positive $\varepsilon' < \varepsilon$ such that $\lambda + n_\lambda(\varepsilon - \varepsilon') - J > 0$ for every one of these finitely many powers. On the physical logarithmic curve put $\ell = t^{\varepsilon - \varepsilon'} \ell'$. The divided remainder is the whole expression

$$t^{\lambda + n_\lambda(\varepsilon - \varepsilon') - J} (\ell')^{n_\lambda} g_\lambda(t^{\varepsilon - \varepsilon'} \ell').$$

Its exponent is positive, so it has a bounded source on fixed strict disks. The already divided higher t -power remainder undergoes the bounded analytic substitution $\ell = t^{\varepsilon - \varepsilon'} \ell'$. This constructs a new representing stage with bounds from the old full coefficients. It does not identify the two independent logarithmic variables on their original disks. Only finitely many such changes are needed by a fixed finite recipe.

Step 4. We control growing truncations. For a root input, Lemma 9.6 and Cauchy's estimate on its origin disk bound the coefficient of physical weight λ by $CD^\lambda \Gamma_E(1 + \lambda/r_i)$. Keep every ℓ and parameter coefficient whole. Let $q_0 > 0$ be the smallest effective root order in the finite recipe. At a truncation threshold N , choose $R_N = cN^{-1/q_0}$ with $c > 0$ fixed and small. On the polydisk which assigns the monomial of weight γ radius R_N^γ , all root polynomials of weight below N are bounded by the original fixed input budgets. The positive parts are uniformly small after c is decreased. Therefore every full analytic operation, unit pullback and inverse contraction remains within its already chosen strict disks, independently of N .

Here is the tail estimate on this polydisk. For generators $\gamma_1, \dots, \gamma_l > 0$ and $|t| \leq R_N/4$, the sum of the monomial bounds of weights at least N is at most

$$(2|t|/R_N)^N \prod_{i=1}^l (1 - 2^{-\gamma_i})^{-1}.$$

The factor N^{N/q_0} in R_N^{-N} is bounded by $D_0^N \Gamma_E(1 + N/q_0)$: integrate the defining gamma integral over $[N/q_0, N/q_0 + 1]$. On the remaining part of any fixed sufficiently small sector, the bounds for the finite polynomial and the whole function give this remainder estimate after its geometric constant is increased. This proves the claimed Gevrey remainder. A fixed normalized division replaces N by $N + J$ and introduces a fixed power of $|t|^{-1}$; its additional gamma factor is absorbed into the geometric constant. For a full inverse, the difference of two fixed points is at most twice the difference of their whole defining maps. It therefore obeys this estimate too.

Finally two choices of an initial-disk ray differ by an integral over a joining arc away from zero. The exponential on that arc is bounded by $\exp(-c|t|^{-r_i})$ on a buffered common sector. The full source norms bound its amplitude. Products, strict analytic operations and the inverse estimate just proved carry this whole bound through the finite recipe. Fixed negative powers introduced by normalized division are absorbed by decreasing c . This proves the assertion about changes of rays. $\square$

Definition 9.21. An ordinary expression is a finite expression in a small positive variable s , with $L = -\log s$, formed from constants, the original parameters, and finitely many $s^\gamma L^j$, $\gamma > 0$,

$j \in \mathbb{N}$, by the following operations. Its canonical inputs are supplied on one common nonreal root direction q , with $\Re q > 0$, positive leading argument scales and compatible root and logarithmic lifts. At each effective order their full integrals have a common strict angular interval of aperture greater than π divided by that order. These common-reference data precede the expression.

- (1) Whole analytic operations on fixed strict disks are allowed.
- (2) A fixed canonical recipe of Definition 9.19 may be evaluated at $x = cs^a \exp H$, $c, a > 0$, and at previously constructed bounded parameter expressions. The expression H is previously constructed and tends to zero. The effective orders and selected canonical determinations are fixed.
- (3) A whole original prepared bank, its correlated inverse, or a finite weak core may be evaluated at $\alpha L + b + H$, $\alpha > 0$, $b \in \mathbb{R}$. Its measures have a positive gap, a linkage $\beta \geq \delta K$ and a summable original power norm. Its independent distribution has locally finite point jets in a finitely generated positive monoid.
- (4) For a fixed $J > 0$, the normalized remainder $s^{-J}(f - P_{\leq J}f)$ is allowed. Here $P_{\leq J}f$ retains all independently computed power/logarithmic terms of weight at most J , with their whole parameter coefficients.
- (5) Suppose a previously constructed whole vector expression $G(s, z; \tau)$ is defined on a fixed larger independent z polydisk, its limiting equation has $G(0, 0; \tau) = 0$, and its Jacobian there has a bounded inverse on the parameter polydisk. Its small actual solution $G(s, z; \tau) = 0$ is allowed. The canonical determinations, structural exponents and original bank labels in G are independent of z .

The independent record $\mathcal{T}_0 f$ uses the canonical records, the original local distributions of the prepared banks, and the unique formal solutions of the regular implicit equations. A generated ordinary Wiener presentation is an original scalar Wiener algebra evaluated at a fixed finite tuple of such expressions lying strictly inside its input radii. Finite exterior modules also permit finitely many explicit real powers of s and integer powers of L .

Proposition 9.22. *For a fixed finite grammar in Definition 9.21, actualization is bounded on a common small sector, and the independent record coefficients are bounded linear maps on each generated ordinary Wiener presentation. The record has locally finite nonnegative power support, with polynomial logarithmic coefficients, and its common kernel is the actual kernel. After a physical parameter specialization, every nonzero real value in a generated presentation or its finite exterior module has a complete leading term $cs^\lambda L^j$, $c \in \mathbb{R} \setminus \{0\}$. Each whole function $s^{-J}f$, with f ordinary, has an exact decomposition*

$$(9.53) \quad s^{-J}f = Q(s, L) + bL + c + R(s),$$

where Q is a finite sum of growing power/logarithmic monomials other than L , and R is an ordinary expression tending to zero. If the coefficients of Q and b in finitely many such phases are real, finite sums of their exponentials times ordinary exterior coefficients have an independent zero test and complete positive heads.

Proof. The proof constructs complete scalar logarithmic sources for canonical approximations, obtains a small positive flatness order from them, and then restores the full canonical integrals one order at a time.

Step 1. We construct scalar logarithmic sources. Use arrays of scalar measures $(m_j)_{j \geq 0}$ supported in $t \geq \eta j$, with norm $\sum_j \int e^{-wt} d|m_j|(t)$, convolution product and actual value $\sum_j L^j \int e^{-tL} dm_j(t)$. The remaining damping is $v = \Re L - w - \eta^{-1} \log_+ |L|$. Lemma 5.4 gives their Banach algebra and whole supported-tail estimate. We retain original scalar leaf measures and their summable tensor integrals; no vector measure is presumed to have a density. Multiplication adds the original carriers and degrees. Since $w \geq \max(w_1, w_2)$ implies $e^{-w(t_1+t_2)} \leq e^{-w_1 t_1 - w_2 t_2}$,

flattening two successive scalar constructions preserves their norm bound with the smaller linkage.

Lemma 5.11 constructs an entire prepared bank at $\alpha L + b + H$ in this algebra. Its exponential uses the full previously constructed source of H . The total-action formula of Lemma 5.10 gives the independent distribution and all its cutoff derivatives. Correlated inverses are the whole banks of Lemma 5.13.

We check two additional operations. For normalized division by s^J , choose a fixed gap $J < B_0 < B_1$ below the next retained power, after the terms through J have been subtracted. A cutoff in this gap has zero compact distribution. Its whole compact actual value is zero, by integration by parts on the compact distribution. The complementary measures have carriers at least B_0 . Translate those carriers by $-J$. The new positive gap is $B_0 - J$, and a valid new linkage is $\eta(1 - J/B_0)$. At a weight increase D its norm is at most

$$(9.54) \quad e^{wJ - D(B_0 - J)} \sum_j \int e^{-wt} d|m_j|(t).$$

The cutoff itself consumes one fixed extra weight, sufficient to pay its binomial lowering sums. This is independent of any later canonical truncation threshold.

For a regular implicit equation, write A_0 for its limiting Jacobian and use $\Phi(z) = z - A_0^{-1}G(z)$. Choose a fixed unknown ball on which the derivative of the scalar limiting Φ is at most $1/4$. The remaining source part has a fixed positive carrier gap. One further weight increase makes its full norm and its derivative norm small enough that Φ maps the ball into itself with contraction ratio at most $1/2$. The original larger unknown disk bounds all Taylor coefficients before the iteration. Its limit is therefore a whole source at this single final stage. The actual homomorphism identifies its small root, and the local distribution of every bounded carrier interval stabilizes under the formal iteration. A perturbation of the complete equation changes its root by at most twice the norm of the preconditioned perturbation.

Step 2. We truncate only the canonical factors. Replace each canonical recipe by its independent physical power/logarithmic polynomial of weight below N , retaining whole parameter coefficients. Apply all the original analytic, prepared and implicit operations to these polynomials. Call the complete result f_N . Every prepared bank and every implicit solution in f_N is still whole. We claim that its scalar logarithmic source has, for constants fixed before N and before an angular cover is chosen,

$$(9.55) \quad w_N = w_* + C_* \log(N + 1), \quad \|m(f_N)\|_{w_N, \eta_*} \leq M,$$

with one positive linkage η_* and fixed positive gaps. Its independent distribution agrees with $\mathcal{T}_0 f$ below $c_0 N - C_0$, for fixed $c_0 > 0$ and $C_0 \geq 0$.

We verify the claim through the finite expression. A canonical term $c_{\lambda j}(Z)x^\lambda(\log x)^j$, at $x = cs^a e^H$, becomes

$$c_{\lambda j}(Z)c^\lambda s^{a\lambda} e^{\lambda H} (-aL + \log c + H)^j.$$

The coefficient bound from Lemma 9.20 is $CD^\lambda \Gamma_E(1 + \lambda/q)$ for an effective canonical order $q > 0$ in its own variable, and $\lambda \geq \varepsilon j$. The whole source costs of $e^{\lambda H}$ and the polynomial in H are exponential in $\lambda + j$. For $\lambda < N$, a choice $aC_* > 1/q$ pays the gamma growth in the factor $e^{-a\lambda w_N}$. The remaining geometric series is uniformly summable, and its positive part becomes as small as needed by increasing w_* . Full analytic operations now have their fixed original strict margins.

The prepared operation is paid by its full affine-source bound, so its new source weight remains of the form (9.55). For a normalized remainder, use (9.54) with an extra weight $D_* \log(N + 1) + D_0$, where $D_* > C_* J/(B_0 - J)$. It retains a uniform bound and merely increases the constant C_* . At an implicit vertex the entire residual and its derivative have the uniform bounds just

obtained; one fixed further weight increase gives the contraction already proved. This proves the whole norm assertion by induction on the finite expression.

For the distributional assertion, proceed through the finite expression. Products add locally finite supports. Strict analytic series have only finitely many positive-degree partitions below a fixed action. A fixed normalized division shifts the accuracy by J ; a power substitution rescales it by its fixed positive exponent. Formal Picard iteration gains a positive power at each step, so an implicit solution on a bounded power window depends only on a finite part of the original distribution. These facts give $c_0N - C_0$ for the finite recipe. They also construct each row independently of actualization and prove its boundedness in the original parameter norms.

Step 3. We prove that a zero record gives whole flatness. Assume $\mathcal{T}_0f = 0$. Choose a cofinal cutoff below $c_0N - C_0$, with transition in a gap near a fixed positive multiple of N . Its compact distribution is zero. Lemma 9.12 and the whole logarithmic source bound give constants $a_0, A_0, A_1, A_2 > 0$ such that

$$(9.56) \quad |f_N(s)| \leq A_0 \exp\{-a_0N\Re L + A_1N \log(N+1) + A_2N \log(1+|L|)\}.$$

All original inverse and parameter degrees have already been summed in this estimate.

On any fixed finite angular cover, the difference $f - f_N$ is bounded by a finite sum of expressions

$$(9.57) \quad C_I D_I^N \Gamma_E(1 + N/q_i) |s|^{a_iN - J_I} (1 + |L|)^{d_I N + d_I}.$$

To obtain this bound, use the whole canonical remainder at each replaced input, then telescope the finite recipe. The prepared inputs have bounded whole derivatives on their strict far logarithmic domains; the implicit roots have the factor-two stability estimate above. Each normalized division contributes only its fixed J_I . Thus (9.57) estimates an entire difference, including the full prepared and inverse tails.

Choose a positive number ϵ so small that $C_*\epsilon < 1/4$, $A_1\epsilon < a_0/4$, and $\epsilon < a_i q_i/4$ for every canonical input. These inequalities are chosen before the angular cover. Put $N = \lfloor \exp(\epsilon \Re L) \rfloor$. On a fixed angular cover $|\Im L|$ is bounded, so $|L| \leq 2\Re L$ eventually. The logarithm of (9.57) is at most $-(a_i - \epsilon/q_i + o(1))N\Re L$. The exponent in (9.56) is also at most $-cN\Re L$. Consequently every whole initial-disk representative satisfies

$$(9.58) \quad |f(s)| \leq C \exp(-c|s|^{-\epsilon})$$

on each strict part of that cover.

Step 4. We restore the full canonical integrals. List the distinct effective canonical orders as $r_1 < \dots < r_m$. Use the common direction supplied in Definition 9.21. For a physical input this choice is furnished by Corollary 8.18 before its reference source is constructed. The full root integrals have a common continued angular interval of width greater than π/r_j at order r_j . Initial-disk integrals, whose endpoint can be chosen inside the origin disk in any lifted direction, give a finite overlapping cover of any prescribed finite angular interval. Their seams at order r_j are bounded by $C \exp(-c|s|^{-r_j})$ by Lemma 9.20. Full prepared factors at $\alpha L + b + H$ are defined on each such fixed logarithmic strip once its real part is large enough.

Start with an initial-disk cover of aperture greater than π/ϵ . The bound (9.58) and Lemma 8.20 raise the flatness to order r_1 . Restore the order- r_1 full integrals on their buffered interval; their difference from the initial-disk integrals is itself r_1 -flat. On that interval the remaining seams have order r_2 . Use its aperture greater than π/r_1 and Lemma 8.20 to raise the estimate to r_2 . Continue through the finite ordered list, with strict buffers chosen at the start. After the last restoration the original whole expression is r_m -flat on an interval of aperture greater than π/r_m . The last assertion of Lemma 8.20 makes it zero. If the recipe has no canonical input, the whole logarithmic source expression itself is defined on every fixed logarithmic strip; apply that last assertion directly to (9.58).

Step 5. We identify complete heads and phases. For a fixed power bound B , the whole logarithmic source cuts and fixed canonical remainders give the entire remainder $O(|s|^B(1 +$

$|L|)^{h_B})$ after all terms below B . The power/logarithmic terms below every bound form a finite set, since all generators are positive. If the record is nonzero, choose its least power and the largest logarithmic degree at that power after the physical parameters have been fixed. Choose a fixed larger power bound in the remainder estimate. The resulting whole remainder is smaller than the selected power/logarithmic term. Conversely an actual zero cannot have this nonzero head. This proves both kernel inclusions and the complete head assertion. Multiplication by a sufficiently large positive power reduces a finite exterior module to this case.

To obtain (9.53), subtract the finite physical record of f through J . Its terms divided by s^J split into Q, bL, c and the terms tending to zero. The remaining normalized remainder is an allowed ordinary expression. Use the negative-exponential convention and write the finite expression explicitly as

$$(9.59) \quad \sum_{\nu=1}^m e^{-P_\nu(s)} g_\nu(s), \quad P_\nu = Q_\nu + b_\nu L + c_\nu + R_\nu, \quad \tilde{g}_\nu = e^{-c_\nu} s^{b_\nu} e^{-R_\nu} g_\nu.$$

Here g_ν is a complete ordinary exterior coefficient, and the decomposition of P_ν is (9.53). This also covers a positive exponential e^{P_ν} by applying the decomposition to $-P_\nu$. Since $L = -\log s$, the expression in (9.59) equals $\sum_\nu e^{-Q_\nu} \tilde{g}_\nu$ exactly. The full unit e^{-R_ν} is ordinary; its convergent exponential uses the whole remainder, and the real power s^{b_ν} is an allowed exterior factor.

At the fixed physical parameter, group equal growing polynomials Q_ν and sum their complete coefficients. The zero test just proved decides whether each group vanishes. Distinct real Q_ν are eventually ordered: their first differing power/logarithmic monomial fixes the sign, and $|Q_\nu - Q_\mu|/L$ tends to infinity. Let Q_* be the least surviving polynomial. Its complete coefficient has a head $c_* s^{\lambda_*} L^{j_*}$. For every later group its entire coefficient has a power/logarithmic head or is zero. The ratio of its contribution to the selected term tends to zero because $\exp[-(Q_\nu - Q_*)]$ dominates that ratio of power/logarithmic scales. The number of groups is finite. Thus the complete head is $c_* e^{-Q_*} s^{\lambda_*} L^{j_*}$, with all remaining groups and the full selected coefficient remainder included. Reality of the leading constant follows by taking the limit of the real actual value divided by its positive scale. $\square$

9.7. Sources for the original physical constructors.

Construction 9.23. Fix a finite literal expression in the alphabets of Definition 9.2, including its prescribed prefixes, inverse branches and whole parameter families. At every occurrence retain its original source measure, root bank or analytic input and its original positive majorant. Products retain the ordered tensor of these inputs before adding carrier coordinates. An inverse retains every ordered forward tensor and its total frequency moment.

An auxiliary source array for the expression has exterior tags γ in a group generated by finitely many real numbers, original nonnegative carrier coordinates ξ , and complete coefficient functions $h_\gamma(\xi; s)$. Write $|\xi|$ for the sum of these coordinates. At a fixed stage its norm and actual value are

$$(9.60) \quad \|m\|_{A,W} = \sum_\gamma W^{|\gamma|} \int e^{-A|\xi|} \|h_\gamma(\xi; \cdot)\| d\nu(\xi), \quad W \geq 1,$$

$$(9.61) \quad \text{Act } m(s) = \sum_\gamma s^\gamma \int h_\gamma(\xi; s) \exp\left(-\sum_l \omega_l(\xi_l) H_l(s)\right) d\nu(\xi).$$

Here ν denotes the retained positive product majorants, with their original face measures and atoms. Coefficient norms are those constructed for the finite canonical and ordinary recipes; an original observed coefficient which has only a buffered holomorphic supremum norm keeps that norm. Each ω_l is its original real frequency or its original cone frequency $q_l x_l + y_l$, with $|q_l| = 1$ and $\Re q_l > 0$. Each H_l is the whole clock at its specified input. The stage has a linkage

$|\gamma| \leq C_{\text{tag}}|\xi|$ on positive sources. The sums include the complete original polynomial, analytic, inverse and parameter degrees.

These arrays and their projective completions are auxiliary source spaces. The presentations to be compared are the original scalar Wiener algebras in a fixed finite tuple of the unfiltered physical expressions, on strict input radii, and their finite exterior modules. The bounded maps from these presentations into the arrays are part of the construction. Original observation triples remain auxiliary modules; an arbitrary observed minor is not adjoined as a new coefficient generator.

Lemma 9.24. *Every finite literal expression in Construction 9.23 has a complete bounded source on a common small sector, with one final choice of source weights and strict domains preceding every total-action cutoff. Every reached strong input has the form*

$$(9.62) \quad W(s) = as^A e^{H(s)}, \quad a, A > 0,$$

where H is a whole bounded source tending to zero. Its zero-positive specialization is an ordinary expression in Definition 9.21. The construction retains the original actual inverse identities and all prescribed original observation full/compact/complement identities.

Proof. We first give the norm operations for positive sources. Then we construct the endpoint inverses, a whole weak core, and the opening and closing of one paired block. The remaining letters and the finite prefix use these explicit operations.

Step 1. We construct positive sources with their original norms. The tag triangle inequality and Fubini's theorem make (9.60) submultiplicative. These are estimates of the retained positive integrals before any carrier cancellation. On a smaller common sector the finitely many clock margins dominate the fixed source weights and the tag loss $C_{\text{tag}}|\xi| |\log |s||$. Thus (9.61) converges absolutely and locally uniformly, with its stated bounded actualization. If a source has an original positive birth gap g , then for $M, D > 0$

$$(9.63) \quad \|s^{-M}m\|_{A+D,W} \leq W^M e^{-Dg} \|m\|_{A,W}, \quad |\gamma - M| \leq (C_{\text{tag}} + M/g)|\xi|.$$

This pays every fixed negative power on the positive ideal and makes its full norm small. Ordinary positive monomials are instead made small in their own Wiener radii; the weight W is not used for that purpose. Every fixed original moment is paid by $t^j e^{-ht} \leq j! h^{-j}$. All these bounds concern complete sources.

Step 2. We retain original labels and canonical inverses. Corollary 8.18 provides one compatible auxiliary reference for the finite endpoint list, with bounded maps from its original source norms. Each compatible analytic original label retains its complete weak replay in that reference source. A separately supplied native implementation keeps its original domain and norm in parallel, outside joint comparison unless its common-ray input is independently supplied. Erasing transition degrees recovers the reference forward and inverse normalizer sources and graph unit, not the entire original endpoint recipe. Its original endpoint gauge remains an independent source. Erasure of original endpoint births alone retains the entire transition source; recovery of the original-label recipe keeps that transition mark. Erasing both sets of positive degrees is the zero-positive specialization in the reference presentation. For clarity the exact two-mark identity is as follows. If the original and reference endpoint gauges are k_o, k_0 , put $G = k_o^{-1}k_0$, $H_G = G/\text{id}$, $D_G = -\log H_G$, and $D_o = -\log(k_o/\text{id})$. With $w = e^{-B_0}$, the physical matching gives $B_o = B_0 + D_G(w)$. Before the two marks are set equal, the full weak correction is

$$(9.64) \quad E(v_t, v_o) = D_G(v_t) + D_o(v_o H_G(v_t)).$$

The one-mark specializations are $E(0, v_o) = D_o(v_o)$ and $E(v_t, 0) = D_G(v_t)$, since $H_G(0) = 1$ and $D_G(0) = D_o(0) = 0$. Thus $E(0, 0) = 0$. On $v_t = v_o = w$ it equals $-\log(k_0(w)/w)$, by $k_o \circ G = k_0$ on the fixed branches. This is Proposition 6.15 with the two substitutions kept independent.

The contraction in Lemma 9.9 applies to this whole two-variable analytic germ too, by replacing both marks by their products with e^{-h} .

At a reached input $X = Cs^{-r}e^{H_1}$, $C, r > 0$, $H_1 \rightarrow 0$, the original root value $K(X)$ has its bounded complete source from Lemma 9.20. For $B(Y) = Y - a_1 \log Y - \log c_1 - \log K(Y)$, solve $B(Y) = X$ by $Y = X(1 + v)$. Its exact equation is

$$(9.65) \quad v = X^{-1}(a_1 \log X + \log c_1) + X^{-1}\{a_1 \log(1 + v) + \log K(X(1 + v))\}.$$

The first term is a bounded small source: $X^{-1} \log X = C^{-1}s^r e^{-H_1}(\log C + rL + H_1)$. The complete relative-root family on a fixed larger v disk has a bounded derivative. Multiplication by X^{-1} makes that derivative smaller than $1/2$ in one final source stage. The whole equation is a contraction there. It constructs the actual inverse on its supplied image domain and its independent regular formal inverse.

We also construct a bounded source for a fixed height increment, without division by an unbounded clock. The expression $I = B^{-1}(B(Y) + h) - Y$ satisfies exactly

$$(9.66) \quad I = h + a_1 \log(1 + I/Y) + \log K(Y + I) - \log K(Y).$$

Keep h independent on a fixed larger disk. The full translated-root family at $Y + I$ is bounded on a fixed I disk, and its derivative is small after the factor Y^{-1} and a fixed root weight reserve are included. The equation is a source contraction with $I = h$ at the scalar limit. It gives the whole bounded inverse family before a positive discrepancy is inserted. Both (9.65) and (9.66) are regular implicit vertices of the ordinary grammar after positive sources have been erased.

Step 3. We construct the full weak core. The original prepared sources of Theorem 5.1 write each weak height map as $\lambda T + b - U(T)$, $\lambda > 0$. A positive analytic weak adapter has this form with $\lambda = 1$ and a degree-zero bank whose carriers are positive integers. Compose these maps using the whole affine-source formula of Lemma 5.11. At every step flatten the original scalar leaf carriers and add their degrees. The linkage is the minimum of the old linkages after their fixed positive scalings; the original positive gaps remain positive. A required degree radius ρ costs at most $\delta^{-1} \log_+(1/\rho)$ of carrier weight. Thus a finite core has the exact form

$$(9.67) \quad J(w) = dw^\Lambda e^{Q(-\log w)}, \quad Q(T) = \sum_{K \geq 0} T^K \int e^{-\beta T} d\nu_K(\beta),$$

with $d, \Lambda > 0$, $\beta \geq g > 0$, $\beta \geq \delta K$, and a summable complete original power norm. The original forward point-jet support is supplied by Proposition 5.2. Its transport by Lemma 5.11 scales the finitely many positive generators, and products add them. A finite core therefore still has finitely generated positive support. The true inverse core reverses its original letters and uses their entire correlated inverse banks from Lemma 5.13. This construction does not introduce an arbitrary new gap bank.

Step 4. We open a paired block. For the current (9.62), put $Y_a = \beta_a/W^{p_a}$ and $w_a = e^{-B_a(Y_a)}$. The exact incoming weak height is $T = B_a(Y_a) + E_a(w_a)$, with the full two-mark version when necessary. Erase the positive births from H , leaving the entire ordinary expression H_0 . Set

$$Y_{a,0} = \beta_a a^{-p_a} s^{-Ap_a} e^{-p_a H_0}, \quad T_0 = B_a(Y_{a,0}), \quad r = Ap_a, \quad \kappa = \beta_a a^{-p_a}.$$

The full difference is

$$(9.68) \quad T = T_0 + \Delta, \quad \Delta = Y_a - Y_{a,0} - a_a \log(Y_a/Y_{a,0}) - \log K_a(Y_a) + \log K_a(Y_{a,0}) + E_a(w_a).$$

Each term of Δ has a positive birth; the unbounded factor in $Y_a - Y_{a,0}$ is paid by (9.63). The relative-root family gives its full coefficient difference before any degree is extracted. Moreover $s^r T_0$ is explicitly a bounded ordinary source tending to κ : expand its definition using $e^{-p_a H_0}$, $s^r L$, $s^r H_0$ and $s^r \log K_a(Y_{a,0})$.

For an original term of (9.67), keep its original β, K and write its exact source as

$$(9.69) \quad s^{-rK} e^{-\beta T_0} (s^r T)^K \exp_*(-\beta m_\Delta) d\nu_K(\beta).$$

It has a new carrier β at order r , whose leading action is $\kappa\beta$. Let $M = \|s^r T\|$, $D = \|m_\Delta\|$. The polynomial and exponential costs are $(W^r M)^K e^{\beta D}$. Including every primary lowering of fixed slope at most κ replaces the polynomial cost by $(W^r M + \kappa)^K$. Therefore the complete source and all those lowerings are paid by one weight

$$(9.70) \quad A_c \geq S + D + \delta^{-1} \log_+(W^r M + \kappa) + C_{\text{enh}},$$

where S is the original bank weight and C_{enh} is a fixed reserve for the finite original tree. This uses the original summable norm before summing over K or integrating β . A further increase makes the new positive source small.

We verify that inverse moments are included in C_{enh} . For one original inverse node let n be its ordered forward degree, K the sum of forward polynomial degrees and B its total original frequency. Suppose $B \geq d_0 K$, let $R_0 > 0$ be its polynomial source radius, and let $a_0 \geq 0$ be the sum of its fixed cutoff slopes. Put $U_0 = R_0 + a_0$. The complete lowering and moment sum is

$$(9.71) \quad \begin{aligned} & \sum_{h=0}^K \binom{K}{h} a_0^h \sum_{j=0}^{\min(K-h, n-1)} \binom{n-1}{j} (K-h)_j B^{n-1-j} R_0^{K-h-j} \\ &= \sum_{j=0}^{\min(K, n-1)} \binom{n-1}{j} (K)_j B^{n-1-j} U_0^{K-j} \\ &\leq B^{n-1} U_0^K (1 + (d_0 U_0)^{-1})^{n-1}. \end{aligned}$$

The first equality follows by factoring $(K)_j$ and summing the remaining binomial; the inequality uses $(K)_j \leq (B/d_0)^j$. With $C_U = 1 + (d_0 U_0)^{-1}$, a weight $d_0^{-1} \log_+ U_0 + q$, $q > C_U N/\lambda$, bounds the entire inverse by $\sum_{n \geq 1} N^n (C_U/q)^{n-1}/(n\lambda^n)$. Here N, λ are its original forward norm and inverse ratio. This convergent series keeps every forward leaf and its total frequency moment. Fixed additional moments use another finite reserve. At an analytic vertex these enhanced norms are put inside its original strict radii. The finite number of vertices proves that C_{enh} can be fixed before any cutoff.

Step 5. We close the paired block. Define

$$(9.72) \quad \begin{aligned} Y_c &= B_b^{-1}(\Lambda B_a(Y_a) - \log d), & w_b &= e^{-B_b(Y_c)}, \\ \zeta &= \Lambda E_a(w_a) - Q(T) - E_b(w_b). \end{aligned}$$

The normalized argument of B_b^{-1} tends to one, so (9.65) constructs its entire source. The mark w_b is a separate endpoint mark; the identity $w_b = d e^{-\Lambda B_a(Y_a)}$ is not used to replace it by an integer power of w_a . All terms of ζ have a new positive birth, and their whole norms fit the fixed inverse-fiber disks after the weight choice already described.

Let $h = h(w_b, \zeta)$ be the whole solution of (9.21) and put $I_b = B_b^{-1}(B_b(Y_c) + h) - Y_c$. The identities (9.23) and (9.72) give

$$B_b(Y_c + I_b) + E_b(e^{-B_b(Y_c + I_b)}) = \Lambda T - \log d - Q(T).$$

The right side is exactly $-\log J(P_a(W))$. Thus the original inverse branch gives $P_b^{-1} J P_a(W) = (\beta_b/(Y_c + I_b))^{1/p_b}$. Its relative output is the entire unit $(1 + I_b/Y_c)^{-1/p_b}$ times the whole baseline $(\beta_b/Y_c)^{1/p_b}$. In (9.62) the leading data therefore change to

$$(9.73) \quad A' = A p_a / p_b, \quad a' = (\beta_b / (\Lambda \beta_a))^{1/p_b} a^{p_a/p_b}.$$

The two endpoint orders remain their original orders. The inverse block uses the exchanged endpoints and the correlated inverse core, on the matched actual inverse domains.

Step 6. We handle strong letters and prepared events. A real-power connector changes the leading data to $A' = \alpha A$, $a' = c a^\alpha$ and gives exactly

$$H' = \alpha H + \log U((a^{\gamma_i} s^{A \gamma_i} e^{\gamma_i H})_i).$$

This is a whole analytic operation in the original positive monomials. Its inverse unit solves its original finite-variable analytic equation with nonzero limiting derivative. The selected positive inverse branch is consequently retained. A prepared strong passage uses its entire bank at $-\log W = AL - \log a - H$, by Lemma 5.11; its inverse uses the whole correlated bank. These are ordinary operations when their positive inputs are erased.

For an original event in (9.4), factor its complete structural height as $F = bs^{-r}U(V(s))$, $U(0) = 1$. The finite tuple V includes the positive powers and logarithmic degrees needed by this factorization and is closed downward in logarithmic degree. The expression $F(s(1+z))/F(s) - 1$ is a full convergent germ in (V, z) , with z derivative $-r$ at zero. Its inverse in z is a whole germ $A(V, w)$ with $A(V, 0) = 0$. Substitution of the complete source $w = \text{Lap } \mu(F)/F$ gives the exact event $s(1 + A(V, w))$. The source has the original positive gap of μ and the bounded factor $s^r U^{-1}/b$. A negative event first uses (9.1) in its original F coordinate, and then this identical structural inverse. Its ordered moments are retained before the changed clock is inserted.

Step 7. We retain the slowdown and seed. We now construct the stipulated normalized densities. Write κ_s for their retained slowdown parameter; it is separate from the leading block coefficient κ in (9.69). Let h_s be the inverse of $T \mapsto T - \kappa_s \log T$, with its original positive branch. In independent small variables $z = U^{-1}$ and $w = U^{-1} \log U$, its whole coefficient equations are

$$(9.74) \quad d_s = \kappa_s \log(1 + \kappa_s w + z d_s), \quad L_s = 1 + \kappa_s w + z d_s, \quad h_s(U) = U L_s = U + \kappa_s \log U + d_s.$$

Choose fixed seed radii $r_z, r_w > 0$ such that $r_z + \kappa_s r_w \leq \eta_s$, where $\eta_s = (4\kappa_s + 4)^{-1}$. On the unit ball for d_s the logarithmic series in (9.74) has norm at most $2\kappa_s \eta_s < 1$, and its derivative has norm at most $\kappa_s r_z / (1 - \eta_s) < 1/4$. It is therefore a contraction in the complete seed Wiener algebra, with $\bar{d}_s := \|d_s\| \leq 1$ and $\|L_s\| \leq \bar{L}_s := 1 + \eta_s$. The solution has zero constant term. Let Q_α be the measure $t^{\alpha-1} dt / \Gamma_E(\alpha)$ for $\alpha > 0$, and let Q_0 be the origin atom. Put $\gamma_E = -\Gamma'_E(1)$. The measure $\omega(dt) = (-\log t - \gamma_E) dt$ has Laplace transform $(\log U)/U$, by differentiating the defining gamma integral at one and substituting t/U . The full seed map J_{seed} sends $z^\rho w^j$ to $Q_\rho * \omega^{*j}$. It is contractive on the fixed positive-power seed radii once $\sigma^{-1} \leq r_z$ and $\|\omega\|_\sigma \leq r_w$. The latter norm is $O((1 + \log \sigma)/\sigma)$ by the substitution $t = v/\sigma$. Its whole analytic series, including those for d_s and L_s , are absolutely summed before any later source integration.

The exact kernel for $h_s(U)^l e^{-th_s(U)}$ is

$$(9.75) \quad K_l(t) = \delta_t * Q_{\kappa_s t - l} * J_{\text{seed}}(L_s^l e^{-t d_s}), \quad t > 0, \quad \kappa_s t - l > 0, \quad K_0(0) = \delta_0.$$

It has support at least t , and its complete absolute norm at weight σ is at most

$$(9.76) \quad \Psi_\sigma(t, l) = e^{-t(\sigma + \kappa_s \log \sigma - \bar{d}_s)} (\sigma \bar{L}_s)^l.$$

Estimate (9.76) applies to the complete positive seed expansion. The beta integral and the whole seed product give $K_l(t) * K_m(v) = K_{l+m}(t+v)$ whenever the displayed orders are positive; the unit handles $(t, l) = (0, 0)$. The majorant is multiplicative in these pairs as well.

Step 8. We retain the original forward-core data. Let the supplied forward core have $n_c \geq 1$ prepared occurrences. At occurrence j retain its incoming height H , incoming section value $v_j > 0$, ratio $\lambda_j > 0$ and actual outgoing coordinate

$$(9.77) \quad y_j = v_j e^{-\lambda_j H + u_j(H)}, \quad u_j(H) = \sum_{K \geq 1} H^K \int e^{-\xi H} d\nu_{j,K}(\xi).$$

By Theorem 5.1 and Proposition 5.2, these are its original complete equation-generated banks. Their positive majorants $\mu_{j,K}$ satisfy $|\nu_{j,K}| \leq \mu_{j,K}$, $\xi \geq \delta_j K$ with $\delta_j = \min(1, \lambda_j)$, and

$$(9.78) \quad N_j = \sum_{K \geq 1} \int e^{-S_j \xi} d\mu_{j,K}(\xi) < \infty$$

at their original fixed weights S_j . For $1 \leq j < n_c$ the original intervening regular map from that outgoing section to the next normalized incoming section is $R_j(y) = a_j y g_j(y)$, with $a_j > 0$ and

$g_j(0) = 1$. Put $\ell_j = \log g_j = \sum_{m \geq 1} \ell_{j,m} y^m$, using the germ zero at zero, and $c_j = -\log(a_j v_j)$. Define the cumulative affine data by

$$(9.79) \quad A_0 = 1, \quad A_j = \lambda_j A_{j-1}, \quad C_0 = 0, \quad C_j = \lambda_j C_{j-1} + c_j \quad (1 \leq j < n_c).$$

The slopes A_j are defined also at $j = n_c$.

Keep independent slots X_j, V_j on fixed positive radii $B_{X,j}, B_{V,j}$. Their intended physical values are $q_j = u_j(H_j)$ and $e_j = e^{-\lambda_j H_j}$. Choose the slot disks so that $v_j B_{V,j} e^{B_{X,j}}$ is strictly inside the original convergence disk of ℓ_j . Before physical substitution define the convergent old-slot expressions and their complete coefficient bounds by

$$(9.80) \quad W_1 = 0, \quad W_{j+1} = \lambda_j W_j - X_j - \ell_j(v_j V_j e^{X_j}),$$

$$(9.81) \quad \overline{W}_1 = 0, \quad \overline{W}_{j+1} = \lambda_j \overline{W}_j + B_{X,j} + \sum_{m \geq 1} |\ell_{j,m}| v_j^m B_{V,j}^m e^{m B_{X,j}}.$$

These finite bounds precede the seed radii and the measure weight. Every W_j has zero constant and uses only slots at earlier occurrences.

Set $H_1 = h_s(U)$. With all earlier slots given their physical values, direct substitution of the original gate gives

$$(9.82) \quad \begin{aligned} H_j &= A_{j-1} h_s(U) + C_{j-1} + W_j(q_1, e_1, \dots, q_{j-1}, e_{j-1}), \\ H_{j+1} &= -\log R_j(v_j e_j e^{q_j}) = \lambda_j H_j + c_j - q_j - \ell_j(v_j e_j e^{q_j}). \end{aligned}$$

The initial conditions and this equality prove the height invariant through the literal forward core, with its original sections and logarithmic branches.

Let

$$(9.83) \quad \delta_c = \min_{1 \leq j \leq n_c} A_{j-1} \delta_j = \min_{0 \leq j \leq n_c} A_j > 0, \quad \kappa_s > 4/\delta_c.$$

The second inequality is the admissibility condition of the supplied forward-core density construction. We retain that original parameter and density. Only seed receiving radii and measure weights are restricted below. For every original $\xi \geq \delta_j K$ and $0 \leq l \leq K$, put $t = A_{j-1} \xi$. Then

$$(9.84) \quad l \leq K \leq \frac{\xi}{\delta_j} \leq \frac{t}{\delta_c}, \quad \kappa_s t - l \geq (\kappa_s - \delta_c^{-1})t > 0.$$

Thus every required $K_l(A_{j-1} \xi)$ is already defined before any weight is increased. The outgoing source uses $(t, l) = (A_j, 0)$ and satisfies the inequality as well.

Step 9. We construct the complete forward-core source. Inductively substitute the constructed earlier sources in W_j and write the resulting whole source as m_{W_j} . For $0 \leq K \leq \xi/\delta_j$ define

$$(9.85) \quad N_{j,K}(\xi) = e^{-C_{j-1}\xi} \sum_{l=0}^K \binom{K}{l} A_{j-1}^l K_l(A_{j-1}\xi) * (C_{j-1}\delta_0 + m_{W_j})^{*(K-l)} * \exp_*(-\xi m_{W_j}).$$

The powers and exponential are complete source operations. The binomial identity and (9.82) give the exact synthesis $H_j^K e^{-\xi H_j}$.

Suppose each earlier slot source, with its positive pre-collection majorant, has norm at most one fourth its fixed slot radius. The entire old-slot coefficient expansion then gives $\|m_{W_j}\|_\sigma \leq \overline{W}_j$. Inserting the majorant (9.76) into (9.85) gives

$$(9.86) \quad \begin{aligned} \|N_{j,K}(\xi)\|_\sigma^{\text{maj}} &\leq e^{-\xi\{A_{j-1}(\sigma + \kappa_s \log \sigma - \overline{d}_s) - |C_{j-1}| - \overline{W}_j\}} (A_{j-1}\sigma \overline{L}_s + |C_{j-1}| + \overline{W}_j)^K \\ &\leq e^{-E_{j,\sigma}\xi}, \\ E_{j,\sigma} &= A_{j-1}(\sigma + \kappa_s \log \sigma - \overline{d}_s) - |C_{j-1}| - \overline{W}_j \\ &\quad - \delta_j^{-1} \log_+(A_{j-1}\sigma \overline{L}_s + |C_{j-1}| + \overline{W}_j). \end{aligned}$$

The notation $\|\cdot\|_\sigma^{\text{maj}}$ denotes the mass of this full retained positive expansion. The estimate precedes integration in ξ and summation in K ; its factor A_{j-1}^l occurs once, outside K_l . All constants in $E_{j,\sigma}$ were fixed on the original slot disks, and $E_{j,\sigma} \rightarrow \infty$. Define the two new sources by

$$(9.87) \quad m_{q_j} = \sum_{K \geq 1} \int N_{j,K}(\xi) d\nu_{j,K}(\xi), \quad m_{e_j} = N_{j,0}(\lambda_j).$$

For $E_{j,\sigma} \geq S_j$, their majorants satisfy

$$(9.88) \quad \|m_{q_j}\|_\sigma^{\text{maj}} \leq N_j e^{-(E_{j,\sigma} - S_j)\delta_j}, \quad \|m_{e_j}\|_\sigma^{\text{maj}} \leq e^{-E_{j,\sigma}\lambda_j}.$$

Each tends to zero. A common sufficiently large σ makes them at most $B_{X,j}/4$ and $B_{V,j}/4$ for every occurrence. Induction using the triangular W_j proves the premise for all earlier slots; no earlier norm increases when the common weight increases. The sources have gaps $A_{j-1}\delta_j$ and A_j , respectively. Fubini under these whole majorants identifies their values with $u_j(H_j)$ and $e^{-\lambda_j H_j}$ on a fixed far sector, and then with the given physical passages and gates on their retained positive branch.

More generally, on the original completed slot algebra

$$(9.89) \quad f = \sum_{\alpha} b_{\alpha}(Z^{(j)})^{\alpha}, \quad \|f\| = \sum_{\alpha} B^{\alpha} \|b_{\alpha}\|_{\text{seed}}, \quad Z^{(j)} = (X_1, V_1, \dots, X_j, V_j),$$

the source is the full sum of $J_{\text{seed}}(b_{\alpha})$ convolved with the ordered slot powers. Its norm is at most $\|f\|$, and its complete tail of slot degree greater than N is at most $4^{-N-1}\|f\|$. All original parameter families use those fixed larger slot disks and their own Cauchy costs before these sums.

For the outgoing normalized height put $T_j^{\text{out}} = H_j - q_j/\lambda_j$ and, before slot substitution, $B_j^{\text{out}} = (W_j - X_j/\lambda_j)/A_{j-1}$. Its supplied normalized residual is the whole expression

$$(9.90) \quad B_j^{\text{out}} - \kappa_s \log(1 + zL_s^{-1}B_j^{\text{out}}),$$

whose exact actual value is $(T \mapsto T - \kappa_s \log T)((T_j^{\text{out}} - C_{j-1})/A_{j-1}) - U$. The coefficient norm of B_j^{out} is fixed by (9.81). A fixed smaller seed radius makes $\|zL_s^{-1}B_j^{\text{out}}\| < 1/2$, so the complete logarithm has its full geometric tail. Its slot constant is zero, and its source keeps a positive birth gap. If an original terminal R_{n_c} is supplied, extend the definitions of c_j, C_j, W_{j+1} to $j = n_c$, with its original strict disk. Then $H_{n_c+1} = A_{n_c}h_s + C_{n_c} + W_{n_c+1}$, and the terminal residual is (9.90) with B_j^{out} replaced by W_{n_c+1}/A_{n_c} . These are the original normalized density recipes. All affine constants, seed factors and the supplied terminal adapter remain in their full sources.

Step 10. We construct the original primary distributions. Let ∂_t denote distributional differentiation in the primary action coordinate, separately from every seed carrier. The local original native distribution is $S_j^{\text{nat}} = \sum_{K \geq 1} \partial_t^K \nu_{j,K}$. For the j th occurrence let S_{W_j} be the old-slot distribution given by the already constructed recurrence. Starting with $S_{W_1} = 0$, define on each original point jet

$$(9.91) \quad \mathcal{T}_j(\partial_t^l \delta_{\xi}) = e^{-C_{j-1}\xi} (A_{j-1} \partial_t \delta_0 + C_{j-1} \delta_0 + S_{W_j})^{*l} * \delta_{A_{j-1}\xi} * \exp_*(-\xi S_{W_j}).$$

Then $S_{q_j} = \mathcal{T}_j S_j^{\text{nat}}$, $S_{e_j} = \mathcal{T}_j \delta_{\lambda_j}$ and

$$(9.92) \quad S_{W_{j+1}} = \lambda_j S_{W_j} - S_{q_j} - \sum_{m \geq 1} \ell_{j,m} v_j^m S_{e_j}^{*m} * \exp_*(m S_{q_j}).$$

Each series is locally finite because its new slots have the positive gaps just proved. Formula (9.91) follows by differentiating $e^{-C_{j-1}\xi} \delta_{A_{j-1}\xi} * \exp_*(-\xi S_{W_j})$ in ξ and using $\partial_{\xi} \delta_{A_{j-1}\xi} = -A_{j-1} \partial_t \delta_{A_{j-1}\xi}$. Integration by parts on a compact action interval gives the identical formula on $\partial_t^K \nu_{j,K}$. It is thus the local distribution of the original source (9.85), rather than a row defined from its scalar value.

Proposition 5.2 places the native knots in $\langle 1, \lambda_j \rangle$. Their scaled actions are in $\langle A_{j-1}, A_j \rangle$; old-slot products add only previously generated actions. Consequently the complete core distribution is supported on $\langle A_0, \dots, A_{n_c} \rangle$, a locally finite positive monoid after its origin is removed. To check every kernel after old-slot expansion, let an old term have action b and degree $p \leq b/\delta_c$. A new term has action $t = A_{j-1}\xi + b$ and degree $l + p$, where (9.84) gives $l + p \leq t/\delta_c$. Products add these inequalities, so the Riesz order remains positive for the whole raw recurrence. On a bounded action window the number of original degrees and positive slot partitions is finite. Their absolute raw sums are bounded by the multiplicative majorant (9.76) and (9.86). The complete seed coefficient is convolved after this primary operation. Total-action cutoffs therefore retain all distributional product derivatives on these original labels, with their whole complementary source.

For a fixed total-action cutoff slope bound Λ_0 , the complete lowering envelope of one original kernel is

$$(9.93) \quad \sum_{h=0}^l \binom{l}{h} \Lambda_0^h \Psi_\sigma(t, l-h) = e^{-t(\sigma + \kappa_s \log \sigma - \bar{d}_s)} (\sigma \bar{L}_s + \Lambda_0)^l.$$

The lowered Riesz order increases by h and remains positive. In (9.86), this replaces $A_{j-1}\sigma\bar{L}_s$ by $A_{j-1}(\sigma\bar{L}_s + \Lambda_0)$. Its logarithmic cost is still $O(\log \sigma)$, so the effective damping still tends to infinity. Apply this bound successively to the enhanced old-slot majorants in the finite recurrence. The fixed original slot radii are therefore respected before summing every lowering, native degree and inverse moment. The cutoff derivative constants are then the separate constants of the total-action product rule; they do not change the retained slowdown.

For a generalized weak unit the original primitive $e^{-\gamma h_s(U)} h_s(U)^j$ uses (9.75) with $(t, l) = (\gamma, j)$. Its Riesz order is positive by its own specified $\kappa_s > j/\gamma$. All original convergent powers are summed at their strict radii. A primary lowering replaces l by $l-h$ in the kernel before its seed factor is synthesized. The bound for the complete lowering sum is obtained by replacing $\sigma\bar{L}_s$ by $\sigma\bar{L}_s + \Lambda_0$, where Λ_0 is the fixed sum of cutoff slopes. The damping still tends to infinity. Thus these are bounds on the stipulated recipes, including all seed, power and inverse degrees.

Their independent records are obtained first from the original local point distributions in the h_s coordinate. At a fixed action t the record of (9.75) is the whole coefficient $U^{l-\kappa_s t} L_s^l e^{-td_s}$ multiplying e^{-tU} . A bounded window contains finitely many primary actions and degrees; the seed is kept whole. The total-action product rule followed by this kernel identity proves its compact square. An arbitrary measure with a positive gap would not supply this independent finite record.

Step 11. We retain finite prefixes and original observations. All letters just constructed preserve (9.62). Their complete relative families are obtained on fixed larger independent input disks before any positive correction is substituted. Choose the input budgets backwards through the finite list. The positive gaps and (9.63) then place every actual substitution inside those budgets. Intersect the finitely many strict sectors and inverse image domains and retain one final source stage. Increasing a positive carrier weight does not increase an earlier positive norm.

For an original observation at a labelled source node, retain the ordered triple $(m, Cm, m - Cm)$. Every later linear kernel acts on these three inputs; products and analytic operations act on the original ordered tensors before their whole degree sums. In an inverse, the observation on the n -fold forward tensor precedes its total frequency moment. Estimate (9.71) bounds the complete observed sums. An observed canonical minor on an old ray is integrated against that ray's original character at the changed whole clock. It is not used as a new holomorphic root minor. The reference presentation is constructed from the whole physical expression using (9.64); its full value agrees with the original full value. The original observation triple remains in the original presentation. Linearity of its later kernel integrals proves each full/compact/complement identity without moving an old filter to a new determination. $\square$

Lemma 9.25. *Consider a presentation constructed in Lemma 9.24. Let $R > 0$ be its greatest positive exponential order, put $X = s^{-R}$, and remove a common exterior growing phase if one is present. At a fixed physical specialization the following assertions hold.*

- (1) *Its order- R action set is contained in a finite union of translates of a monoid generated by finitely many positive real numbers. Each action has an independently constructed finite coefficient expression over the complete lower-order graph. The expression keeps all ordinary canonical vertices and every prepared bank at logarithmic argument.*
- (2) *The row maps into these coefficient presentations are bounded and linear, and are defined before any actual zero test. A total-action cut in a gap has actual value $\sum_{\beta < k} e^{-\beta X} \text{Act } C_\beta F$. Its complete complement satisfies*

$$(9.94) \quad |\text{Act } F - \text{Act}_\chi F| \leq CH_\chi \|F\|_{\text{orig}} e^{(-a_0 k + C_0)\Re X}.$$

Here $a_0, C > 0$, $C_0 \geq 0$, the source stage and the small sector are fixed before the cutoff, and H_χ is the cost in (9.28). The uncut value is bounded by $C\|F\|_{\text{orig}} e^{C_0 \Re X}$ on that sector.

- (3) *If no positive exponential order remains, the full coefficient expression belongs to the ordinary grammar or a finite module of the phases in (9.53) with real growing and logarithmic coefficients.*

The bounds use the full original parameter norms before numerical action collection at the specified physical specialization.

Proof. We describe the finite coefficient operation at each kind of vertex, then prove the bound for the whole parent source.

Step 1. We obtain finite coefficients with whole lower inputs. Erase only the births of order R from the finite expression graph. At an analytic vertex retain its entire lower input u_0 . The coefficient of action β in $G(u_0 + u_+)$ is the sum of $G^{(n)}(u_0)u_{\beta_1} \cdots u_{\beta_n}/n!$ over the partitions $\sum \beta_i = \beta$. Each removed birth has a fixed positive gap, so only finitely many n and partitions occur. The outer derivative is taken at the full lower input, including every one of its positive degrees. This coefficient rule also applies to an original Wiener input: its derivatives at the lower input are bounded by Cauchy's estimate on its original strict radius.

At an implicit or inverse vertex differentiate its whole equation. The coefficient of the unknown at action β occurs linearly with the full lower Jacobian. This is the invertible Jacobian of the source contraction constructed in Lemma 9.24. Its inverse is its convergent whole Neumann source. All other terms use strictly smaller positive partitions. Thus the row is defined independently and is a finite expression over the retained lower graph. In an original time inverse, this is equivalently extraction from its ordered forward tensors before their total frequency moments.

For a prepared bank of order less than R , retain the whole original bank at its lower argument. Only its argument increments of order R are expanded. A fixed row therefore contains finitely many full argument derivatives of that bank. Their original moment reserves were fixed in (9.71). For a bank of order R , first use its original local distribution. Its finite positive generation is the equation-generated property in Proposition 5.2, preserved by the full correlated inverse and the fixed positive affine transports. A finite native action ω gives the finite expression

$$\sum_j c_{\omega,j} T^j e^{-\omega T}.$$

Expand its other order- R increments by the finite partition rule just given. The lower increments remain whole. For a normalized density from a slowdown the corresponding coefficient is the whole seed in (9.75); its fixed derivatives and ordered powers stay inside the finite lower coefficient.

The ordinary part of the clock has the form $T_0 = \kappa s^{-R} A_0(s)$, with $A_0 \rightarrow 1$ ordinary. Apply (9.53) to this entire expression. Its growing coefficients, including its coefficient of L , are real. To

see this without imposing reality on a selected normalizer, use (9.68): the physical T is real and $T - T_0$ has a positive exponential gap, hence is smaller than every ordinary power. The finite ordinary asymptotic record therefore has real growing coefficients. Removing $\omega \kappa s^{-R}$ leaves a finite real growing phase of strictly smaller order, real exterior powers, and a whole ordinary residual. The factor T^j is a finite exterior power times a bounded whole source. Fixed negative powers on lower positive increments are paid by (9.63).

All these operations are bounded on the original input presentation, with a constant allowed to depend on the fixed action window. They define linear row maps because the variable original Wiener input enters linearly; every other source and every evaluation kernel has already been fixed. The resulting finite lower presentation can use a later coefficient stage for that row. This row-dependent bound is never used to choose the parent stage in (9.94). The action generators are the finitely many positive scaled native generators and endpoint births in the original finite expression. Finite exterior phases of order R contribute only finitely many real translates. Numerical coincidences are collected after the physical specialization.

Step 2. We make one total cutoff on the original source. On an ordered source tensor write $\Omega_R = \sum_l \lambda_l a_l$ for the sum of its order- R actions, with $\lambda_l > 0$ the fixed leading clock coefficients. For one exterior translate ζ apply $\chi(\zeta + \Omega_R)$ once to the total local distribution. By Lemma 9.10 its original tensor formula contains every derivative $\chi^{(|J|)}$, with its full chain factor λ^J . All original derivative degrees have a linkage to Ω_R . Choose $\delta_* > 0$ below the finite list of scaled linkages. Then their sum is at most Ω_R/δ_* . A nonzero cutoff derivative has order at most $\lfloor (k^+ + |\zeta|)/\delta_* \rfloor$.

The enhanced majorants (9.25), (9.70) and (9.71) pay all their polynomial, binomial, analytic and inverse-degree sums at the fixed parent stage. The remaining derivative factor is bounded by H_χ . The expression with $1 - \chi$ is the entire complement and has $\zeta + \Omega_R \geq k$ in every surviving term. The original distributional product rule, tested first on finite tensors and then extended by these absolute majorants, proves the exact compact square with the previously constructed rows. At a seed node use the literal kernel identity (9.75) after the product rule. Thus the compact square holds on the original sources, before the lower coefficient is tested for zero.

On a common smaller sector, each whole order- R clock has real part at least $(1 - \epsilon_0)\lambda_l \Re X$ for a fixed $\epsilon_0 \in (0, 1)$. Its lower growing terms are $o(X)$; all its bounded and positive source increments have already been paid in the parent norm. The tag loss is linear in carrier times $|\log |s|| = o(\Re X)$. Split the remaining exponential damping into fixed fractions. One fraction pays the original source weights, moments and all linked polynomial degrees; another gives $a_0 \Omega_R \Re X$ for a fixed $a_0 > 0$. The slower sources retain their full positive margins in their own clocks. The finite exterior offsets and phases below R cost at most $e^{C_0 \Re X}$. Integration of the complete positive majorant now proves (9.94). Integration without a lower carrier threshold gives the uncut bound. The sector, weights and constants were chosen before k, k^+ or any row degree.

Step 3. We identify the last lower coefficient. When every positive exponential birth has been erased, each paired block still contains its entire canonical expression

$$\left(\frac{\beta_b}{B_b^{-1}(\Lambda B_a(\beta_a/W^{p_a}) - \log d)} \right)^{1/p_b}.$$

Its inverse and translated fibers are given by (9.65) and (9.66). Every strong prepared bank remains whole at $AL - \log a - H$. The real-power connectors are the whole analytic operations already displayed. A prepared event has its whole structural inverse in the finite positive-power/log variables, and its zero-positive value is the identity. All these vertices meet exactly the ordinary grammar. A single logarithmic disk parameter can be chosen below the finite effective order list before this finite lower recipe is formed; no previously supplied independent

disk is silently changed. The finite phase decomposition used above produces precisely its stated exterior modules. This proves the last assertion. $\square$

Proof of Theorem 9.3. Lemma 9.24 constructs every original letter, inverse and prefix, and its bounded map from the original Wiener input. We prove equality of the kernels by induction on the finite set of positive exponential orders in that constructed expression. The base retains all ordinary coefficients, and each induction step retains the complete lower row.

Step 1. We treat the ordinary base. With no positive exponential order, the last assertion of Lemma 9.25 and Proposition 9.22 give the exact zero test and the complete positive head. This includes the strong prepared passages and their whole implicit compositions. No prepared bank has been replaced by a finite asymptotic expansion.

Step 2. We treat a positive order. Let R be the largest remaining order. For every order- R action, the first assertion of Lemma 9.25 gives a finite complete lower presentation, with strictly fewer positive orders. The induction hypothesis gives its faithful comparison quotient and its complete head. Its logarithmic scale is $o(X)$, $X = s^{-R}$: each of its positive orders is strictly smaller than R , its finite growing phases also have smaller order, and the last ordinary power/logarithmic factor has logarithmic magnitude $O(\log X + \log \log X)$. The induction thus provides both injectivity and the quantitative lower-head hypothesis of Setup 9.11.

The independently constructed row followed by this lower quotient is bounded and linear. Its target is a Banach quotient of the complete lower presentation, with injective actual realization. Lemma 9.25 provides the exact compact square and the whole uniform tail on the fixed parent sector. Proposition 9.13 therefore proves equality of the order- R row kernel and the entire actual kernel. Its zero branch uses the cofinal cutoffs and the full parent remainder, so it includes all infinite sums and every cancellation. Its nonzero branch selects a least surviving action only after the lower row has been tested as a whole function.

More explicitly, for the surviving action β_* choose one fixed cutoff with $k > \beta_*$ and $a_0 k - C_0 > \beta_* + 1$. The complete complement is then smaller than its selected head. The later compact actions form a finite set, and their positive exponential gaps dominate the logarithmic scales of their entire lower coefficients. This gives the complete head at order R , with no assertion about the sign of an uncollected summand.

Finite exterior modules are handled within this induction by first grouping equal growing phases faster than R . Distinct real such phases are eventually ordered; their differences dominate X and hence the uncut exponential bound. Terms of order exactly R give the finite action translates in the cutoff lemma. The remaining phases belong to the lower coefficients. At the ordinary base their comparison is the final assertion of Proposition 9.22. This supplies the stated complete heads also for these finite modules.

Step 3. We retain the original quotient and families. For an algebra presentation the actual map is a bounded unital homomorphism on one fixed small domain. Its kernel is therefore a closed ideal. The kernel equality just proved makes the quotient faithful, and the norm is the infimum of the original input norms. All bounded rows and source operations descend with those same bounds by Proposition 4.20. The source constructions used full norms on the original larger parameter disks; their Cauchy costs precede coefficient extraction and numerical action collection. At a fixed physical parameter the first nonzero row can change or disappear; the argument selects it anew at that parameter. If the resulting whole value is real, its leading constant is real by division by its positive scale and passage to the limit. This is Definition 2.3.

The ordered inverse formulas, complete lowering sums and cutoff-independent parent estimates are respectively (9.1), (9.71) and (9.94). They were constructed before the comparison induction and hence do not depend on the kernel equality they prove. $\square$

9.8. The model scalar initialization. We construct the scalar input at an identity strong word. The relative endpoint source is kept on its original disks. Its comparison is particularly explicit because the model graph has only its original unit atom.

Lemma 9.26. *Fix $0 < \varepsilon_0 < a < a_+ < \pi/2$ and $R > 1$. Put $q_0 = 1$, $q_+ = \exp(i(\pi/2 - a + \varepsilon_0))$, $q_- = \overline{q_+}$ and $Q(u) = u_0 + q_+u_+ + q_-u_-$ for $u \in [0, \infty)^3$. Let $\mathcal{M}_{2R}$ be the convolution algebra of measures with norm $\int e^{-2R|u|} d|\nu|(u)$, where $|u| = u_0 + u_+ + u_-$. If f is holomorphic on a neighborhood of $\{|z| \geq R, |\arg z| \leq a\}$ inside $\{|\arg z| < a_+\}$ and $|f(z)| \leq M|z|^{-\epsilon}$ there, with $\epsilon > 0$, there exists a linear contour construction J_R , independent of M and ϵ , such that*

$$(9.95) \quad \int e^{-Q(u)z} d(J_R f)(u) = f(z), \quad \|J_R f\|_{2R} \leq MR^{-\epsilon} \left(\frac{a}{\pi} + \frac{1}{\pi\epsilon \sin \varepsilon_0} \right)$$

on $\Re(q_0 z), \Re(q_+ z), \Re(q_- z) > 2R$. The source consists of three separately labelled axis densities, has no origin atom, and has a full positive contour majorant with the displayed bound. Uniform bounds on original larger parameter disks give Banach holomorphy and coefficient bounds with their strict Cauchy costs.

Proof. Use the following densities on the three axes, for $t > 0$:

$$(9.96) \quad \begin{aligned} j_0(t) &= \frac{1}{2\pi i} \int_{-a}^a e^{tRe^{i\phi}} f(Re^{i\phi}) iRe^{i\phi} d\phi, \\ j_+(t) &= \frac{q_+ e^{ia}}{2\pi i} \int_R^\infty e^{q_+ t r e^{ia}} f(r e^{ia}) dr, \\ j_-(t) &= -\frac{q_- e^{-ia}}{2\pi i} \int_R^\infty e^{q_- t r e^{-ia}} f(r e^{-ia}) dr. \end{aligned}$$

The last two integrals converge because $\Re(q_+ e^{ia}) = \Re(q_- e^{-ia}) = -\sin \varepsilon_0$. The first density has weighted norm at most $MR^{-\epsilon} a/\pi$. Each of the others has norm at most

$$\frac{M}{2\pi} \int_R^\infty \frac{r^{-\epsilon}}{2R + r \sin \varepsilon_0} dr \leq \frac{MR^{-\epsilon}}{2\pi\epsilon \sin \varepsilon_0}.$$

Taking absolute values inside the complete contour integrals gives the claimed positive majorant.

For each axis integrate its exponential kernel first in t . On the corresponding contour this gives $1/(z - w)$. The three terms are the inner circular arc, the outward upper boundary, and the inward lower boundary, with denominator $z - w$. For (9.96), Cauchy's formula on a truncated sector then gives $f(z)$: the outer circular contribution has modulus $O(ML^{-\epsilon})$ as its radius L tends to infinity. The absolute estimates above justify Fubini and this limit. They also give a norm bound $ML^{-\epsilon}/(\pi\epsilon \sin \varepsilon_0)$ for the omitted infinite-contour tails. These bounds hold on the full larger parameter disks. Limits of the finite contour integrals are therefore Banach holomorphic; their Cauchy coefficient and derivative formulas commute with the complete source construction. $\square$

Proposition 9.27. *Consider the real analytic model*

$$(9.97) \quad \dot{u} = u^2, \quad \dot{v} = -v,$$

with incoming section $v = b > 0$, outgoing section $u = a > 0$, incoming coordinate $\sigma = u$, and outgoing coordinate $v/(b \exp(1/a))$. Use the identity strong word, its original endpoint observation, and a separately labelled reciprocal mark, both with output germ $w \mapsto w$. Their full relative input, with smoothing integer 3, is

$$(9.98) \quad Z(\sigma, X) = \sigma^{-1}(1 + \sigma^3 X)^{-1}, \quad v_X = \exp(-Z(\sigma, X)), \quad t_X = \exp(-Z(\sigma, X)).$$

The variable X ranges over a fixed strict disk inside its original larger disk. All model sections and positive logarithm branches are fixed.

For any prescribed positive radii $\rho = (R_S, R_E, R_V, R_T)$ and $\mathbf{r}$ in a finite tuple τ , retain the original scalar Wiener algebra

(9.99)

$$\mathcal{P} = \left\{ F = \sum_{\alpha, i, j, m, n} a_{\alpha i j m n} \tau^\alpha S^i E^j V^m T^n : \|F\|_{\mathcal{P}} := \sum_{\alpha, i, j, m, n} |a_{\alpha i j m n}| \mathbf{r}^\alpha R_S^i R_E^j R_V^m R_T^n < \infty \right\},$$

where all indices are nonnegative integers. There exists a bounded unital source construction on this entire algebra, on a sufficiently small sector about the positive σ ray and the fixed parameter disks, whose whole synthesis is

$$(9.100) \quad \text{Act } F(\sigma, X, \tau) = F(\tau, \sigma, \sigma \log \sigma, v_X, t_X).$$

It retains the two endpoint and mark labels separately, their original native origin atoms, their compensated sources and complete relative source lifts. Every mark power is an ordered power of one fixed mark source. Every original source window retains its full complementary source and its own norm. For some $C > 0$ and $0 < \theta < 1$, the source norm of the complete outer tail of total (S, E, V, T) degree greater than N is at most $C\theta^{N+1}\|F\|_{\mathcal{P}}$. The constants precede all outer coefficients; all parameter bounds use the original norms or the fixed strict Cauchy costs on larger disks.

Write $a_{ijmn}(\tau) = \sum_{\alpha} a_{\alpha i j m n} \tau^\alpha$. The independent complete comparison rows are the bounded linear maps

$$(9.101) \quad D_{k\ell j} F(\tau) = (-1)^j \sum_{m+n=k} a_{\ell-j, j, m, n}(\tau), \quad k, \ell \geq 0, \quad 0 \leq j \leq \ell.$$

Their common kernel equals the whole analytic kernel of (9.100). It is a closed ideal, and the quotient is faithful in the infimum of the original norm. At a fixed parameter τ , if the realized value is nonzero, let k_* be the least index with a nonzero row, ℓ_* the least ordinary degree at that index, and j_* the greatest nonzero logarithmic degree at (k_*, ℓ_*) . Uniformly on each fixed strict X disk, on the positive σ ray,

$$(9.102) \quad \text{Act } F(\sigma, X, \tau) = D_{k_* \ell_* j_*} F(\tau) \exp(-k_*/\sigma) \sigma^{\ell_*} (-\log \sigma)^{j_*} (1 + o(1)).$$

The remainder includes the whole original input. Its coefficient is real when the whole realized value is real. At $X = 0$ the actual endpoint passage and its supplied inverse are $C(\sigma) = \exp(-1/\sigma)$ and $C^{-1}(w) = (-\log w)^{-1}$. The literal next height furnished by the mark is

$$(9.103) \quad H_*(\sigma) = 1/t_0 = \exp(1/\sigma), \quad \log H_* = 1/\sigma, \quad \sigma = (\log H_*)^{-1}.$$

No new native bank or nonidentity physical word is part of this scalar initialization.

Proof. We first identify the physical model data and construct their original compensated sources. We then transport the two labelled sources on the full relative disk, complete the original scalar Wiener input, and prove its independent kernel and whole remainder.

Step 1. We construct the actual model and its original atoms. Along an orbit of (9.97), $dv/du = -v/u^2$. Integrating from (σ, b) to the outgoing section gives

$$v_{\text{out}} = b \exp(1/a - 1/\sigma).$$

The prescribed section coordinate therefore gives $C(\sigma) = e^{-1/\sigma}$, with the stated positive inverse. The normalizing coordinates and both section corrections are the identity for this equation. At the identity strong word all positive discrepancy sources and all preceding prefix corrections vanish. The relative parameter is still independent: replacing σ by $\sigma(1 + \sigma^3 X)$ gives exactly (9.98) on its whole disk. The integer 3 is fixed here; it exceeds both physical orders if the order-two model is also prelisted. It is unrelated to any later native root order.

In the endpoint source the nonconstant graph measure is zero, the whole graph is δ_0 , and the weak output germ is the identity. Thus its original ordered weak coefficients are $\mu_1 = \delta_0$ and

$\mu_n = 0$ for $n \neq 1$. This is the original native atom, with native carrier zero. For the mark the output unit is also one and its clock and root integers are both one. Its single original source has primary action one, native carrier zero and derivative index zero. All its powers have their ordered factor labels and integer primary action equal to their degree. The endpoint and mark remain separately labelled implementations throughout.

Fix any $\kappa > 0$ and let $h(U)$ be the far positive inverse of $Y - \kappa \log Y$. Put $L_0(U) = h(U)/U$ and $d(U) = \kappa \log L_0(U)$. Use the complete convergent seed realization J_{seed} constructed in Lemma 9.24. For the original endpoint and, in a separate labelled copy, for the original mark, the compensated measure is

$$(9.104) \quad \eta = \delta_1 * Q_\kappa * J_{\text{seed}}(L_0^{-\kappa}), \quad Q_\kappa(dy) = \frac{y^{\kappa-1}}{\Gamma_E(\kappa)} dy.$$

It is supported at $y \geq 1$ and retains the independent native origin atom. Its synthesis and its ordered powers are

$$(9.105) \quad \text{Lap } \eta(U) = e^{-h(U)}, \quad \eta^{*n} = \delta_n * Q_{\kappa n} * J_{\text{seed}}(L_0^{-\kappa n}) \quad (n \geq 1).$$

Since $h = U + \kappa \log U + \kappa \log L_0$, Laplace multiplication gives the first equality. The second follows by absolute convolution and the beta integral for the Q factors. It is an identity of the complete measures after the ordered seed coefficients are pushed forward. The original ordered labels can still be retained before that pushforward. At a sufficiently large weight A ,

$$\|\eta\|_A \leq e^{-A} A^{-\kappa} \|J_{\text{seed}}(L_0^{-\kappa})\|_A.$$

The last factor tends to one as A increases; the nonconstant seed norm tends to zero. Every fixed exponential moment in y is paid by a further increase of A . These estimates control the full compensated measure, including its unbounded tail.

Step 2. We construct the full relative source. Write $z = 1/\sigma$ and

$$Z(z, X) = z(1 + z^{-3}X)^{-1}, \quad U(z, X) = Z(z, X) - \kappa \log Z(z, X).$$

Choose a slightly larger X disk than the disk to be used, still strictly inside the original disk. On a fixed buffered proper z sector, uniformly on that larger disk, $Z/z \rightarrow 1$ and $U/z \rightarrow 1$. Consequently $\Re U(z, X) \geq c|z|$ for some $c > 0$ on fixed inner contours, beyond a common radius. Fix $0 < d_* < c/4$. For $y \geq 1$ define the complete cone kernel

$$(9.106) \quad N_y(X) = \delta_{(d_*, 0, 0)} * J_R[z \mapsto e^{d_* z - y U(z, X)}],$$

using Lemma 9.26. All its angles and its radius R are fixed before y . The bound

$$|z| |e^{d_* z - y U(z, X)}| \leq \frac{4}{ce} \exp(-cyR/2) \quad (|z| \geq R, y \geq 1)$$

follows by reserving $cy|z|/2$ for the displayed exponential and maximizing $xe^{-cx/4}$ on $x > 0$. Thus the complete positive contour norm of N_y is at most $K_R \exp(-cyR/2)$, uniformly on the larger X disk. The kernels are measurable in y by their contour formulas. Their norm limits on compact y intervals and their full dominating tail also justify Banach integration against η .

Increase R once so that $cR/2$ exceeds the weight of any prescribed original exponential enhancement of η . Integrating (9.106) against that original measure gives two separate source copies

$$(9.107) \quad A_v(X) = \int N_y(X) d\eta_v(y), \quad A_t(X) = \int N_y(X) d\eta_t(y).$$

The subscripts on η record the endpoint and mark labels; their scalar measures have the formula (9.104). Each copy keeps its primary action one, native carrier zero, generated carrier y , contour label and cone carrier. Fubini and (9.105) give the exact synthesis $e^{-h(U(z, X))} = e^{-Z(z, X)}$. All branches agree on the positive model and then on the connected chosen complex domain.

These sources are supported in the first cone coordinate $u_0 \geq d_*$. Once constructed they remain fixed as the cone weight $B \geq 2R$ increases, and their whole positive norms decrease by at least $e^{-(B-2R)d_*}$. Their norms on any fixed strict X disk therefore tend to zero. For the mark power we use exactly $A_t(X)^{*n}$, with all ordered factors. A separate application of J_R to the n th scalar power is not used as its source. The construction of J_R was linear, and does not assert multiplicativity.

Every originally prescribed window is multiplication on its own retained labels before the integrations above. Splitting that multiplier into its compact and complementary parts gives an exact whole-minus-window identity by linearity. Its norm is the integral of its own absolute multiplier against the full positive majorant. The exponential reserves pay each prescribed polynomial moment before integration. The primary action, the generated y and the target cone carrier are distinct coordinates in these statements.

Step 3. We complete the source in the original scalar norm. The first-axis source for $S = 1/z$ is $Q_1(dy) = dy$. The source for $E = -\log z/z$ is $(\log y + \gamma_E) dy$, where $\gamma_E = -\Gamma'_E(1)$. The gamma-integral derivative used in Lemma 9.24 gives these exact syntheses. Their weighted norms tend to zero as $B \rightarrow \infty$, with bounds $1/B$ and $O((1 + \log B)/B)$. Together with (9.107) these are four complete source arguments.

To measure the relative family in a Banach space, choose $\rho_X < \rho'_X$ within its original larger disk and take the Wiener norm in X at ρ_X . A supremum bound M_X at ρ'_X gives the full norm bound $M_X/(1 - \rho_X/\rho'_X)$. The preceding support and seed bounds allow one fixed larger B to place each of the four source arguments inside θ times its originally prescribed radius, for some $0 < \theta < 1$. Taking their ordered products coefficientwise in (9.99) gives

$$(9.108) \quad \left\| \sum_{i+j+m+n>N} \sum_{\alpha} a_{\alpha i j m n} \tau^{\alpha} A_S^{*i} * A_E^{*j} * A_v^{*m} * A_t^{*n} \right\| \leq \theta^{N+1} \|F\|_{\mathcal{P}}.$$

The estimate precedes every original coefficient sum. The complete products retain each endpoint and mark copy and every parameter and contour coordinate. The scalar synthesis is bounded on a farther common cone; absolute summation proves (9.100) on the entire original input. The chosen source of the unit is the separate origin atom.

For a family with a supplied holomorphic supremum bound on larger τ disks, Cauchy's formula first gives the full coefficient sum on strict disks, with the factor $\prod_j (1 - r_j/r_j^+)^{-1}$. A fixed derivative has its separate factorial and radius cost. The positive contour majorants are uniform on those larger disks, so these operations precede every source and outer degree. Relative coefficient tails satisfy the additional geometric factor $(\rho_X/\rho'_X)^{N+1}$ with that full larger-disk bound. At the physical identity prefix the relative correction is zero, so its insertion is simply evaluation at $X = 0$. The full relative family was constructed before this insertion.

Step 4. We identify independent rows and the whole kernel. The original two labels both have primary action one and primary derivative index zero. At total action k their ordered finite fiber is exactly $m + n = k$. Its complete ordinary coefficient is

$$(9.109) \quad f_k(\sigma, \tau) = \sum_{i,j \geq 0} \sum_{m+n=k} a_{i j m n}(\tau) \sigma^{i+j} (\log \sigma)^j.$$

This is an absolutely convergent ordinary power/logarithmic series on the chosen small sector. Its coefficient at $\sigma^{\ell} (-\log \sigma)^j$ is precisely (9.101). Each extraction is bounded from the original norm: the finite sum has $i = \ell - j$, fixed j and $m + n = k$, and costs at most

$$R_S^{-(\ell-j)} R_E^{-j} \max_{m+n=k} (R_V^{-m} R_T^{-n}).$$

This defines the rows before evaluating any function. The ordering is increasing k , then increasing ℓ , then decreasing j within that finite logarithmic fiber.

For the exact relative input set

$$(9.110) \quad B_0(\sigma, X) = \exp\left(\frac{\sigma^2 X}{1 + \sigma^3 X}\right).$$

It is a whole convergent unit, tends uniformly to one on the fixed X disk, and satisfies $e^{-Z} = e^{-1/\sigma} B_0$. For example its original Wiener norm on radii r_σ, r_X with $r_\sigma^3 r_X < 1$ is bounded by $\exp(r_\sigma^2 r_X / (1 - r_\sigma^3 r_X))$. Absolute summation in the original input norm gives the whole identity

$$(9.111) \quad \text{Act } F = \sum_{k \geq 0} e^{-k/\sigma} B_0(\sigma, X)^k f_k(\sigma, \tau).$$

It checks the synthesis of the retained source; it does not replace its labels or source windows.

For precision, write $r_* = \min(R_V, R_T)$ and shrink the fixed sector so that $|\sigma| \leq R_S$, $|\sigma \log \sigma| \leq R_E$ and $r_*^{-1} e^{-\Re Z} < 1$, uniformly on the fixed disks. The complete action tail of (9.111) obeys

$$(9.112) \quad \left| \sum_{k > N} e^{-k/\sigma} B_0^k f_k \right| \leq \|F\|_{\mathcal{D}} (r_*^{-1} e^{-\Re Z})^{N+1}.$$

This follows term by term from the original norm weights: each $V^m T^n$ contributes a ratio at most $(r_*^{-1} e^{-\Re Z})^{m+n}$. Thus the estimate includes all ordinary, parameter and positive degrees, before any cancellation.

If every row vanishes at a fixed τ , each collected ordinary coefficient in (9.109) vanishes. The entire expansion is then zero for all X . Conversely choose the first nonzero row at that parameter. For its fixed k_* , the finite degree- ℓ_* polynomial has leading logarithmic term $D_{k_* \ell_* j_*} F \sigma^{\ell_*} (-\log \sigma)^{j_*}$. The full higher ordinary-degree tail is smaller. After factoring its first degree, it is bounded by a constant times $|\sigma|^{\ell_*+1} (1 + |\log \sigma|)^{\ell_*+1}$ on a strict smaller ordinary radius. Dividing by the displayed leading term makes this tend to zero. The lower ordinary degrees vanish by their finite coefficient collection, and the smaller logarithmic powers at degree ℓ_* are also negligible.

All actions below k_* vanish identically. Bound (9.112) with $N = k_*$ for all the larger actions. On the positive ray $Z = 1/\sigma + O(\sigma^2)$ uniformly on the full X disk. The resulting factor $e^{-(k_*+1)/\sigma}$ times a fixed constant is smaller than the chosen action- k_* head. Also $B_0^{k_*} = 1 + o(1)$. These facts prove (9.102) with the entire original remainder. The positive scale there proves nonvanishing on the tail. If the whole value is real, division by that scale and passage to the limit make its coefficient real. This proves the kernel equality at every fixed parameter. For the full parameter family a nonzero analytic row has a nonzero value at some interior parameter, so the family kernel is exactly the common kernel of the bounded row maps as well.

Finally, actualization is a bounded unital algebra map into bounded holomorphic functions on a fixed small domain. Its kernel is a closed ideal. The independent kernel equality therefore gives the faithful Banach quotient in the infimum of the original norm, by Proposition 4.20. Taking $X = 0$ in the one original mark gives (9.103); its logarithm and inverse use the fixed positive branch. This completes the scalar initialization from its actual identity word and original endpoint data. $\square$

10. RETAINED SOURCE FORESTS

The source class below records the whole data used at each fold. Its closure retains inactive branches and original source norms. Two orders are used separately. Completed lower native levels supply the parent charts, reciprocal sources and bounded inclusions before transport at the next native level. Within a fixed level, strictly slower weak classes are completed at their own drivers before transport, whereas comparable classes retain their whole residuals and are compared jointly. Neither order is the physical chronology of births. In the parameter calculations below, $D_\mu = \partial_\lambda^\mu$ denotes differentiation in the fixed original parameters at their

zero section. Coefficient notation $h_\mu = [\lambda^\mu]h$ is distinct from $D_\mu h = \mu!h_\mu$. The polynomial $\mathcal{B}_\mu$ denotes the finite polynomial determined by $D_\mu \exp J = (\exp J)\mathcal{B}_\mu((D_\nu J)_{0 < \nu \leq \mu})$.

Definition 10.1 (Coordinate records). Fix a finite source construction, its original norms and branches, and its increasing order of actual growth classes. This order is distinct from the ancestry depth and from the execution order of physical maps. At an actual driver u , write $u_x = x + i\eta(x)$, where $\eta, \eta' \rightarrow 0$. A coordinate record consists of the following data.

- (1) Before adjoining the weak class comparable to u , each fixed nonzero element of the field formed from strictly slower classes has its complete head with logarithmic magnitude $o(x)$, its correct-coordinate differentiated numerator maps, and its actual transported domain. No newly adjoined u -weak belongs to this field.
- (2) A finite structural list of coordinate maps, normalized phases, protected units and denominator clearers is fixed before a variable source input, together with a finite derivative order k . Its subprincipal members have actual buffered disks of radius $\exp(-q(x))$, with $q = q_{\mathcal{L},k}(x) = o(x)$, on which the selected nonzero denominators are zero-free and the required values, reciprocals and derivatives have bounds $\exp(C_k q(x))$. Explicit exponential clocks retain their clock factors: their logarithms, rather than their absolute derivatives, enter this list.
- (3) Every fixed nonzero completed readout retains, at its own driver v , its full law on fixed buffered disks:

$$(10.1) \quad \frac{f(v+z)}{f(v)} = e^{-\lambda_f z}(1 + o(1)), \quad \frac{f'(v)}{f(v)} \rightarrow -\lambda_f, \quad \lambda_f \in \mathbb{R}.$$

Fractions retain the difference of the actions. Their absolute inverse costs are not included in q unless their logarithmic magnitudes are subprincipal at the driver under consideration. For example, $f(v) = e^{-\tau v}$ has inverse logarithmic magnitude $\tau \Re v$.

- (4) Each appropriate whole Banach stage has its original-norm realization on every sufficiently far tube $|\Im u - \eta(\Re u)| < \exp(-a \Re u)$, for each fixed $a > 0$. The domain and operator constant precede the variable input. Original native damping, relative radii, analytic disks, inverse disks and complete tails remain separate data. An ordinary coordinate change also retains the actual fixed-disk displacement and whole pulled-domain hypotheses of Lemma 11.21 (proved below independently of forest reconstruction).

The function q is not an ancestry profile. The inequalities for A_h, b_h and the first-removed-factor reserves remain independent records. Full sources involving faster factors keep their original separate domains. These assertions concern each fixed finite list and order, not one domain or bound for all field elements or all orders.

Lemma 10.2 (Finite structural buffers). *Let $u_x = x + i\eta(x)$ with $\eta, \eta' \rightarrow 0$. Fix finitely many fractions whose numerators and denominators belong to finitely many already constructed Banach stages. Assume these stages have whole original-norm bounds on every far tube of width $\exp(-a \Re u)$, $a > 0$. Assume every selected nonzero numerator and denominator has logarithmic magnitude $o(x)$ on u_x . For each prescribed finite order k , there exists a positive $q(x) = o(x)$ such that the selected fractions, their requested inverses and derivatives through order k have bounds $\exp(C_k q(x))$ on buffered disks of radius $\exp(-q(x))$ about u_x , with every selected denominator zero-free. The list and k precede q ; no estimate uniform over variable inputs or over the whole fraction field is asserted.*

Proof. Fix $a > 0$. Since $\eta' \rightarrow 0$, a disk of radius $c \exp(-ax)$ about u_x lies in the given tube for a small fixed $c > 0$. The whole-stage bounds give a constant upper bound for each selected numerator and denominator there. Cauchy's estimate gives $|d'| \leq C_a \exp(ax)$ on the half disk for every selected denominator d . Choose $r(x) = o(x)$ so that all their central moduli, and those of numerators whose inverses are requested, are at least $\exp(-r(x))$. On a disk of radius

$c_a \exp(-ax - r(x))$, integration of the derivative bound gives $|d(u) - d(u_x)| \leq |d(u_x)|/2$. Thus d is zero-free and $|d^{-1}| \leq 2 \exp(r(x))$. Products give bounds for the finitely many fractions; Cauchy on a further strict disk gives derivatives through order k with cost $\exp(O_k(ax + r(x)))$.

Perform this construction for $a = 1/n$. Choose increasing thresholds so that the finitely many constants are absorbed and $r(x)/x$ is less than $1/n$ beyond the n th threshold. On consecutive threshold intervals choose the corresponding radii, shrinking once more for a fixed buffer. Their negative logarithms have a positive sublinear majorant q . A continuous larger majorant can be used without changing $q = o(x)$. This proves the assertion from the old whole-tube bounds, without using a horizontal law of a new source. $\square$

Lemma 10.3 (Transport through exponential charts). *Suppose an already constructed inverse of $P = \exp J_P(u)$ has, on every fixed bounded P disk, $|\Delta u| = O(|P|^{-1}) = O(\exp(-\gamma x + o(x)))$, where $\gamma > 0$. For a finite structural list with the buffers and derivative bounds of Definition 10.1, every listed coordinate v with $|v'_u| \leq \exp(C_k q(x))$ on its selected buffered disk, where $q(x) = o(x)$, satisfies $|\Delta v| = O(\exp(-\gamma x + o(x)))$. Fixed readout denominators are transported by their complete own-driver relative laws, not by an absolute inverse estimate.*

Let $X = \exp J_X(u)$ be an old strictly slower principal. Assume $X/P \rightarrow 0$, $J'_P(u_x) \rightarrow \gamma$, and $J'_X(u_x)$ is bounded. Assume also that these logarithms are holomorphic on each far exponentially thin u tube and bounded there by $C_a(1+x)$. Then, for every fixed $R > 0$,

$$(10.2) \quad \frac{dX}{dP} = \frac{X}{P} \frac{J'_X}{J'_P}, \quad \sup_{|z| \leq R} |X(P+z) - X(P)| = O(X/P) \rightarrow 0.$$

Consequently an old completed X stage with a fixed own-driver disk, whole norm bound and complete relative law supplies a transported subprincipal field at P . The whole bound is pulled back first; Lemma 10.2 then supplies the new finite structural buffers. The assertion holds also for $H_X = X - a \log X - \log c$ with fixed $a \in \mathbb{R}$, $c > 0$.

Proof. We prove the structural and sibling displacement estimates under the stated analytic hypotheses, then pull back the completed own- X law.

Step 1. We transport the finite structural coordinates.

The displacement Δu lies in every selected $\exp(-q(x))$ buffer, since $q = o(x)$. Integration of the structural derivative gives the first assertion. For a structural denominator both its derivative and its reciprocal enter the bound, so its relative change has modulus $O(\exp(-\gamma x + o(x)))$. A full own-driver denominator instead uses (10.1) with the already proved displacement in its driver.

Step 2. We prove the sibling displacement estimate under the stated tube hypotheses.

For the sibling assertion, Cauchy on an old tube gives $J''_X = O((1+x) \exp(2ax))$ on a buffered disk. Choose $2a < \gamma$. On the inverse displacement $O(P^{-1})$ the variation of J'_X is $o(1)$; hence it stays bounded there. The inverse construction also keeps J'_P bounded away from zero. Integration gives $J_X(u(P+z)) - J_X(u(P)) = O(P^{-1})$ uniformly for $|z| \leq R$. Exponentiation proves the second assertion of (10.2); the first is the exact chain rule. Since $dH_X/dX = 1 - a/X$, the normalized driver has the required displacement too.

Step 3. We pull back the completed own- X law and the whole operator bounds.

Only now insert (10.2) in the old own- X law. Its exponential tends to one, its complete relative error tends to zero, and a larger fixed P disk gives the derivative law by Cauchy. The old whole operator bound restricts to this actual pullback without normalization of any variable input. Its fixed heads have logarithmic magnitude $O(X) = o(P)$. They and the pulled whole-stage bounds meet Lemma 10.2 at P . For finitely many stages and denominators, diagonalizing fixed disk radii $1, 2, \dots$ gives a slowly widening common horn. In particular both $X = \exp(\beta u)$, $0 < \beta < \gamma$, and $X = P/\log P$ satisfy this displacement estimate; neither requires X'_u to be subexponential. For an ordinary change such as $P = u^2$, the fixed-disk inlet must instead be supplied directly: its displacement $O(1/u)$ need not lie in a buffer of radius $\exp(-\sqrt{u})$. $\square$

Lemma 10.4 (Stored-principal transport inlet). *Consider a native level- j receiving step in Set-up 10.6, or in the mixed construction of Definition 11.26 below. Assume that all lower native levels through $j - 1$ have completed records. These include the already constructed current- u inverse chart, its whole proper strong realization on every far exponentially thin tube, and the bounded inclusions and cuts carrying the earlier reciprocal sources. The corresponding old-parent stages have their whole tube bounds and complete differential-field heads in their own coordinates. For distinct root parents at the minimum height, retain the actual model or basal charts.*

Let $u_x = x + i\eta(x)$, with $\eta, \eta' \rightarrow 0$. Suppose the new $P = \exp J_P(u)$ already has its genuine inverse with $|\Delta u| = O(|P|^{-1})$ on every fixed bounded P disk, $\Re J_P(u_x) = \gamma x + o(x)$ and $J'_P(u_x) \rightarrow \gamma > 0$. Assume also that, for every $a > 0$, J_P is holomorphic on the selected sufficiently far current- u tube of width $\exp(-a\Re u)$ used in Lemma 10.3, and satisfies $|J_P(u)| \leq C_a(1 + \Re u)$ there for some $C_a > 0$. Let X be a stored principal with retained full clock

$$(10.3) \quad \log X = \gamma_X V + \ell_X \log V + c_X + \rho_X(V),$$

where $\gamma_X > 0$, the constants and logarithmic branch are fixed, and ρ_X is a bounded old proper strong source. Assume $X = O(P)$ on the physical curve and that the completed current- u differential field contains V and u . Then $V = O(u)$, $V/u \rightarrow L$ and $V'_u \rightarrow L$ for a finite L . The function $J_X(u) = \log X(V(u))$ has bounded physical derivative and is holomorphic on every far exponentially thin current- u tube, with $|J_X(u)| \leq C_a(1 + \Re u)$ there. If $X = o(P)$ and its own- X stage has already been completed in growth order, Lemma 10.3 therefore gives (10.2) and transports its complete relative law and whole original-norm bound. These conclusions also apply to its retained normalized driver $X - a \log X - \log c$, with $a \in \mathbb{R}$, $c > 0$.

Proof. We use only completed lower native levels to obtain the clock hypotheses, and use the completed own- X law only after proving complex displacement.

Step 1. We treat the identity parent and the physical derivative bounds. If $V = u$, use the identity chart at every height. Then $V/u = V'_u = 1$. Lemma 8.22(2) at this old parent gives $\rho'_X \rightarrow 0$, and its retained whole tube bound gives the asserted bound for J_X directly. Its displacement is the already constructed $O(|P|^{-1})$ displacement of u . No reciprocal construction is needed in this case.

Assume henceforth that $V \neq u$. Keep the entire stored logarithm. When $X = O(P)$, its positive physical leading terms give $V = O(u)$. In the completed current- u differential field, Lemma 8.22(2) gives $V/u \rightarrow L$ and $V'_u \rightarrow L$. In the old differential field with derivative d/dV , boundedness of ρ_X and that lemma give $\rho'_X(V) \rightarrow 0$. These are two different derivative coordinates. The exact chain rule is $J'_X = (\gamma_X + \ell_X/V + \rho'_X)V'_u$, so J'_X is bounded.

Step 2. We obtain the reciprocal source from completed lower native levels. The new principal P has native level j , and all uses of lower charts and reciprocal constructions in this step are at levels at most $j - 1$. A higher-depth old principal cannot satisfy $X = O(P)$ by (10.17). In the equal-level case, u and V have depth $d + j - 1$. If V has an old birth, that birth has level $j - 1 \geq 1$. Denote its principal by S . The retained birth identity (10.10) gives

$$(10.4) \quad \begin{aligned} S &= K_v \exp(e_{j-1} \Xi_v), \\ V &= S - a_v \log S - \log c_v, \\ d_V &:= \frac{1}{V} = S^{-1} \left(1 - \frac{a_v \log S + \log c_v}{S} \right)^{-1}. \end{aligned}$$

The exact original first root supplied in the ordered-source data of Set-up 10.6 gives K_v^{-1} by taking its original integer power. The full projected correction has its positive and negative Banach exponential series at this completed lower height, so $S^{-1} = K_v^{-1} \exp(-e_{j-1} \Xi_v)$ is a whole source. The normalized logarithmic slot and strict inverse unit in (10.4) are the whole-source reciprocal construction of Lemma 11.27. Every ingredient is fixed by the cut (10.18) at $j - 1$. Thus d_V belongs to a finite previous-height proper strong stage, without either the old

X -weak or the new P -weak. For a lower-depth old principal whose parent has an old birth, its corresponding reciprocal source is already retained by this cut.

The carrying chart is the already completed current- u chart, not merely the native V chart. The whole parent-tube data of Set-up 10.6 restrict by its old cuts and bounded inclusions to the stage containing d_V . In the mixed construction these data are (11.64), carried by the genuine principal or normalized chart of Lemmas 11.16 and 11.28 at the already completed height that produced u . The retained-forest construction keeps these earlier charts by Proposition 10.28. These references are used one height lower in the induction, before the current level- j extension. They do not use the present X -to- P transport or a future weak law.

On the physical current- u curve, $V \rightarrow +\infty$ in its retained normalized sense and $|V| \leq Cx$. Hence

$$|\log |d_V(u_x)|| = O(\log x) = o(x).$$

Apply Lemma 10.2 to the particular numerator 1 and denominator d_V on this current- u whole-tube stage. For each fixed finite order at least two it gives buffered disks of radius $\exp(-q_u(x))$, with $q_u = o(x)$, on which d_V is zero-free and V, d_V and the required u derivatives have bounds $\exp(C_k q_u(x))$. No other fraction denominator is needed. The inverse motion of size $O(|P|^{-1})$ lies inside these disks. Integrating the bound for V_u'' along that motion gives

$$|V'_u(u) - V'_u(u_x)| \leq O(|P|^{-1}) \exp(C_k q_u(x)) = o(1).$$

The bounded physical derivative V'_u already obtained therefore stays bounded throughout the motion, and $|\Delta V| = O(|P|^{-1})$.

Step 3. We retain the base charts and the whole current-coordinate tubes. At the minimum height, distinct root parents use the original model inlet rather than a preceding birth. With $s = \sigma^{-1}$, the whole scalar sources $z = 1/s$ and $l = \log(1/s)/s$, their derivative maps, and the literal logarithm, positive power of a logarithm, square-root and exponential-of-square-root coordinate changes are those of Lemma 10.30. Proposition 10.31 retains their completed atlas and whole pullback bounds. A normalized basal coordinate uses the literal inverse ψ and its whole-source pullback in Lemma 11.15. These supplied base charts give the current-coordinate inlet; no arbitrary change between root parents is inferred.

For every fixed $b > 0$, the second-derivative buffer argument also keeps V'_u bounded across a current- u tube of width $\exp(-bx)$, and gives $|\Delta V| = O(\exp(-bx))$. Choose an old V -tube exponent $a > 0$ with $aC < b$, where $|V(u_x)| \leq Cx$. The current- u tube then maps strictly inside that old V tube on a far tail. The entire old source ρ_X and the fixed logarithm branch pull back there. Consequently the full clock

$$J_X(u) = \gamma_X V(u) + \ell_X \log V(u) + c_X + \rho_X(V(u))$$

is holomorphic and bounded by $C_b(1+x)$ on each such tube. This supplies the complex-domain hypotheses for (10.2) before its Cauchy estimate and before the completed own- X law is used. Prepared normalized clocks are first reduced to their retained principal chart with their full correction.

If $X = o(P)$, the clock hypotheses of Lemma 10.3 have now been obtained from completed lower native data. That lemma proves the complex-disk displacement before using the already completed own- X law, and then gives the asserted whole-source transport and normalized-driver conclusion. No current X law has been deduced from a future P law. $\square$

Definition 10.5 (Original-parameter positive-jet record). Fix the original parameters λ , their correlations and norms, the physical branches, observations and exterior modules of a finite raw construction. Hold its positive structural rates and original leading data fixed. For every fixed finite set of multi-indices, with maximum order k , an original-parameter positive-jet record consists of the following separate data.

- (1) At each actual own driver u , specify the normalized structural quantities used by the finite recipe, including logarithmic clocks, normalized relative increments and protected units. After retaining all complete faster coefficient targets separately, each newly exposed pure structural multiplier A_μ has its established pre-field, correct-coordinate derivatives and actual complex domain, and satisfies

$$(10.5) \quad \log^+ |A_\mu| = o(\Re u).$$

Zero terms are omitted. This requirement does not concern all parameter derivatives of arbitrary complete field readouts.

- (2) Retain the exact finite product, chain and Bell-polynomial formulas in the original parameters before physical specialization. They give bounded linear maps from the whole carried input into finite marked targets, using the original projective norm for multilinear raw factors. For $h = \Phi(h; \lambda)$, retain its original buffered unknown disk and a contraction constant $\vartheta_{\text{inv}} < 1$; its coefficient recursion is

$$(10.6) \quad (1 - \partial_h \Phi(h_0; 0))h_\mu = [\lambda^\mu] \Phi(h_{<|\mu|}; \lambda), \quad \|(1 - \partial_h \Phi)^{-1}\| \leq (1 - \vartheta_{\text{inv}})^{-1}.$$

Only lower total degrees of the unknown occur on the right.

- (3) Every newly growing factor has a specified paying positive factor at its actual fold. A positive native Euler term is divided as a whole by its exact original first root T_e^{-1/P_e} before its structural multiplier is attached. A positive weak monomial retains one occupied weak factor. A prepared derivative retains its whole original carrier, gap, linkage, ordered inverse convolution and proper-subtree moments, using spare carrier weight for fixed moments. A cut derivative retains the first-removed-factor decomposition before differentiation; extra fixed reference powers are paid by the weighted cut at that removed height. Faster old coefficient and observation targets stay whole at their later folds. Each admitted positive mark has its actual source decomposition into finitely many terms of form

$$(10.7) \quad c e^{-av+r(v)} U, \quad a > 0, \quad r = o(v), \quad \Im r \rightarrow 0,$$

where $c \neq 0$ is fixed, r belongs to the already constructed pre-field of that fold, and U is a complete bounded coefficient or unit on its retained original domain. Scalar decay alone is not this source decomposition.

- (4) These finite marks are received before the relevant fold with all their derivative and horizontal inlets. Their fixed-order cofinal rectangle costs have the form $\exp(C_k T \log(eT))$. For weak total degree $n \geq 1$ a strict ratio $0 < \theta < 1$ pays

$$(10.8) \quad \sup_{n \geq 1} (1+n)^k \theta^{n/2} < \infty,$$

leaving $\theta^{n/2}$ for the entire tail. All original Euler, division, Cauchy and inclusion costs precede fresh receiving radii. Completed own-driver laws are proved in increasing growth order before transport to a later class.

- (5) Retain the finite strict intersection of the literal original root, logarithm, relative, analytic, damping and inverse domains. Every far physical point has a local chart $\lambda = r_t \eta$, whose positive radius is constant on a smaller driver disk, with exact identity

$$(10.9) \quad [\lambda^\mu] F = r_t^{-|\mu|} [\eta^\mu] F(r_t \eta).$$

No positive lower bound for r_t and no subexponential estimate for $r_t^{-|\mu|}$ is required. Independent original coefficient maps come from item (2), not from suppressing this recovery factor. Cuts commute with the fixed original parameter labels. Different multi-indices retain separate targets and norms, which are not summed.

Neither geometric coordinate buffers nor a positive reserve for one old root implies (10.5). For $0 < \sigma < \delta$, the product $e^{-\delta u} e^{\sigma u}$ still decays, but $e^{\sigma u}$ cannot be absorbed by an arbitrarily small future positive reference action. The normalized-jet bound is therefore an independent part of the receiving data.

Set-up 10.6. Fix a finite original normed source presentation $\mathcal{P}$, its bounded analytic realization on a connected neighborhood of a physical tail, and bounded linear comparison maps R_λ defined independently of that realization. Assume $\ker \text{Act} = \bigcap_\lambda \ker R_\lambda$ and put $Q = \mathcal{P} / \ker \text{Act}$, with its original infimum norm. The following are supplied data on this presentation.

Ancestry. A finite labelled rooted forest has base x_0 , current vertex v , and parent $p(b)$ for each nonbase birth b . Its actual positive section germs satisfy the exact identities

$$(10.10) \quad \begin{aligned} t_b &= D_b x_{p(b)}^{\eta_b} e^{-G_b}, & T_b &= \beta_b t_b^{-m_b} = K_b e^{\Xi_b}, & K_b &= C_b B_{e(b)}^{\alpha_b}, \\ q_b &= c_b e^{-T_b} T_b^{a_b}, & x_b &= d_b q_b e^{-\ell_b(T_b)} A_b(q_b e^{-\ell_b(T_b)}), \\ P_b &= K_b e^{\rho_{i_b} \Xi_b}, & u_b &= P_b - a_b \log P_b - \log c_b, & i_b &= l(e(b)). \end{aligned}$$

Here $D_b, \beta_b, C_b, c_b, d_b > 0$, $\eta_b, \alpha_b \in \mathbb{Q}_{>0}$, $m_b \geq 1$ is integral, $a_b \in \mathbb{R}$, and $A_b(0) = 1$. Every displayed function, logarithmic lift, original native correction, analytic disk and actual inverse is retained. The maps ρ_i are specified below. The current is $s = D x_v^\kappa e^{-G}$, $D > 0$, $\kappa \in \mathbb{Q}_{>0}$. Each stem e has level $l(e) \geq 1$ and its own parent coordinate u_e . At a nonbasal parent,

$$(10.11) \quad \log B_e = \gamma_e u_e + \log b_e + \log V_e(e^{-u_e}), \quad \gamma_e, b_e > 0, \quad V_e(0) = 1,$$

on its original strict analytic disk. At the base the supplied exact data are $x_0 = C_x e^{-P_0} P_0^{a_0} W_0$, its nonvanishing bounded whole source unit W_0 , and each basal reference $B_e = \beta_e x_0^{-r_e}$ with its exact root and logarithm:

$$(10.12) \quad B_e^{-1/P} = \beta_e^{-1/P} C_x^{r_e/P} e^{-r_e P_0/P} P_0^{a_0 r_e/P} W_0^{r_e/P}.$$

The integers r_e, P and all branches are supplied. No analyticity of W_0 as a function of e^{-P_0} is assumed. Birth labels with equal values remain distinct, including inactive births.

Ordered sources. The inventory of $\mathcal{P}$ is a finite composition of completed projective tensors of the original root banks and coefficient Wiener spaces, for each finite coordinate count d_c ,

$$(10.13) \quad \mathcal{W}_\rho(E) = \left\{ \sum_{\nu \in \mathbb{N}^{d_c}} f_\nu z^\nu : \sum_\nu \|f_\nu\|_E \rho^\nu < \infty \right\}.$$

It records every repeated physical label and parameter. Every structural map is a specified bounded analytic substitution, original relative-bank substitution, divided native remainder, or genuine inverse on stated strict domains. Scalar limiting functions of families are retained whole. The finite growth-class order of the comparison folds is part of the data; the physical recipes separately retain their execution order. At each native fold all the data of Set-up 9.14 and Construction 9.15 are supplied on the already constructed lower algebra: its independent rows and whole cofinal estimate, original-norm widening-horn bound, logarithmic clocks, exact first roots and positive action reserves, and the compatible original joint ray. The tensor uses the full original banks.

At each weak fold supply a differential field $\mathcal{K}$ of holomorphic germs in its actual driver T , containing $\mathbb{C}$, and faithful commutative Banach stages E_s with bounded coherent inclusions and $\mathcal{K} = \text{Frac}(\bigcup_s E_s)$. Each whole stage is bounded on its specified real tail and on compact subsets of its connected domain. For every s there exist a later u , fixed $c_j \in \mathcal{K}$, and bounded linear $B_j : E_s \rightarrow E_u$ such that $Da = \sum_j c_j B_j(a)$ for every $a \in E_s$. Supply $H \in \mathcal{K}$, $h = \Re H \rightarrow \infty$, $\Im H \rightarrow 0$, and, for each $a \in \mathcal{K}^\times$, a complete head $a = d_a \theta_a(1 + o(1))$, where $d_a \neq 0$, $\theta_a > 0$ and $|\log \theta_a| = o(h)$. For a finite weak-coordinate list of length d_w , the weak variables are exactly

$$(10.14) \quad w_j = e^{-\tau_j H + P_j}, \quad \tau_j > 0, \quad P_j \in \mathcal{K}, \quad P_j = o(h), \quad \Im P_j \rightarrow 0,$$

with specified original radii. For their finite fibers retain the witness lattice

$$(10.15) \quad L = \{k \in \mathbb{Z}^{d_w} : \exists b \in \mathcal{K}^\times, Db/b = k \cdot DP\}, \quad \beta_k = e^{k \cdot P(T_*)} b/b(T_*),$$

where T_* avoids the finitely many denominators used in a row. For $\tau \cdot \nu = t$, partition the finite fiber into classes modulo L , choose $\nu_C \in C$, and retain the bounded row maps

$$(10.16) \quad R_t F = \left(d_t \sum_{\nu \in C} f_\nu \beta_{\nu - \nu_C} \right)_C.$$

Here a nonzero fixed denominator clearer d_t and one later Banach target are chosen before F . Whenever the next construction uses a complex own-driver horn, also supply its actual chart with $H = x$: each fixed finite coefficient and denominator list and one whole stage have representatives on a widening zero-free horn, bounded there in the original stage norm; and $a(x + iy)/a(x) \rightarrow 1$ uniformly on $|y| \leq A$ for every fixed $a \in \mathcal{K}^\times$ and $A < \infty$. These maps precede the weak extension. Roots needed at the next native fold are formed as positive weak marks before that fold.

Depth and atlas. Fix the original depth offset d . A level- i native root has parent depth $d + i - 1$ and its weak variables have depth $d + i$. For each occupied depth h , let b_h be the minimum of its finite positive principal equivalents and let A_h be one plus their sum. Adjacent occupied depths satisfy

$$(10.17) \quad A_{h-1} = o(b_h), \quad \log A_h = O(A_{h-1}) = o(b_h).$$

At the bottom the supplied basal scale replaces A_{h-1} . For every retained parent u of depth $d + i - 1$, its appropriate whole strong stage E_i has a bounded original-norm realization on each fixed tube of width $e^{-a\Re u}$, $a > 0$, about the curve $u = x + i\eta(x)$, where $\eta, \eta' \rightarrow 0$. Retain the coordinate record of Definition 10.1, separately from (10.17). Its finite subprincipal structural lists have buffers $\exp(-q(u))$, $q = o(\Re u)$; completed readout denominators use their full own-driver relative laws. The proper strong own-parent inventory and explicit logarithmic clocks are distinguished from subsequently reinserted weak siblings. Retain positive ratio heads, correct-coordinate differentiated numerator maps and genuine strong inverse source maps. Deferred faster sources keep their separate original domains.

Original-parameter positive jets. Supply Definition 10.5 for every fixed finite original parameter order. In particular, its normalized structural subprincipality is an input independent of the coordinate atlas. Complete faster coefficient targets remain at their own folds; all positive actions, original norms, correlations and recovery factors are retained.

Cuts. On the original raw factors and every stored recipe let p_i retain native levels at most i , replace later native banks by their scalars, and erase positive weak and auxiliary labels of depth at least $d + i$. This truncation is performed in each coefficient and each argument of a recipe. Supply independently faithful retained targets E_i and contractive unital maps

$$(10.18) \quad \rho_i : Q \rightarrow E_i, \quad \iota_i : E_i \rightarrow Q, \quad \rho_i \iota_i = \text{id}, \quad e_i = \iota_i \rho_i, \quad e_i e_j = e_{\min(i,j)},$$

realizing these raw operations and commuting with old inclusions. The whole raw difference has a finite first-removed-factor decomposition: a term has one distinguished positive weak factor, or one native first root obtained by bounded whole division of the scalar-zero native factor; its remaining coefficient map is bounded in the original projective/Wiener norm before all degrees. The distinguished factors have strictly positive reserves at their removed depths. In particular

$$(10.19) \quad |\text{Act } f - \text{Act } e_i f| \leq C_i \|f\|_Q e^{-c_i b_{d+i}}, \quad c_i > 0,$$

on a common tail, with zero error if the removed list is empty.

Positive certificates. Every G_b and G is physically real, scalar zero, and has a finite exact identity, at its own parent u (at P_0 for the base),

$$(10.20) \quad G = \sum_{\mu=1}^N e^{-t_\mu u} u^{h_\mu} A_\mu, \quad t_\mu > 0, \quad h_\mu \in \mathbb{R}.$$

Each A_μ is a complete bounded coefficient source, possibly with later factors. Each scalar-zero Ξ_b has a finite exact certificate made from positive root or weak factors, bounded whole native division, analytic differences, and already constructed cut discrepancies. The certificate lists all whole coefficient and inclusion costs and the fresh receiving copies that give any prescribed small norm after these costs are fixed. Original input radii are unchanged.

Families and observations. Each original family keeps its full original norm and domain. Composed domains are the connected physical components of the literal inverse images of those domains, with strict source, damping, analytic disk and inverse margins; their parameter map is the restricted identity. They contain a neighborhood of every sufficiently far physical point at parameter zero. For every fixed original multi-index ν supply a bounded whole-input linear map J_ν into its separately faithful marked target, and its finitely many derivative targets, with $\text{Act } J_\nu f = [\lambda^\nu] \text{Act } f$. All Cauchy, derivative-factorial, inverse, degree and exterior costs are included. These target norms are not summed. Every observation, window, ordered moment and full complement keeps its original labelled node, root, ray, lift, domain, norm and codomain. Finite exterior modules retain their separate norms; p_i is an algebra operation and is not applied to them.

Proposition 10.31 supplies this set-up for the constructed model base. The explicitly bounded receiving operations below retain its original data and construct the required output records. The mixed extension is stated in Theorem 11.2 using Definition 11.26; that definition allows positive real ancestry exponents and is not an identification with the rational-ancestry assumptions of Set-up 10.6.

10.1. Whole native tensors.

Definition 10.7. Fix an integer $d \geq 1$, a complex number q with $|q| = 1$ and $\Re q > 0$, and its logarithm with imaginary part in $(-\pi/2, \pi/2)$. For $1 \leq e \leq d$, fix an integer $P_e \geq 1$, constants $r_e, g_e > 0$, and $0 < \vartheta_e < \pi/2 - |\arg q|$. Put

$$(10.21) \quad D_e = \{|\zeta| < r_e\} \cup \{\zeta \neq 0 : |\arg \zeta| < \vartheta_e/P_e\}.$$

The native bank $\mathcal{B}_e$ consists of pairs $v = (v_0, \phi)$, where $v_0 \in \mathbb{C}$, ϕ is holomorphic on D_e , $\phi(0) = 0$, and the following original norm is finite:

$$(10.22) \quad \|v\|_e = |v_0| + 8P_e g_e^{-1/P_e} \sup_{\zeta \in D_e \setminus \{0\}} \frac{e^{-g_e |\zeta|^{P_e}} (1 + g_e |\zeta|^{P_e})^{1+1/P_e} |\phi(\zeta)|}{|\zeta|}.$$

Its realization, for $\Re(qY) > g_e$, and its coefficient maps are

$$(10.23) \quad \begin{aligned} S_e(v)(Y) &= v_0 + \int_0^\infty e^{-qY\xi} \phi(\xi^{1/P_e}) \frac{d\xi}{\xi}, \\ b_{e,0}(v) &= v_0, \\ b_{e,j}(v) &= \Gamma(j/P_e) q^{-j/P_e} [\zeta^j] \phi \quad (j \geq 1), \end{aligned}$$

where $\Gamma(t) = \int_0^\infty e^{-\xi} \xi^{t-1} d\xi$ for $t > 0$. The integral uses the positive root of ξ and the specified logarithm of q . Every bank retains its own $r_e, g_e, P_e, \vartheta_e$. Its product and Banach algebra structure are those of Set-up 9.4 and Lemma 8.2.

Set-up 10.8. Let w be continuous and nondecreasing, with $w(x) \rightarrow \infty$ and $w(x) \leq \sqrt{x}$ for $x > x_0 > 1$. Fix

$$(10.24) \quad H_w = \{x + iy : x > x_0, |y| < w(x)\}, \quad \gamma(x) = x + i\eta(x), \quad \eta(x) \rightarrow 0,$$

where η is continuous. Let E be a unital commutative Banach algebra, with a bounded injective unital algebra homomorphism $A: E \rightarrow \text{Hol}(H_w)$ satisfying $\sup_{H_w} |Af| \leq L\|f\|_E$. Let Λ be the additive monoid generated by finitely many strictly positive real numbers, so $0 \in \Lambda$. For every $\lambda \in \Lambda$, fix a Banach space V_λ and a bounded linear map $C_\lambda: E \rightarrow V_\lambda$ with common kernel zero. Assume that the first nonzero row of each nonzero $f \in E$ determines $c_f \neq 0$ and $\theta_f(x) > 0$ such that

$$(10.25) \quad Af(\gamma(x)) = c_f e^{-\lambda_f x} \theta_f(x) (1 + o(1)), \quad |\log \theta_f(x)| = o(x),$$

where $\lambda_f = \min\{\lambda : C_\lambda f \neq 0\}$. Assume also that constants $C_0, C_1, c_1 > 0$ and $T_0 \geq 1$, independent of f and T , satisfy

$$(10.26) \quad [C_\lambda f = 0 \text{ for all } \lambda \leq T] \implies |Af(z)| \leq C_0 \|f\|_E e^{C_1 T \log(eT) - c_1 T \Re z} \quad (z \in H_w, T \geq T_0).$$

Supply holomorphic clocks K_e on H_w , constants $k_e > 0$, and elements $t_e \in E$ such that

$$(10.27) \quad \sup_{\substack{e, z \in H_w \\ \Re z \geq X}} |K'_e(z) - k_e| \longrightarrow 0, \quad \Im K_e(\gamma(x)) \longrightarrow 0, \\ W_e = e^{K_e}, \quad At_e = e^{-K_e/P_e}, \quad a_e = k_e/P_e \in \Lambda.$$

The roots must satisfy the complete reserve condition

$$(10.28) \quad C_\lambda \left(f \prod_{e=1}^d t_e^{j_e} \right) = 0 \quad \text{if} \quad \lambda < \sum_{e=1}^d a_e j_e, \quad f \in E, \quad j \in \mathbb{N}^d.$$

All these data precede the addition of the banks in Definition 10.7.

Definition 10.9. Under Set-up 10.8, the original native source space is

$$(10.29) \quad \mathcal{P} = E \hat{\otimes}_\pi \mathcal{B}_1 \hat{\otimes}_\pi \cdots \hat{\otimes}_\pi \mathcal{B}_d.$$

The projective norm is the completion of the infimum of $\sum_\ell \|f_\ell\|_E \prod_e \|v_{\ell,e}\|_e$ over finite representations $\sum_\ell f_\ell \otimes v_{\ell,1} \otimes \cdots \otimes v_{\ell,d}$. For $T \geq 0$, define on simple tensors

$$(10.30) \quad Q_T \left(f \otimes \bigotimes_{e=1}^d v_e \right) = \sum_{\substack{0 \leq j_e \leq \lfloor T/a_e \rfloor \\ 1 \leq e \leq d}} \left(\prod_{e=1}^d b_{e,j_e}(v_e) \right) f \prod_{e=1}^d t_e^{j_e}.$$

Lemma 10.10. Each Q_T extends to a bounded linear map $\mathcal{P} \rightarrow E$. For $\lambda \in \Lambda$, the map

$$(10.31) \quad \mathcal{R}_\lambda = C_\lambda Q_T, \quad T \geq \lambda,$$

is independent of T . Constants $C, C' > 0$, depending only on the fixed data, satisfy

$$(10.32) \quad \|Q_T\| \leq C \exp(C'T \log(e + T)) \quad (T \geq 1).$$

Proof. Fix $0 < \rho_e < r_e$. The coefficient integral on $|\zeta| = \rho_e$, together with (10.22), gives

$$|b_{e,j}(v)| \leq C_e \rho_e^{-j} \Gamma(j/P_e) \|v\|_e \quad (j \geq 1).$$

For $u \geq 1$, bounding $t^u e^{-t/2}$ by its maximum and integrating $e^{-t/2}$ gives $\Gamma(u+1) \leq 2(2u/e)^u$. The finitely many indices with $j/P_e < 1$ contribute another fixed constant. Thus the coefficient bound is at most $C_e (D'_e)^j j^{j/P_e} \|v\|_e$ for a fixed $D'_e > 0$. Submultiplicativity in E bounds each term of (10.30) by $\|f\|_E \prod_e (|b_{e,j_e}(v_e)| \|t_e\|_E^{j_e})$. The rectangle contains at most $\prod_e (1 + T/a_e)$ terms, and every index is $O(T)$. Summation proves (10.32) in the projective norm and hence proves continuous extension.

If $T' > T \geq \lambda$, every term added to the rectangle has $j_e > \lfloor T/a_e \rfloor$ for at least one e . Its total action exceeds T . Condition (10.28) kills its C_λ row. This proves stabilization on simple tensors, and boundedness extends the identity to $\mathcal{P}$. $\square$

Lemma 10.11. *Under Set-up 10.8, the whole map*

$$(10.33) \quad \text{Act} \left(f \otimes \bigotimes_e v_e \right) (z) = Af(z) \prod_e S_e(v_e)(W_e(z))$$

extends boundedly from the original projective tensor to holomorphic functions on a sufficiently far fixed strip containing the physical curve. If $F \in \mathcal{P}$ has a nonzero row, let $\lambda_ = \min\{\lambda : \mathcal{R}_\lambda F \neq 0\}$. Then its first complete lower row gives $c_* \neq 0$, $\theta_*(x) > 0$, and $\epsilon_F(x) \rightarrow 0$ such that*

$$(10.34) \quad \text{Act } F(\gamma(x)) = c_* e^{-\lambda_* x} \theta_*(x) (1 + \epsilon_F(x)), \quad |\log \theta_*(x)| = o(x).$$

This equality includes the entire F . If its realization is real, c_ is real. In particular,*

$$(10.35) \quad \ker \text{Act} \subseteq \bigcap_{\lambda \in \Lambda} \ker \mathcal{R}_\lambda.$$

Proof. We first bound the complete integrals, then estimate a finite rectangle against the entire tensor, and finally use the first surviving row.

Step 1. The complete integrals are bounded. Integrating $K'_e - k_e$ along the real axis gives $K_e(x) = k_e x + o(x)$. Integration along bounded vertical segments gives $K_e(x + iy) = K_e(x) + ik_e y + o(1)$ on each fixed strip. The segment from x to $\gamma(x)$ has length tending to zero, so (10.27) also gives $\Im K_e(x) \rightarrow 0$. Choose a fixed strip narrow enough that $\Re(qW_e(z)) \geq \delta |W_e(z)|$ for every e , with $\delta > 0$. Its far tail lies in H_w and contains γ . The defining norm bounds the integrand by a constant times

$$\|v_e\|_e e^{-(\delta |W_e| - g_e)\xi} \xi^{1/P_e - 1} (1 + g_e \xi)^{-1 - 1/P_e}.$$

This is integrable, uniformly for sufficiently large $|W_e|$. It bounds each full S_e before any tensor sum. Multiplication by Af and taking projective infima proves boundedness of (10.33). The integrals and their absolutely convergent source sums are holomorphic on the strip.

Step 2. The entire omitted part has a strict action gap. For one bank fix $0 < r'_e < \rho_e < r_e$ and an integer $n \geq 0$. On $0 \leq \xi \leq (r'_e)^{P_e}$, the complete Taylor remainder of its minor is bounded by $C_{e,n} \|v_e\|_e \xi^{(n+1)/P_e}$. Its integral has bound $C'_{e,n} \|v_e\|_e |W_e|^{-(n+1)/P_e}$. On $\xi \geq (r'_e)^{P_e}$ the full minor has exponential damping. The subtracted finite polynomial has its own exponentially small integral on this interval. Integration of each monomial over $(0, \infty)$ gives the coefficient in (10.23), by the change of scale in the defining gamma integral and holomorphy in the right half-plane. Consequently

$$\left| S_e(v_e)(W_e) - \sum_{j=0}^n b_{e,j}(v_e) e^{-jK_e/P_e} \right| \leq C_{e,n} \|v_e\|_e \left(|W_e|^{-(n+1)/P_e} + e^{-c_{e,n}|W_e|} \right).$$

All constants precede the variable bank input. Telescope the finite product, keeping every other factor whole or equal to its bounded finite polynomial. With $n_e = \lfloor T/a_e \rfloor$, the result on γ is

$$(10.36) \quad |\text{Act } F(\gamma(x)) - A(Q_T F)(\gamma(x))| \leq C_T \|F\|_\pi e^{-b_T x + o(x)}, \quad b_T = \min_e (n_e + 1)a_e > T.$$

The error includes the whole minor above each splitting point and the whole subtracted-polynomial integral. These integral estimates hold uniformly on the fixed thin strip chosen above. In particular, for every fixed buffered disk $|\zeta| \leq r$ in that strip,

$$(10.37) \quad \sup_{|\zeta| \leq r} |\text{Act } F(x + \zeta) - A(Q_T F)(x + \zeta)| \leq C_{T,r} \|F\|_\pi e^{-b_T x + o(x)}.$$

Integrating (10.27) gives $K_e(x + \zeta) - K_e(x) = k_e \zeta + o(1)$ uniformly there, so every damping margin and remainder bound used in the integral proof is uniform. The $o(x)$ and all constants

precede F . Projective completion preserves both estimates because they were proved before the sum of simple tensors.

Step 3. The first row determines the full head. A monoid generated by finitely many positive numbers has finite intersection with every bounded interval: each generating multiplicity is bounded separately. Thus λ_* exists. Fix $T \geq \lambda_*$. Lemma 10.10 identifies the first nonzero lower row of $Q_T F$ with λ_* . Apply (10.25) to $Q_T F$. Dividing (10.36) by $e^{-\lambda_* x} \theta_*(x)$ gives an error tending to zero, since $b_T > \lambda_*$ and $|\log \theta_*| = o(x)$. This proves (10.34). For a real realization the ratio by the positive scale has real limit, so $c_* \in \mathbb{R} \setminus \{0\}$. A nonzero row therefore prevents a zero realization, which proves (10.35). $\square$

Remark 10.12. Lemma 10.11 does not prove the reverse inclusion in (10.35). If all rows vanish, (10.36) and (10.26) give estimates of arbitrarily large fixed exponential order on the physical curve. Such estimates alone do not imply that a holomorphic germ is zero. For example, $\exp(-\exp z)$ is nonzero and is smaller than $\exp(-Tx)$ on the positive real axis for every fixed $T > 0$. The zero-row implication requires the quantitative reconstruction on widening domains and the corresponding finite angular continuation argument.

10.2. Strong cuts and original quotient norms.

Lemma 10.13. *Let $\mathcal{P}$ and $\mathcal{P}_i$ be unital Banach algebras with bounded unital algebra homomorphisms Act and Act_i to analytic functions on a common physical tail. Suppose each has independently defined bounded rows whose common kernel equals its analytic kernel; denote these closed ideals by K and K_i . Let $p_i: \mathcal{P} \rightarrow \mathcal{P}_i$ and $j_i: \mathcal{P}_i \rightarrow \mathcal{P}$ be contractive unital homomorphisms such that*

$$(10.38) \quad p_i j_i = \text{id}, \quad \text{Act } j_i = \text{Act}_i.$$

Assume $b(x) \rightarrow \infty$ and $C, c > 0$ satisfy, for every $F \in \mathcal{P}$,

$$(10.39) \quad |\text{Act } F(x) - \text{Act}_i p_i F(x)| \leq C \|F\|_{\mathcal{P}} e^{-cb(x)}.$$

Assume also that every nonzero $\text{Act}_i G$ is eventually nonvanishing and $|\log |\text{Act}_i G(x)|| = o(b(x))$. Then p_i and j_i induce contractive unital maps $\rho_i: \mathcal{A} \rightarrow \mathcal{A}_i$ and $\iota_i: \mathcal{A}_i \rightarrow \mathcal{A}$, where $\mathcal{A} = \mathcal{P}/K$ and $\mathcal{A}_i = \mathcal{P}_i/K_i$ have their original infimum norms. They satisfy $\rho_i \iota_i = \text{id}$, and ι_i is injective. The estimate (10.39) descends with $\|F\|_{\mathcal{P}}$ replaced by the quotient norm.

If $j_i p_i j_h p_h = j_{\min(i,h)} p_{\min(i,h)}$ on the raw source, the endomorphisms $e_i = \iota_i \rho_i$ satisfy

$$(10.40) \quad e_i e_h = e_{\min(i,h)}.$$

Every commuting square of such raw maps descends to the quotients.

Proof. Take $F \in K$. If $\text{Act}_i p_i F$ were nonzero, its logarithmic estimate would give $|\text{Act}_i p_i F(x)| \geq e^{-cb(x)/2}$ eventually. This contradicts (10.39), whose right side is $C \|F\|_{\mathcal{P}} e^{-cb(x)}$. Thus $p_i K \subseteq K_i$. The realization identity gives $j_i K_i \subseteq K$. Both quotient maps are therefore defined. Taking infima over all original representatives proves their norm bounds. The raw identity $p_i j_i = \text{id}$ descends and proves injectivity of ι_i . The left side of (10.39) is independent of the representative, so taking its infimum proves the quotient estimate. The product identity and each commuting square pass to classes by applying their raw identities to a representative. No choice of representatives with a bounded selection map is used. $\square$

Lemma 10.14. *In the setting of Lemma 10.13, fix a scalar function h on the physical tail. Let B be a Banach algebra with injective analytic realization Act_B . Suppose finitely many bounded linear maps $D_\ell: \mathcal{P} \rightarrow B$ and fixed $u_\ell \in B$ satisfy*

$$(10.41) \quad \text{Act}_B \left(\sum_{\ell=1}^m u_\ell D_\ell F \right) = h(\text{Act } F - \text{Act}_i p_i F) \quad (F \in \mathcal{P}).$$

Then the formula on the left induces a bounded linear map $W: \mathcal{A} \rightarrow B$, with

$$(10.42) \quad \|W\| \leq \sum_{\ell=1}^m \|u_\ell\|_B \|D_\ell\|.$$

Separately, if $\log^+ |h(x)| = o(b(x))$, then

$$(10.43) \quad |h(x)(\text{Act } f(x) - \text{Act } e_i f(x))| \leq C \|f\|_{\mathcal{A}} e^{-cb(x)/2}$$

on a sufficiently far tail.

Proof. The finite sum in (10.41) has norm at most the right side of (10.42) times $\|F\|_{\mathcal{P}}$. If $F \in K$, cut descent makes its realization zero. Injectivity in B makes the sum zero, so it descends; taking original infima proves the norm bound. For the scalar assertion, choose a tail with $|h(x)| \leq e^{cb(x)/2}$ and multiply the whole cut estimate. The scalar assertion supplies no elements u_ℓ or maps D_ℓ . To obtain a receiving norm at most ε , their construction must first bound all D_ℓ in the original input norm and then arrange $\sum_\ell \|u_\ell\| \|D_\ell\| \leq \varepsilon$. $\square$

10.3. Exact ancestry at a return.

Definition 10.15. A retained ancestry record has finitely many labeled positive section germs x_b , a base x_0 , and a parent $p(b)$ for each nonbase label, with no cycle in the parent relation. For each birth retain the exact identities

$$(10.44) \quad \begin{aligned} t_b &= D_b x_{p(b)}^{\eta_b} e^{-G_b}, & T_b &= \beta_b t_b^{-m_b} = K_b e^{\Xi_b}, \\ q_b &= c_b e^{-T_b} T_b^{a_b}, & x_b &= F_b(q_b e^{-\ell_b(T_b)}), \end{aligned}$$

where $D_b, \beta_b, c_b > 0$, $\eta_b \in \mathbb{Q}_{>0}$, $m_b \geq 1$ is integral, $a_b \in \mathbb{R}$, and $F_b(z) = d_b z A_b(z)$ with $d_b > 0$ and $A_b(0) = 1$. The original logarithms, powers, native corrections ℓ_b , and analytic maps F_b are part of the record, on their specified domains. The current section has the form

$$(10.45) \quad s = D x_v^\kappa e^{-G}, \quad D > 0, \quad \kappa \in \mathbb{Q}_{>0}.$$

Changing v to its parent does not delete any birth or source in the record. An ancestry record alone specifies no Banach presentation.

Lemma 10.16. For the data of Definition 10.15, a positive analytic map $I(s) = d_I s A_I(s)$, with $A_I(0) = 1$, changes the current record to

$$(10.46) \quad (D, \kappa, G) \mapsto (d_I D, \kappa, G - \log A_I(s)).$$

If $v = b$ is a nonbase birth and $Z_R > 0$ is the actual inverse height of a supplied reverse passage of order $r \geq 1$ and constant $\beta_R > 0$, put $Y_o = \kappa T_b$. Before the inverse incoming analytic map, its positive inner output $(\beta_R/Z_R)^{1/r}$ has parent record

$$(10.47) \quad \begin{aligned} D_R &= \left(\frac{\beta_R}{\kappa \beta_b} \right)^{1/r} D_b^{m_b/r}, \\ \kappa_R &= \eta_b m_b / r, \\ G_R &= (m_b/r) G_b + \frac{1}{r} \log \frac{Z_R}{Y_o}. \end{aligned}$$

For a direct re-entry at v with a stored outgoing birth b , order $M \geq 1$ and constant $\beta > 0$, its exact new height is

$$(10.48) \quad \begin{aligned} \theta &= \frac{\kappa M}{\eta_b m_b}, & C &= \beta D^{-M} (\beta_b D_b^{-m_b})^{-\theta}, \\ K_{\text{new}} &= C K_b^\theta, & \Xi_{\text{new}} &= \theta \Xi_b + M G - \theta m_b G_b, \\ \beta s^{-M} &= K_{\text{new}} e^{\Xi_{\text{new}}}. \end{aligned}$$

All identities use the original compatible logarithmic lifts. These algebraic update identities also hold for positive real ancestry and current exponents whenever the displayed full birth and current identities and compatible logarithmic lifts are supplied; their validity does not require rationality of those exponents.

Proof. Substitution of (10.45) into I gives (10.46). For the reverse, use the stored $T_b = \beta_b t_b^{-m_b}$ to write

$$\left(\frac{\beta_R}{Z_R}\right)^{1/r} = \left(\frac{\beta_R}{\kappa\beta_b}\right)^{1/r} t_b^{m_b/r} \exp\left(-\frac{1}{r} \log \frac{Z_R}{Y_o}\right).$$

Inserting the complete t_b identity gives (10.47). In particular the exponent at the returned parent is $\eta_b m_b / r$; the current κ has already entered Y_o and the positive constant. For a re-entry, taking the positive power θ gives

$$T_b^\theta = (\beta_b D_b^{-m_b})^\theta x_v^{-\kappa M} e^{\theta m_b G_b}.$$

Multiply by $C e^{MG - \theta m_b G_b}$ and use $T_b = K_b e^{\Xi_b}$ to obtain (10.48). These substitutions keep every term of G , G_b , and Ξ_b . $\square$

Lemma 10.17. *Suppose a completed direct output with trailing analytic maps has the exact form $y = F(e^{-u} e^{-E})$, where $F(z) = DzA(z)$, $D > 0$, $A(0) = 1$, on fixed disks with compatible logarithms. For $M \geq 1$ integral and $\beta > 0$, set*

$$(10.49) \quad z = e^{-u}, \quad w = ze^{-E}, \quad N = \beta F(z)^{-M}, \quad \Delta = E + \log A(z) - \log A(w).$$

Then, for every original integer root order $P \geq 1$,

$$(10.50) \quad \begin{aligned} \beta y^{-M} &= N e^{M\Delta}, \\ N^{-1/P} &= \beta^{-1/P} D^{M/P} e^{-Mu/P} A(e^{-u})^{M/P}, \\ \log N &= Mu + \log(\beta D^{-M}) - M \log A(e^{-u}). \end{aligned}$$

If $\log A(t) = ta_A(t)$ on the original disk, its full analytic difference is

$$(10.51) \quad \log A(z) - \log A(w) = z(a_A(z) - e^{-E} a_A(ze^{-E})).$$

Proof. The exact quotient $F(w)/F(z) = e^{-E} A(w)/A(z)$ is $e^{-\Delta}$. Taking its $(-M)$ -th power proves the height identity. The formula $F(z) = De^{-u} A(e^{-u})$ gives the root and logarithm with the stipulated lifts. Substitution of $w = ze^{-E}$ into $\log A(t) = ta_A(t)$ proves (10.51) without truncating either analytic function. In a source construction the exponential in the root has positive parent action M/P ; the bounded coefficient and the full difference in (10.51) must still be received in their original norms. $\square$

10.4. Whole weak extensions.

Lemma 10.18. *At a weak fold with the field, stages, whole differentiation maps and phase data of Set-up 10.6, the displayed maps R_t are bounded and have common kernel equal to the analytic kernel of the whole coefficient Wiener input. Their infimum quotient is a faithful Banach algebra. Every nonzero whole input has a first finite action fiber and a complete head*

$$(10.52) \quad \text{Act } F = d e^{-t^* h} \theta(h) (1 + o(1)), \quad d \neq 0, \quad \theta > 0, \quad |\log \theta| = o(h).$$

For each fixed $0 < \eta < 1$, on one tail before F and B ,

$$(10.53) \quad \left| \sum_{\tau \cdot \nu \geq B} f_\nu w^\nu \right| \leq C_s \|F\|_{s,\rho} e^{-\eta B h}.$$

The quotient field of the directed receiving stages is differential. Its whole-stage derivatives have bounded numerator maps with finitely many fixed field multipliers after a strict receiving restriction. Its nonzero elements have complete ratio heads with logarithmic magnitude $O(h)$.

Proof. The weak-fold data in Set-up 10.6 supply precisely Set-up 8.21, including its whole-stage differentiation maps and complete subprincipal field heads. The lattice, witness normalization and denominator-cleared finite rows coincide with those of Theorem 8.23. That theorem gives the asserted complete kernel, first fiber, whole tail and differentiated numerator maps with the original norms. Its fraction-field conclusion gives the stated ratio heads. $\square$

Lemma 10.19. *Suppose the additional own-driver horn and horizontal-head data in Set-up 10.6 hold, with $H = x$. Let $\mathcal{K}_{\text{res}} = \mathcal{K}(e^{P_1}, \dots, e^{P_{d_w}})$, the field of fractions of finite Laurent sums $\sum_j a_j e^{j \cdot P}$ with $a_j \in \mathcal{K}$. Every nonzero $f \in \mathcal{K}_{\text{res}}$ is horizontally stable, meaning*

$$(10.54) \quad \frac{f(x + iy)}{f(x)} \longrightarrow 1 \quad \text{uniformly on } |y| \leq A \quad \text{for each fixed } A > 0, \quad f'/f \longrightarrow 0.$$

Each whole weak stage admits a widening horn on which its whole tail satisfies

$$(10.55) \quad \left| \sum_{\tau \cdot \nu \geq B} f_\nu(z) w(z)^\nu \right| \leq C \|F\| e^{-\eta B \Re z}.$$

If the rows through B vanish, this bound also holds for the whole value. Its first surviving fiber has the horizontal form $e^{-t_ z} g(z)(1 + o(1))$, with $g'/g \rightarrow 0$ on each fixed strict strip. All bounds use the original input norm.*

Proof. The additional own-driver chart, zero-free horn representatives, whole-stage norm bound and fixed horizontal ratios in Set-up 10.6 are exactly the extra hypotheses of Theorem 8.25. Apply it after Theorem 8.23. Its entire action-tail estimate and original quotient infima give the bound also when all rows through the chosen cutoff vanish. For the first row retain its exact exponential factor and use its finite coefficient's horizontal ratio and logarithmic derivative. $\square$

Lemma 10.20. *The native input in Set-up 10.8 has*

$$(10.56) \quad \ker \text{Act} = \bigcap_{\lambda \in \Lambda} \ker R_\lambda, \quad R_\lambda = C_\lambda Q_T \quad (T \geq \lambda),$$

with its original projective and quotient norms, and the complete head of Lemma 10.11. If its lower field has whole differentiated numerator maps and each W'_e/W_e belongs to that field, the receiving field has such maps as well. Suppose in addition that each nonzero lower source a has its complete horizontal head, in the precise form

$$(10.57) \quad \frac{Aa(x + \zeta)}{Aa(x)} = e^{-\lambda_a \zeta} (1 + o(1))$$

uniformly on each fixed buffered disk in its strip, where λ_a is its first nonzero row. Then for each fixed nonzero native value $f = \text{Act } F$, on every sufficiently small fixed buffered disk,

$$(10.58) \quad \frac{f(x + \zeta)}{f(x)} = e^{-\lambda_* \zeta} (1 + o(1)), \quad \frac{f'(x + \zeta)}{f(x + \zeta)} = -\lambda_* + o(1).$$

The assertions hold for fractions with the difference of their first actions. The relative errors are for each fixed nonzero input, and the whole remainder bounds retain the original input norm.

Proof. The displayed lower setup is exactly Set-up 9.14; its rectangles and rows are those of Construction 9.15. The kernel equality, including the all-zero direction, therefore follows from Theorem 9.17. That use requires its whole cofinal bound and compatible continuations; fixed-order native expansions alone would not supply the conclusion.

For a bank of order P the whole Euler operator is

$$\mathcal{E}_P(v_0, \phi) = (0, -\zeta \phi'(\zeta)/P), \quad S(\mathcal{E}_P v)(W) = W \partial_W S(v)(W).$$

Cauchy on a strict minor domain and a larger weight makes this a bounded map between the specified original and receiving banks. On a tensor, differentiate one factor at a time and

multiply the native derivative by its fixed logarithmic derivative W'_e/W_e . Together with the supplied lower maps this is a finite sum of bounded whole numerator maps after one common denominator is cleared. Actual differentiation annihilates the old kernel, so faithfulness and infimum norms give descent. Fractions then obey the quotient rule.

On a sufficiently thin fixed strip all native damping margins remain positive. Fix a nonzero f and choose $T \geq \lambda_*$. The lower element $Q_T F$ has first row λ_* . Its supplied complete law (10.57) has a nonzero denominator on a far tail, and $|A(Q_T F)(x)| = \exp(-\lambda_* x + o(x))$ times a fixed positive constant. The entire complex remainder (10.37), divided by this quantity, tends to zero uniformly on the buffered disk because $b_T > \lambda_*$. Thus the complete native value has the first law in (10.58). This deduction uses the full tensor remainder, not just the physical-curve conclusion of the native reconstruction theorem.

For clarity, at a weak lower fold the required complete lower law follows from Lemma 10.19: its first finite fiber has exact factor $e^{-\lambda_* z}$ and horizontally stable residual coefficient; every later fixed fiber has a positive gap. Choose a fixed action cutoff with $\eta B > \lambda_*$ and use the whole action-tail bound for all remaining degrees. The finitely many retained fibers and that entire tail give the law uniformly on that buffered disk. At an already native lower fold use its previously constructed law.

Apply the derivative integral on the boundary of a buffered disk to the holomorphic normalized functions $f(x + \zeta)/f(x)$ on a slightly larger fixed disk. Their uniform limit is $e^{-\lambda_* \zeta}$, which is nonzero. Their derivatives converge uniformly on smaller disks; division gives the second law in (10.58). Taking quotients gives both assertions for fractions. $\square$

10.5. Original relative families and finite reconstruction.

Lemma 10.21. *For every original bank of order P choose strict relative radii $0 < r_0 < r_1 < 1$ and a receiving bank on its original ray such that*

$$(10.59) \quad \mathcal{D}_\eta(v_0, \phi) = (v_0, \phi((1 + \eta)^{-1/P} \zeta)) = \sum_{n \geq 0} \mathcal{D}_n(v) \eta^n, \quad \|\mathcal{D}_n\| \leq L r_0^{-n}.$$

Its whole actual value at Y is $S(v)(Y(1 + \eta))$. Let $\mathcal{B}'$ be this receiving bank and let $i : \mathcal{B}' \rightarrow \hat{Q}$ be a bounded unital homomorphism into a faithful Banach algebra, with $L_i = \|i\|$ and the actual square $\text{Act } i(b) = S(b)(Y)$ at the reference clock Y used above. If $\eta \in \hat{Q}$ has $\|\eta\| = a < r_0$, on the common strict actual domain its entire relative substitution is

$$(10.60) \quad \mathcal{R}_\eta v = \sum_{n \geq 0} i(\mathcal{D}_n v) \eta^n, \quad \text{Act } \mathcal{R}_\eta v = S(v)(Y(1 + \text{Act } \eta)), \quad \|\mathcal{R}_\eta v\| \leq \frac{L_i L \|v\|}{1 - a/r_0}.$$

The degree greater than N tail is at most this bound times $(a/r_0)^{N+1}$. For finitely many bank factors and a lower coefficient map of norm L_0 , the complete projective tensor bound is

$$(10.61) \quad L_0 \prod_e \frac{L_{i,e} L_e}{1 - a_e/r_{0,e}}.$$

The complement of the rectangle $n_e \leq N_e$ is bounded by this constant times the original tensor norm and $\sum_e (a_e/r_{0,e})^{N_e+1}$. A bounded whole first-division map is also available: $\mathcal{N}_1$ into a later bank with

$$(10.62) \quad S(v)(Y) - v_0 = Y^{-1/P} S(\mathcal{N}_1 v)(Y).$$

The maps commute with original absolutely summed coefficient families before their extraction.

Proof. Lemma 8.7, with $\delta = 1/P$, constructs the receiving domain, weight and bounded operator family and proves its entire actual identity on the original ray. Its coefficient bound is the displayed bound for $\mathcal{D}_n$. Apply i before multiplication by η^n . The complete norm sum is $L_i L \|v\| \sum_n (a/r_0)^n$, and its omitted part is the stated geometric tail. The actual square for i identifies the whole sum with the original relative family at Y . Multiplying these bounds on

simple tensors and taking projective infima proves the tensor assertion, with the lower coefficient-map norm included. The complement of a degree rectangle is a union of its individual degree tails; summing their bounds proves the asserted complement estimate.

Here is a whole formula for the division, including the small-root case $P = 1$. Put $\alpha = 1/P$, choose an integer $k > \alpha$, and set

$$\chi(\zeta) = \frac{\phi(\zeta) - [\zeta]\phi\zeta}{\zeta}, \quad I(\zeta) = \frac{1}{\Gamma(k - \alpha)} \int_0^1 t^{\alpha-1} (1-t)^{k-\alpha-1} \chi(\zeta t^{1/P}) dt.$$

With $D_0 = \zeta \partial_\zeta / P$, put

$$\mathcal{N}_1 v = (b_1(v), q^{-\alpha} D_0(D_0 + 1) \cdots (D_0 + k - 1) I).$$

The beta integral shows that the coefficient of ζ^j of its minor, multiplied by $\Gamma(j/P)q^{-j/P}$, is $b_{j+1}(v)$. The subtracted numerator vanishes to order two; hence $\chi(0) = 0$. On a smaller disk the coefficient formula bounds the quotient by the original supremum divided by the fixed disk-radius gap. On the sector use the original exponential bound; integration in t preserves it with a fixed polynomial cost. The finitely many Euler derivatives are bounded after a strict angular restriction and a larger weight. This proves a whole original-bank operator bound. Integrating the beta formula against the Laplace kernel, justified by these absolute bounds, gives the displayed actual identity. All operators are linear and bounded before the original coefficient sum, which proves the family assertion. $\square$

Lemma 10.22. *Let $A : \mathcal{P} \rightarrow Q$ be an original bounded map with $\ker A = \ker \text{Act}$, and let $B : \mathcal{P} \rightarrow \hat{Q}$ be a bounded map into an independently faithful receiver with $\widehat{\text{Act}}B = \text{Act}$. Then B induces a bounded injective map $Q \rightarrow \hat{Q}$ with its original infimum bound. Finite projective products and coefficient Wiener completions may be reordered before constructing these maps, without changing their norms or identifying correlated labels.*

Proof. The actual square and independent faithfulness give $\ker B = \ker \text{Act} = \ker A$. The induced map is therefore injective. Taking the infimum of $\|BF\| \leq \|B\| \|F\|_{\mathcal{P}}$ over every original representative proves its norm bound. Projective tensor symmetry is isometric on simple tensors and their finite sums by its defining infimum. Completion preserves the equality. Also $\mathcal{W}_\rho(E)$ is the weighted ℓ^1 sum of copies of E . Expanding a projective tensor into summable representatives proves, in both directions without changing the norm, its commutation with this sum. These operations permute slots; an originally shared coefficient is still one coefficient and an originally repeated parameter is still one parameter. $\square$

Lemma 10.23. *Under Set-up 10.6, one may jointly add finitely many compatible full native banks at CB_e^α , $C > 0$, $\alpha \in \mathbb{Q}_{>0}$, and finitely many positive marks whose phases in a supplied own-driver chart X are $-\tau X + P$, $\tau > 0$, with the supplied lower-field, horn and derivative data for $P = o(X)$, $\Im P \rightarrow 0$. The original whole source has a bounded injective inclusion into the resulting independent receiver. All original factors are reconstructed in increasing actual growth classes, retaining physical chronology inside their raw recipes and tied classes, including faster weak factors at their own later folds. The full original input bounds, cofinal remainders, differentiated numerator maps and horizontal heads persist.*

Proof. We construct the changed folds on the original raw factors and then propagate their output through the finitely many later folds.

Step 1. We construct the new native reference. For a nonbasal reference $M = CB_e^\alpha$ at parent u , the exact data are

$$\log M = \alpha \gamma_e u + \log(Cb_e^\alpha) + \alpha \log V_e(e^{-u}), \quad M^{-1/P} = C^{-1/P} b_e^{-\alpha/P} e^{-\alpha \gamma_e u/P} V_e(e^{-u})^{-\alpha/P}.$$

The first root has positive action $\alpha \gamma_e / P$. At the base use the displayed basal root formula in the setup, including its logarithmic power and whole unit. Add these finitely many positive marks

before the native fold. Their complete analytic units add only nonnegative actions; for degree j their original strict-disk coefficient norm is at most e^{Cj} , by the entire exponential of the bounded logarithm. Thus the rectangle cost stays $\exp(CT \log(eT))$ and the exact root reserve persists. Lemmas 10.18 and 10.19 construct the lower rows and cofinal estimate before the new native tensor. The exact logarithmic derivative tends to $\alpha\gamma_e > 0$; the analytic term has derivative tending to zero on a strict horn. The supplied basal atlas gives the corresponding assertion for basal references. The common original ray and Lemma 10.20 now give the entire joint native fold, including every old bank at that fold. Its complete own-parent law is (10.58), obtained from the lower horizontal law and the whole complex rectangle remainder. The real-tail head alone is not used for this conclusion.

On a sufficiently thin parent strip the common original ray and strict native damping margins bound each entire original bank. The full lower-stage bound and projective summation therefore give the whole original-norm bound for the new tensor, also on every far exponentially thin parent tube. This bound precedes the choice of any newly enlarged finite structural list. The complete heads and this whole bound give the hypotheses of Lemma 10.2 for its newly selected subprincipal members. Explicit clocks retain their exact factors and logarithms. In particular, the exact root $M^{-1/P}$ has positive parent action $\alpha\gamma_e/P$; the logarithmic magnitude of its inverse is $(\alpha\gamma_e/P)\Re u + o(\Re u)$. This cost is not included in a subprincipal buffer at u . The root retains its full action law under Definition 10.1(3); only at a genuinely faster receiving driver can its inverse cost be subprincipal. The basal case retains its separately supplied exact root and reserve.

Step 2. We transport it to later drivers. The order and the lower-field hypotheses persist as follows. At the changed parent u the resulting whole head on a fixed thin strip is

$$(*) \quad f(u) = e^{-\lambda u} g(u)(1 + \epsilon(u)), \quad |\log |g(u)|| = o(\Re u), \quad g'/g \rightarrow 0, \quad \epsilon \rightarrow 0.$$

Its whole original-norm remainder is the fixed-rectangle bound from Lemma 10.20. A later distinct parent X has $u = o(X)$. For an ordinary stored change, retain its actual fixed-disk displacement on the original whole pullback domain, as required in Lemma 11.21 (proved below independently of forest reconstruction). A derivative limit on the physical curve alone does not provide this domain. For a new exponential chart, first use Lemma 10.25 (proved below from raw tube bounds and a positive derivative limit) and its fixed-disk estimate (10.65). The exact logarithm and positive derivative required for the new reference were obtained in Step 1. Structural own-parent coordinates then use Lemma 10.3. For a reinserted slower principal, Lemma 10.4 supplies its logarithmic-clock hypotheses from completed lower native levels, and (10.2) gives its motion on that genuine inverse chart. Thus the small complex-disk displacement is proved before a new horizontal source law is invoked.

Insert this displacement in (10.58), or in the completed weak law of Lemma 10.19. The exact factor $e^{-\lambda \Delta u}$ tends to one. The full relative error and a larger-disk Cauchy estimate give the transported ratio and logarithmic derivative. The head has logarithmic magnitude $O(u) = o(X)$. Fixed denominators are zero-free by their own-driver laws; their absolute inverse costs are not charged to the old structural buffer. Pull back the whole original-stage bound on its actual domain before applying Lemma 10.2 to a new finite subprincipal fraction list at X . Whole numerator maps transform by the fixed coordinate derivative and its finite denominator clearers. A slowly widening horn is chosen for the finite stage and denominator list, not for all variable readouts. This constructs the next lower inlet without using its new weak class.

Step 3. We control comparable classes and whole tails. Comparable positive principal clocks are treated together. Their ratios belong to the common strong differential field before these weak variables. Lemma 8.22(1) gives a finite limit or directional escape for each ratio. The positive principal equivalents make each nonzero finite ratio limit positive. This orders the finite clock list into comparable classes, with zero or infinite ratios between distinct classes. At an anchor X , a clock with ratio limit $\ell > 0$ has the exact phase $-\tau \ell X + [-\tau(Y - \ell X) + a \log Y + \log c]$.

The bracket is a sub- X field residual on the supplied actual chart; all of it is retained. Distinct classes are processed in increasing order. The number of classes is bounded by the fixed finite clock list. Each projected nonzero weak column has a strictly positive leading rate. Its bounded-action index sets are finite; zero columns remain complete coefficient factors. The witness rows of Lemma 10.18 collect exact relations at each finite fiber. One fixed denominator per row is sufficient. The original absolute Wiener norm bounds the sum of the unnormalized fiber norms; no sum of denominator-cleared norms is taken.

For completeness, tails after the first surviving fiber still have one whole bound. The upward closed set of later nonnegative indices has finitely many minimal indices. Any infinite sequence of nonnegative integers has an infinite nondecreasing subsequence. Applying this successively to the finitely many coordinates would give two ordered elements of a proposed infinite set of minimal indices, a contradiction. Split the later set among these finitely many minima and factor their positive gap. The remaining degrees use their original absolute norm. The first gap dominates the complete lower head, while the fixed-rectangle estimate controls its current complement. If all rows vanish, continue each unnormalized zero fiber to the original connected physical component and sum there. This proves the complete zero and first-head assertions without an intersection of infinitely many fiber-dependent horns.

Repeat these operations through the remaining finite ordered list. An empty weak list requires no weak extension. If a native list is empty, retain the constructed lower receiver; the native reconstruction theorem is invoked only for its stipulated nonempty bank lists. Every old clock is still its exact original function; every old positive reserve and matching witness is retained. The new positive marks have arbitrary receiving radii, since restricting farther makes their physical values smaller than any such radius. Unit tensoring does not enlarge the old coefficient inclusion norm. Finite exterior factors stay outside the unphased algebra: sort their finite phases by this procedure and apply the constructed coefficient rows to each tied finite sum. All actual old-source squares are unchanged, so Lemma 10.22 proves bounded injective old inclusion with original infimum norms. $\square$

Lemma 10.24. *In the setting of Lemma 10.23, for every finite list $M = CB_e^\alpha$, $\alpha \geq 0$, and $\varepsilon > 0$, the weighted complements at the cuts $l(e)$ have simultaneous whole receiving maps of norm at most ε . The positive certificates in the setup remain arbitrarily small in one common further receiving stage after all fixed inclusion costs.*

Proof. Use the supplied first-removed-factor decomposition on the raw input. A removed weak coordinate is extracted using its original input radius; a removed native factor is replaced by its first root times the whole divided bank from Lemma 10.21. Thus the full cut difference is a finite sum $\sum_\mu w_\mu D_\mu F$, with each D_μ bounded before every degree and each w_μ a distinguished positive factor at a removed depth. The remaining degrees and every later factor remain in $D_\mu F$. For the weight M^α , the depth gap gives $\log M = O(A_{d+l(e)-1}) = o(b_{d+l(e)})$. At the parent of each removed factor, $M^\alpha w_\mu$ therefore has the original strictly positive primary action as w_μ , with the exact additional lower phase $\alpha \log M$. Its finite phase, derivative and denominator list is one of the permitted buffered lists of the source atlas. Lemma 10.23 constructs a fresh copy of this entire positive mark in the joint receiver.

Fix all coefficient, division, relative and old inclusion costs first, including each actual bank map i_e and the products in (10.61). Let C_μ bound the whole map of D_μ into that stage. Only then choose the fresh receiving mark norms r_μ with $\sum_\mu r_\mu C_\mu \leq \varepsilon$, simultaneously for the finite list of weights. No old degree is extracted using r_μ . The sum of the whole source maps is bounded by $\varepsilon \|F\|_{\mathcal{P}}$. The receiver is independently faithful and the scalar identity is the entire weighted cut discrepancy. Lemma 10.14 gives descent. The old positive certificates have this finite factor structure. Replaying them with fresh copies after their fixed costs gives their joint source-smallness assertion. Analytic differences are received as whole differences on the original strict disks; their finite coefficient bounds precede the new radii. $\square$

10.6. Inverse charts and original coefficient targets.

Lemma 10.25. *Let $u_x = x + i\eta(x)$, with $\eta, \eta' \rightarrow 0$, and let E be a Banach source with a bounded whole realization on every far tube*

$$(10.63) \quad \mathcal{T}_a = \{u : \Re u > x_a, |\Im u - \eta(\Re u)| < e^{-a\Re u}\}, \quad a > 0.$$

Suppose a holomorphic logarithm J on these tubes satisfies, for some $c > 0$,

$$(10.64) \quad |J(u)| \leq C_a(1 + \Re u), \quad \Re J(u_x) = cx + o(x), \quad J'(u_x) \rightarrow c.$$

Put $P = e^J$, and suppose $\Re P(u_x) \rightarrow +\infty$, $\Im P(u_x) \rightarrow 0$, on the logarithmic lift with $\Im J(u_x) \rightarrow 0$. Then, for every $0 < \nu < 1$ and $b > 0$, P has a genuine inverse on a far strict horn $|\Im w| < b(\Re w)^\nu$. Its image lies in one of the tubes (10.63), and the pullback of the entire E is bounded in its original norm. Writing χ for this inverse, for every fixed $R > 0$, on a far tail with $w = P(u_x)$ and $|z| \leq R$,

$$(10.65) \quad |\chi(w+z) - \chi(w)| \leq \frac{4}{3c} |\log(w+z) - \log w| = O(|w|^{-1}).$$

The logarithms and inverse values use the prescribed common branch. For fixed $a_0 \in \mathbb{R}$ and $c_0 > 0$, the coordinate $P - a_0 \log P - \log c_0$ has the corresponding same-branch inverse and whole source bound. These conclusions use only the displayed raw analytic bounds, without a family or forest admission hypothesis.

Proof. We construct local inverse disks, glue them, and compare their size with the prescribed horn.

Step 1. We construct buffered inverse disks. Fix a small $a > 0$. The derivative integral on circles of radius comparable to e^{-ax} in $\mathcal{T}_a$ bounds J'' by $C(1+x)e^{2ax}$ on a buffered smaller tube. Integrating along vertical segments of length at most e^{-3ax} shows that $J' \rightarrow c$ uniformly there. Set $r_x = \delta e^{-4ax}$, where $\delta > 0$ is fixed and small. On a far tail, $|J'(u) - c| \leq c/4$ for $|u - u_x| \leq 2r_x$. For $|Z - J(u_x)| < cr_x/16$ consider

$$(10.66) \quad u \mapsto u_x + \frac{Z - J(u_x)}{c} - \frac{J(u) - J(u_x) - c(u - u_x)}{c}.$$

On the closed r_x disk its displacement is at most $5r_x/16$ and its contraction constant is at most $1/4$. Its inverse solution is holomorphic and has displacement at most $r_x/12$. The fixed-point inequality also gives $|u - u_x| \leq 4|Z - J(u_x)|/(3c)$.

Step 2. We glue the local inverses. The derivative of $\Re J(u_x)$ with respect to x tends to c , since $du_x/dx = 1 + i\eta'(x)$. If two logarithmic image disks intersect, their real centers differ by at most $c(r_x + r_y)/16$, so $|x - y| \leq (r_x + r_y)/8$ on a far tail. Consequently $r_y/r_x \rightarrow 1$ uniformly for such pairs, and $\eta' \rightarrow 0$ makes their source centers equally close. Both inverse candidates lie in a common buffered source disk. There the segment integral of $J' - c$ gives $|J(z) - J(w) - c(z - w)| \leq (c/4)|z - w|$; hence J is injective and the candidates agree. They therefore glue in the original component.

Step 3. We obtain the inverse on the whole horn. A displacement $|\Im w| < b(\Re w)^\nu$ gives a logarithmic displacement $O(\exp(-(1-\nu)cx + o(x)))$ near the corresponding real center. The offset of that center is smaller: $\Im J(u_x) = o(1/|P(u_x)|)$ by the absolute hypothesis on $\Im P$. Choose $4a < (1-\nu)c$. The increasing real logarithmic centers cover a far half-line, so their disks cover the logarithm of a farther strict horn. The glued inverse maps it into $\mathcal{T}_a$. Its whole E bound is the original bound on that tube, pulled through this fixed inverse before the carried input is chosen. For a fixed disk $|z| \leq R$ about $w = P(u_x)$, $\log(w+z) - \log w = O(|w|^{-1}) = O(\exp(-cx + o(x)))$. Choose $4a < c$. This logarithmic disk lies inside the local image disk of radius $cr_x/16$ on a far tail. The fixed-point bound in **Step 1** proves (10.65), without a transport or source-law assumption. Finally $w \mapsto w - a_0 \log w - \log c_0$ has derivative tending to 1. Its correction is $O(\log |w|)$, which is smaller than the buffer of every fixed positive power horn. The disk contraction (10.66), with

limiting derivative 1 on a slightly larger horn, gives its inverse and the asserted composition. All inverse branches have the prescribed physical anchor. $\square$

Lemma 10.26. *Fix finitely many original projective and coefficient Wiener inputs, with physical driver t and original parameters $\lambda \in \mathbb{C}^p$. Suppose their old families have bounded maps for each fixed original coefficient into separate faithful marked targets, with their original norms and finite exterior modules. Consider a finite composition of raw source operations with the following data, fixed before the variable carried input f .*

Every structural substitution is holomorphic on its original buffered domain. Products and analytic substitutions have their whole larger-disk bounds. Every native factor is received at its reference clock by a specified bounded map i_e with its actual square, and its relative argument belongs to a fixed strict ball of Lemma 10.21. Every implicit structural inverse $h = \Phi(h; \lambda)$ is a whole analytic contraction with constant $\vartheta_{\text{inv}} \in (0, 1)$ on a fixed buffered ball in one receiving Banach algebra. The carried maps are linear in f ; multilinear maps on its raw factors use the original projective norm.

Supply the finite physical order of raw operations and, separately, the increasing growth-class reconstruction order of their native and weak comparison folds, with the lower inlets of Set-ups 9.14 and 8.21, including the horizontal horn data of Theorem 8.25. The old coefficient targets keep their whole ordered presentations. Independently of geometric chart holomorphy, supply the normalized structural-jet bound and exact original-parameter formulas of Definition 10.5(1)–(2) through each requested finite order. Its complete faster coefficient targets are not structural multipliers at an earlier parent. For each fixed finite derivative order, assume the following positive reserves at their actual folds. Each constructor must establish these hypotheses from its exact formulas before invoking this lemma; structural chart holomorphy alone does not supply them.

- (i) *After the scalar-zero Euler part of a differentiated native factor at T_e is divided, each newly exposed structural multiplier A has $T_e^{-1/P_e} A$ as a positive parent mark. Its phase is $-au + P$, where $a > 0$, P belongs to the pre-weak field, $P = o(u)$ and $\Im P \rightarrow 0$, after a fixed nonzero leading constant is taken outside. Its complete bounded coefficient and exact root unit retain their original norms.*
- (ii) *In a differentiated positive weak monomial, each newly exposed structural multiplier times one occupied weak factor has this same positive-phase form at that weak's own driver. Remaining faster coefficient targets are retained at their own later folds.*

For prepared factors and cut discrepancies, retain also the carrier, ordered inverse, linkage and first-removed-factor requirements of Definition 10.5(3). Fixed carrier moments use reserved positive weight slack, and extra reference powers in a differentiated discrepancy use its weighted-cut map at the removed height. These finite extra marks have the supplied whole derivative and horn inlets; their analytic units add nonnegative actions and their fixed rectangle costs are at most $\exp(C_k T \log(eT))$. Keep original observation maps and their complements as bounded maps to their separate codomains.

Then the composed families exist on the connected physical components of the literal pulled domains. Each far physical point has a local chart $\lambda = r_t \eta$, with $r_t > 0$ constant on a buffered driver disk, and a whole bound $L\|f\|_{\text{original}}$ for $|\eta_j| \leq 2$. On $|\eta_j| \leq 1$ their degree at least N tail is bounded by

$$(10.67) \quad 3^p L\|f\|_{\text{original}}(3/4)^N, \quad [\lambda^\mu]F = r_t^{-|\mu|}[\eta^\mu]F(r_t \eta).$$

Separately, for every fixed μ there exists a bounded linear map J_μ from the entire original raw input into a finite independently faithful marked physical target such that

$$(10.68) \quad \text{Act } J_\mu f = [\lambda^\mu] \text{Act } f, \quad \|J_\mu f\| \leq C_\mu \|f\|_{\text{original}}.$$

The constants retain all bank inclusion, original Cauchy, derivative, root, degree and exterior costs. The targets and their norms are separate for different μ . Their independent rows have common kernel exactly family zero on the connected pulled domain. These conclusions require no family record for the new output.

Proof. We first construct local charts and then construct the physical coefficient targets directly in the original parameters.

Step 1. We retain literal domains and local charts. Take the inverse images of every original root, logarithm, analytic, relative and inverse domain under its actual structural argument. Intersect their strict damping, source-norm and contraction margins. The whole operator bounds are finite products of the given costs, including (10.61); they precede all carried source degrees. Strictness at parameter zero gives a neighborhood of each far physical point. Take its connected component.

On a buffered driver disk D_t choose a preliminary parameter polydisk $|\lambda_j| \leq 2r_*(t)$ in that domain. Let V_k be the finite analytic functions or Banach-valued operators whose norm inequalities express all the margins, and let $\delta_k > 0$ be strict margins at its center. Uniform norms include the entire auxiliary disks of inverse and relative operators. Put

$$(10.69) \quad B_{k,t} = \sum_{j=1}^p \sup_{D_t, |\lambda_j| \leq 2r_*(t)} \|\partial_{\lambda_j} V_k\|, \quad r_t = \min \left\{ \frac{r_*(t)}{4}, \min_k \frac{\delta_k}{8(1+B_{k,t})} \right\}.$$

Shrink D_t until each zero-parameter variation is less than $\delta_k/4$. For $|\eta_j| \leq 2$ the parameter-segment integral bounds the additional variation by $2r_t B_{k,t} \leq \delta_k/4$. All original margins persist on this one product chart. An implicit inverse contributes its exact derivative with the full factor $(1 - \partial_h \Phi)^{-1}$, of norm at most $(1 - \vartheta_{\text{inv}})^{-1}$, to these Jacobian bounds. A native clock contributes its entire logarithmic derivative, including every change of its physical input. The scale r_t is constant on D_t ; no uniform lower bound is used. All charts are restrictions of the literal composed analytic map and therefore agree on overlaps.

Cauchy on the radius-2 polydisk gives coefficient bound $L\|f\|_{\text{original}} 2^{-|\mu|}$. For $|\mu| \geq N$ use $2^{-|\mu|} \leq (3/4)^N (2/3)^{|\mu|}$ and sum the geometric product. This proves (10.67), including its exact recovery factor. That factor is used only in the local chart.

Step 2. We keep the original coefficients separate. Fix μ and put $k = |\mu|$. Take the finitely many whole old coefficient targets through order k . Differentiate each finite structural recipe in the original λ variables. Whole analytic operations use their original larger-disk Cauchy constants. For an implicit inverse let $h_0 = h(0)$ and let $h_{<k}$ be its entire Taylor polynomial through total degree $k-1$. Its degree- μ equation is exactly

$$(10.70) \quad (1 - \partial_h \Phi(h_0; 0))h_\mu = [\lambda^\mu] \Phi(h_{<k}; \lambda).$$

The right side involves finitely many lower-degree structural jets. The inverse multiplier is the complete geometric unit with norm at most $(1 - \vartheta_{\text{inv}})^{-1}$. Induction constructs these fixed jets before the variable input f and its source coefficients.

A native derivative is a finite sum of whole Euler derivatives and original bank-parameter derivatives, multiplied by products of exact jets of $\log T_e$. Its positive Euler terms have scalar zero. Whole division in Lemma 10.21 extracts T_e^{-1/P_e} before any growing multiplier is adjoined. The reserved marked roots in the hypotheses therefore receive all such terms with their full divided bank. Each relevant relative operator carries its actual inclusion norm $L_{i,e}$. A scalar term uses its old coefficient map; it has no logarithmic clock derivative.

For a weak monomial of total degree $n > 0$, the k th derivative has at most a fixed polynomial of degree k in n , times the finite structural jets and the differentiated whole coefficients. Partition monomials by their first occupied weak label and extract one copy of that label to carry these structural factors. Choose the fixed strict substitution ratio $\theta < 1$ for the remaining factors.

Extraction costs its original radius and a fixed power of θ^{-1} , independent of n . For this fixed derivative order and strict ratio,

$$(10.71) \quad C_{k,\theta} = \sup_{n \geq 1} (1+n)^k \theta^{n/2} < \infty$$

leaves $\theta^{n/2}$ for the entire degree tail. Several slots use total degree and that first-occupied-label partition. Degree zero uses the whole old coefficient target. All faster old targets stay as whole later factors, rather than being extracted as earlier multipliers.

For this fixed μ , insert the finitely many reserved marks before the relevant raw fold. Apply Theorems 8.23 and 8.25 to their supplied pre-weak fields. Apply Theorem 9.17 only after its whole lower estimate and exact root reserve have been constructed. Rebuild later comparison folds in increasing actual growth order, carrying every old marked target and its physical recipe correlations. Prove each enlarged class's complete own-driver law before transporting it to the next class. The stated rectangle bounds and positive reserves are the lower inlets at these folds. This constructs an independent marked receiver from the raw operators, before quotient descent and without an assumption about new coefficient targets. Finite exterior generators retain their separate norms and their actual derivative factors in their finite modules.

Fix every coefficient, Euler, division, Cauchy, inclusion and exterior cost in this finite construction. Fresh positive receiving radii are chosen only afterwards. Products of these constants and $C_{k,\theta}$ give a finite C_μ in (10.68). Linearity in the carried input and projective continuity give the whole map J_μ . Its actual square follows from the exact derivative identities, not from the local factor $r_t^{-|\mu|}$.

Step 3. We identify the family kernel. Compose the independent rows of each marked receiver with its bounded J_μ . If all these rows vanish, each original Taylor coefficient is zero as a physical germ. It is holomorphic along the connected zero section, so the identity theorem continues that zero through the whole zero-section component. At each point a positive parameter polydisk remains, where the entire Taylor series is zero. Holomorphic continuation then gives family zero on the connected pulled domain. Conversely a zero family has all these coefficients zero. This uses neither a common horn for countably many μ nor a sum of their target norms. Observation and complement maps compose with these literal restrictions, preserving their own bounded codomains throughout. $\square$

Theorem 10.27. *Suppose Set-up 10.6 holds. For each finite list $M = CB_e^\alpha$ with $C > 0$ and $\alpha \in \mathbb{Q}_{>0}$, each finite list of real $a \geq 0$ and each $\varepsilon > 0$, there exists a common independently faithful receiving algebra $\hat{Q}$ containing the whole old Q by a bounded injective unital map, receiving every prescribed compatible original full native bank at M , and admitting bounded linear maps*

$$(10.72) \quad W_{M,a} : Q \rightarrow \hat{Q}, \quad \text{Act } W_{M,a} f = M^a (\text{Act } f - \text{Act } e_{l(e)} f), \quad \|W_{M,a}\| \leq \varepsilon.$$

All original input norms and radii remain fixed.

For a stored birth b , put

$$(10.73) \quad M = K_b, \quad Z = \Xi_b, \quad Y = T_b, \quad P = K_b e^{\rho_{i_b} \Xi_b}.$$

There exists a joint receiver for every prescribed finite whole coefficient Wiener input at $c_j e^{-\tau_j T_b} T_b^{a_j}$, where $c_j, \tau_j > 0$ and $a_j \in \mathbb{R}$. It contains source-small representatives

$$(10.74) \quad B = T_b - P = W_{K_b,1}(e^{\Xi_b}), \quad L = \Xi_b - e_{i_b} \Xi_b = \log(T_b/P).$$

The discrepancy B tends to zero absolutely, and $\Im P \rightarrow 0$. The exact P and $P - a \log P - \log c$ charts have genuine same-branch inverses carrying the appropriate whole strong source on each fixed strict power horn of exponent in $(0, 1)$.

These reference-reception and stored-birth-normalization operations extend the coordinate and original-parameter positive-jet records of Definitions 10.1 and 10.5, retaining the old data unchanged under their bounded inclusions. Each complete comparison kernel is the complex

analytic kernel; every nonzero whole value has a complete head $c\Theta(1 + o(1))$, $c \neq 0$, $\Theta > 0$, including the entire remainder, and c is real for a real value. All old inclusions use the original infimum norms. All original families, separate coefficient targets, cuts and observations retain the data of Set-up 10.6. These operations do not execute a new physical reverse or a re-entry.

Proof. We receive the finite reference list, normalize the stored birth, and construct its cut and original-parameter outputs from the old records.

Step 1. We receive references and normalize the stored birth. The native assertion and the maps $W_{M,a}$ are Lemmas 10.23 and 10.24. For each newly received reference fold, retain the complete own-parent law (10.58) and the whole original-norm bound of Lemma 10.20. The lower horizontal law, entire complex rectangle remainder and strict native damping margins give these data before Lemma 10.2 is used for a newly enlarged subprincipal structural list. Exact roots retain their positive actions and full reciprocal costs separately from that list. The original-parameter extension is verified independently in **Step 3**, using (10.75); it is not inferred from the coordinate buffers. Write $i = i_b$, $Z_0 = e_i Z$ and keep the entire strong source Z_0 . The exact identities

$$P = Me^{Z_0}, \quad Y - P = W_{M,1}(e^Z), \quad \log(Y/P) = Z - Z_0$$

follow by multiplicativity of the cut. The first-removed-factor construction makes the last two elements source-small in one target. It also gives their absolute scalar decay, even after multiplication by M . Since Y is physically positive, $\Im P = -\Im(Y - P) \rightarrow 0$ absolutely.

The logarithm of P , in the actual parent u of its stem, is its positive linear term plus the retained whole bounded source Z_0 and the original lower logarithmic and unit terms. Its derivative tends to that positive coefficient: Lemma 8.22(2), in the completed old strong differential field with derivative d/du , makes the derivative of each bounded correction tend to zero. The supplied parent tubes carry the entire strong source and bound $\log P$ by $C(1 + \Re u)$. Its real part is a positive multiple of $\Re u$ plus $o(\Re u)$, and the amplified cut estimate has already given the absolute limit $\Im P \rightarrow 0$ on the prescribed logarithmic lift. Lemma 10.25 therefore constructs the genuine P and $P - a \log P - \log c$ charts with their whole original source bounds.

To insert a weak at Y , first add the baseline $v_j = c_j e^{-\tau_j P} P^{a_j}$ at its actual principal class. Its exact phase has a positive primary rate and lower-field subprincipal residual. The just constructed chart and whole source bounds give its domain inlet. For the finite appropriate proper-strong own-parent inventory, the structural bounds in Definition 10.1 and Lemma 10.3 give displacement of modulus $O(\exp(-\gamma \Re u + o(\Re u)))$. A subsequently reinserted slower weak principal $X = o(P)$ instead uses Lemma 10.4 at its completed lower native levels, followed by the exact identity and uniform complex displacement (10.2). A large native clock retains relative variation through its logarithm; its absolute derivative need not be subexponential at u . The complete own-driver laws then give horizontal ratio one and zero-free fixed denominators after pullback. Whole source bounds are pulled through the actual inverse independently of these fixed-readout normalizations. Lemma 10.2 supplies the new finite structural buffers only after that whole bound.

For comparable old principals the clocks already belong to the projected strong fraction field: their received exact roots give K_b and every retained reference by inverse integer powers, and the bounded projected corrections have full exponential series there. Thus P , an old comparable X , and the entire difference $P - \lambda X$ belong to that field before their weak classes, with all fixed correct-coordinate derivatives. Use their complete old heads and actual chart for phase comparison; do not discard the difference merely because it is $o(P)$. Reconstruct all tied and later classes by Lemmas 10.18, 10.19 and 10.20, in the order of Lemma 10.23. The actual weak is exactly $v_j U_j$, $U_j = \exp(-\tau_j B + a_j L)$. Its whole coefficient norm is fixed before the baseline receiving radius. Choose that radius so $\|v_j\| \|U_j\| \leq \theta \rho_j$, with $\theta < 1$ and the original ρ_j fixed.

Then the whole input map is

$$\sum_{\nu} f_{\nu} z^{\nu} \mapsto \sum_{\nu} I(f_{\nu}) \prod_j (v_j U_j)^{\nu_j}, \quad \left\| \sum_{|\nu| \geq N} \dots \right\| \leq \|I\| \theta^N \sum_{\nu} \|f_{\nu}\| \rho^{\nu}.$$

This proves the claimed entire weak receiving map and tails. The stored original native relative margins remain part of the input.

Step 2. We retain every cut and original observation. Raw cuts act in every coefficient and argument of the reference and normalization recipes. They commute with complete analytic sums and with retained original inverse recipes on their buffered disks. Construct each retained strong target independently by Lemma 10.23. The first-removed-factor decomposition keeps its distinguished positive root or weak and its entire remaining coefficient. The original depth gap gives its whole exponential error bound. Lemma 10.13 then gives descent, $e_i e_j = e_{\min(i,j)}$, and the old commuting squares. The stored ancestry is unchanged by normalization.

For the finite reference powers use their actual exponential birth formulas. At a nonbasal occupied depth the finite positive rates give $\log b_{\text{new}} \geq c b_{\text{parent}}$ for some $c > 0$, whereas $\log A_{\text{parent}} = o(b_{\text{parent}})$. Hence $A_{\text{parent}} = o(b_{\text{new}})$; the upper bound follows from the exact logarithmic clocks. At the base retain the displayed basal powers and unit. No enlargement of a depth profile to bound arbitrary denominators is used. Original observations and full complements compose with the bounded raw maps at their original labelled nodes; exterior modules retain their separate norms.

Step 3. We construct the original coefficients without assuming the new record. Fix an original parameter order k . For each reference $M = CB_e^{\alpha}$, with fixed C, α , its exact normalized jets are $D_{\mu} \log M = \alpha D_{\mu} \log B_e$ for $|\mu| > 0$. For the stored normalization use (10.75)

$$\log P = \log K_b + e_i \Xi_b, \quad D_{\mu} P = P \mathcal{B}_{\mu}((D_{\nu} \log P)_{0 < \nu \leq \mu}), \quad D_{\mu} \log T_b = D_{\mu}(\log K_b + \Xi_b),$$

where $\mathcal{B}_{\mu}$ is the finite multivariate exponential Bell polynomial. The old normalized-jet record, finite sums and products, and derivatives of protected units give (10.5) for the newly exposed normalized factors. Complete old coefficient jets remain in their separate marked targets. This bound does not follow from the atlas.

A positive Euler derivative of a native factor has scalar zero. Divide its whole bank by the original first root before attaching the finite Bell polynomial. The explicit root of CB_e^{α} has positive action $\alpha \gamma_e / P_e$, and its structural multiplier has logarithmic size $o(u)$, so that action remains positive. The complete divided bank and relative unit are retained. Differentiate B and L through their full first-removed-factor expressions, not through separately large values of T_b and P . Any fixed positive power of K_b exposed at order k is paid by a weighted-cut map: $\log K_b = o(X)$ at the removed height X . The complete coefficient and its original derivative target remain attached to that factor.

For $H = P - a \log P - \log c$, its jets contain P times finite normalized structural polynomials. Their logarithmic size is $O(u) + o(u) = o(H)$, at the new weak's driver, not at u . One occupied weak factor pays these multipliers. The degree polynomial is bounded by (10.8), leaving a geometric bound for the full tail. Original implicit recipes use (10.6) with their retained contraction constants. The coordinate inverse has only the new factors $P(\log P)'$ and $1 - a/P$; the first logarithmic derivative is bounded away from zero on its inverse disks, and $|a/P| < 1/2$ on a far tail. Their fixed jets use the old structural buffers.

These exact formulas verify the hypotheses of Lemma 10.26 before it is invoked. Receive the finite marks at their folds, prove each completed own-driver law there, and replay the later growth classes with every old target whole. All Euler, division, relative, Cauchy and inclusion costs precede the fresh radii. The lemma gives the bounded original coefficient maps, literal connected domains and local parameter charts. Their exact $r_t^{-|\mu|}$ recovery is retained and is

not used as an unproved subprincipal multiplier. This proves the original-parameter positive-jet output and its independent family kernel, as well as the observation conclusions, for the two operations in the statement. $\square$

Proposition 10.28. *Suppose Set-up 10.6 holds and the current is an actual direct birth b followed by positive analytic maps. On its retained strong chart write their complete composite, including the next incoming analytic map, as*

$$(10.76) \quad y = F(e^{-u}e^{-E}), \quad F(z) = DzA(z), \quad D > 0, \quad A(0) = 1.$$

Supply a canonical direct of order $m \geq 1$ and constant $\beta > 0$, with its literal physical matching, positive inverse branch and interval, and all original native, positive monomial/logarithmic, coefficient, relative, divided-remainder and prescribed inverse recipes on their strict buffered domains. Retain each full original bank, its norm, integer root order P_e , ray and root/logarithm lift, with compatible joint ray for the simultaneous tensor. Retain the bounded bank inclusions and their actual squares, every required relative or divided bank enlargement, and all original parameter correlations, domains, true recipe inverses and protected inverse margins. Every additional native clock is $\beta_e y^{-m_e}$ with its original integer $m_e \geq 1$; its reference is $N_e = \beta_e F(e^{-u})^{-m_e}$. These auxiliary references and their original full root orders are not replaced by the principal reference or by the passage order. There exists a bounded injective original-norm inclusion into a new independently faithful receiver with all the data of Set-up 10.6, retaining all old births and adding a distinct new stem

$$(10.77) \quad N = \beta F(e^{-u})^{-m}, \quad \Delta = E + \log A(e^{-u}) - \log A(e^{-u}e^{-E}), \quad \beta y^{-m} = Ne^{m\Delta}.$$

At original root order P , its first root and logarithm are exactly

$$(10.78) \quad N^{-1/P} = \beta^{-1/P} D^{m/P} e^{-mu/P} A(e^{-u})^{m/P}, \quad \log N = mu + \log(\beta D^{-m}) - m \log A(e^{-u}).$$

The entire original native and weak lists are received, with their original norms and strict whole-degree tails. All existing cuts extend and commute with the old inclusion; the new cut retains the parent weak sources and the new native banks. All finite prescribed weighted cut complements have arbitrarily small receiving operator norm. The new principal $P_+ = Ne^{m\rho_j\Delta}$, $j = l(e(b)) + 1$, has $\Im P_+ \rightarrow 0$, its genuine strong inverse atlas, and source-small whole discrepancies $\beta y^{-m} - P_+$ and $m(\Delta - e_j\Delta)$. The new current is the exact original direct output, and the new birth has its actual parent, parent exponent one and the incoming analytic correction $-\log A_{\text{in}}(x_b)$. Every original family and observation persists as in the setup.

Proof of Proposition 10.28. We construct the new native inlet at the actual parent, extend every cut, and then receive the whole outgoing weak source. Normalize the last direct birth by the stored-birth assertion of Theorem 10.27, specifically (10.73)–(10.74). Its exact data are

$$T = Ke^{\Xi}, \quad P = Ke^{\rho_i\Xi}, \quad u = P - a \log P - \log c, \quad E = (T - P) - a \log(T/P) + \ell(T).$$

Compose the trailing analytic maps and the next incoming map at their literal nodes to obtain the stated $F(z) = DzA(z)$. Lemma 10.17 proves the identities for N, Δ and every exact first root. In particular the new root has action $m/P > 0$ in the actual parent u .

Step 1. We construct the joint native fold. First rebuild the raw presentation through that parent. The pre-weak field at u has its supplied sub- u heads, whole differentiated numerator maps and true inverse source chart. Prelist the new roots $e^{-m_e u/P_e}$, all positive monomial and logarithmic marks required by the original canonical slots, and the full analytic factors $A(e^{-u})^{m_e/P_e}$. Use Lemmas 10.18 and 10.19 for their entire lower Wiener input, together with all old weak variables at that class. The analytic unit's expansion adds only nonnegative u actions, and the norm of its k th power is at most e^{Ck} on a fixed larger disk. Thus the new roots have the stated reserve and the finite rectangle bound remains $\exp(CT \log(eT))$. Also $\partial_u \log N_e \rightarrow m_e$ uniformly on a strict widening horn and $\Im \log N_e \rightarrow 0$ on the physical curve. Every hypothesis of Theorem 9.17 has now been constructed before adding these new native factors. Apply it

to all old tied banks and all new banks simultaneously, on their compatible original ray. At a later parent X , first establish the actual complex-disk motion. Ordinary stored changes use the fixed-disk inlet of Lemma 11.21; a newly exponential change first uses its genuine inverse and the fixed-disk estimate of Lemma 10.25. Lemma 10.3 then treats its proper strong structural list. For reinserted weak siblings, Lemma 10.4 supplies the clock hypotheses from completed lower native levels before transport. Only after that motion is proved does the completed own- u law supply the sub- X horizontal head. Pull back the whole stage bound, then obtain finite new structural buffers by Lemma 10.2. Whole differentiated numerator maps retain their fixed coordinate denominators. Reconstruct each subsequent original fold in increasing growth order, keeping physical chronology within its recipes and correlated slots. Original quotient descent is taken only after this joint kernel proof.

Step 2. We control cuts and complete relative sources. At a raw cut below level $j = i + 1$, scalarize the new banks and erase their positive root marks. At cuts at least j , retain their whole formulas. Construct the retained targets independently by these same folds. The difference has the old first-removed-factor decomposition plus one whole native division for each newly removed factor. Its positive reserve and the depth gap give the whole cut bound, so Lemma 10.13 gives every cut, its nested identities and commuting old squares. For a multiplier N^a , its logarithm is $O(u)$, strictly below the first removed principal depth. Multiply each distinguished removed root or weak by this fixed subprincipal phase and adjoin its full positive mark. Choose that mark's receiving radius after the entire coefficient cost. Lemma 10.14 then gives all the new weighted maps with any prescribed norm.

The complete Δ has a finite smallness certificate. The old amplified discrepancy and logarithmic discrepancy already have theirs. The scalar-zero $\ell(T)$ uses whole native division. For the analytic difference use the exact factorization

$$\log A(z) - \log A(ze^{-E}) = z\{a_A(z) - e^{-E}a_A(ze^{-E})\}, \quad \log A(z) = z a_A(z).$$

Fix the full coefficient and relative costs, including the actual map i_e from each receiving bank at N_e and the product (10.61). Work first with the uniform bound $a_e \leq r_{0,e}/2$ in each relative ball. Then choose fresh positive copies for these finitely many distinguished factors. This makes every $\exp(m_e\Delta) - 1$ strictly smaller than its original relative radius in this joint receiver. Lemma 10.21 now maps the entire original banks at N_e to the actual heights $\beta_e y^{-m_e}$, with all their native and relative tails. Each original canonical monomial or logarithmic slot has a positive parent factor and a bounded whole unit. Give only its receiving copy a norm below a fixed fraction of its original slot radius. The complete projective/Wiener norm then gives its entire slot tail. Prescribed finite canonical inverse and remainder operations are received as their full original bounded source maps under this single common substitution.

Step 3. We construct the new principal and outgoing source. Put

$$P_+ = Ne^{m\rho_j\Delta}, \quad B_+ = W_{N,1}(e^{m\Delta}), \quad L_+ = m(\Delta - e_j\Delta).$$

Their actual identities are $B_+ = \beta y^{-m} - P_+$ and $L_+ = \log(\beta y^{-m}/P_+)$. They are source-small, and the whole amplified cut bound makes $B_+ \rightarrow 0$ absolutely. Positivity of βy^{-m} gives $\Im P_+ \rightarrow 0$. In the parent coordinate, $\log P_+$ has derivative tending to m . The entire new strong stage on each tube $e^{-a\Re u}$ is bounded from its original kernels. Old finite structural buffers control the normalized arguments. Large native clocks keep their explicit factors; small variation of their logarithms preserves a fixed part of their angular damping. Full denominators retain their own-driver relative laws and are not assigned subexponential absolute inverse costs. The strict relative and analytic radii bound the entire coefficient sums before their degrees. These original-norm bounds precede any application of Lemma 10.2 to new fractions. The exact formula

$$(10.79) \quad \log P_+ = mu + \log(\beta D^{-m}) - m \log A(e^{-u}) + m\rho_j\Delta$$

is bounded by $C(1 + \Re u)$ on those tubes and has derivative tending to m on their physical curve. Its real part is $m\Re u + O(1)$, and the amplified estimate has given the absolute limit $\Im P_+ \rightarrow 0$ on the prescribed lift. Lemma 10.25 now gives the genuine P_+ and $P_+ - a_+ \log P_+ - \log c_+$ inverse charts carrying that whole strong stage. This uses the constructed raw bounds before any family admission of the new stem. Earlier parent charts retain their appropriate strong stages by their exact pullbacks; later factors remain on their separate domains.

The literal new weak and output are

$$q_+ = e^{-H_+} e^{-B_+ + a_+ L_+}, \quad H_+ = P_+ - a_+ \log P_+ - \log c_+, \quad x_+ = F_+(e^{-H_+} e^{-B_+ + a_+ L_+ - \ell_+(\beta y^{-m})}).$$

Receive the baseline weak at P_+ by its pre-weak field and the just proved chart. For the fixed proper-strong own-parent inventory, Lemma 10.3 gives $|\Delta X| = O(\exp(-m\Re u + o(\Re u)))$ on bounded P_+ disks. For a reinserted slower same-depth weak class, Lemma 10.4 supplies the logarithmic-clock inlet from completed lower native levels. Its displacement is then $\Delta X = O(X/P_+) = o(1)$ by (10.2). The complete old own-driver laws give horizontal ratio one only after these domain estimates. Their fixed denominators remain zero-free, their complete heads have logarithmic size $O(A_{d+j-1}) = o(P_+)$ for the proper strong inventory, and their whole bounds pull back in the original norm. The exact roots and bounded projected correction exponentials put every comparable principal and its full difference in the old strong fraction field before its weak fold. Compare the complete comparable class jointly and then replay faster classes. Apply Lemma 10.2 after each new lower whole bound, and establish each completed own-driver law before further transport. Multiplication by the displayed complete unit gives the original weak map; choose its receiving radius after that unit's norm. Its total-degree tails follow from the full absolute Wiener sum. The new birth has parent b and incoming identity $I_{\text{in}}(x_b) = d_{\text{in}} x_b A_{\text{in}}(x_b)$, hence parent exponent one and correction $-\log A_{\text{in}}(x_b)$ in the positive parent ideal. Store N as a distinct stem even if an older one happens to have an equal physical value. The exact logarithm of N is its new stem formula. This proves all ancestry, source, atlas, cut and positivity records.

Step 4. We retain original families. Use Lemma 10.26 with the whole old coefficient targets and the raw native, weak and inverse maps just constructed. The old targets are taken through each fixed requested derivative order, in their original norms. The new exact jets are

$$(10.80) \quad \partial_{\lambda_k} \log(\beta_e y^{-m_e}) = -m_e \frac{\partial_{\lambda_k} y}{y}, \quad \log(\beta_e y^{-m_e}) = \log N_e + m_e \Delta.$$

At its parent u , use the independent original-parameter record of Definition 10.5, not geometric holomorphy. The exact formula $\log N_e = m_e u + \log(\beta_e D^{-m_e}) - m_e \log A(e^{-u})$ and the full identity for Δ show that each newly normalized jet is a finite sum and product of old normalized jets and derivatives of the protected analytic units. Complete later coefficient targets are retained separately. These structural parts satisfy (10.5) before a particular root reserve is used. For a cut discrepancy, differentiate its entire first-removed-factor expression. Extra fixed powers of N_e are paid at the removed height X , where $\log N_e = O(u) = o(X)$, by the weighted maps, rather than classified as subexponential at u . The new P_+ and H_+ jets contain P_+ times finite normalized polynomials, with logarithmic size $O(u) + o(u) = o(H_+)$ at the new weak driver. These statements retain the full exact parameter formulas before any family coefficient is extracted. The divided native root is N_e^{-1/P_e} times the strict relative unit. Its exact formula has action $m_e/P_e > 0$. Multiplication by those structural jets keeps that reserve; derivatives of Δ with later factors remain whole old coefficient targets in their actual later folds. The analytic factor $A(e^{-u})^{m_e/P_e}$ and its fixed derivatives are entire on their original strict disk and add only nonnegative actions. Thus their degree costs preserve the lower cofinal rectangle bound.

For a new weak $c_e e^{-\tau_e T_e} T_e^{a_e}$, a fixed derivative uses powers of T_e , its logarithmic jets and whole derivatives of the correction unit. The new structural factors have logarithmic size $o(P_e^{\text{pr}})$ at the principal driver P_e^{pr} of T_e . One occupied baseline weak carries them with its positive

action τ_e . All these finite marked phases have the pre-weak field and actual horn just proved for that principal driver. The remaining weak degrees obey the original strict ratio. For each original inverse recipe the strict single-stage contraction has a full geometric inverse derivative; its fixed coefficient recursion is (10.70). It involves these same structural marks and old targets.

The preceding joint fold construction applies to the finite extra marks before quotient descent and rebuilds every later factor. All its coefficient and bank inclusion costs precede fresh receiving radii. The lemma therefore constructs the new literal family domains, full-Jacobian local charts and each bounded original J_μ . It also preserves the separate observation codomains and complete complements. This proves the original-family and observation data of Set-up 10.6 for the new stem, including the original-parameter positive-jet record, from the old records and the constructed raw maps. $\square$

Construction 10.29. Fix $q = e^{i\theta}$, $|\theta| < \pi/2$, its indicated logarithm, an even integer $R \geq 2$, $r, g > 0$, and $0 < \vartheta < \min(\pi/4, \pi/2 - |\theta|)$. At each variable native label take a separate full bank $\mathcal{B}(R, g, r, \vartheta, q)$ of Set-up 9.4. Fix every scalar, weak and coefficient-family radius before the construction, within the stated analytic recipe domains. Repeated original coefficient labels remain correlated.

Use the passages of $u^{m+1}\partial_u - mv\partial_v$, $m = 1, 2$, from $v = b > 0$ to $u = a_m > 0$, in outgoing coordinate $v/(be^{a_m^{-m}})$:

(10.81)

$$C(z) = e^{-1/z}, \quad H(z) = e^{-1/z^2}, \quad C^{-1}(w) = (-\log w)^{-1}, \quad H^{-1}(w) = (-\log w)^{-1/2}.$$

Start with the identity strong word in σ and retain the original scalar variables $\sigma, \sigma \log \sigma$, the endpoint $C(\sigma)$ and a separately labelled reciprocal mark with that same value. Put

$$(10.82) \quad H_* = e^{1/\sigma}, \quad K = e^{H_*}, \quad Z_0 = e^K, \quad t = e^{-Z_0} = C^{\circ 4}(\sigma).$$

Adjoin the full original banks at H_*, K, Z_0 , the prescribed weak lists at these clocks, and their complete scalar and canonical coefficient factors. The selected tree parent is K , its opening is e^{-K} , its child is Z_0 , and its source increment is zero. At the next opening retain the labelled forward tree

$$(10.83) \quad y_1 = t, \quad R_1 = t^{-1}; \quad y_2 = C(t), \quad R_2 = y_2^{-1}; \quad \tilde{y}_1 = t, \quad \tilde{R}_1 = t^{-2}.$$

The third vertex is a distinct sibling of the first whose original recipe coefficients equal those of the first vertex. Retain all full native, canonical and prescribed weak factors at these vertices; its additional weak list may be empty. Apply the literal C^{-1} at $C(t)$ and then H at t , with identity adapters. Write $x_1 = H(t)$ and $T_1 = t^{-2} = \tilde{R}_1$. Apply H to obtain $x_2 = H(x_1)$, with $T_2 = x_1^{-2}$, and then its genuine H^{-1} to return to x_1 . Call the resulting complete raw presentation $\mathcal{P}_*$ and its independent quotient Q_* ; retain x_2 , all its full banks, and the whole original inverse families. From this same Q_* , re-enter by H to create a distinct $x'_2 = H(x_1)$, and apply H once more to obtain $x_3 = H(x'_2)$. No further physical suffix is used.

At every listed canonical source take its complete finite original presentation: the full primitive native banks, independent slots y^γ and $y^\epsilon \log y$ on their original disks, and each prescribed bounded analytic operation, lifted inverse, derivative and divided-remainder map with its full receiving presentation. The raw source is the completed projective product of these labelled factors and the original scalar factors, followed by the complete coefficient Wiener sums in every prescribed weak and parameter label. The operators are applied before any quotient. Original coefficient and parameter identifications are made only by their literal common substitution. Selected structural corrections are zero and connectors are the identity; variable banks, inverse families and source slots are not specialized to their scalar values.

10.7. The prescribed model bootstrap.

Lemma 10.30. *For the fixed full-bank data of Construction 10.29, take any finite lists of native clocks βe^{mu} , $\beta, m > 0$, at its listed parents u , and any finite weak lists $ce^{-\tau u}u^a$ at its listed positive principal clocks. All their full original projective/Wiener sources admit independent comparison, whole first heads and bounded differentiated numerator maps in the following order of drivers:*

$$(10.84) \quad \begin{aligned} s &= \sigma^{-1}, & H_* &= e^s, & K &= e^{H_*}, & Z_0 &= e^K, \\ R_1 &= e^{Z_0}, & T_1 &= e^{2Z_0}, & R_2 &= e^{R_1}, & T_2 &= e^{2T_1}, & T_3 &= e^{2T_2}. \end{aligned}$$

Equal-valued labels retain separate raw factors. Native banks are inserted at their logarithmic parent; a weak is inserted at its own principal driver. The model's full finite canonical recipes and inverse families have bounded maps into these receivers with original input norms and separate original parameter domains.

Proof. We verify the first lower field explicitly, propagate whole fold data in the displayed order, and receive the additional recipes.

Step 1. We initialize the scalar presentation. Use Proposition 9.27 on its entire original scalar source. It supplies the identity strong word, the separate endpoint and reciprocal-mark labels, their actual source maps, independent rows and original relative family, including their origin atoms, compensated sources and ordered mark powers. Its physical specialization has $s = \sigma^{-1}$ and $H_* = e^s$. The physical coefficient field needed for the first fold is generated from the coefficient Wiener sources in $z = \sigma$ and $l = \sigma \log \sigma$. Their complete scalar degree/logarithmic heads are the degree part of that proposition's rows. They have logarithmic size $O(\log s) = o(s)$ and horizontal ratio heads. On a sufficiently slow widening s horn their original absolute series are bounded uniformly; their exact derivatives

$$\partial_s z = -z^2, \quad \partial_s l = -z(l + z)$$

give whole numerator maps on strict receiving radii. The fraction field contains s and $\log s$.

The original relative family of that proposition is also received whole. Its smoothing is fixed at 3, and its exact bounded relative correction is

$$Z(\sigma, X) = \frac{s}{1 + \sigma^3 X}, \quad e^{-Z(\sigma, X)} = e^{-s} \exp\left(\frac{\sigma^2 X}{1 + \sigma^3 X}\right).$$

Keep its original larger X disk and take its stated strict restriction. The displayed correction is a convergent whole scalar source there with its original coefficient bound. Thus both distinct endpoint and mark factors use their own original weak copy multiplied by this correlated entire unit. Strict substitution gives the whole original outer-degree tail before any X coefficient is selected. Every fixed coefficient has its separate bounded target from the proposition. These are the initial old coefficient targets for Lemma 10.26; its hypotheses at each later fold are verified below. Original windows keep their own maps and complete complements. The new physical heights use $X = 0$; the original relative family remains on its given domain in parallel. Adjoin the endpoint and reciprocal labels as two distinct weak slots of value e^{-s} , together with the finite positive marks needed for their prescribed coefficient targets. Their joint weak rows collect their equality only after the original norms have been used. At s add each exact first native root $\beta^{-1/R} e^{-ms/R}$ needed for the full banks at βe^{ms} , including H_* . Lemmas 10.18 and 10.19 give the independent lower rows and a whole tail $\exp(-cT\Re s)$. Multiplication by the first root has reserve m/R . The exact logarithmic clock is $ms + \log \beta$, with derivative m and physical imaginary part zero. Thus Theorem 9.17 applies to the whole original tensor, with its prescribed q and full order- R banks. No assertion of surjectivity of a root refinement is used.

Step 2. We construct the next driver and its whole source. The next-driver inlet follows explicitly from the constructed fold. If $u < v$ are consecutive displayed drivers, their actual

relation $u = u(v)$ is a logarithm, a fixed positive power of a logarithm, a positive square root, or an exponential of a square root of a logarithm. The transition from R_2 to T_2 is $R_2 = \exp(\sqrt{\frac{1}{2} \log T_2})$. In all cases

$$u = o(v), \quad \partial_v u = o(1),$$

uniformly on fixed strict v strips. All are positive with their displayed actual inverses. The preceding whole head is $e^{-\lambda u} g(u)(1 + o(1))$ on a fixed thin u strip, with $g'/g \rightarrow 0$. Its pullback has logarithmic size $O(u) = o(v)$ and is horizontally stable, since a bounded v displacement changes u by $o(1)$. More explicitly, the complete buffered-disk law at u has factor $e^{-\lambda \zeta}$ as in (10.58); after the actual displacement $u(v + \zeta) - u(v) = o(1)$ its factor tends to one. The uniform remainder and Cauchy derivative bound hold for each fixed nonzero input and its fractions. A sufficiently slow widening v horn maps into that fixed strip and carries the entire preceding Banach stage with its original norm bound. Its fixed nonzero denominators are zero-free there after restriction. The whole numerator maps are transported by the exact coordinate derivative, whose denominator is a fixed field element. The field contains the next driver: an exact inverse root from a bank gives its clock by taking an integer inverse power; the listed logarithms and powers give the remaining relations. It also contains $\log v$, by the displayed driver identities.

Every original weak at v has residual $a \log v + \log c$, which is in this preceding field, is $o(v)$ and has vanishing physical imaginary part. The weak and horizontal lemmas supply its whole field and row data. Native clocks at this fold have exact logarithms $mv + \log \beta$; their root marks are inserted before their whole native tensor. Lemma 10.20 gives the independent kernel, head, horizontal bounds and full numerators needed at the next fold. This proves the finite induction.

In particular the banks at R_1 and T_1 are inserted together at Z_0 , the banks at R_2 are inserted at R_1 , and the banks at T_2 are inserted at T_1 . The old weak e^{-R_2} is inserted only at the later driver R_2 . The returned and re-entered copies at T_2 are distinct factors in one joint native tensor at T_1 . The T_3 banks occur at T_2 . This order accounts for the entire inactive $C(t)$ and x_2 branches. Isometric reordering of the original raw factors, as in Lemma 10.22, realizes this order without changing any input norm.

Step 3. We retain canonical recipes and original inverse families. At an opening $y = e^{-u}$, the exact canonical slots are $y^\gamma = e^{-\gamma u}$ and $y^\epsilon \log y = -ue^{-\epsilon u}$. They are the positive weak marks just constructed, on arbitrary fixed original slot radii. A finite complete canonical recipe is received on its final finite native/slot presentation. Products and analytic operations are entire absolute sums on strict disks; differentiation uses the whole numerator maps. A normalized native remainder is the whole division operator, iterated the finite prescribed number of times; positive monomial remainders use the finite first-occupied-index partition. Thus no polynomial replaces the original remainder. Lifted analytic inverses are solved in one strict receiving Banach stage around their whole scalar limiting function: its nonzero derivative has a bounded inverse on the original buffered disk, and the remaining positive slots can be made source-small. The resulting Picard map has derivative norm less than $1/2$. Its complete uniform limit is the genuine branch by the original scalar inverse identity. These constructions give bounded maps of the original recipe input before its coefficients are summed.

For the particular model reverse at a clock Y , its whole logarithmic inverse family is explicitly

$$Z(\zeta) = Y + \zeta, \quad s_{\text{rev}}(\zeta) = Y^{-1/m}(1 + \zeta/Y)^{-1/m}.$$

On any prescribed finite ζ disk, make the receiving copy of Y^{-1} sufficiently small. The full binomial series then has a strict geometric bound. Every new reference bank is evaluated by its entire relative operator at $\eta = \zeta/Y$. If its receiving bank map at Y is i , the full bound is $\|i\|L\|f\|_{\text{original}}/(1 - \|\zeta/Y\|/r_0)$, with the entire relative tail of (10.60). The lower inclusion and all tensor factors contribute the full product (10.61). Fix those costs before choosing a fresh receiving copy of Y^{-1} ; all old bank arguments stay fixed. The doubled-order outgoing reference after the C reverse is $(R_1 + \zeta)^2 = T_1(1 + \zeta/R_1)^2$. Its complete native relative increment is

$2\zeta/R_1 + \zeta^2/R_1^2$, inside the original relative ball. Its physical specialization is $\zeta = 0$; the full inverse family remains on its own original domain.

Step 4. We retain the atlas and coefficients at every model fold. The whole stage constructed at a parent u is bounded on its strict fixed strip and therefore on every far tube $|\Im u| < e^{-a\Re u}$. For each new principal clock βe^{mu} its logarithm is $mu + \log \beta$. Thus the hypotheses of Lemma 10.25 hold with $c = m$ and $\eta = 0$, before any new family record. It gives the genuine principal and normalized strong charts with the entire stage. The explicit logarithms and positive roots identify these branches with the model's original inverses.

For the raw construction at the ordered drivers in (10.84), first verify the independent normalized structural-jet bound of Definition 10.5. The verification below keeps the stronger bound $\log^+ |A_\mu| = o(u)$ before using a particular positive root reserve. Only after it and the exact finite parameter formulas have been supplied do we invoke Lemma 10.26. At the scalar start every fixed jet of $s/(1 + \sigma^3 X)$ at $X = 0$ is $(-1)^k k! s \sigma^{3k}$; its whole old target is supplied by Proposition 9.27. At a native fold $\log T_e = \log \beta_e + m_e u$, so

$$(10.85) \quad \partial_\lambda^\mu \log T_e = m_e \partial_\lambda^\mu u \quad (|\mu| > 0), \quad \partial_{\lambda_j} \log(e^{-\tau u} u^a) = (-\tau + a/u) \partial_{\lambda_j} u.$$

Differentiation of each exponential, positive root or logarithm in the literal model gives finite products of the active and earlier heights and their reciprocal powers, together with the whole previously constructed coefficient targets. For a fixed derivative order these structural products have logarithmic size $o(u)$ in the parent u : powers and reciprocal powers of any of the finitely many earlier positive heights have logarithmic size $o(u)$, and a power of u contributes only $O(\log u)$. The displayed driver relations place all those logarithms in the pre-weak field at u . Inactive faster targets stay in their later folds and are never extracted as such structural products. The complete pulled horizontal heads have their exact exponential factors retained, so these fixed structural phases have the required derivative and horn bounds.

Consequently a differentiated native factor exposes its whole first root $\beta_e^{-1/R} e^{-m_e u/R}$ times a structural product of logarithmic size $o(u)$. It has positive action m_e/R . At a weak's own driver, one occupied $e^{-\tau u} u^a$ similarly absorbs every new fixed structural product with action $\tau > 0$. The polynomial degree cost is precisely (10.71); the whole remaining coefficients retain their original strict ratio. At the two physical reverses the additional explicit coefficients are

$$(10.86) \quad [\zeta^k]_{s_{\text{rev}}}(\zeta) = \binom{-1/m}{k} Y^{-1/m-k}, \quad [\zeta^k] S(f)(Y + \zeta) = Y^{-k} S(\mathcal{D}_k f)(Y).$$

These are positive marked powers with the complete original relative operator, including its bank inclusion cost. The outgoing squared height after the C reverse has only the additional relative jets $2/R_1$ and $1/R_1^2$. Subsequent exponential compositions differentiate these exact jets at their new own driver, where the preceding reserve argument applies again. Lifted analytic inverses have the strict geometric derivative multiplier already constructed, so their fixed jets use (10.70).

Each fixed coefficient order therefore adds only finitely many positive marks with the conditional lower inlets of the fold induction. Their complete analytic units add nonnegative actions; their fixed derivative powers are subprincipal. The independent raw folds receive those marks before the full native and weak degree sums. All their original inclusion costs are fixed before the new mark radii. This verifies every hypothesis of Lemma 10.26 at s, H_*, K, Z_0 , at the joint R_1, T_1 native fold, at the separate R_2, T_2 folds in their stated order, at both reverse families, and at the re-entry and T_3 folds. It constructs each separate original coefficient target along with the literal composed parameter domains and their local Jacobian charts. A later exponential may shrink those domains; the entire older inverse family stays on its own domain in parallel. Original windows, exterior modules and complete complements keep their specified bounded maps. This proves the recipe and family assertions from the original scalar targets and the raw operators. $\square$

Proposition 10.31. *Construction 10.29 has a bounded whole original-source realization into a faithful complete quotient at each of its stages, with independently defined bounded comparison rows whose common kernel is its complex analytic kernel. It exports the particular FirstReturn presentation $\mathcal{P}_* \rightarrow Q_*$ at x_1 and the complete depth-three state at x_3 . Both retain the full $\mathcal{P}_*$ input map, every inactive source at x_2 , and its original inverse and coefficient families. The depth-three output satisfies Set-up 10.6, with base t , active path t, x_1, x'_2, x_3 , and distinct inactive birth x_2 . For its structural H births,*

$$(10.87) \quad \begin{aligned} D_b = \eta_b = \beta_b = c_b = d_b = 1, \quad m_b = 2, \quad a_b = G_b = \Xi_b = \ell_b = 0, \quad A_b = 1, \\ T_b = K_b = P_b = u_b = x_{p(b)}^{-2}. \end{aligned}$$

Every nonzero whole value has its complete positive scale and remainder, with real leading coefficient for real values. All original norms, domains, cuts, separate coefficient targets, observations, basal, tree, forward-tree and sibling sources persist. For each prescribed readout radius $\rho > 0$ the output has a bounded unital map $E_\rho : \mathcal{W}_\rho(\mathbb{C}) \rightarrow Q_3$ with

$$(10.88) \quad \text{Act } E_\rho f = f(x_3), \quad \|E_\rho f\| \leq L_\rho \|f\|_\rho, \quad \|E_\rho f_{\geq N}\| \leq L_\rho \vartheta_\rho^N \|f\|_\rho, \quad 0 < \vartheta_\rho < 1,$$

where the receiving stage, L_ρ and ϑ_ρ precede f and N . Its positive scalar map $\sigma \mapsto x_3$ has the actual inverse obtained from the displayed C and H inverses.

Proof of Proposition 10.31. We construct the physical model, receive its whole labelled input, and verify the full output data of Set-up 10.6.

Step 1. We identify the physical passages. The leaf equation of the supplied model is $d \log v / du = -mu^{-m-1}$. Integration from the incoming coordinate z to a_m gives $v(a_m) = b \exp(a_m^{-m} - z^{-m})$. The specified outgoing coordinate is therefore $\exp(-z^{-m})$, with the indicated positive inverse. Multiplying its vector field by $-1/m$ and taking $x = \zeta_m u$, $\zeta_m^m = -1/m$, gives $x^{m+1} \partial_x + v \partial_v$. Its structural normalizer and genuine fiber inverse are the identity, its residue is zero, and its graph correction is zero. Their complete minors therefore match the prescribed original q ; the structural orders 1 and 2 divide the prescribed even R . The variable bank at each label is nevertheless the full order- R bank in the original norm.

Step 2. We construct the complete original source. Apply Lemma 10.30 to the literal finite inventory in the construction. It constructs the original banks at H_*, K, Z_0 , the trace at the parent K with zero source increment, the full forward tree and its doubled-order sibling, and the two model reverses with their full inverse families. The common- y maps on every canonical recipe are the explicit slot and native substitutions in that lemma. Every prescribed bounded original recipe operator is applied to its entire input first, so its norm multiplies the whole receiving bound. A canonical quotient relation vanishes under the specified literal coordinate pullback; independent target faithfulness then annihilates it. Infima give the original quotient bound without a section of the quotient map. The full scalar physical identities are

$$C^{-1}(C(t)) = t, \quad x_1 = H(t), \quad H^{-1}(H(x_1)) = x_1, \quad x'_2 = H(x_1), \quad x_3 = H(x'_2).$$

The raw stage after the first H^{-1} is exactly $\mathcal{P}_*$: all earlier factors and all factors at x_2 are still present. It maps boundedly, with its unchanged actual square, to the depth-three receiver. Lemma 10.22 gives the bounded injective Q_* inclusion.

Step 3. We retain the full source records at depth three. We verify the data of Set-up 10.6 for the depth-three output. The ancestry and stem equations are the displayed structural identities, with $G_b = \Xi_b = 0$; $\log T_1 = 2Z_0$ and $\log T_j = 2T_{j-1}$ for $j = 2, 3$. The model fold lemma gives every weak and native inlet before its quotient, including differentiated numerator maps, full original bank norms, witness fibers and original common rays. For the grade gaps, at each depth let m_h, M_h be the minimum and maximum of the finite model clocks. Their exact exponential formulas give $m_{h+1} \geq e^{c_h m_h}$ and $M_{h+1} \leq e^{C_h M_h}$. Starting with $\log M_h = o(m_h)$ at the first displayed depth, these inequalities imply both $M_h = o(m_{h+1})$ and $\log M_{h+1} = o(m_{h+1})$. They

apply to the whole finite tree, including its sibling. For the coordinate record, the whole stage at each model driver is bounded on its already constructed fixed strip, independently of a readout. The explicit driver changes in (10.84) have uniformly vanishing displacement on fixed disks: in particular $R_1 = T_1^{1/2}$, $T_1 = (\log R_2)^2$, $R_2 = \exp(\sqrt{\log T_2}/2)$, and $T_2 = (\log T_3)/2$. Their actual branches pull a sufficiently slow widening horn into the old strip. Insert these displacements in the complete old own-driver laws and retain the whole norm bounds by restriction. Fixed subprincipal fractions then satisfy Lemma 10.2 at the new driver. This is the coordinate record of Definition 10.1; its buffers need not be controlled by the previous ancestry depth.

For example, at $u = T_2$ the retained value $a = \exp(-R_2)$ has $\log|a^{-1}| = R_2 = o(T_2)$, although R_2 is not $O(T_1)$. The derivative $R'_2(u) = R_2/(4u\sqrt{\log u/2})$ tends to zero on fixed disks, so $a(u+z)/a(u) \rightarrow 1$ uniformly there. Its finite derivatives are products of powers of u^{-1} , $\log u$ and R_2 ; a finite sublinear buffer bound absorbs their costs and the reciprocal cost. At T_1 this factor is still faster and is deferred. No ancestry profile is enlarged. The exact logarithms, roots and model inverse families retain the genuine charts and all original whole bounds.

Define the raw cuts on all the listed labels and recipes synchronously. Native first division and extraction of the first removed weak degree give their entire exponential error bound in the original norm. All retained heads have logarithmic size below the first removed principal depth. Independently construct the retained target by Lemma 10.30 and apply Lemma 10.13. This proves contractivity, all nested identities and old commuting squares. Each nonzero correction in a retained parameter recipe is an exact positive root or weak factor times a whole bounded analytic coefficient; for example the model inverse uses $\log(1 + \zeta/Y) = (\zeta/Y)a(\zeta/Y)$. The full division, analytic-difference and fresh-copy construction gives the finite positive certificates with original radii fixed. The structural corrections themselves have the empty certificate. Lemma 10.30 constructs the original relative and inverse families on larger disks and checks the exact jet and positive-reserve hypotheses of Lemma 10.26 at each fold. That application begins with the independent scalar coefficient targets of Proposition 9.27. It gives the literal domains, local full-Jacobian charts, separate bounded J_ν and original observation complements before asserting Set-up 10.6 for the model. These supply its original-family and observation data. The finite structural products and exact inverse coefficients in (10.85)–(10.86) verify Definition 10.5: normalized structural jets are subprincipal at their actual driver; native roots and occupied weaks then pay their fixed-order products; old faster targets remain whole. The literal local parameter charts retain their exact recovery factors and their separate original coefficient maps. This verifies all records for this particular full presentation; no admission of an arbitrary initial presentation is used.

Step 4. We identify the original readout. Include a last independent weak copy with actual value x_3 at a receiving radius at most $\theta\rho$, $\theta < 1$. Tensoring every other original factor with its unit embeds the entire original scalar series f into that slot. The resulting map satisfies $\|E_\rho f\| \leq L_\rho \|f\|_\rho$ and $\|E_\rho f_{\geq N}\| \leq L_\rho \theta^N \|f\|_\rho$ by its original absolute coefficient sum. This choice is made before f or N , and a finite list of prescribed ρ is handled simultaneously. The actual positive inverse $\sigma \leftarrow t \leftarrow x_3$ is the composition of the stated genuine inverses, so it identifies the physical tail without altering a source. $\square$

Construction 10.32. Use the fixed q, R , original banks, original driver σ , labelled raw factors, correlations, inverse families and strict domains of Construction 10.29. Stop that literal construction after $x_1 = H(t)$, the genuine child $x_2 = H(x_1)$, and its first physical return $H^{-1}(x_2) = x_1$, before the distinct re-entry birth x'_2 . The input of this first return is the entire original completed presentation through x_2 , including the inactive forward-tree branch at $C(t)$, the doubled-order sibling, and every original native, weak, canonical and parameter factor. Its output is the particular $\mathcal{P}_* \rightarrow Q_*$ of Proposition 10.31, with that whole input map in its original infimum norm.

The stored parent coordinate is $u = T_1 = t^{-2}$. In the notation of a complete direct-ending opening $F(z) = DzA(z)$ and a reverse of order r , its correlated normalized data are, in this construction,

(10.89)

$$D = \beta = \beta_R = 1, \quad m = r = 2, \quad A(z) = 1, \quad N = \beta F(e^{-u})^{-m} = e^{2u} = T_2, \quad V(\zeta) = 1 + \zeta/N.$$

For every prescribed finite original logarithmic inverse radius $\rho > 0$, the whole family on $|\zeta| < \rho$ is

$$(10.90) \quad s(\zeta) = Ce^{-\lambda u} A(e^{-u})^\lambda V(\zeta)^{-1/r}, \quad \lambda = m/r = 1, \quad C = \beta_R^{1/r} \beta^{-1/r} D^{m/r} = 1.$$

It is constructed by the full relative and binomial operators at the stored N , before $\zeta = 0$, and has actual value $(-\log x_2 + \zeta)^{-1/2}$. Each new original inverse-reference bank is received as $S_q(f)(N + \zeta)$ in its full original norm on that same family. The physical current is $s(0) = x_1$. The units A and V in this notation are these correlated constructed sources; no independent replacement units are allowed. The return retains every full bank and source at x_2 , all original inverse and coefficient families, the original driver and all old actual arguments. Its complete rows, original-norm bounds, true inverse domains and separate original coefficient targets are those proved for this $\mathcal{P}_* \rightarrow Q_*$. It also retains the coordinate and original-parameter positive-jet records of Definitions 10.1 and 10.5, supplied by the explicit model folds and inverse coefficients, with every complete old target and correlation. This construction supplies a particular first-return history and does not designate an arbitrary Banach presentation as such a history.

11. MIXED SOURCES AND EXTENSIONS FROM AN ARBITRARY CURRENT

We retain the chronological order of the physical passages, including the sections and branches on which their inverses are defined. The identities for a current are obtained by following this order. The estimates for its complete sources require additional constructions. In comparisons and replay, an earlier class means a strictly slower growth class, not an earlier birth. Physical occurrence order and ancestry depth retain their separate meanings.

Definition 11.1. An initial mixed receiving history starts with the entire first-return presentation at x_1 of Construction 10.32, including its original whole inverse family and every retained raw factor. It performs the following finite chronological construction.

At the literal returned inner strong section, insert one supplied original prepared passage or its correlated original inverse, and then its next supplied canonical direct passage. After this first connector and direct, allow a finite sequence of canonical reverse, prepared forward or correlated inverse, and canonical direct triples at that recorded parent. All positive analytic changes occur in their original positions. Retain the entire presentation of the first connector and direct when the subsequent sequence is empty.

End this alternation at a permitted returned strong prefix, or execute its next supplied canonical reverse to reach that strong section. Perform the whole parent-source handoff there. A finite suffix of positive analytic changes and canonical returns to recorded older parents is allowed. No already executed reverse is executed again, and no return goes below the retained base.

From this whole handoff output, allow positive analytic changes, direct re-entry along a recorded outgoing birth, and reverse passage to a recorded parent. Fresh supplied prepared passages and their correlated original inverses may be inserted at the literal current. Re-entry retains its recorded stem; this initial history does not yet add a genuinely new exponential stem.

The history includes every original raw input map, its norm, parameter domain, physical section and branch, and every active or inactive birth. Analytic slots use their original Wiener norms, prepared slots their original whole measure norms, and native slots their original root-bank norms. Repeated slots retain their common parameters and original multiplication; only independent slots form projective tensor products. An inverse prepared slot is the ordered inverse

of its recorded forward source. Its leaves and proper-subtree moments keep their original carriers. Original observations retain their literal domains and complete complements. At each receiving step the whole chronological raw maps are reconstructed before their independent comparison quotient is taken.

Theorem 11.2. *The entire receivers constructed from the histories of Definition 11.1, and their whole outputs under finite iteration of the extensions below, admit joint prepared-saddle sources and their same-source inverse, and admit a new canonical direct passage from every represented non-base positive real current*

$$(11.1) \quad S = Dx_v^\kappa e^{-G}, \quad D, \kappa > 0,$$

whose full source and physical phase data have been retained. If $x_v = d_v e^{-u} W_v$, a new passage of order p and scale β uses

$$(11.2) \quad N = \beta D^{-p} d_v^{-\kappa p} e^{\kappa p u}, \quad \Xi = pG - \kappa p \log W_v, \quad Z = N e^\Xi.$$

The whole transported error, its amplified exponential, all old sources and all new native factors are reconstructed before quotienting. No restriction that the current end in a direct passage or have rational κ is imposed. The extended presentation again has bounded independent comparison rows with analytic kernel and the complete leading-term conclusion. Every reverse passage retains and returns to the recorded parent rather than deleting its birth data. All assertions include whole analytic families in their original input norms. Every output retains the mixed receiving records of Definition 11.26 below, including the separate coordinate and original-parameter positive-jet records of Definitions 10.1 and 10.5.

The constructed entire receivers also retain the following positive-factor receiving operation. At a non-base current $s = Dx_v^\kappa e^{-G}$ with $x_v = d_v e^{-u} W_v$, put $c = -\log D - \kappa \log d_v$ and $h = G - \kappa \log W_v$. For the fixed degree-zero prepared bank $U_*(T) = \sum_{m \geq 1} e^{-mT}/m$, the whole translated source satisfies

$$(11.3) \quad B = U_*(\kappa u + c + h) = e^{-(\kappa/2)u} B_{\kappa/2}, \quad B_{\kappa/2} = \sum_{m \geq 1} \frac{e^{-mc}}{m} e^{-(\kappa m - \kappa/2)u} e^{-mh}.$$

The complete residual has its bounded original coefficient source and strict insertion margins before any new receiving mark is chosen. After these coefficient costs and the bounded costs of its forward prepared passage and same-source inverse are fixed, every $\delta > 0$ admits a further whole receiving extension in which the retained B has norm less than δ . The old quotient has a bounded injective algebra inclusion preserving actual values. All original input radii and norms remain fixed. The extension retains the whole native, prepared and inverse sources, the independent exact comparison kernel, the complete-head conclusion, the original input readouts and the separate marked-family targets. It remains an input for every subsequent literal operation with its original witnesses on the corresponding strict physical pullback domains.

11.1. The chronological current.

Definition 11.3. Consider a finite composition of supplied positive passages and analytic coordinate changes, on their literal matching domains. Start with a section x_0 and current x_0 . A direct canonical passage at the current section creates a new occurrence b , with parent the current label v , input t_b , and original data

$$(11.4) \quad T_b = \beta_b t_b^{-m_b}, \quad q_b = c_b e^{-T_b} T_b^{a_b}, \quad x_b = F_b(q_b e^{-\ell_b(T_b)}), \quad F_b(z) = d_b z A_b(z).$$

Here m_b is a positive integer, $\beta_b, c_b, d_b > 0$, $a_b \in \mathbb{R}$, and $A_b(0) = 1$. The functions, their original domains, and all logarithms in this formula are recorded with b . The logarithm ℓ_b is determined by the original normalizing factor, whose value at t is $\exp(-\ell_b(\beta_b t^{-m_b}))$. The new current is x_b . A reverse passage at a nonbase occurrence b changes the current label to its recorded parent. Its

original positive inverse height is denoted by Z_R , so that its inner output is $(\beta_R/Z_R)^{1/r}$, where r is its original positive integer order. The record of b remains in the list.

An analytic coordinate change or a prepared passage leaves the current label fixed. Distinct occurrences have distinct records, even when their actual heights agree. Throughout the composition all the maps are evaluated on their original connected matching components.

The birth equation in Definition 11.3 uses the supplied factorization of Theorem 6.1. It records an actual map and makes no assumption about a completed source space containing that map.

Lemma 11.4. *Every finite composition in Definition 11.3 has recursively determined data $D, \kappa > 0$ and a holomorphic correction G such that its current is $S = Dx_v^\kappa e^{-G}$. The correction is real on the positive physical interval. The initial data are $(D, \kappa, G) = (1, 1, 0)$. The updates at a fixed current label are as follows:*

$$(11.5) \quad \begin{aligned} I(t) = dtA(t) : (D, \kappa, G) &\mapsto (dD, \kappa, G - \log A(S)), \\ D_{\text{pre}}(t) = at^\eta e^{U(-\log t)} : (D, \kappa, G) &\mapsto (aD^\eta, \eta\kappa, \eta G - U(-\log S)), \\ D_{\text{pre}}^{-1} : (D, \kappa, G) &\mapsto ((D/a)^{1/\eta}, \kappa/\eta, G/\eta + J((- \log S + \log a)/\eta)). \end{aligned}$$

Here $d, a, \eta > 0$, $A(0) = 1$, and $(\text{id} - U/\eta)^{-1}(w) = w + J(w)$ is the inverse belonging to the original prepared passage. At a new direct birth, record the current data as (D_b, η_b, G_b) and reset the current data to $(1, 1, 0)$ at b .

Suppose the current label is a nonbase birth b , whose recorded input was $t_b = D_b x_v^{\eta_b} e^{-G_b}$. The positive-real-exponent identity of Lemma 10.16 gives, before the inverse incoming analytic coordinate change, the following data for a reverse of order r :

$$(11.6) \quad \begin{aligned} D_R &= \left(\frac{\beta_R}{\kappa\beta_b} \right)^{1/r} D_b^{m_b/r}, \quad \kappa_R = \frac{\eta_b m_b}{r}, \\ G_R &= \frac{m_b}{r} G_b + \frac{1}{r} \log \frac{Z_R}{\kappa T_b}. \end{aligned}$$

The incoming analytic coordinate change then uses the first line of (11.5).

Proof. We check the updates by substitution and then induct on the number of executed letters. An analytic coordinate change gives $I(S) = dDx_v^\kappa \exp(-G + \log A(S))$. For a prepared forward passage, raising S to η and multiplying by $a \exp U(-\log S)$ gives the second line of (11.5). The inverse formula in Theorem 5.1 gives the third line, with its original J and its original input argument.

At a direct birth, (11.4) defines the new section itself, so the reset is an identity. For a reverse, the positive original root satisfies

$$(11.7) \quad \left(\frac{\beta_R}{Z_R} \right)^{1/r} = \left(\frac{\beta_R}{\kappa\beta_b} \right)^{1/r} t_b^{m_b/r} \exp \left(-\frac{1}{r} \log \frac{Z_R}{\kappa T_b} \right).$$

Inserting the recorded formula for t_b proves (11.6). In particular, the current κ occurs in the scale and the height ratio; the exponent at the returned parent is $\eta_b m_b / r$.

Each update preserves positivity of D and κ . On the physical interval, $S/(Dx_v^\kappa) > 0$; the continued logarithm anchored there gives $G = -\log(S/(Dx_v^\kappa)) \in \mathbb{R}$. The assertion follows by finite induction. This argument proves the identities without asserting source bounds for G . $\square$

Lemma 11.5. *Let a current obtained in Lemma 11.4 be at a nonbase section $x_v = d_v e^{-u} W_v$, where $d_v > 0$ and W_v has its recorded logarithm. For an original direct height $Z = \beta S^{-p}$, with $\beta > 0$ and positive integer p , define*

$$(11.8) \quad \gamma = \kappa p, \quad N = \beta D^{-p} d_v^{-\kappa p} e^{\gamma u}, \quad \Xi = pG - \kappa p \log W_v.$$

Then $Z = Ne^{\Xi}$. If P_e is any supplied positive integer root order for a native factor at this reference, its original root is

$$(11.9) \quad N^{-1/P_e} = \beta^{-1/P_e} D^{p/P_e} d_v^{\kappa p/P_e} e^{-\kappa p u/P_e}.$$

Its rate at the parent is the positive real number $\kappa p/P_e$. If $u = x + i\eta(x)$ with $\eta(x) \rightarrow 0$, then $\text{Im} \log N \rightarrow 0$ on the recorded branch.

Proof. The exact current identity gives $S = D d_v^{\kappa} e^{-\kappa u} W_v^{\kappa} e^{-G}$. Taking its inverse p th power proves (11.8). Taking the recorded P_e th root proves (11.9). The logarithm of its positive real prefactor is real, so $\text{Im} \log N = \kappa p \eta(x)$. $\square$

11.2. Whole prepared translations and residuals.

Definition 11.6. Let E be a complex Banach space. For a list of E -valued measures ν_K on $[0, \infty)$, dominated in variation by positive measures μ_K , write

$$(11.10) \quad B(T) = \sum_{K \geq 0} T^K \int e^{-\beta T} d\nu_K(\beta), \quad M_{S_0} = \sum_{K \geq 0} \int e^{-S_0 \beta} d\mu_K(\beta).$$

We use this representation where the displayed series converges absolutely. Suppose $M_{S_0} < \infty$ and the measures have support $\beta \geq g$ and $\beta \geq \delta K$, for fixed $g, \delta > 0$. For a receiving list σ_l with carrier α and positive numbers b, r , its carrier norm is

$$(11.11) \quad \|(\sigma_l)\|_{b,r} = \sum_{l \geq 0} r^l \int e^{-b\alpha} d|\sigma_l|(\alpha).$$

This norm is a norm on the measure lists. It makes no assertion that their realized germs determine the lists.

Lemma 11.7. Use the data of Definition 11.6. Fix $\lambda > 0$, $c \in \mathbb{R}$, and $R, r > 0$. Choose $b > 0$ and $s_0 > 0$ such that

$$(11.12) \quad b\lambda \geq S_0 + |c| + R + \delta^{-1} \log_+(\lambda r + |c| + R) + s_0, \quad \log_+ t = \max(0, \log t).$$

The whole translated family $B(\lambda u + c + z)$, $|z| < R$, has a holomorphic receiving measure list of norm at most M_{S_0} . The assertion holds in the original norm of the input list. Inserting an old coefficient map of norm L_0 changes this bound to $L_0 M_{S_0}$.

For any $0 < \epsilon < \lambda g$, a whole residual list $B_\epsilon(u, z)$ exists and satisfies

$$(11.13) \quad B(\lambda u + c + z) = e^{-\epsilon u} B_\epsilon(u, z), \quad \|B_\epsilon(\cdot, z)\|_{b,r} \leq e^{b\epsilon} M_{S_0}.$$

Its carrier α has gap $\lambda g - \epsilon$ and linkage $\alpha \geq q\lambda\delta l$, where $q = 1 - \epsilon/(\lambda g) > 0$ and l is its log-power degree. All these bounds precede the z coefficients and log-power sums. Each fixed additional carrier moment is bounded using a positive part of the slack s_0 .

Proof. We construct the translated list first and then shift its total carrier.

Step 1. We translate the entire list. For each original K , expand only the polynomial factor:

$$(11.14) \quad (\lambda u + c + z)^K e^{-\beta(\lambda u + c + z)} = \sum_{l=0}^K \binom{K}{l} \lambda^l u^l (c + z)^{K-l} e^{-\beta(c+z)} e^{-\lambda\beta u}.$$

Push each measure forward by $\beta \mapsto \lambda\beta$. The norm of the sum of these lists is bounded by

$$(11.15) \quad \sum_K \int e^{-b\lambda\beta + (|c|+R)\beta} (\lambda r + |c| + R)^K d\mu_K(\beta) \leq \sum_K \int e^{-S_0\beta - s_0\beta} d\mu_K(\beta).$$

The inequality follows from $K \leq \beta/\delta$ and (11.12). Thus the majorant is at most M_{S_0} . Local uniform convergence in the receiving norm proves Banach holomorphy in z . Applying a bounded coefficient map inside the integrals multiplies this entire majorant by its norm.

Step 2. We treat one residual shift. In the translated list replace its carrier $\lambda\beta$ by $\alpha = \lambda\beta - \epsilon$, leaving its coefficient unchanged. This proves (11.13) on the original convergence domain, and continuation preserves the identity on its connected component. The receiving weight gains the fixed factor $e^{b\epsilon}$. Since $\beta \geq g$ and $l \leq K$,

$$(11.16) \quad \alpha \geq \lambda g - \epsilon > 0, \quad \alpha \geq q\lambda\beta \geq q\lambda\delta l.$$

For a fixed integer $h \geq 0$ and $0 < s_1 < s_0$, $\sup_{\beta \geq 0} \beta^h e^{-s_1\beta} < \infty$. This factor bounds the added moment in (11.15), while the remaining slack preserves the whole convergence bound. $\square$

Lemma 11.8. *Let Γ be generated additively by a nonempty finite list of positive real numbers. For $g_0 > \epsilon > 0$, the set $\{\omega - \epsilon : \omega \in \Gamma, \omega \geq g_0\}$ is contained in an additive monoid generated by finitely many positive real numbers. Every bounded interval meets that monoid in a finite set.*

Proof. Let $a_{\max}$ be the largest generator. Given a representation of $\omega \geq g_0$, add its generators until the sum first reaches g_0 . The partial sum μ lies in $[g_0, g_0 + a_{\max})$. Only finitely many values of μ are possible: the coefficient of each positive generator is bounded by $(g_0 + a_{\max})$ divided by that generator. Adjoin the finitely many numbers $\mu - \epsilon > 0$ to the old generators. Then $\omega - \epsilon = (\mu - \epsilon) + (\omega - \mu)$ belongs to the resulting monoid. Bounding the coefficient of each of its positive generators proves the final assertion. $\square$

Lemma 11.9. *Suppose a scalar forward bank U has the representation in Definition 11.6, gap $g > 0$, linkage $K \leq \beta/\delta$, and $M = \sum_K \int r^K e^{-S_1\beta} d\mu_K < \infty$ for $r > 0$. Let $\eta > 0$, put $c_r = 1 + 1/(\delta r)$, and choose $A > c_r M/\eta$. For the ordered products of this bank in*

$$(11.17) \quad J(T) = \sum_{n \geq 1} \frac{\partial_T^{n-1}(U(T)^n)}{n! \eta^n},$$

the whole receiving norm at carrier weight $S_1 + A$ and log radius r is at most

$$(11.18) \quad \frac{M}{\eta} \sum_{n \geq 1} \frac{q_0^{n-1}}{n}, \quad q_0 = \frac{c_r M}{A\eta} < 1.$$

The part with $n > N$ is bounded by $(M/\eta)q_0^N/(1 - q_0)$ for every integer $N \geq 0$. All leaves remain occurrences of the original forward source. Any residual factor from Lemma 11.7 is taken after the total convolution.

Proof. For an ordered n -tuple of leaves let β and K be their total carrier and total log degree. Then $\beta \geq ng$ and $K \leq \beta/\delta$. At degree n , differentiating $T^K e^{-\beta T}$ gives the coefficients $\binom{n-1}{l}(K)_l(-\beta)^{n-1-l}$ for $0 \leq l \leq \min(K, n-1)$, where $(K)_l = K!/(K-l)!$. Their absolute receiving sum is at most

$$(11.19) \quad \frac{\beta^{n-1} r^K}{n! \eta^n} \left(1 + \frac{1}{\delta r}\right)^{n-1}.$$

Retaining the $(n-1)$ st term of the exponential series gives $\beta^{n-1} e^{-A\beta} \leq (n-1)! A^{-(n-1)}$. After integration of the ordered product majorant, its n th norm is therefore at most $(M/\eta)q_0^{n-1}/n$. Summing proves (11.18) and the stated tail.

For the actual prepared inverse, Lemma 5.13 identifies (11.17) with its original inverse source. Ordered leaves and any proper-subtree moments in that source keep their original carriers in this estimate. A residual changes the total carrier $\lambda\beta$ to $\lambda\beta - \epsilon$ once; applying that change to each of n leaves would instead subtract $n\epsilon$ and would give a different source. $\square$

Lemma 11.10. *Let E be a unital commutative Banach algebra with unit of norm one. Fix $\kappa > 0$, $c \in \mathbb{R}$, $S_0 > 0$, $0 < r < R$, and $h \in E$ with $\|h\| \leq r$. For the original degree-zero bank U_**

in (5.48), put

$$(11.20) \quad M_* = \sum_{m \geq 1} \frac{e^{-S_0 m}}{m}, \quad \epsilon = \kappa/2, \quad b\kappa > S_0 + |c| + R.$$

The whole residual family at an independent additive argument z , $|z| < R$, has receiving norm at most $e^{b\epsilon} M_*$. Its complete insertion at h has norm at most

$$(11.21) \quad C_* = \frac{e^{b\epsilon} M_*}{1 - r/R}.$$

These bounds use the entire original atomic measure $\sum_{m \geq 1} m^{-1} \delta_m$ and precede all m and additive-argument degrees.

Suppose the following whole maps have been constructed: a bounded linear map from the entire inserted-source space into a receiving algebra Q with operator norm at most L_0 ; a unital inclusion $I_{\text{pair}} : Q \rightarrow Q_{\text{pair}}$ of norm L_{pair} , retaining the original forward and correlated inverse sources; and, for $0 < \sigma < \sigma_0$, unital inclusions $I_\sigma : Q_{\text{pair}} \rightarrow Q_\sigma$ of norm at most L_1 independent of σ . Suppose Q_σ is faithfully realized on the original connected component, all the inclusions preserve actual values, and Q_σ contains a marked element t_σ with $\text{Act } t_\sigma = e^{-\epsilon u}$ and $\|t_\sigma\| \leq \sigma$. If Q_{pair} also retains the whole element $B = U_*(\kappa u + c + h)$, then its retained image satisfies

$$(11.22) \quad \|I_\sigma B\|_{Q_\sigma} \leq \sigma L_1 L_{\text{pair}} L_0 C_*.$$

Consequently it has norm less than any prescribed $\delta > 0$ when $\sigma < \min(\sigma_0, \delta/(L_1 L_{\text{pair}} L_0 C_*))$, if this denominator is positive. If that denominator is zero, the retained image of B is zero.

Proof. Apply Lemma 11.7 with $\lambda = \kappa$, $K = 0$, $g = 1$ and $\epsilon = \kappa/2$. The remaining action of the m th atom is $\kappa m - \kappa/2 \geq \kappa m/2 > 0$. Its coefficient is the original $m^{-1} e^{-mc} e^{-mz}$. The full majorant is

$$(11.23) \quad \sum_{m \geq 1} \frac{1}{m} e^{-b(\kappa m - \kappa/2) + m(|c| + R)} \leq e^{b\kappa/2} M_*.$$

The larger-disk coefficient estimate, followed by insertion of the whole h , gives the full sum

$$(11.24) \quad e^{b\epsilon} M_* \sum_{n \geq 0} (r/R)^n = C_*.$$

This proves the exact residual in (11.3) with its entire coefficient. Lemma 5.13 supplies the correlated inverse of this degree-zero bank; its ordered original leaves are included in the prescribed pair map.

Let $C \in Q$ denote the image of the inserted residual. Its norm is at most $L_0 C_*$. The element $t_\sigma I_\sigma I_{\text{pair}} C$ has actual value B by the exact factorization. Faithfulness of Q_σ identifies it with the retained $I_\sigma B$. Multiplying the three fixed whole-map bounds and the marked radius proves (11.22). All these bounds are fixed before the final choice of σ . $\square$

11.3. Total action and complete discrepancies.

Lemma 11.11. *Let $\nu_1, \dots, \nu_s$ be locally finite measures on the carrier variables $\beta_1, \dots, \beta_s$, and let $K_1, \dots, K_s$ be nonnegative integers. Put $\Omega = \sum_a d_a \beta_a - \epsilon_0$ with $d_a > 0$. For a smooth compactly supported cutoff χ in total action and a smooth test function F , wherever the following pairings are defined,*

$$(11.25) \quad \left\langle \chi(\Omega) \bigotimes_a \partial_{\beta_a}^{K_a} \nu_a, F \right\rangle = \sum_{0 \leq l_a \leq K_a} (-1)^{\sum_a K_a} \left(\prod_a \binom{K_a}{l_a} d_a^{l_a} \right) \int \chi(\sum_a l_a)(\Omega) \partial_{\beta}^{K-l} F d(\bigotimes_a \nu_a).$$

Here $\partial_\beta^{K-l} = \prod_a \partial_{\beta_a}^{K_a-l_a}$. For $F = \prod_a e^{-\beta_a T_a}$, with T_a independent of the carriers, the summand has sign $(-1)^{\sum_a l_a}$ and its last derivative factor is $\prod_a T_a^{K_a-l_a} e^{-\beta_a T_a}$. Carrier-dependent coefficients inside F are differentiated in full.

Proof. The definition of a distributional derivative gives the sign $(-1)^{\sum K_a}$ and applies ∂_β^K to $\chi(\Omega)F$. The product rule, together with $\partial_{\beta_a}\chi(\Omega) = d_a\chi'(\Omega)$, gives (11.25). For the displayed exponential test function, $\partial_\beta^{K-l}F = (-1)^{\sum(K_a-l_a)} \prod_a T_a^{K_a-l_a} e^{-\beta_a T_a}$, which gives the asserted sign. Absolute majorants permit the identity to pass to convergent sums of such pairings. $\square$

In a prepared inverse, the variables in Lemma 11.11 are the original carriers of the ordered product. A compact cut acts on their total action after convolution and the single residual shift. When $\sum K_a \leq T_1/\delta_*$ on the support of the cut, only derivatives of χ through order $\lceil T_1/\delta_* \rceil$ occur. This observation retains every derivative in the compact square. A comparison argument also needs a cofinal estimate for the entire complement, uniform on its prescribed common source domain.

Lemma 11.12. *Let E be a unital commutative complex Banach algebra with submultiplicative norm. Let $B(z) = \sum_{n \geq 0} b_n z^n$ be holomorphic for $|z| < R$, with $\sup_{|z| < R} \|B(z)\| \leq M$. For $h, k \in E$ with $\|h\|, \|k\| \leq r < R$, its complete substitution satisfies*

$$(11.26) \quad B(h) - B(k) = (h - k)D_B(h, k), \quad \|D_B(h, k)\| \leq \frac{M}{R(1 - r/R)^2}.$$

If Φ and Φ_c are holomorphic maps on a convex ball in E , $h = \Phi(h)$ and $h_c = \Phi_c(h_c)$ lie in that ball, and $\|\Phi'(z)\| \leq q < 1$ there, then

$$(11.27) \quad h - h_c = (\text{id} - A)^{-1}(\Phi(h_c) - \Phi_c(h_c)), \quad A = \int_0^1 \Phi'(h_c + t(h - h_c)) dt, \quad \|(\text{id} - A)^{-1}\| \leq (1 - q)^{-1}.$$

Here A is a bounded operator on E .

Proof. Cauchy's coefficient bound gives $\|b_n\| \leq MR^{-n}$. Use the exact identity $h^n - k^n = (h - k) \sum_{j=0}^{n-1} h^j k^{n-1-j}$ and sum its absolute majorant $\sum_{n \geq 1} MR^{-n} nr^{n-1}$ to obtain (11.26). For the second assertion, integrate Φ' along the segment from h_c to h . Subtract the two fixed-point equations and invert $\text{id} - A$ by its norm-convergent geometric series. $\square$

Lemma 11.13. *Let $V, E_1, \dots, E_s, Q$ be Banach spaces, with Q a commutative Banach algebra realized multiplicatively by holomorphic functions on a common domain. Suppose bounded linear maps $D_a : V \rightarrow E_a$ and $I_a : E_a \rightarrow Q$ have norm products $L_a = \|I_a\| \|D_a\|$. Suppose a linear difference δf of realized functions has the entire decomposition*

$$(11.28) \quad \delta f = \sum_{a=1}^s m_a \text{Act}(I_a D_a f).$$

Here m_a are fixed holomorphic functions on that domain. Let M_0 be another fixed holomorphic function. If $b_a \in Q$ have realizations $M_0 m_a$ and norms at most r_a , then

$$(11.29) \quad Wf = \sum_{a=1}^s b_a I_a D_a f, \quad \text{Act}(Wf) = M_0 \delta f, \quad \|W\| \leq \sum_{a=1}^s r_a L_a.$$

Thus an operator bound ϵ follows by fixing the L_a first and then receiving these finitely many marked factors with $\sum_a r_a L_a \leq \epsilon$.

Proof. Linearity follows from the displayed definition of W . Multiplicativity of realization proves its actual identity, and submultiplicativity of the norm proves the operator bound. $\square$

For a source cutoff, the first removed factor in a monomial gives a decomposition of the form (11.28) whenever its coefficient maps have already been bounded. An erased prepared term has the whole residual in (11.13) as its coefficient. For a retained prepared family, first project its coefficients, then apply (11.26) to its entire increment. A retained inverse uses (11.27) on the full projected equation. In each case the coefficient cost precedes the choice of the marked radius. Lemma 11.13 concerns a linear map into a receiving algebra; it does not require M_0 to be a bounded multiplier of the original algebra.

11.4. The amplified correction.

Lemma 11.14. *Let Q be a unital commutative Banach algebra with $\|1\| = 1$ and a unital multiplicative realization Act . Let $e : Q \rightarrow Q$ be a bounded unital algebra projection. Let N be a nonvanishing holomorphic function on the realization domain. Suppose a unital commutative receiving Banach algebra $\tilde{Q}$, also with $\|1\| = 1$, includes Q by a bounded unital homomorphism I preserving realization, and a bounded linear map $W : Q \rightarrow \tilde{Q}$ satisfies*

$$(11.30) \quad \text{Act}(Wf) = N \text{Act}(f - ef), \quad \|W\| \leq \epsilon.$$

For $\Xi \in Q$, put

$$(11.31) \quad Y = N \exp(\text{Act } \Xi), \quad P = N \exp(\text{Act}(e\Xi)), \quad B = W(\exp \Xi), \quad L = I(\Xi - e\Xi).$$

Then $\text{Act } B = Y - P$, $\text{Act } L = \log(Y/P)$ on the continued branch, and $\|B\| \leq \epsilon e^{\|\Xi\|}$. For $c, \tau > 0$ and $a \in \mathbb{R}$ the exact weak factor is

$$(11.32) \quad \begin{aligned} ce^{-\tau Y} Y^a &= ce^{-\tau P} P^a \text{Act}(\exp(-\tau B + aL)), \\ \|\exp(-\tau B + aL)\| &\leq \exp(\tau \epsilon e^{\|\Xi\|} + |a| \|L\|). \end{aligned}$$

If Y is positive on the physical interval and $\text{Act } B \rightarrow 0$ there, then $\text{Im } P \rightarrow 0$ absolutely.

Proof. A bounded unital homomorphism commutes with the convergent exponential series. In particular, $e(\exp \Xi) = \exp(e\Xi)$, so (11.30) proves the first identity. The logarithm of Y/P continued as $\text{Act}(\Xi - e\Xi)$ proves the second. The exponential norm estimate gives the bound for B . Substituting $Y = P + \text{Act } B$ and $Y^a = P^a \exp(a \text{Act } L)$ gives (11.32) with its entire exponential and its norm bound. Finally, $\text{Im } P = -\text{Im } \text{Act } B$ on the positive physical interval. $\square$

In the canonical output of residue a_+ and weak constant c_+ , set $H = P - a_+ \log P - \log c_+$. If the original native logarithm $\ell_+(Y)$ has already been received, the complete endpoint formula is

$$(11.33) \quad E_+ = \text{Act } B - a_+ \text{Act } L + \ell_+(Y), \quad x_{b_+} = F_+(e^{-H} e^{-E_+}).$$

We keep the strong part of Ξ intact in (11.33). For example, with $N = e^{\gamma u}$ and $\Xi = e^{-\sigma u}$, where $0 < \sigma < \gamma$, the quantity $N(e^\Xi - 1)$ grows like $e^{(\gamma - \sigma)u}$. The fact that $\Xi \rightarrow 0$ does not allow this term to be discarded.

For an original Wiener source of radius ρ , fix the norm of the correction unit in (11.32) before receiving $ce^{-\tau P} P^a$. If their product of norms is at most $\theta\rho$, with $0 < \theta < 1$, Lemma 4.5 gives the whole substitution and its degree greater than M tail $\theta^{M+1} \|F\|_{\mathcal{W}_\rho}$. For repeated source slots the original multiplication is performed before the single map of that slot.

11.5. Charts carrying whole sources.

Lemma 11.15. *Fix $a \in \mathbb{R}$ and the logarithm on the right half-plane. Let w be a positive continuous nondecreasing function tending to infinity, with $w(x) \leq \sqrt{x}$ for large x . The map $H(P) = P - a \log P$ has a holomorphic inverse $P = \psi(H)$ on a far horn*

$$(11.34) \quad \text{Re } H > x_0, \quad |\text{Im } H| < \frac{1}{4} w(\text{Re } H/2).$$

Its image lies in the original horn $|\operatorname{Im} P| < w(\operatorname{Re} P)$, after increasing x_0 . It satisfies (11.35)

$$\psi(H) = H + a \log \psi(H), \quad \psi(H) = H + O(\log \operatorname{Re} H), \quad \psi'(H) = \left(1 - \frac{a}{\psi(H)}\right)^{-1} \longrightarrow 1.$$

Any bounded whole source map on the original horn pulls back through ψ with its original operator bound. In particular, an exact action term becomes

$$(11.36) \quad e^{-\omega P} L_\omega(P) = e^{-\omega H} \psi(H)^{-a\omega} L_\omega(\psi(H)).$$

Proof. Write $x = \operatorname{Re} H$ and take $C = 4(1 + |a|)$. On $|P - H| \leq C \log x$ and for far x , we have $\operatorname{Re} P \geq x/2$, $|\log P| \leq 2 \log x$, and $|a/P| \leq 2|a|/x < 1/2$. The map $P \mapsto H + a \log P$ is a strict contraction of this disk into itself. Its iterates converge locally uniformly in H , giving the holomorphic solution and its unique branch. Differentiating its equation proves (11.35).

The imaginary part satisfies $|\operatorname{Im} \psi| \leq |\operatorname{Im} H| + 2|a| |\operatorname{Im} \psi|/x$. For large x , this gives $|\operatorname{Im} \psi| \leq 2|\operatorname{Im} H| < w(x/2)/2 \leq w(\operatorname{Re} \psi)$. It proves the domain assertion and hence the whole pullback bound. Finally $\psi - H = a \log \psi$ proves (11.36). $\square$

In an uncut carrier integral, the factor $\psi(H)^{-a\omega}$ in (11.36) remains in the original kernel. Its possible growth is $\exp(O(\omega \log \operatorname{Re} H))$. A fixed positive fraction of the carrier damping $\exp(-c\omega \operatorname{Re} H)$ absorbs this cost on a far horn, uniformly in ω . Only after a compact total-action cut may its finitely many surviving values be placed in individual lower coefficients.

Lemma 11.16. *Let $u_x = x + i\eta(x)$ with $\eta \in C^1$, $\eta(x), \eta'(x) \rightarrow 0$, and let $\gamma > 0$, $c \in \mathbb{R}$. Suppose ρ is holomorphic on a domain containing, for each fixed $a > 0$ and sufficiently large x , the disks $|u - u_x| < e^{-ax}$. Suppose ρ is bounded uniformly on each such union of disks and $\rho(u_x), \rho'(u_x) \rightarrow 0$. Define*

$$(11.37) \quad \Phi(u) = \gamma u + c + \rho(u), \quad P(u) = \exp \Phi(u),$$

and suppose $\operatorname{Im} P(u_x) \rightarrow 0$ absolutely. Then P has an inverse on every far strict power horn $|\operatorname{Im} Z| < b(\operatorname{Re} Z)^\nu$, where $b > 0$ and $0 < \nu < 1$, on its branch along u_x . If a Banach space E has a whole realization bound $\sup |\operatorname{Act} f| \leq L_a \|f\|_E$ on each of the above unions of disks, this inverse carries that entire source to the power horn with a fixed original-norm bound. The assertion also holds for $P - a_+ \log P - \log c_+$, with $a_+ \in \mathbb{R}$ and $c_+ > 0$. The two image physical curves, reparameterized by their real parts, have imaginary parts and imaginary derivatives tending to zero.

Proof. We construct local inverses with quantitative radii, glue their branches, and compare these radii with a power-horn displacement. Fix $a > 0$. Cauchy's estimate on the larger disks gives $|\rho''(u)| \leq C_a e^{2ax}$ on disks of half the radius about u_x . Thus $\Phi'(u) \rightarrow \gamma$ uniformly when $|u - u_x| \leq e^{-3ax}$. Put $r_x = e^{-4ax}$. For all far x , $|\Phi'(u) - \gamma| \leq \gamma/4$ on $|u - u_x| \leq 2r_x$.

For $|v - \Phi(u_x)| < \gamma r_x/16$, solve

$$(11.38) \quad u = u_x + \frac{v - \Phi(u_x)}{\gamma} - \frac{\Phi(u) - \Phi(u_x) - \gamma(u - u_x)}{\gamma}.$$

On the r_x disk the right side has displacement at most $5r_x/16$ and contraction constant at most $1/4$. The local inverse is holomorphic and has displacement at most $r_x/12$. Moreover, $d\operatorname{Re} \Phi(u_x)/dx \rightarrow \gamma$. If two image disks intersect, this lower derivative bound gives $|x - y| \leq (r_x + r_y)/8$. The radii are then asymptotically equal and $\eta' \rightarrow 0$. Both inverse points lie in one buffered disk, where integrating $\Phi' - \gamma$ on their segment proves injectivity. The inverse branches therefore agree and glue.

Since $\rho(u_x) \rightarrow 0$, the real parts of $P(u_x)$ grow like $e^{\gamma x + c}$ and $\operatorname{Im} \Phi(u_x) \rightarrow 0$. The stronger assumption $\operatorname{Im} P(u_x) \rightarrow 0$ implies $\operatorname{Im} \Phi(u_x) = o(e^{-\gamma x})$. For a point of a strict power horn, its logarithmic distance from the corresponding center is $O(e^{-(1-\nu)\gamma x}) + o(e^{-\gamma x})$. Choose $4a <$

$(1 - \nu)\gamma$. This distance is smaller than $\gamma r_x/16$ for all far x , so the glued inverse covers the horn. Its image lies in the disks carrying the original E bound.

Finally the equation $Z - a_+ \log Z - \log c_+ = H$ is solved by $Z = H + a_+ \log Z + \log c_+$ on a disk of radius $C \log \operatorname{Re} H$ for a fixed sufficiently large C . Its derivative in Z has size $O(1/\operatorname{Re} H)$. The solution belongs to a slightly larger strict power horn. Composing it with the inverse of P constructed by (11.38) gives the final inverse and the whole source bound. Along the original curve, $dP(u_x)/dx = P(u_x)(\gamma + o(1))(1 + i\eta'(x))$. Its real part is eventually positive, and the ratio of its imaginary and real parts tends to zero. This proves the assertion about the reparameterized P curve. Since $dH/dP = 1 - a_+/P \rightarrow 1$ and $\Im \log P \rightarrow 0$, it proves the assertion for H too. $\square$

11.6. Local parameter domains.

Lemma 11.17. *Let V, E be complex Banach spaces, let D_t be a driver disk, and let $h \geq 1$. Suppose a whole linear operator family $\mathcal{T}$ maps V to holomorphic E -valued functions on $D_t \times \{|z_a| < 2\}_{a=1}^h$ and satisfies $\|\mathcal{T}(f)(t, z)\|_E \leq C\|f\|_V$. Then, uniformly on D_t , its total degree at least n tail on $|z_a| \leq 1$ has norm at most*

$$(11.39) \quad 3^h C \|f\|_V (3/4)^n.$$

If $\mathcal{T}(f)(t, z) = F(f)(t, r_t z)$ for an original family in parameters $\lambda_a = r_t z_a$, where $r_t > 0$ is constant on D_t , its exact coefficient recovery is

$$(11.40) \quad \begin{aligned} [\lambda^\mu] F(f)(t, \lambda) &= r_t^{-|\mu|} [z^\mu] \mathcal{T}(f)(t, z), \\ \partial_\lambda^\mu F(f)(t, 0) &= \mu! [\lambda^\mu] F(f)(t, \lambda). \end{aligned}$$

Here $|\mu| = \sum_a \mu_a$ and $\mu! = \prod_a \mu_a!$. The coefficient estimate includes $r_t^{-|\mu|}$.

Proof. For $1 < R < 2$ the coefficient is the contour integral

$$(11.41) \quad [z^\mu] \mathcal{T}(f)(t, z) = \frac{1}{(2\pi i)^h} \int_{|\zeta_1|=R} \cdots \int_{|\zeta_h|=R} \frac{\mathcal{T}(f)(t, \zeta)}{\zeta_1^{\mu_1+1} \cdots \zeta_h^{\mu_h+1}} d\zeta_h \cdots d\zeta_1.$$

Estimating these integrals and then letting R increase to 2 gives $\|[z^\mu] \mathcal{T}(f)\|_E \leq C\|f\|_V 2^{-|\mu|}$. For $|\mu| \geq n$, $2^{-|\mu|} \leq (3/4)^n (2/3)^{|\mu|}$. The sum of $(2/3)^{|\mu|}$ over $\mathbb{N}^h$ is 3^h , which proves (11.39). Taylor substitution proves (11.40). $\square$

The required radius can be chosen from the whole structural constraints. Let a finite list of holomorphic functions $V_b(t, \lambda)$ take values in Banach spaces, and let O_b be their open admissible sets. Suppose $\operatorname{dist}(V_b(t_0, 0), O_b^c) \geq \delta_b > 0$. On a larger driver disk and $|\lambda_a| \leq 2r_0$ assume bounded derivatives, and put

$$(11.42) \quad J_{b,t} = \sum_a \sup \|\partial_{\lambda_a} V_b\|, \quad r_t = \min \left\{ \frac{r_0}{4}, \min_b \frac{\delta_b}{8(1 + J_{b,t})} \right\}.$$

For inverse constraints the supremum also ranges over their buffered unknown disk. Restrict D_t until the zero-section variation is less than $\delta_b/4$ for every b . On $|z_a| \leq 2$, integration along the parameter segment bounds the remaining variation by $2r_t J_{b,t} \leq \delta_b/4$. Each constraint then has margin at least $\delta_b/2$. The radius is constant on D_t ; it need not have a positive lower bound as the driver tends to its end.

Example 11.18. A later exponential passage can destroy a fixed unscaled parameter disk. Fix $0 < \rho < 1$ and let $R(w) = w/(1-w) = \sum_{j \geq 1} w^j$, an original Wiener source with norm $\rho/(1-\rho)$. For real $T > 0$ consider

$$(11.43) \quad q(T, \zeta) = \exp(-\exp((T + \zeta)^2)), \quad R(q(T, \zeta)).$$

Set $T_n = \sqrt{\log(2\pi n)}$ and $\zeta_n = \sqrt{T_n^2 + i\pi/2} - T_n$, using the square root with positive real part. Then

$$(11.44) \quad |\zeta_n| = \frac{\pi/2}{|\sqrt{T_n^2 + i\pi/2} + T_n|} \leq \frac{\pi}{4T_n} \rightarrow 0, \quad q(T_n, \zeta_n) = 1.$$

At this point $\partial_\zeta q = -4\pi i n \sqrt{T_n^2 + i\pi/2} \neq 0$. Consequently $R(q)$ has a pole in every fixed ζ disk for arbitrarily large drivers.

For a scaled disk, put $\zeta = z/T$ and fix $|z| \leq c < \pi/4$. For any a_0 with $2c < a_0 < \pi/2$, sufficiently large T gives

$$(11.45) \quad \operatorname{Re} \exp((T + z/T)^2) \geq (\cos a_0) \exp(T^2 - 2c - c^2/T^2).$$

To verify it, expand $(T + z/T)^2 = T^2 + 2z + z^2/T^2$; its imaginary part has absolute value at most $2c + c^2/T^2 < a_0$, and its real part is at least $T^2 - 2c - c^2/T^2$. The full original series for $R(q)$ therefore converges on this disk for large T . Its degree greater than M tail is bounded by $(|q|/\rho)^{M+1} \rho/(1 - \rho)$.

An estimate for a scaled chart is distinct from a uniform estimate for an original Taylor coefficient. The latter must include the recovery factors in (11.40). If each prescribed multi-index has its own bounded marked target, its norm is used only for that target. These separate norms supply no summed family bound without an additional summability estimate.

11.7. Whole receiving maps.

Lemma 11.19. *Let $\mathcal{B}_A(D)$ be a root bank of Definition 8.1, with order P , and let D' have a strictly smaller origin disk and angular sector. There exist a later weight $A' > A$ and a bounded linear map $\mathcal{N}_1 : \mathcal{B}_A(D) \rightarrow \mathcal{B}_{A'}(D')$ such that*

$$(11.46) \quad \mathcal{S}_q f(Z) - b_0(f) = Z^{-1/P} \mathcal{S}_q(\mathcal{N}_1 f)(Z).$$

The scalar part of $\mathcal{N}_1 f$ is the entire coefficient $b_1(f)$, with the coefficient maps of (9.11). This operator bound holds on every absolutely summed original coefficient family, including every omitted subset.

Proof. Put $\delta = 1/P$, choose an integer $k > \delta$, and write $f = (c, \phi)$. On the original continued domain define

$$(11.47) \quad \begin{aligned} \chi(z) &= \frac{\phi(z) - \phi_1 z}{z}, \\ I(z) &= \frac{1}{\Gamma(k - \delta)} \int_0^1 t^{\delta-1} (1-t)^{k-\delta-1} \chi(zt^\delta) dt, \\ \psi(z) &= q^{-\delta} \prod_{j=0}^{k-1} \left(\frac{z}{P} \partial_z + j \right) I(z), \quad \mathcal{N}_1 f = (b_1(f), \psi). \end{aligned}$$

The singularity at zero is removable and $\chi(0) = 0$. Fixed-circle Cauchy bounds control ϕ_1 and $\chi(z)/z$ on a smaller disk. Outside it, the spare exponential weight at any fixed $A_1 > A$ absorbs division by z and the subtracted monomial. Thus $\|\chi\|_{A_1} \leq C_1 \|f\|_A$. For $A_2 \geq 2A_1$, the defining weight gives

$$(11.48) \quad M_{A_2}(\chi(zt^\delta)) \leq t^\delta M_{A_1}(\chi) \sup_{v \geq 0} e^{-v/2} (1+v)^{1+\delta}.$$

The resulting beta integral is finite. On the strict sector, Cauchy disks of radius $\epsilon|z|$ control each of the finitely many Euler derivatives in (11.47); on the small disk use a fixed larger circle. A further weight $A' > A_2(1 + \epsilon)^P$ absorbs their polynomial costs. This proves one bound $\|\mathcal{N}_1 f\|_{A'} \leq C \|f\|_A$ on the entire continued minor.

The coefficient of z^n , $n \geq 1$, in ψ is $q^{-\delta} \Gamma((n+1)/P) \phi_{n+1} / \Gamma(n/P)$. Consequently its independent formal map is the first ordinary remainder of the formal map of f . The injective

algebra map in Lemma 8.2 proves the source identity after multiplication by the exact element representing $Z^{-1/P}$. Its actualization proves (11.46). Every operation above is linear with the one bound C . Summing that bound in the original coefficient norm proves the family assertion. No operation in this construction acts on a nonanalytic window. $\square$

Lemma 11.20. *Assume Setup 9.14 and Construction 9.15. In addition, suppose every fixed nonzero $g \in E$ satisfies*

$$(11.49) \quad \frac{Ag(\gamma(x) + z)}{Ag(\gamma(x))} = e^{-\lambda_g z}(1 + o(1))$$

uniformly on every fixed buffered disk in a sufficiently small common strip about the physical curve. The index λ_g is its first row in that setup. There exists $r > 0$, fixed by the source domains and clocks, with the following properties. For each fixed $T \geq 0$ put

$$(11.50) \quad n_e = \lfloor T/a_e \rfloor, \quad b_T = \min_e (n_e + 1)a_e > T.$$

For every $\epsilon > 0$ there exist $C_{T,\epsilon}$ and $x_{T,\epsilon}$, preceding $F \in \mathcal{P}$, such that

$$(11.51) \quad \sup_{|z| \leq 2r} |\text{Act } F(\gamma(x) + z) - A(Q_T F)(\gamma(x) + z)| \leq C_{T,\epsilon} \|F\|_\pi e^{-(b_T - \epsilon)x}, \quad x \geq x_{T,\epsilon}.$$

This remainder includes the complete original minors, their high-carrier tails and all subtracted-polynomial tails. If the whole native value $f = \text{Act } F$ is nonzero and has first row λ_ , then, for that fixed f ,*

$$(11.52) \quad \frac{f(\gamma(x) + z)}{f(\gamma(x))} = e^{-\lambda_* z}(1 + o(1)), \quad \frac{f'(\gamma(x) + z)}{f(\gamma(x) + z)} = -\lambda_* + o(1), \quad |z| \leq r.$$

The first ratio law holds also on $|z| \leq 2r$ after reducing r once at the outset. The errors are uniform on the indicated disks for each fixed value, not uniformly over nonzero inputs. A fraction of two such values has the corresponding difference of actions and is zero-free on these buffered disks on a sufficiently far tail.

Proof. We estimate a fixed native rectangle in the original norms, then compare it with the complete first row and differentiate.

Step 1. We bound the whole rectangle error. Choose r so small that the prescribed positive carrier ray has one strict damping margin for every bank at $\gamma(x) + z$, $|z| \leq 4r$. The clock derivative condition and the physical matching give, uniformly there,

$$(11.53) \quad K_e(\gamma(x) + z) = k_e x + o(x) + k_e z + o(1), \quad \Im K_e(\gamma(x)) \rightarrow 0.$$

Since $\Re q > 0$, the choice $4r \max_e k_e < (\pi/2 - |\arg q|)/2$ suffices on a far tail. Thus the original integral (9.10) has $\Re(qe^{K_e}) \geq c_e |e^{K_e}|$ throughout these disks. No ray or root determination is changed.

For one bank choose $0 < r_0 < \rho < r_e$ and subtract the Taylor polynomial of its minor through degree n . Cauchy's formula in the original disk bounds the remainder on $|\zeta| \leq r_0$ by $C_n \|v\|_{B_e} |\zeta|^{n+1}$. Integration with $\zeta = \xi^{1/P_e}$ gives

$$(11.54) \quad \left| S_e(v)(Z) - \sum_{j=0}^n b_{e,j}(v) Z^{-j/P_e} \right| \leq C_n \|v\|_{B_e} \left(|Z|^{-(n+1)/P_e} + e^{-c_n |Z|} \right).$$

Here the small-carrier integral is bounded by $C_n \|v\|_{B_e} \int_0^\infty e^{-c|Z|\xi} \xi^{(n+1)/P_e - 1} d\xi$. Above $r_0^{P_e}$ the original minor envelope has damping $\exp[-(c|Z| - g_e)\xi]$; its entire integral is at most $C \|v\|_{B_e} e^{-c'|Z|}$. Every term of the subtracted polynomial has its full integral on that interval as well. For fixed n its polynomial factor is absorbed by a further fixed fraction of this damping. This proves (11.54) with all tails.

The whole native factors and each fixed polynomial factor are bounded by constants times their bank norms on these disks. Telescope their product using (11.54), and multiply by the bound $L\|g\|_E$ for the entire lower coefficient. For a simple tensor the resulting error is at most

$$C_T\|g\|_E \prod_e \|v_e\|_{B_e} \sum_e \left(e^{-(n_e+1)\Re K_e/P_e} + e^{-c_e|e^{K_e}|} \right).$$

The clock estimate (11.53) makes this at most $C_{T,\epsilon}\|g\|_E \prod_e \|v_e\|_{B_e} e^{-(b_T-\epsilon)x}$. Sum in a projective representation and take its infimum. Continuity gives (11.51) on the completed tensor, including complete inverse and coefficient families already in its lower factor.

Step 2. We prove the relative estimate. Fix a nonzero f . Choose one $T \geq \lambda_*$ and then $0 < \epsilon < b_T - \lambda_*$. Stabilization of the rows in Construction 9.15 makes $Q_T F$ a nonzero element of E with first row λ_* . Its physical head has modulus $\exp(-\lambda_* x + o(x))$. Apply (11.49) to this one complete coefficient. It is bounded below, relative to its central value, on the fixed disk. Dividing (11.51) by that central value gives a uniformly vanishing error. It gives a vanishing error at the center too, so division by $f(\gamma(x))$ proves the first law in (11.52).

The holomorphic function $e^{\lambda_* z} f(\gamma(x) + z)/f(\gamma(x))$ tends uniformly to one on the larger disk. Cauchy's estimate makes its derivative tend to zero on the smaller disk. It is bounded away from zero there, proving the second law and the zero-free assertion. Division of two fixed laws proves the assertion about fractions. $\square$

Lemma 11.21. *Let $u_x = x + i\eta(x)$ be an own-parent physical curve, with $\eta(x) \rightarrow 0$. Suppose a realized source has a bounded whole map in its original norm on a fixed buffered strip around this curve. Suppose each fixed nonzero field readout there satisfies the own-parent law $f(u_x + z)/f(u_x) = e^{-\lambda_f z}(1 + o(1))$, with $\lambda_f \in \mathbb{R}$, uniformly on fixed sufficiently small buffered disks, and has $\log |f(u_x)| = O(x)$.*

Let $X_y = y + i\eta_X(y)$ be a strictly faster physical coordinate with $\eta_X(y) \rightarrow 0$ and an actual inverse chart $u = \chi(X)$. Require $\chi(X_y) = u_{x(y)}$, $x(y) \rightarrow \infty$, $x(y) = o(y)$, and, for every fixed $A > 0$,

$$(11.55) \quad \sup_{|z| \leq A} |\chi(X_y + z) - \chi(X_y)| \rightarrow 0$$

on the literal pulled-back buffered domains. Then each fixed nonzero transported readout $\tilde{f} = f \circ \chi$ satisfies

$$(11.56) \quad \frac{\tilde{f}(X_y + z)}{\tilde{f}(X_y)} \rightarrow 1, \quad \frac{\tilde{f}'(X_y + z)}{\tilde{f}(X_y + z)} \rightarrow 0, \quad |z| \leq A, \quad \log |\tilde{f}(X_y)| = o(y).$$

The derivative law uses a larger fixed buffered disk. For any fixed finite list of field denominators and whole source stages there exists one slowly widening X horn in the pulled-back domains, on which those denominators are nonzero and those stages retain their whole original-norm bounds. No choice of this horn depends on a variable input in one of the stages.

Proof. Insert the displacement $\chi(X_y + z) - u_{x(y)}$ in the own-parent law. It tends uniformly to zero by (11.55), while the original law is uniform on a fixed larger disk. Both its exponential and its relative error tend to one and zero, respectively. This proves the first assertion of (11.56). Cauchy's estimate on a larger fixed X disk differentiates the normalized holomorphic value. Its denominator is bounded away from zero there for this fixed readout, which proves the derivative assertion. The head-size assertion follows from $x(y) = o(y)$. Taking $z = -i\eta_X(y)$ first transfers the central value to the positive real axis. The uniform law on a larger disk then gives the ratios centered at that axis required in (8.114).

Each of the finitely many fixed denominators is nonzero on a sufficiently far union of the original buffered u disks: its relative law is bounded away from zero. Restriction through χ preserves this fact. It also preserves the original absolute operator bound for an entire stage,

without dividing by a variable readout. For each positive integer m , choose a threshold after which all displacements with $|z| \leq m$ stay in these buffers and all finite domain restrictions hold. Choose a continuous nondecreasing width tending to infinity sufficiently slowly through these thresholds, and bounded by $\sqrt{y}$. Shrink it once to allow for the vanishing imaginary part of the physical curve. This gives the asserted horn. The choices use only the finite fixed denominators, stages and charts. Estimates for an additional fixed readout may require a farther tail on this horn; no uniform relative estimate over variable inputs or countable intersection of their zero-free domains is used. $\square$

Definition 11.22. For a fixed finite history, its receiving raw presentation is constructed in increasing parent-height classes and, within a class, in the recorded chronological order. Begin with the original pre-native coefficient stage at that class. Independent native slots form its completed projective tensor product with the banks of Setup 9.4. Positive analytic slots form $\mathcal{W}_R(E)$, with norm $\sum_\nu R^\nu \|f_\nu\|_E$. Correlated slots are multiplied in their one original algebra before this map is applied.

A prepared slot retains its original measures, weights, positive gap, degree linkage, locally finite point-jet distribution and ordered inverse leaves. Its receiving map is the full affine map of Lemma 11.7; its norm is the original weighted absolute sum multiplied by the displayed receiving and coefficient costs. Independent such slots use the projective norm. They are not identified as measures by their actual values.

Stored analytic and relative recipes are their entire convergent series on strict smaller receiving disks. A stored regular implicit recipe is its complete Banach fixed point on the recorded buffered disk. Its full scalar limiting solution is retained. A recipe whose increment involves a later factor first retains its complete baseline coefficient family; the increment is inserted after that later factor has been received. The actual domain is the connected component containing the far physical section of the literal pullback of all these original domains and strict inequalities.

Each raw strong cut keeps native levels at most a , takes the scalar of later native factors, and erases positive weak, prepared and auxiliary factors of depth at least $d + a$, where d is the original depth offset. This mask is applied to every coefficient, argument and implicit equation. Its retained raw presentation is constructed by these rules in its own right. A level- i native root has depth $d + i - 1$; its weak output has depth $d + i$. A prepared baseline at a level- i parent has depth $d + i$.

The next lemma describes the extra comparison step required by prepared sources. Its input is the original chronological presentation, with its supplied own-parent charts; a field of scalar germs alone does not supply that input. For equation-generated prepared banks, the original finite action monoid and point-jet identity are supplied by Proposition 5.2. The degree-zero bank uses its explicit atomic distribution in (5.48). These are distribution identities in addition to the whole measure estimates.

Lemma 11.23. *Take a finite chronological raw presentation with its original bounded coefficient maps, buffered own-parent charts, differential numerator maps and independent earlier comparison rows. Suppose the earlier classes have complete horizontal heads and the native data of Setup 9.14. At a parent u , adjoin a finite list of original affine prepared families and their correlated inverses, positive roots, and positive marks e^{-tu+P} with $t > 0$, $P = o(u)$, $\Im P \rightarrow 0$, in the pre-native differential field. Require the fixed phases and their derivatives on the supplied buffered own-parent domains. Require each original prepared distribution to be point-jet supported on a monoid generated by finitely many positive actions, with its original positive gap and degree linkage. Keep the full original kernels at comparable parents and replay all later factors.*

The raw presentation of Definition 11.22 then has bounded independent comparison rows whose common kernel is its complex analytic kernel. Its quotient is faithful in the original infimum norm. Every nonzero whole value has a complete head. Every prescribed finite derivative has a bounded whole numerator map with one fixed nonzero denominator. Unchanged old raw

maps descend to bounded injective inclusions. The assertions hold for each retained strong presentation and for finite lists of the supplied marks in one receiver.

More precisely, at each prepared or weak pre-native comparison class u , let $\mathcal{P}_u$ be its complete original projective and coefficient Wiener input, before native factors of that class or faster factors are adjoined. There exist a monoid Γ_u generated by finitely many positive real numbers and bounded independent maps $C_\lambda : \mathcal{P}_u \rightarrow E_\lambda$, $\lambda \in \Gamma_u$, where each E_λ is a finite Banach direct sum of already constructed coefficient stages for that row. The row includes the entire coefficient at action λ , all its finite phase relations and its fixed denominator clearers. Its common kernel is the full actual kernel. There exist a slowly widening own-driver horn $\mathcal{H}_u$ in the actual buffered source domains and constants $C_2, A, c > 0$, fixed before $f \in \mathcal{P}_u$ and $T \geq 1$, such that

$$(11.57) \quad [C_\lambda f = 0 \text{ for every } \lambda \leq T] \implies |\text{Act } f(z)| \leq C_2 \|f\|_{\mathcal{P}_u} \exp(AT \log(eT) - cT \Re z), \quad z \in \mathcal{H}_u.$$

This is an estimate for the entire original input. It descends by its original quotient norm infimum. It does not put the entire final receiver, including faster native and weak factors, on $\mathcal{H}_u$.

Proof. We construct the compact rows and their whole complement first, then reconstruct the native and later classes. At a fixed pre-native class write $\mathcal{P} = \mathcal{P}_u$.

Step 1. We construct compact rows of the prepared factors. Write an old comparable parent as $X = c_X u + Z_X$, $c_X > 0$, $Z_X = o(u)$. Its actual chart and complete horizontal heads give a slowly widening common horn on which this estimate is uniform and all original coefficient and damping margins persist. The finite restriction precedes the carrier indices. Keep

$$(11.58) \quad (\lambda X + c + v)^K e^{-\beta(\lambda X + c + v)}$$

as one kernel. In particular $e^{-\lambda \beta Z_X}$ is not taken out as a uniformly bounded coefficient. The full translated norm and reduced linkage in Lemma 11.7 absorb the cost $\exp(o(\beta \Re u) + O(\beta \log \Re u) + O(\beta))$ into a fixed fraction of $\lambda c_X \beta \Re u$. This estimate applies to the full inverse by Lemma 11.9. At the base use the actual $P_0 = \psi(H_0)$ of Lemma 11.15; every carrier-dependent power of ψ stays inside its kernel.

For a product, denote the leading scale of leaf a by d_a , its original carrier by β_a , and its derivative order by K_a . If the already formed residuals have shifts ϵ_a , its numerical action is

$$(11.59) \quad \Omega = \sum_a d_a \beta_a - \sum_a \epsilon_a + \sum_b \gamma_b n_b.$$

The final sum contains the positive atomic root and weak actions. Multiply the full product distribution by one $\chi(\Omega)$. Lemma 11.11 gives every derivative of this product, including derivatives of its complete coefficients. The shifts are constants, so their carrier derivatives are zero. Proper-subtree moments retain their original carriers.

After the compact cut, the supplied point-jet identities, as in Lemma 8.13, give a finite list of jets. Their residual phases belong to the preceding differential field. Apply Theorem 8.23 to that finite list: form its witness lattice, collect every tied numerical action, and clear the finitely many witness denominators in one fixed coefficient stage for this row. Follow with that stage's independent lower rows. Coefficient extraction, finite multiplication and the compact distribution maps are bounded. This defines the rows before any zero decision for the variable whole input.

Step 2. We control the complete complement. Scaling finitely many positive generator lists and adjoining finitely many roots gives a finitely generated positive monoid. Each total residual shift preserves this property by Lemma 11.8. Hence bounded action intervals contain only finitely many actions, even when their generators are irrational. There exists a fixed reduced linkage $\delta_* > 0$ for the finite positive inventory. If χ is one through T_0 and zero past T_1 , at most

$J = \lceil T_1/\delta_* \rceil$ carrier derivatives occur on its transition. The binomial factors in the full product rule, the linked polynomial factors and the reserved damping give

$$(11.60) \quad \begin{aligned} |\text{Act } f - \text{Compact}_\chi f| &\leq CH_\chi \|f\|_{\mathcal{P}} e^{-cT_0 \Re u}, \\ H_\chi &= 2 + C_*^J \max_{1 \leq l \leq J} \|\chi^{(l)}\|_\infty. \end{aligned}$$

The maximum is zero for $J = 0$. To see the uniformity, apply the unlowered majorant of (11.58) outside the transition and the product rule inside it. Each lowering uses the already reserved carrier slack; the sum of its binomial coefficients is at most an exponential in J . Degree-zero lower coefficients keep their complete original norms and are not charged to δ_* . The inverse-degree sum has already converged in its original majorant. Strict analytic series are summed in their original coefficient norms. Products use the projective norm. These bounds prove (11.60) on one horn, before f , T_0 , T_1 and all degrees.

Lemma 8.16 supplies cuts with $T \leq T_0 < T_1 \leq 2T$ and $H_\chi \leq C_1 \exp(AT \log(eT))$; replacing 2^J by C_*^J costs only $\exp(O(T))$. Thus

$$[\text{all rows through } 2T \text{ vanish}] \implies |\text{Act } f| \leq C_2 \|f\|_{\mathcal{P}} \exp(AT \log(eT) - cT \Re u).$$

Replace T by $T/2$ and increase the fixed constants on the bounded interval $1 \leq T \leq 2$. This proves (11.57) with rows through T on the original horn. If all rows vanish, choose T comparable to $\exp(b \Re u)$ with $b > 0$ sufficiently small. The right side is at most $C \|f\|_{\mathcal{P}} \exp(-c' \exp(b \Re u))$. The horn contains a fixed strip of total width greater than π/b . Under $w = e^{-u}$, viewed on its logarithmic cover, the single-function assertion of Lemma 8.20 forces zero. The identity theorem continues this zero on the original connected component.

If a row survives, choose its least action γ . Theorem 8.23 gives the complete subprincipal head of its finite coefficient. Choose a fixed compact cut with $cT_0 > \gamma$. (11.60) makes the entire complement negligible relative to this head. The finitely many later rows inside the cut have a positive action gap. Their entire relative remainders tend to zero. This proves both kernel inclusions and the complete-head assertion for the prepared fold.

Step 3. We identify the horizontal head of the new prepared source. Fix a nonzero whole value f and keep its complete first action γ . Write u_x for the physical curve in the own-parent coordinate. In the finite compact sum, the earlier horizontal input and Theorem 8.25 give the complete first fiber p the law

$$p(u_x + z)/p(u_x) = e^{-\gamma z}(1 + o(1))$$

on fixed buffered complex disks in the original horn. Here the full tied phase sum is taken before its head. For its subprincipal coefficient, the bridge supplies uniform vertical ratios and a logarithmic derivative tending to zero. Integrating that derivative gives the ratios for bounded real shifts. Combining the two gives the law on a complex disk, including disks centered at u_x since $\Im u_x \rightarrow 0$. Its modulus at the center is $\exp(-\gamma \Re u_x + o(\Re u_x))$. Polynomial and logarithmic factors remain in this finite coefficient; their bounded-displacement ratios tend to one. Every fixed residual phase is in the earlier differential field, to which the horizontal bridge has been applied before the present source is formed.

Choose the fixed cut farther if necessary so that $cT_0 > \gamma + \delta$ for some $\delta > 0$. The bound (11.60) holds on the larger buffered disks, not just on the physical curve. Division by $p(u_x)$ therefore makes the entire complement tend uniformly to zero there. The finitely many later fibers inside the cut have a positive action gap; their own complete horizontal heads make these later fibers uniformly negligible. Consequently

$$(11.61) \quad \frac{f(u_x + z)}{f(u_x)} = e^{-\gamma z}(1 + o(1)), \quad \frac{f'(u_x + z)}{f(u_x + z)} = -\gamma + o(1)$$

on a smaller fixed disk. For the second assertion, apply Cauchy's estimate to $e^{\gamma z} f(u_x + z)/f(u_x) - 1$ on the larger disk. Its uniform convergence also gives a zero-free disk. Fractions retain the

difference of the two actions. These are assertions about each fixed nonzero readout; the absolute complement estimate remains in the whole original norm. In particular every fixed element of the new pre-native algebra satisfies (11.49).

Step 4. We retain native folds through growth-ordered replay. Adjoin the exact roots before the native tensor. A root e^{-K_e/P_e} with $K'_e = k_e + o(1)$ has action $k_e/P_e > 0$. Multiplying by roots of total action r shifts all uncollected actions by at least r . Collection can remove a row but cannot create a row below r . This proves the reserve in (9.35). The compact-row construction supplies the independent lower rows, whole bound and cofinal estimate. A strictly slower parent requires uniform small motion on actual complex disks, not just a physical derivative limit. For an unchanged ordinary chart, use its retained fixed-disk displacement. For an exponential principal chart, first retain its genuine inverse. Structural derivative bounds are used only for the proper strong coordinate inventory in Lemma 10.3. For a stored slower sibling, use Lemma 10.4 from completed lower native levels and retain the exact factor X/u in the resulting displacement. After this motion is established, the old complete own-driver law and whole pullback bound supply the lower inlet. Fixed subprincipal lists then receive buffers by Lemma 10.2. Comparable clocks retain their complete residuals in the old field before their joint weak comparison. The common rays and full banks are the original supplied ones. This checks all data of Setup 9.14, so Theorem 9.17 applies to the full tensor of old and new banks in that fold.

The physical head from that theorem is now extended to buffered disks by Lemma 11.20. Its additional lower horizontal hypothesis is precisely (11.61), including the exact roots already placed in the pre-native algebra. For a fixed nonzero native readout choose one root rectangle beyond its first action and use (11.51). That entire original-projective-norm remainder is exponentially smaller than the complete coefficient on the buffered complex domain. Thus the newly enlarged native stage and its fractions have the exact own-parent horizontal law and logarithmic derivative, before any later replay.

At a later parent X with $u = o(X)$ the derivative argument gives $\partial_X u \rightarrow 0$ on the physical curve. For this old coordinate change, $u(X)$ and its fixed derivative numerator and denominator are unchanged structural members of the original proper field at X . Its supplied horizontal inlet applies to these fixed members. Thus $\partial_X u \rightarrow 0$ uniformly on every fixed buffered X disk on its original inverse chart. Integration on each disk gives (11.55). This use of the old structural inlet involves none of the newly added variable source. Lemma 11.21 then carries the new whole stage and each of its fixed fraction readouts to X . The finite original denominators remain nonzero on the restricted pulled domains, and whole operator bounds keep the product of their fixed original inclusion costs. One slower widening X horn is selected for this finite data. It is here, after the small displacement, that the new lower horizontal ratio tends to one. Its logarithmic size is $O(u) = o(X)$.

A later prepared fold repeats the compact construction above; a native fold uses its full pre-native tensor and Lemma 11.20; a weak fold uses Theorems 8.23 and 8.25 with its actual own-driver inlet just checked. A newly created principal chart is treated by Lemma 11.21 after its actual inverse and displacement bounds have been constructed. Future positive weaks never enter their own lower field. Complete baseline families whose increments involve those later factors are now inserted, with bound $LM/(1 - r/R)$ and tail factor $(r/R)^{n+1}$. Complete implicit recipes use their original buffered contractions, including their zero solutions.

At each faster fold a zero lower row is an identity for its entire unnormalized germ. Continue that identity on its own connected domain, then sum the original absolute majorant. This uses neither a sum of normalized heads nor a countable intersection of head-dependent horns. The finite sequence of bounded raw maps has the product of its original costs. Unchanged old raw inputs keep their actual values; their old kernel therefore maps into the new independently

proved kernel. Taking infima of original raw norms proves descent, and old faithfulness proves injectivity.

For differentiation, native Euler operators follow by differentiating Lemma 8.7 on a strict disk; their scalar derivative is zero. Prepared differentiation inserts finitely many carrier moments and polynomial lowerings, paid by fixed extra slack. Wiener differentiation has the bounded degree factor $\sup_n n\theta^{n-1}$ on strict radii. Differentiate an implicit equation using its entire inverse Jacobian. Clear the finitely many original coordinate denominators after these whole numerator maps have been built. This proves the differentiated source inlet at the next fold and every fixed finite iteration. The proof applies unchanged to each separately constructed retained raw presentation. $\square$

Lemma 11.24. *In a receiver constructed by Lemma 11.23, retain the original depth profiles and buffered own-parent atlas. At each occupied depth h , let b_h be the minimum of the finitely many positive principal equivalents and let A_h be one plus their sum. Suppose the adjacent profiles satisfy $A_{h-1} = o(b_h)$ and $\log A_h = O(A_{h-1})$ with the original basal convention. Then all synchronous raw masks in Definition 11.22 descend to compatible strong cuts $e_a = i_a \rho_a$, with $\rho_a i_a = \text{id}$ and $e_a e_b = e_{\min(a,b)}$.*

For every fixed original raw stage $\mathcal{P}$ and cut a , there exist finitely many coefficient Banach stages E_ν , bounded linear maps $C_\nu : \mathcal{P} \rightarrow E_\nu$ with costs L_ν , and fixed distinguished positive factors v_ν in a common receiving raw presentation, such that, using its fixed inclusions for all terms,

$$(11.62) \quad f - e_a^{\text{raw}} f = \sum_{\nu=1}^m v_\nu C_\nu f, \quad \|C_\nu f\|_{E_\nu} \leq L_\nu \|f\|_{\mathcal{P}}.$$

Each v_ν is an extracted weak factor, an original first native root, or the positive factor of one whole prepared residual at an erased depth. It has a specified strictly positive numerical action at its own parent. Each C_ν retains its entire coefficient, including all later factors, divided banks, residual sources and inverse Jacobians. These stages, maps and costs precede f and all its source and parameter degrees. All terms retain one literal pulled source component, fixed before f . If b_a^{rem} is the first occupied removed profile, then on its far physical tail their values satisfy $|\text{Act}(f - e_a^{\text{raw}} f)| \leq C_a \|f\|_{\mathcal{P}} e^{-c_a b_a^{\text{rem}}}$ for fixed $C_a, c_a > 0$. The decomposition and difference are zero if the removed list is empty.

For every finite list of references $M = CR^\alpha$ of level a , with $C, \alpha > 0$ and $t \geq 0$, and every $\varepsilon > 0$, there exists a common receiving refinement and a bounded whole linear map

$$(11.63) \quad \text{Act } \mathcal{W}_{M,t} f = M^t \text{Act}(f - e_a f), \quad \|\mathcal{W}_{M,t}\| \leq \varepsilon.$$

Its construction multiplies only the distinguished factor in (11.62) by M^t . Each added phase $t \log M$ is subprincipal at that factor's removed height and leaves a positive action reserve. Fix all original coefficient, division, extraction, residual and old inclusion costs, uniformly below fixed provisional receiving upper radii, before choosing the fresh marked radii to give the bound ε . All old inclusions, original norms, selected cuts and complete coefficient families are retained. The exponent α may be any positive real number.

Proof. We verify the domains for the enlarged strong stages, then factor their complete cut differences.

Step 1. We choose the appropriate whole strong charts. At a retained parent u_0 , fix the old structural list and its finite derivative order before the carried input. Use Definition 10.1: its subprincipal structural members have derivative bounds $\exp(C_k q(\mathcal{R}u_0))$ on actual buffered disks of radius $\exp(-q(\mathcal{R}u_0))$, where $q = o(\mathcal{R}u_0)$. Complete readout denominators are separate; they retain their full own-driver laws and explicit actions. An exponential native clock is kept with its exact logarithmic clock and is controlled by relative variation, not an absolute subexponential derivative. These are old records of the recipe, not consequences of the new whole-tube bound

to be proved. For each fixed $c_0 > 0$, a curved tube of width $\exp(-c_0 \Re u_0)$ eventually lies in these disks. Integration across the tube changes each structural argument by at most $\exp(-c_0 \Re u_0 + o(\Re u_0)) = o(1)$. The original logarithmic branches, relative radii and analytic margins persist. For a native height $Z_e = \exp K_e(v)$, the controlled quantity is ΔK_e , so $Z_e(\text{new})/Z_e(\text{old}) = \exp(\Delta K_e) = 1 + o(1)$. This preserves a fixed part of its original positive angular damping uniformly in its carrier; ΔZ_e need not tend to zero. Fixed full denominators stay zero-free by their own-driver relative laws, with their actual exponential actions retained. For prepared kernels, the carrier multiplying this variation stays inside (11.58); a fixed fraction of its positive damping absorbs the variation uniformly in that carrier. Linkage controls the log powers. Thus, for the entire appropriate strong stage,

$$(11.64) \quad \sup_{\text{tube}} |\text{Act } f| \leq L_{a,c_0} \|f\|_{\mathcal{P}}.$$

Native integrals, full relative substitutions and strict Wiener sums have been estimated before their degrees. Projective summation and the independent strong kernel descend (11.64) by infima. Deferred faster factors keep their separate domains. Each stored principal inverse pulls a fixed strict power-horn displacement back to an exponentially small parent displacement; hence it carries this entire enlarged appropriate source. Lemma 11.16 gives that assertion for a newly constructed principal. Horizontal transport uses the actual chart and the correct-coordinate derivative, not a change to an unproved real-part coordinate. Once the new whole-stage bounds and its complete real heads have been obtained, apply Lemma 10.2 to each requested finite subprincipal structural fraction list. This proves its new buffer without using that buffer to obtain its own whole-stage premise. For a later stored weak sibling, the logarithmic-clock hypotheses are supplied by Lemma 10.4 at completed lower native levels. Its genuine inverse chart and Lemma 10.3 then prove the actual complex displacement. An ordinary change instead uses an independently retained ordinary fixed-disk chart.

Step 2. We distinguish one removed factor. For an original Wiener sum, partition the terms according to the first erased positive label and remove one occurrence of that label. Its coefficient map costs the inverse of that label's original radius. For an erased native factor use Lemma 11.19. For an erased prepared baseline use its complete residual $e^{-\epsilon X} B_\epsilon$ from Lemma 11.7. All remaining factors stay in the coefficient.

For a retained analytic family first telescope changes in its coefficients, then use Lemma 11.12 for its entire argument difference. Its coefficient cost is $M/[R(1-r/R)^2]$, on the original buffered disks. For a retained implicit recipe, solve the entire projected equation on that same disk and use the full inverse Jacobian identity of that lemma, with cost $(1-q)^{-1}$. This retains its whole zero solution. Strict substitutions introduce only $\sup_n n\theta^{n-1} < \infty$. Induction through the finite recipe list produces (11.62). The list comprises finitely many structural types; each C_ν includes its entire coefficient sum. The L_ν are fixed before f and every infinite degree. Each distinguished v_ν has positive reserve at a removed depth. The depth inequalities, reduced prepared linkage and native root reserve give

$$(11.65) \quad |\text{Act}(f - e_a^{\text{raw}} f)| \leq C_a \|f\|_{\mathcal{P}} e^{-c_a b_a^{\text{rem}}},$$

where b_a^{rem} is the first occupied removed profile. If the removed list is empty the difference is zero. A nonzero retained strong head has logarithmic magnitude $o(b_a^{\text{rem}})$. It cannot cancel (11.65). Therefore an actual-zero input has actual-zero projected image. The independent retained kernels descend the masks. Their raw contractivity in the absolute norms, unit inclusions and nested masks prove all cut identities; uniqueness of the entire projected fixed point proves these identities for implicit recipes.

Step 3. We bound the whole weighted differences. For a removed factor at height X , multiply just that factor by M^t . The additional phase $t \log M$ is $o(X)$, has imaginary part tending to zero, and has correct- X derivative tending to zero. The function $\log M$ is its recorded lower

logarithmic clock with a fixed positive real coefficient; the adjacent profiles give the first assertion, and Lemma 8.22(2) in the completed differential field with derivative d/dX gives the derivative assertion. For a prepared factor the new mark is $\exp(-\epsilon X + t \log M)$ and its coefficient is the complete B_ϵ . For a native factor mark only the exact root in (11.46). All coefficients and inverse Jacobians in (11.62) remain entire.

Apply Lemma 11.23 to these finitely many positive marks with every old and later factor. Choose the common receiving coefficient stages first, and fix their inclusion, division, residual, extraction and analytic costs. For each marked factor fix a provisional positive upper radius within every required strict recipe disk. All later analytic and implicit estimates use these upper radii, so the coefficient-map costs are uniform when the fresh marked radii decrease. Choose only its fresh receiving radius r_ν so that $\sum_\nu r_\nu L_\nu \leq \varepsilon$. The coefficient stages precede the new marked slots; their costs therefore do not depend on these radii. Lemma 11.13 proves the whole raw linear bound. Its actual identity is (11.63). Since the cut already descends, actual-zero inputs have actual-zero weighted image. The independently faithful target kernel and original norm infima give the claimed quotient map. A finite list uses the union of the marked types and a common distribution of the radius budget. The construction does not require bounded multiplication by M^t on the old algebra. $\square$

Definition 11.25. A positive parent record for a current $s = Dx_v^\kappa e^{-G}$ consists of a finite exact expression

$$(11.66) \quad G = \sum_{\nu=1}^m e^{-t_\nu u} u^{h_\nu} C_\nu, \quad t_\nu > 0, \quad h_\nu \in \mathbb{R},$$

where each C_ν is an entire bounded coefficient source with its original norm and its receiving inclusion cost. The coefficients may contain later factors. At the base replace u by its recorded P_0 , retaining the complete basal powers and units. Each recorded birth also retains its complete unit W_b and the finite source factorizations of its transported error, native correction and analytic unit. These factorizations specify the positive factors and the complete coefficient maps to which Lemma 11.24 applies; a scalar limit is not a positive parent record.

Definition 11.26 (Mixed receiving records). The records of a mixed receiver comprise its particular raw construction in Definition 11.22 and the following data. This terminology does not assert that an arbitrary raw presentation has these data.

Its finite labelled forest has the exact birth formulas (11.4), the stored parent input $t_b = D_b x_{p(b)}^{\eta_b} e^{-G_b}$ and reference $T_b = K_b e^{\Xi_b}$, and current $s = Dx_v^\kappa e^{-G}$. Constants and exponents D_b, η_b, D, κ are positive real numbers; passage orders and bank root orders are their original positive integers. All active and inactive births remain.

Each parent retains the split coordinate record of Definition 10.1: its finite subprincipal structural buffers, completed own-driver relative laws with their exact actions, and independent whole original-norm maps on the actual domains. The adjacent ancestry profiles remain those of Lemma 11.24; they are not absolute inverse-denominator bounds. Each principal has its genuine inverse on a strict power horn, carrying the entire appropriate strong stage. The proper-strong head bound $O(A_{d+i-1}) = o(P)$ is retained when that projected inventory is pulled to a new principal P . Physical recipes follow execution order, whereas comparison and replay use increasing actual growth classes.

Each class has independently constructed bounded comparison rows, the complex analytic kernel in its original infimum norm, complete positive heads and the whole cofinal bound (11.57). Native classes use every datum in Setup 9.14. All strong cuts commute with old inclusions and have the whole first-removed-factor decomposition (11.62). Each finite list of positive real reference powers has the weighted maps (11.63), after all old coefficient and inclusion costs.

The current and stored corrections have the positive records of Definition 11.25. Every source keeps its original physical domain, branch, root, compatible ray, input norm and parameter

correlation; every prepared inverse keeps its forward leaves and proper-subtree moments. Original observations keep their whole complements at their own nodes. The receiver separately retains the original-parameter positive-jet record of Definition 10.5, including its normalized structural subprincipality, exact finite original coefficient maps, protected inverse Jacobians, own-scale positive factors and complete later coefficient targets. Structural positive rates and original leading data are fixed during these assertions. For each finite parameter order the model supplies the input record; each constructor proves its new normalized-jet bounds from its exact formulas, then its positive reserves, and only then constructs the independent new coefficient targets. Coordinate buffers alone do not supply this record.

Lemma 11.27. *Let a constructed receiver have the data of Definition 11.26. At a nonbase current $s = Dx_v^\kappa e^{-G}$, the full correction $\Xi = pG - \kappa p \log W_v$ can be received inside every prescribed finite list of original relative source disks. The full original native maps at $Y = \beta s^{-p} = Ne^\Xi$ and the original positive-power and logarithmic strong slots are then bounded whole maps.*

Proof. Write the stored height as $T_v = K_v e^{\Xi_v}$, with principal P_v , and put

$$(11.67) \quad E_v = (T_v - P_v) - a_v \log(T_v/P_v) + \ell_v(T_v), \quad W_v = e^{-E_v} A_v(e^{-u} e^{-E_v}).$$

Step 1. We express the full correction through positive factors. The first two terms of E_v are the complete weighted and unweighted cut discrepancies. They have the small receiving maps of Lemma 11.24, including after every already fixed coefficient inclusion. Lemma 11.19 gives

$$(11.68) \quad \ell_v(T_v) = K_v^{-1/P_v^{\text{root}}} e^{-\Xi_v/P_v^{\text{root}}} D_v(T_v),$$

where P_v^{root} denotes the original integer root-bank order and D_v is its whole divided source. The notation P_v^{root} is distinct from the principal height P_v . The exact old root has positive reserve at its recorded parent. Receive its fresh copy after the entire coefficient and relative costs. On the original analytic disk write $\log A_v(z) = z a_v^{\text{an}}(z)$. Then exactly

$$(11.69) \quad \log W_v = -E_v + e^{-u} e^{-E_v} a_v^{\text{an}}(e^{-u} e^{-E_v}).$$

Its last coefficient is whole and bounded. A fresh e^{-u} copy makes this term small after that cost. Likewise (11.66) is a finite list of positive factors with whole coefficient costs. In one common receiver,

$$(11.70) \quad \|\Xi\| \leq p \sum_{\nu} r_{\nu} L_{\nu} + \kappa p (\epsilon L_{\text{cut}} + r_{\text{root}} L_{\text{nat}} + r_z L_{\text{an}}).$$

Step 2. We receive the correction inside the original relative disks. All L 's and the original relative disks precede the receiving sizes. Choose them so that $\|e^{\Xi} - 1\| \leq r < \rho_0$ in every original disk. Lemma 8.7 now gives the entire original-bank map

$$(11.71) \quad f \mapsto \sum_{n \geq 0} I(\mathcal{D}_n f) (e^{\Xi} - 1)^n, \quad \|\text{map } f\| \leq \frac{L_I C}{1 - r/\rho_0} \|f\|_{\text{original}}.$$

Its degree greater than n_0 tail gains $(r/\rho_0)^{n_0+1}$. Products of finitely many banks use the projective norm and the sum of the corresponding rectangular complement bounds.

Step 3. We receive the original positive-power and logarithmic strong slots. Put $V = D d_v^\kappa W_v^\kappa e^{-G}$. On its protected original logarithmic branch,

$$(11.72) \quad s^\gamma = e^{-\kappa \gamma u} V^\gamma, \quad s^\delta \log s = u e^{-\kappa \delta u} V^\delta (-\kappa + \log V/u).$$

Here $\gamma, \delta > 0$ are fixed. The normalized reciprocal $1/u$ is the original source $P_v^{-1} [1 - (a_v \log P_v + \log c_v)/P_v]^{-1}$ on its strict disk. All coefficients in (11.72) are therefore whole bounded sources. After their costs, receive the displayed positive factors within the original strong slot radii

and apply the strict substitution estimate. No inverse of a zero-augmentation element of the completed algebra is taken. $\square$

Lemma 11.28. *Suppose a constructed receiver has the data of Definition 11.26. Let $Y = Me^\Xi$ be a positive actual height, with an admitted original reference M at level j and its full bounded correction Ξ , whose retained scalar limit is $\xi_0 \in \mathbb{R}$. In its stored parent u , suppose the physical curve is $u_x = x + i\eta(x)$ with $\eta \in C^1$ and $\eta, \eta' \rightarrow 0$. Suppose $\log M = \gamma u + c + \rho_0$, where $\gamma > 0$, $c \in \mathbb{R}$, and ρ_0 is a bounded whole proper strong source tending to zero, with the supplied complete differential heads and numerator maps. Fix $a_+ \in \mathbb{R}$ and $c_+ > 0$; for a canonical output these are its original residue and weak constant. Then it receives the entire original Wiener source at every finite list $c_\nu e^{-\tau_\nu Y} Y^{a_\nu}$, $c_\nu, \tau_\nu > 0$, $a_\nu \in \mathbb{R}$. Its principal and weak chart are*

$$(11.73) \quad P = Me^{e_j \Xi}, \quad H = P - a_+ \log P - \log c_+.$$

Both have their actual inverse charts carrying the entire appropriate strong stage. In particular $\Im P \rightarrow 0$ absolutely on the physical interval. Before adjoining these weak variables, the proper strong field has horizontal ratios tending to one on fixed buffered P and H disks, with its finite fixed denominators nonzero and its whole source stages bounded in their original norms on a slowly widening horn. The new independent comparison and growth-ordered replay retain all old factors, including inactive births. For each retained coefficient stage E and original polyradius R , there exist a common receiving algebra $\tilde{Q}$ and a bounded linear map $\mathcal{T} : \mathcal{W}_R(E) \rightarrow \tilde{Q}$ with fixed $0 < \theta < 1$ and $C_{\text{in}} < \infty$ such that, for $F = \sum_n f_n Z^n$ and $N \in \mathbb{N}$,

$$(11.74) \quad \|\mathcal{T}F\| \leq C_{\text{in}} \|F\|_{\mathcal{W}_R(E)}, \quad \left\| \mathcal{T} \left(\sum_{|n| > N} f_n Z^n \right) \right\| \leq C_{\text{in}} \theta^{N+1} \|F\|_{\mathcal{W}_R(E)}.$$

The complete coefficient and correction-unit costs precede the fresh weak radii, the variable F and N . The logarithmic-reference hypothesis is part of this statement: it does not include a nonzero multiple of $\log u$ in ρ_0 .

Proof. We receive the complete cut error, construct the two actual charts, check their lower horizontal inlet, and then sum the original weak sources before replay. Apply (11.63) with $(M, t) = (M, 1)$ to e^Ξ , and with exponent zero to Ξ . In a common receiver put $B = \mathcal{W}_{M,1}(e^\Xi)$ and $L = (1 - e_j)\Xi$. Their complete norms can be made small after their coefficient costs. (11.65) survives this fixed weight because $\log M$ is below the first removed profile. Hence $\text{Act } B \rightarrow 0$ absolutely. Lemma 11.14 gives $\text{Act } B = Y - P$, $\text{Act } L = \log(Y/P)$ and $\Im P \rightarrow 0$.

The appropriate strong stage has (11.64) for every fixed positive tube exponent. The retained scalar limit and positive correction data give $\rho = \rho_0 + e_j \Xi - \xi_0 \rightarrow 0$. This convergence precedes the derivative assertion. In the completed old strong differential field with derivative d/du , Lemma 8.22(2) then gives $\rho' \rightarrow 0$ along the actual parent curve. Lemma 11.16 therefore applies to $\log P = \gamma u + (c + \xi_0) + \rho$ and supplies its genuine inverse on a fixed strict power horn, carrying the whole strong bound. Its final assertion supplies the inverse of H .

Step 1. We construct the lower horizontal inlet on the new chart. This domain assertion is supplemented by a relative estimate. Put $\Phi = \log P$. On every fixed buffered P disk the actual inverse just constructed satisfies

$$(11.75) \quad \frac{du}{dP} = \frac{1}{P\Phi'(u)} = O(P^{-1}), \quad \frac{du}{dH} = \frac{1}{P\Phi'(u)(1 - a_+/P)} = O(P^{-1}).$$

The estimates are uniform there: its image lies in the disks on which $|\Phi' - \gamma| \leq \gamma/4$. In particular bounded P or H displacements move u by $O(P^{-1})$.

Fix the finite structural profiles and denominators of an appropriate proper strong stage, before choosing a variable input in that stage. An earlier own-parent coordinate v in this list has its original buffer of radius $\exp[-o(\Re u)]$ and its structural derivative bound $|\partial_u v| \leq \exp(o(\Re u))$ on a smaller buffer. These are the structural bounds of Definition 10.1, obtained for new finite

lists by Lemma 10.2 after their whole-stage bounds; for $v = u$ the derivative is one. This list consists only of own-parent coordinates of factors retained by the proper strong cut. A level- j sibling native clock may be an inverse root power in the fraction field, but its native bank has own parent u ; its depth- $d + j$ weak and level- $(j + 1)$ native factors are erased by e_j . Thus that clock is not added to this structural own-parent list merely by field membership. Fixed full denominators retain (10.1), not a subexponential absolute inverse bound. Comparable profiles retain their entire residuals in their original charts. Since $P = \exp(\gamma \Re u + o(\Re u))$, the displacement in (11.75) stays inside each such buffer and moves v by at most $\exp(-\gamma \Re u + o(\Re u))$. Also $v = o(P)$ for this proper strong inventory.

Each fixed source readout at its own v has the complete horizontal head in the receiving records. At an enlarged native stage this is the law of Lemma 11.20, whose full rectangle remainder was checked before the stage was replayed. Apply Lemma 11.21 to the actual P inverse and to the H inverse. It gives the relative horizontal ratios, their logarithmic derivatives and the $o(P)$ logarithmic sizes of the proper strong field. Its zero-free argument applies to the fixed denominators, and its restriction argument retains the whole stage bound. Thus one slowly widening horn carries each finite required list in the original norms. The variable source is never normalized by its own first coefficient to obtain this absolute bound.

For a comparable old principal X , first prove membership of the full residual in the old projected field. Comparable principals have one ancestry level by the adjacent depth separation and hence use one cut e_j . The already admitted reference bank contains its exact first root M^{-1/P_e} , so M is its inverse integer power in the strong fraction field. The complete bounded source $e_j \Xi$ has its entire positive and negative exponential series in a later strong stage. Thus $P = M \exp(e_j \Xi)$ belongs to that field before P -weak. The stored expression $X = K_X \exp(e_j \Xi_X)$ has its exact reference root and bounded projected correction there too. A finite common stage consequently contains $P, X, P - \lambda X$ as fractions and contains their correct-coordinate derivatives. Use the old complete heads and the actual charts for this entire residual, including its imaginary part. The retained bound is $|\log \Theta_f| = O(A_{d+j-1}) = o(P)$, not a universal $O(u)$ bound for the entire proper strong field.

Strictly slower same-generation weak classes, erased by the cut, are inserted separately in increasing growth order. For $X = o(P)$ use its stored logarithmic clock and the completed lower-native data of Lemma 10.4. With the genuine P inverse already constructed, that inlet and Lemma 10.3 prove $\sup_{|z| \leq R} |X(P + z) - X(P)| = O(X/P) \rightarrow 0$ on actual complex disks. Only then use its old completed own- X law and whole source pullback. Its logarithmic head size is $O(X) = o(P)$, and Lemma 10.2 supplies the resulting finite pre- P structural buffers after the whole bound. This is distinct from the bound $|\Delta v| = O(\exp(-\gamma \Re u + o(\Re u)))$ for the proper-strong structural list. Faster classes remain on their own domains until their later replay. The complete comparable class is compared jointly. These steps supply the hypotheses of Theorems 8.23 and 8.25 before the new weak is inserted; its whole cofinal tail is obtained in the original norm.

Step 2. We retain whole weak sources through replay. At each new principal weak use the exact identity (11.32). First fix the entire norm of $\exp(-\tau_\nu B + a_\nu L)$ and the old coefficient inclusions. Then choose fresh principal weak radii so their products with these costs are strictly smaller than the original Wiener radii. For the whole original input $\sum f_n Z^n$ the bound is $C_{\text{in}} \sum \|f_n\| r^n$, and its total-degree greater than N tail is at most $C_{\text{in}} \theta^{N+1} \sum \|f_n\| R^n$, where $\theta = \max_\nu r_\nu / R_\nu < 1$. This is (11.74). The horizontal and whole-source inlets for this weak fold have now both been checked, so Lemma 11.23 proves its independent kernel and reconstructs later factors, with the new whole horizontal laws proved before each further transport. The full $e_j \Xi$ has remained in P throughout. $\square$

Proposition 11.29. *From a constructed receiver with the data of Definition 11.26, a supplied canonical reverse at any nonbase current has a whole receiving construction. It returns to*

the recorded parent and has the data (11.6). It retains the entire birth and inverse sources and preserves all these records. Positive analytic changes and direct re-entry along a recorded outgoing birth preserve these records.

Proof. We construct the full inverse equation and its normalized returned source; the current formula then follows from Lemma 11.4.

Step 1. We solve the original inverse equation. At a current $s = Dx_b^\kappa e^{-G}$ let $t_b = D_b x_v^{\eta_b} e^{-G_b}$ be its recorded input. Include the outgoing inverse adapter as $I_w(s) = d_w s C_w(s)$. Set

$$(11.76) \quad \begin{aligned} Y_o &= \kappa T_b, & a_o &= \kappa a_b, \\ c_o &= d_w D d_b^\kappa c_b^\kappa \kappa^{-\kappa a_b}, \\ E_o &= \kappa \ell_b(T_b) + G - \kappa \log A_b(q_b e^{-\ell_b(T_b)}) - \log C_w(s). \end{aligned}$$

Then $I_w(s) = c_o e^{-Y_o} Y_o^{a_o} e^{-E_o}$ on the original branch. Write $T_o(X) = X - a_o \log X - \log c_o$ and $T_R(X) = X - a_R \log X - \log c_R$. The actual near-identity map $J_R = T_R^{-1} T_o$ is $J_R(X) = X(1 + \chi(X))$, where

$$(11.77) \quad \chi = \frac{(a_R - a_o) \log X + \log c_R - \log c_o}{X} + \frac{a_R}{X} \log(1 + \chi).$$

This is a strict whole analytic contraction in the original normalized slots $1/X$ and $\log X/X$.

Jointly receive all original reverse banks at κK_b , with their original integer root orders, relative disks and common-ray witnesses. Define the complete bank family $L_f(z) = \mathcal{S}_q f(J_R(Y_o + z))$. Its relative increment is $e^{\Xi_b}(1 + z/Y_o)(1 + \chi(Y_o + z)) - 1$. Lemma 11.27, applied to the stored factorizations, supplies the original strict relative margins on a fixed buffered z disk. The exact derivative $J'_R = (1 - a_o/X)/(1 - a_R/J_R)$ shows that the relative derivative of this family gains $1/(\kappa K_b)$. Its entire bank and coefficient cost precedes the fresh root used to realize that gain.

For the original inverse parameter ζ , solve

$$(11.78) \quad h = E_o + \zeta + a_o \log(1 + h/Y_o) - L_{\ell_R}(h), \quad Z_R = J_R(Y_o + h).$$

Choose a fixed unknown disk larger than twice the full forcing bound, the original ζ radius and the bounded normalizer term. Receive the normalized roots after these costs. The logarithmic derivative and the relative derivative just estimated make the right-hand derivative less than $1/2$ there. Its whole Picard series converges, with geometric remainder and inverse Jacobian at most 2. The defining equation is precisely the original inverse matching equation; at $\zeta = 0$, uniqueness identifies its solution with the supplied positive physical inverse. Every projected equation retains its entire forcing and is solved on that buffered disk.

Step 2. We identify the returned positive record. At a nonbasal recorded parent write $x_v = d_v e^{-u} W_v$ and set

$$(11.79) \quad \begin{aligned} \alpha &= \eta_b m_b, \\ C_Y &= \kappa \beta_b D_b^{-m_b} d_v^{-\alpha}, \\ U_Y &= \exp(m_b G_b - \alpha \log W_v), \\ r_Y &= C_Y^{-1} e^{-\alpha u} U_Y^{-1}, \\ s_Y &= r_Y(\alpha u + \log C_Y + \log U_Y). \end{aligned}$$

Then $r_Y = 1/Y_o$ and $s_Y = (\log Y_o)/Y_o$ have positive parent action α . At the base use the recorded $x_0 = C_x e^{-P_0} P_0^{a_0} W_0$, retaining its exact powers and unit. The full normalized identities are

$$(11.80) \quad \frac{\log(Y_o + h)}{Y_o + h} = \frac{s_Y + r_Y \log(1 + r_Y h)}{1 + r_Y h}, \quad \frac{Z_R}{Y_o} = (1 + r_Y h)(1 + \chi(Y_o + h)).$$

They show that $H_R = \log(Z_R/Y_o)$ lies in the analytic ideal (r_Y, s_Y) before choosing any parameter coefficient. Analytic division in the two normalized variables of (11.77) and (11.78) gives

$$(11.81) \quad \log(Z_R/Y_o) = Y_o^{-1}A_1 + \frac{\log Y_o}{Y_o}A_2,$$

with whole bounded A_1, A_2 , including the entire inverse family. The factor h in $\log(1 + h/Y_o)$ belongs to its bounded coefficient. Expressing Y_o through the recorded t_b makes both normalized factors positive parent factors at v , retaining basal powers when v is the base. (11.6) now proves the new positive parent record. In particular the returned exponent is $\eta_b m_b/r$. The current κ has already entered Y_o and its leading constant and is not inserted a second time into that exponent.

The new inverse denominators are explicit. The units $1 + z/Y_o$, $1 + h/Y_o$ and $1 + \chi$ lie in fixed strict disks about one; $1 - a_R/J_R$ and $1 - a_o/X$ are eventually bounded away from zero. The full inverse Jacobian $1 - \Phi_h$ has inverse norm at most 2. The reciprocal clock is the explicit positive mark r_Y , not an arbitrary field inverse. Old field denominators retain their own-driver laws and original actual domains.

Project every coefficient and normalized argument and solve the whole projected equation on its original buffered unknown disk. If a cut erases r_Y, s_Y , formula (11.80) gives $\chi = H_R = 0$; no raw reciprocal clock is evaluated at zero. The original strict relative margins and the contraction bound persist. On each appropriate strong chart, native damping is preserved by logarithmic relative clock variation, while the whole relative operator and Picard bounds give the new whole thin-tube estimate. Deferred faster factors keep their separate literal domains. Reconstruct the raw factors by Lemma 11.23, including the complete own-driver horizontal law, and only then use Lemma 10.2 for a new finite structural list. Lemma 11.24 supplies the synchronous cuts and weighted maps. The returned parent has not changed its actual coordinate. The now inactive birth and its entire inverse family remain.

Step 3. We handle analytic changes and recorded re-entry. For a positive analytic change, $\log A(s) = s a^{\text{an}}(s)$ proves its positive record directly. For direct re-entry from v along a recorded outgoing birth b , the positive-real-exponent identity of Lemma 10.16 gives

$$(11.82) \quad \begin{aligned} \theta &= \frac{\kappa M}{\eta_b m_b}, & C &= \beta D^{-M} (\beta_b D_b^{-m_b})^{-\theta}, \\ K_{\text{new}} &= CK_b^\theta, & \Xi_{\text{new}} &= \theta \Xi_b + MG - \theta m_b G_b. \end{aligned}$$

Substitution gives $Y_{\text{new}} := \beta s^{-M} = K_{\text{new}} e^{\Xi_{\text{new}}}$ exactly. Set $i = i_b$, the common native level of the recorded birth and the new birth along this retained stem, not the parent vertex level. The positive power $K_{\text{new}} = CK_b^\theta$ creates no new ancestry level. Identify all projected quantities through the receiving inclusion for this common cut $e_i = \iota_i \rho_i$ of Lemma 11.24. Put $F = MG - \theta m_b G_b$ and $\sigma_F = e_i F$. For the principals,

$$(11.83) \quad P_{\text{new}} = CP_b^\theta \exp(\sigma_F), \quad P_b = K_b \exp(e_i \Xi_b).$$

Write $C_* = C \exp(\lim \sigma_F) > 0$, retaining that scalar limit.

First receive the exact roots and whole native data at the common parent. For $\log K_b = \gamma_b u + c_b + \log V_b(e^{-u})$ and each original integer order P_e , the exact root is

$$(11.84) \quad K_{\text{new}}^{-1/P_e} = C_*^{-1/P_e} e^{-\theta c_b/P_e} e^{-\theta \gamma_b u/P_e} V_b(e^{-u})^{-\theta/P_e}.$$

Adjoin these positive marks before the whole joint tensor of old and new banks. The strict analytic unit contributes $\exp(O(n))$ at fixed degree n , leaving the cofinal rectangle cost $\exp(CT \log(eT))$. The original integer orders and joint ray are unchanged. At the base use the original basal root formula. The complete Ξ_{new} certificate gives its original relative ball after all bank inclusion costs.

After this reception, inverse integer powers of the exact old and new roots give K_b, K_{new} in the common projected strong fraction field. The bounded projected corrections have their full positive and negative Banach exponential series there. Thus P_b, P_{new} , their differences and all fixed derivatives belong to this enlarged strong fraction field before either weak class is inserted. No membership of K_b^θ in the unextended input field is inferred for an arbitrary real θ .

Weighted cuts now give $B_{\text{new}} = Y_{\text{new}} - P_{\text{new}} \rightarrow 0$ absolutely and $L_{\text{new}} = (1 - e_i)\Xi_{\text{new}}$, without discarding H_R . The old birth retains $B_b = T_b - P_b \rightarrow 0$ absolutely. Both Y_{new} and T_b are real positive on the physical interval, so Lemma 11.14 gives $\Im P_{\text{new}} \rightarrow 0$ and $\Im P_b \rightarrow 0$ separately. The enlarged whole strong-tube bound is obtained before constructing the new principal inverse. The logarithmic slopes are γ_b and $\theta\gamma_b$; the bounded projected corrections have derivatives tending to zero by Lemma 8.22(2). The inverse construction protects $P_{\text{new}}(\log P_{\text{new}})'_u$ and $1 - a_{\text{new}}/P_{\text{new}}$ on its genuine disks.

We next reconstruct the weak classes in growth order. If $0 < \theta < 1$, construct and complete the new weak class before replaying the old P_b stage. Its subsequent transport to that faster stage uses the retained genuine P_b inverse. If $\theta > 1$, use the new genuine P_{new} inverse just constructed before transporting the already completed old P_b stage into its pre-field. In both cases the exact identity is $dP_a/dP_c = (P_a/P_c)(\log P_a)'_u/(\log P_c)'_u$ for the slower and faster principal. The completed lower-native inlet of Lemma 10.4 and Lemma 10.3 give uniform complex-disk motion $O(P_a/P_c) = o(1)$ on the genuine faster inverse before the slower own-driver law is transported. Other old classes are sorted by their actual growth too.

If $\theta = 1$, keep the entire comparable residual

$$(11.85) \quad R = P_{\text{new}} - C_*P_b = C_*P_b(\exp(\sigma_F - \lim \sigma_F) - 1).$$

It belongs to the common projected strong fraction field constructed above. Put $\delta = \sigma_F - \lim \sigma_F$. Then $\delta \rightarrow 0$, and the exact derivative is

$$(11.86) \quad \frac{dR}{dP_b} = C_*(e^\delta - 1) + C_*e^\delta \frac{(\sigma_F)'_u}{(\log P_b)'_u} \rightarrow 0.$$

Here $(\sigma_F)'_u \rightarrow 0$ follows from its bounded old strong source and Lemma 8.22(2), and $(\log P_b)'_u \rightarrow \gamma_b > 0$. The absolute imaginary limit has a different reason:

$$\Im R = -\Im B_{\text{new}} + C_*\Im B_b \rightarrow 0.$$

It follows from the complete weighted-cut discrepancies and the real physical heights, not from $R = o(P_b)$ or from the derivative limit. Compare the complete tied class jointly.

For a pure reverse followed immediately by a direct, $F = \theta H_R$. The retained term $(\log Y_o)/Y_o$ in H_R can contribute $K_b^{\theta-1} \log K_b$ after multiplication by the new scale. It stays in the full principal and in every comparable residual even when $\theta \geq 1$.

The exact exponential births give $\log b_{\text{new}} \geq cb_{\text{parent}}$ for the finite positive-rate list, and $\log A_{\text{parent}} = o(b_{\text{parent}})$ gives the required adjacent separation. The upper logarithmic bound follows from the stored clocks; the abstract profile inequalities alone are not raised to arbitrary powers.

After the actual slower-stage pullbacks, establish complete horizontal laws, then finite new buffers. Receive the exact weak $[ce^{-\tau P_{\text{new}}} P_{\text{new}}^a] \exp(-\tau B_{\text{new}} + aL_{\text{new}})$ with the complete unit and all inclusion costs fixed before its fresh radius. Its original Wiener norm bounds every degree and geometric tail. Replay the remaining faster classes only after this completed class. The output has a distinct birth label; the old stem, old birth and all original source domains remain.

Step 4. We propagate the original-parameter positive-jet record. Fix a finite original parameter order k , with all rates and leading data fixed. Retain every old coefficient target through order k . Put $J = \log Y_o$ and $q_\mu = D_\mu J$. Its exact identity is

$$(11.87) \quad q_\mu = D_\mu(\log \kappa + \log K_b + \Xi_b) = D_\mu(\log C_Y + \alpha u + \log U_Y).$$

The old normalized jets of $u, G_b, \log W_v$ and the protected inverse unit give finite sums and products satisfying (10.5). Derivatives of complete old coefficients stay in their own marked targets, including faster ones. The normalized reciprocal identities are

$$(11.88) \quad D_\mu(Y_o^{-1}) = Y_o^{-1}\mathcal{B}_\mu((-q_\nu)_\nu), \quad D_\mu(JY_o^{-1}) = Y_o^{-1}\mathcal{C}_\mu(J, (q_\nu)_\nu),$$

where $\mathcal{C}_\mu$ is the finite polynomial obtained by the product rule. Hence differentiating (11.81) retains Y_o^{-1} in every term, with the complete coefficient targets of A_1, A_2 . The factor $J = O(u)$ has multiplicative logarithmic cost $O(\log u)$; the other newly exposed structural factors have cost $o(u)$. Thus the parent action $\alpha > 0$ remains positive.

We verify that the implicit differentiation does not hide an unpaired large Y_o . Put $W = Y_o + h$, $Z_R = J_R(W)$. For a first original parameter derivative, with $q_\lambda = D_\lambda \log Y_o$,

$$(11.89) \quad \begin{aligned} D_\lambda \log(1 + h/Y_o) &= \frac{D_\lambda h - hq_\lambda}{Y_o + h}, \\ D_\lambda \log(Z_R/Y_o) &= \left(\frac{W J'_R(W)}{Z_R} - 1 \right) q_\lambda + \frac{J'_R(W)}{Z_R} (D_\lambda h - hq_\lambda), \\ \frac{W J'_R(W)}{Z_R} - 1 &= \frac{W - Z_R + a_R - a_o}{Z_R - a_R} = O(\log Y_o/Y_o), \quad \frac{J'_R(W)}{Z_R} = O(Y_o^{-1}). \end{aligned}$$

Here $Z_R - W = W\chi(W) = O(\log W)$ on the protected unknown disk. In a differentiated native factor the full relative derivative is $J'_R(Y_o + z)/(\kappa K_b)$. The factor Y_o in $D_\lambda Y_o = Y_o q_\lambda$ cancels into the bounded relative unit $Y_o/(\kappa K_b)$. Higher fixed derivatives of these normalized identities give finite Bell and product formulas involving $q_\nu, h_\nu/Y_o$, explicit reciprocal factors and the protected units, not unpaired powers of Y_o in a native logarithmic jet.

The exact recursion for the inverse coefficients is (10.6), applied to (11.78). Its inverse multiplier has norm at most 2. Its right side uses only lower inverse degrees, whole old forcing and bank coefficient targets, and the normalized slots just checked. This proves normalized structural subprincipality inductively, before spending any particular root reserve. The explicit inverse parameter contributes $h_\zeta = (1 - \Phi_h)^{-1}$ and $(\log Z_R)_\zeta = O(Y_o^{-1})$; higher degrees follow that recursion.

For the returned current, differentiate the complete equality $G_R = (m_b/r)G_b + H_R/r$. Its new positive terms are precisely those in (11.88). Its normalized section identity is $\log S_R = \log D_R + \kappa_R \log x_v - G_R$. An analytic adapter uses $\log A(S_R) = S_R a^{\text{an}}(S_R)$; every argument derivative retains S_R times a finite polynomial in the normalized section jets, with bounded original Cauchy coefficients. This preserves their zero positive exponential order.

For re-entry, the full identities give

$$(11.90) \quad D_\mu \log K_{\text{new}} = D_\mu \log C + \theta D_\mu \log K_b, \quad D_\mu \log Y_{\text{new}} = D_\mu (\log C + \theta \log K_b + \Xi_{\text{new}}).$$

For $|\mu| > 0$, $D_\mu \log C = 0$, since the original leading constants and structural rates in (11.82) are fixed. All newly normalized structural multipliers still have logarithmic size $o(u)$. Thus every fixed positive action $\theta \gamma_b/P_e$, however small, pays them after full scalar-zero Euler division. The scalar bank coefficient uses its old target and has no clock derivative. Derivatives of the amplified error are taken in the full first-removed-factor expression. A fixed power of K_{new} is paid by the weighted map at that faster removed height, not by an $o(u)$ estimate. Jets of the new principal and H_{new} have a factor P_{new} times a finite polynomial of normalized jets. Their logarithmic size is $O(u) + o(u) = o(H_{\text{new}})$, so one occupied new weak pays them. The remaining degree polynomial is bounded by (10.8) with its original strict ratio. The output unit is exactly $\exp(-E_{\text{new}})A_{\text{new}}(\exp(-H_{\text{new}} - E_{\text{new}}))$, where $E_{\text{new}} = B_{\text{new}} - a_{\text{new}}L_{\text{new}} + \ell_{\text{new}}(Y_{\text{new}})$. Its cut, native-root and analytic positive factors give the reusable record at the new driver.

For each finite order, receive these marked factors before their fold, retaining the full divided banks, coefficients and original projective norms. Complete the own-driver horizontal law before later growth-ordered replay. All operator costs precede fresh radii and have the fixed-order

cofinal form $\exp(C_k T \log(eT))$. The literal strict intersection includes the old root and logarithm branches, native relative/damping domains, analytic disks and the buffered inverse unknown disk. Their finite Banach Jacobians give local parameter charts by (11.42), with radius constant on a smaller driver disk. The scaled family and its geometric tail are bounded before extraction. Independently, Lemma 10.26 applied to the exact finite formulas above constructs each original coefficient map. It retains $r_t^{-|\mu|}$ without assigning it subprincipal growth. Cuts act on fixed labels and commute with these derivatives. This proves the full record of Definition 10.5, and hence the receiving records of Definition 11.26, without using the later summary of coefficient targets as an input. $\square$

11.8. The prepared entry and parent handoff.

Proposition 11.30. *Start with the entire first-reverse presentation in Definition 11.1, on its original returned inner section. One supplied prepared forward passage or its correlated original inverse, followed by the next supplied canonical direct passage, has a bounded whole raw receiving map and a faithful independent quotient. These conclusions hold after finitely many reverse, prepared and direct triples at this parent, and at each permitted terminating reverse or prepared prefix.*

All original banks, inverse families, input norms, physical branches and inactive births are retained. The new direct births have their whole native and weak sources, principal charts and born strong cut. The inclusion of the entire first-reverse quotient is bounded and injective. Each output retains Definition 11.26, including its independent original-parameter positive-jet record.

Proof. We construct the connector at the literal returned section, then its direct height and the complete alternating recurrence.

Step 1. We construct the original connector. The first-reverse data give

$$(11.91) \quad \begin{aligned} u &= H_{k-1}, \quad z = e^{-u}, \\ s &= C e^{-\lambda u} A(z)^\lambda V^{-1/r}, \quad \lambda = m/r > 0, \\ b &= -\log C, \quad h = -\lambda \log A(z) + r^{-1} \log V, \quad -\log s = \lambda u + b + h. \end{aligned}$$

Here A and V are the entire original analytic and inverse units, with the original strict inverse and parameter disks. Any analytic adapter before the connector is included once in these exact constants and units. At the particular inlet of Construction 10.32, $m = r = 2$, $C = 1$, $A = 1$ and $V(\zeta) = 1 + \zeta/N$ on its prescribed original inverse disk. These values are inserted in the full families in (11.91); the variable banks and the whole ζ family remain. The displayed notation also records the subsequent supplied passages. For the supplied prepared passage $D_{\text{pre}}(t) = at^\eta e^{U(-\log t)}$, retain its original inverse J with $(\text{id} - U/\eta)^{-1} = \text{id} + J$. Proposition 5.2 supplies the original forward and inverse point-jet distributions on the monoid generated by 1 and η . Construct both baseline families at

$$(11.92) \quad (\lambda, b), \quad (\lambda/\eta, (b + \log a)/\eta).$$

Lemma 11.7 supplies the complete original coefficient bound on every fixed additive disk with a larger buffer; Lemma 11.9 retains the full ordered same-source inverse. Adjoin these families jointly with all old parent actions and native roots by Lemma 11.23. Receive every later old factor and then insert the entire h or h/η . With baseline bound M and increment norm $r_h < R_h$, the full cost is $LM/(1 - r_h/R_h)$ and its degree greater than N tail gains $(r_h/R_h)^{N+1}$. The exact outputs are

$$(11.93) \quad \begin{aligned} D_{\text{pre}}(s) &= a C^\eta e^{-\eta \lambda u} A(z)^{\eta \lambda} V^{-\eta/r} e^{U(\lambda u + b + h)}, \\ D_{\text{pre}}^{-1}(s) &= (C/a)^{1/\eta} e^{-\lambda u/\eta} A(z)^{\lambda/\eta} V^{-1/(\eta r)} e^{-J((\lambda u + b + \log a + h)/\eta)}. \end{aligned}$$

All factors are entire sources on their literal matching domains. Applying the original inverse identity in Theorem 5.1 shows that these are correlated inverses wherever their compositions are defined.

Step 2. We treat the first direct passage. Use the first-reverse native reference $N = \beta_k[D_0 z A(z)]^{-m}$. Its exact returned section can also be written $s = \beta_R^{1/r} N^{-1/r} V^{-1/r}$. After the connector its current is $q = C_q N^{-\theta} \Omega$, where

$$(11.94) \quad \begin{array}{c|cc} & \theta, C_q & \Omega \\ \hline \text{forward} & \eta/r, a\beta_R^{\eta/r} & V^{-\eta/r} e^{U(-\log s)} \\ \text{inverse} & 1/(\eta r), (\beta_R^{1/r}/a)^{1/\eta} & V^{-1/(\eta r)} e^{-J((- \log s + \log a)/\eta)} \end{array}$$

For the supplied incoming analytic map $I(q) = dq A_s(q)$ and direct order p , the exact height is

$$(11.95) \quad Y = K e^{\Xi}, \quad K = \beta(dC_q)^{-p} N^{p\theta}, \quad \Xi = -p \log \Omega - p \log A_s(q).$$

The prepared logarithms have the complete positive residual factorizations already constructed. The analytic logarithm factors as $q a_s^{\text{an}}(q)$. The old inverse unit keeps its original smallness certificate. Fix all these coefficient costs, then receive positive copies so that $e^{\Xi} - 1$ belongs to the original relative disks. The exact roots of K have positive actions $mp\theta/P_e$ in the parent u ; their full $A(z)$ units are retained.

Rebuild the entire pre-native fold and apply Theorem 9.17 as in Lemma 11.23. The clock slope $mp\theta$ is positive for every $\eta > 0$. Synchronous masks and the complete first-removed-factor construction in Lemma 11.24 give the born cut and its weighted error. Lemma 11.28 constructs the full new weak source and its genuine principal chart. The original canonical factorization then defines the new birth, retaining all its banks. Its new profile is an exponential of a positive multiple of this parent; hence the adjacent profile inequalities used for that cut follow directly.

Step 3. We treat alternating prefixes. At the fixed parent retain the underlying reference child height $T_* = N e^{m\Delta}$ of the first-reverse input. It is not replaced by the most recently created Y_h . Write each subsequent complete direct height as

$$(11.96) \quad Y_h = C_h T_*^{\alpha_h} e^{E_h}, \quad E_h = \log U_h - \alpha_h m \Delta, \quad E_{-1} = 0, \quad \alpha_{-1} = C_{-1} = 1.$$

Here U_h is its complete relative unit, not just its scalar part. A reverse at Y_h is constructed by the full identity (11.78), with $H_R = \log(Z_R/Y_h)$. That construction uses the height Y_h and its original native banks and therefore applies before a parent handoff. (11.81) expresses H_R as $Y_h^{-1} A_1 + (\log Y_h/Y_h) A_2$, with entire coefficients. Both normalized factors have positive parent marks. This also constructs the whole terminating reverse prefix without a future direct passage.

Let $I_-(s) = d_- s A_-(s)$ and $I_+(q) = d_+ q A_+(q)$ be its literal adapters. For a prepared sign $\sigma \in \{1, -1\}$ put $\tau = \eta$ for $\sigma = 1$ and $\tau = 1/\eta$ for $\sigma = -1$, and let B be the corresponding full U or J insertion. A subsequent direct of order p gives $\mu = p\tau/r$ and exactly

$$(11.97) \quad \begin{aligned} \alpha_{h+1} &= \mu \alpha_h, \\ E_{h+1} &= \mu E_h + \mu H_R - p\tau \log A_-(s) - p\sigma B - p \log A_+(q). \end{aligned}$$

The corresponding formula for $\log U_{h+1}$ has first term $\mu \log U_h$; subtracting $\alpha_{h+1} m \Delta$ cancels the entire slow term identically. No positive-ideal assertion about Δ is needed. Every remaining term has its complete positive parent factorization: the inverse from the normalized equation, the analytic terms from $\log A(t) = t a^{\text{an}}(t)$, and the prepared term from its single total-carrier residual. This proves the positive record for E_h by finite induction.

At each step use the reference split, growth ordering and original-parameter verification in the following step for the whole translated source, native reconstruction and principal construction. Their input is the entire preceding raw receiver, so all inactive births and parameter correlations persist. A terminating prepared prefix uses (11.93) at its actual positive rate and needs no later

direct. The independent target kernels then descend the old raw maps by original norm infima.

Step 4. We prove the common output records and original-parameter bounds. For every constructed child retain the reference/correction split

$$(11.98) \quad K_h = C_h N^{\alpha_h}, \quad \Xi_h = \alpha_h m \Delta + E_h = \log U_h, \quad Y_h = K_h e^{\Xi_h}.$$

With $N = C_N e^{mu} A(e^{-u})^{-m}$ its exact root is

$$(11.99) \quad K_h^{-1/P_e} = C_h^{-1/P_e} C_N^{-\alpha_h/P_e} e^{-m\alpha_h u/P_e} A(e^{-u})^{m\alpha_h/P_e}.$$

The slope is $\gamma_h = m\alpha_h > 0$. The analytic unit has its original strict disk, and every original integer bank order and joint ray is retained. The initial model is exactly Construction 10.32, with $\Delta = 0$; (11.97) records the full cancellation but does not extend this inlet to an arbitrary nonzero bounded Δ . Such an extension would require whole source smallness in each newly requested relative disk.

For each finite prefix, receive the prepared baseline and inverse source, the exact root marks and all original native banks at the parent u before its child weak classes. The prepared inverse is formed from its full ordered convolution; a residual is shifted only after that convolution. A translated family on $|z| < R$ with full bound M has insertion cost $LM/(1 - r_h/R)$ at $\|h\| \leq r_h < R$ and entire tail factor $(r_h/R)^{n+1}$. Its fixed carrier moments use $\sup_{\beta \geq 0} \beta^k e^{-s_1 \beta} < \infty$ with $0 < s_1 < s_0$ in the original translation slack. The inverse degree majorant has strict ratio $q_0 < 1$; a fixed polynomial in the number of leaves is absorbed by a stricter geometric ratio. Linkages and proper-subtree moments remain attached to their original carriers. These estimates precede all degrees and give the whole raw bounds.

At the common cut e_j define

$$(11.100) \quad P_h = K_h e^{e_j \Xi_h}, \quad B_h = \mathcal{W}_{K_h,1}(e^{\Xi_h}) = Y_h - P_h, \quad L_h = (1 - e_j) \Xi_h, \quad H_h = P_h - a_h \log P_h - \log c_h.$$

The weighted first-removed-factor estimate gives $B_h \rightarrow 0$ absolutely, hence $\Im P_h \rightarrow 0$. The whole strong stage is bounded on old thin tubes before the inverse chart is formed. Its logarithm is $\gamma_h u + c_h^* + \rho_h$, with ρ_h a bounded whole strong correction with retained scalar limit. Lemma 8.22(2) in the old strong differential field gives $\rho_h' \rightarrow 0$; tube Cauchy estimates and contraction then give the principal inverse. Its exact denominators are $P_h(\gamma_h + \rho_h')$ and $1 - a_h/P_h$ for the H_h normalization. They are protected by $|\rho_h'| < \gamma_h/4$ on inverse disks and $|a_h/P_h| < 1/2$. This verifies the logarithmic and chart hypotheses of Lemma 11.28, rather than assuming its new horizontal conclusion.

Set $Q_{\text{ref}} = N \exp(me_j \Delta)$ in the following clock formula. Then $P_h = C_h Q_{\text{ref}}^{\alpha_h} \exp(e_j E_h)$. The sequence $\alpha_{h+1} = (p\tau/r)\alpha_h$ need not increase. Reconstruct weak classes in increasing α_h , retaining birth labels and physical execution order in the raw recipes. For $\alpha_a < \alpha_b$, the exact identity and its uniform consequence are

$$(11.101) \quad \frac{dP_a}{dP_b} = \frac{P_a}{P_b} \frac{m\alpha_a + \rho_a'}{m\alpha_b + \rho_b'}, \quad \sup_{|z| \leq R} |P_a(P_b + z) - P_a(P_b)| = O(P_a/P_b) \rightarrow 0.$$

The inverse moves u by $O(1/P_b)$; Cauchy on its old exponentially thin tubes makes the displayed logarithmic derivatives uniform there. Lemma 10.3 proves the complex displacement before the old own-driver law is used. The normalized H_a has derivative $1 - a_a/P_a$ and obeys this motion too. For equal exponents retain the full difference

$$(11.102) \quad P_a - (C_a/C_b)P_b = C_a Q_{\text{ref}}^{\alpha_a} (\exp(e_j E_a) - \exp(e_j E_b)).$$

Exact roots put K_a, K_b in the projected strong fraction field; the complete exponentials of $e_j \Xi_a, e_j \Xi_b$ put both principals, their difference and fixed derivatives there before either weak fold. Keep the proper-strong head bound $O(A_{d+j-1}) = o(P_b)$ and the full imaginary and derivative data. Compare the equal-rate class jointly. Faster old factors are not coefficients of a newly

slower weak; they wait for their own reconstruction. Each completed class gets its full own-driver law from the compact cut and whole native rectangle before transport to the next class. Its original whole-stage bound is pulled back first; only then does Lemma 10.2 give new finite buffers.

We now verify the independent parameter record, holding all original leading constants and positive structural rates fixed as required by Definition 10.5. For every positive multi-order ν , this convention gives $D_\nu \log D_h = D_\nu \log C_h = 0$ for the constants in (11.104): D_h is a product of fixed leading constants and fixed-rate powers, and $C_{h+1} = D_h C_h^\mu$ starts with fixed $C_{-1} = 1$. It also gives zero positive-order derivatives of the re-entry constant C , the returned constant D_{ret} , and $C_N = \beta D^{-p} d_v^{-\kappa p}$ in (11.8). The leading-constant derivative terms retained in the displays are therefore zero; no varying-leading-data assertion is made. Fix k and put $R_{h,\mu} = D_\mu \log Y_h$ for $0 < |\mu| \leq k$. For a baseline argument $T = \lambda u + b + h$, use exactly $D_\mu T = \lambda D_\mu u + D_\mu b + D_\mu h$. The new h consists of logarithms of protected units. Initially, $V = 1 + \zeta/N$ and each positive coefficient of $\log V$ retains N^{-k} ; mixed derivatives use the normalized jets of $\log N$. For later returns use (11.89) and the triangular coefficient recursion (10.6). Its inverse norm is at most 2. Every derivative of $H_R = Y_h^{-1} A_1 + (\log Y_h/Y_h) A_2$ retains Y_h^{-1} times finite normalized polynomials and complete coefficient targets. Native derivatives inside A_1, A_2 use full Euler maps, so their clock factor is a logarithmic jet, not an unscaled $D_\mu Y_h$.

Prepared argument derivatives are the finite chain polynomials in these normalized jets. The reserved moments just bounded and one fixed extraction $0 < \epsilon < \lambda g$ keep the entire expression $\exp(-\epsilon u)$ times its full coefficient. Analytic adapters use $\log A(t) = t a^{\text{an}}(t)$ and $D_\mu t = t \mathcal{B}_\mu((D_\nu \log t)_\nu)$, with bounded larger-disk derivatives. For the returned section and prepared output,

$$(11.103) \quad \begin{aligned} \log s &= \text{constant} - r^{-1} \log Y_h - r^{-1} H_R, \\ \log q &= \text{constant} - (\tau/r) \log Y_h - (\tau/r) H_R + \tau \log A_-(s) + \sigma B. \end{aligned}$$

Their newly exposed normalized structural jets have logarithmic size $o(u)$ by the old record and these finite exact operations. The explicit s and q factors carry positive actions γ_h/r and $\gamma_h \tau/r$.

In the notation of the alternating step, put $c_\sigma = a$ for $\sigma = 1$ and $c_\sigma = a^{-1/\eta}$ for $\sigma = -1$. The exact constant and jet recursions are

$$(11.104) \quad \begin{aligned} D_h &= \beta_+ [d_+ c_\sigma (d_- \beta_R^{1/r})^\tau]^{-p} > 0, \\ C_{h+1} &= D_h C_h^\mu, \\ R_{h+1,\nu} &= D_\nu \log D_h + \mu R_{h,\nu} + \mu D_\nu H_R \\ &\quad - p\tau D_\nu \log A_-(s) - p\sigma D_\nu B \\ &\quad - p D_\nu \log A_+(q). \end{aligned}$$

This follows by differentiating the entire cancellation identity with fixed μ, α_h . No derivative of a structural rate times Δ is introduced. Each new term has the positive decomposition just proved, with newly growing structural multipliers of logarithmic size $o(u)$ and complete faster coefficients stored separately. Thus the normalized structural record remains subprincipal without spending a fixed amount of an old root action. Every future fixed $\gamma_{h+1}/P_e > 0$ can therefore pay the required native jets.

Unscaled $D_\nu Y_h = Y_h \mathcal{B}_\nu((R_{h,\mu})_\mu)$ is not subprincipal at u . For new weak jets its principal factor is paid at $H_h \sim P_h$, where $\log P_h = O(u) = o(H_h)$. For $D_\nu B_h$, differentiate the whole removed-factor expression; fixed powers of K_h are paid at its faster removed height by the weighted maps. One occupied weak pays its fixed structural derivatives, and (10.8) pays the degree polynomial. Degree zero and every faster old coefficient keep their whole targets.

All actual domain constraints are the finite strict pullbacks of the root/logarithm, relative, prepared additive, carrier damping, analytic and inverse disks just used. They give the local full-Jacobian charts with exact coefficient recovery. The fixed-order formulas verify the input of Lemma 10.26; its application now constructs each separate original coefficient target and full local family. A terminating reverse has already obtained these records from its normalized inverse equations, and a terminating prepared prefix from its entire translated source and correlated inverse. Neither uses a future direct. This proves the output records of Definition 11.26 at every finite prefix, with its original norms and observations. $\square$

Proposition 11.31. *The entire terminating strong receiver of Proposition 11.30 admits a whole receiving inclusion at its recorded parent, with all strong cuts and the weighted maps (11.63) for each finite list of positive real references. An already executed terminal reverse is not repeated.*

From this output, any finite literal sequence of positive analytic changes, returns to recorded older parents, re-entry along recorded outgoing births, and fresh supplied prepared passages or their correlated inverses has a constructed receiver with positive parent records and all the preceding whole source, comparison, cut and atlas assertions. Returns never go below the retained base. The empty return sequence is allowed. Every output has the records of Definition 11.26, including its independent original-parameter positive-jet record.

Proof. Let T_* be the retained underlying reference child height in (11.96), with stored input $t_* = D_* x_v^{\eta_*} e^{-G_*}$ and $T_* = \beta_* t_*^{-m}$. It is not the most recently created Y_h . For the initial handoff of Construction 10.32, the actual data are $u = T_1$, $x_v = t_* = x_1 = e^{-u}$, $D_* = \eta_* = d_v = 1$, $G_* = 0$, $W_v = 1$, and $T_* = N = e^{2u} = T_2$, with $m = 2$ and $\Delta = 0$. The general current notation below retains the exact algebra used by subsequent operations; it does not enlarge the model-only prepared-entry inlet to arbitrary bounded nonzero Δ . At a terminal reverse of order r , use the already constructed $Z_R = Y_h e^{H_R}$ in (11.96).

Step 1. We identify the returned inner section and its positive record. Its exact returned inner section has data

$$(11.105) \quad \begin{aligned} D_{\text{ret}} &= \left(\frac{\beta_R}{C_h \beta_*^{\alpha_h}} \right)^{1/r} D_*^{m\alpha_h/r}, \\ \kappa_{\text{ret}} &= \eta_* m\alpha_h/r, \\ G_{\text{ret}} &= (m\alpha_h/r)G_* + E_h/r + H_R/r. \end{aligned}$$

This follows by taking the original positive root of Z_R , then substituting t_* ; thus it uses the reverse exactly once. If it has not yet been executed, the terminating-reverse construction of Proposition 11.30 first executes that one supplied passage. The complete positive representations proved there and the old parent record for G_* give a positive record for G_{ret} . The literal incoming analytic adapter and a terminating prepared prefix use (11.5) on this record.

Step 2. We construct the all-parent inclusion and retain the cuts. Construct the all-parent inclusion directly on the chronological raw presentation. At each old parent retain its old pre-native coefficient stage, receive the positive factors just displayed, and receive every full affine prepared baseline and its ordered inverse. At comparable parents keep (11.58); at the base keep the actual ψ chart. Reconstruct every old native tensor with the new lower source and every later class in order by Lemma 11.23. This is a new receiving map on the whole source. It does not identify two source presentations from the equality of their current sections.

Physical chronology specifies the original raw recipes and shared parameters; comparison classes are rebuilt in increasing actual growth. For a strictly slower principal $X = o(P)$, use the completed lower-native inlet of Lemma 10.4 and the genuine faster inverse to prove its complex-disk motion by (10.2) before using its completed own- X law. A comparable difference belongs to the old projected strong fraction field through the already received exact reference roots and

the complete exponentials of bounded projected corrections. Retain the entire difference and its derivatives before adjoining the joint weak class. At a comparable prepared parent retain the whole kernel (11.58), including its carrier-dependent residual. Faster factors wait on their own domains.

At each enlarged class, the whole original damping, relative, Wiener and Picard bounds precede the new finite-list buffer. Joint comparison and its complete cut or native rectangle give the own-driver law; only afterwards is that law transported. The pullback preserves the whole original norm independently of a fixed readout. Apply Lemma 10.2 to its new subprincipal list. The only new inverse factors are protected units $1 + \chi$, $1 + h/Y_h$, $1 - a_R/J_R$ and the complete inverse Jacobian of norm at most 2, in addition to explicit reciprocal clocks. No inverse of a positive correction is used.

The adjacent profile assertion uses the exact exponential births. For a finite list of fixed positive powers, $\log X = \gamma V + O(\log V) + O(1)$ with $\gamma > 0$ gives $\log b_{\text{new}} \geq cb_{\text{parent}}$ for some $c > 0$. The old $\log A_{\text{parent}} = o(b_{\text{parent}})$ implies $A_{\text{parent}} = o(b_{\text{new}})$; the upper logarithmic bound follows from the stored clock formula. At the base keep the full basal powers and unit. Positive residual shifts keep a strict portion of their original carrier gap and linkage. This does not infer power stability from the two abstract profile inequalities alone. Apply Lemma 11.24 at every old level, including below the born cut, using one mask for every prepared increment. This proves all cuts, old inclusion squares and the finite list of weighted maps. Their coefficient costs are fixed before their new receiving radii.

Step 3. We construct the subsequent return suffix and fresh prepared letters. An older-parent reverse and a recorded-edge re-entry are now the constructions of Proposition 11.29. They preserve these records. The empty return word is the whole receiving inclusion just constructed. For a fresh prepared letter at the current $s = Dx_v^\kappa e^{-G}$, use exactly

$$(11.106) \quad -\log s = \kappa u - \log D - \kappa \log d_v + G - \kappa \log W_v.$$

Receive both baseline families and their complete inverse at the rates κ and κ/η . Its full increment is inserted after all later coefficients as in the first connector. The residual provides its positive parent record. The current updates are exactly (11.5). At the base use instead the recorded $x_0 = C_x e^{-P_0} P_0^{a_0} W_0$ and $H_0 = P_0 - a_0 \log P_0$; the affine carrier family is built in H_0 and pulled back by the genuine ψ . Its residual is $e^{-\epsilon P_0} P_0^{a_0 \epsilon}$ times its whole coefficient. Thus every carrier power and basal unit remains.

Joint comparison, all masks and their whole differences are reconstructed once more on these full sources. Strict original analytic and inverse margins persist by the coefficient estimates, and the whole strong-tube estimate applies. Finite induction proves the stated suffix construction.

Step 4. We retain the independent original-parameter record. Fix a finite order k and keep every old original coefficient target through that order. Differentiating the exact returned correction gives

$$(11.107) \quad D_\nu G_{\text{ret}} = \frac{m\alpha_h}{r} D_\nu G_* + \frac{D_\nu E_h + D_\nu H_R}{r}.$$

At a nonbasal parent, the normalized logarithmic identities are

$$(11.108) \quad \begin{aligned} D_\nu \log Y_h &= \alpha_h D_\nu \log T_* + D_\nu E_h, \\ D_\nu \log T_* &= m\eta_* D_\nu u - m\eta_* D_\nu \log W_v + m D_\nu G_*. \end{aligned}$$

The leading constants and rates are fixed, so $D_\nu \log C_h = D_\nu \log D_{\text{ret}} = 0$ for $|\nu| > 0$. This accounts for the absence of a leading-constant term in (11.108). Jets of $\log W_v$ use only finite products of unit jets and powers of the protected bounded W_v^{-1} . Complete old coefficient derivatives are left whole in their original targets; the newly normalized structural factors have logarithmic size $o(u)$ by the old independent record. Writing $J = \log Y_h$, derivatives of Y_h^{-1} and JY_h^{-1} are Y_h^{-1} times finite polynomials in J and its normalized jets. Thus the full identity

$H_R = Y_h^{-1} A_1 + JY_h^{-1} A_2$ retains positive parent action $\eta_* m \alpha_h$ after every fixed derivative. The targets of A_1, A_2 come from the triangular inverse recursion with full inverse-Jacobian norm at most 2. The normalized cancellations in (11.89) account for each native logarithmic derivative, so no bare large Y_h is hidden in this structural bound.

The returned normalized section is $\log S_{\text{ret}} = \log D_{\text{ret}} + \kappa_{\text{ret}} \log d_v - \kappa_{\text{ret}} u + \kappa_{\text{ret}} \log W_v - G_{\text{ret}}$. The added leading-constant term has zero positive-order original parameter derivatives under the fixed-leading-data convention. Its finite jets therefore have the required normalized decomposition before any later positive reference power is requested. An analytic change adds only $-S_{\text{ret}} a^{\text{an}}(S_{\text{ret}})$ to the correction; finite chain differentiation keeps that occupied positive section factor and bounded original analytic Cauchy maps.

For a fresh prepared letter the exact argument is $T = \kappa u + c + h$, $h = G - \kappa \log W_v$. Its jets are $D_\nu T = \kappa D_\nu u + D_\nu h$ with the fixed leading data. The inverse uses $(T + \log a)/\eta$, with positive rate κ/η and its correlated original J . Choose $0 < \epsilon < \kappa g$, or $0 < \epsilon < \kappa g/\eta$ for the inverse, and keep a strict part of the gap after extraction. Each argument derivative adds a fixed carrier moment or polynomial lowering, paid by a positive part of the original translation slack. Complete the ordered inverse convolution first, retaining every proper-subtree moment, and shift its total carrier only once. The remaining structural products have logarithmic size $o(u)$, while the explicit $e^{-\epsilon u}$ pays them. Whole faster coefficient targets remain at their later folds. The two updated corrections $\eta G - U(T)$ and $G/\eta + J((T + \log a)/\eta)$ therefore retain both their positive source decomposition and the stronger normalized structural bound.

At the base the actual $P_0 = \psi(H_0)$ is retained. Its new derivative factor $(1 - a_0/P_0)^{-1}$ is a protected unit. The residual is the entire $e^{-\epsilon P_0} P_0^{a_0 \epsilon}$ times its full coefficient; fixed powers of P_0 and $\log P_0$ have logarithmic size $o(P_0)$. Thus the basal operation has the required own-scale reserve and actual domain without replacing ψ by an asymptotic expression.

During all-parent replay, a native positive Euler term is divided by its whole original first root before its finite normalized multiplier is attached. Its exact positive reference action pays the multiplier. A weak derivative retains one occupied factor at its own driver and uses (10.8) for all remaining degrees. Weighted discrepancies are differentiated in their full removed-factor expressions; fixed extra reference powers are paid at the removed height. For each nonzero admitted multiplier keep its old field head, separate its fixed leading constant, and retain its entire residual phase with vanishing imaginary part. This provides the source mark (10.7), not merely scalar decay.

The finite strict domain list includes W_v , its logarithm and inverse, the normalized time-map units, every original native relative ball, analytic adapter disk, prepared additive/carrier domain and the whole inverse unknown disk. Their complete bounded operators give the local parameter charts of (11.42) and the geometric full-family bound of Lemma 11.17. Independently, the exact finite formulas in this step verify Lemma 10.26 before invoking it. It constructs the original coefficient maps with all original Cauchy, Euler, division, inclusion and exterior costs. Their exact local recovery factors are not declared subprincipal and their target norms are not summed. The raw cuts use fixed parameter labels, so their identities commute with coefficient extraction. This proves Definition 10.5 at the handoff and after each fresh prepared letter or empty suffix, and completes the output record of Definition 11.26. $\square$

Corollary 11.32. *Let Q_* be the entire first-reverse output exported by Proposition 10.31, for the model child $x_2 = H(x_1)$ followed by its genuine inverse, with $H(t) = \exp(-1/t^2)$. At its current x_1 , the auxiliary sequence $I_1; H_{\text{aux}}; H_{\text{aux}}^{-1}; I_2$, where I_1, I_2 are identity prepared linear passages and H_{aux} is a distinct original order-two model occurrence, has a bounded whole raw receiving map and an injective inclusion of Q_* . Its current is x_1 , and its entire output is a receiver of Proposition 11.31. Every old bank and the new auxiliary birth and inverse family remain in that output.*

Proof. Use $a = \eta = 1$ and $U = J = 0$ in the connector of Proposition 11.30. Use its direct construction for H_{aux} and its terminating-reverse construction for H_{aux}^{-1} . Apply Proposition 11.31 at this already returned prefix, then its empty older-return suffix and its fresh identity prepared insertion I_2 . For the complete auxiliary family put $u = -\log x_1 = T_1$ and $Y = x_1^{-2} = e^{2u}$. Its auxiliary direct output is $\exp(-Y)$. This clock equals the old x_2 clock, but its label is distinct; compare their full native banks jointly, with ratio one and residual zero. Its exact order- P_e root is $\exp(-2u/P_e)$, and its logarithmic slope is two. The inverse $u = (\log Y)/2$ supplies the actual domain directly. Both prepared identity letters have $U = J = 0$ and contribute no carrier or additional denominator.

Retain the entire original additive inverse parameter ζ :

$$\begin{aligned} s_{\text{ret}}(\zeta) &= (Y + \zeta)^{-1/2} = x_1(1 + \zeta/Y)^{-1/2}, \\ [\zeta^k]s_{\text{ret}} &= \binom{-1/2}{k} Y^{-1/2-k}, \\ (11.109) \quad G_{\text{ret}}(\zeta) &= \tfrac{1}{2} \log(1 + \zeta/Y), \quad [\zeta^k]G_{\text{ret}} = \frac{(-1)^{k+1}}{2k} Y^{-k} \quad (k \geq 1). \end{aligned}$$

The strict normalized disk $|\zeta/Y| < \rho < 1$ protects the full binomial, logarithm and inverse units. Each positive coefficient of G_{ret} has explicit parent action $2k$. Mixed old derivatives use $D_\mu \log Y = 2D_\mu u$ and finite Bell polynomials, with normalized structural cost $o(u)$ and complete old coefficient targets kept separately. A positive Euler derivative of an auxiliary variable bank still uses whole native division; an identity structural normalizer does not specialize that bank. The original-norm relative and binomial bounds precede coefficient extraction and give Definition 10.5. At $\zeta = 0$ the actual composition is identity on its literal overlap by the original inverse identities. Each step is a bounded map on the entire preceding raw presentation; its independent kernel descends that map. The equality of currents is used only for the stated actual value. It does not delete either the old inactive x_2 birth or the auxiliary birth, and zero structural corrections do not specialize any variable bank. $\square$

11.9. A new stem at the current.

Proposition 11.33. *Let Q be an entire receiver constructed in Proposition 11.31, or a whole output of finitely many applications of the present construction and its intervening operations. Retain its data of Definition 11.26. Let its current be nonbase, $s = Dx_v^\kappa e^{-G}$, with $D, \kappa > 0$ and its derived positive parent record. A supplied original canonical direct passage of integer order p and scale $\beta > 0$ has a bounded whole source construction at this current. Its exact reference is*

$$(11.110) \quad N = \beta D^{-p} d_v^{-\kappa p} e^{\kappa p u}, \quad \Xi = pG - \kappa p \log W_v, \quad Y = Ne^\Xi.$$

It creates a distinct labelled birth at v . Its entire output retains all old and new comparison, cut, weighted-map, positive-parent and atlas records, and admits the intervening operations of Proposition 11.31. The entire output retains Definition 11.26, including the separate coordinate and original-parameter positive-jet records.

Proof. We construct the reference receiver before inserting the full correction; we then construct the principal and verify the reusable output.

Step 1. We construct independent reference sources. Lemma 11.5 proves (11.110). If v has level i , the new native level is $j = i + 1$. For each original integer bank order P_e , adjoin the exact root

$$(11.111) \quad N^{-1/P_e} = \beta^{-1/P_e} D^{p/P_e} d_v^{\kappa p/P_e} e^{-\kappa p u/P_e}.$$

Its numerical reserve is $\kappa p/P_e > 0$, without any rationality assumption. Include every original receiving bank required by relative substitution and first-root division. At the parent u , join the

entire old prepared and inverse sources, keeping their original carrier kernels and all comparable-parent phases. Lemma 11.23 gives the pre-native rows, whole cofinal estimate, bounded horn maps and differentiated numerators. The clock $\log N$ has exact derivative κp , and $\Im \log N \rightarrow 0$ on the recorded physical curve. (11.111) proves the row reserve before any native degree. The original joint ray, root lift and full bank are supplied by the actual passage. Thus the full tensor satisfies Setup 9.14. Apply Theorem 9.17 and reconstruct every later old fold.

For an old or new reference CR^α , $\alpha > 0$, the exact root at its own nonbasal parent, if $\log R = \gamma u + c + \log V(e^{-u})$, is

$$(11.112) \quad C^{-1/P} e^{-\alpha c/P} e^{-\alpha \gamma u/P} V(e^{-u})^{-\alpha/P}.$$

The whole analytic unit is retained and its reserve is $\alpha \gamma / P$. At the base retain its full original power and source-unit formula. This constructs all finite old or new reference requests independently, without a power relation between unrelated stems. Lemma 11.24 now supplies every old cut, the new cut at j , and all their weighted whole maps. Their domains are the appropriate strong tubes; faster factors keep their separate original domains.

Step 2. We retain the entire transported correction. Lemma 11.27 uses the full current and birth certificates to put $e^\Xi - 1$ in every original relative disk after the complete inclusion costs. (11.71) therefore supplies the entire native maps at the actual $Y = \beta s^{-p}$, including all coefficient, native and relative remainders. (11.72) supplies every original canonical strong slot. These give the full source inputs in the supplied canonical factorization.

At the cut j , put

$$(11.113) \quad P = N e^{e_j \Xi}, \quad B = \mathcal{W}_{N,1}(e^\Xi), \quad L = (1 - e_j) \Xi, \quad H = P - a_+ \log P - \log c_+.$$

Lemma 11.28 proves $B = Y - P$, the absolute limit $\Im P \rightarrow 0$, the genuine P and H inverse charts, and the entire new weak source. The full $e_j \Xi$ remains in the principal; no small term has been discarded before multiplication by N .

Step 3. We obtain the reusable birth. The actual output is

$$(11.114) \quad E_+ = B - a_+ L + \ell_+(Y), \quad x_+ = F_+(e^{-H} e^{-E_+}).$$

This is the original canonical passage on its recorded physical branch, by Theorem 6.1. Store its parent v , its exact input (D, κ, G) , its original order and scale, and all data in (11.110) and (11.113). The current is reset to x_+ with data $(1, 1, 0)$. The first two terms of E_+ are whole cut discrepancies. The third has the complete native first-root factorization. The new unit $W_+ = e^{-E_+} A_+(e^{-H} e^{-E_+})$ therefore has exactly the certificate (11.69). The coordinate and original-parameter exit records, including the adjacent profile inequality, are verified next from these exact formulas.

Step 4. We prove the coordinate and original-parameter exit records. Write $\gamma = \kappa p$ and $C_N = \beta D^{-p} d_v^{-\kappa p}$. The exact logarithmic reference is $\log N = \gamma u + \log C_N$. Its absolute size is not charged to a structural buffer. For $\phi = \log P$ the inverse denominators are exactly

$$(11.115) \quad \frac{du}{dP} = \frac{1}{P\phi'}, \quad \frac{du}{dH} = \frac{1}{P\phi'(1 - a_+/P)}, \quad \frac{1}{H} = P^{-1} \left(1 - \frac{a_+ \log P + \log c_+}{P} \right)^{-1}.$$

The source-tube inverse construction keeps ϕ' within $\gamma/4$ of γ and $|a_+/P| < 1/2$ on its actual disks. The final bracket is a protected geometric unit. The finite old proper-strong coordinate list pulls back with displacement of modulus $O(\exp(-\gamma \Re u + o(\Re u)))$; full denominators use their own-driver laws. Reinserted slower weak siblings use the completed lower-native inlet of Lemma 10.4, then (10.2) on the genuine faster inverse. Comparable residuals belong to the old strong fraction field by their exact roots and bounded projected exponentials before the joint weak fold. Complete each own-driver law before later growth-ordered replay. Whole original operator bounds pull back on the actual domains first; the new finite structural buffers then

follow from Lemma 10.2. Full B, L and their units with faster factors remain on their separate original domains.

Fix a finite original parameter order k . Native logarithmic jets are given exactly by

$$(11.116) \quad D_\mu \log Y = \gamma D_\mu u + D_\mu \Xi + D_\mu \log C_N, \quad D_\mu \Xi = p D_\mu G - \kappa p D_\mu \log W_v.$$

For $|\mu| > 0$, the fixed-leading-data convention gives $D_\mu \log C_N = 0$ in (11.116). Derivatives of $\log W_v$ are finite products of its old jets and powers of the protected bounded inverse unit W_v^{-1} . Keep complete later coefficient targets separate. The newly exposed normalized structural factors in (11.116) are finite sums and products of the old independent normalized-jet list, so their logarithmic size is $o(u)$. This proves their stronger normalized property before using a native reserve. For the native chain rule, fix the original ray, logarithmic lift and integer root order P_e , and let $f(\lambda)$ be a whole bank input. Write S_e for its synthesis in (10.23) and put $J = \log Y$. The whole Euler operator and its actual identity are

$$(11.117) \quad \begin{aligned} \mathcal{E}_{P_e}(c, \phi) &= (0, -\zeta \phi'(\zeta)/P_e), \\ S_e(\mathcal{E}_{P_e} f)(Y) &= Y \partial_Y S_e(f)(Y). \end{aligned}$$

For a nonzero multi-index μ and $1 \leq j \leq |\mu|$, define

$$(11.118) \quad Q_{\mu,j} = \frac{\mu!}{j!} [z^\mu] \left(\sum_{0 < \nu \leq \mu} \frac{D_\nu J}{\nu!} z^\nu \right)^j.$$

Here z is an independent parameter increment, the inequalities between multi-indices are componentwise, and $\binom{\mu}{\nu} = \mu! / (\nu! (\mu - \nu)!)$. Set $Q_{0,0} = 1$, $Q_{\kappa,0} = 0$ for $\kappa \neq 0$, and $\mathcal{E}_{P_e}^0 = \text{id}$. Expanding in $J(\lambda + z) - J(\lambda)$ proves the finite chain rule: its j th logarithmic-clock derivative is $S_e(\mathcal{E}_{P_e}^j f)(Y)$ by (11.117). The increment has zero constant term, so only $j \leq |\mu|$ and derivatives with componentwise index at most μ contribute. Coefficient extraction gives exactly the factorials in (11.118).

Each bank $\mathcal{E}_{P_e}^j g$ with $j \geq 1$ has scalar zero. Whole first division therefore gives the exact identity

$$(11.119) \quad \begin{aligned} S_e(\mathcal{E}_{P_e}^j g)(Y) &= Y^{-1/P_e} S_e(\mathcal{N}_1 \mathcal{E}_{P_e}^j g)(Y) \\ &= N^{-1/P_e} e^{-\Xi/P_e} S_e(\mathcal{N}_1 \mathcal{E}_{P_e}^j g)(Y). \end{aligned}$$

Each positive-Euler term contains one original first root, not a j -fold root. For parameter-dependent input the finite Leibniz rule, followed by this whole division, gives for $\mu \neq 0$

$$(11.120) \quad \begin{aligned} &D_\mu [S_e(f(\lambda))(Y(\lambda))] \\ &= \sum_{\nu \leq \mu} \binom{\mu}{\nu} \sum_{j=0}^{|\mu-\nu|} Q_{\mu-\nu,j} S_e(\mathcal{E}_{P_e}^j D_\nu f)(Y) \\ &= S_e(D_\mu f)(Y) \\ &\quad + N^{-1/P_e} e^{-\Xi/P_e} \sum_{\substack{\nu \leq \mu \\ \nu \neq \mu}} \binom{\mu}{\nu} \sum_{j=1}^{|\mu-\nu|} Q_{\mu-\nu,j} \\ &\quad \cdot S_e(\mathcal{N}_1 \mathcal{E}_{P_e}^j D_\nu f)(Y). \end{aligned}$$

The derivatives $D_\nu f$ are original bank-input derivatives at fixed clock. The only $j = 0$ term is the whole old coefficient derivative $S_e(D_\mu f)(Y)$, not merely its scalar coefficient. It retains its separately supplied target, including any later raw factors; no first-root assertion is made for it. If the input has no explicit parameter dependence, only $\nu = 0$ remains. At $\mu = 0$ we retain the original whole source map. Original correlations are handled by this finite product rule before projective completion, not by introducing independent copies of a shared parameter.

For fixed $k = |\mu|$, choose finitely many strict receiving minor domains and larger weights for all Euler powers through k and their required first divisions. The whole operator estimates give, for each relevant original coefficient bank $\mathcal{B}_0$,

$$(11.121) \quad \begin{aligned} \|\mathcal{E}_{P_e}^j g\|_{\mathcal{B}_j} &\leq C_{E,j} \|g\|_{\mathcal{B}_0}, \\ \|\mathcal{N}_1 \mathcal{E}_{P_e}^j g\|_{\tilde{\mathcal{B}}_j} &\leq C_{D,j} C_{E,j} \|g\|_{\mathcal{B}_0}. \end{aligned}$$

The constants and banks may depend on the fixed old coefficient target. These bounds use strict Cauchy restriction for the Euler maps and the whole division of Lemma 10.21; the ray and original integer order remain unchanged.

Every enlarged divided bank also needs its own bounded reference inclusion i_j at N , with its actual square $\text{Act } i_j(b) = S_e(b)(N)$. Fix these inclusions and the strict relative operators before choosing the final small correction. With $\eta = e^\Xi - 1$, $a_j = \|\eta\| < r_{0,j}$, the entire relative map satisfies

$$(11.122) \quad \|\mathcal{R}_\eta \mathcal{N}_1 \mathcal{E}_{P_e}^j g\| \leq \frac{L_{i,j} L_{\text{rel},j}}{1 - a_j/r_{0,j}} C_{D,j} C_{E,j} \|g\|_{\mathcal{B}_0}.$$

Its relative-degree greater than M tail has the additional factor $(a_j/r_{0,j})^{M+1}$. This is the full divided bank evaluated at the actual $Y = Ne^\Xi$, with the relative unit in (11.119) still attached. For $g = D_\nu f$, retain the original derivative-map cost, including $\nu!$, and the binomial factor in (11.120). Original Taylor coefficients are recovered by division by $\mu!$. Finite projective products include the lower coefficient inclusion and the product of all these whole-bank bounds. The finite sum is bounded by the sum of its term costs; taking the maximum of its finitely many strict relative ratios preserves a complete geometric tail.

The parent action $\gamma/P_e > 0$ remains positive after multiplication by each finite polynomial in the normalized structural jets. Its logarithmic cost is $o(u)$ by the supplied normalized-jet record. This scalar bound preserves the numerical reserve only. The actual positive-mark decomposition, pre-field membership, correct-coordinate derivatives and complex domains remain the separate inlets of Definition 10.5; they are not deduced from scalar decay. Complete faster coefficient targets stay whole at their later folds. Fix all Euler, division, domain, weight, reference-inclusion and original coefficient costs, using provisional bounds $a_j \leq r_{0,j}/2$, before choosing fresh positive receiving radii. No original input norm or parameter radius is renamed. The construction is for each fixed finite order and does not sum the separate targets over all parameter orders.

The coordinate jets obey

$$(11.123) \quad D_\mu P = P \mathcal{B}_\mu((D_\nu \phi)_\nu), \quad D_\mu H = D_\mu P - a_+ D_\mu \phi - D_\mu \log c_+.$$

Thus their new structural parts have logarithmic size $\log P + o(u) = O(u) + o(u) = o(H)$. The factor P is subprincipal at H , where the new weak uses it, not at u . For the cut errors differentiate their complete first-removed-factor expressions, with fixed cuts and parameter labels. Each term retains one removed positive factor, a whole coefficient map and finitely many normalized jets; any fixed additional positive power of N is received by the weighted map at that removed height X , where $\log N = O(u) = o(X)$. This pays $D_\mu B$ without subtracting separately large jets of Y and P . The unweighted expression pays $D_\mu L$.

For $Q_\nu = c_\nu e^{-\tau_\nu P} P^{a_\nu}$ retain the whole unit $U_\nu = \exp(-\tau_\nu B + a_\nu L)$. Differentiation of an occupied original weak gives that weak times finite polynomials in the coordinate and unit jets. The structural coordinate factors are subprincipal at P , and the unit jets have the whole cut targets just constructed. One positive weak pays these factors. Its full degree sum uses (10.8); degree zero uses the whole old coefficient map.

Put $z_+ = \exp(-H - E_+)$. The exact output identity is

$$(11.124) \quad \log W_+ = -E_+ + z_+ a_+^{\text{an}}(z_+), \quad E_+ = B - a_+ L + \ell_+(Y).$$

The native term uses (11.120). Every argument derivative of the analytic term retains an occupied $\exp(-H)$, with finite Bell factors from (11.123). Original larger-disk Cauchy maps bound its complete coefficients. The section map and its derivative are

$$(11.125) \quad \begin{aligned} F_+(z) &= d_+ z A_+(z), & d_+ &> 0, \\ F'_+(z) &= d_+ (A_+(z) + z A'_+(z)), & F'_+(0) &= d_+. \end{aligned}$$

The retained strict source, image and inverse-image disks protect the unit A_+ and the derivative F'_+ , not A'_+ . Lemma 4.18 applies to F_+ . Its retained compositional inverse is the actual branch F_+^{-1} ; the logarithm and multiplicative reciprocal $1/A_+$ belong to the unit A_+ and are different operations. For the whole normalized fiber inverse retain the original larger-disk norms and complete parameter algebra, together with the strict source/image radii and contraction inequalities of Lemma 8.6. The whole contraction and its inverse-jet recipe remain part of these data. Scalar nonvanishing of F'_+ alone supplies no whole-family operator bound. For a retained implicit recipe use (10.6) and its original strict contraction constant, not an inverse of a scalar asymptotic expression.

The finite first-root and weak marks just verified have their pre-field, derivative and complex-domain inlets before their folds. Their original operator and cofinal costs are fixed before fresh radii. The full intersection of original root, logarithm, relative, analytic, damping and inverse domains supplies the local parameter charts and full geometric family tail. Lemma 10.26 now constructs each independent original coefficient map from the finite formulas, retaining its exact recovery and factorial costs. This proves Definition 10.5 for the new birth, separately from its coordinate record.

The ancestry separation also follows from the exact exponential birth: for the fixed finite reference list, $\log b_{\text{new}} \geq c b_{\text{parent}}$ with $c > 0$, whereas $\log A_{\text{parent}} = o(b_{\text{parent}})$. Hence $A_{\text{parent}} = o(b_{\text{new}})$, and the upper logarithmic bound follows from the explicit clock. These statements keep the original depth profiles independent of the structural buffer functions.

Step 5. We retain the intervening operations.

Positive analytic changes use their exact analytic difference. Fresh prepared letters use (11.106) with the new whole W_+ and its certificate, their original correlated inverse and one total residual shift. The joint comparison and all cut maps are reconstructed on the now enlarged inventory by the construction of Proposition 11.31. An older return or a recorded-edge re-entry uses Proposition 11.29 on the current records just established. Thus each intervening operation preserves the inputs for this construction. Distinct births remain distinct even if their actual clocks agree. Finite induction proves the assertion about subsequent whole outputs. $\square$

Lemma 11.34. *Every finite construction in Propositions 11.30, 11.31, 11.29 and 11.33 preserves complete original analytic families on their literal connected pullback domains. At each far physical point there exist local scaled parameter charts with the whole bound and tail of Lemma 11.17. For each fixed original Taylor multi-index μ there exists a separate bounded linear whole-input map to a finite independently faithful marked target whose actual value is that original coefficient. All original recovery, derivative and exterior factors are retained. The norms for distinct μ are not summed. Vanishing of all these independent coefficient records is equivalent to zero of the entire analytic family on its connected component.*

Proof. We summarize the finite induction already supplied by the model and the four stated constructor results. Their outputs include Definition 10.5; its normalized structural bound was proved from their exact recursions before the corresponding root and weak reserves. The following construction uses those verified inputs to Lemma 10.26, not an assumption that the new coefficient targets already exist. We first bound the whole local family, then construct each original coefficient without treating its recovery factor as a bounded source multiplier.

Step 1. We construct the whole local family. Take the literal pullbacks of the original argument, root, logarithmic and image domains, and intersect the strict damping, source, relative, analytic and inverse inequalities. Inverse inequalities hold uniformly on their buffered unknown disks. At parameter zero the preceding whole estimates give strict margins at every far physical point. The finite full constraint functions and their Banach Jacobians therefore give precisely the local radius in (11.42). It is constant on a sufficiently small driver disk. For a new height, for example, the exact parameter derivative is $-p \partial_\lambda s/s$ for $\partial_\lambda \log Y$, including the whole G and W_v . The full root integrals, translated carrier majorants, relative series, Wiener sums and Picard bounds give one $C\|f\|$ on the doubled scaled polydisk. Lemma 11.17 proves its entire tail and exact recovery formula. Maps on overlapping local charts agree because they restrict the original functions and equations.

Step 2. We fix an original coefficient. Fix one μ and differentiate all recipes in the original parameters before physical specialization. Retain the finitely many old coefficient and derivative targets through degree $|\mu|$, each in its original norm. Analytic recipes use their whole buffered Cauchy maps. For $h = \Phi(h; \lambda)$, the coefficient equation, when $|\mu| > 0$, is

$$(11.126) \quad (1 - \partial_h \Phi(h_0; 0))h_\mu = [\lambda^\mu] \Phi(h_{<|\mu|}; \lambda).$$

The right side contains only lower total-degree coefficients of h ; its inverse multiplier is the entire protected geometric unit. Thus finitely many complete earlier targets suffice.

A prepared derivative inserts fixed finite carrier moments, polynomial lowerings and full argument jets. Form the original inverse convolution before these insertions, retain all proper-subtree moments, and reserve a positive fraction of the gap for a distinguished $e^{-\epsilon X}$. Fixed additional moments are paid by extra carrier slack in Lemma 11.7. The newly exposed normalized structural products specified by Definition 10.5 have logarithmic size $o(X)$ at this actual prepared driver. Its exact finite source decomposition, full phase and derivative inlet let the distinguished factor receive them as a positive mark. This bound does not concern arbitrary faster coefficient targets. An old faster observation is retained as its entire old target.

A native derivative uses the full relative Euler map, whose scalar is zero; extract its first root by Lemma 11.19. Its remaining divided bank and relative unit remain whole. The exact normalized logarithmic jets of the constructor, with whole faster coefficient targets separated, satisfy (10.5) at this root's own parent. Their finite Bell polynomials therefore leave a positive root reserve. An unscaled new height jet is instead paid at its weak driver; fixed reference powers in differentiated cut discrepancies are paid by the first removed factor through the whole weighted-cut map. For a weak source of total degree $n \geq 1$, extract one occupied weak factor to carry the finite growing structural factors. For a strict original degree ratio $\theta < 1$,

$$(11.127) \quad \sup_{n \geq 1} (1+n)^{|\mu|} \theta^{n/2} < \infty.$$

The other $\theta^{n/2}$ bounds its entire remaining degree sum. The degree-zero term uses the old coefficient map.

These are finitely many marked types for this fixed μ . Lemmas 11.23 and 11.24 receive them jointly with every complete old target, before the variable input and any of its degrees. Clear their finitely many fixed coordinate denominators; exterior modules retain their original finite shifted records. The resulting map is linear in the whole original family input, because every structural source and inverse solution is fixed independently of that input. All original Cauchy factors, inverse costs and derivative factorials remain in these independently constructed coefficient maps. The local identity also retains its exact $r_t^{-|\mu|}$ recovery; that factor is not used to construct the original target or asserted to have subprincipal growth. No sum over μ has been taken.

Step 3. We identify the complete family kernel. If every such target record vanishes, each whole unnormalized Taylor coefficient is the zero germ by its independent kernel. Continue this zero

identity along the common zero-section driver component, even if its marked comparison used a farther tail. At each fixed point a positive parameter polydisk remains; its entire Taylor series is therefore zero. Continue this zero identity on the connected pullback component. Conversely a zero family has all coefficients zero. This argument needs no common countable horn. $\square$

Original observations also remain at their original nodes. If M_a is an observation, L_a its permitted raw window, I_a the original receiving map and Pull its literal matching pullback, the exact two-component identity is

$$(11.128) \quad \begin{aligned} & (M_{\text{new}}I_af, -\text{Pull } M_aL_af) \\ &= (M_{\text{new}}I_af, -\text{Pull } M_af) + (0, \text{Pull } M_a(f - L_af)). \end{aligned}$$

Only the first pair uses its original matching identity. The second is its entire original complement, with all mixed terms, norms, codomains, carrier moments and root lifts. A new bank at Y does not move an old observation to Y . Nonanalytic windows remain in their raw spaces. A retained gauge keeps its genuine analytic inverse on its literal inverse domain.

Proof of Theorem 11.2. We induct on the finite physical history, retaining at each output Definition 11.26, including the separate coordinate and original-parameter positive-jet records. The model input is Construction 10.32. At each constructor, exact normalized jets are checked first, their own-scale positive reserves next, and independent original coefficient maps last. Growth-ordered reconstruction does not change physical execution order or any original correlation.

Step 1. We assemble the receiving operations and their comparison properties. Proposition 11.30 constructs the connector, direct and alternating prefixes in Definition 11.1. Proposition 11.31 constructs their entire all-parent handoff, including the empty return suffix, subsequent recorded-edge re-entry and fresh prepared letters. Their positive current records are derived by (11.97) and (11.105). Thus the input definition has now been supplied with actual bounded raw maps and independent quotients.

Proposition 11.33 constructs a new direct passage at every nonbase current of each such whole receiver and verifies all records again. Its formula is exactly the one in the theorem. Its positive root reserve uses only $\kappa > 0$; it requires neither a direct-ending current nor rational κ . Proposition 11.29 constructs every recorded-parent return with its whole inverse equation. Hence finite iteration preserves every claimed source and comparison property. After the constructor-specific normalized-jet and positive-factor checks, Lemma 11.34 summarizes the independently constructed full original families. It is not used to establish a constructor's own parameter inlet. For any real nonzero whole value, its positive scale and relative remainder give a real nonzero leading coefficient by taking their quotient limit on the original physical interval.

Step 2. We receive the nonzero degree-zero prepared bank. It remains to verify the stated receiving operation for the nonzero degree-zero prepared bank. The original bank in (5.48) has finite weighted mass $\sum_{m \geq 1} e^{-Sm}/m$ on every positive weight, gap one and degree zero. Lemma 5.13 supplies its correlated whole inverse. At the current, (11.106) gives the exact argument $\kappa u + c + h$. Choose the complete additive family disk first, and insert the full h after the costs of all old coefficients. Its residual with $\epsilon = \kappa/2$ is exactly (11.3). Lemma 11.10 gives its full residual coefficient cost C_* .

Step 3. We bound the full prepared pair after its coefficient costs. Perform the original forward prepared passage and its same-source inverse on their literal overlap using the fresh construction already proved. This has one fixed entire inclusion cost L_{pair} after the original coefficient cost L_0 . Reconstruct a receiver for the complete residual coefficient and all these old factors by Lemma 11.23. Those coefficient stages and their whole inclusion cost L_1 are fixed before adjoining a fresh copy of the distinguished positive mark $e^{-(\kappa/2)u}$ at radius σ . No coefficient-stage cost depends on σ . Its total cost is therefore at most $\sigma L_1 L_{\text{pair}} L_0 C_*$. Choose σ

to make this less than the prescribed δ . The mark's phase is zero and its action is $\kappa/2 > 0$, so the joint reconstruction applies with every retained native, prepared and inverse factor. Independent kernel descent and original norm infima give the bounded injective old inclusion. The all-cut, atlas and coefficient-target constructions apply to this same receiver, so it remains an input for each subsequent literal operation. $\square$

12. LITERAL FINITE WORDS AND COMPLETE ACTUALIZATION

We construct a complete auxiliary source presentation before applying any letter of the physical word. The construction first retains a model forest and then admits its whole original input to the prepared calculus. The final induction uses this common presentation at every prefix.

12.1. Auxiliary passages and the original input.

Lemma 12.1. *For $m \in \{1, 2\}$ and sufficiently small $a, b > 0$, the polynomial vector field*

$$(12.1) \quad V_m = u^{m+1}\partial_u - mv\partial_v$$

has passage $E_m(z) = \exp(-z^{-m})$ from the section $v = b$, with coordinate $z = u$, to the section $u = a$, with coordinate $v/[b \exp(a^{-m})]$. Its positive inverse is

$$(12.2) \quad E_m^{-1}(w) = (-\log w)^{-1/m}.$$

For $n \geq 3$, put $C = E_1$, $H = E_2$ and $X_n(\sigma) = H^{\circ n}(C^{\circ 4}(\sigma))$. Then X_n maps a sufficiently small positive interval analytically and increasingly onto a positive interval ending at zero. If $t = (E_2^{-1})^{\circ n}(s)$, its inverse is

$$(12.3) \quad X_n^{-1}(s) = \frac{1}{\log(\log(\log(-\log t)))}.$$

Every root and logarithm in these formulas is the positive determination on a sufficiently small positive interval.

Proof. Along a positive orbit of (12.1),

$$\frac{d \log v}{du} = -mu^{-m-1} = \frac{d(u^{-m})}{du}.$$

The orbit starting at (z, b) therefore reaches $u = a$ with $v = b \exp(a^{-m} - z^{-m})$. It reaches that section in finite positive time for $0 < z < a$, since $du/d\tau = u^{m+1} > 0$. The vector field is transverse to the two sections: its relevant components are $-mb$ and a^{m+1} . The stated outgoing coordinate gives E_m . Taking the positive logarithm and root proves both inverse identities in (12.2). Moreover,

$$E'_m(z) = mz^{-m-1} \exp(-z^{-m}) > 0, \quad \lim_{z \rightarrow 0^+} E_m(z) = 0.$$

Each E_m thus has the asserted positive interval image and inverse. Their finite compositions have these properties as well.

To check the number of logarithms in (12.3), put

$$H_* = \exp(1/\sigma), \quad K = \exp(H_*), \quad Z_0 = \exp(K), \quad t = \exp(-Z_0).$$

Then $t = C^{\circ 4}(\sigma)$ and $-\log t = Z_0$, $\log Z_0 = K$, $\log K = H_*$, $\log H_* = 1/\sigma$. This proves the formula and checks each branch. In particular $X_n^{-1}(s) \rightarrow 0$ as $s \rightarrow 0^+$. $\square$

Lemma 12.2. *The linear saddle $\dot{x} = x$, $\dot{y} = -y$ admits a passage and its inverse both equal to the identity between positive analytic section coordinates. For $H = E_2$, the auxiliary sequence consisting of such an identity passage, then H , then its genuine inverse, then a second identity passage has composition equal to the identity on its matching holomorphic domain.*

Proof. Take the incoming section $y = b$ with coordinate $s = x$, and the outgoing section $x = a$ with coordinate $(a/b)y$. The solution from (s, b) reaches the latter at time $\log(a/s)$ with $y = bs/a$. Both section components of the field are nonzero, and the resulting coordinate map is $s \mapsto s$. Its inverse has that formula as well. For this linear equation the nonlinear prepared correction is zero. Its inverse equation has zero correction too.

Lemma 12.1 supplies the genuine inverse of H . Substituting its logarithmic expression shows $H^{-1}(H(s)) = s$ on the connected determination containing the positive interval. The two outer identity maps preserve that equality. It also holds after any holomorphic substitution for s whose image lies in this domain. $\square$

12.2. Whole maps and original quotient norms. For the next lemma, a raw comparison datum consists of a supplied unital complex Banach algebra $\mathcal{P}$, a unital algebra homomorphism $\text{Act}: \mathcal{P} \rightarrow \mathcal{H}$ into an algebra of holomorphic germs, and bounded linear maps $\mathcal{R}_\alpha$ into Banach spaces. The maps $\mathcal{R}_\alpha$ are supplied independently of Act . We assume that their common kernel is an ideal and that

$$\bigcap_{\alpha} \ker \mathcal{R}_\alpha = \ker \text{Act}.$$

The common kernel is closed, so its quotient has the original infimum norm. These are hypotheses on supplied maps; they do not construct a raw presentation or its comparison maps.

Lemma 12.3. *Let $(\mathcal{P}_j, \text{Act}_j, (\mathcal{R}_{j,\alpha})_\alpha)$, $j = 0, 1$, be raw comparison data, and let $\mathcal{A}_j = \mathcal{P}_j / \bigcap_{\alpha} \ker \mathcal{R}_{j,\alpha}$. Suppose $T: \mathcal{P}_0 \rightarrow \mathcal{P}_1$ is bounded and linear, with $\|T\| \leq L$, and*

$$(12.4) \quad \text{Act}_1(Tr) = \text{Act}_0(r) \quad (r \in \mathcal{P}_0)$$

as germs on a common connected domain. Suppose also that T preserves the unit after realization. Then T induces a bounded injective unital algebra map $I: \mathcal{A}_0 \rightarrow \mathcal{A}_1$, and

$$(12.5) \quad \|If\|_{\mathcal{A}_1} \leq L\|f\|_{\mathcal{A}_0}.$$

Proof. If $r \in \ker \text{Act}_0$, then (12.4) gives $Tr \in \ker \text{Act}_1$. Hence $I[r] = [Tr]$ is well defined. Conversely, $[Tr] = 0$ implies $\text{Act}_0(r) = 0$, hence $[r] = 0$. This proves injectivity. For $r, s \in \mathcal{P}_0$, multiplicativity of both realizations gives

$$\text{Act}_1(T(rs) - (Tr)(Ts)) = \text{Act}_0(rs) - \text{Act}_0(r) \text{Act}_0(s) = 0.$$

The target comparison kernel therefore makes I multiplicative. The unit assertion follows by the assumed unit identity. The quotient estimate in Proposition 4.20 applies to the bounded raw map T and gives (12.5). No choice of a bounded representative map is needed. $\square$

In particular, replay of an original model presentation requires a bounded map T on every original raw factor and a jointly constructed target comparison before Lemma 12.3 can be used. Equality of two terminal currents does not supply (12.4) for the other factors.

Lemma 12.4. *Let B be the root bank of Setup 9.4, with data (P, g, r, ϑ, q) and synthesis S_q . Choose $0 < \rho_0 < \rho_1 < 1$, and choose $r', \vartheta', G > 0$ such that*

$$(12.6) \quad \begin{aligned} \vartheta' + \sup_{|\eta| \leq \rho_1} |\arg(1 + \eta)| &< \vartheta, & |\arg q| + \sup_{|\eta| \leq \rho_1} |\arg(1 + \eta)| &< \pi/2, \\ r' &< r(1 - \rho_1)^{1/P}, & G &> g/(1 - \rho_1). \end{aligned}$$

Let B' be the root bank with data $(P, G, r', \vartheta', q)$. With the branch equal to one at $\eta = 0$, define

$$(12.7) \quad D_\eta(c, \phi) = (c, \phi((1 + \eta)^{-1/P} \cdot)), \quad L = G(1 + \rho_1)/g.$$

This is a holomorphic bounded-operator family $B \rightarrow B'$ for $|\eta| < \rho_1$. Its coefficients $D_j = [\eta^j]D_\eta$ satisfy

$$(12.8) \quad \|D_\eta\| \leq L, \quad \|D_j\| \leq L\rho_0^{-j}.$$

On their common continued germ,

$$(12.9) \quad S_q(D_\eta v)(W) = S_q(v)((1 + \eta)W).$$

For the independent native coefficients of that setup, $b_k(D_\eta v) = (1 + \eta)^{-k/P} b_k(v)$ for every $k \geq 0$.

Proof. Apply Lemma 8.7 with $\delta = 1/P$, $A = g$ and $B = G$. Its domain, damping and weight hypotheses are precisely (12.6). It gives the norm, holomorphy and actual substitution assertions. For the independent coefficient, multiplication of the native k th Taylor coefficient by $(1 + \eta)^{-k/P}$ gives

$$\Gamma(k/P)q^{-k/P}[\zeta^k]\phi((1 + \eta)^{-1/P}\zeta) = (1 + \eta)^{-k/P}b_k(c, \phi), \quad k \geq 1.$$

The scalar coefficient at $k = 0$ is unchanged. $\square$

Lemma 12.5. *Let B, B' be Banach spaces, with B carrying its original source norm, and let A be a unital Banach algebra with submultiplicative norm and $\|1\| = 1$. Suppose bounded linear maps $D_j: B \rightarrow B'$ and $i: B' \rightarrow A$ satisfy $\|D_j\| \leq L\rho_0^{-j}$ and $\|i\| \leq L_i$. If $\eta \in A$ and $\|\eta\| \leq a < \rho_0$, then*

$$(12.10) \quad T_\eta b = \sum_{j \geq 0} i(D_j b) \eta^j, \quad \|T_\eta b\| \leq \frac{L_i L}{1 - a/\rho_0} \|b\|_B.$$

The entire sum over $j > M$ has the latter bound multiplied by $(a/\rho_0)^{M+1}$. For finitely many independent original factors, with a bounded linear lower coefficient map into A of norm L_0 , the corresponding multiplication map on their projective tensor product has bound

$$(12.11) \quad L_0 \prod_e \frac{L_{i,e} L_e}{1 - a_e/\rho_{0,e}}.$$

The complement of all restrictions $j_e \leq M_e$ has this bound multiplied by $\sum_e (a_e/\rho_{0,e})^{M_e+1}$.

Proof. The norm of the j th summand is at most $L_i L \|b\|_B (a/\rho_0)^j$. Summing this geometric series proves convergence and (12.10); summing from $M+1$ proves the tail estimate. Multiplication in A and the defining projective norm give (12.11) first on simple tensors, then on their completion. Every multi-index outside the rectangle violates at least one restriction. Summing the corresponding geometric bounds, with all other factors kept whole, proves the last assertion. $\square$

Here the inclusion i and the small element η must belong to one already constructed target. In a native application $\eta = \exp(\Xi) - 1$, so a scalar estimate $\Xi \rightarrow 0$ alone does not verify the norm hypothesis of Lemma 12.5. If this target has a multiplicative realization with bounded evaluation χ at a fixed point, then $|\chi(\eta)|^k \leq \|\chi\| \|\eta\|^k$ for every positive integer k . Taking k th roots gives $|\chi(\eta)| \leq \|\eta\|$. Thus the strict source-norm bound places every actual relative parameter in the disk of Lemma 12.4. Bounded evaluation of the convergent whole series then gives its actual height substitution. Descent from the original raw factors still requires the complete square (12.4).

Lemma 12.6. *Let B be a Banach space and A a unital commutative Banach algebra with submultiplicative norm and $\|1\| = 1$. Let $I: B \rightarrow A$ be bounded and linear with norm at most C_0 . Fix a positive integer d , a positive radius vector $\rho = (\rho_1, \dots, \rho_d)$ and elements $z_i \in A$ with $\|z_i\| \leq r_i$ and $0 < r_i < \rho_i$. On the entire original coefficient space*

$$(12.12) \quad \mathcal{W}_\rho(B) = \left\{ \sum_{\nu \in \mathbb{N}^d} f_\nu Z^\nu : \sum_\nu \rho^\nu \|f_\nu\|_B < \infty \right\},$$

the map $F \mapsto \sum_\nu I(f_\nu) z^\nu$ has norm at most C_0 . Its complete total-degree tail $|\nu| \geq M$ has norm at most

$$(12.13) \quad C_0 \theta^M \|F\|_{\mathcal{W}_\rho(B)}, \quad \theta = \max_i (r_i/\rho_i) < 1.$$

Proof. Apply Proposition 4.20 first to the coefficient map I , whose induced map on the original Wiener space has norm at most C_0 , and then to substitution at the commuting elements z_i . Its strict substitution estimate gives (12.13) for $M \geq 1$; the whole norm bound gives the case $M = 0$. The coefficient inclusion cost C_0 occurs once, outside the sum. $\square$

Lemma 12.7. *Fix a finite list of positive radii $\rho_1, \dots, \rho_l$ before constructing any receiving space. Let Q_{seed} be an original model quotient with realization Act_{seed} and supplied algebra readouts $E_{*,\rho} : \mathcal{W}_\rho(\mathbb{C}) \rightarrow Q_{\text{seed}}$, one for each listed radius. Suppose $\text{Act}_{\text{seed}} E_{*,\rho} f = f(X_n)$ and*

$$(12.14) \quad \|E_{*,\rho} f\| \leq A_\rho \|f\|_\rho, \quad \|E_{*,\rho} f_{\geq M}\| \leq A_\rho \eta_\rho^M \|f\|_\rho, \quad 0 < \eta_\rho < 1.$$

Here $M \geq 0$ and the constants precede f and M . Let $I_0 : Q_{\text{seed}} \rightarrow Q_0$ be a bounded algebra inclusion for the initial handoff, and let $I_j : Q_{j-1} \rightarrow Q_j$, $1 \leq j \leq N$, be bounded algebra inclusions along the word. Suppose each preserves actual realization and has original-norm bound $\|I_j\| \leq C_j$. Then

$$(12.15) \quad E_{j,\rho} = I_j \circ \dots \circ I_0 \circ E_{*,\rho}$$

has actual value $f(X_n)$ and satisfies

$$(12.16) \quad \|E_{j,\rho} f\| \leq L_{j,\rho} \|f\|_\rho, \quad \|E_{j,\rho} f_{\geq M}\| \leq L_{j,\rho} \theta_{j,\rho}^M \|f\|_\rho$$

with

$$(12.17) \quad L_{j,\rho} = A_\rho \prod_{k=0}^j C_k, \quad \theta_{j,\rho} = \eta_\rho.$$

In particular, if the model quotient is a unital commutative Banach algebra with submultiplicative norm and unit of norm one, and a supplied element x_* satisfies $\text{Act}_{\text{seed}} x_* = X_n$ and $0 < \|x_*\| \leq a < \min_i \rho_i$, its power-series evaluation gives the required seed readouts with $A_\rho = 1$ and $\eta_\rho = a/\rho$.

Proof. Compose the actual squares of the supplied maps to obtain $\text{Act}_j E_{j,\rho} f = f(X_n)$. Composition preserves the algebra property. Apply the original operator norm of each inclusion to $E_{*,\rho} f$ and then to the whole element $E_{*,\rho} f_{\geq M}$; this gives (12.17) and both asserted bounds. Each inclusion cost is paid once, after the complete series has been evaluated in the model quotient. For the last assertion, apply Lemma 12.6 with scalar coefficients, the unit inclusion, and substituted element x_* . The strict ratio a/ρ is positive and less than one at every prescribed radius. This conclusion requires the stated source-norm bound on x_* ; its scalar limit alone does not provide that bound. $\square$

12.3. The exact current after a reverse passage.

Lemma 12.8. *Fix a retained birth with positive parent coordinate x and whole germ identities*

$$(12.18) \quad t_b = D_b x^{\eta_b} e^{-G_b}, \quad T_b = \beta_b t_b^{-m_b},$$

$$(12.19) \quad q_b = c_b e^{-T_b} T_b^{a_b}, \quad x_b = d_b q_b e^{-\ell_b(T_b)} A_b(q_b e^{-\ell_b(T_b)}).$$

Here $D_b, \eta_b, \beta_b, c_b, d_b > 0$, m_b is a positive integer, a_b is real, and A_b is an analytic unit with $A_b(0) = 1$. Let the current after the inverse outgoing regular map be $S_w = D_w x_b^\kappa e^{-G_w}$, with $D_w, \kappa > 0$. Suppose the new canonical core has inverse endpoint germ h_R , with inverse k_R , and forward expression

$$(12.20) \quad y \mapsto k_R(c_R e^{-Z} Z^{a_R} e^{-\ell_R(Z)}), \quad Z = \beta_R y^{-r},$$

where $\beta_R, c_R > 0$, r is a positive integer and a_R is real. Normalize $h'_R(0) = k'_R(0) = 1$ and use all logarithms on the fixed matching branches. Put $C_h = h_R/\text{id}$ and

$$(12.21) \quad Y_o = \kappa T_b, \quad a_o = \kappa a_b, \quad c_o = D_w d_b^\kappa c_b^\kappa \kappa^{-\kappa a_b},$$

$$(12.22) \quad E_o = \kappa \ell_b(T_b) + G_w - \kappa \log A_b(q_b e^{-\ell_b(T_b)}) - \log C_h(S_w).$$

Then $h_R(S_w) = c_o e^{-Y_o} Y_o^{a_o} e^{-E_o}$.

Write $T_o(z) = z - a_o \log z - \log c_o$, $T_R(z) = z - a_R \log z - \log c_R$, and $J_R = T_R^{-1} \circ T_o$ on their matching inverse domains. Suppose an entire source construction supplies the solution of

$$(12.23) \quad h = E_o + \zeta + a_o \log(1 + h/Y_o) - \ell_R(J_R(Y_o + h)).$$

At $\zeta = 0$, let $Z_R = J_R(Y_o + h)$ be on the specified positive inverse branch and set $L_R = \log(Z_R/Y_o)$. The output before the inverse incoming regular map is

$$(12.24) \quad y_R = D_R x^{\eta_R} e^{-G_R},$$

$$(12.25) \quad D_R = \left(\frac{\beta_R}{\kappa \beta_b} \right)^{1/r} D_b^{m_b/r}, \quad \eta_R = \frac{\eta_b m_b}{r}, \quad G_R = \frac{m_b}{r} G_b + \frac{L_R}{r}.$$

Proof. We first retain the whole old birth in the inverse forcing, then substitute the resulting height in the old parent identity. Using (12.19) and $h_R(S_w) = S_w C_h(S_w)$ gives

$$h_R(S_w) = D_w d_b^\kappa c_b^\kappa e^{-\kappa T_b} T_b^{\kappa a_b} \exp\left(-\kappa \ell_b(T_b) - G_w + \kappa \log A_b(q_b e^{-\ell_b(T_b)}) + \log C_h(S_w)\right).$$

Substitution of $Y_o = \kappa T_b$ proves the claimed identity. In particular neither analytic unit has been replaced by its constant term.

By the definition of J_R ,

$$T_R(Z_R) = T_o(Y_o + h) = T_o(Y_o) + h - a_o \log(1 + h/Y_o).$$

Using (12.23) therefore gives

$$T_R(Z_R) + \ell_R(Z_R) = T_o(Y_o) + E_o + \zeta.$$

At $\zeta = 0$, exponentiation shows that the weak argument of k_R in (12.20) equals $h_R(S_w)$. Hence the specified positive root $y_R = (\beta_R/Z_R)^{1/r}$ is the genuine inverse output. Finally, the positive-real-exponent identity of Lemma 10.16, on these supplied full birth identities and compatible logarithms, gives

$$\left(\frac{\beta_R}{Z_R} \right)^{1/r} = \left(\frac{\beta_R}{\kappa \beta_b} \right)^{1/r} t_b^{m_b/r} e^{-L_R/r}.$$

Inserting (12.18) proves (12.25). The factor κ has entered Y_o and D_R ; it introduces no additional factor in η_R . $\square$

The hypothesis supplying (12.23) is a whole-source hypothesis. For example, if its right-hand side is an analytic self-map of a closed ball in one complete source algebra and has contraction constant $q_c < 1$, its iterates satisfy

$$\|h - h_j\| \leq \frac{q_c^j}{1 - q_c} \|h_1 - h_0\|.$$

This estimate follows by summing the bounds $\|h_{j+1} - h_j\| \leq q_c^j \|h_1 - h_0\|$. The iterates are therefore Cauchy. Completeness of the ball and continuity of the operator give a fixed point, and the contraction bound makes it unique in that ball. It retains the entire inverse remainder once the operator, its ball and its original input bounds have been constructed. Pointwise inversion of the scalar current does not construct that operator.

12.4. Recuts and selected endpoint labels. The algebra of the recut clocks is also exact. For a direct passage write the incoming current as $y = D x_v^\kappa e^{-G}$ and the stored birth as $x_v = d_v e^{-u_v} W_v$. Then

$$(12.26) \quad K = \beta D^{-p} d_v^{-\kappa p} e^{\kappa p u_v}, \quad \Xi = pG - \kappa p \log W_v, \quad Z = \beta y^{-p} = K e^\Xi.$$

For an original regular recut $J_e(z) = d_e z A_e(z)$, with $d_e > 0$ and $A_e(0) = 1$, let its own clock be $Y_e = \beta_e J_e(z)^{-m_e}$. In the direct case $z = y$, so

$$\theta_e = m_e/p, \quad M_e = \beta_e d_e^{-m_e} \beta^{-\theta_e} K^{\theta_e},$$

$$(12.27) \quad \Xi_e = \theta_e \Xi - m_e \log A_e(y), \quad Y_e = M_e e^{\Xi_e}.$$

In the reverse case $z = y_R$, use $T_b = K_b e^{\Xi_b}$ and $Z_R = \kappa T_b e^{L_R}$ from Lemma 12.8. They give

$$(12.28) \quad \begin{aligned} \theta_e &= m_e/r, & M_e &= \beta_e d_e^{-m_e} \beta_R^{-\theta_e} \kappa^{\theta_e} K_b^{\theta_e}, \\ \Xi_e &= \theta_e \Xi_b + \theta_e L_R - m_e \log A_e(y_R), & Y_e &= M_e e^{\Xi_e}. \end{aligned}$$

These identities follow by substituting $y^{-p} = Z/\beta$ or $y_R^{-r} = Z_R/\beta_R$ in the defining clock. Their native application still requires the full original bank inclusion at M_e and a source representative with $\|e^{\Xi_e} - 1\| < \rho_{0,e}$. Lemma 12.5 then gives the original bank bound, including its entire relative-degree tail.

Lemma 12.9. *Consider one chronological prepared or weak pre-native stage, or its full native extension, with the original raw presentation of Definition 11.22. Supply its buffered own-parent domains, original whole coefficient bounds, fixed differentiated numerator maps, and the horizontal lower-field inlet of Theorem 8.25. Prepared factors have their original positive gaps and degree linkages and the point-jet distributions of Proposition 5.2; their correlated inverses retain the full ordered leaf convolution. At a native fold supply all the data of Setup 9.14. Its coefficient algebra is such a constructed pre-native stage; the prepared list is allowed to be empty. Use the independent rows of Lemma 11.23 for a prepared stage, and of Construction 9.15 for a native stage.*

For each fixed nonzero whole readout f , with least nonzero action λ_f , there exists a fixed buffered complex disk in its own parent coordinate u on which

$$(12.29) \quad \frac{f(u+z)}{f(u)} = e^{-\lambda_f z} (1 + o(1)), \quad \frac{f'(u)}{f(u)} = -\lambda_f + o(1),$$

uniformly for z in each smaller closed disk as $\Re u \rightarrow \infty$ along the physical parent curve. For a fixed fraction f/g , the action is $\lambda_f - \lambda_g$. The relative estimate is for each fixed readout; whole absolute estimates retain the original raw norm.

Let $u = \psi(v)$ be an actual later inverse-coordinate map whose buffered domains carry that entire stage and its fixed denominators. If $u = o(v)$ and, uniformly for every fixed bounded z , $\psi(v+z) - \psi(v) \rightarrow 0$, then the transported fixed fractions have horizontal ratio tending to one in v . A sufficiently slowly widening common horn retains the whole-stage bound and any prescribed finite denominator list.

Proof. We keep a finite complete row and bound its entire complement on a buffered complex domain. For native inputs we additionally bound a full root rectangle.

Step 1. We construct prepared rows and control the entire cut complement. Choose a finite positive generating list for the original actions, including the scaled and residual actions, and a reduced linkage $\delta_* > 0$ for the finite inventory. On the supplied own-parent horn keep each comparable kernel in the form

$$(bX + c + v)^K \exp[-\beta(bX + c + v)], \quad X = d_X u + Z_X(u), \quad d_X, b > 0, \quad Z_X = o(u).$$

The full residual phase stays in this kernel. The fixed strict damping absorbs $\exp(o(\beta \Re u) + O(\beta \log \Re u))$ uniformly in the carrier and linked degree. This follows from $K \leq \beta/\delta_*$ and the original weighted absolute majorant, on a farther horn fixed before the input. For the inverse use its already summed original ordered source and make its residual shift only afterwards.

Take a smooth cutoff χ in a gap of the full numerical action set, equal to one through T_0 and zero beyond T_1 . Apply it to the total action of the convolved product. Lemmas 8.12 and 8.15 give the full distributional product rule. In an affine kernel, one carrier derivative inserts the complete factor $-(bX + c + v)$; its polynomial powers remain subject to the reserved damping of the affine kernel. At most $J = \lceil T_1/\delta_* \rceil$ derivatives meet the transition. The binomial factors from all finitely many slots sum to at most C_*^J times their original absolute majorant. Outside

the transition use the unlowered majorant. Thus the compact operator and its entire complement satisfy, on a common buffered complex strip,

$$(12.30) \quad |\text{Act } r(u) - \text{Compact}_\chi r(u)| \leq CH_\chi \|r\|_{\mathcal{P}} e^{-cT_0 \Re u}, \quad H_\chi = 2 + C_*^J \max_{1 \leq j \leq J} \|\chi^{(j)}\|_\infty.$$

The maximum is zero for $J = 0$. The constants precede r ; for this fixed cutoff H_χ is finite. Whole zero-action coefficients retain their original norms and are not charged to δ_* . Strict analytic and Wiener slots are summed in their full original norms. Products use the projective norm; inverse slots use their full absolute majorant. These facts extend the displayed estimate from finite inputs to the completion. Every derivative of the total-action cutoff has been included. Lemma 8.16 provides gap cutoffs with arbitrarily large T_0 .

By Lemma 8.13, the compact part is exactly a finite sum $\sum_{\lambda < T_0} e^{-\lambda u} a_\lambda(u)$. Each a_λ is its whole finite coefficient after all phase relations are collected and its fixed denominators cleared. The earlier horizontal bridge gives $a_\lambda(u+z)/a_\lambda(u) \rightarrow 1$ for each nonzero such coefficient on smaller buffered disks. For real displacements this follows by integrating its logarithmic derivative; for imaginary displacements it is the supplied horizontal law. Its logarithmic size is $o(\Re u)$.

For a fixed nonzero $f = \text{Act } r$, choose a fixed gap cutoff with $cT_0 > \lambda_f + 1$. All rows below λ_f vanish as whole germs on their domains. The finitely many larger rows in the compact part have a positive action gap. Formula (12.30) makes the entire complement relatively small on the fixed buffered disks of (12.29): its ratio to the first row is at most $C_{f,\chi} \exp(-(cT_0 - \lambda_f)\Re u + o(\Re u))$. This proves the first identity in (12.29) for the full prepared readout. The cutoff may depend on that fixed readout's first action; the absolute bound and original norm preceding this choice do not.

Step 2. We bound the whole native rectangle remainder. For one original order- P bank, a fixed integer $J \geq 0$ and a strict sector $\Re(qZ) \geq \delta_q |Z| > 0$, its complete realization obeys

$$(12.31) \quad \left| S(v)(Z) - \sum_{j=0}^J b_j(v) Z^{-j/P} \right| \leq C_J \|v\|_B |Z|^{-(J+1)/P}$$

on a far sector. Here b_j are the original coefficient maps of Setup 9.4. To verify this whole bound, fix a circle strictly inside the original root disk. Cauchy's formula bounds the minor Taylor remainder on the half disk by $C_J \|v\|_B |\zeta|^{J+1}$. Integrating this part against $e^{-qZ\xi} d\xi/\xi$, with $\zeta = \xi^{1/P}$, gives the right side by the gamma integral. On the remaining positive ray the original minor bound has exponential weight $e^{g\xi}$. Once $\Re(qZ) > 2g$, the integral of that entire tail is bounded by $C \|v\|_B \exp(-c_0 |Z|)$. The tails of the finitely many subtracted monomials satisfy the bound $C \|v\|_B \exp(-c_0 |Z|)$ with a fixed polynomial factor. They are absorbed in the right side of (12.31). Thus this estimate retains the whole minor outside the root disk as well as the polynomial subtraction.

Now use the original clocks $Z_e = e^{K_e(u)}$ and $a_e = k_e/P_e > 0$. Their uniform derivative limits and physical anchoring give $\Re K_e(u) = k_e \Re u + o(\Re u)$ uniformly on a fixed buffered strip. Choose that strip sufficiently narrow that every qZ_e remains in a strict right sector; all finitely many original angular margins remain positive. Put $J_e = \lfloor T/a_e \rfloor$ in the rectangle Q_T of Construction 9.15. Telescoping the product of the full native values and their truncated polynomials, then using (12.31), gives, for every fixed $T > 0$ and $0 < \epsilon < 1$,

$$(12.32) \quad |\text{Act } r(u) - A(Q_{T\epsilon})(u)| \leq C_{T,\epsilon} \|r\|_{\mathcal{P}} e^{-(1-\epsilon)T \Re u}.$$

Here A is the original pre-native actualization. For a simple tensor each term is bounded by the original coefficient norm times the product of all original bank norms. Taking the projective infimum and then completion proves the estimate for the entire input. In particular the constant is fixed before that input. The strict inequality $a_e(J_e + 1) > T$ supplies its positive remainder reserve.

For a fixed nonzero native readout with first action λ_f , choose $T \geq \lambda_f$ so that $(1-\epsilon)T > \lambda_f + 1$. The row reserve makes the first complete row of $Q_T r$ equal to that of r . It is one fixed element of the whole pre-native coefficient stage, with bounded map Q_T in the original projective norm. The prepared argument just proved applies to that complete coefficient. Estimate (12.32) is uniformly smaller on its buffered disk than $\exp(-\lambda_f \Re u + o(\Re u))$. It proves (12.29) for the full native value, including every original coefficient and minor tail. No relative bound uniform over all variable r is asserted.

Step 3. We retain derivatives and fractions in later coordinates. On a smaller disk, Cauchy's formula differentiates the holomorphic error in the first identity of (12.29); its derivative tends to zero. This proves the second identity. The full relative bound for $f(u+z)/f(u)$ keeps any fixed nonzero denominator free of zeros on those disks far enough out. Dividing the two laws proves the fraction law, with the difference of their actions. Whole numerator maps still use their supplied fixed denominators; Cauchy differentiation on a strict inner buffer pays a fixed radius cost before the variable input.

For $u = \psi(v)$ substitute the uniformly small displacement $\psi(v+z) - \psi(v)$ into the own-parent law. Its exact exponential tends to one, as does its complete relative error. The original whole stage bound pulls back on the actual domain; no normalization by a variable input is used for this bound. For finitely many fixed denominators, take farther tails for successively larger bounded v disks. A sufficiently slowly widening horn lies in all the required inverse images and keeps the denominators nonzero. This is the stated horizontal inlet at the later coordinate. $\square$

Lemma 12.10. *Let Q be a constructed mixed receiver with the records of Definition 11.26. Retain a birth b , whether active or inactive, and its strong cut e_i . Fix $C, \theta > 0$ and a fixed bounded whole source F , real on the physical tail, and put*

$$(12.33) \quad M = CK_b^\theta, \quad \zeta = \theta \Xi_b + F, \quad Y = CT_b^\theta e^F = Me^\zeta > 0.$$

Assume the full original reference banks at M have already been received. In the retained parent coordinate u , with $u = P_0$ at the base, retain the exact reference formula

$$(12.34) \quad \text{Log } M = \gamma u + \ell \text{Log } u + c + \rho_0, \quad \gamma > 0, \quad \ell \in \mathbb{R},$$

where ρ_0 is a bounded whole proper strong source with its retained scalar limit. Its correct-coordinate derivatives and those of $e_i \zeta$ belong to the supplied proper strong differential field. Keep that stage's whole parent-tube maps, finite structural buffers, horizontal heads and earlier comparison records. Original prepared factors have the equation-generated point-jet records of Proposition 5.2, including their complete correlated inverse sources.

For the original-parameter assertions only, assume additionally that the original finite recipes of F and $e_i F$ retain, at every fixed finite parameter order, their independently constructed separate whole coefficient targets, normalized structural jets and protected units, with the positive paying factors, pre-fields, phase, derivative and actual-domain inlets of Definition 10.5 at the relevant folds. Complete faster coefficient targets remain separate from the newly exposed structural multipliers. All original norms, parameter correlations, factorial and recovery costs are retained, without summing the target norms over different multi-indices. This inlet concerns the existing recipes of F and $e_i F$, not the new recut output.

For every finite original list $w_\nu = c_\nu e^{-\tau_\nu Y} Y^{a_\nu}$, with $c_\nu, \tau_\nu > 0$ and $a_\nu \in \mathbb{R}$, there exists a bounded whole receiving map from its original coefficient Wiener space, with a geometric tail in total weak degree. Its strong principal is $P = M \exp(e_i \zeta)$. For every fixed $a_+ \in \mathbb{R}$, $c_+ > 0$, both P and $H = P - a_+ \text{Log } P - \log c_+$ have genuine inverse charts on far strict power horns, carrying the entire appropriate strong stage in its original norm. One may take $a_+ = a_\nu / \tau_\nu$ and $c_+ = c_\nu^{1/\tau_\nu}$ for a chosen weak variable. Independent joint comparison, all later factors, strong cuts, weighted maps and original parameter domains are retained. The non-parameter output records of Definition 11.26 hold for the stated general bounded whole F . All

original-parameter conclusions, including the parameter part of that receiving definition and its independent original-parameter positive-jet record, are conditional on the additional inlet for F and $e_i F$. This assertion supplies weak reception; a native relative map at Y still requires its own strict source-norm margin.

Proof. We construct the strong chart before adding a weak variable, check its lower-field inlet, and then receive the whole weak input together with the later original factors.

Step 1. We bound the absolute cut error. The whole weighted-cut maps of Lemma 11.24, applied to the fixed inputs e^ζ and ζ , give in a common receiver

$$(12.35) \quad B = Y - P = \mathcal{W}_{M,1}(e^\zeta), \quad L = \zeta - e_i \zeta = \mathcal{W}_{M,0}(\zeta).$$

These equalities concern actualized sources; the independent kernel identifies their representatives. Their norms are arbitrarily small after the two entire input costs and all fixed inclusions. In its first-removed-factor decomposition, $\log M$ is subprincipal at every removed height. Multiplying that factor by M therefore leaves a strictly positive action reserve. The whole scalar cut estimate gives $B \rightarrow 0$ absolutely. Since $Y > 0$, it follows that $\Im P = -\Im B \rightarrow 0$. Fix $\log P$ by continuation from the positive logarithm of Y through the segment from Y to $Y - B$ on a far tail. This is compatible with the stored logarithms up to a fixed additive multiple of $2\pi i$, which does not alter any derivative or source bound. The bounded correction in (12.34) gives $\log P/u \rightarrow \gamma$; its scalar limit is retained throughout.

Step 2. We construct the genuine principal chart. Write the actual parent curve as $u_x = x + i\eta(x)$, with $\eta, \eta' \rightarrow 0$, and put $\phi(u) = \log P(u)$ on its continued branch. The coordinate, the full ϕ and its first two derivatives belong to the already constructed strong differential field with complete heads. Since $\phi/u \rightarrow \gamma$, Lemma 8.22(2) gives $\phi'_u(u_x) \rightarrow \gamma$ in the actual parent coordinate. The derivative along the physical curve remains $d[\phi(u_x)]/dx = \phi'_u(u_x)(1 + i\eta'(x))$; its curve factor tends to one and has not been discarded.

For fixed $a > 0$ the bounded part of ϕ has its whole source bound on buffered disks of radius e^{-ax} . Cauchy's estimate bounds its second derivative by $C_a e^{2ax}$ on half-sized disks. The explicit term $\ell \log u$ satisfies that bound too; it has not been absorbed into a remainder tending to zero. Hence $\phi' \rightarrow \gamma$ uniformly on disks of radius e^{-3ax} about u_x . Put $r_x = e^{-4ax}$ and go far enough that $|\phi' - \gamma| \leq \gamma/4$ on the doubled r_x disks. For $|v - \phi(u_x)| < \gamma r_x/16$, the equation

$$(12.36) \quad u = u_x + \frac{v - \phi(u_x)}{\gamma} - \frac{\phi(u) - \phi(u_x) - \gamma(u - u_x)}{\gamma}$$

maps the closed r_x disk into its interior and contracts by at most $1/4$. Its successive substitutions converge uniformly on smaller v disks, giving a holomorphic inverse. Its displacement from u_x is at most $r_x/12$. The real center images have derivative tending to γ . Intersecting image disks therefore have centers at distance $O(r_x + r_y)$, with comparable radii, and both inverse values lie in a common buffered disk. Integration of $|\phi' - \gamma| \leq \gamma/4$ on the segment between them proves uniqueness there. The inverse branches glue on the original connected branch.

For a horn $|\Im Z| < b(\Re Z)^\nu$, $0 < \nu < 1$, choose $4a < (1 - \nu)\gamma$. Its logarithmic displacement from the matching center is $O(\exp(-(1 - \nu)\gamma x + o(x)))$, which lies inside the image disk above. The center's imaginary displacement is $o(1/|P(u_x)|)$, by the absolute bound for $\Im P$. The real center images cover a far tail, so this gives the actual inverse on the entire far horn. Pullback through this inverse uses the whole parent-tube estimate $C_a \|f\|_{\text{original}}$ for every f in the stage. The constant is chosen before f , rather than from a pointwise continuation of one element.

The equation $Z - a_+ \log Z - \log c_+ = H$ has a second inverse by contraction on $|Z - H| \leq C \log \Re H$, with fixed sufficiently large C . Its derivative in the fixed-point equation is a_+/Z . It maps a strict power horn into a slightly larger one, so composing the two inverses retains the entire source bound. Reparametrization of the physical curve by $\Re P$ gives imaginary part and

imaginary derivative tending to zero, since $dP(u_x)/dx = P\phi'(u_x)(1+i\eta')$ has argument tending to zero.

Step 3. We identify the pre-weak field on this chart. Use the projected proper strong presentation at level i , before any weak variable with principal comparable to P is adjoined. Denote its fraction field, pulled through the new inverse, by $K_P^{(0)}$. Retain the finite structural buffers of Definition 10.1, with $q = o(\Re u)$ for its specified structural list, and retain complete readout denominators by their own-driver relative laws with explicit actions. The whole original-norm tube maps remain separate from both estimates. No bound by the previous ancestry profile is asserted for arbitrary inverse denominators. The stored strong heads and adjacent depth profiles give $|\log \Theta_f| = O(A_{d+i-1}) = o(P)$ for each of its nonzero elements; at the bottom use the supplied basal profile. The preceding profiles satisfy $\log A_{d+i-1} = O(A_{d+i-2}) = o(u)$, whereas $\log P = \gamma u + o(u)$. Thus $A_{d+i-1} = \exp(o(u)) = o(P)$, also after any fixed positive reference power. Complete heads are unchanged by the actual pullback. The whole differentiated numerator maps transform by

$$\frac{d}{dP} = \frac{1}{P\phi'(u)} \frac{d}{du}.$$

The displayed Jacobian is a fixed nonzero field factor on the buffered chart. It gives one fixed denominator for each old whole numerator map, independently of its input.

We check horizontal transport on the actual inverse disks before using an output weak law. The exact inverse satisfies $|u(P+z) - u(P)| = O(|P|^{-1}) = O(\exp(-\gamma x + o(x)))$ uniformly for bounded z . On the finite proper-strong structural list this lies inside every old buffer, and Lemma 10.3 gives the required absolute $o(1)$ motion of its own-parent coordinates. Explicit large native reference heights are treated through their logarithms: $\Delta \log Z_e = o(1)$ gives relative variation $1 + o(1)$, preserving their original strict angular damping uniformly in carrier. Their absolute variation need not vanish. Full fixed readout reciprocals are not classified as subprincipal structural multipliers.

For every already constructed whole prepared or native source, apply Lemma 12.9 before this transport. Its earlier-field inlet is the retained one at that source's own parent. Its action gap, degree linkage and original whole inverse majorant are the original ones; at each native stage the receiving records supply all the clocks, roots, reserves and full banks in Setup 9.14. The buffered total-cut estimate (12.30) and the full root rectangle estimate (12.32) retain every carrier, native and coefficient tail. Hence the source has its exact own-parent law $f(u+z)/f(u) = e^{-\lambda_f z}(1+o(1))$ and its logarithmic-derivative limit on a strict smaller buffer. Under the displayed $o(1)$ displacement the exponential tends to one. It follows, for each fixed $f \in K_P^{(0)} \setminus \{0\}$, that

$$\frac{f(P+iy)}{f(P)} \rightarrow 1 \quad \text{uniformly on every fixed } |y| \leq A.$$

The almost-real physical curve can be replaced by the real P axis within these same charts. For a finite list of denominators and phases choose successively farther tails for $A = 1, 2, \dots$. A sufficiently slowly increasing width then gives a common widening zero-free horn. The whole stage bound already holds on the larger power horn and does not depend on this list's variable source inputs. This supplies the inlet of Theorem 8.25, as well as every stage and derivative condition in Setup 8.21.

Before the class of P , reinsert the already constructed strictly slower weak classes in increasing growth order. For a principal $X = o(P)$ retain its original parent V and exact logarithmic formula $\log X = \gamma_X V + \ell_X \log V + c_X + \rho_X(V)$. The completed current- u differential field contains V/u and its required derivatives before this transport. Positivity gives $V = O(u)$, so Lemma 8.22(2) gives $V/u \rightarrow L$ and $V'_u \rightarrow L$. At the old V driver, boundedness of ρ_X and that lemma give $d\rho_X/dV \rightarrow 0$. Thus $(\log X)'_u$ is bounded. Lemma 10.4, using completed lower native levels and the identity chart when $V = u$, supplies the whole current- u tube hypotheses.

With the genuine P inverse already constructed, Lemma 10.3 makes the estimate uniform on its complex disks. Therefore

$$(12.37) \quad \frac{dX}{dP} = \frac{X (\log X)'_u}{P \phi'(u)}, \quad \sup_{|z| \leq R} |X(P+z) - X(P)| = O(X/P) \rightarrow 0.$$

For a normalized or prepared driver first use its retained principal chart and full logarithmic correction. Only after this actual complex-disk motion is proved does the old own- X law transfer. The complete old stage pulls back on the intersection of its original domain and the new chart, with its original norm and whole tails. Its heads have logarithmic size $O(X) = o(P)$; its numerator maps use the exact chain rule. These whole bounds and heads precede the application of Lemma 10.2 to new finite subprincipal fractions at P . A slow widening horn for the finite stage and denominator list then supplies the enlarged K_P . Comparable and faster weak classes have not supplied this field.

Step 4. We establish joint comparison and control the original weak tail. For a comparable old principal X , the adjacent ancestry gaps force its level to be i , so its stored expression uses the cut e_i . Before any new P -weak, the hypothesis that the full bank at M has already been received supplies $t_M = M^{-1/P_e}$ and hence $M = t_M^{-P_e}$ in $\text{Frac}(E_i)$. The bounded proper strong source $e_i\zeta$ has its full positive and negative exponential series in a later E_i stage. Thus $P = M \exp(e_i\zeta)$ and P^{-1} belong to that old fraction field. The old expression $X = K_X \exp(e_i\Xi_X)$ retains its exact reference root and complete projected correction there too. A finite common strong stage contains both clocks and $r = P - \lambda X$, with every fixed correct-coordinate derivative, before their weak folds. This proves membership; the scalar bound $r = o(P)$ alone would not do so. Its complete nonzero head keeps $O(A_{d+i-1}) = o(P)$, not a universal $O(u)$ bound. If $r = 0$, omit it from nonzero denominator lists. Retain the entire residual; its imaginary part tends to zero absolutely. The complete heads of its correct- X derivatives and the integration argument above give $d(P - \lambda X)/dX \rightarrow 0$. The common charts just constructed retain this phase and its derivatives. Apply Theorems 8.23 and 8.25 in this pre-weak field to the finite old and new weak list, with phases $a_\nu \log P + \log c_\nu$ and the retained comparable differences. Their full witness lattice collects each finite numerical-action fiber before any original infinite input is evaluated. Noncomparable classes keep their established order. The finite list of positive real action generators is locally finite, without a rationality assumption.

Keep every prepared distribution on its original carrier with its positive gap and linkage. Its finite generated action support and that of its correlated inverse are the ones in Proposition 5.2. An affine change scales these positive generators; after a sufficiently small residual shift, removing one occupied generator expresses the shifted support using a finite positive list again. The whole source and the new independent rows therefore give precisely the inputs of Lemma 11.23 for ordered replay of every later native and weak factor. Its cofinal estimate is used on each pre-native stage before the faster factors are adjoined. After each newly enlarged prepared or full native stage, apply Lemma 12.9 on that stage's own buffered domain before transporting it to a later coordinate. The appropriate fixed rectangle is chosen beyond the fixed readout's first action; its original-norm remainder was bounded before that readout. Actual inverse-coordinate transport uses the small earlier-coordinate motion and the fixed denominator buffers just checked. Thus the new source's horizontal inlet is proved before the next fold. The new weak variables supplied none of their own lower-field data.

Finally the exact identity is

$$(12.38) \quad w_\nu = [c_\nu e^{-\tau_\nu P} P^{a_\nu}] \exp(-\tau_\nu B + a_\nu L).$$

Fix the complete correction-unit norms and all old coefficient and replay costs below provisional upper receiving radii. Choose the fresh principal weak radii so that the norm of each corrected variable is at most ϑR_ν , $0 < \vartheta < 1$, where R_ν is its original radius. For the entire original

coefficient input $f = \sum_{\mu} f_{\mu} w^{\mu}$ this gives

$$\|Tf\| \leq L_* \sum_{\mu} \|f_{\mu}\|_{\text{original}} R^{\mu}, \quad \left\| T \sum_{|\mu| \geq N} f_{\mu} w^{\mu} \right\| \leq L_* \vartheta^N \sum_{\mu} \|f_{\mu}\|_{\text{original}} R^{\mu}.$$

The bounds precede all weak, carrier and parameter degrees. The independent target kernel is already constructed, so the raw map descends by original quotient infima. Synchronous masks erase the new weak factors at their own strong cut and project their whole phases and coefficients at higher cuts. The first-removed-factor decomposition and weighted maps of Lemma 11.24 consequently apply on the replayed source, with the costs fixed above. Every family uses the retained connected component of the literal original-domain pullbacks and the retained strict margins. The argument depends on the stored birth, not on whether it is the current birth.

Step 5. We retain the original-parameter positive-jet record. Assume the additional original-parameter inlet for both F and $e_i F$ in Lemma 12.10. Fix a finite original order k before the carried input. Put $J_M = \log M$, $\sigma = e_i \zeta$ and $\phi = J_M + \sigma$. The exact normalized formulas are

$$(12.39) \quad D_{\mu} J_M = D_{\mu} \log C + \theta D_{\mu} \log K_b, \quad D_{\mu} \zeta = \theta D_{\mu} \Xi_b + D_{\mu} F, \quad D_{\mu} \phi = D_{\mu} J_M + e_i D_{\mu} \zeta.$$

Cuts use the original fixed labels and commute with these derivatives. Separate every complete old coefficient target, including later ones, from newly exposed normalized structural multipliers. The latter are finite sums and products of the old receiver's input record, the additionally assumed existing-recipe data for both F and $e_i F$ in Lemma 12.10, and protected unit jets. These three inputs give (10.5) at u . The assertion for arbitrary bounded whole F alone does not supply these parameter data. A nonzero fixed leading constant is separated, and each residual phase retains its old field membership, correct derivatives and vanishing imaginary part. No u -weak source provides its own pre-field.

The new inverse factors are only $P\phi'(u)$ and $1 - a_+/P$, in addition to old fixed denominators. The exponential makes P nonzero; $\phi' \rightarrow \gamma > 0$ uniformly on its inverse disks and $|a_+/P| < 1/2$ protect the remaining factors. Higher fixed chart jets use finite chain and quotient rules on these actual buffers. Whole original implicit recipes retain their contraction inverse and (10.6). Their coefficient recursion has no new unprotected inverse.

For original native factors at M , positive Euler terms first undergo whole scalar-zero division. Their exact roots and relative units remain attached to the entire divided bank. Each new normalized structural polynomial is subprincipal at the root's own parent, preserving its fixed positive action. Differentiate B and L through (11.62), including all coefficient and implicit difference maps. Fixed powers of M appearing in a derivative are paid by the existing weighted-cut maps at the removed height X , where $\log M = o(X)$. They are not subprincipal at u by assertion. The complete faster coefficients of B, L and $\exp(-\tau_{\nu} B + a_{\nu} L)$ keep their original domains and later folds, outside K_P until their actual reception.

For P and H use the exact Bell formulas $D_{\mu} P = P \mathcal{B}_{\mu}((D_{\nu} \phi)_{\nu})$ and $D_{\mu} H = D_{\mu} P - a_+ D_{\mu} \phi - D_{\mu} \log c_+$. The new structural cost is $O(u) + o(u) = o(H)$ at the weak driver. One occupied weak factor pays these jets, and (10.8) pays the entire remaining degree sum. Prepared derivatives during replay insert only fixed carrier moments and polynomial lowerings, using spare original translation weight and a strict part of the carrier gap. The full ordered inverse and its proper-subtree moments are retained before any residual shift. All old derivative and observation targets remain whole in their separately bounded spaces.

Thus, for this finite order, every additional marked factor is admitted before its fold with its whole phase and domain inlet. The fixed cofinal costs remain $\exp(C_k T \log(eT))$ and every Euler, division, Cauchy and old-inclusion cost precedes fresh radii. Apply Lemma 10.26 to these exact formulas and the actual strict domain intersection. It constructs the whole local scaled families and their independent original coefficient maps, retaining $r_t^{-|\mu|}$ exactly and not summing distinct target norms. The physical identity and old input squares remain unchanged. This proves the

complete output record of Definition 11.26, including Definitions 10.1 and 10.5, for recut and later replay. $\square$

Lemma 12.11. *Let Φ_0 be a source-equipped reference normalizer of a fixed prepared equation of order p and residue λ , and let Φ_σ be a compatible analytic label of Definition 6.2. Both satisfy $X\Phi_j = (1 + \lambda x^p)\Phi_j$ and tend uniformly to the identity on every retained smaller fixed fiber disk. Retain their genuine fiber inverses, strict composition domains and a connected matched overlap containing the physical tail, with a common continued root/logarithm lift. Fix an actual passage between common inner sections on that component, including its endpoint limit. No native source is required for Φ_σ . Write the incoming graph as $z = g(x)$, where $g(0) = g_0 \neq 0$, and the outgoing section as $x = x_w$, $z = z_* + b_w t$, where $b_w \neq 0$ is its fixed coordinate scale and z_* is the actual limiting endpoint. Put*

$$(12.40) \quad E(x) = \exp(\lambda \operatorname{Log} x - 1/(px^p)), \quad d = b_w \partial_z \Phi_0(x_w, z_*), \quad \alpha(x) = \frac{E(x_w)}{dE(x)}.$$

Define the endpoint germs by $h_j(t) = \Phi_j(x_w, z_* + b_w t)/d$ and $k_j(w) = (\Psi_j(x_w, dw) - z_*)/b_w$. Then $d \neq 0$ is also the endpoint derivative for Φ_σ , and these are inverse germs tangent to the identity. Set $C_\sigma = h_\sigma \circ k_0$. On the common strict fiber composition domains,

$$(12.41) \quad \alpha(x)\Phi_\sigma(x, z) = C_\sigma(\alpha(x)\Phi_0(x, z)).$$

On the physical tail $\alpha = V_{\text{phys}}/g_0$, where $V_{\text{phys}} = ce^{-Z}Z^{\lambda/p}$ is the weak value of the original core. In particular this identity holds for all fiber variables in those domains, before graph substitution.

Proof. We determine the whole fiber transition from its differential equation and then identify its endpoint normalization.

Step 1. We determine the whole fiber transition from its differential equation. On the retained base lift, $E'/E = (1 + \lambda x^p)/x^{p+1}$. Thus (6.27) implies that Φ_j/E is constant along each original leaf segment. The transition $g_\sigma(x, w) = \Phi_\sigma(x, \Psi_0(x, w))$ satisfies

$$x^{p+1} \partial_x g_\sigma + (1 + \lambda x^p) w \partial_w g_\sigma = (1 + \lambda x^p) g_\sigma.$$

For the reference pair (Φ_0, Ψ_0) , the inverse identities and identity limits are supplied by Proposition 6.7. For the general pair $(\Phi_\sigma, \Psi_\sigma)$, the fixed fiber domains, genuine inverse identities and uniform identity limits are supplied by Definition 6.2 and the hypotheses of Lemma 12.11. Restricting these finitely many retained domains gives a fixed smaller w disk on a far part of this overlap. Expand there as $g_\sigma = \sum_{j \geq 0} a_j(x) w^j$. Coefficient differentiation gives

$$x^{p+1} a'_j(x) = (1 - j)(1 + \lambda x^p) a_j(x), \quad a_j(x) = c_j E(x)^{1-j},$$

with constants c_j on the connected overlap. On the physical lift $-1/(px^p) = Z \rightarrow +\infty$, so $|E(x)| \rightarrow \infty$. Uniform convergence $g_\sigma - w \rightarrow 0$ on the fixed smaller disk, and the Cauchy estimate for its linear coefficient, give $c_0 = 0$ and $c_1 = 1$. At any fixed base point the coefficient series defines a convergent germ $h_*(v) = \sum_j c_j v^j$. Consequently $g_\sigma(x, w) = E(x) h_*(w/E(x))$ wherever this expression and the transition are both defined. Connected analytic continuation gives the identity $g_\sigma(x, w) = E(x) h_*(w/E(x))$ on the specified matched component. This argument does not require a uniform saturated first-integral disk on the whole overlap.

Step 2. We identify the outgoing normalization and its genuine inverse. For the outgoing normalization, constancy of Φ_j/E along the actual passage gives

$$\Phi_j(x_w, z_* + b_w P_0(s)) = \frac{E(x_w)}{E(x(s))} \Phi_j(x(s), g(x(s))).$$

The right side tends to zero; endpoint continuity implies $\Phi_j(x_w, z_*) = 0$. The true inverse gives $d \neq 0$. Since $c_1 = 1$, differentiating the transition at zero shows $b_w \partial_z \Phi_\sigma(x_w, z_*) = d$ as well. The germs

$$\hat{h}_j(t) = \frac{\Phi_j(x_w, z_* + b_w t)}{d}, \quad \hat{k}_j(v) = \frac{\Psi_j(x_w, dv) - z_*}{b_w}$$

are therefore mutual inverses tangent to the identity. The preceding passage identity says $\hat{h}_j(P_0(s)) = \alpha(x(s))g_0 G_j(s)$. Here $\alpha(x(s))g_0 = ce^{-Z}Z^{\lambda/p}$ for a fixed nonzero constant c , independently of j , because E, d, g_0 are fixed. The germs just constructed are exactly the h_j, k_j in the statement. Since $G_j \rightarrow 1$ and $k'_j(0) = 1$, the actual positive passage divided by $e^{-Z}Z^{\lambda/p}$ tends to c . Thus $c > 0$ and it is the unchanged physical leading constant. This proves $\alpha = V_{\text{phys}}/g_0$ directly from the actual passage, without a native-source hypothesis for the other label.

Step 3. We establish matching on the common fiber disks. At x_w we obtain $g_\sigma(x_w, dv) = dC_\sigma(v)$. Substitution in the formula with h_* determines its germ and yields $g_\sigma(x, w) = \alpha(x)^{-1}C_\sigma(\alpha(x)w)$. Putting $w = \Phi_0(x, z)$ proves (12.41) on the actual common fiber disks. Every continuation and composition uses their original connected matched component. $\square$

Lemma 12.12. *Let h_0, k_0 and h_σ, k_σ be pairs of inverse analytic germs fixing zero and tangent to the identity. Put $C_\sigma = h_\sigma \circ k_0$ and define the removable germs $H_\sigma = C_\sigma / \text{id}$ and $H_\sigma^- = C_\sigma^{-1} / \text{id}$. Let $\Phi_0(x, \cdot)$ have inverse $\Psi_0(x, \cdot)$ and fix $g_0 \neq 0$. On strict common composition domains, the two germs*

$$(12.42) \quad \begin{aligned} \Phi_\sigma^{[0]}(x, z) &= \Phi_0(x, z)H_\sigma(V\Phi_0(x, z)/g_0), \\ \Psi_\sigma^{[0]}(x, w) &= \Psi_0(x, wH_\sigma^-(Vw/g_0)) \end{aligned}$$

are inverse in their second variables. If $G_0 = \Phi_0(x, g(x))/g_0$, they satisfy

$$(12.43) \quad k_\sigma(VG_0H_\sigma(VG_0)) = k_0(VG_0).$$

Proof. The formulas (6.61) and (6.62) have an algebraic meaning for arbitrary mutually inverse analytic fiber maps. We verify that meaning here without imposing the source-norm hypotheses of Lemma 6.16. Set $\alpha = V/g_0$. For $\alpha \neq 0$, the first expression in (12.42) is $\alpha^{-1}C_\sigma(\alpha\Phi_0(x, z))$. Its inverse is $\Psi_0(x, \alpha^{-1}C_\sigma^{-1}(\alpha w))$, which is exactly the second expression. Both expressions extend analytically to $\alpha = 0$ with values Φ_0 and Ψ_0 . Their compositions therefore remain the identity there. Since $k_\sigma \circ C_\sigma = k_0$, substitution of VG_0 proves (12.43). $\square$

For the corresponding norm estimate, suppose the complete reference endpoint family has norm M_Φ on its original larger disk, and $H_\sigma(z) = \sum_{j \geq 0} h_j z^j$. At a receiving weak radius r_V the whole forward majorant is

$$(12.44) \quad M_\Phi \sum_{j \geq 0} |h_j| \left(\frac{r_V M_\Phi}{|g_0|} \right)^j.$$

This follows by multiplying the complete absolute series. A strict ratio to the original gauge radius bounds the complete tail after each weak-degree cutoff by a geometric factor. For the inverse one first sums the absolute series of H_σ^- and verifies that $wH_\sigma^-(Vw/g_0)$ lies in the buffered domain of Ψ_0 . The original norm of Ψ_0 then bounds its whole composition. This use of the genuine inverse germ C_σ^{-1} cannot be replaced by the reciprocal of H_σ . Lemma 6.16 states the whole forward norm and geometric tail, and bounds the inverse first on a larger weak radius satisfying (6.64). Restriction gives its complete tail by the ratio of the two weak radii. For an original coefficient Banach algebra use $\|g_0^{-1}\|$, not a pointwise nonzero scalar value. A larger holomorphic parameter domain with uniform inverse margins supplies that norm on a prescribed smaller disk by Cauchy. At $V = 0$ the reference pair is recovered. Identifying the physical recipes

with another label uses Lemma 12.11; it gives no map on an arbitrary independent native bank on another ray.

Lemma 12.13. *Let a complete original family have, on a buffered local driver disk, a realization $F(r_z\eta)$ holomorphic on a neighborhood of the closed doubled polydisk in h parameters, where $r_z > 0$ is constant on that driver disk. Suppose its norm on $|\eta_i| \leq 2$ is at most $L\|f\|_{\text{original}}$. Then its entire parameter tail of total degree at least M on $|\eta_i| \leq 1$ is bounded by*

$$(12.45) \quad 3^h L\|f\|_{\text{original}}(3/4)^M.$$

For every fixed multi-index μ its original coefficient and derivative are recovered by

$$(12.46) \quad [\lambda^\mu]F = r_z^{-|\mu|}[\eta^\mu]F(r_z\eta), \quad \partial_\lambda^\mu F(0) = \mu![\lambda^\mu]F.$$

Proof. With no parameters the family is constant and both assertions follow directly. For $h \geq 1$, apply Lemma 11.17 to scalar multiples of the given whole family on the supplied driver disk. Its norm constant for this scalar input is $L\|f\|_{\text{original}}$. Take its total degree $n = M$, receiving parameter $z = \eta$ and original scale $r_t = r_z$. Its whole-tail estimate gives (12.45), and its two exact coefficient identities give (12.46). The factor $r_z^{-|\mu|}$ and the derivative factorial are retained for each fixed coefficient; target norms for different marked coefficients are separate estimates. $\square$

12.5. The complete model and its prepared admission.

Proposition 12.14. *Fix $q = \exp(i\theta)$, $|\theta| < \pi/2$, with $\text{Log } q = i\theta$, an even positive integer R , and $r, g > 0$, $0 < \vartheta < \min\{\pi/4, \pi/2 - |\theta|\}$. At each original variable native-bank label take a separately labelled full bank of Definition 8.1, with root order R , weight g , domain*

$$(12.47) \quad D = \{|\zeta| < r\} \cup \{\zeta \neq 0 : |\arg \zeta| < \vartheta/R\},$$

and its original norm. Fix these banks before their variable inputs; retain every prescribed correlation of original coefficients and parameters. Different occurrences remain distinct bank labels. Fix an integer $n \geq 3$ and finitely many positive scalar, weak and coefficient-family input radii inside their original analytic recipe domains. Prescribe also a finite list of new positive readout radii.

Use the model passages $C(s) = \exp(-1/s)$ and $H(s) = \exp(-1/s^2)$, with their genuine positive inverses, and put

$$(12.48) \quad H_* = \exp(1/\sigma), \quad K = \exp(H_*), \quad Z_0 = \exp K, \quad t = \exp(-Z_0) = C^{\circ 4}(\sigma).$$

Apply the original scalar initialization and the model forest bootstrap of Proposition 10.31 to these data, with its complete tree, opening, sibling and inverse inputs. Retain its intermediate whole first-return presentation Q_* at $x_1 = H(t)$ and its distinct inactive birth $x_2 = H(x_1)$. Continue the bootstrap through its distinct re-entry $x'_2 = H(x_1)$ and restored child $x_3 = H(x'_2)$. For $j = 4, \dots, n$, apply the direct-ending child construction of Proposition 10.28 to H , with the identity incoming and outgoing regular maps, and write $x_j = H(x_{j-1})$, $X_n = x_n$.

The chronological raw presentation $\mathcal{P}_{n,q,R}$ uses the original projective norm at each native tensor and the prescribed original coefficient Wiener norm at each weak or analytic series. The whole original bank, relative, inverse and receiving inputs of each stated constructor are part of this presentation. These operations construct bounded independent comparison maps on $\mathcal{P}_{n,q,R}$ and a bounded raw quotient map $A_n : \mathcal{P}_{n,q,R} \rightarrow Q_{n,q,R}$, where $Q_{n,q,R}$ is the quotient by their common kernel with the original infimum norm. That kernel equals the complete complex analytic kernel. The entire output satisfies Setup 10.6. Its active ancestry is $t, x_1, x'_2, x_3, \dots, x_n$. It retains x_2 and every earlier basal, tree, forward-tree, sibling, inverse and source factor.

At each selected structural H birth b the exact data are

$$(12.49) \quad D_b = \eta_b = \beta_b = c_b = d_b = 1, \quad m_b = 2, \quad a_b = G_b = \Xi_b = \ell_b = 0, \quad A_b = 1, \\ T_b = K_b = P_b = u_b = x_{p(b)}^{-2}.$$

These equalities specify the structural model values; all original variable banks and whole recipe families remain in the presentation. Every nonzero realized element has a positive leading scale with its complete remainder, and a real nonzero leading coefficient when its actual value is real. Every original source inclusion is bounded and injective in its original infimum norm.

The original σ, t, X_n readouts persist. For each prescribed new radius $\rho > 0$ there exists a bounded algebra map $E_{,\rho} : \mathcal{W}_\rho(\mathbb{C}) \rightarrow Q_{n,q,R}$ with actual value $f(X_n)$ and constants $A_\rho < \infty$, $0 < \eta_\rho < 1$ such that*

$$(12.50) \quad \|E_{*,\rho}f\| \leq A_\rho \|f\|_\rho, \quad \|E_{*,\rho}f_{\geq M}\| \leq A_\rho \eta_\rho^M \|f\|_\rho \quad (M \geq 0).$$

These constants precede f and M ; the receiver may depend on the finite list of new radii. All original input norms and radii stay fixed. The strong cuts, weighted complement maps, parent charts, whole parameter families and separate original coefficient targets of the retained-forest setup persist on their prescribed domains. The map $\sigma \mapsto X_n(\sigma)$ has a positive cofinal analytic inverse germ. This construction executes no letter of an unrelated physical itinerary.

Proof. We instantiate the bootstrap on the prescribed full banks, continue only at its direct-ending outputs, and obtain the input readout from the last original weak factor.

Step 1. We retain original banks and model incidence. Lemma 12.1 constructs the two passages and their specified inverses from the displayed polynomial fields. Their normalizing and inverse corrections are zero, and the incoming and outgoing coordinates in that lemma give unit leading constants and zero residues. These actual identities hold on the prescribed q : the normalizing map and its fiber inverse are the identity, so their constant sources synthesize those maps on every allowed receiving domain. This verifies the model's local incidence and matching without a change of ray.

The domain D is radial, contains an origin disk and the positive ray, and has a strict angular reserve for q . Thus Definition 8.1 and Lemma 8.2 apply to each full bank in the statement. A new copy is supplied at every occurrence required by the bootstrap and each subsequent child. We do not obtain these copies from the image of an order-two refinement. In particular, the element

$$v_R = \left(0, \frac{q^{1/R}\zeta}{\Gamma(1/R)}\right)$$

belongs to each suitable original bank and synthesizes to $Z^{-1/R}$ on the specified lift. The synthesis integral is the gamma integral after multiplication by $q^{1/R}$, with the compatible logarithm of q . The weighted norm is finite at zero and at infinity. At native degree k its independent coefficient is $\Gamma(k/R)q^{-k/R}[\zeta^k]\phi$. Hence the full native coefficient and exact root data used by each constructor are the ones on the chosen R and q .

Step 2. We construct the depth-three input and its common return. For the original scalar inlet apply Proposition 9.27 with the identity strong word, its identity endpoint observation and its separately labelled identity reciprocal mark. Its model equation is the order-one model in Lemma 12.1. Choose its finite original scalar and parameter radii first and keep the full family $Z(\sigma, X) = \sigma^{-1}(1 + \sigma^3 X)^{-1}$ on the fixed strict X disk. The initializer supplies its bounded original Wiener map, independent rows and whole degree tail. At $X = 0$ it gives $C(\sigma) = e^{-1/\sigma}$ and the literal mark height $H_* = e^{1/\sigma}$. This evaluation is performed after constructing the entire family; the endpoint and reciprocal-mark inputs remain separately labelled. These are exactly the original scalar factors of Construction 10.29. Apply Proposition 10.31 with the banks just specified, zero structural corrections and identity regular connectors. Its whole original realization starts at the identity strong word, and its successive actual clocks are H_* , K and Z_0 . The tree is rooted at K ; its selected opening is $\exp(-K)$ and its retained child clock is Z_0 . Thus its final weak value is $t = \exp(-Z_0)$.

The bootstrap retains both forward clocks t^{-1} and $C(t)^{-1}$, and the separate doubled-order sibling t^{-2} with the original coefficient recipe of the first-stage input. Its original reverse C^{-1} followed by H gives $x_1 = H(t)$. Its next H birth gives $x_2 = H(x_1)$, and its genuine inverse returns to x_1 . Denote this entire intermediate output by Q_* , with its raw presentation and quotient map as supplied by the bootstrap. It retains all sources of x_2 and the whole inverse family. Grading is performed on this particular Q_* . The recorded-edge re-entry creates the distinct x'_2 , after which the restored child gives x_3 . The empty further suffix therefore supplies the depth-three state of Setup 10.6. All of these assertions use the whole-source conclusion of the bootstrap, including its native reconstruction, weak-fold and parent-chart inputs. The equalities of the scalar model clocks only verify its specified structural data.

Step 3. We establish continuation and independent comparison. At x_3 the current ends at an actual direct birth, and its next incoming regular map is the identity. These are precisely the additional inlet conditions of Proposition 10.28. Apply it with the order-two model and its next full root- R bank. Its entire output again satisfies Set-up 10.6 and ends at the new direct birth. This application of Proposition 10.28 can therefore be repeated $n-3$ times. No arbitrary-current real extension is used in this induction.

The current and every stored structural record follow by substitution in the direct birth formula. At the first birth $K_1 = \exp(2Z_0)$; at each later parent $\text{Log } K_b = 2u_{p(b)}$. Since the incoming map is the identity, the input exponent is one, and $T_b = x_{p(b)}^{-2}$. The zero residue, zero normalizing correction and identity outgoing map give $P_b = u_b = T_b$ and all the remaining displayed constants. The parent of x'_2 is x_1 , and the parent of x_3 is x'_2 . The inactive x_2 has not been removed.

At each native step retain the full projective tensor over the already constructed lower source, and at each weak step retain its original coefficient Wiener space. The bootstrap and child theorem construct their independent comparison on this whole input, before any descent through old relations. Their bounded original source maps compose: a finite tensor step multiplies the finitely many factor bounds, and a coefficient step pays its whole coefficient bound once. The final raw quotient map A_n is therefore bounded by the finite product of the constructor bounds, independently of the variable inputs and their degrees.

The final constructor gives equality of its independent comparison kernel with its entire complex analytic kernel. The common kernel is closed, since each row is bounded, and the original raw quotient is complete. Its algebra structure, the original-norm inclusions and the complete leading scale are the conclusions of the bootstrap and child constructors on the verified inlets. They concern the whole completed source, including all native and weak tails. Real realizations have real leading coefficients by that comparison.

Step 4. We identify the original input readout. The original scalar monomial gives σ , and the retained tree weak factor gives t . At the last structural birth the original weak argument is exactly

$$\exp(-T_n) = H(x_{p(n)}) = X_n.$$

For each of the finitely many prescribed radii ρ , include $\mathcal{W}_\rho(\mathbb{C})$ as an original coefficient input at this last weak factor, with units in the other original slots. These units have norm one. The weak receiving conclusion of the bootstrap or last child, according as $n = 3$ or $n > 3$, chooses its strict receiving ratio after the whole coefficient costs have been fixed. For the finite list the restrictions can be made jointly. It supplies a bounded coefficient-stage inclusion and a whole weak argument of norm $r_\rho < \rho$ on that receiver.

Apply Lemma 12.6 to the complete original coefficient series at that factor. With its original coefficient bound and the finite subsequent raw-map costs included in A_ρ , it gives both asserted inequalities, with $\eta_\rho = r_\rho/\rho \in (0, 1)$. Thus the tail estimate is obtained before quotienting, for the entire original series; taking quotient infima can only decrease its bound. The algebra

property and the actual identity $E_{*,\rho}f = f(X_n)$ follow from the original weak substitution and its independent target kernel. The input norm is still $\sum_k |f_k|\rho^k$.

All domain, cut and parameter conclusions pass through the finite list of constructors defining these maps. For a new composition use its strict connected pullback component containing the positive physical tail; the old family and its original domain are retained in parallel. Each fixed original coefficient uses its own previously supplied bounded marked target. The finite products of its recovery and derivative costs do not involve a sum over different targets. Finally Lemma 12.1 gives the positive cofinal inverse of X_n . This completes the auxiliary construction before any physical itinerary letter is used. $\square$

Proposition 12.15. *Take the entire original construction of Proposition 12.14, including its intermediate first-return presentation Q_* , all original recipe families and the prescribed finite list of new readout radii. At that same Q_* insert the auxiliary sequence*

$$(12.51) \quad I_1; H_{\text{aux}}; H_{\text{aux}}^{-1}; I_2.$$

Here I_1, I_2 are actual identity passages of the linear saddle in positive section coordinates. The map $H_{\text{aux}}(s) = \exp(-1/s^2)$ is a new model occurrence with its genuine positive inverse and separately labelled whole banks of the originally chosen q, R . Keep its complete inverse family, although its auxiliary inverse parameter is zero on the actual model. The two prepared corrections and their correlated inverse corrections are zero.

The prepared entry constructors, with the reverse already executed in this sequence and with empty subsequent return suffix, construct an entire prepared receiver at x_1 . Replay the original x'_2 by the recorded-edge direct operation, and replay $x_3, \dots, x_n$ by the real new-stem construction of Theorem 11.2. This gives an entire reusable prepared receiver $\tilde{Q}_n$, constructed with its independent comparison before quotient descent, and bounded maps

$$(12.52) \quad B_n : \mathcal{P}_{n,q,R} \rightarrow \tilde{Q}_n, \quad I_n : Q_{n,q,R} \rightarrow \tilde{Q}_n$$

such that I_n is an injective unital algebra map and

$$(12.53) \quad B_n = I_n A_n, \quad \widetilde{\text{Act}} B_n = \text{Act } A_n, \quad \|B_n r\| \leq L_n \|r\|_{\mathcal{P}_{n,q,R}}, \quad \|I_n f\| \leq L_n \|f\|_{Q_{n,q,R}}.$$

The single finite constant L_n precedes every original variable input, native, relative, weak and parameter degree. The equality of actualizations is equality of the whole original germ on its connected physical pullback component.

The active current is the original X_n , with its original ancestry. All original inactive births, the additional auxiliary birth, all their full banks and inverse families, and every original observation and complete complement remain. For each original strong cut $e_a = i_a \rho_a$ there exist corresponding target cuts $\tilde{e}_a = \tilde{i}_a \tilde{\rho}_a$ and bounded strong inclusions $I_{n,a}$ with

$$(12.54) \quad \tilde{e}_a I_n = I_n e_a, \quad \tilde{\rho}_a I_n = I_n \rho_a, \quad \tilde{i}_a I_{n,a} = I_n i_a.$$

All selected-height weighted complements retain their own cuts and bounded maps from their original inputs. The readouts $I_n E_{*,\rho}$ have the bounds of Proposition 12.14 multiplied by L_n , with unchanged full-degree factors. Whole original families retain their original domains and norms; each fixed original coefficient and derivative has its separate bounded target with its original recovery factors. The entire output, including these original and auxiliary inputs, is an input to every subsequent eligible prepared, regular, direct or reverse operation of Theorem 11.2.

Proof. We construct the prepared receiver from the common first-return input, replay the finite original raw recipe, and descend only after checking its whole actual square in the independently compared target. We then verify the strong-cut and family squares.

Step 1. We construct the literal prepared receiver. Take exactly the Q_* retained in Proposition 12.14, before its grading operation. Its particular first-return input is Construction 10.32:

with $u = T_1$ and $N = e^{2u} = T_2$, the whole inverse family is

$$V(\zeta) = 1 + \zeta/N, \quad s(\zeta) = e^{-u}V(\zeta)^{-1/2}.$$

Every original inverse-reference bank is received at $N + \zeta$ on this same full family, before $\zeta = 0$ gives x_1 . Thus the input includes the original x_2 source and its correlated inverse family. Lemma 12.2 constructs the two identity prepared passages and the positive inverse of H_{aux} . Their composition is the identity in the old input, and in every old parameter, on their literal overlap.

Apply Corollary 11.32 on this particular whole input. Its inlet and order are verified as follows. The entry is Proposition 11.30, followed by Proposition 11.31. The first connector receives the entire Q_* and I_1 . Its next direct constructor receives H_{aux} with its full separately labelled bank; the alternating construction performs its genuine reverse and ends at x_1 . Its parent handoff receives this terminating reverse prefix without another reverse operation. The return constructor has empty further suffix. The final prepared occurrence is I_2 . Its zero forward correction and correlated zero inverse correction leave the actual current unchanged, while giving the entire prepared output required by the real continuation theorem. Denote this output by Q_*^p .

The original outgoing edge from x_1 to the inactive x_2 is still recorded. Proposition 11.29 replays x'_2 by its recorded-edge direct operation. It creates a distinct birth at the original stem at x_1 . From that point use Proposition 11.33 for $x_3, \dots, x_n$. These are allowed at their nonbase currents and retain Definition 11.26. They include the original finite native lists and the additional receiving copies needed by the original model recipe. The resulting target has been constructed, with its independent comparison, before a map from the completed original $Q_{n,q,R}$ is asserted. All the original and supplemental inactive sources are part of this target.

Step 2. We retain original raw factors and their arguments. We define the map from $\mathcal{P}_{n,q,R}$ in its chronological order. The scalar, logarithmic, relative, early native and weak inputs, the tree and sibling inputs, and the first child and return inputs map by the whole inclusion from Q_* to Q_*^p and then by the constructed later inclusions. Their actual arguments remain their original ones.

At x'_2 use the original reference powers of the recorded edge, the original native minors, the complete relative and positive-power/logarithmic slots, and every coefficient weak input. The old and prepared recorded-edge formulas have identical parents, input heights and equations. Each original inverse family is replayed on its original branch with its complete original inverse equation. At each later child, the real new-stem constructor receives the corresponding original native and weak lists, analytic units and inverse families. It permits all the rational positive powers occurring in the original recipe, without altering any integer native root order.

The empty-suffix grading and forest stages have added no unrelated physical letter. Their extra raw inputs are their specified finite native receiving refinements, positive marks, divided differences and weighted complement targets. The prepared receiving construction supplies these at unchanged retained stem powers. Apply the original bounded native restriction, first-root division and relative maps to the original bank inputs. An original observation, window, moment or exterior module stays at its own labelled raw factor and codomain; it is not converted to a new algebra generator.

One change of reference needs an explicit check. Write an original preceding section, with its complete unit and correction, as

$$F(e^{-u}e^{-E}), \quad F(z) = dzA(z), \quad A(0) = 1.$$

For an original child of order M and scale β its reference and relative source are

$$(12.55) \quad \begin{aligned} N_{\text{old}} &= \beta d^{-M} e^{Mu} A(e^{-u})^{-M}, \\ \Xi_{\text{old}} &= M(E + \log A(e^{-u}) - \log A(e^{-u}e^{-E})). \end{aligned}$$

The prepared new-stem reference uses

$$N_{\text{pure}} = \beta d^{-M} e^{Mu}, \quad \Xi_{\text{pure}} = ME - M \log A(e^{-u} e^{-E}).$$

Consequently, for the whole original unit,

$$(12.56) \quad \chi = -M \log A(e^{-u}), \quad N_{\text{old}} = N_{\text{pure}} e^{\chi}, \quad \Xi_{\text{pure}} = \Xi_{\text{old}} + \chi.$$

These are identities of the full analytic recipes on their original branches, including the parameter arguments.

The removable function $\log A(z)/z$ has a bounded whole analytic coefficient source on its prescribed strict disk. Thus χ is a positive weak factor times that whole coefficient source. Choose the retained cut e_a immediately before this weak factor is adjoined. Its synchronous raw mask erases the factor, so $e_a \chi = 0$, including when its coefficient contains later factors. After fixing that complete coefficient norm and the subsequent inclusion costs, apply Lemma 11.24 with any retained level- a reference M and exponent zero. Its whole map obeys

$$\text{Act } W_{M,0} \chi = \text{Act } \chi, \quad \|W_{M,0} \chi\| \leq \varepsilon \|\chi\|_{\text{original}}.$$

Target faithfulness identifies this representative with the included χ . The finite list of such maps has one common receiver, with their desired small bounds chosen after all finite inclusion costs. For an original native factor at CN_{old}^α use $\eta = \exp(\alpha\chi) - 1$ at CN_{pure}^α . Choose a common receiver with $\|\eta\| < \rho_0$ for the finite list of original relative radii. Lemmas 12.4 and 12.5 give its full original-norm map and the bound

$$(12.57) \quad \|T_\eta b\| \leq \frac{L_i L}{1 - \|\eta\|/\rho_0} \|b\|_{\text{original}}.$$

They also give the entire relative-degree tail.

Since $N_{\text{old}} \exp(\Xi_{\text{old}}) = N_{\text{pure}} \exp(\Xi_{\text{pure}})$, every subsequent relative and inverse recipe still uses the actual height $N_{\text{old}} \exp(\Xi_{\text{old}}) = N_{\text{pure}} \exp(\Xi_{\text{pure}})$. Successive relative substitutions multiply their whole units on buffered intermediate balls. Their logarithms use the complete χ , and every positive-power/logarithmic slot retains its original product radii. The exact root reserve at this child is M/R , or $\alpha M/R$ at the displayed positive power. It is positive and has not changed the integer root order. At the structural value $A = 1$ the increment is zero; the family map just proved keeps the original variable unit. For a recipe with nonzero limiting scalar, the bounded analytic composition used for the relative and inverse recipes is centered at that scalar with its full parameter dependence and strict original buffers.

Step 3. We construct a single bounded raw map. The preceding list exhausts the original finite recipe. At an original tensor factor use the bounded lower map and the finitely many whole native maps. Their product bounds the resulting map in the original projective norm. At an original coefficient Wiener step write $F(Z) = \sum_\nu f_\nu Z^\nu$, with fixed input radii ρ . If the entire coefficient map has bound C_0 , the prepared receiver gives the required source arguments with $\|z_i\| \leq r_i < \rho_i$ after that cost is fixed. Then Lemma 12.6 gives

$$\left\| \sum_\nu I(f_\nu) z^\nu \right\| \leq C_0 \sum_\nu \|f_\nu\| r^\nu \leq C_0 \|F\|_{\mathcal{W}_\rho},$$

and its full total-degree tail gains $\max_i (r_i/\rho_i)^M$. A shared original coefficient slot is multiplied or convolved in that original slot before its single whole map is applied. Only distinct original slots are independent tensor factors.

Each original bounded analytic, inverse or receiving recipe contributes its fixed whole operator bound. In particular, an inverse uses the complete source-stage contraction and its inverse Jacobian, including the full Picard remainder, before any coefficient is extracted. Only finitely many such operations occur. Their finite products and sums give a constant L_n on finite original raw inputs, independent of their variable degrees.

Every completion in $\mathcal{P}_{n,q,R}$ is one of the prescribed native, projective or Wiener completions. The bounds just proved send its norm-Cauchy approximants to Cauchy sequences in the complete target. Their limits define a bounded linear map B_n on the entire original raw presentation, independently of the approximants.

On a finite raw input each old factor has its original argument: this holds on Q_* by its whole inclusion, on the supplement by its literal identity, at re-entry by the original recorded edge, and at later children by (12.56). The original implicit inverse equation has the original anchored positive solution on the strict overlap. These identities give $\widetilde{\text{Act}}B_n = \text{Act}A_n$ on finite raw inputs. Bounded evaluation on compact subsets of the common connected domains extends the identity to the complete inputs. For holomorphic families it is the restricted identity of the whole original family on that physical component. Windows retain their original bounded raw maps on their specified domains.

Step 4. We identify the independent kernel and original quotient. The prepared entry and each later constructor apply Lemma 11.23 to their verified inputs and the full original and supplemental list. Here the only prepared passages in the neutral supplement have $U = J = 0$, so their original point-jet distributions are zero. Their action-support and degree-linkage conditions are vacuous; one may use the positive monoid generated by 1. The other entries are the finite original root and weak lists of the model, on their retained buffered charts. Each positive mark has its fixed positive rate and its whole subprincipal phase. The constructors retain the differential numerator maps and earlier horizontal heads of these same stages. Thus the cofinal estimate (11.57) applies at each pre-native class to its entire original input, with its constants chosen before the input and cutoff. Later and faster factors are replayed in their own classes; they are not put on that pre-native horn. They have already constructed the target's independent records. Their common kernel is its complex analytic kernel. In particular this comparison did not use the completed original $Q_{n,q,R}$ as an unproved lower coefficient space at a new native fold. The actual square now gives

$$B_n r = 0 \iff A_n r = 0.$$

For the forward implication use the original seed's faithfulness; for the reverse implication use the independently established target faithfulness. Thus B_n descends to an injective map I_n . The actual values of products and units are preserved, so target faithfulness makes I_n a unital algebra map. The quotient estimate of Lemma 12.3, applied to the bounded raw map into the faithful target, gives (12.53) by the infimum over original raw representatives. No bounded choice of representatives or bounded inverse on the image is used.

Step 5. We retain cuts, weighted complements and families. Align the strong cuts by the original retained parent labels. The supplemental birth is an additional branch above x_1 ; it changes none of the original parent-depth assignments. On each original raw factor the two cut orders agree. A retained native factor keeps its full relative recipe; a later native factor keeps its scalar part. Every erased positive weak or root argument is set to zero in the entire original analytic recipe, together with its dependent coefficients and units. In particular a cut applied to (12.56) keeps the entire projected unit, not just the structural value $A = 1$.

An analytic recipe commutes with these unital source projections by its full convergent series. For an implicit inverse, projecting the entire equation and its original arguments gives the projected equation; uniqueness in the original buffered contraction ball identifies its solution. These commutation identities hold for complete native relative series by their norm convergence. Thus replay followed by cut and cut followed by replay agree on every finite raw input. Their bounded extensions agree on the completion. Descending through the independently known kernels gives the three strong-cut squares in the statement.

Retain also the original first-removed-label partition (11.62), whose finite coefficient maps are bounded on the entire original raw stage before its input or parameter degrees are selected. At a removed native factor its whole first-root division supplies the positive root and the

bounded remaining source. At an analytic factor use its full divided difference times the removed argument. At an inverse factor use the complete inverse Jacobian times the projected forcing difference. Every such factor and its original coefficient cost is among the replayed raw maps. Lemma 11.24 supplies the corresponding weighted maps on all retained stem powers. For a weight involving N_{old}^a , (12.56) gives $N_{\text{old}}^a = N_{\text{pure}}^a \exp(a\chi)$. The last exponential is a bounded whole coefficient source. Fix its bound and the old inclusion cost before choosing the prescribed small norm of the receiving N_{pure}^a -weighted map. This constructs the original weighted complement with its actual old cut. The maps are linear from the original inputs; no unbounded height multiplier is put on the old algebra.

The receiving constructors retain every old coordinate record of Definition 10.1, including the separate structural buffers, full own-driver relative laws, curved tube bounds and genuine inverse charts. They also retain the independent original-parameter positive-jet record of Definition 10.5, proved from the model formulas and propagated by the exact constructor recursions before new original coefficient targets are formed. The exact height identity in (12.56) identifies the old chart argument. Every new composition uses the strict connected pullback domain containing the physical tail and a neighborhood of each sufficiently far physical point at parameter zero. Old families remain on their original domains in parallel.

On each buffered driver disk choose the receiving parameter chart from the finite list of full source and phase Jacobians, after their original coefficient costs. Its radius is constant on that disk. The whole map is bounded on the doubled chart; hence Lemma 12.13 gives the complete parameter tail and the exact original coefficient and derivative factors. Lemma 11.34 supplies each fixed original coefficient's own marked target, including its native, inverse and exterior costs. Those target norms are not summed. Observations and windows retain their original raw definitions and codomains.

The original X_n readout is one of these complete maps. Applying I_n to its whole series and its whole tail multiplies both bounds by at most L_n . The constructed receiver has every prepared input record supplied by its final constructor, including the supplemental inactive data. It is therefore reusable before the first unrelated physical letter. $\square$

12.6. Admission of the original physical equations.

Proposition 12.16. *Fix $q = \exp(i\theta)$, $0 < \theta < \pi/2$, with its specified logarithm, before constructing the compared native banks. Let Q be an entire output of the prepared construction and its reusable operations in Theorem 11.2, whose compared native folds use that reference. At a nonbase recorded birth v , let its current be $S = Dx_v^\kappa \exp(-G)$, where $D, \kappa > 0$ are real and G is the whole generated positive-parent source. Retain every stored birth identity*

$$(12.58) \quad \begin{aligned} t_b &= D_b x_{p(b)}^{\eta_b} e^{-G_b}, & T_b &= \beta_b t_b^{-m_b} = K_b e^{\Xi_b}, \\ q_b &= c_b e^{-T_b} T_b^{a_b}, & x_b &= d_b q_b e^{-\ell_b(T_b)} A_b(q_b e^{-\ell_b(T_b)}), \end{aligned}$$

with its original positive constants, real exponent and residue, integer order m_b , analytic unit $A_b(0) = 1$, and full original source, inverse and branch data.

Fix a finite list of actual real canonical passage occurrences with their original equations, physical sides, sections, true inverse branches and compatible analytic sectorial labels of Definition 6.2 containing the physical tail. The independently compared native inputs are the source-equipped common- q references and the original compatible banks on that q ; other labels enter by their whole analytic weak recipes. Include their finitely many original inner cores and analytic recipe families, with original norms and strict equation, endpoint, graph and inverse-image domains. Repeated original data stay correlated. Structural leading constants are fixed before these families.

For either direction of any chosen next occurrence, its complete equation-derived source package is admitted on the entire Q . In the direct direction, its actual regular section maps

give $P_{\text{out}} = R_{\text{out}} \circ P_0 \circ R_{\text{in}}$, where

$$(12.59) \quad P_0(y) = k_0(ce^{-Z}Z^aG_0(Z)), \quad Z = \beta y^{-p}, \quad p \geq 1, \quad \beta, c > 0.$$

The reference normalizer, graph factor G_0 , true fiber inverse and endpoint inverse germs are those constructed from the original equation on q . After R_{in} the new-stem rule applies to the exact current, including its whole stored unit.

In the reverse direction, after R_{out}^{-1} write the current as $S_w = D_w x_b^k e^{-G_w}$, with $b = v$. If the new reverse order and scale are r, β_R , and Z_R is its whole source inverse height, put $Y_o = \kappa T_b$, $L_R = \text{Log}(Z_R/Y_o)$. Before R_{in}^{-1} the exact output is

$$(12.60) \quad \left(\frac{\beta_R}{Z_R}\right)^{1/r} = \left(\frac{\beta_R}{\kappa\beta_b}\right)^{1/r} D_b^{m_b/r} x_{p(b)}^{\eta_b m_b/r} \exp\left(-\frac{m_b}{r}G_b - \frac{L_R}{r}\right).$$

This is the genuine positive inverse branch. The departed birth and its whole sources remain; the new reverse equation need not be the equation that created that birth.

Every original recut $J_e(z) = d_e z A_e(z)$, $d_e > 0$, $A_e(0) = 1$, keeps its full clock $\beta_e J_e(z)^{-m_e}$ and original integer native root order. The full original native and coefficient weak families at those clocks have bounded source maps on a common receiver, with strict geometric relative- and weak-degree tails. All fixed coefficient and inclusion costs precede the small receiving marks. Old quotient inclusions use a single actual coordinate pullback on every old factor and coefficient.

For each original selected scalar label σ , require the core's graph constant $g_0 \neq 0$. For an original coefficient-family occurrence, require g_0 to be a unit in its coefficient Banach algebra and retain its actual inverse norm, as in Lemma 6.16 and the original family data of Definition 6.2. If the input is instead supplied as bounded holomorphic data on a larger original parameter polydisk, retain uniform nonvanishing and inverse-image margins there; Cauchy on the prescribed strictly smaller polydisk supplies the Wiener inverse norm. This is a separate analytic-input route, not an equivalence between pointwise nonvanishing and invertibility in an arbitrary Banach algebra. With these respective hypotheses, put $C_\sigma = h_\sigma \circ k_0$, $H_\sigma = C_\sigma/\text{id}$, $H_\sigma^- = C_\sigma^{-1}/\text{id}$. Its whole weak-coordinate recipes are

$$(12.61) \quad \Phi_\sigma^{[0]}(x, z) = \Phi_0(x, z)H_\sigma(V\Phi_0(x, z)/g_0), \quad \Psi_\sigma^{[0]}(x, w) = \Psi_0(x, wH_\sigma^-(Vw/g_0)).$$

They have the original physical values on the specified matched diagonal, are true fiber inverses, and preserve the whole endpoint identity and ordered matching. Independent free native banks on other original rays may retain their parallel original implementations, but are not inputs to this jointly compared receiver. No bounded map of such arbitrary banks is implied by the gauge identity.

The entire output is again a reusable prepared output, with bounded injective original-norm inclusions, independent comparison with exact analytic kernel, and the complete positive-scale head for every nonzero realized element. It retains all strong cuts, weighted maps, parent charts, inactive sources, whole prepared forward and correlated inverse distributions, original domains and separate marked-family targets. In particular its output carries Definition 11.26, with the separate coordinate and original-parameter positive-jet records of Definitions 10.1 and 10.5. No source admission for an arbitrary scalar germ or rotation of an old complete bank is used.

Proof. We first obtain a common-reference source from each original equation. We then check the current and every recut in both directions, and finally transport the selected endpoint labels and whole families.

Step 1. The reference is chosen before the sources. Apply Corollary 8.18 to the finite list of original equations and inner cores, with the q fixed in the statement. For an order- p core its root coordinate is $x = \omega \exp(-\text{Log } Z/p)$, where $\omega^p = -1/p$. The logarithms obey $\text{Log}_+(qZ) = \text{Log } q + \text{Log } Z$ and are continued together on the specified physical lift. This is the root coordinate used in its full normalizer and inverse-normalizer source. It is not chosen independently at a later occurrence.

The corollary supplies the whole coefficient families of the reference fiber map Φ_0 and its genuine inverse Ψ_0 , on strict original parameter and graph disks. For an incoming graph g with $g_0 = g(0) \neq 0$, the graph factor and its correction are the complete substitutions

$$G_0(Z) = \frac{\Phi_0(x, g(x))}{g_0}, \quad \ell_0 = -\text{Log } G_0.$$

The constant term is one. The graph disk and the disk for the logarithm are restricted before the source weights are selected. The complete graph coefficient map has a fixed original-norm bound; the scalar-zero part is then received within the chosen logarithm disk. Thus the logarithm is a bounded whole source operation, including its entire degree tail. This order of endpoint, logarithm and weight choices applies to Ψ_0 , the endpoint inverse, and all prescribed analytic recipe families.

The actual endpoint germs and section maps furnished by Proposition 6.13 give $P_{\text{out}} = R_{\text{out}} \circ P_0 \circ R_{\text{in}}$ on their connected physical composition domain. Their positive interval inverses specify the reverse branch. All constants in this factorization are those of the original equation and sections. In particular the complex reference normalizer is not assumed to be a convergent real normalizer; its matched whole expression has the required real physical value.

If a later fold uses a common multiple of the finitely many native root orders, Lemma 8.3 gives the bounded map from each original bank, with its original norm. Its constant depends on the two orders. A new full bank at the common order is supplied separately whenever the constructor requires it. No surjectivity of a root-refinement map is used.

Step 2. We construct the direct and reverse currents. Write $R_{\text{in}}(z) = d_{\text{in}} z A_{\text{in}}(z)$. Its exact regular update replaces (D, G) by $(d_{\text{in}} D, G - \log A_{\text{in}}(S))$ and leaves κ unchanged. Inserting the retained whole identity $x_v = d_v e^{-u_v} W_v$ then gives (12.26). The nonbase condition and the retained positive-parent certificate are the inlet conditions of Proposition 11.33, with the records of Definition 11.26. Its root of original order P has exponential reserve $\kappa p/P > 0$ at u_v . Neither this reserve nor that operation requires κ to be rational. Apply the operation with the whole equation-derived package just obtained. Its actual output is $P_0(R_{\text{in}}(S))$; the final regular update by R_{out} completes the direct passage.

For a reverse, first apply R_{out}^{-1} on its specified positive domain. Use the stored birth $b = v$ and retain all the terms of its identity, including A_b and ℓ_b . In the notation of Lemma 12.8, the forcing E_o includes these two entire terms, the current correction G_w , and the whole inverse endpoint unit. The reverse operation of Proposition 11.29 constructs its buffered inverse (12.23). Its unknown, operator and solution belong to the complete receiving source, before any coefficient or derivative is taken. The original inverse margin bounds the Jacobian inverse and the entire iterative remainder. In particular its value at the inverse parameter zero is the specified physical inverse height Z_R .

Lemma 12.8 now gives (12.60). It applies to the stored old birth and the supplied new reverse equation; it does not identify their orders or their endpoint maps. It also shows that κ occurs in Y_o and the leading constant, and does not multiply the returned parent exponent $\eta_b m_b/r$. Applying R_{in}^{-1} gives the final regular update. The old birth remains as an inactive labelled source with its original inverse families.

Step 3. We retain all original recut clocks. For a direct recut the exact principal and perturbation are (12.27); for a reverse recut they are (12.28). Thus in each case the clock has the form $Y_e = M_e e^{\Xi_e}$ with M_e a positive power of the new stem or of the recorded old stem. The powers are m_e/p and m_e/r , respectively. The unit A_e is included in Ξ_e as a whole analytic function. On the reverse side the full L_R is included too.

The native operations in Propositions 11.33 and 11.29 receive the prescribed full bank at M_e . It uses that bank's original integer root order and ray, and the fixed original bound for its coefficient map. The generated perturbation Ξ_e has the finite certificate of Definition 11.25:

the current certificate, whole analytic differences, native divided remainders and the inverse discrepancy are exactly the retained sources in that operation. After their finite coefficient and inclusion costs have been fixed, take a common receiver with

$$a_e = \|e^{\Xi_e} - 1\| < \rho_{0,e}$$

for the finite recut list. This is a source-norm choice provided by those certificates. Lemma 12.5 gives the whole map from the original relative bank, with bound $L_i L_e / (1 - a_e / \rho_{0,e})$. Its complete tail of relative degree at least M has the additional factor $(a_e / \rho_{0,e})^M$. Finite products have the corresponding union-tail bound before the original coefficient input is chosen.

The weak receiver requires a new principal chart at each of these exact clocks. After the native maps just constructed, we verify the additional original-parameter inlet of Lemma 12.10 for its two actual sources before using its parameter conclusion. In the direct case its birth is the newly stored one, its power is m_e/p , and its additional source is $F_e = -m_e \log A_e(y)$. (12.26) gives the logarithmic-reference slope $\gamma = \kappa m_e > 0$ and $\ell = 0$. The full correction $pG - \kappa p \log W_v$ is in Ξ , including its projected strong part.

In the reverse case the birth is the departed recorded b , the power is $\theta_e = m_e/r$, and $F_e = \theta_e L_R - m_e \log A_e(y_R)$. Both T_b and the genuine inverse height Z_R are positive on the physical interval. Thus this full F_e is real there. It is a bounded whole source by the buffered inverse and analytic estimates above; its relative-source smallness was checked separately.

Fix a finite original parameter order k , keeping all leading data and structural rates fixed. On the retained zero-free analytic disk write $\log A_e(z) = z a_e(z)$. The whole larger-disk coefficient and derivative maps of Lemma 4.3 and Proposition 4.20 apply before specialization. Every explicit original coefficient derivative of $\log A_e$ still vanishes at zero, since $A_e(0) = 1$ is fixed. For the direct argument the exact identities are

$$\log y = \log(Dd_v^\kappa) - \kappa u_v + \kappa \log W_v - G, \quad D_\alpha y = y \mathcal{B}_\alpha((D_\nu \log y)_{0 < \nu \leq \alpha}) \quad (0 < |\alpha| \leq k).$$

The finite ordinary chain and product sums for $D_\mu \log A_e(y)$ therefore retain a factor y in every term. Higher powers of y remain inside the complete analytic coefficient target on its strict disk. After separating the complete old coefficient targets, the pure normalized structural polynomials have logarithmic cost $o(u_v)$ by the input record. The exact factor $y = Dd_v^\kappa e^{-\kappa u_v} W_v^\kappa e^{-G}$ retains its positive parent action κ , its unit and its original phase and domain.

For the reverse source, (11.81) gives the full identity

$$L_R = Y_o^{-1} A_1 + (\log Y_o) Y_o^{-1} A_2,$$

with the entire correlated inverse family in A_1, A_2 . Put $J = \log Y_o$ and $q_\mu = D_\mu J$. (11.87) and (11.88) show that every finite Leibniz term of $D_\mu L_R$ retains Y_o^{-1} , a polynomial in J, q_ν , and a complete original coefficient target of A_1 or A_2 . Here $J = O(u)$ and the pure normalized q_ν products have logarithmic cost $o(u)$, so the parent action $\alpha = \eta_b m_b > 0$ of Y_o^{-1} persists. The cancellation in (11.89) accounts for the apparent large derivative of Y_o . The original recursion (10.6) for (11.78) supplies the independent whole coefficient targets with inverse multiplier at most 2. Its strict disks protect $1 + h/Y_o$ and $1 + \chi$. Moreover $\log y_R = (\log \beta_R - J - L_R)/r$. The identity $\log A_e(y_R) = y_R a_e(y_R)$ and its finite ordinary chain sums retain y_R with its already constructed action α/r . At a basal parent these calculations retain the original basal powers and whole unit. No term of L_R is discarded.

The synchronous masks of Lemma 11.24 act on these complete analytic and inverse recipes and on every original coefficient formula. They commute with fixed-label D_μ and preserve the zero-constant analytic factors and the normalized inverse ideal in (11.80). An erased positive slot gives zero ideal contribution; no reciprocal is evaluated at a fictitious zero clock. Thus $e_i F_e$ retains the separated coefficient targets and positive-mark data along with F_e . The analytic and implicit divided differences of Lemma 11.12 give (11.62) before differentiation, with full coefficient and inverse-Jacobian targets. For the amplified discrepancy, the exact finite Leibniz

sum is

$$D_\mu(M_e(1 - e_i)\exp(\Xi_e)) = \sum_{\nu \leq \mu} \binom{\mu}{\nu} (D_\nu M_e)(1 - e_i) D_{\mu-\nu} \exp(\Xi_e).$$

Every whole exponential-derivative target in this sum is retained, not just $\exp(\Xi_e)$. Differentiating the complete first-removed-factor expression keeps its coefficient targets. Any old differentiated native bank uses its whole Euler maps and Lemma 11.19, retaining the exact original first root. Extra fixed powers of M_e are paid by (11.63) at the removed height X , where $\log M_e = O(u) = o(X)$, not at u .

All finite Cauchy, factorial, Euler, division, weighted-map and inclusion costs are fixed in the original norms before fresh receiving radii. Different parameter multi-indices keep separate targets, and (10.9) retains its exact local recovery factor. These existing analytic and return recipes supply the additional inlet for F_e and $e_i F_e$ without using the new recut's parameter output. At a nonbasal old stem the retained reference is either the pure exponential of (11.110), or its original analytic-unit reference. For the analytic-unit reference

$$\log K_b = \alpha_b \gamma_b u + c_b^* + \alpha_b \log V_b(e^{-u}), \quad \alpha_b, \gamma_b > 0, \quad V_b(0) = 1.$$

This is the exact nonbasal reference record of Setup 10.6, retained through the intervening real constructions. Taking the positive power θ_e gives (12.34) with $\gamma = \theta_e \alpha_b \gamma_b > 0$, $\ell = 0$, and the full bounded analytic-unit remainder. The pure reference has zero such remainder.

If the stored parent is the base, retain instead $x_0 = C_x e^{-P_0} P_0^{a_0} W_0$ and $K_b = C_b (\beta_b^* x_0^{-r_b^*})^{\alpha_b}$. Writing the positive prefactor of M_e as C_e^* gives the exact formula

$$\log M_e = \gamma P_0 - \gamma a_0 \log P_0 + c_e^* - \gamma \log W_0, \quad \gamma = \theta_e \alpha_b r_b^* > 0.$$

Here c_e^* collects only the fixed constants. The retained branch of $\log W_0$ is a whole bounded strong source; its scalar part may be included in that constant. The basal proper strong chart, differential heads and full source-unit bounds are the original records retained by the receiving constructions. Thus the local lemma applies with $u = P_0$ and $\ell = -\gamma a_0$. In particular no basal logarithmic power has been treated as a vanishing remainder.

For both directions all parent-tube, differential and finite structural data are retained by Definition 11.26. The prepared entries are the original equation-generated ones, whose whole point-jet support is supplied by Proposition 5.2. The new lemma constructs $P_e = M_e \exp(e_i \Xi_e)$ and its genuine weak-coordinate inverse charts before the weak fold, establishes the pre-weak horizontal inlet, and jointly replays every later original factor. Its identity (12.38) receives the entire original weak list with strict geometric degree tails after all old costs. This applies at the departed birth even when it is now inactive.

Step 4. We retain selected labels and ordered matching. For each original selected endpoint label, apply Lemma 12.11 to the equation's reference normalizer, the original selected normalizer and their genuine inverses. The common connected overlap and outgoing section were retained in the physical core above. The lemma gives (12.41) for every fiber variable, with $\alpha = V_{\text{phys}}/g_0$. Inverting its actual germ C_σ and then Φ_0 identifies the original Ψ_σ on the strict inverse image domain. Lemma 12.12 therefore gives the two displayed whole recipes and proves their inverse identities also for the independent weak variable. The inverse uses C_σ^{-1}/id . Graph substitution agrees with Proposition 6.15 and gives the whole endpoint identity. The reference ray is unchanged.

The forward whole norm is bounded by (12.44). Choose its weak radius so that the argument has a strict ratio to the original gauge disk. For a full parameter Wiener family carry these formulas in its original coefficient Banach algebra, retaining the inverse norm of g_0 and the protected endpoint derivative. If bounded holomorphic data are supplied instead, use their uniform nonvanishing and inverse-image margins on a larger original polydisk; Cauchy applied to $1/g_0$ and the complete gauge and inverse families gives their Wiener norms on prescribed strictly

smaller radii, with the explicit product radius loss of Lemma 6.16. A pointwise lower bound alone is not substituted for that inverse Wiener norm. These whole costs precede the receiving weak radius. The full fiber identity holds there as a holomorphic identity, with unchanged retained endpoint and inverse-image margins. For the inverse, the complete absolute series of H_{σ}^{-} must first send the whole receiving disk into the original buffered disk of Ψ_0 ; its original norm then bounds the composition. These choices precede the weak coefficients and give geometric tails of every degree. They also apply to the supplied full analytic families, with their original parameter norms. For successive labels the actual transition maps satisfy

$$(h_{\tau} \circ k_{\sigma}) \circ (h_{\sigma} \circ k_0) = h_{\tau} \circ k_0.$$

Associativity on the strict common domains proves the ordered matching. Bounded composition of the finite whole maps proves it on the completed sources. An independently variable native bank originally selected on another ray remains a parallel original implementation, outside the tensor whose faithful common- q comparison is asserted here. Only the fixed equation-derived matched source and its whole analytic recipe family are replayed. The forward and inverse norm and geometric-tail bounds are those of Lemma 6.16. At zero transition degree they recover the reference forward and inverse sources and graph unit. The original endpoint germs remain in the whole endpoint identity; zeroing the transition variable does not identify the original and reference endpoint gauges.

Step 5. We control the entire output in its original norms. All the operations just used are the raw native, weak, analytic and inverse operations of the reusable prepared construction. Each is made on the entire preceding input. In particular each old native factor, coefficient, unit and inverse recipe is pulled back by one common actual coordinate map. On simple tensor inputs the actualization square follows from these exact substitutions. The full original coefficient estimates, the strict relative and weak tails, and the buffered inverse estimate extend it to the completion. Lemma 11.23, on the inlets of those constructors, gives independent comparison on that completion with its complex analytic kernel. Lemma 12.3 therefore gives the bounded injective old quotient inclusion with its original infimum norm. Its hypotheses are now supplied by the whole constructor, rather than by a scalar change of a quotient representative.

The constructor retains the ancestry and each inactive label. Cuts act on the original raw labels and their whole recipes; the receiving maps commute with them. Weighted complements are the original first-removed-factor maps at their stated cuts, followed by the bounded whole inclusion, so their coefficient costs are paid before their new small marks. The strong and inverse parent charts have the exact clocks just checked. Their structural buffers and full own-driver denominator laws are the separate records of Definition 10.1. The direct, reverse and recut constructors verify normalized original parameter jets from their exact logarithmic formulas before constructing new coefficient targets. Native Euler jets keep the original first root; cut-error reference powers are paid at the removed height; weak height jets are paid at their new principal driver. Their protected inverse Jacobians and full later targets remain in Definition 10.5. The analytic label recipes add only strict whole analytic substitutions and their actual compositional inverses, with the original coefficient inverse norms just fixed. Thus the adapter exports this identical receiving invariant rather than deriving parameter estimates from its coordinate chart alone. Prepared distributions, their correlated inverse and their proper subtree carriers are among the unchanged original inputs of the full construction.

For each family use the connected physical component of the literal inverse images of its original domains, with all original strict margins. Its structural leading data and positive rates are fixed as in Definition 11.26. A neighborhood of every sufficiently far physical point at the original parameter zero lies in this component. Local doubled parameter polydisks give the original coefficient recovery and the factorial bounds of Lemma 12.13; Lemma 11.34 supplies each fixed coefficient and derivative's separate target. Original windows and observations keep their own domains and complete complements. These finite pullbacks and receiving restrictions

preserve eligibility for the next literal operation. Finally the independent comparison theorem for the resulting entire prepared output gives its complete positive-scale head and remainder. This proves all assertions for both passage directions. $\square$

12.7. Finite-word assembly.

Proposition 12.17. *Let $W = L_N \circ \cdots \circ L_1$ be a fixed word of positive passage germs as in Theorem 7.5, and write $W_0 = \text{id}$, $W_j = L_j \circ \cdots \circ L_1$ for its prefixes. Prescribe any finite list of new positive scalar readout radii. Suppose the following source constructions have been supplied, with all structural constants fixed before their variable inputs.*

- (1) *For the chosen $n \geq 3$, there exists a complete original model presentation at X_n with active chain of depth n above t . It includes every earlier auxiliary ancestor, the two distinct births x_2, x'_2 , and their entire original native, weak, inverse and coefficient-family inputs. Before the first letter there exists a bounded raw map from this whole presentation to a mixed presentation, preserving all actual arguments and original norms. The target has independent comparison with actual kernel. The original X_n has a bounded source there, together with its whole Wiener readout at each prescribed new radius. These model readouts have bounds A_ρ and strict geometric whole-degree factors η_ρ as in Lemma 12.7, with their original input norms.*
- (2) *Each reached presentation retains its actual parents and a current $S = Dx_v^\kappa e^{-G}$, with $D, \kappa > 0$, the whole unit $x_v = d_v e^{-u_v} W_v$, and all original source data. These include the prepared forward bank and its correlated inverse, all active and inactive native and weak inputs, original analytic domains, branches, strong cuts, weighted complement maps, derivative targets and independent comparison maps.*

On this entire presentation, supplied constructor theorems apply to every next regular or prepared letter and to each canonical letter at a nonbase vertex. They preserve the preceding data and give an entire presentation of their output. A canonical direct letter creates a child with height (12.26); a canonical reverse constructs the full inverse (12.23) and returns by (12.25) to the recorded parent. The reverse equation may differ from the equation that created the departed birth.

- (3) *All compared native banks use a compatible reference chosen before their construction. The next original equation supplies its full local source and matching package at that reference, including every original recut and the whole selected-label forward and inverse recipes. The whole operators and their strict domains have bounds in the original input norms. The new independent comparison is constructed before quotient descent, has kernel equal to actual zero, and gives a complete positive-scale leading term for every nonzero real realization. These remain hypotheses of this conditional proposition. For the constructed receivers, the independent comparison, complex analytic kernel, complete heads and whole cofinal bound are supplied by Lemma 11.23; the original physical equations and matching data are supplied by Proposition 12.16.*
- (4) *Old source inclusions preserve actual values and are bounded and injective in the original infimum norms. In any reached presentation, each fixed finite list of current readouts admits a common further receiver with source norms strictly below any prescribed positive analytic input radii, after their coefficient and inclusion costs have been fixed. These refinements preserve the data in items (1)–(3). Whole analytic families keep their original norms and domains; each fixed original Taylor coefficient has its separately bounded target with its original recovery and derivative costs.*

Let d_j, r_j count canonical direct and reverse letters in the first j positions, and put

$$(12.62) \quad B = \max_{0 \leq j \leq N} (r_j - d_j).$$

If $n \geq B + 3$, every prefix has the stated entire presentation, with exact current $W_j(X_n)$ and active depth

$$(12.63) \quad h_j = n + d_j - r_j \geq 3.$$

All prefix readouts have a common complete original-coefficient Wiener map with a strict geometric total-degree tail. The retained original-input readouts satisfy (12.16) at every prescribed radius, with each whole inclusion cost paid once. The displacement $W(X_n) - X_n$ is either zero or has a positive leading term with complete remainder. In the original coordinate, $W - \text{id}$ has the corresponding alternative under pullback by X_n^{-1} , without a derivative factor.

Proof. We keep the whole source presentation at each prefix, use the depth deficit to check every reverse input, and compare the final output and the original input in one presentation.

Step 1. We construct the initial presentation and original inclusions. The first hypothesis supplies the entire mixed presentation before any letter of W . Its current is X_n , with record $D = \kappa = 1$, $G = 0$ at the last model birth. Its active depth is n . The original source inclusion descends by Lemma 12.3. All inactive births and their complete sources are part of this input; the scalar cancellation of an auxiliary inverse pair has removed none of them.

Step 2. We treat regular and prepared letters. Suppose the construction has reached the j th prefix. For a regular map $R(s) = dsA(s)$, its exact new record is

$$D_{\text{new}} = dD, \quad \kappa_{\text{new}} = \kappa, \quad G_{\text{new}} = G - \log A(S).$$

The prescribed full analytic substitution constructs this logarithm on a strict original disk. In particular the identity $\log A(S) = SB_A(S)$ uses the removable analytic function $B_A(z) = \log A(z)/z$ with all its coefficients.

For a prepared passage, write $D_{\text{pre}}(s) = as^\lambda \exp U(-\log s)$. Put $L = -\log S$ and define its correlated inverse correction by $(\text{id} - U/\lambda)^{-1}(w) = w + J_U(w)$. The formulas in Theorem 5.1 give

$$\begin{aligned} D_{\text{new}} &= aD^\lambda, & \kappa_{\text{new}} &= \lambda\kappa, & G_{\text{new}} &= \lambda G - U(L) \quad \text{in the forward direction,} \\ D_{\text{new}} &= (D/a)^{1/\lambda}, & \kappa_{\text{new}} &= \kappa/\lambda, & G_{\text{new}} &= G/\lambda + J_U((L + \log a)/\lambda) \quad \text{in the inverse direction.} \end{aligned}$$

The full logarithmic argument at the nonbase birth is

$$L = \kappa u_v - \log D - \kappa \log d_v + G - \kappa \log W_v.$$

The assumed constructor receives the entire translated bank at this argument. It uses the original forward-leaf convolution for J_U , with any residual translation applied to the total carrier after convolution. It retains every derivative of the total-carrier cutoff and its entire complement. An inverse letter can therefore be the first physical occurrence of that saddle: the inverse bank comes from its local forward equation, rather than an earlier executed letter. These three operations keep the vertex and its depth.

Step 3. We treat the two canonical directions. For a direct letter, first execute its actual incoming regular map. The current then has (12.26); its complete source constructor creates a child with actual parent v . The outgoing regular map completes exactly the original physical letter. For a reverse letter, execute its inverse outgoing regular map and use the full supplied inverse operator. Lemma 12.8 proves that its positive output has the record (12.25) at the stored parent. The inverse incoming regular map completes the original reverse letter. The departed birth and all its banks remain in the presentation. The original reverse order r and departed birth order m_b enter the exponent as $\eta_b m_b/r$; no equality of these orders is required.

The strict relative source maps for the original recuts use (12.27) or (12.28), with Lemma 12.5. The full endpoint inverse recipes use Lemma 12.12 on their original matching domains. Their existence as jointly compared sources, and the availability of their small receiving representatives, are exactly the constructor hypotheses of this proposition. Both directions preserve all prior

source data by those hypotheses. A direct letter increases active depth by one; a reverse decreases it by one.

Step 4. We retain the depth reserve and common readouts. Before an operation at prefix j the depth is $n + d_j - r_j$. The regular, prepared and canonical cases prove this identity at the next prefix whenever the operation is defined. The numerical bound $n + d_j - r_j \geq n - B \geq 3$ holds at every prefix. Thus a reverse always starts at a nonbase birth and lands at a retained nonbase parent, and a new direct stem has its required nonbase input. Induction proves all prefixes and (12.63). Prepared inverses have not contributed to r_j .

The exact source squares identify the new current with $W_{j+1}(X_n)$, and the original-norm inclusions carry every earlier prefix source into it. The set of arrows is finite; their whole operator bounds have a finite product. The receiving hypothesis then puts all prescribed prefix readouts in one target with norms $r_i < \rho_i$. Lemma 12.6 gives their whole coefficient map and the entire total-degree tail. Every constant precedes the variable coefficient sequence and its degrees. Original input radii remain the radii in (12.12).

For the original input X_n , the model readout is carried through the initial handoff and every subsequent whole inclusion. Lemma 12.7 gives (12.16), with the explicit cost in (12.17). Any receiving refinements are included among those whole maps. The finite list of new radii was fixed before constructing this receiver; the proof does not assert one receiving norm for all positive radii.

Every old family is carried on the literal connected component of its original pullback domain containing the physical tail. Finite intersections of the positive tail intervals remain positive tail intervals. The strict complex-domain conditions are part of the constructor hypotheses; positivity alone has not supplied them. The separate original coefficient targets, raw windows, moments and whole complements remain at their recorded domains and norms.

Step 5. We identify the displacement in its original coordinate. In the common final presentation form the difference of the whole sources for $W(X_n)$ and X_n . Independent comparison therefore applies to this difference after all cancellations. If its realization is zero, then $W(X_n(\sigma)) = X_n(\sigma)$ on a positive tail. Lemma 12.1 shows that the image is a full positive interval, so $W(s) = s$ there.

Otherwise the assumed complete comparison gives

$$W(X_n(\sigma)) - X_n(\sigma) = a\Theta_{\text{model}}(\sigma)(1 + \epsilon(\sigma)), \quad a \in \mathbb{R} \setminus \{0\}, \quad \Theta_{\text{model}} > 0, \quad \epsilon \rightarrow 0.$$

Reality follows because W and X_n are the actual real maps on the fixed physical branches. Set

$$\Theta_{\text{actual}}(s) = \Theta_{\text{model}}(X_n^{-1}(s)), \quad \epsilon_{\text{actual}}(s) = \epsilon(X_n^{-1}(s)).$$

The cofinal inverse preserves positivity and the vanishing limit of the entire remainder. Substitution gives $W(s) - s = a\Theta_{\text{actual}}(s)(1 + \epsilon_{\text{actual}}(s))$. Only the input of this displacement has changed, so this identity contains no derivative factor. It has the meaning of Definition 2.3. Finally $B \leq N$, and hence the bound $n \geq N + 3$ suffices. $\square$

Theorem 12.18. *Let $L_N \circ \dots \circ L_1$ be any literal physical word supplied by Theorem 7.5. For $n \geq N + 3$, define the positive cofinal analytic reparametrization*

$$(12.64) \quad C(t) = e^{-1/t}, \quad H(t) = e^{-1/t^2}, \quad X_n(\sigma) = H^{\circ n}(C^{\circ 4}(\sigma)).$$

There exists one entire original source presentation admitting every prefix of this word on X_n , including all inactive retained sources before its first letter. All prefix operations preserve the full mixed-source construction, the physical branches and the original analytic family norms. Independent complete comparison has kernel equal to analytic realization. The entire displacement $(L_N \circ \dots \circ L_1)(X_n) - X_n$ is either zero or has a positive leading term with complete remainder. Pullback by X_n^{-1} gives the corresponding identity or complete leading term in the original section coordinate, without a derivative factor.

Before constructing the initial receiver, prescribe any finite list of new scalar readout radii $\rho > 0$. At each prefix let Q_j denote the comparison quotient in its original infimum norm. For every prescribed ρ , the construction retains a bounded algebra readout $E_{j,\rho} : \mathcal{W}_\rho(\mathbb{C}) \rightarrow Q_j$ with actual value $f(X_n)$ and constants $L_{j,\rho} < \infty$ and $0 < \theta_{j,\rho} < 1$ such that

$$(12.65) \quad \|E_{j,\rho}f\|_{Q_j} \leq L_{j,\rho}\|f\|_\rho, \quad \|E_{j,\rho}f_{\geq M}\|_{Q_j} \leq L_{j,\rho}\theta_{j,\rho}^M\|f\|_\rho \quad (M \geq 0).$$

Here $f_{\geq M}$ is the complete part of degree at least M . The constants precede f and M . Every whole inclusion along the word carries these readouts with its bounded inclusion cost. The receivers and constants may depend on the prescribed new radii; all original source input norms and radii remain fixed.

Proof. We construct the auxiliary input and its prepared handoff before the first letter, then verify the four construction hypotheses of Proposition 12.17.

Step 1. We choose one original model before executing the word. Fix $q = \exp(i\pi/16)$ with logarithm $i\pi/16$ before constructing any of the compared native banks or reference normalizers. The word contains finitely many original canonical equations, inner cores and recuts. Corollary 8.18 gives their full equation-derived native reference sources on this q , with the original matching domains and true inverse branches. Every other compatible analytic label is retained by the whole forward, inverse and graph weak recipes of Lemma 6.16. Only common- q native inputs enter the jointly compared tensor; independently free other-ray banks are not transferred by a label identity. Choose an even common multiple R of their finitely many native root orders and the two model orders. The restrictions of the finitely many angular domains have a positive common reserve. Fix also the finite original coefficient radii and the prescribed new readout radii before the receiving stages. These choices are independent of every later variable input and degree.

Proposition 12.14 constructs the entire original depth- n model from separate full banks at this q, R . Its original current is X_n , and it retains the earlier tree, the inactive birth x_2 , the active re-entry x'_2 and their entire inputs. Proposition 12.15 constructs the prepared receiver at the intermediate Q_* and replays this whole original model into it. Its map is bounded and injective in the original infimum norm and preserves every original actual argument. The target comparison has its analytic kernel independently of that descent. The original readouts $E_{*,\rho}$ and all their whole-degree tails are carried by this map. This proves the first hypothesis of the assembly proposition.

Step 2. We establish closure at each literal current. The prepared receiver has current $Dx_v^\kappa e^{-G}$ with $D = \kappa = 1$, $G = 0$ at its last birth. Propositions 11.31, 11.29 and 11.33 return entire outputs with Definition 11.26. Its prepared forward and inverse operations receive the whole source of Theorem 5.1, including the correlated inverse on its original carriers. To check the action hypothesis in Lemma 11.23, apply Proposition 5.2 separately to each original equation of exponent $\lambda > 0$. Its forward and correlated inverse distributions have the complete point-jet support $\{M + \lambda N : M, N \in \mathbb{N}, M + N > 0\}$, with the original gap and degree linkage. They are the equation-generated distributions used in this word, with repeated occurrences still correlated.

For completeness, this property survives the actual receiving operations. A positive affine slope b replaces the two positive action generators by b and $b\lambda$. For a residual shift ϵ choose $0 < \epsilon < b \min\{1, \lambda\}$, as allowed by the strict gap in the translated residual map. Each shifted action $b(M + \lambda N) - \epsilon$ is the sum of one of $b - \epsilon, b\lambda - \epsilon$ and nonnegative multiples of $b, b\lambda$: remove one occupied original generator. Hence these four positive numbers generate a monoid containing the full shifted support. The argument using the four positive generators applies to any finite union of the retained action lists. Ordered inverse convolution is completed before the single residual shift, so it still has the original additive support. Translation and the shift retain the strictly positive gap and the rescaled degree linkage of the original residual map.

A finite word adds only finitely many such positive generators and fixed positive weak rates. Their minimum is positive; an action bound T bounds every multiplicity by T divided by that minimum. The resulting action set is locally finite, also for irrational λ .

The supplied own-parent charts and whole differentiated numerator maps are retained at each of these classes. Thus (11.57) applies to the full original pre-native input with constants independent of its coefficients and of $T \geq 1$. The finite coefficient maps in (11.62) likewise precede every input and retain all later factors in their coefficients. These are the whole estimates used in the reconstruction and cut steps; no truncation of the action support is substituted.

A positive regular change is its whole analytic operation. Their current data satisfy Lemma 11.4: the initial data are $(1, 1, 0)$, each regular unit has $A(0) = 1$ and positive derivative, and each prepared exponent and leading constant is positive. Its inverse is the correlated original J . Thus (11.5) gives every fixed-label update, and (11.6) gives the recorded-parent update. These identities preserve $D, \kappa > 0$; the cited receiving constructors preserve the generated positive-parent certificate.

For a canonical letter at a nonbase birth, apply Proposition 12.16. All its equation and common-reference hypotheses were fixed above. It supplies the actual section adapters, every original recut and the full forward and inverse endpoint recipes. The direct operation adds a genuine new stem even after a prepared passage with irrational exponent. The reverse operation uses its whole inverse equation and returns to the recorded parent with exponent $\eta_b m_b / r$. It retains the departed birth. The resulting entire output is again a permitted inlet to the prepared construction. This proves the second and third hypotheses, including their physical branches, independent kernels and complete remainders.

For completeness, the small readouts in the fourth hypothesis use the positive receiving certificates, not merely the scalar limits of the currents. At any retained nonbase current write its exact source as

$$S = e^{-\kappa u_v} [Dd_v^\kappa \exp(\kappa \log W_v - G)].$$

The bracket is its whole bounded coefficient source on the recorded strict domain. The mark $e^{-\kappa u_v}$ has positive reserve κ . Receive it with that entire coefficient and choose the retained strong cut e_a of the stem that created v . Its native reference is $M = K_v$. This cut precedes v 's weak class and its synchronous raw mask erases the mark, hence $e_a S = 0$. A current retained after its parent becomes inactive retains the cut e_a and the displayed factorization of S . Fix its original raw norm N_S and all finite subsequent inclusion costs. Lemma 11.24, at this retained level- a reference M with exponent zero, gives a receiving map with

$$\text{Act } W_{M,0} S = \text{Act } S, \quad \|W_{M,0} S\| \leq \varepsilon N_S.$$

The independent target kernel identifies $W_{M,0} S$ with the included current. For a prescribed radius ρ and subsequent inclusion bound L , choose $\varepsilon N_S L < \rho$. The finitely many prefix readouts use the joint receiving choice of Lemma 11.24; their costs are fixed before these finitely many choices of ε . Thus every current lies strictly inside its prescribed radius in its original source norm. Lemma 12.6 gives the whole joint coefficient map and its geometric total-degree tail. The whole inclusions of the three propositions and the mixed-source theorem preserve actual values, cuts and original family domains; their original quotient norms are those in Lemma 12.3. The separate marked targets retain the original coefficient and derivative factors. This establishes the fourth hypothesis.

Step 3. We establish finite depth and complete comparison. For any prefix, the number of canonical reverses minus the number of canonical directs is at most N . Consequently $n \geq N + 3$ supplies the depth reserve required by Proposition 12.17. Its induction now applies to the actual constructors just supplied. It identifies every current with the literal prefix on X_n and retains all inactive inputs. The finitely many original-norm inclusions place all prefix presentations in the final entire presentation.

If their successive operator bounds, including the initial handoff and any receiving refinements, are $C_0, \dots, C_j$, then Lemma 12.7 gives

$$E_{j,\rho} = I_j \cdots I_0 E_{*,\rho}, \quad L_{j,\rho} = A_\rho \prod_{k=0}^j C_k, \quad \theta_{j,\rho} = \eta_\rho.$$

This proves (12.65) with the whole original input norm and the inclusion costs paid once. The constants precede the input and its truncation degree.

Finally the assembly proposition compares the entire final displacement, after all cancellations, with its independent records. Equality of the analytic and comparison kernels gives the zero alternative; its complete-head conclusion gives the other alternative with real nonzero coefficient and positive scale. Lemma 12.1 supplies the positive cofinal inverse of X_n . Composing the whole identity or the complete remainder with that inverse gives the claimed conclusion in the original section coordinate. This is substitution in a displacement, and therefore introduces no derivative factor. $\square$

13. CLOSED FORMAL COMPARISON OF THE PHYSICAL RETURN

The formal displacement must compare the final physical current with the original input in one source space. Separate records for these two germs do not perform that comparison. We first describe the closed quotient attached to the independent collected rows. We then construct a joint readout, keeping its entire input series and its original norm, and explain how this construction controls displacement, changes of section and marked families.

Theorem 13.1. *For every itinerary of Definition 2.2, its literal physical construction admits complete original Banach source presentations $\mathcal{P}_a$ and bounded independent comparison maps $\mathcal{R}_a$. The kernels $K_a = \ker \mathcal{R}_a$ are closed; with their original quotient infimum norms, $A_a = \mathcal{P}_a/K_a$ are Banach comparison algebras. Actual realization factors faithfully through these quotients. Multiplication, strict analytic substitution, the literal direct and inverse operations, and the stated parameter and marked-family operations descend with their proved bounds.*

The original base coordinate and final current admit a single joint readout before quotienting. Its formal displacement $\hat{d}_\Gamma$ satisfies

$$(13.1) \quad \hat{d}_\Gamma = 0 \iff \Delta_\Gamma = \text{id}.$$

If $\hat{d}_\Gamma \neq 0$, then on the original positive section

$$(13.2) \quad \Delta_\Gamma(x) - x = a\Theta(x)(1 + \epsilon(x)), \quad a \in \mathbb{R} \setminus \{0\}, \quad \Theta(x) > 0, \quad \epsilon(x) \rightarrow 0,$$

with the entire post-cancellation remainder. The construction retains divergent normalizers and general compatible non-real matching data. For an analytic positive change of section coordinate J , the transformed displacement is given by the full divided difference of J ; the zero test and positive leading-term conclusion persist. Suppose two depth realizations are supplied by bounded certificate maps from one complete original recipe presentation with one fixed original norm. Assume their actual values, for every original input, are pullbacks of one common original germ by positive cofinal analytic reparametrizations with positive tail inverses, and their independent comparison kernels equal their analytic kernels. Then their pulled comparison kernels on that original presentation coincide, and its identity induces an isometry between the two comparison quotients in the original infimum norm.

The quantitative input to the construction. The original-input readout in Theorem 12.18 supplies a geometric bound for all powers of the retained base coordinate. The positive-factor receiving operation in Theorem 11.2 supplies a small source for the reached current after its complete coefficient costs have been fixed. We use these two estimates to construct one whole readout in both variables.

Definition 13.2. Let $\mathcal{P}$ be an original unital commutative Banach source algebra, with its specified norm, and let Act be its algebra map to analytic germs on the specified connected physical domain. Let $(r_i)_{i \in I}$ be its independently constructed bounded scalar comparison rows. Choose bounds $|r_i(p)| \leq c_i \|p\|_{\mathcal{P}}$ before choosing p . Its *collected comparison map* and *original comparison quotient* are

$$(13.3) \quad \mathcal{R}p = \left(\frac{r_i(p)}{1 + c_i} \right)_{i \in I} \in \ell^\infty(I), \quad K = \ker \mathcal{R}, \quad \mathcal{A} = \mathcal{P}/K, \quad \|[p]\|_{\mathcal{A}} = \inf_{k \in K} \|p + k\|_{\mathcal{P}}.$$

The rows in this definition are the joint collected rows of the entire presentation. A native coefficient projection or a prepared carrier projection alone is not substituted for them.

The raw spaces here are the chronological completions constructed in Theorems 11.2 and 12.18. Their inputs retain their original coefficient norms, independent variable radii and occurrence labels. Definition 13.2 specifies a quotient of those spaces; it does not construct new raw spaces by prescribing a desired zero kernel. The normalization of the rows only supplies a common Banach codomain for $\mathcal{R}$ and does not change their common kernel.

Lemma 13.3. Suppose that the data of Definition 13.2 satisfy

$$(13.4) \quad \bigcap_{i \in I} \ker r_i = \ker \text{Act}.$$

Then K is a closed ideal, $\mathcal{A}$ is a Banach algebra in the norm (13.3), and Act induces an injective algebra map on $\mathcal{A}$. If $T : \mathcal{P}_a \rightarrow \mathcal{P}_b$ is a bounded linear source map whose actual square sends actual zero to actual zero, and both endpoints satisfy (13.4), then T descends and its quotient norm is at most $\|T\|$.

For a multilinear or analytic source operation with a prescribed strict inlet, equality of formal inputs implies equality of formal outputs whenever its actual operation has that property on the prescribed domains and branches. Its established source estimates remain estimates on that inlet; this assertion does not extend the operation to additional inputs.

Proof. Each r_i is continuous, so their kernel intersection is closed. Moreover, $p \in K$ and $q \in \mathcal{P}$ give $\text{Act}(pq) = \text{Act}(p) \text{Act}(q) = 0$. Thus (13.4) gives $pq \in K$.

Lemma 4.7, applied to this closed ideal, gives completeness and the submultiplicative original quotient norm. The kernel equality (13.4) proves injectivity of the induced actual map.

If $p \in K_a$, the actual square gives $\text{Act}_b(Tp) = 0$, and the target kernel equality gives $Tp \in K_b$. The quotient estimate of Lemma 4.7 therefore gives

$$(13.5) \quad \|[Tp]\|_{\mathcal{A}_b} \leq \|T\| \|[p]\|_{\mathcal{A}_a}.$$

For the remaining operations, formal equality first implies equality of actual inputs. The specified actual operation gives equal outputs on its fixed domain and branch, and the target kernel equality gives formal equality. Original convergent source series can be passed through every row by continuity. For a multilinear bound one takes infima over the finitely many lifts; for a strict analytic bound one uses lifts within the available strict margins. No linear selection of representatives is required. $\square$

Set-up 13.4. Fix a literal physical word of length N , an integer $n \geq N + 3$, and a finite list of new positive base-readout radii before constructing the chronological presentations of Theorem 12.18. At a reached prefix write $X = X_n$ for the retained original input and s for its current. Use the retained non-base vertex v and the full current sources of Theorem 11.2, with notation

$$(13.6) \quad s = Dx_v^\kappa e^{-G}, \quad x_v = d_v e^{-u} W_v, \quad D, d_v, \kappa > 0.$$

The whole unit W_v , its specified logarithm, G , their original coefficient sources, the prepared sources and their ordered inverse sources are retained. All rates in this display are fixed when

family estimates are asserted. Let Q be its original comparison quotient and $B_{\text{old}} : \mathcal{P}_{\text{old}} \rightarrow Q$ its bounded unital algebra source map, with bound C_{old} . For one prescribed radius ρ_X , denote the readout supplied by (12.65) by $E_X : \mathcal{W}_{\rho_X}(\mathbb{C}) \rightarrow Q$, and its constants by $L_X < \infty$ and $0 < \eta_X < 1$. The receiving algebras and constants may depend on the prescribed radii. Every whole inclusion retains the readout with its bounded inclusion cost, while every original input norm and radius remains fixed.

For the positive-factor receiving operation, use the fixed bank

$$(13.7) \quad U_*(T) = \sum_{m \geq 1} \frac{e^{-mT}}{m}.$$

Write $L_s = -\log s = \kappa u + c + h$, where $c = -\log D - \kappa \log d_v$ and $h = G - \kappa \log W_v$. Denote the complete residual source of (11.3) by

$$(13.8) \quad B = U_*(L_s) = e^{-\varepsilon u} B_\varepsilon, \quad \varepsilon = \kappa/2,$$

All original coefficient sources and strict insertion margins belong to this retained source. In particular, the small receiving norm will be obtained by the stated operation in Theorem 11.2, after its full coefficient and inclusion costs.

Lemma 13.5. *In Setup 13.4, for every new $\rho_s > 0$ there exists a finite receiving extension Q^+ , with a bounded injective algebra inclusion $I : Q \rightarrow Q^+$ preserving actual values, that contains a whole element $\mathbf{S}$ of actual value s such that*

$$(13.9) \quad \|\mathbf{S}\|_{Q^+} < \rho_s/2.$$

The construction preserves the current, its retained vertex and the entire old presentation, and its constants are chosen before a readout input.

Proof. We first exhibit a genuine prepared pair. We then use its complete positive residual to obtain the norm estimate.

Step 1. We construct the prepared passage and its actual inverse. On small positive sections take

$$(13.10) \quad J_*(t) = \frac{t}{1-t}, \quad J_*^{-1}(w) = \frac{w}{1+w}.$$

These maps are realized by the polynomial saddle $\dot{x} = x$, $\dot{y} = -y$: between $y = a$, $x = at$ and $x = a$, $y = at$, the first integral xy makes the passage the identity in t . Use J_* as the outgoing analytic coordinate. Its prepared correction is precisely (13.7), since $J_*(t) = t \exp U_*(-\log t)$.

This bank has degree zero and the full measure $\sum_{m \geq 1} m^{-1} \delta_m$, with positive carrier gap one. At every positive weight A its norm is $N_A = \sum_{m \geq 1} e^{-Am}/m < \infty$. In Lemma 5.13 take $\nu_0 = \sum_{m \geq 1} m^{-1} \delta_m$ and $\nu_K = 0$ for $K > 0$. Its hypotheses (5.47) hold with gap and action-degree constant both equal to one. At logarithmic radius one and $\lambda = 1$, the inverse constants in (5.29) are $c_1 = 0$, $C_1 = 2$ and $q > 2N_A$. Thus its entire ordered inverse has a finite majorant at weight $A + q$, and its full part with more than M forward leaves is bounded by

$$(13.11) \quad N_A \frac{\vartheta^M}{(M+1)(1-\vartheta)}, \quad \vartheta = \frac{2N_A}{q} < 1.$$

Every fixed carrier moment is finite after further positive weight slack. The inverse is therefore the ordered forward-leaf inverse of Lemma 5.13, which applies to degree zero. Its actual correction is

$$(13.12) \quad R_*(w) = \log(1 + e^{-w}), \quad (\text{id} - U_*)(w + R_*(w)) = w.$$

For $z = e^{-w}$, $\exp[-w - R_*(w)] = z/(1+z)$, which verifies the second identity on the chosen branch. Uniqueness of the actual prepared inverse identifies this formula with the actualization of the original inverse source. We keep that source and all its ordered leaves.

Applying J_* at s supplies $B = -\log(1-s)$ and changes G to $G-B$. Applying its same-source inverse supplies $R_*(-\log J_*(s)) = \log(1+J_*(s)) = B$ and restores G and s . Both operations preserve the retained vertex. Their whole inclusions and auxiliary source occurrences remain in the receiver.

Step 2. We identify the positive residual and the new mark. The complete translated source is (11.3). Its shifted carrier satisfies $\kappa m - \varepsilon \geq \kappa m/2 > 0$. The shift is taken once from the total carrier; all original inverse leaves and their moments remain those of the ordered inverse construction. Theorem 11.2 bounds the whole residual and its coefficient sources before any new receiving mark is chosen.

Choose $\delta > 0$ with $e^\delta - 1 < \rho_s/2$. Apply the positive-factor receiving operation of Theorem 11.2 after fixing the residual bound and the full costs of the forward and inverse inclusions. It gives $I : Q \rightarrow Q^+$ with $\|B\|_{Q^+} < \delta$, preserving the entire old presentation, its original input norms and radii, its independent exact kernel, complete heads and separate marked targets. The resulting receiver remains valid for later original letters. In that algebra set

$$(13.13) \quad \mathbf{S} = 1 - e^{-B}.$$

The entire exponential series gives $\|\mathbf{S}\| \leq e^{\|B\|} - 1 < \rho_s/2$ and $\text{Act } \mathbf{S} = s$. These choices involve the original current and the new radius only, and precede every coefficient of a later readout input. $\square$

Lemma 13.6. *In Setup 13.4, take one prescribed retained radius ρ_X and fix a new $\rho_s > 0$. There exists a bounded algebra readout $\mathcal{C}$ on the entire original coefficient Wiener algebra*

$$(13.14) \quad F(U, Z) = \sum_{m,j \geq 0} p_{mj} U^m Z^j, \quad \|F\|_{\rho_X, \rho_s} = \sum_{m,j \geq 0} \|p_{mj}\|_{\mathcal{P}_{\text{old}}} \rho_X^m \rho_s^j < \infty,$$

whose actual value is $\sum_{m,j} (\text{Act } p_{mj}) X^m s^j$. There exist constants $C < \infty$ and $0 < \theta < 1$, fixed before F , such that

$$(13.15) \quad \|\mathcal{C}(F)\|_{Q^+} \leq C \|F\|_{\rho_X, \rho_s}, \quad \|\mathcal{C}(F_{\geq M})\|_{Q^+} \leq C \theta^M \|F\|_{\rho_X, \rho_s}.$$

Here $F_{\geq M}$ includes every term with $m+j \geq M$. Pulling back the independent rows of Q^+ gives independent bounded rows on this entire input, with common kernel equal to its actual kernel. Its comparison quotient is complete in the infimum of (13.14). Each nonzero real output has the complete head of the receiving construction.

Proof. Let $\mathbf{X} = E_X(U) \in Q$, and take $\mathbf{S}$ from Lemma 13.5. Applying (12.65) to U^m gives

$$(13.16) \quad \|\mathbf{X}^m\|_Q \leq L_X \eta_X^m \rho_X^m.$$

The readout is an algebra map. Equivalently, its source for U^m and the product $\mathbf{X}^m$ have equal actual values and hence coincide in the faithful quotient. Estimate (13.16) also includes $m = 0$.

Let L_I bound the whole inclusion $I : Q \rightarrow Q^+$. Define

$$(13.17) \quad q_j = \sum_{m \geq 0} B_{\text{old}}(p_{mj}) \mathbf{X}^m \in Q, \quad \mathcal{C}(F) = \sum_{j \geq 0} I(q_j) \mathbf{S}^j \in Q^+.$$

Every summand is bounded by

$$(13.18) \quad L_I C_{\text{old}} L_X \|p_{mj}\|_{\mathcal{P}_{\text{old}}} \rho_X^m \rho_s^j \eta_X^m 2^{-j}.$$

Thus both sums converge absolutely. The inclusion cost L_I is paid once, outside the whole power $\mathbf{X}^m$, rather than once per factor. Take $C = L_I C_{\text{old}} L_X$ and $\theta = \max(\eta_X, 1/2)$. For $m+j \geq M$ the last two factors in (13.18) are at most θ^M . Summation proves both estimates in (13.15). Multiplication follows first for polynomials and then for the entire Wiener algebra by absolute convergence. Bounded actualization gives the claimed actual square.

For every independently constructed target row r , take the bounded row $r\mathcal{C}$ on the input. These rows are fixed before F . The faithful target gives

$$(13.19) \quad (r\mathcal{C}(F) = 0 \text{ for every } r) \iff \mathcal{C}(F) = 0 \iff \text{Act } \mathcal{C}(F) = 0.$$

Lemma 13.3 proves the quotient assertion. It concerns the original Wiener input norm; it does not assert that the image of $\mathcal{C}$ is closed in Q^+ .

If the coefficients are originally specified in the old quotient, lift them with weighted norm errors whose total is less than any prescribed $\alpha > 0$. Applying (13.18) and then letting α tend to zero gives the quotient coefficient estimate. Old kernel terms vanish by their actual square. This argument needs no bounded simultaneous selection of lifts. Finally, the complete-head theorem retained in Q^+ applies to the whole element $\mathcal{C}(F)$, including all coefficient tails and all cancellations between degrees. $\square$

Literal source operations. At each original occurrence we use its entire forward source and its specified genuine inverse. Theorems 11.2 and 12.18 supply their actual squares and their independent collected kernels. Lemma 13.3 then gives the corresponding operations on the original quotients. A repeated equation or parameter remains a shared input of the recipe; distinct occurrence indices do not create independent replacement data.

For prepared passages, the inverse is the complete ordered convolution inverse in Theorem 5.1. Products of source distributions, total-carrier cutoffs and the complete complement maps remain at their original source nodes. An equality of scalar quotient classes does not identify two raw distributional windows. For a canonical passage, both the actual fiber inverse and the inverse height branch remain part of its literal operation. In particular, taking a reciprocal of a normalizing factor is not a construction of that inverse passage.

The following local identity explains how the non-real matching data are retained by this descent.

Lemma 13.7. *Let h_0, h_σ be normalized invertible analytic fiber germs, let k_0, k_σ be their genuine inverses, and put $C_\sigma = h_\sigma \circ k_0$. Suppose $C_\sigma(0) = 0$ and $C'_\sigma(0) = 1$. Define the analytic germs*

$$(13.20) \quad H_\sigma(w) = \frac{C_\sigma(w)}{w}, \quad H_\sigma^-(w) = \frac{C_\sigma^{-1}(w)}{w}.$$

Let $\Phi_0(x, \cdot)$ and $\Psi_0(x, \cdot)$ be mutually inverse analytic fiber maps on specified strict image disks. For a nonzero constant g_0 and an independent weak variable V , the full replay and its fiber inverse are

$$(13.21) \quad \Phi_\sigma^{[0]}(x, z) = \Phi_0(x, z)H_\sigma(V\Phi_0(x, z)/g_0),$$

$$(13.22) \quad \Psi_\sigma^{[0]}(x, w) = \Psi_0(x, wH_\sigma^-(Vw/g_0)).$$

These identities extend analytically to $V = 0$. If Φ_0 is a whole source of norm at most M and $H_\sigma(w) = \sum_{j \geq 0} h_j w^j$, their forward replay has the full majorant

$$(13.23) \quad M \sum_{j \geq 0} |h_j| \left(\frac{r_V M}{|g_0|} \right)^j$$

whenever the displayed sum converges within the original strict disks. The inverse assertion holds with the corresponding H_σ^- majorant and the prescribed Ψ_0 image margin.

Proof. For $\alpha = V/g_0 \neq 0$, the forward fiber map is $z \mapsto \alpha^{-1}C_\sigma(\alpha\Phi_0(x, z))$. Solving this equality for z gives $\Psi_0(x, \alpha^{-1}C_\sigma^{-1}(\alpha w))$, which is (13.22). Both displayed expressions have convergent Taylor series in V with nonnegative degrees, so the identity extends to $V = 0$. The Banach algebra product estimate, followed by the absolute sum of all weak degrees, gives (13.23). The inverse expression uses its own strict image disk before this summation.

For an endpoint graph set $G_0(x) = \Phi_0(x, g(x))/g_0$. Before any weak diagonal is imposed, the full identities are

$$(13.24) \quad G_\sigma^{[0]} = G_0 H_\sigma(VG_0), \quad k_\sigma(VG_\sigma^{[0]}) = k_0(VG_0).$$

They follow by applying k_σ to $VG_\sigma^{[0]} = C_\sigma(VG_0)$. For a third label τ , the exact cocycle identity is $h_\tau \circ k_0 = (h_\tau \circ k_\sigma) \circ (h_\sigma \circ k_0)$. Thus both the inverse and the ordered cocycle persist as whole identities. $\square$

Lemma 13.7 is applied to the actual compatible equation packets retained by Theorem 12.18; it is an identity for those packets, not a bounded change between arbitrary free banks on different rays. In particular H_σ^- is the factor of a compositional inverse, not $1/H_\sigma$. Every positive weak degree in (13.21) remains. At $V = 0$ the reference forward and inverse pair and graph unit are recovered, not an identification of the original endpoint gauge with the reference gauge. The original endpoint germs remain in (13.24). The source-equipped divergent reference normalizers remain whole root sources; other compatible analytic normalizers retain their entire weak recipes, not fictitious native constant terms. The original-norm forward and inverse bounds, including the larger weak-radius inverse margin and coefficient Banach inverse norm, are those of Lemma 6.16. Reality will be used only for the fully matched physical displacement.

Definition 13.8. For a literal closed word with final current $s = \Delta_\Gamma(X)$, use the original input (13.14) and the pulled independent rows of Lemma 13.6. Its *formal displacement* is the class

$$(13.25) \quad \hat{d}_\Gamma = [Z - U] \quad \text{in} \quad \mathcal{W}_{(\rho_X, \rho_s)}(\mathcal{P}_{\text{old}}) / \bigcap_r \ker(r\mathcal{C}).$$

Its faithful image in the receiving algebra is $\mathcal{C}(Z - U) = \mathbf{S} - I(\mathbf{X})$.

Proposition 13.9. *For a literal closed physical word realized by Theorem 12.18, the formal displacement in Definition 13.8 is zero if and only if $\Delta_\Gamma = \text{id}$ on the original positive tail. If it is nonzero, that original displacement has a positive leading term with its entire post-cancellation remainder.*

Proof. By the actual square of the joint readout,

$$(13.26) \quad \text{Act } \mathcal{C}(Z - U)(\sigma) = \Delta_\Gamma(X(\sigma)) - X(\sigma).$$

Both occurrences of X are the original input of the literal recipe. The equivalence (13.19) equates formal zero with complex actual zero. The positive cofinal map X has a positive inverse from a full positive x -tail to a full positive σ -tail. Hence actual zero in (13.26) implies $\Delta_\Gamma(x) = x$ on that x -tail. Conversely an identity there gives zero on an interval inside the connected complex realization domain; the analytic identity theorem gives complex zero, and (13.19) gives formal zero.

For a nonzero class, apply the receiver's complete-head conclusion to the whole element $\mathcal{C}(Z - U)$. On the positive model section it gives

$$(13.27) \quad \Delta_\Gamma(X(\sigma)) - X(\sigma) = a\Theta_0(\sigma)(1 + e_0(\sigma)), \quad \Theta_0 > 0, \quad e_0 \rightarrow 0.$$

The left side is real because the entire physical word is real on its chosen positive sections. Dividing by Θ_0 and taking the limit shows $a \in \mathbb{R} \setminus \{0\}$. The exact equality then makes e_0 real as well. No corresponding reality assertion for an individual sectorial normalizer was used.

Set $\Theta(x) = \Theta_0(X^{-1}(x))$ and $\epsilon(x) = e_0(X^{-1}(x))$. The cofinal inverse preserves positivity and the limit of the entire error. Substitution into (13.27) proves the asserted formula for $\Delta_\Gamma(x) - x$. This is composition of a scalar equality in its input, so it introduces no derivative factor. The complete-head estimate was applied after summing both complete sources and after every cancellation; its error therefore contains the entire displacement. $\square$

Lemma 13.10. *Let $\mathcal{P}$ be one complete original recipe presentation, with one fixed original norm and an actual germ $a(p)(x)$ for each $p \in \mathcal{P}$. Suppose two bounded certificate maps $B_i : \mathcal{P} \rightarrow Q_i$, $i = 1, 2$, have actual values $a(p)(X_i)$, where X_i are positive cofinal analytic reparametrizations with positive tail inverses. Suppose each Q_i has independent bounded rows whose common kernel is its analytic kernel. Then the two pulled row kernels on $\mathcal{P}$ are equal. The map induced by the identity on $\mathcal{P}$ identifies their comparison quotients isometrically in their original infimum norms.*

Proof. For each i and each fixed p , vanishing of the pulled rows is equivalent to $a(p)(X_i) = 0$ as a complex germ. By the positive inverse and the analytic identity theorem on the connected realization domain, this is equivalent to $a(p)(x) = 0$ on the original positive tail. Denote that common subspace of $\mathcal{P}$ by K . Both quotient norms are $\inf_{k \in K} \|p + k\|_{\mathcal{P}}$. $\square$

This statement presupposes the common original recipe presentation. It does not construct an identification between unrelated completed sources. In particular it requires neither equality of the target row sets nor a bounded inverse of either certificate map. For two complete heads $a_i \Theta_i(1 + e_i)$ of one nonzero physical displacement, their exact equality gives $\Theta_1/\Theta_2 \rightarrow a_2/a_1 > 0$.

Lemma 13.11. *Let Γ be a physical itinerary and let J be real analytic near zero, with $J(0) = 0$, $J'(0) > 0$, and its genuine local inverse. On sufficiently small new readout disks, the whole analytic divided difference*

$$(13.28) \quad B_J(U, Z) = \sum_{k \geq 1} j_k \sum_{i=0}^{k-1} Z^{k-1-i} U^i, \quad J(z) = \sum_{k \geq 1} j_k z^k,$$

and its reciprocal belong to the scalar Wiener algebra. For the joint readout of Lemma 13.6 constructed with these newly prescribed radii, the changed displacement is the formal product of the original displacement with the unit represented by B_J . Formal zero is preserved and reflected.

If the original complete head is $a\Theta(1 + \epsilon)$ in the original coordinate, the changed return $J \circ \Delta_\Gamma \circ J^{-1}$ has head coefficient $aJ'(0)$, positive scale $\Theta \circ J^{-1}$, and the entire error $(\epsilon + u + \epsilon u) \circ J^{-1}$, where $u(x) = B_J(x, \Delta_\Gamma(x))/J'(0) - 1 \rightarrow 0$.

Proof. If J is analytic on a disk of radius R , every $0 < r < R$ gives

$$(13.29) \quad \|B_J\|_{r,r} \leq \sum_{k \geq 1} k |j_k| r^{k-1} < \infty.$$

Since $B_J(0, 0) = j_1 > 0$, choose r smaller so that $\|B_J/j_1 - 1\|_{r,r} < 1/2$. The reciprocal is then the entire geometric series $j_1^{-1} \sum_{\ell \geq 0} (-1)^\ell (B_J/j_1 - 1)^\ell$ in this Wiener algebra. Polynomial factorization at each degree and (13.29) give the full identity

$$(13.30) \quad J(Z) - J(U) = (Z - U)B_J(U, Z).$$

Applying the joint algebra readout also realizes the reciprocal, so the second factor is a unit. The zero equivalence follows.

The required disks are new readout disks. Theorem 12.18 permits prescribing them at the initial receiver and carries their whole original-input readouts through the finite word. At the final current, Lemma 13.5 supplies the new Z -disk. Earlier input radii remain fixed. A comparison with a realization at another depth requires two bounded certificate maps supplied on one complete original presentation with one fixed original norm. When both maps realize each original input as the pullback of its prescribed original germ by a positive cofinal reparametrization with a positive tail inverse, and each target's independent comparison kernel equals its analytic kernel, Lemma 13.10 identifies their pulled comparison quotients isometrically. The new readout construction alone does not supply such a common presentation.

In actual coordinates $x' = J(x)$, (13.30) says

$$(13.31) \quad (J \circ \Delta_\Gamma \circ J^{-1})(x') - x' = (\Delta_\Gamma(x) - x)B_J(x, \Delta_\Gamma(x)).$$

Both arguments of B_J tend to zero, so $u(x) \rightarrow 0$. Multiplication of the two exact factors $1 + \epsilon$ and $1 + u$ gives the full error $\epsilon + u + \epsilon u$. Pullback by J^{-1} proves the assertion in the changed section. $\square$

Whole families and marked coefficients. Independent exterior Wiener parameters can be summed outside (13.18). The constant in that estimate does not depend on their indices. Proposition 4.20 then gives the corresponding full degree tails at strict parameter ratios. For physical parameters one must instead retain their actual buffered domains and their prescribed local scaling. The following bound records the recovery factors explicitly.

Lemma 13.12. *Let E be a Banach space and let $F(\lambda)$ be an E -valued analytic family in h parameters. On a fixed buffered driver disk, suppose a constant $r_* > 0$ gives a holomorphic family $F(r_*z)$ on a neighborhood of $|z_i| \leq 2$, with norm at most M . Write $F(r_*z) = \sum_{\mu \in \mathbb{N}^h} a_\mu z^\mu$. Then*

$$(13.32) \quad \|a_\mu\|_E \leq M 2^{-|\mu|}, \quad \sum_{|\mu| \geq N} \|a_\mu\|_E \leq 3^h M (3/4)^N,$$

and the exact original parameter recovery is

$$(13.33) \quad [\lambda^\mu]F = r_*^{-|\mu|} a_\mu, \quad \partial_\lambda^\mu F(0) = \mu! r_*^{-|\mu|} a_\mu.$$

Suppose additionally that the scalar actualization $f(t, \lambda)$ is holomorphic on a connected domain $\Omega \subset D \times \mathbb{C}^h$, where D is a connected driver domain and $D \times \{0\} \subset \Omega$. For each fixed μ , suppose its bounded coefficient map to a faithful marked target realizes $[\lambda^\mu]f$ on a nonempty open part of D . Then vanishing of every complete marked record is equivalent to $f = 0$ on Ω .

Proof. Lemma 4.23, applied to $F(r_*z)$ with radius two in every variable, gives the first estimate in (13.32). For $|\mu| \geq N$, $2^{-|\mu|} \leq (3/4)^N (2/3)^{|\mu|}$, and summing the product geometric series gives $\sum_{\mu \in \mathbb{N}^h} (2/3)^{|\mu|} = 3^h$. Comparison of coefficients under $\lambda = r_*z$ proves (13.33).

For a fixed μ , faithfulness of its marked target equates record zero with actual zero of $[\lambda^\mu]f$ on a nonempty open subset of D . This coefficient is holomorphic on D , so the analytic identity theorem extends its zero to D . Applying this argument to each μ makes every Taylor coefficient vanish. At each point of $D \times \{0\}$ the convergent parameter Taylor series is therefore zero on a neighborhood. Analytic continuation gives $f = 0$ on Ω . Conversely, $f = 0$ makes each coefficient zero, and the exact kernel of its marked target makes its record zero. This reasoning does not choose one tail by intersecting countably many coefficient tails, nor sum the norms of distinct marked targets. $\square$

In this application M is the proved whole source bound on the buffered chart. For each fixed μ , the receiving target retains its Cauchy, derivative, inverse and source recovery costs in addition to (13.33). The chart radius is constant on its driver disk; it is not a driver-dependent holomorphic substitution. The domains are the connected physical components of the original-domain pullbacks with the new strict source and inverse margins. The construction does not place all fixed-index marked targets in one Banach sum or assert a uniform nonzero head through a parameter specialization where the displacement vanishes.

Proof of Theorem 13.1. We construct the quotients before testing any displacement, descend the literal operations, and then use the common input/current readout.

Step 1. We construct independent quotients and literal operations. For the given itinerary Γ , Theorem 7.5 supplies its literal physical word of length N . Choose $n \geq N + 3$ and prescribe the finite list of new readout radii required for the displacement and any specified analytic section change. Apply Theorem 12.18 to this word and these radii, with all original physical branches, shared parameters and source norms retained. For each actual chronological raw algebra, use its independently constructed collected rows in Definition 13.2. The exact-kernel conclusions of Theorems 11.2 and 12.18, followed by Lemma 13.3, give closed ideals, complete original

quotient norms and faithful actualization. The original source maps at each letter have their full actual squares, so their linear, multilinear and strict analytic operations descend by that lemma. Their actual inverse identities and cocycles descend with them. Lemma 13.7 retains all compatible weak matching coefficients; no convergent normalizer or individually real sectorial factor is required.

Step 2. We identify the original displacement and its coordinate changes. In the retained ancestry, a regular or prepared letter preserves the current vertex, a canonical direct letter adds one edge, and a canonical reverse returns to its recorded parent. Starting at depth n , the depth after j original letters is therefore at least $n - j \geq 3$. Every reached current has the non-base retained vertex used in Setup 13.4. The neutral pair in Lemma 13.5 preserves that vertex and the actual current. Its receiving extensions remain valid inputs for later letters by Theorem 11.2. The degree-zero inverse needed by this pair is provided by Lemma 5.13, with its hypotheses checked in Lemma 13.5. Lemma 13.6 therefore applies at the final current. Its rows are defined on the entire original input before quotienting and before using $Z - U$. Proposition 13.9 gives the exact zero test and complete physical head. Lemma 13.11 gives the full divided-difference transformation. For the final comparison assertion, take the two supplied bounded certificate maps on the single original presentation with its fixed norm. Their common original germs, positive cofinal reparametrizations with positive tail inverses, and independent exact analytic kernels are precisely the hypotheses of Lemma 13.10. That lemma gives equality of the pulled kernels and the claimed isometry in the original infimum norm.

Step 3. We retain whole parameter operations. The retained whole-family estimates and Proposition 4.20 apply before any coefficient extraction. Lemma 13.12 gives the exact local recovery, derivative factors and zero test for the separate marked targets. These assertions supply the stated parameter operations with their original bounds and complete the construction. $\square$

14. FINITENESS OF ISOLATED PERIODIC ORBITS

In this section, we combine the complete physical displacement of Theorem 13.1 with Proposition 3.20 to prove the finiteness theorem.

Proof of Theorem 1.1. We obtain the complete displacement on an auxiliary input, transport its zero or strict-sign alternative to the original section, and apply the global reduction. Fix a real planar polynomial vector field V . If $V = 0$, it has no nonconstant periodic integral curve. We may therefore assume that $V \neq 0$. To apply Proposition 3.20, fix an arbitrary itinerary Γ of Definition 2.2. Write x for its original positive base coordinate and Δ_Γ for its physical return. All equations, sections, paths, branches, and repeated occurrences are fixed with Γ .

Step 1. We obtain the complete original displacement. By Theorem 7.5, the return has an exact literal factorization $W = L_N \circ \cdots \circ L_1 = \Delta_\Gamma$ in this coordinate. Choose one integer $n \geq N + 3$ and use the construction of Theorem 12.18 with its input $X_n(\sigma)$. Every prefix, the original input X_n , and the final current $W(X_n)$ retain their complete sources. Apply Theorem 13.1 to this literal construction and its joint readout of the original input and final current. The resulting formal displacement $\hat{d}_\Gamma$ represents exactly

$$(14.1) \quad d_n(\sigma) := W(X_n(\sigma)) - X_n(\sigma) = \Delta_\Gamma(X_n(\sigma)) - X_n(\sigma).$$

The auxiliary input and the receiving extensions execute no additional physical passage. Thus this readout retains the original base section and every lap of Γ .

If $\hat{d}_\Gamma = 0$, Theorem 13.1 gives $\Delta_\Gamma = \text{id}$ on a positive interval in the original coordinate. Suppose that $\hat{d}_\Gamma \neq 0$. Faithful realization makes (14.1) nonzero. The complete comparison in Theorem 12.18 then gives

$$d_n(\sigma) = a_\Gamma \theta_n(\sigma)(1 + \eta_n(\sigma)), \quad a_\Gamma \in \mathbb{R} \setminus \{0\}, \quad \theta_n(\sigma) > 0, \quad \eta_n(\sigma) \rightarrow 0.$$

Here η_n is real and includes the entire remainder after all cancellations in the displacement. Reality concerns this physical displacement; the source construction of Theorem 13.1 includes divergent normalizers and compatible non-real matching data.

Step 2. We transport the displacement to the physical section. The positive cofinal inverse of X_n gives germs on the original section by

$$\Theta_\Gamma(x) = \theta_n(X_n^{-1}(x)), \quad \epsilon_\Gamma(x) = \eta_n(X_n^{-1}(x)).$$

Since $X_n^{-1}(x) \rightarrow 0$ as $x \rightarrow 0^+$, we have $\Theta_\Gamma(x) > 0$ and $\epsilon_\Gamma(x) \rightarrow 0$. Substitution into (14.1) yields the exact equality

$$(14.2) \quad \Delta_\Gamma(x) - x = a_\Gamma \Theta_\Gamma(x) (1 + \epsilon_\Gamma(x)).$$

This substitution changes only the input of the displacement, so it introduces no derivative factor.

For completeness, an analytic change $x' = J(x)$ of the base coordinate, with $J(0) = 0$ and $J'(0) > 0$, has the separate transport formula from Theorem 13.1. For $\Delta'_\Gamma = J \circ \Delta_\Gamma \circ J^{-1}$, it is

$$\Delta'_\Gamma(J(x)) - J(x) = (\Delta_\Gamma(x) - x) B_J(x, \Delta_\Gamma(x)), \quad B_J(0, 0) = J'(0) > 0,$$

where B_J is the full analytic divided difference of J . Writing $B_J(x, \Delta_\Gamma(x)) = J'(0)(1 + b(x))$, we have $b(x) \rightarrow 0$. The complete relative error becomes $\epsilon_\Gamma + b + \epsilon_\Gamma b$, which tends to zero, and the coefficient becomes $a_\Gamma J'(0)$. Thus the zero and strict-sign alternatives persist under the section changes retained in the construction.

Choose $\delta_\Gamma > 0$ so that $|\epsilon_\Gamma(x)| < 1/2$ for $0 < x < \delta_\Gamma$. By (14.2),

$$|\Delta_\Gamma(x) - x| \geq \frac{|a_\Gamma|}{2} \Theta_\Gamma(x) > 0, \quad 0 < x < \delta_\Gamma,$$

and the displacement has the sign of a_Γ throughout this interval. The choice of interval is made for each fixed itinerary and fixed physical data. We have therefore established the premise of Proposition 3.20 for every itinerary it requires.

Step 3. We derive the global finiteness conclusion. The geometric alternatives in that proposition use Lemma 3.6 for a regular periodic limit, including the two-circuit return $P \circ P$ when one-circuit holonomy reverses the transverse sides. For resolved limits, the periodic lifts avoid the entire exceptional divisor. Lemma 3.7 first excludes its transverse regular points from the limit. With the invariant-or-transverse divisor condition of Proposition 3.16, that proposition then supplies complete connecting orbits and their singular endpoints. The resulting itinerary in Proposition 3.17 is a physical first return covered by the dichotomy just proved. Its identity case makes the accumulating periodic leaves nonisolated, whereas its strict-sign case excludes their fixed points tending to zero. Applying Proposition 3.20 proves finiteness for V , including when its polynomial coefficients have a common factor. $\square$

APPENDIX A. DETAILS OF THE USE OF GENERATIVE AI

The results of this paper were obtained using Danus [Liu⁺26], an automated mathematical reasoning system built on Rethlas [Ju⁺26]. For transparency we record the experiment in detail: how the system was run (§A.1), the problem exactly as it was posed (§A.2), and the guidance the authors gave during the run (§A.3).

A.1. How the system was run. The retained research history includes work from predecessor projects. The migrated project record was created on 6 September 2026 at 04:39:55 UTC, initially without a research launch. The authorized continuing research session began on that host on 6 September 2026 at 09:34:27 UTC. It used `gpt-6-astra`, with the main session at `ultra`, four research workers at `max`, three at `xhigh`, and the research verifier at `max`. The earliest start of all inherited research is not established by these timestamps. Some predecessor records name a different model; they are not attributed retrospectively to this configuration.

The research system recorded acceptance of its terminal submission on 22 September 2026 at 11:46:41.852 UTC. The elapsed interval from the continuing session's creation was 16 days, 2 hours, 12 minutes, and 14.852 seconds, including operational pauses. This is a wall-clock interval for the recorded session, not a measure of uninterrupted work or of the duration of all inherited research. The seven research workers were stopped that day, and the manuscript stage did not restart them. No controlled comparison experiment is established by the retained records.

The manuscript was written on the host of the continuing research session using the configured Responses route, with `gpt-6-astra` and separate section-writing and reviewing sessions at `max`; the main session retained `ultra`. Twelve technical sections were assigned, while the introduction and notation were maintained by the main session. The source records remained read-only throughout writing. Mathematical corrections were made in the manuscript, including the two geometric points described below. Separate section and whole-manuscript reviews raised objections that were examined against the arguments and addressed before the subsequent writing stages. These automated checks are not certificates of mathematical truth, and the records do not establish a complete independent human verification of the manuscript.

The mathematical work was textual. File extraction, reference checks, hashes, and compilation were mechanical checks; no mathematical conclusion in the proof rests on a numerical, symbolic, solver, or proof-assistant computation.

A.2. The problem as posed. The following is the complete posed problem, including its historical migration-only paragraph. The dated authorizations recorded below subsequently lifted that initial pause and later authorized writing.

```
# Dulac individual finiteness by a complete Écalle route
This project consolidates the Écalle-route research from Host_v2's V3
`dulac`, [predecessor host 2]'s V2 `[predecessor project 2]`, and the
associated local Codex/Fable reconstruction. The separate `[other
project]` project pursues failure of degree-uniform finiteness; it is
not this project's theorem.
## Terminal objective
Give a complete, referee-checkable proof that every real planar
polynomial vector field has only finitely many isolated periodic orbits.
Complete the Écalle route: construct a precise formal/transseries and
analytic actualization framework for the complete physical return maps,
prove the required closure and faithfulness, and establish identity or
an actual first nonzero term with a controlled complete remainder.
After compactification and finite resolution, consider every fixed
one-sided monodromic elementary cyclic itinerary. Repeated occurrences
of a singularity or sector are allowed. Fix all physical sections,
orientations, root/log branches and lateral choices. For its actual
return germ Delta_Gamma, prove:
- If its complete formal displacement is zero, Delta_Gamma is the
identity. - Otherwise Delta_Gamma(x)-x = a Theta_Gamma_actual(x)(1+o(1))
on the physical positive side, where a is real and nonzero and
Theta_Gamma_actual is positive. Control the entire remainder after all
actual cancellations.
Prove the global accumulation-to-itinerary reduction and transport this
dichotomy back to the original return. Identity yields a period annulus,
not isolated cycles; eventual strict sign excludes accumulating fixed
points. The quantifier is over every relevant finite itinerary, not one
selected certified word.
Include divergent normalizers and general non-real Martinet-Ramis data,
with physical reality obtained by proved matching. Convergent,
superreal, fully hyperbolic, single-passage, finite Euler or same-parent
```

results are useful intermediate theorems. None substitutes for the target. No uniform bound in polynomial degree is required.

What may change in the route

Écalles's formalization/actualization architecture is binding; every particular 1992 auxiliary category or averaging device is not. An exact replacement, corrected domain, different proved averaging construction, or direct complete-source first-head theorem is allowed when its relation to the physical return is explicit. The stronger full transfinite asymptotic expansion need not be reconstructed if a fully proved first-term dichotomy suffices. A defect in an auxiliary lemma neither disproves Dulac finiteness nor ends this project.

Inherited proof discipline

Inside the Dulac analytic corridor—including resurgent asymptotics, summation, acceleration, composition, first-head transfer and quasianalyticity—literature is a roadmap, not a terminal proof black box. State and prove the exact inputs needed in the native graph. Outside that corridor, stable textbook analysis, ODE, topology and planar desingularization may be registered as explicit T1 facts with precise statements and references. Reuse existing valid registrations.

Attribution-only external_refs do not pay a proof dependency.

The migrated graph contains unchanged native facts with complete retained source dependencies. Its starting ledger records exact scopes and review limits. Historical revoked chains and newly held false interfaces remain archived; their old completion claims cannot be reused. These restrictions concern identified claims, not a permanent ban on proving a correct replacement. Local notes, migration audits, proof clarifications and proposed strategies are references/directives only, never inherited verified facts.

Mathematical research is text-only: do not run numerical, symbolic, solver, search or proof-assistant computations. Mechanical file, hash, dependency and deployment checks are separate. All main, worker, verifier and helper roles use gpt-6-astra at the deployment's configured reasoning tiers.

This migration creates a paused, unstarted project. It does not launch agents, services, backend probes or a paper workflow.

A.3. Guidance given during the run. The dated guidance retained with the research is recorded below. Operational resumptions preserved the original mathematical objective. The independent literature comments remained advice until their exact claims were examined; their delivery alone did not make them proof inputs.

- (1) On 6 September 2026 at 09:33 UTC, the operator authorized continuation of the transferred Écalles-route project after the host prerequisites, and then instructed the main session to read its contract and begin. The original objective and retained source boundaries were preserved. At 09:37 UTC, the operator required subsequent helpers to use separate recorded native homes and the seven workers to be allocated after the initial ledger review. The resulting allocation and helper-routing adoption were recorded without changing the mathematical strategy.
- (2) On 7 September at 02:38 UTC, the operator resumed the session begun on 6 September after an operational pause and required bounded recovery without duplicate launches or concurrent bursts. At 05:51 UTC, following a host task-cap incident, recovery was restricted to demonstrably failed or stalled actors after checking committed state. At 06:06 UTC, the operator reported released verifier requests and specified the four remaining stalled worker

loops. These were runtime instructions; they preserved the existing tasks, research scope, models, and text-only discipline.

- (3) On 12 September at 05:06 UTC, the operator again authorized the complete Écalle-route objective stated in §A.2, explicitly retaining divergent normalizers, general compatible nonreal data, all seven workers, and the existing source and revocation boundaries. The resumed research continued from its preserved checkpoint.
- (4) A literature comment delivered on 12 September at 08:18 UTC proposed a finite phase-prefix argument under a one-sided sign condition. The main session read it and retained its conditional scope in the ongoing analysis. It was not taken as a proof of general physical admission or as a restriction to one-sided phase data.
- (5) A comment delivered on 13 September at 04:02 UTC identified a scope problem in a stated convolution majorant in the 1992 source and proposed larger weighted norms with whole-family bounds. The main session read the report and commissioned a bounded source/dependency check. No automatic revocation or rejection of the finiteness theorem followed from the comment. The manuscript supplies its required complete analytic estimates directly.
- (6) A comment delivered on 13 September at 10:54 UTC qualified a probabilistic interpretation in a later original source. The main session read it and distinguished that issue from the independently proved algebraic recursion then under consideration. It did not start a new averaging campaign or treat the comment as a physical counterexample.
- (7) On 20 September at 02:48 UTC, the operator resumed the original session after another host pause, retaining the full target, workers, records, and control schedule. The recorded adoption included the seven native worker starts. This instruction did not yet authorize a manuscript.
- (8) A comment delivered on 20 September at 02:51 UTC proposed an optional convex real-Laplace first-term argument and distinguished it from a complex inverse-Borel contour. The subsequent recorded check at 03:03 UTC found that four old source identifiers mentioned in the dated advice belonged to revoked history. Their old acceptance was not reused; only separately established replacements could serve as current inputs.
- (9) On 20 September at 15:52 UTC, the operator initially requested truthful intake of migrated advisory reports, including historical items. The later explicit correction, recorded at 16:13 UTC, cancelled both the historical reading task and any delegated screening. The retained reports became a reference archive for specific needs, with genuine earlier reading receipts preserved. This correction remained in force; completion of an unread historical quota was not required.
- (10) On 22 September, the operator resumed the transferred Écalle-route project after the 21 September operational stop, preserving its theorem and source boundaries. The continuing literature adviser was to remain closed, and the cancelled historical reading task was not revived. The original seven-worker configuration was restored before the recorded research closure later that day.
- (11) The writing authorization recorded on 22 September at 13:01 UTC requested a complete checked proof manuscript on the host of the continuing research session with existing credentials, while keeping the fact graph read-only and the research workers stopped. It required two explicit manuscript repairs: the two-circuit return in a Möbius neighborhood, and exclusion of transverse regular divisor limits before proving completeness and endpoints of lifted edges. The manuscript incorporates both repairs and their actual applications, as described in Lemmas 3.6 and 3.7 and Proposition 3.16. The paper reviews checked these arguments without backfilling the research graph.

- (12) The operator's relayed source notice on 22 September, labelled approximately 15:10 UTC, supplied Roussarie's published chapter and images of printed pages 349–351. The actual pages were examined, including Proposition 1.1 and its proof. A printed index slip in Definition 1.4 was recorded, and the published-page locator check was closed. This source-access clarification did not by itself establish the correctness of the manuscript's geometric application.

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